

Studies in Linguistics and Philosophy 100

Louise McNally
Zoltán Gendler Szabó *Editors*

A Reader's Guide to Classic Papers in Formal Semantics

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Volume 100

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Editors

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Preface

Formal semantics began 50 years ago with the famous pronouncement by Richard Montague that there is “no important theoretical difference between natural languages and the artificial languages of logicians.”¹ It is hard to appreciate the boldness of this claim today, but in 1970 it flew in the face of widespread and well-entrenched opinion. Those who formulated the artificial languages that Montague adverts to – Boole, Frege, Russell, Carnap, and Tarski, among others – undertook the effort precisely because they thought natural languages suffer from deep defects which make them unsuitable for conducting scientific inquiry. These languages were *made* to be different from the ones we already have.

The pre-Montagovian assumption can be seen in Russell and Strawson’s dispute over the theory of descriptions. Strawson contends that Russell’s theory is mistaken because it characterizes the existence and uniqueness conditions of definite descriptions as entailments of the sentences containing those descriptions, when they should instead be viewed as presuppositions of speakers using those sentences. In his reply, Russell says that he has never been concerned with ordinary language, agrees with Strawson that “ordinary language has no exact logic,”² and characterizes his theory of description as a “linguistic novelty,” intended for the “modification of common speech.”³

Montague showed how natural languages (or at least, an interesting fragment of one natural language) could be interpreted model-theoretically. No aspect of his semantics is uncontroversial, but two of its fundamental assumptions gradually gained wide acceptance among philosophers and linguists. The first is that some of the expressions of natural languages are truth-evaluable and that we can interpret them by specifying the conditions under which they are true. The second is that this should be done by assigning semantic values to their smallest parts and then

¹ Montague (1970): 398.

² Strawson (1950): 344.

³ Russell (1957): 387–389.

building up their interpretation step-by-step parallel to their syntactic structure. These assumptions facilitated the formalization of semantic theory.

The decades after Montague saw an explosion of formal approaches to a wide range of semantic phenomena, both on topics he addressed, such as anaphora, quantification, and modification, and expanding into new areas, such as aspect, plurality, genericity, presupposition, comparatives, negative polarity, questions, and much else. There have also been many extensions and alternatives developed for the specific type-theoretic approach Montague advocated, including flexible type semantics, dynamic semantics, and variable-free semantics.

Studies in Linguistics and Philosophy is a book series that has tracked the development of formal semantics from the early years to the present, publishing its first volume in 1978 and reaching its 100th volume with this one. We, co-editors of the series, together with Yael Sharvit, when this project was undertaken, have celebrated this occasion by identifying some of the landmark articles in the field and soliciting commentaries on them from distinguished experts. We hope that this book can help our colleagues (primarily linguists, but also philosophers) teach these classic articles in their courses. The chapters published here make the target articles more accessible by providing background, modernizing the notation, contributing critical commentary, explaining the afterlife of the proposals, and compiling a useful bibliography for further study.

Together with Yael, who was involved in the initial stages of the preparation of this volume, we selected 21 target articles based on three main considerations. We wanted to include papers that have been widely cited and read, we wanted to cover most of the important subfields of formal semantics, and we wanted to include no more than one paper from any given author. In selecting the authors, we paid attention to prominence, proximity in research focus, and diversity. The latest paper is from 1999 – we did not want to suggest that anything from the present century is already a classic. We tried to impose limits on the length of the chapters, but we did not try to enforce further uniformity. Some of the chapters here are more exegetical, some more focused on the afterlife of the target article. Each is a valuable companion piece.

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Chapter 1

On Abusch's "Sequence of Tense and Temporal *de re*"



Ana Arregui

Abstract This chapter provides an overview of Dorit Abusch's 1997 paper "Sequence of tense and temporal *de re*", reporting both on the ideas presented in the paper and its influence in the field. The paper has had a lasting impact, both in terms of its interaction with literature at the time of publication, and in terms of how it continues to shape our debates today. The chapter aims to elucidate that by tracing how various aspects of Abusch's paper resurface in later debates.

Keywords Sequence of tense · Semantics of tense · Embedded tense · Temporal *de re* · Double access readings · Upper limit constraint · Propositional attitudes

Introduction

In "Sequence of Tense and Temporal *de re*", Abusch investigates the interpretation of tense in intensional contexts. The paper has had a lasting impact, both in terms of its interaction with literature at the time of publication, and in terms of how it continues to shape our debates today. It remains an obligatory point of reference for theories of the interpretation of embedded tense, providing profound insights regarding the resolution of temporal reference, and a rich toolbox for extending the inquiry to cross-linguistic variation. The paper begins with a set of puzzles that is given an analysis within an *independent* theory of tense. This is followed by a set of problematic examples, and a new analysis in terms of *sequence-of-tense*. A special principle, dubbed the *Upper Limit Constraint*, is invoked to explain restrictions in tense interpretation in attitude complements. Finally, the paper addresses the interpretation of present tense embedded under past attitude verbs, bringing together all strands of the proposal.

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The theory is empirically grounded on English, and key data includes examples like (1) and (2), illustrating ‘shifted backwards’ readings and ‘simultaneous’ readings of embedded past tense:

(1) Shifted backward reading

The defendant Past₂ was actually at home watching ‘The Simpsons’ at the time of the crime₂. But after hearing the testimony of the first eye-witness, the jurors clearly Past₃ believed that he Past₂ was in the laboratory building. (Abusch 1997a: 2)

(2) Simultaneous reading

Mary Past₂ believed it Past₂ was raining. (Abusch 1997a: 3)

In cases like (1), the interpretation of past embedded under *believed* appears shifted towards the past of the matrix-clause time. In cases like (2), the eventualities appear co-temporal. Examples like (2), in which a past embedded under past generates an intuition of simultaneity, have traditionally been classified as examples of ‘sequence of tense’¹ (I will informally talk about sequence of tense *effects* to avoid theoretical bias). A first puzzle, therefore, is to spell out a theory of embedded tense that captures intuitions about (1) and (2). In addition, Abusch (1997a) focuses on the specific issues raised by presents embedded under past, known to give rise to intricate interpretations:

(3) Present-under-Past

John believed that Mary is pregnant. (Abusch 1997a: 39)

In addressing such data, Abusch (1997a) provides a synthesis of earlier work (Abusch, 1988, 1991, 1994) and enters a dialogue with contemporaneous literature (e.g. Ogihara, 1995b, 1996; Stowell, 1993, 1995, 1996; Kratzer, 1998; von Stechow, 1995).

The interpretation of tense in embedded contexts constitutes both a long-standing domain of inquiry² and a lively current agenda. In contemporary linguistics, it provides a testing ground for views about interfaces (morphosyntax vs. semantics vs. pragmatics), theories of agreement and feature transmission, views

¹ After the traditional grammatical term applied to similar effects in Latin: *consecutio temporum*.

² General overviews of the semantics of tense that can be useful to situate the issues in a broader context include Kuhn and Portner (2002), Grønn and von Stechow (2016), Ogihara (2011), and Ogihara and Sharvit (2012) provide a crucial critical assessment focused on embedded tenses with a cross-linguistic perspective. General descriptive overviews of tense interaction in subordinate clauses in English include a.o. Quirk et al. (1985) and Huddleson and Pullum (2002). Comrie (1985) provides a good first introduction to sequence of tense phenomena (Chap. 5), while Binnick (1991) encompasses more complex data descriptions and a history of the problem (including a brief review of the facts in Latin). A short review of discussions of sequence of tense in English within more traditional grammars can be found in Ogihara (1996: 68ff.). Early discussions and treatments of sequence of tense effects in English can be found in Ladusaw (1977), Smith (1978), Dowty (1982), and Comrie (1986). Finally, though not focused on embedded clauses, the annotated online bibliography in Binnick (2017) provides very useful general references.

about logophoricity, indexicality and the role of the context parameter, differences between sentence-level phenomena vs. discourse-level phenomena, and a rich empirical domain in which to study cross-linguistic variation. As we will see, one of the important responses to Abusch's work has been to examine it in light of cross-linguistic variation. The result has been a much more nuanced understanding of typology in this domain, dismissing the hypothesis of a two-way distinction between languages with, or without, sequence of tense effects.

My presentation focuses on some key ingredients of Abusch (1997a) and follows closely the structure of the original paper, while establishing links to related work. The structure is as follows: Section "[An independent theory of tense](#)" provides an overview of Abusch's independent theory of tense, section "[The upper limit constraint](#)" discusses the Upper Limit Constraint, section "[Capturing tense dependencies](#)" focuses on the mechanisms of tense transmission that capture tense dependencies, and section "[Present-under-past](#)" concludes with a discussion of double access readings for present-under-past.

An Independent Theory of Tense

In Abusch's characterization, an *independent* theory of tense is a referential theory in which the interpretation of tense is always anchored on the speech time/time of the context. Following Partee (1973), tenses are treated as variables that refer to times. The reference time for the interpretation is taken to always be the speech time, regardless of whether tenses are embedded or not. Abusch's original views were reconstructed in Heim (1994) as the presuppositional treatment of past (Past) and present (Pres) tense in (4), which has become part of the standard repertoire in the literature.³

- (4) a. $[[\text{Past}_i]]^{g,c}$ is defined only if $g(i) < t_c$, in which case
 $[[\text{Past}_i]]^{g,c} = g(i)$
 b. $[[\text{Pres}_i]]^{g,c}$ is defined only if $g(i) o t_c$, in which case
 $[[\text{Pres}_i]]^{g,c} = g(i)$ (Heim, 1994: 144)
 (Where '<' means *wholly precedes*, 'o' means *overlaps*, t_c is the time of the context, and an utterance is only felicitous if it has a well-defined truth-value in the context.)

The independent theory seeks to derive the interpretations of (1) and (2) in terms of differences in the anaphoric relation established by embedded past. We could speculate that past-shifted readings obtain when the embedded tense is anaphoric to a tense before the matrix believing time (as in (1)), and that simultaneous readings

³ Abusch (1997a) only includes present and past in the category of tense. A semantics for *will* is not explicitly spelled out (but see Abusch, 1998, 2004, as well as Heim, 1994 and Condoravdi, 2002, 2003).

obtain when it is anaphoric to the past matrix time (as in (2)). In dealing with counterexamples to this simple proposal, Abusch puts forward a *de re* analysis of embedded tense that remains an exciting and influential aspects of Abusch's work.⁴ A first type of problematic example is illustrated in (5), which crucially lacks the forward-shifted reading for $Past_2$ represented below in the complement of *thought*:

- (5) John $Past_1$ found an ostrich in his apartment yesterday. Just before he $Past_2$ opened the door, he $Past_3$ thought that a burglar $Past_2$ attacked him.

The problem is that the independent theory does not seem to predict the absence of such forward shifted reading. As long as the embedded tense presupposition is satisfied and $Past_2$ is past with respect to the time of the context, it should not matter whether or not $Past_2$ is future with respect to $Past_3$.

Abusch began by investigating whether a *de re* analysis of *believe*, extended to a temporal *res*, could provide the insights needed to differentiate permitted anaphora in (1) and (2) from forbidden anaphora in (5). The account brings together the independent theory of tense with a *de re* analysis of attitude verbs, building on Lewis (1979) and Cresswell and von Stechow (1982): *de re believe* combines with a *res*-time and a property of times in the form of a structured proposition. An illustration is provided with (6a) (=2), (simultaneous reading):⁵

- (6) a. Mary believed it was raining.
 b. [Mary $Past_2$ believed [[$Past_2$] λt_3 [it t_3 was raining]]] (see Abusch 1997a: 13)

De re tense is in an extensional position and the independent theory predicts that it will be interpreted as preceding the speech time (t_c in (4a)). The semantics of *believe* is spelled out in terms of quantification over centered worlds, centered on both an individual and, crucially, a time (Abusch uses the notation $\langle x_{self}, t_{now}, w \rangle$ for centered worlds to emphasize the interpretation of the variables as corresponding to a 'self' and the self's 'now'). A salient acquaintance relation is responsible for identifying the *res* 'counterpart' in the belief-worlds quantified over. In (6), the salient acquaintance relation can simply identify the time perceived by Mary to be her 'now' in a belief world. The truth of (6) will require that it be raining at that time in the centered worlds corresponding to Mary's beliefs in the evaluation world at $g(2)$ ($=t_2$). The *de re* semantics also requires that Mary bear the relevant acquaintance relation to t_2 at t_2 in the evaluation world (i.e. that she perceive it as her 'now'). The result will be a simultaneous reading. The analysis for past-shifted

⁴ A *de re* analysis of attitudes with consequences for temporal relations was also put forward independently by Ogihara (1989, 1995a, 1996, etc.), targeting a broader empirical base including data from Japanese.

⁵ I aim to present Abusch's ideas in broad strokes. For formal details, the reader is referred to Abusch (1997a) and the critical reconstruction offered in Heim (1994). There are well known problems at the syntax-semantics interface for *de re* attitudes, independent of the issues under discussion here, which I set aside.

cases like (1) is basically the same as for the simultaneous case, with Past₂ receiving an independent interpretation in relation to the speech time. The difference with (6) is that the acquaintance relation will now identify the time in the belief worlds before t_{now} at which the witness experienced certain events (causally related to the witness's description of what happened). This gives rise to a past-shifted reading. The explanation for the disallowed forward-shifted option in (5) lies in constraints on possible/salient acquaintance relations. In (1) and (6), the acquaintance relations establish a perceptual or causal link between the individual and the *res* in the base worlds, but no such relation is available for (5). In the future-shifted example, the *res* time t_2 is a future time with respect to the matrix attitude time, and does not stand in a perceptual nor causal relation to the holder of the attitude at the belief-time in the evaluation world.

Abusch's hypothesis is that acquaintance relations have to be either causally or perceptually grounded, ruling out future-shifted readings for embedded past.

As we will see in section "[The upper limit constraint](#)", a further series of problematic examples eventually leads Abusch to conclude that constraints on acquaintance relations in *de re* attitudes are not sufficient to explain constraints on the interpretation of embedded tenses. However, the idea that we can shed light on temporal interpretation by examining the kind of descriptive content that seeps into composition when we refer to times remains an exciting and promising aspect of Abusch's proposal. It opens the door to empirical research on the types of acquaintance relations that are available to resolve reference across modal contexts, as well as characterizing times as an empirical base for testing hypotheses about how such descriptive content is identified and manipulated in composition.

While Abusch (1997a) did not take properties of a *de re* analysis to provide an explanation for the absence of forward-shifted readings for embedded past, a *de re* construal remained an open possibility (indeed, as we will see in section "[Present-under-past](#)", the only possibility for present under past as in (3)). Past under past examples like *Mary believed it was raining* were considered structurally ambiguous, allowing embedded past to be interpreted as dependent on the matrix past or *de re*. More recent literature has explored this as a possible locus for variation, across and within language. Ogihara and Sharvit (2012), for example, establish differences between English, Japanese, Russian and Hebrew on the basis of differences in how movement is regulated to derive the LFs leading to *de re* interpretations (see also Bar-Lev, 2015); Grønn and von Stechow (2010) consider that factive verbs in Russian obligatorily receive a *de re*-style interpretation while other embedding verbs do not.

The Upper Limit Constraint

Abusch discusses three main types of data that militate against the view that constraints on acquaintance relations in *de re* attitudes explain the absence of future-shifted readings for embedded past. One is the observation that future shifted

readings are not obtained even in cases in which the anaphoric antecedent for past is found within the belief context:

- (7) a. Last Monday John Past₂ believed that he Past₃ was in Paris on Tuesday₃.
 b. John Past₂ believed he Past₃ was in Paris at some time₃. (Abusch 1997a: 16, 17)

(7a) lacks a forward-shifted reading in which *Tuesday* is interpreted as following the believing time last Monday (even though in principle *Tuesday* can refer to future times, as in *I will be in Paris on Tuesday* (Abusch 1997a: 16). A similar point arises with (7b). Abusch's concern is that in these examples the temporal adverbial is part of the attitude, and not a characterization ascribed by the speaker. A *de re* analysis of the adverbials does not seem the right way to go, and so the constraints on the acquaintance relation between the time (t₃) and the holder of the attitude that had been claimed to rule out (5) would not be expected to be relevant. A second type of data, which in a sense mirrors (7), concerns examples like (8):

- (8) Leo will go to Rome on the day of Lea's dissertation. Lia_i believes [that she_i will go to Rome with him then.] (Abusch 1997a: 26)

In the salient reading, *then* is anaphoric to *the day of Lea's dissertation*. As a case of anaphora in intensional context, Abusch posits that *then* should receive a *de re* reading in this example.⁶ But this would require allowing an acquaintance relation between the holder of the attitude and the future time referred to by *then*. This should not be possible assuming the constraint on acquaintance relations barring acquaintance with future times.

A third set of data, maybe more worrying, corresponds to examples like (9):

- (9) Sue Past₃ believed that she Past₃ would marry₂ a man who Past₂ loved her.

We interpret the t₂ 'marrying' time in (9) as later than the t₃ 'believing' time, and undetermined relative to the speech time. In the salient interpretation, loving is simultaneous with marrying. Even though tense in the relative clause is past (Past₂), loving is not ordered in relation to the speech time at all. The independent theory does not seem able to account for this.

Problems like the ones noted above led Abusch to abandon the independent theory of tense and the view that constraints on the interpretation of embedded past tense comes from properties of acquaintance relations. Abusch's new solution to block forward-shifted readings, however, appeals to an intuition that she considers rather similar (but only briefly sketches): time branches towards the future and the times towards the future of the now of an attitude holder are not sufficiently 'determinate' to refer to (in contrast with what happens in the past). This intuition is cashed out in the *Upper Limit Constraint* (ULC), which states that the local

⁶ For a discussion of the role of adverbials in shifting the interpretation of embedded tense from a cross-linguistic perspective, see Ogihara and Sharvit (2012).

evaluation time serves as an upper limit for the evaluation of tenses (Abusch 1997a: 25).⁷ The reason why Past₂ in (5) cannot be shifted to the future of Past₃ (the 'thinking' time) is that the 'now' corresponding to the epistemic alternatives relevant for the interpretation of *think* serves as an upper limit for the denotation of the embedded tense.⁸

In spite of concerns regarding the justification of the ULC, it has become a standard point of reference in the literature and continues to be a focus of debate. It has been evaluated from a cross-linguistic perspective by Ogihara and Sharvit (2012), who adopt the ULC and incorporate it into a modified analysis; it has been re-examined in light of challenging data from attitude verbs like *hope*, e.g. *Bill hoped it rained*, which has led some authors to reject the stipulation of the ULC as a uniform principle (e.g. Altshuler & Schwarzschild, 2013a; Klecha, 2016), and it has been part of debates regarding the interpretation of present-under-past (to be addressed in section "[Present-under-past](#)").

Capturing Tense Dependencies

As Abusch notes, the ULC solves only part of the problem raised by tenses in intensional contexts. Designed to account for the absence of forward-shifted readings of embedded past, the ULC does not explain cases in which the reference of past tense does not seem to lie in the past at all. Abusch offers (10), after a famous French example by Kamp and Rohrer (1984):

- (10) John Past₃ decided a week ago that in ten days at breakfast he Past₃ would say₄ to his mother that they Past₄ were having their last meal together.
(Abusch 1997a: 27)

The most deeply embedded past in (10) (Past₄) does not precede the speech time (or any other salient reference time).

Examples like this motivate the view that tense morphology can dissociate from semantics: there are instances of past tense morphology that do not univocally associate with a past meaning. They are instead the reflection of another past tense. In Abusch's theory, the dissociation is facilitated by the fact that tense information encompasses a variety of ingredients: a temporal variable that refers to a time,

⁷ In her discussion of Abusch's work, Heim (1994) spells out a presuppositional account of the ULC, which is endorsed by Abusch (1997a). Tense nodes are claimed to carry the presupposition that the interval they refer to does not come after the local evaluation time. This presupposition is added to other constraints on reference coming from tense itself. See also related constraints termed *Abusch's Constraint* in von Stechow (1995), Kratzer (1998).

⁸ Shifting towards the future in embedded clauses is possible with *will/would*. However, Abusch considers that these forms spell out both tense and an intensional operator (see section "[Capturing tense dependencies](#)" for some discussion).

a relation between the reference of that variable and a local evaluation time, a constraint on relations between times (either *precedence* or *non-precedence*), and a specification of the relations that can be affected by that constraint. Sequence of tense effects are captured in terms of a tense transmission mechanism that makes higher tenses ‘visible’ (accessible) to tenses in attitude complements, allowing the temporal relations corresponding to higher tenses to interact with temporal constraints associated with tenses below. The transmission mechanism depends on intensionality, with the arguments of intensional operators characterized as properties of times. A temporal binder in the embedded clause binds a time that corresponds to the ‘now’ of the attitude holder and serves as the local evaluation time for the embedded tense (in matrix contexts, this is the speech time).⁹

Abusch’s transmission-account aims to capture the fact that in examples like (10), *Past_t* is legitimate in the embedded clause even if the time of the meal is not past with respect to the local evaluation time (or the speech time). This is because that embedded past is taken to inherit the temporal relations from higher tenses, and the precedence constraint associated with *Past_t* can be fulfilled by one of those other temporal relations (e.g. the one corresponding to the matrix tense). In principle, this allows the embedded tense variable to be interpreted as simultaneous with the saying time (identified with the local evaluation time) while the temporal constraints associated with past are fulfilled via tenses higher up in the clause. In the case of past-shifted readings of embedded past, the relation between the embedded tense variable and its local evaluation time is one of precedence (Abusch 1997a does not work out such an example, but points in this direction on page 39).

While Abusch (1997a) presents several examples illustrating the types of structures and the transmission mechanism she has in mind, there is considerable vagueness regarding aspects of the implementation. This has been critically taken up in literature in comments such as Heim (1994) and von Stechow (1994), which offer alternative exegeses that have been very influential in subsequent work. Heim (1994), for example, spells out a reformulation of Abusch’s account that does not rely on transmission of relations. Heim’s proposal introduces instead a Tense Licensing Condition that allows past morphemes to be licensed if they are in the domain of at least one affix encoding precedence. This allows several instances of past morphology to be licensed by a single ‘semantic’ past (see also Stowell, 1993).

In Abusch (1997a), the transmission of temporal relations is mediated by intensional operators that take properties of times as arguments. This includes propositional attitude verbs like *believe*, but also *will/would* and nouns describing psychological states such as *desire*. Illustrations are provided in (11) and (12) below. Sequence of tense effects are not observed in extensional contexts, with an example by Abusch in (13):

- (11) a. She will marry a man she met recently. (Dowty, 1982)
 b. He will buy a fish that is alive. (Ogihara, 1989, 1996)

⁹ See discussion in von Stechow (1995), Kratzer (1998), also Ogihara (1989, 1996).

The temporal interpretation of the relative clauses in (11) is shifted towards the future, with a meeting time that precedes the (future) marriage time in (11a), and a fish alive at the (future) time of purchase in (11b). *Will* is responsible for this shift. Even though she does not provide a semantic proposal, Abusch considers *will* as the morphological spell out of PRES + *woll*, where *woll* is an intensional operator and thus gives rise to sequence of tense effects via transmission. As we have seen, e.g. (9), (10), *would* (PAST + *woll*) also gives rise to sequence of tense effects, noted also in a past version of (11b): *He said he would buy a fish that was alive* (see Ogihara, 1989, 1996). The example in (12) illustrates a case of sequence of tense effects associated with the temporal argument of a noun:

- (12) I know that Mary was a strange child. But her desire to marry a man who resembled her is really bizarre. (Ogihara, 1996, attributed to I. Heim)

The context in (12) sets up the temporal location of Mary's desire in the past, and this is sufficient to license past morphology in the relative, allowing us to interpret the common resemblance as contemporary with the (future) marrying time. In the extensional example in (13), however, both instances of past morphology are associated with past semantics, with the corresponding eventualities ordered before the speech time, and unordered with respect to each other.

- (13) Last week John met a woman who was in the next room (#now). (Abusch 1997a: 28)

The restriction of tense transmission to intensional contexts is an important feature of Abusch's theory. With variation, alternative (more or less) contemporary accounts such as Enç (1987), Ogihara (1989, 1996), Stowell (1993, 1996, 2007), von Stechow (1995, 2009), Kratzer (1998), Kusumoto (1999, 2005), and Schlenker (1999, 2003) defended views linking sequence of tense effects more clearly to structural configurations rather than to a (purely) semantic characterization of the environment¹⁰ (it is interesting to see that in his brief introduction to the problem, Ogihara (1996: 71ff.) remarks on a similar dichotomy already present in traditional grammars). A comparison between the two types of perspectives depends on how intensional operators are identified, and on views about the nature and role of temporal arguments associated with syntactically tenseless predicates (e.g. the noun in (12)). While Abusch does not offer independent discussion of such matters, some issues have been taken up in subsequent literature. For example, a discussion of similarities/differences in the temporal arguments associated with verbs vs. other categories can be found in Kusumoto (1999, 2005) (for a discussion of temporal arguments of nouns more generally see a.o. Enç 1986, Musan 1995). As noted earlier, the technicalities of Abusch's transmission proposal raised concerns, with

¹⁰ I will not be able to do justice here to the tradition that provides analyses of sequence of tense effects on the basis of more specifically syntactic theories, without a main focus on compositional interpretation. In addition to cited work by Enç and Stowell, syntactically-oriented accounts include a.o. Zagona (1990, 2014), Uribe-Etxebarria (1994), Demirdache and Uribe-Etxebarria (2004), and Hornstein (1990).

alternative accounts often appealing to more familiar mechanisms: Ogiwara (1989, 1996) and von Stechow (1995), for example, appeal to a tense deletion rule Stowell (1993, 1996, 2007), spells out syntactically-based analyses that build on the idea of polarity as relevant for licensing Kusumoto (1999, 2005), spells out a semantic theory building on Stowell's polarity proposal, and Kratzer (1998) and Schlenker (1999) appeal to morphological feature agreement. Subsequent literature has also re-examined the scope of the problem, both in terms of the types of elements that enter into a dependency and the characterization of the domain.¹¹ Both Kratzer (1998) and Schlenker (1999, 2003, 2004), for example, have provided unified accounts of 'transmission' across temporal and pronominal domains in terms of features and logophoricity, with Schlenker arguing for 'sequence of person' and 'sequence of mood' effects analogous to sequence of tense. Sharvit (2003, 2004, 2008) has pointed to parallelisms between sequence of tense in embedded clauses and tense behaviour across sentential boundaries in Free Indirect Discourse, suggesting that the latter may be explained by means of an operator functioning similarly to an attitude verb.

It seems worthwhile to make two more points here. One is that Abusch (1997a) is not concerned with cross-linguistic variation, examining exclusively data from English. Sequence of tense effects, however, are known to vary robustly across languages and in order to arrive at a complete picture, Abusch's proposal would need to be re-examined in light of a broader empirical base. As an illustration, two examples from languages that have traditionally been claimed to lack sequence of tense effects are presented below, and can be compared with the English example in (6). In these cases, the simultaneous reading that we noted for embedded past in (6) is obtained with an embedded present tense:

(14) Japanese

Bernhard-wa Junko-ga byooki-da-to it-ta
Bernhard-TOP Junko-NOM be-sick-PRES say-PAST
'Bernhard said that Junko was sick' (Kusumoto, 1999)

(15) Russian

On skazal, čto živet pod Moskvoy.
He say-PAST-PF that be-PRES around Moscow
'He said he was living just outside Moscow' (Grønn & von Stechow, 2010)

¹¹ There are accounts in the literature that have actually moved away from analyzing sequence of tense effects in terms of structural properties properly associated with embedded tense (e.g. Altshuler, 2008 for Russian). Some recent proposals have argued that pragmatic considerations actually play a central role in our interpretation of tense in such contexts, e.g. Altshuler and Schwarzschild (2013a, 2013b), Klecha (2018), with Gennari (2003) an early exponent of this view. Concerns have been raised, however, regarding how pragmatics handles cross-linguistic variation (e.g. Ogiwara & Sharvit, 2012; Smirnova, 2009; Kusumoto, 1999). See also Bochnak (2017) for discussion of sequence of tense effects in Washo, an optional tense language, that supports a structural rather than pragmatic account.

Proposals that tackle cross-linguistic data have accounted for variation in several ways, including cross-linguistic differences in tense deletion rules, in the characterization of the original tense inventory, in terms of constraints on tense movement, etc.

A related second point is that a cross-linguistic perspective has led to a more fine-grained picture of sequence of tense effects, which is arguably pertinent for our understanding of English data as well (see e.g. Ogiwara & Sharvit, 2012; Sharvit, 2003). The domains relevant for sequence of tense effects have been shown to vary significantly across languages, as well as the range of interpretations available to the different members of the tense inventory. Substantial differences have been found, for example, in cross-linguistic comparisons that have examined both tenses embedded in attitude complements and in temporal adjunct clauses (a.o. Arregui & Kusumoto, 1998; Kubota et al., 2009; Grønn & von Stechow, 2013a, 2013b; Sharvit, 2014), as well as in relative clauses (e.g. Ogiwara & Sharvit, 2012; Grønn & von Stechow, 2013a, 2013b). English data has become part of a more ambitious conversation.

Present-Under-Past

The interpretation of present tense clauses embedded under past tense intensional operators is known to raise a number of intricate issues, and Abusch's solution relies on bringing together the different strands in her work.¹² Abusch's original example is provided in (16):

(16) John believed that Mary is pregnant. (Abusch 1997a: 39)

(16) has been associated with the intuition that Mary's pregnancy overlaps both the past time of John's belief and the speech time, leading to the term *double access reading* (DAR). There is more to it than that. The sentence can be true even if there is no actual pregnancy overlapping both times. Here is an alternative version:

(17) John said two weeks ago that Mary is pregnant but actually she has just been overeating for the last three months. (Abusch 1997a: 40)

When John made the assertion reported in (17), his commitment was only regarding what was going on at that time, not at the future time when (17) is uttered. The truth of (17) requires a link to the speech time thus not predicted in terms of John's original commitment. In Abusch's words: "What seems to license the Pres\Past in

¹² An early description of the puzzle can be found in Smith (1978), who noted a distinction between factive verbs/verbs of saying vs. other kinds. In Smith's terms, the descriptive generalization for the first group is that when present is embedded under past, the speaker is 'responsible' for the complement being true or relevant at the speech time (Smith, 1978: 66). See also Enç (1987) for a mix of examples with factive verbs and verbs of saying/hearing.

this example is that the speaker is interested in the explanation for Mary's symptoms in an interval spanning the two times. In fact, [(16)] seems inappropriate if Mary's symptoms, her big belly, and so forth, do not persist at the utterance time" (Abusch 1997a: 40).

Importantly, DAR is the only reading available for this type of examples. It is forced, in Abusch's system, by the interaction between the tense transmission mechanism operative in intensional contexts and the temporal constraints associated with present. When tense is present, the associated non-precedence constraint affects all accessible temporal relations. None of them can be precedence relations.¹³ Since this constraint is incompatible with the relation inherited via tense transmission from the matrix past, the immediate prediction is that present tense will not be possible in its base position within the complement clause. The solution is to appeal to *res* movement, placing the embedded present tense in a syntactic position within the scope of the attitude verb but outside the scope of the binder in the embedded clause (and leaving behind a simple trace). In this configuration, present does not inherit the temporal relation of the matrix past, and the non-precedence constraint does not lead to contradiction. Having broken the binding relation, the interpretation of present in this structure will be established in relation to the local evaluation time of the matrix (the speech time), with (16) receiving a structure as sketched in (18) (including *res* movement and the λt_0 binder associated with the argument of *believed* corresponding to the believer's 'now'):

(18) [John Past₁ believed [[Pres₂] λt_2 [λt_0 [Mary t_2 is pregnant]]]]

The non-precedence constraint will ensure that the reference of present overlaps the speech time (the local evaluation time for the matrix clause). Why do we have the intuition that it also overlaps John's Past₁ belief-time? The answer, suggests Abusch, lies in the acquaintance relation that identifies the *res* time in John's belief worlds. A plausible acquaintance relation picks out the maximal interval overlapping John's 'now' in which Mary has pregnancy symptoms (like a big belly). For (18) to be true, the *de re* analysis requires that in all the centered worlds $\langle x_{\text{self}}, t_{\text{now}}, w \rangle$ compatible with John's beliefs at the Past₁ belief-time in the base world, Mary be pregnant in w at the maximal interval overlapping t_{now} at which Mary has pregnancy symptoms (note: (i) no commitment regarding the speech time, (ii) according to Abusch, the ULC blocks future-shifted acquaintance relations which could identify times in John's belief worlds that completely follow John's 'now'). The *de re* analysis also imposes a constraint on the evaluation world: the Pres₂ *res* time is the maximal interval overlapping John's Past₁ belief-time (counterpart of John's 'now' in his belief worlds) in which Mary has pregnancy symptoms. The interpretation of Pres₂ is thus subject to two temporal constraints: overlap

¹³ There is an asymmetry in this respect regarding past tense, which simply requires that at least one accessible relation be of precedence.

with the speech time (non-precedence) and overlap with John's past belief-time (acquaintance relation), giving rise to the double access reading.¹⁴

In Abusch's proposal, the *res* of belief in present-under-past cases is a temporal interval and constraints on the evaluation world (e.g. that Mary's symptoms persist throughout a time that overlaps both the speech time and John's past belief-time) are derived via the salient acquaintance relation. Ogihara (1989, 1995a, 1996, 1999) has independently argued for a *de re* analysis of present under past examples, with the view that the *res* of belief is not a time but a state (see Abusch, 1997b for a reply to Ogihara 1995a; see also Kratzer, 1998 for a discussion of double access readings in terms of *de re* attitudes towards eventualities).

Double access readings continue to prove fertile ground for debate. Amongst recent proposals, a line of research has focused on the role of pragmatics in generating such readings, arguing that our theories of tense in these contexts could actually be simplified once pragmatic considerations were taken into account (e.g. Altshuler & Schwarzschild, 2013a; Klecha, 2018; see Smirnova, 2009 for a comparison on the basis of Albanian). Current discussions of double access readings have also called into question the empirical adequacy of the ULC, with sentences like *The stewardess told me you have my bags* counting as the report of an original statement *The assistant at the baggage counter will have your bags* (Altshuler & Schwarzschild, 2013a; Altshuler et al. 2015; Bary & Altshuler, 2014). Flipside examples can be found in Anand and Hacquard (2007), who discuss cases in which presents embedded under past need not refer to the speech time, such as *The medieval monarch King Richard said that he would let his daughter marry any knight who comes back from the Third Crusade*. Finally, let me note that double access readings are hard. Ogihara (1995a, 1996) provides extensive discussion of the special conditions in which present-under-past readings are appropriate. Bary et al. (2018) present experimental research designed to test the conditioning of present under past reports.

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¹⁴ I hope to have provided a faithful short overview of Abusch's main ideas about DAR here. The technical presentation in Abusch (1997a) is rather brief regarding some of these issues. The reader is also referred to the reconstruction provided in Heim (1994), as well as comments in von Stechow (1994).

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Chapter 2

On Bach's "The Algebra of Events"



Susan Rothstein

Abstract Emmon Bach's short 1986 paper was a milestone in the study of aspect, exploring the parallels between aspectual properties of verbs and verb phrases on the one hand, and the mass/count distinction on the other. As he himself points out, he was not the first to notice these parallels, but his paper played a crucial role in bringing the topic to the attention of semanticists and syntacticians. The paper identifies the issues which need to be explored, focusing attention on the nature of the relation between telicity and countability.

Keywords Events · Count-mass distinction · Cumulativity · Homogeneity · Telicity · Boundedness

The Content of the Paper

Emmon Bach's short 1986 paper was a milestone in the study of aspect, exploring the parallels between aspectual properties of verbs and verb phrases on the one hand, and the mass/count distinction on the other. As he himself points out, he was not the first to notice these parallels, but his paper played a crucial role in bringing the topic to the attention of semanticists and syntacticians. The paper identifies the issues which need to be explored, focusing attention on the nature of the relation between telicity and countability.

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Bach's basic aim is stated very clearly: he wants to elucidate the parallel between the "thing – stuff" relation and the "event – process" relation.¹ In the same way that things, or objects, are discrete, countable individuals made up of stuff, discrete events are countable individuals made up out of processes. He draws four parallels between the nominal and the verbal domain. They involve, respectively, quantity expressions, homogeneity, coercion and partitivity.

Bach's first point is that there is a common pattern of distribution of quantity expressions (expressions denoting numbers or amounts) in the nominal and the verbal domain. Mass and count nouns can be distinguished by their interaction with quantity expressions, as shown in (1). (Bach is concerned here only with 'non-coerced contexts' – he recognizes that the examples marked "#" are possible in certain contexts, but notes that in these cases "we have to do a certain amount of work to impose a special interpretation" (p. 6).²)

- (1) *Mass nouns:* a lot of mud, much mud, #three muds, #many muds
Count nouns: #a lot of dog, #much dog, three dogs, many dogs

Parallel verbal modifiers divide verbs into two groups, with group 1 patterning with mass nouns and group 2 patterning with count nouns. The examples are adapted from Bach's own examples.

- (2) a. *Group 1:* sleep/walk a lot, (didn't) sleep/walk much, #slept/walked three times, #slept/walked many times.
 b. *Group 2:* #found a unicorn a lot, #(didn't) find a unicorn much, found a unicorn three times, found a unicorn many times; #built a house a lot, #(didn't) build a house much, built a house three/many times.

Bach points out that, as had already been noted by numerous writers (he mentions Allen, 1966; Taylor, 1977; Mourelatos, 1978; Carlson, 1981; Hoepelman & Rohrer, 1980), these contrasts seem to reflect the relevance of **boundedness** in the interaction between predicates and quantity expressions. Singular count nouns such as *dog* denote sets of bounded elements or individuals and the sum of two entities in the denotation of *dog* does not fall under the denotation of *dog*. Mass nouns such as *mud* denote unbounded elements and are **homogeneous** in two directions. They are **upwardly homogeneous** or **cumulative** since the sum of two accumulations of mud falls under the denotation of the predicate *mud*, and they are **downwardly homogeneous** or **divisive** since parts of a quantity of mud are also mud. Boundedness is associated with countability, while **homogeneity** is associated with mass nouns. Comparison between (1) and (2) suggests that the verbal predicates in (2a) denote unbounded entities which Bach calls **processes**.

¹ Note that Bach's use of 'event' diverges from a common usage in which 'event' is contrasted with 'state' and 'events' include Vendler's (1957) activities, achievements, and accomplishments.

² All page references without name and date are to E. Bach, 1986 *Linguistics and Philosophy* 9, 5–16.

These are Vendler's **activities**. Process predicates are similar to mass predicates as they are homogeneous in both directions. A walking event denoted by *walk* can be part of a bigger event in *walk*, while a part of a walking event (above a minimal threshold) will also count as a walking event. In contrast, the predicates in (2b), which fall under Vendler's definition of achievement and accomplishment predicates and which Bach terms "event" predicates are bounded.³ Intuitively, the idea is that just as count nouns denote individuals which can be thought of as discrete countable portions of "stuff", event predicates denote "bounded bits of processes" which are also countable (p. 8). So, the count noun *apple* denotes a set of bounded chunks of apple-stuff, while *build a house* denotes a set of bounded chunks of building processes. In neither case is the boundedness *ad hoc*. Apples are non-arbitrary apple units, and build-a-house events are non-arbitrary building events which end at what Verkuyl (1972) calls a **set terminal point** (sometimes called the 'telic point'), i.e. the point where the house is built.

Bach's second contribution is to show that this parallelism can be expressed using Link's (1983) mereological theory of noun denotations, applying the mereological analysis to verbal denotations too.⁴ Link proposes that singular count nouns denote sets of atomic individuals. Plural count nouns denote atomic join semi-lattices, the result of closing the denotation of the singular count noun under sum.⁵ Plural count nouns can be modified by cardinals, as in *three cats*, because they are generated by a set of atoms. Mass nouns denote non-atomic join semi-lattices which can be non-atomic. The mereological analysis expresses the commonality between plural count and mass nouns, namely that they both have unbounded or (upwardly) homogeneous denotations, as well as their differences. While plural count nouns can be modified by cardinals because their denotations form atomic semi-lattices, mass nouns are not countable because their denotations are not generated by a set of atoms.

Link's theory allows the relation between the mass and the count domain to be expressed: a homomorphism maps an entity in the count domain onto the stuff in the mass domain which it consists of. The stuff making up an individuated entity is distinct from the individuated entity itself, and the two can have separate properties. For example, a ring can be new, while the gold that it is made up of is old. The

³ Note that Bach's taxonomy of verb classes as expressed in the tree on page 6 of his paper does not allow for processes and states as a natural class. Bach thus parts company with Verkuyl (1972), Dowty (1979), Rothstein (2004) and many others who treat states and activities/processes as homogeneous. See also Levin (this volume) and Filip (this volume).

⁴ For discussion see Champollion and Brasoveanu (this volume).

⁵ Link's view on plurals is exclusive, that is, he does not include atoms in the denotation of plurals. This is problematic because it yields the wrong prediction for sentences like (i) and (ii):

- (i) Do you have children? Yes, one.
- (ii) I have had students who didn't major in mathematics. At least one.

Today most semanticists opt for the inclusive interpretation of plurals. Chierchia (1998) argues for the exclusive view, but in later papers, e.g. Chierchia (2010), he reverts to the more standard analysis. See Dayal (this volume) on Chierchia (1998).

inverse mapping is not a homomorphism since stuff can appear in the count domain under a number of different descriptions. Bach's example is that the stuff making up an individual called Jones could either be mapped onto an atom in the denotation of *man/men* or a sum in the denotation of *cells*. There is a sense in which these items would be "the same" item, since they occupy the same spatio-temporal location. However, individuation is relative to a particular description, and the choice of description determines the properties than an entity has. Thus *that man is walking* is clearly true in a way that *those cells are walking* is not. Similarly, the same piece of fabric can be described as a shawl, a baby-carrier, a wrap, or a quantity/piece of fabric, and this will similarly determine what properties can be truly predicated of it – as a shawl it is elegant but too big to be useful, but as a baby carrier it is just the right size (but not elegant). The model distinguishes between three different "part-of" relations. The relation between the denotation of *three rings* and the individual rings of which it consists is expressed in the join part-of structure of *rings*. The relation between a ring and the gold of which it consists is expressed in the homomorphism from the count domain to the mass domain. A third relation, for example between a chain and the links in the chain, is not expressed, since the model does not allow atoms to have other atoms as parts.

Bach shows that Link's model can be used in the verbal domain, making precise the parallelism shown in (1) and (2). Plural event predicates denote atomic join semi-lattices, and are thus countable, and process predicates denote non-atomic join semi-lattices. Plural event predicates denote sums of individual events, for example *Mary sprained her ankle* denotes an event which is an atomic part of the plurality of events denoted by *Mary fell and Mary sprained her ankle*. Singular event predicates can be seen as bounded chunks of processes, as in the case of *build a house*, which denotes a bounded chunk of (house-)building. As in the nominal domain, a unidirectional homomorphism maps from the event domain to the process domain since an event can be constituted of only one process-chunk, but this does not guarantee a mapping from a chunk of process onto a single event. For example, a single process denoted by *John walked* can be referred to in a particular context as an individual, bounded event using a variety of different predicates: *John walked to the corner/walked to the store/walked away from the house/walked to the bus stop* and so on.

Bach's third parallel concerns coercion. He points out that both mass and count domains allow packaging (coercing a mass predicate into a count interpretation) and grinding (coercing a count predicate into a mass interpretation). *Ice cream*, though normally mass, can have a count interpretation, while *dog* can have a mass interpretation (3):

- (3) a. How many ice creams should I buy?
- b. There was dog splattered all over the road.

Similarly, the process verb *look for* can be used to denote a bounded event, while event predicates like *read a book* can be interpreted as homogenous processes:

- (4) a. John finished looking for the book.
- b. John read a book for an hour.

Since grinding is a linguistic correlate of the homomorphism operation mapping an individual onto the stuff it is made of, there is an asymmetry between the grinding and packaging operations. Packaging stuff and then grinding it gives you back the original stuff, but grinding an object and then packaging it does not necessarily give you the original object back.

Finally, Bach suggests that the well-known imperfective paradox has a correlate in the semantics of partitives in the nominal domain. Dowty (1979) showed that *John was crossing the street* must have an intensional interpretation since it does not entail *John crossed the street* as in (5):

(5) John was crossing the street when he was run over by truck.

(5) asserts that there was a process which was part of a cross-the-street event, although the complete cross-the-street event did not exist in our world. Bach claims that, similarly, (6) asserts a part-of relation between an existing object and a non-existent object.

(6) This is part of a paper on natural language metaphysics which I never finished.

In sum, Bach claims that "events and processes are disjoint" (p. 15) and that this disjunction parallels the disjunction between individuals and stuff.

Impact of the Paper

Bach's paper was a major factor in encouraging linguists to apply Link's model to the domain of events, which became a crucial tool in the formal analysis of aspect. Much work on aspect in the thirty-odd years following Bach's paper has explored the role of homogeneity in distinguishing between activities and processes, i.e. atelic predicates, from accomplishments and achievements, i.e. telic predicates (see Levin, [this volume](#); Filip, [this volume](#)). Krifka (1989, 1992, 1998) focusses on the role of the incremental theme in distinguishing homogeneous and non-homogeneous verbal predicates.⁶ Krifka argues that events are non-homogeneous and telic, because the theme argument is affected incrementally in the course of the event.⁷ Given an event, *e*, and its theme *x*, each proper part of *x* is the theme of one and only one proper part of *e*. Since each proper part of *x* is the (sub)-theme of one and only one part of *e*, the duration of *e* cannot be indefinitely extended. When *x* is "used up", *e* is over.

⁶ 'Incremental theme' is a term is due to Dowty (1991) and is standard today. Krifka acknowledges that his own terminology is intended to capture essentially the same notion as Dowty's. See Levin ([this volume](#)) for additional discussion of incremental themes.

⁷ Since not all telic predicates have incremental themes, this cannot be the whole story. See Rothstein (2004) for a discussion of telic uses of accomplishments with non-incremental themes such as *repair the computer* and *grow up*.

Events therefore are bounded and have a set terminal point or telic point.⁸ *Build a house* denotes a set of events with a set terminal point, since the extent of each individual event is determined by when its incremental theme, a house, is ‘used up’ i.e. completely affected or acted on – in this case, completely built. Process predicates do not have an incremental theme, either because they have no theme at all, as in *John ran*, or because the theme is not an incremental argument of the verb, as in *John pushed the cart*.

A different approach to explaining boundedness is explored in Hay et al. (1999), Kennedy (2012), and Rappaport Hovav (2008). They suggest that the process/event distinction derives from the fact that events are inherently scalar, and implement a certain measure of change, while processes are non-scalar. The incremental theme is one of a number of mechanisms for determining the measure of scalar change that an event involves. Unlike Krifka’s theory, this scalar theory does not maintain the parallelism between atelic verbs and mass nouns and between event predicates and count nouns. Scalarity is a property of verbal predicates inherently related to their dynamicity. Dynamic predicates involve change and scalar dynamic predicates (event predicates) quantify how much change is involved. Since count nouns are not dynamic, the individuability of events and the individuability of count noun denotations don’t come from the same source.

Bach’s observation that the imperfective paradox can be seen as an instance of more general partitivity phenomena has also had impact. Thus White (1993) points out that, in addition to finding the imperfective paradox with progressives, we find it with aspectual verbs. *John started running* entails that John ran, while *John started building a house* does not entail that John built a house. But there are crucial differences between partitivity in the verbal and the nominal domain. It has been convincingly argued that the progressive (and the semantics of aspectual verbs) is intensional, relating a partial event to its completed whole. While theories differ as to what this intensional relation is, many researchers, including Dowty (1979) and Landman (1992) have argued that the partial event must be related to a ‘close’ enough completed event. For Dowty, the completion of the event must occur in an inertia world, i.e. a normal continuation of the actual world in which nothing unexpected occurs. For Landman, the completion of the event must occur in a plausible continuation of the actual world. In both accounts, *I am writing a story about ghosts* is true if (i) a partial event of writing a ghost story has occurred or is occurring, and (ii) there is good reason to assume that it will expand into a complete event of writing a ghost story in a close enough possible future. However, partitives in the nominal domain are not intensional in this way. *This is part of a story about ghosts*, if true, is not necessarily true because of its intensional relation

⁸ Note that events have a set terminal point or telic point. Event predicates are telic in grammatical contexts which require that telic point to be reached. Thus the event predicate in *Mary built a house* is telic. The event predicate in *Mary was building a house* is used in a non-telic context. Rothstein (2009) shows that mood can have the same effect. *Ezra ate his soup in ten minutes* has a natural telic interpretation while the imperative “*Eat your soup!*” has a natural atelic use as an instruction to get on with eating your soup!

with a plausible complete story about ghosts. Suppose I am a writer who collects good ideas, and I hear an interesting anecdote about ghosts. I might well write the anecdote down and file it away to use (possibly) at some unidentified point in the future, saying "This is clearly part of a story about ghosts, maybe I can use it one day." Crucially, *I am writing a story about ghosts* entails *I have written part of a story about ghosts* (as long as the event is passed the planning stage and some actual writing has taken place). However, *I have written part of a story about ghosts* does not entail *I am writing a story about ghosts* since having written part of the story says nothing about the possibility or probability of my taking the event through to its conclusion.⁹ Landman (1992) makes the same point in a different way, showing that a person may construct an object and truly announce "this is part of an Escher cube (as much of it as one can physically construct)" while "I am building an Escher cube" is false.¹⁰

Beyond its immediate impact, however, Bach's paper raises two central questions:

- (i) Is homogeneity as defined in the nominal domain the relevant property with which to distinguish event (or telic) predicates from process (or atelic) predicates?
- (ii) Is there good evidence that the event/process distinction (or the atelic/telic distinction) should be characterized in the same way as the count/mass distinction?

In the following sections, I will consider both of these issues.

Homogeneity

Krifka, 1989, 1992, 1998 develops what is probably the most comprehensive and technically complete exposition of parallels between homogeneity in the nominal and the verbal domain. He adopts a mereological framework in which predicates denote join semi-lattices (i.e. lattices without the null element), within which the relevant properties are defined. Homogeneity is divided into upward homogeneity, or cumulativity, as defined in (7), and downward homogeneity or divisiveness, as defined in (8).

⁹ A reviewer suggests that *This is part of story about ghosts* is intensional because it is part of a possible story about ghosts. But the crucial thing about the truth of the progressive *I am writing a story about ghosts* is that it requires a relatively predictable relation between the event part which has already taken place and the completed event, and this entails a relation between the part of the story and the whole story. However, *I wrote part of a story about ghosts* can be true without there being any conception of the path which will lead to a completed story, and without my having any intention or plan for how to complete this story.

¹⁰ Similarly, "I am counting as many of the prime numbers as I can before I die" may be true, but "I am counting the prime numbers" was judged false by my informants.

- (7) A predicate P is **cumulative**, or upwardly homogeneous, iff
 $\forall x \forall y [P(x) \wedge P(y) \rightarrow P(x \sqcup y)]$
 “ P is cumulative just in case whenever two things are in the extension of P , so is their sum.”
- (8) A predicate P is **divisive**, or downwardly homogeneous iff
 $\forall x \forall y [P(x) \wedge y \sqsubset x \rightarrow P(y)]$
 “ P is divisive just in case whenever something is in the extension of P , so are all its parts.”¹¹

Cumulativity plays a crucial role in the categorization of predicates. Mass nouns such as *mud* and plural count nouns such as *books* are cumulative since the sum of two quantities of mud is in the denotation of *mud*, and similarly for *books*. Singular count nouns are not cumulative, since the sum of two individuals in the denotation of *book* is not in the denotation of *book*. Divisiveness, however, does not distinguish between count nouns and mass nouns: count nouns are clearly not divisive, but mass nouns are not divisive either since entities in *water* e.g. a single water molecule, cannot be divided into two parts which are water (Link, 1983; Landman, 2011).

Krifka proposes a second property, quantization, as a positive property characterizing non-cumulative predicates:

- (9) A predicate is **quantized** iff:
 $\forall x [P(x) \rightarrow \neg \exists y [y \sqsubset x \wedge P(x)]]$
 “ P is divisive just in case everything in the extension of P is the sum of two non-overlapping things in the extension of P .”

Krifka argues that singular count nouns are quantized, since any individual in the denotation of *book* does not have a proper part which is also in *book*.¹² Plural count nouns and mass nouns are not quantized, since there are sums in the denotation of *books* and *mud*, which may have proper parts which are books and mud.

Krifka argues that these definitions seem to be relevant in the verbal domain too. He assumes that verbs denotes sets of events which can be assigned a mereological structure parallel to the join semi-lattice structures assigned as denotations of nominal predicates. State and activity predicates, which are categorized as atelic –

¹¹ Sometimes semanticists employ a stronger definition of divisiveness: where $\otimes$ is the relation of material overlap.

(8') A predicate P is **divisive**, or downwardly homogeneous iff
 $\forall x [P(x) \rightarrow \forall y \forall z [x = y \sqcup z \wedge y \otimes z \wedge P(y) \wedge P(z)]]$
 “ P is divisive just in case everything in the extension of P is the sum of two non-overlapping things in the extension of P .”

Unlike Krifka's definition, this rules out that there should be anything in the extension of a divisive predicate that lacks proper parts.

¹² Rothstein (2004) raises some problems with using quantization to define count nouns. For example, puzzle cubes may have cubes as parts. But this is orthogonal to the issue that we are focusing on here.

such as *love* and *run* – are non-quantized and cumulative: eventualities in the denotation of *love* and *run* have smaller eventualities of the same type as proper parts, and so are non-quantized. He argues that *run* is cumulative since two events of running can be summed as in expressed in (10):

- (10) Mary ran from 9.00 to 10.00 and from 10.00 to 11.00. So she ran from 9.00 to 11.00.

Parallel examples can be created with *love*. In contrast accomplishment and achievement predicates, which are telic, are quantized and non-cumulative. No event in the denotation of *build a house* or *arrive* has a build-a-house event or an arrival as its proper part, while two consecutive events of building a house or arriving cannot be characterized as *build a house* or *arrive*.

However, while these observations are accurate as far as they go, it is not clear that nominal and verbal 'homogeneity' are expressions of the same phenomenon. Cumulativity, as defined in (7), is a property of a predicate that is closed under summing, and is thus a property which does not distinguish between minimal individuals and pluralities of these individuals. Quantization as defined in (9) is a property of predicates which denote only minimal individuals.¹³ These are precisely the relevant contrasts in the nominal domain, where the goal is to distinguish between the properties of singular count nouns, plural count nouns and mass nouns, and to explain why plural count nouns and mass nouns share so many grammatical properties. However, the atelic/telic contrast is not a result of the interaction of verbal predicates with summing or pluralization. In fact, the examples in (11) suggest that both telic and atelic predicates are cumulative.

- (11) a. John ran. Mary ran. So (both) John and Mary ran.
b. John built a house. Mary built a house. So (both) John and Mary built a house.

In each case, the third sentence, with the plural subject, is ambiguous between a collective and a distributive interpretation. The distributive reading shows that the predicates *run* and *build a house* are closed under sum. If e_1 is the event of John building a house and e_2 is the event of Mary building a house, then (11b) shows that the sum of these two events is also in the denotation of the predicate *build a house*, and the same is true for *run* as shown in (11a).¹⁴

The contrast between atelic *run* and telic *build a house* is not then a contrast in cumulativity, and in fact, (10) on its intended interpretation is not an expression of cumulativity or closure under sum at all. What (10) shows is that single events in *run* can be turned into **bigger** singular events of the same event-type under certain

¹³ We assume that "minimal individual" is determined contextually. For the purposes of this discussion, predicates like *fence* are quantized, even though what counts as one fence in context c_1 may count as a sum of several fences in context c_2 (Rothstein, 2010).

¹⁴ This is independent of any analysis of conjunction reduction, though space concerns prevent me from discussing this here.

conditions (most usually, when the agent remains the same and there is no temporal gap between the original event and its extension.) Rothstein (2004) tries to capture this by proposing that atelic predicates allow an operation in which two temporarily adjacent events can be fused into a single event when the end point of the first is the initial point of the second.¹⁵ The condition of temporal adjacency prevents the operation from applying to telic predicates and thus correctly distinguishes activities and states from achievements and accomplishments. Landman (2008) and Landman and Rothstein (2012a, b) show that a better way to define atelicity is in terms of incremental temporal extension: a singular event in the denotation of an atelic predicate can be temporally extended while staying “the same event”.¹⁶ Assume an event e in the denotation of *run* which has a run time i , an interval, and starts at t , the initial point of i . We call any j which is a subinterval of i which begins at point t an *initial interval* of i . During the first (short) initial interval of i , the onset, we may say that the event of running has begun, but we are not yet ready to say that an event of running has occurred. Dowty (1979) explains this in his discussion of the imperfective paradox: if John has only lifted one of his legs, then while we may agree that e , the event of running, has begun, we are not yet ready to say that an event of running is underway, and thus we cannot yet assert “John is running”. After the onset of this event, hosted by the initial minimal interval of i , “John is running” is true and, because the predicate is atelic, there is also an event of running to justify the assertion “John has run”. From this point on, every instant in the running time of e is itself the final point of a growing initial interval beginning at t , which hosts an event of running. Until the event has stopped, each such point is also the non-final point in a slightly bigger initial interval. Thus as e continues, we accumulate more and more points of this kind, and thus extend the event of running continuously.

Landman (2008) and Landman and Rothstein (2012b) call an event of this kind *incrementally homogeneous* with respect to the V/VP *run*. It is this incremental homogeneity which makes the predicate atelic: there is no set terminal point, since an event in its denotation can be indefinitely extended and still remain an instance of the same predicate. In contrast, events in the denotation of telic predicates are not incrementally homogeneous. Take an event e in the denotation of *build a house*, with a running time i starting at t . Landman and Rothstein argue that the onset of e is determined by the V *build*. After an initial minimal interval hosting the onset of e , we are able to assert that “John is building a house” and that there has been an event of building, though, crucially, not an event of building a house.¹⁷ The event e is incrementally homogeneous with respect to the predicate *build*, since the event of building can be incrementally extended. However, it is not incrementally

¹⁵ For details, see Rothstein (2004), Chaps. 1 and 8.

¹⁶ While intuitively clear, the formal definition of when x and y count as the “same event” is non-trivial, and beyond the scope of this paper. Landman (2008) and Landman and Rothstein (2012b) define a notion of cross-temporal identity which is designed to deal with this issue.

¹⁷ Since “building” conventionally involves a variety of kinds of work including digging foundations and etc., it may well be the case that the onset of a building a house event may not be an event in the denotation of *build part of a house*.

homogeneous with respect to *build a house*: an event in *build a house* cannot be extended and has a single individuating final point.

While this is a somewhat rough outline of the concept of incremental homogeneity, I have included it here in order to try and make precise the contrast in the sense in which mass nouns and process verbs are homogeneous. Mass nouns and plural count nouns in English are homogeneous predicates because their denotations are closed under sum. Singular count nouns are not homogeneous because their denotations are not closed under sum. Homogeneity in the verbal domain is not related to the singular/plural distinction, but to a very different property, the property of being indefinitely extendable in time.

Note that homogeneity is relevant in distinguishing between mass nouns and count in English only because count nouns are morphologically marked as singular or plural. In a language such as Dëne Sųliné (Wilhelm, 2008) which has a mass/count distinction, there is no morphological marking of singular vs. plural count nouns, and all nouns, whether mass or count, are morphologically bare. They are also cumulative in the sense of (7), since all are closed under sum. Neither mass nouns nor count nouns are quantized, since there are no nominal predicates which consist only of minimal elements. Thus it seems that homogeneity in the sense of (7) may be a result of the singular/plural distinction in English, and not fundamental to the mass/count distinction.

On a closer look, then, mass nouns and atelic verbs do not share the property of being homogeneous since properties put down to homogeneity are in fact expressions of very different phenomena in the nominal and verbal domain.

This leads to our second question. Assume that a single definition of homogeneity cannot characterize both the mass/count and the atelic/telic distinction. Is there still good reason to argue for the second parallelism, namely that telic predicates, like count nouns, are countable while process predicates, like mass nouns, are not? I examine this question in the next section.

Are Telic Predicates Countable?

Bach uses the data in (2) above as evidence to suggest that atelic predicates, like mass nouns, are not countable while telic predicates, like count nouns are. As (2) shows, *much* and *a lot* easily modify atelic predicates as in *I sleep a lot*, or *I don't sing much anymore* but not telic predicates as in *#John didn't find a unicorn much*. Conversely, *three times*, *many times* easily modify telic predicates, and are often less natural with atelic ones. However, the distribution is very flexible. (12) shows telic predicates which are easily modified by *n times*, but (13) gives examples of atelic predicate which are equally felicitous with these modifiers:

- (12) a. John jumped three times. (semelfactive)
 b. Mary has written a book three times. (accomplishment)
 c. I have landed in *São Paulo* airport three times (achievement)
- (13) a. I have been in Brazil three times. (state)
 b. I haven't slept more than three times this week! (activity)
 c. Bill has drunk whisky three times. (activity)
 d. I only smoked once in my life! (activity)

A similar point is made in (14), illustrating modification by *often*. (14a) asserts that there was a plurality of events of being in Brazil or of running events which occurred within a relatively short time frame. (14b), which has a telic VP, asserts the same about events of landing at *São Paulo* airport.

- (14) a. John was in Brazil/ran often.
 b. John has landed at *São Paulo* airport often.

Of course, Bach is aware of these data. He suggests that the examples in (13/14a) are the result of coercion, similar to the kind of packaging operation which allows us to shift mass nouns like *coffee* to countable portion readings as in *two coffees*. So while a verb like *be in Brazil* is an atelic predicate, in (13a/14a) it has shifted to a telic reading. Note that the shifting operation in the verbal domain is very different from shifting in the nominal domain in English. Packaging in the nominal domain in English is felicitous only in 'restaurant' type situations, and the packaged portions have to be portions which are standard in a restaurant context. Neither of these constraints is necessary to count either states or activities, as (13/14a) show: all that is required is some contextually relevant way of determining what counts as 'one' single event. Once this is determined, predicates denoting both events and processes can be modified by cardinal modifiers like *three times* and *often*. In fact, shifting from process/atelic to event/telic interpretations in the verbal domain is much more like nominal mass-to-count shift in Brazilian Portuguese and Hungarian. This is much freer than in English, as described in Rothstein (2017a). Thus (13/14) do not disprove Bach's generalizations about countability.

Clear evidence that telic and atelic predicates behave differently with quantity modifiers comes from examples like (15):

- (15) a. John ran a lot.
 b. John has landed at *São Paulo* airport a lot.

In (15b), where the verb is telic, the VP only has a plural reading and the sentence asserts that there were multiple times of John landing at the airport. In (15a), where the verb is inherently atelic, the sentence can have two readings. On its atelic interpretation, it has a cumulative reading and asserts that John ran for a long time. If the verb has shifted to a telic interpretation, it has a plural reading and asserts that there were lots of events of John running. (Note that when *a lot* modifies states, the interpretation is only an expression of high intensity as in *John loves Mary a lot*.) Thus *a lot* can either be interpreted as cardinality predicate similar to *often*, in

which case it induces a telic interpretation of process predicates or it can modify process predicates with a singular atelic interpretation, when it asserts that the event under discussion lasted a long time on the temporal axis. *A little* like *a lot* can modify an atelic predicate as in *John worked a little yesterday*, but does not have a cardinality interpretation shown by the infelicitous *#John has landed at São Paulo airport a little*. So there is a real difference between singular atelic predicates, which can be modified by imprecise predicates of extent of duration, and telic predicates denoting events with a set terminal point, which cannot. These telic predicates can be modified by precise predicates of duration, as in *John built a house in three weeks* and by the imprecise predicate *a lot*, but only if it is interpreted as a cardinal modifier. *A little* can never be interpreted as a cardinal modifier.

The question remains whether this contrast is an expression of countability vs. non-countability, and it seems to me that this is still an open issue.

We have learned from research over the last twenty years (Gillon, 1992; Chierchia, 1998; Rothstein, 2010) that countability is a grammatical property distinct from individuability. What counts as an individual in a count noun denotation may be contextually determined. 'Object' mass nouns, such as *furniture* may denote sets of discrete individuals. Furthermore, object mass nouns, while not countable, may be evaluated in terms of cardinality. *A lot of furniture* naturally refers to a lot of pieces of furniture, while *more furniture* and *most furniture* may in some contexts be synonymous to *more/most pieces of furniture* (Barner & Snedeker, 2005; Bale & Barner, 2009; Landman, 2016; Rothstein, 2017a, 2021). This means that modification by a (non-numeral) predicate which operates on a cardinal dimension is not itself evidence that a predicate is count. It is possible that telic predicates are count, but it is also possible that telic predicates denote sets of individual events which, like those in the denotation of object mass nouns, are not countable.¹⁸ If that is the case then the telic/atelic distinction does not correspond to the count/mass distinction. Rather, both kinds of verbal predicate are 'mass-like' predicates, but the telic predicates are object-mass predicates and denote sets of singularities.

There is some tentative evidence that this last possibility is the correct one. First, as (12) and (13) show, neither telic nor atelic predicates can be directly modified by numerals, but rather require the use of *times* (or *twice*, which is equivalent to *two times*). Landman (2004) has argued that *times* is a classifier in the verbal domain. Classifiers can occur with both mass nouns and plural count nouns, so are not themselves evidence that telic predicates are not count. But the absence of any direct numeral modification of verbal predicates does suggest that countability is not grammaticalized in telic predicates.

Second, both telic and atelic predicates are modified by *less* which, in the nominal domain, is a comparative predicate associated with non-count predicates.

¹⁸ This is the claim made in Rothstein, 2017b, although I now think that, as phrased there, the claim is possibly too strong.

- (16) a. John has less/*fewer furniture/wine than Mary.
- b. John has fewer/*less books than Mary.
- (17) a. John ran less than Mary/fewer times than Mary.
- b. John landed at *São Paulo* less than Mary/fewer times than Mary.

The fact that count *fewer* occurs with the classifier *times* while *less* directly modifies the verb phrase suggests that the verbal predicate may not be directly countable.

In sum, Bach's observation that telic, or event, predicates correspond to count nouns and atelic or process predicates correspond to mass nouns may well be too strong. It is clear that telic predicates denote individuable events, where the criterion for what counts as an individual is given either explicitly or contextually. Whether individuality in the verbal domain corresponds to grammatical countability is still an open question.

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Chapter 3

On Barwise and Cooper's "Generalized Quantifiers and Natural Language"



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Abstract I review "Generalized Quantifiers and Natural Language", 1981, by Jon Barwise and Robin Cooper. I focus on the flurry of activity triggered by their paper in the early 1980s and then focus on what I consider the two major innovations in their paper: direct interpretation of English, and hypothesized language universals. I document some of the related work done in the decades following their paper.

Keywords Quantifiers · Generalized quantifiers · Natural language · Semantic interpretation · Semantic universals · Conservativity · (Co-)interjective · Monotone

The article mentioned above, henceforth B&C, was the one that kick-started the explicit study of generalized quantifiers in natural language. It is divided into five sections with three appendices. The article is a classic, though it hardly needs a classical review: It was well read at the time of publication and is still well cited today, 40 years later. Following shortly upon its publication there was an explosion of work on natural language quantification, some (built on work)ⁿ built on B&C, the rest independently motivated. We mention some of this work here, as it forms a consistent whole.

Early History Beginning with B&C, 1981, the first half of the 1980s saw extensive joint participation by linguists and mathematicians in the study of generalized quantifiers in natural language: Barwise a mathematician, Cooper a linguist; Higginbotham & May, 1981, a philosopher and linguist respectively, develop a section on restricted quantification for which they acknowledge the aid of mathematician Howard Stein. Mathematician Johan van Benthem and linguist Frans

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Zwarts actively cooperated in this period; van Benthem, 1982/1983, 1982/1984¹ refer to Zwarts, 1981. Mathematician Westerståhl, 1982/1984, 1982/1985 responded quickly to the B&C stimulus. Also Keenan, 1983; Keenan & Moss, 1984a, b with Keenan the linguist and Moss the mathematician. Finally Keenan & Stavi (K&S), 1981/1986, Stavi the mathematician, circulated at this time.

Already by the mid to later 1980s there were two collections of articles on generalized quantifiers: van Benthem & ter Meulen, 1984; Gärdenfors, 1987. Later collections of relevance are Kanazawa & Piñón, 1994; van der Does & van Eijck, 1996; van Benthem & ter Meulen, 1997. Some works that include overviews of this history are Westerståhl, 1995; Peters & Westerståhl, 2006; Keenan, 2011 and, the most up to date, Szabolcsi, 2010.

During this period an independent line of research on adverbial quantifiers (A-quantifiers; Partee, 1987) was initiated by David Lewis, 1975: *Ed always / usually, / never walks to work*. B&C mention the possibility of temporal quantifiers but are concerned only with Determiner quantifiers (D-quantifiers) like *every* which combine with common noun (phrases) to form NPs like *every boy*, and quantify over the domain of discourse (of a model). We return to A-quantifiers shortly.

Preliminary Notions: Sections 1–3

Section 1 of B&C introduces the concept of *generalized quantifier*, GQ, and referring to Mostowski, 1957, later to Lindström, 1966, illustrates ones in English not definable in first order logic, in particular not definable in terms of the standard universal and existential quantifiers. Their early examples are ones built from *only a finite number of*, *more than half the/of*, and *most (of)*.²

Other points to note in this section include: “quantifiers”, as B&C use the term, are GQs, denoted by phrases such as *every student*, *most teachers*, etc., which they call “Noun-Phrases” (NPs), a usage we follow here for faithfulness of presentation (today some would prefer DP, for Determiner Phrase). Given a *universe* E of objects, Nouns, like *woman*, *man*, etc. (extensionally) denote *properties*, subsets of E, and Determiners (Dets), like *all* and *most*, denote functions **all** and **most**, mapping properties to sets of properties, GQs. Thus *every doctor* denotes the set of properties

¹ Occasionally, to better capture the flurry of activity immediately around the publication date of B&C I use dates of the form 19xx/yy to mean that the publication date was 19yy but that the piece was circulated and fed the literature in 19xx. E.g. Westerståhl, 1982/1984, 1982/1985 did not formally appear until 1984 and 1985, but in fact were drawn on right after B&C in 1982. The same holds for van Benthem, 1982/1983, 1982/1984. The gap is even bigger for K&S, 1981/1986, cited as 1981 in Westerståhl, 1982/1984, 1982/1985; Keenan & Moss, 1984a, b; Thyse, 1984; van Benthem, 1982/1983, 1982/1984 cite it as 1982.

² This point is not uncommonly ignored in the linguistic literature. E.g. studies of quantifier scope ambiguities are often illustrated with just first order ones: *allevery/each, some/a, no, one, two, ..., at least one, ...*

that every doctor has. It is the value that the function **every**, denoted by *every*, maps the property **doctor**, denoted by *doctor*, to. Proper nouns also denote property sets: $[Harry]_{NP}$ denotes the set of properties that the entity Harry has (the set of subsets of E that the entity Harry is a member of).

In sections 2 and 3 B&C follow the Montagovian paradigm of defining a fragment of English, translating it into $L(GQ)$, a language whose expressions are directly interpreted in a model structure. B&C point out that in distinction to Montague's PTQ, 1974, their translation from English into $L(GQ)$ is almost trivial: Dets like *every* are directly interpreted as functions, like **every**, from properties to property sets. **Direct interpretation** is the first of the two major innovations in B&C which we discuss here, **language universals** is the second.

Direct Interpretation

The quantifiers that B&C explicitly define are: *every/all/each*, *some*, *no*, (*at least*) *n*, *n* a numeral, *exactly n*, *the n*, *both*, and *neither*, given respectively by: **every**(A) = {B | $A \subseteq B$ }. Equivalently, **every**(A)(B) = 1 if and only if (iff) $A \subseteq B$. **some**(A)(B) = 1 iff $A \cap B \neq \emptyset$, **no**(A)(B) = 1 iff $A \cap B = \emptyset$, **n**(A)(B) = 1 iff $|A \cap B| \leq n$, **n!**(A)(B) = 1 iff $|A \cap B| = n$, **(the n)**(A)(B) = **every**(A)(B) if $|A| = n$, undefined otherwise, **both**(A)(B) = **every**(A)(B) if $|A| = 2$, undefined otherwise; and **neither**(A)(B) = **no**(A)(B) if $|A| = 2$, undefined otherwise.

Note that **the n**, **both** and **neither** are properly partial functions. Most of the later literature ignores partiality, van Eijck, 1996 being a major exception. See also Muskens, 1995. Also *each*, *every*, and *all* are treated as synonyms.

Note further that these definitions do not use variable binding. The representation of *Every man sneezed*, p. 165, is simply **(every man)(sneeze)**. Variable binding and quantification are independent. *Every poet admires himself* is interpretable as **(every poet)($\lambda x.x$ admire x)**, which is true iff **poet** $\subseteq$ {b | b admire b}. So variable binding is done by the lambda operator. B&C are just concerned with quantification.

There are several immediate advantages to direct interpretation (Keenan, 1983). For one, semantic representation matches the audible structure of expressions better than approaches which translate English into standard first order logic. *Some man sneezed* and *Every man sneezed* have isomorphic semantic structures – **(D man)(sneeze)**, where **D** = **some** or **every**. In first order logic these representations are not isomorphic: $\exists x(\text{man}(x) \ \& \ \text{sneeze}(x))$ and $\forall x(\text{man}(x) \rightarrow \text{sneeze}(x))$.

A second, far reaching advantage of direct interpretation, is that using normal mathematical definitions and reasoning we can analyze known linguistic issues semantically, drawing on the lexical interpretation of quantifiers. For example B&C define a Det **D** to satisfy the intersection condition if $B \in \mathbf{D}(A)$ iff $B \in \mathbf{D}(A \cap B)$. Some intersective **D** are: **some**, **no**, **at least/at most/exactly five**, **several**. Some non-intersective **D** are **every**, **the five**, **most**, **both**. Given this sample we might guess that quantified NPs which occur naturally in Existential There (ET) contexts – *There are D students in the class* – are ones built from intersective Dets. This is not

exactly what B&C propose – they use a more complicated definition of *strong* vs *weak* Dets – but it illustrates the kind of hypothesis their semantic analysis makes possible. And crucially, the semantic nature of the definition enables us to generalize to Dets B&C did not consider. For example, **between five and ten** is provably intersective and the NPs it builds do, as predicted, occur in ET contexts: *There were between five and ten students in the building*.

Going beyond B&C slightly, observe that the acceptability of [Det+N] in an ET context is decided by the Det. If *several students* occurs naturally in ET contexts then so does *several firemen*. But holding the N constant and changing Dets may change acceptability: *There were several/*most cats in the yard*, an observation consistent with treating NPs as DPs (Determiner Phrases).

Another example: B&C define a Det to be *definite* iff for all A for which **D**(A) is defined there is a non-empty set B such that **D**(A) = {X|B⊆X}, the *principal filter* generated by B. Of Dets they consider, **both** and **the n** are definite. Of linguistic interest here is that if an NP denotation is a principal filter then there is exactly one property which generates it, so given the principal filter **D**(A) we can refer to the B which generates it, which allows us to compositionally interpret partitives like **two of the ten cats**. **The ten cats** is the principal filter generated by **cat** if |**cat**| = 10, so we interpret *of* as a function mapping a principal filter to its generator, so *of the ten cats* denotes **cat**, so **two of the ten cats** denotes the same as **two cats** if |**cat**| = 10, undefined otherwise. (B&C acknowledge that they have no explanation for **one of both students* but *one of the two students* is fine).

Again we generalize to new cases: *John's ten cats* is the filter generated not by **cat** but by **cat which John has**, a subset of **cat**, and so is definite.

A last example which also played an important role later is the definition of *monotonicity* properties. A GQ F is (*monotone*) *increasing* iff whenever $A \subseteq B$ and $F(A) = 1$ then $F(B) = 1$. F is (*monotone*) *decreasing* iff whenever $A \subseteq B$ and $F(B) = 1$ then $F(A) = 1$. Otherwise F is *non-monotonic*. For example, **John, every man, at least/more than five men, both men** are increasing. **No man, neither man, and fewer than five men** are decreasing, and **exactly five men** is non-monotonic.

Ladusaw, 1983, 1996 observed that being monotone decreasing (downward entailing) is a property that sentential negation (*John isn't singing* ⇒ *John isn't singing loudly*) and “negative” NPs (**nobody, no man**, etc.) have in common. So he generalized from Klima's (1964) observation that NEGATION LICENSES NEGATIVE POLARITY ITEMS (NPI's) – *He hasn't ever been to Paris*, but **He has ever been to Paris* – to DECREASING CONTEXTS LICENSE NPI'S: *No one here has ever been to Paris*, but **Someone here has ever been to Paris*.

Lastly, an obvious corollary of direct interpretation of quantificational expressions is that we can reason rigorously about them, expressing the results of our study as theorems. This is in practice very helpful in deciding whether a claim we want to make follows from what we've already established or not.

B&C's second innovation, which had significant repercussions in later linguistic work concerns several hypothesized linguistic universals. Their clarity of formulation enabled linguists to support or reject them. B&C themselves did not draw on any language other than English for support.

Hypothesized Universals

U1. NP-Quantifier universal: *Every natural language has syntactic constituents (called noun-phrases) whose semantic function is to express generalized quantifiers over the universe of discourse.*

Given that expressions like *John, you, this doctor* are NPs and denote GQs, U1 is surely true, but I suspect, more vacuously than B&C intended. But their third hypothesized universal U3, strengthens it considerably.

U3. Determiner universal³ *Every natural language contains basic expressions, (called determiners) whose semantic function is to assign to common count noun denotations (i.e., sets) A a quantifier that lives on A.*

Linguists have interpreted U1+U3 to imply that all languages have NPs of the form [Det+Noun]_{NP}, order irrelevant. We consider *lives on* shortly.

Today we are better positioned to assess the validity of U1+U3 than were B&C, thanks to several cross language studies of quantifiers: Bach et al., 1995; Matthewson, 2008; Keenan & Paperno, 2012; Paperno & Keenan, 2017 (KP and PK, respectively). See also Matthewson, 1998 on Salish.

KP&PK plus Suihkonen, 2007 for Finnish cover 36 languages from diverse areas and genetic affiliations. They include a dozen languages with little written tradition and fairly few speakers, as well as Russian and American sign language. The articles responded to a questionnaire prepared by the editors. All 36 languages presented some NPs of the form [Det+N], per U1+U3. So at least we can say that GQ denoting expressions of this form are well attested in the world's languages and not limited to a few areas or genetic groupings. The Overview article in PK summarizes the findings.

U1+U3 has provoked some dissension among linguists. A first, minor, issue concerns B&C's syntactic claim in U3 that the Det in [Det+N]_{NP} is "basic". Presumably basic expressions are not morpho-syntactically derived and compositionally interpreted, though B&C do not state this explicitly. Is *all* in French basic? It contains a root+ agreement morphemes for gender and number: *tout, toute, tous, toutes*? And certainly we find non-basic Dets in English: *hardly any, almost all, (not) more than five, at most half (of)*, etc. Early on B&C seem to treat *the two* and *exactly two* as non-basic.

A language whose Dets were all non-basic would be a counter-example to U1+U3. Matthewson, 2001 claims that Lilloet Salish (British Columbia, Canada) is such a language. (1) is illustrative (Note that the plural Det is discontinuous, adding to the complexity of the structure).

³ B&C's U2 is: Dislocated expressions associated with variable binding always include NPs. U2 of a different character than the other universals B&C suggest. The truth of U2 cannot be decided at this time. Derivation by movement is ubiquitous and whether it is associated with variable binding is not clear. Moreover B&C provide no analysis of variable binding.

- (1) [tákem [i smelhmúlhats-a]DP]QP
 all DET.PL woman. pl-DET
all the women

In English just how “basic” a Det is, is simply unclear. In KP&PK only about six of the languages presented an apparently monomorphemic *most*, more used a complex form like *thela majority of* (e.g. *la plupart de* in French). *La plupart de* is clearly syntactically composite, but I suspect is learned like an unanalyzed lexical item. Many apparently simplex quantifiers are historically complex: *always* = *all* + *way* (<*weg*) + *s*(genitive); *most* is historically bimorphemic, the *st* being a reflex of the superlatives *first*, *last* and *next* of *fore*, *late*, and *nigh*.

So let us weaken U1+U3 by eliminating “basic”. The result claims that all languages present GQ denoting expressions of the form [Det+N]_{NP}, where Det may be complex. This claim still produces interesting challenges.

In addition to D-quantifiers as in B&C and most later mathematically oriented work, all 36 languages studied in KP&PK presented A-quantifiers, as did several of the articles in Bach et al., 1995 as well as Matthewson, 2008.

A-quantifiers form complex predicates – adverbially, or by verbal affixing, reduplication, co-verbs, etc. They appear more diverse than D-quantifiers and their naive semantics is more complicated (e.g. involving event structure), less well understood, and probably not uniform even within a given language. Some adverbial ones are illustrated in (2) from English.

- (2) a. Ed always/usually/frequently/sometimes/rarely/never walks to work
 b. Quadratic equations usually/almost always have two solutions
 c. Always when a child feeds a dog it bites him
 d. Ned watched that film twice/n times/over and over
 e. Tom always/often/ . . . brushes his teeth before going to bed

The Ss in (2) have been treated as quantification over events, “cases” (Lewis, 1975), “select tuples” (Schwarzchild, 1989), “situations” (Berman, 1989; von Stechow, 1994) and “times” (de Swart, 1996). Sometimes A-quantifiers seem to assimilate to D-quantifiers, as in (3) adapted from Bittner & Trondhjem, 2008:

- (3) a. *All* good restaurants are expensive
 Good restaurants are *always* expensive
 b. *Most* women voted for Reagan
 Women *mostly* voted for Reagan

Note that U1+U3 in no way denies the existence of A-quantification, as the English data show. Equally Bittner, 1995; Evans, 1995 exhibit enlightening examples of A-quantifiers in verbal morphology in Eskimo and Mayali (Australia), languages for which they also attest D-quantifiers. And Bicevskis et al., 2017 cite both D- and A-quantifiers in Gitksan (N.W. American coast; areally and typologically related to Salish languages, cited (below) as lacking D-quantifiers).

But to my knowledge, no author cites languages lacking A-quantifiers. So it would seem possible that a language have only A-quantifiers. But are there any clear examples? We consider four candidates that linguists have proposed.

Lee, 2008 doubts U1+U3 in San Lucas Quiavini Zapotec (SLQZ). Lee's data show that several pronominal quantifiers such as numerals and the universal quantifier may, but do not always, carry verbal morphology, such as irrealis, habitual, and future perfect. But their meanings change from those they have on verbs: irrealis now codes exhaustivity, habitual now marks numerals as ordinals, etc. Some borrowed quantifiers, such as *caad* (< Sp. *cada* 'each') do not take verbal morphology at all.

I do not count Lee's observations as sufficient counterexamples to U1+U3, but they do serve to make it more explicit: U1+U3 should not be taken to imply that Determiners in all languages look or behave the same syntactically. Any *x* that combines with a property denoting expression to form a property-set denoting one will do. [Det+N]s in the closely related English, German, Russian and Italian differ in case selection and person/number/gender agreement. So the fact that several Dets in SLQZ are verbally derived does not invalidate U1+U3. But it does suggest the basic nature of A-quantifiers in that language.

It is also helpful to realize that within English, distinct expressions of the form [Det N]_{NP} exhibit different syntactic behavior: Only some occur naturally in Existential There Ss (*There are several/*most cats on the mat*), only some follow of in plural count partitives (*Two of [the ten men/*no men]*), only some host negation (*Not all/*Not some students like poetry*). B&C provide some semantic analysis in these cases. So the GQ picture in B&C is "rough and ready", open to much refinement.

Jelinek, 1995; Vieira, 1995 argue against U1+U3 for Salish languages (N.W. coast, N. America) and Asurini do Trocará (Tupi-Guarani, Brazil) respectively, claiming that basic quantification in each case is A-quantification. Both claim that their languages are of the pronominal-argument (pro-arg) type, established by Jelinek. The defining criterion of that type is that in nuclear Ss the verbs carry referential affixes that satisfy their argument requirements. Verbs with these affixes can stand alone as truth bearing expressions when the referents of the affixes are clear from context, as in *She likes him*. Full NPs are not directly arguments of predicates but optional adjuncts that bind or corefer with these affixes. Faltz, 1995 suggests that the fundamental interpretative mechanisms in pro-arg languages – matching referents, might differ from those of English-like languages which involve "filling gaps", i.e. providing arguments for functions. Baker, 1995 carried this idea further (below), illustrating his claims with Mohawk. We consider these claims in more detail.

To get a feeling for pro-arg languages, consider first more familiar examples from French below, clearly not a pro-arg language, despite the predicate in (4a) being marked for third person singular feminine:

- (4) a. Marie est_{3.sg} heureuse_{fem.sg} "Mary is happy"
 b. *[_S est heureuse] "She is happy"

But there are S-types in French with extensive Right Dislocation which seem pro-arg in character, as (5) from Céline's *Voyage au Bout de la Nuit*:

- (5) a. Elle l'aime bien Marie sa mère b. Elle l'aime bien
 She her'likes well Mary her mother She her'likes well
 Mary likes her mother *She likes her*

Were it not for the phonology (5a) could pass for Salish, (6):

- (6) q'əq'nəɫ = Ø 'aɫ š-š ə t-ŋ-Ø cə s'eləx^w (Jelinek, 1995)
 slow = 3abs Conj sub-walk-middle-3abs Dem elder
 He_i is slow when he_i walks the old man_i

Now Jelinek claims that quantification in Salish is A-quantification. Strong quantifiers are adverbial, linked to a predicate-argument structure by an overt linker, as in (7a). Weak quantifiers are simply predicates, as in (7b). (The terms *strong* and *weak* are from B&C: D is *strong* iff $A \in \mathbf{D}(A)$, all relevant A, or $A \notin \mathbf{D}(A)$, all relevant A. D is *weak* iff D is not strong).

- (7) a. mək'^w = ɪ 'əw' ŋa-t-Ø cə sceenəx^w
 all = 1p.nom link eat-tr-abs dem fish
 We ate all the fish or *We all ate the fish* or *We ate the fish all up*
 b. ŋən' = Ø cə leŋ-n-ən
 many = 3abs dem see-tran-saw
 They are many, the (ones) I saw

Jelinek considers that Ss like (7a) exemplify unselective binding (Lewis, 1975). Vieira makes comparable claims for Asurini. Jelinek considers five other quantifier(-like) expressions, translating *again*, *always*, *almost*, *very/too*, and *never/not yet*, implying by absence that no others are attested. Vieira mentions about ten quantifiers for Asurini.

How strong is the case that Straits Salish and Asurini are counterexamples to U1+U3? I have one mundane reservation (below), but overall Jelinek and Vieira have shown us what a language might be like if it lacked [Det+N] constructions but could still express quantification. And they document languages which use productively this type of quantification. Jelinek draws on B&C and understands what she is providing an alternative to. Von Fintel & Matthewson, 2008 cite these cases and some others as well. So I regard this as a nice case where formal work has prodded linguists to support or refute an understandable claim and it has led us to notice a novel structure type.

The reservation I have is simply that as a matter of good science, we need more documentation glossed and translated so as to be accessible to the general linguistic community. Recall: U1+U3 are fully consistent with the existence of A-quantifiers, they only claim that all languages have Det+N quantifiers. We'd like to know for example what happens in Salish when we try more complex quantifiers: *between*

five and ten, most but not all, all but two, more than/less than/exactly/about ten/half, only a few or else almost all, seven out of ten, more male than female, etc.? If these are new to speakers will they generalize the A-quantifier pattern? (We expect that they would).

Linguists (present company excluded) tend to treat quantifiers as a small closed class. Yet many, as just illustrated, are morphosyntactically complex. Keenan, 1983, 1996 lists 13 classes of English Dets, 12 of which are infinite.

Equally, we'd like an explicit compositional semantic interpretation for A-quantifiers. When we interpret *every* as **every** we are being clear and explicit. We see, easily, that the GQs it builds are monotone increasing. Those built from **no** and **fewer than five** are decreasing, whence they license NPIs.

Can we treat A-quantification comparably? If we interpret A-quantifiers as adjuncts coreferential with pronouns do we have in effect expressive power comparable to that of D-quantifiers? We need an explicit formal semantic analysis. Jelinek suggests unselective binding à la Lewis, 1975 but nothing is worked out. Later work however, e.g. Berman, 1987, 1989; von Stechow, 1994, drawing on Heim, 1982, 1990; Rooth, 1985; Schubert & Pelletier, 1987; Kratzer, 1989 inter alia, moved away from unselective binding in favor of quantification over "situations" (naïvely understood, not as in situation semantics à la Barwise & Perry, 1983). Many linguistically insightful observations were made concerning how to determine the domain of quantification in sentences with elements in focus, and variation in the domain of quantification with indefinite subjects and differing temporal adverbs. This work is linguistically far more fine grained than that in B&C and its successors. A little later yet Schwarzschild, 1989; de Swart, 1991, 1996 provide A-quantifiers with expressive power comparable to D-quantifiers. Both treat them as generalized quantifiers, over tuples in Schwarzschild, times in de Swart, not as coreference (or binding) between pronouns and adjunct nouns. They both treat temporal subordinate clauses with *when*, and de Swart with *before* and *after*: *Before he goes to bed Paul always takes a shower*. They show convincingly that several of the properties of D-quantification – conservativity, monotonicity, intersectivity apply in the temporal domain as well.

Lastly, let us consider Baker's strong claim that the absence of properly quantified NPs in pro-arg languages is principled, that is, it follows from the nature of the language as pro-arg.

Baker initially considers a constraint to the effect that pronouns cannot be locally bound by a quantifier ("presumably at LF"). The motivation (my examples, originals from Italian) comes from sentences like (8).

- (8) a. He_i's out of work again, my father;
 b. *He_i's out of work again, everyone_i / nobody_i / someone_i

So a referential NP like *my father* can corefer with a pronoun in dislocation contexts but a properly quantified NP cannot. Verb internal referential affixes in pro-arg languages are treated as comparable to pronouns in dislocation structures in English (Italian, ...).

The contrast in (8a, b) is enticing, but Baker eventually abandons this line of attack for reasons related to ellipsis and to binding by indefinite NPs, reasonably considered existentially quantified. He opts ultimately for something he calls “the so called ‘pragmatic coreference’ holding between pronouns and NPs that does not depend on c-command and is not limited to a single sentence”. It is illustrated by pairs like (9a, b).

- (9) a. The woman who saw John_i / someone_i in the garage said he_i ran away
 b. *The woman who saw everyone_i / no one_i in the garage said he_i ran away

The judgments in (9) are good, but the “pragmatic coreference” condition is unsatisfyingly open ended. Baker considers that it covers cases like (10):

- (10) a. Juma_i a_i.li.m_j.piga Haruna_j b. [S alimpiga] Swahili
 Juma he.pst.him.hit Haruna He hit him
Juma hit Haruna

But this pattern seems not to generalize in several pro-arg Bantu languages. The examples in (11), from Zerbian & Krifka, 2008 who also provide comparable ones from N. Sotho, show that quantified NPs, including universal ones, both plural *all* and singular distributive ones *every/each*, can bind verb internal referential affixes. (The odd numbered noun class markers are singular, the even numbered ones plural. A Bantu language will have 12–20 such class markers. The ppx gloss refers to a pre-prefix, which plays some (language dependent) role in definiteness.)

- (11) a. Kila m-toto a-na-cheza na mpira wake Swahili
 every cl.1-child sc1-pres-play prep cl.3.ball cl.3-his
Every child is playing with his/her ball
 b. Wa-toto w-ote wa-na-cheza nje
 cl.2-child ppx2-all sc2-pres-play outside
All children are playing outside
 c. Wa-toto kadhaa wa-na-cheza nje
 cl.2-child some cl.2-pres-play outside
Some/Few children are playing outside
 d. Idadi kubwa y-a wa-toto wa-na-cheza nje
 cl.9.part cl.9.big ppx9-of cl.2-child cl.2-pres-play outside
The majority of children are playing outside

We note that both *kila* and *-ote* have distributive uses, but only *-ote* occurs with collective predicates, such as *play together*. Zerbian and Krifka also cite A-quantifiers in Swahili with properly adverbial functions.

The Bantu data then block any easy inference that pro-arg languages must lack universal type D-quantifiers. I’m inclined to think that languages may vary more or less continuously from prominent A-quantification as in Salish and Asurini to the relatively more prominent D-quantification in French and English. Bantu languages

are plausibly intermediate though closer to English than Salish. Faltz, 1995 suggests that Navaho and Lakhota are also intermediate, if closer to the Salish/Asurini endpoint.

Properties of Dets

B&C's U3 says that Every natural language contains basic Dets which assign to a property A a quantifier that *lives on* A. That is, for A, B properties, the truth of $D(A)(B)$ just depends on (1) A and (2) the As that are Bs: $D(A)(B) = D(A)(A \cap B)$. This property sounds algebraic, but it just says that to check whether Det poets daydream we need only consider individuals who are poets and which of them daydream (from which we can compute which don't). We can ignore daydreamers who are not poets. So the role of the Noun property is to determine the local domain of evaluation. Thus (12a) and (12b) are logically equivalent.

- (12) a. Det As are Bs b. Det As are As that are Bs

The redundancy in (12b) results from the fact that it repeats information given in the noun. That is, the Det lives on its argument.

Most of us who worked on quantifier semantics at this period noticed, in one way or another, the *lives on* property. Higginbotham & May, 1981 called it *intersectivity*. Keenan, 1981 called it *conservativity*, the term in general use today.⁴ Keenan & Stavi went on to support that over a finite universe all conservative functions are denotable (They count as Dets expressions like *most female* in *All male and most female doctors* and *John's five* as in *John's five dogs*) so no stronger constraint can obtain.

Worth noting is that conservativity might (wrongly) appear trivial – Does it not merely guarantee that *Det birds fly* if and only if *Det birds are birds that fly*? But in a universe E of n entities, the number of functions from pairs of subsets of E into {0,1} is 2 raised to the power $2^n \times 2^n$. So if $|E| = 2$, there are $2^{16} = 65,536$ functions from pairs of properties to truth values. Only 2 to the $3^n = 512$ are conservative! So for even small universes conservativity rules out most logically possible Det denotations.

Conservativity is a candidate for a non-trivial semantic universal. Are there no Dets that fail to be conservative? One example (the clearest) was noted early on (Johnsen, 1987), namely *only*. **Only** is the converse of **all** (de Swart, 1996): $\text{only}(A)(B) = \text{all}(B)(A)$. *Only poets daydream* iff *all people who daydream are poets*. If *only* were conservative then for all A, B $\text{only}(A)(B) = \text{only}(A)(A \cap B) = \text{all}(A \cap B)(A)$, which is always true since $A \cap B \subseteq A$ (*All people who are poets and daydream are poets*). But *Only poets daydream* is not

⁴ Keenan & Faltz, 1985; Peters & Westerståhl, 2006; Bastiaanse, 2013; Romero, 2015; Ahn & Sauerland, 2017; von Fintel & Keenan, 2018; Zuber & Keenan, 2019.

logically true. Ahn & Sauerland, 2017 find bare proportionality quantifiers applied to the object argument of two place predicates to be systematic counterexamples to conservativity. *GM hired fifty percent women (last year)* is illustrative.

We discuss this generalization shortly. Here we note only that Zuber & Keenan, 2019 find a weakening of conservativity which covers these cases and is still very strong. We say that **d** is *weakly conservative* iff **d** is cons_1 : $\mathbf{d}(A)(B) = \mathbf{d}(A \cap B)(B)$, or **d** is cons_2 : $\mathbf{d}(A)(B) = \mathbf{d}(A)(A \cap B)$. So cons_2 is *conservativity* as previously used. So *weak conservativity* is just *conservative on at least one argument*. It turns out that *only* and bare proportionality quantifiers are provably cons_1 .

Weak conservativity is still a very strong constraint. E.g. of the 65,536 functions from pairs of subsets of E into {0,1}, only 512 were conservative (cons_2) and only 1008 are weakly conservative.

Conservativity (cons_2) also enables us to see how interrogative Dets (not discussed in B&C) such as *which?* and *how many?* behave like declarative Dets. An answer to *Which students passed?* is also an answer to *Which students are students who passed?*, and conversely, so questions can be logically equivalent in the sense of requesting the same information. And the interrogative Dets *which?* and *how many?* are conservative (cons_2).

B&C's U4

B&C's U4 makes two claims: (1) if a Det function D is not defined for all sets A, then for those A for which it is defined, D(A) is not trivial (true for all B or false for all B). And (2) if D is not defined for all A, then there is another Det defined for all A which agrees with D on the A for which D is defined.

U4 is of limited semantic import: it does yield that if a language has a *both* then it has its completion *all*, but, so far, all languages with D-quantifiers have at least one universal quantifier so the claim is true but has little empirical bite. And U4 also yields that the presence of *neither* implies that of *no*. But very little is known cross linguistically about *neither* as a Det (*Neither boy cried*). From KP&PK we see that few languages have a lexical *no* (only Germanic Ls: English, German, Danish, ... with Romance borderline⁵).

⁵ In French *aucun* when used as a subject of a complete sentence still requires an accompanying *ne* and so is not (?yet) a stand alone Det like *tout* 'all'.

Aucun étudiant n'est arrivé à l'heure
no student NEG'is arrived at the'hour
No student arrived on time

In contrast Spanish *ningún* 'no' is used alone with a preverbal subject (but still requires a verbal negation when used postverbally). Thanks to Louise McNally and Nancy Gutierrez for the observation.

Also here B&C claim here that no Det is trivial (i.e. $D(A)(B)$ is always true or always false). But this is false. (**Fewer than zero**)(A)(B) is always false and (**at least zero**)(A)(B) is always true, though these Dets are not among those B&C would call basic.

Weak vs Strong NPs

B&C's next two proposed universals build on the weak/strong distinction noted above and monotonicity properties, which, in my opinion, run quite deep. These properties are purely semantic, defined in terms of the (mathematical) objects that Dets and NPs denote.

B&C use their weak/strong distinction to support an analysis of existential *There* (ET) sentences of the form *There is/are Det+N*, which they hold to be true iff the existence property E is in the property set denoted by the NP (the N may be complex, e.g. *There are five dogs in the yard*). Now if the Det is positive strong, then the sentence will be a tautology since $\mathbf{d}$ is assumed conservative so $E \in \mathbf{d}(A)$ iff $E \cap A = A \in \mathbf{d}(A)$, which is so since $\mathbf{d}$ is positive strong. By similar reasoning $A \notin \mathbf{d}(A)$ if $\mathbf{d}$ is negative strong, so the ET sentence is logically false. But if $\mathbf{d}$ is weak then the sentence might be true or it might be false, hence non-trivial. This was a novel analysis of the definiteness effect in ET sentences (see Milsark, 1977), and didn't rely on an undefined notion of "indefinite".

In a similar spirit B&C here also characterize *definite* Dets in terms of properties of their denotations: $\mathbf{d}$ is *definite* iff for each A for which $\mathbf{d}$ is defined there is a non-empty set B such that $\mathbf{d}(A)$ is the set of supersets of B. NPs built from definite Dets occur naturally in contexts like *two of ____* and *of* can now be interpreted as a function mapping such DPs to the unique property B which generates them. This B will often be the overtly given argument of the Det in simple cases like *two of the five boys*. But in possessive cases, not considered in B&C, like *two of John's five cats* it will be **cat which John has**. Definite Dets are positive strong ($A \in \mathbf{d}(A)$), but the converse may fail. *Every* is positive strong but not definite since $\emptyset \in \mathbf{every}(\emptyset)$, so there is no non-empty B such that $\mathbf{every}(\emptyset)$ is the set of supersets of B.

Ningún estudiante vino

no student came

No student came

Monotonicity

B&C were the first to import monotonicity from logic to a linguistic context, though Fauconnier's, 1975, 1978 *scale reversal*, and Ladusaw's, 1979 *upward and downward entailing*, were rediscovering the concept linguistically.

Van Benthem, 1984 generalizes these notions to all arguments of a Det. **all** and **no** are $\downarrow$ on their first argument: *All/No poets daydream* implies *all/no romantic poets daydream*. But they differ in that **all poet** is $\uparrow$ but **no poet** is $\downarrow$. So **all** can be classified as $\downarrow\text{all}\uparrow$ and **no** as $\downarrow\text{no}\downarrow$. B&C call *persistent* (*anti-persistent*), a quantifier that is increasing (decreasing) on its first argument.

Monotonicity properties have played a significant role in the linguistic discussion of negative polarity items (Ladusaw, 1983, 1996). Subject NPs which are decreasing license NPI's in the predicate: *No students/Fewer than five students here have ever been to Pinsk*. Replacement by increasing NPs is ungrammatical: **Some/*All students here have ever been to Pinsk*. Further, the monotonicity of [Det N] is decided by the Det. If [D student] is decreasing so is [D teacher], a fact which supports treating NPs as DPs, Determiner Phrases. And we note finally, since **no** and **every** are decreasing on their argument they should license NPIs there, and they do: *No/every tourist who has ever been to Pinsk has loved it*.

B&C's U5

*There is a simple NP which denotes the mon $\downarrow$ quantifier $\sim Q$ iff there is a simple NP with a weak non-cardinal Det that denotes the mon $\uparrow$ quantifier Q . ($\sim Q = \text{def } \{B \subseteq E | B \not\subseteq Q\}$). We can think of $\sim$ as denoted by negation (*not*). B&C give as an instance of U5 that the existence in English of *no man* implies that of *some man*. The implication holds but *some* is cardinal: *Some boy cried* iff *At least one boy cried*. **Some**(A) = $\{B \subseteq E | |A \cap B| \geq 1\}$. The truth of **some**(A)(B) is decided by the cardinality of $A \cap B$. Also, contrary to what they claim, *neither man* does not denote the negation of *a/some man*. *Either man* would be a better choice (though its distribution is limited: *Either student could do it,? *Either student did it*).*

On the other hand their claim that U5 predicts the absence of basic Dets meaning **not most**, **not every**, **not the** is good since the corresponding NPs built by the unnegated Dets are strong. Why should this be so? Furthermore, later (p. 197) B&C offer another generalization concerning negative NPs:

- (13) Only increasing NPs host negation (*not*)

Going beyond the NPs B&C consider, we have: *not every/not a single/not one student, not all/not more than five/students passed the exam*. But **not few student(s), *not at most two students, *not no students (passed)*. B&C note that not all increasing NPs do host negation: **not some student(s), *not each student*. B&C

intend (13) to hold for simple NPs and there are some complicating factors when more complex ones are considered.

Again worthy of note (not present in B&C) is that whether an NP of the form [Det+N] hosts *not* is decided by the Det. Given that *not every poet* is grammatical, we can replace *poet* by pretty much any singular count N: *not every teacher*, *not every blond ski instructor*, etc. preserving grammaticality. But we cannot do this freely with the Det. This is supportive of the view that what B&C call NPs are headed by the Det, not the N.

B&C's U6

The simple NP's of any natural language express monotone quantifiers or conjunctions of monotone quantifiers. This is clear and to the best of my knowledge, true. It rules out simple NPs with e.g. the denotation of *either very few or very many students*. And we can almost eliminate "*conjunctions of monotone quantifiers*". But some consider that bare numerals, as in *two men*, mean the same as *exactly two men*. This latter is not monotone but means the same as the conjunction *at least two men and not more than two men* (NP in the second conjunct is not "simple"). I would prefer to define a *meet* operation, $\wedge$, on NP denotations: $(F \wedge G)(B) = F(B) \wedge G(B)$. Here the meet on the right is just the meet of truth values, given by the truth table for **and**. Note that in stating U5 B&C rightly distinguished the syntactic notion of negation from the $\sim$ operation on NP denotations; now we further distinguish syntactic conjunction from semantic meet. U6 can then be given as the stronger *The simple NPs of a language express (meets of) monotone quantifiers.* (Perhaps B&C would also allow joins of monotone quantifiers).

B&C's U7

Positive strong Dets are increasing, negative strong ones are decreasing. B&C point out that this predicts that Ss like (14a) are judged valid, whereas those in (14b) are not – their truth varies with the model.

- (14) a. Most/Both/The three men that love Mary love Mary
b. No/Some/At least three men that love(s) Mary love(s) Mary

We can improve slightly on the force of U7. B&C intend it to be an empirical claim, and give an example of a function of the right type which is positive strong but not increasing, namely $\mathbf{d}(A)(B) = 1$ iff $|B - A|$ is finite and even. However $\mathbf{d}$ is not conservative (For $B = \{a, b\}$ and $A = \{a\}$, $\mathbf{d}(A)(B) = 0$ but $\mathbf{d}(A)(A \cap B) = 1$). So it is not obvious if U7 is a new empirical constraint or if it follows from conservativity. In fact it is a new empirical constraint. The function $\mathbf{D}$ given by: $\mathbf{D}AB = 1$ iff $A \subseteq B$

or $(A = E \ \& \ B = \emptyset)$ is, provably positive strong, conservative, and not increasing (for E with at least two elements a, b we have $\mathbf{D}(E)(\emptyset) = 1$ but $\mathbf{D}(E)(\{a\}) = 0$).

Similarly B&C's **U8** *all persistent determiners are $\uparrow$ and weak* is empirically significant. To see this consider the functions denoted by **at least n of the m or more**, $n \leq m$. Consistent with B&C's earlier usage such functions are undefined if $|A| < m$. Such functions are persistent, conservative and increasing but not weak, indeed they are positive strong. B&C would presumably not accept them as Dets, but it does show that there are functions which are persistent but not weak, so U8 is empirically significant.

Duals and Negations

For F, G quantifiers, F is the *dual* of G iff for all sets B , $F(B) = \sim G(\sim B)$, and we write $F = \sim G \sim$. F is *self dual* iff $F = \sim F \sim$. Basically just individuals (e.g. proper noun denotations) are self dual (Zimmermann, 1993). B&C's **U9** claims that decreasing NPs and self dual ones do not accept overt negation: *not every/not one man left*, but **not no man/*not John/*not the one man left*. **Some**(A) and **every**(A) are duals, and by extension so are **some** and **every**. B&C's **U10** claims that **some** and **every** seem to be the only duals expressible by basic Dets. If F and F' are duals so are $\sim F$ and $\sim F'$, so **no** and $\sim$ **every** are duals, though *not every* is not a basic Det (see Horn, 1989: 256). B&C also note that **at least half** and **more than half** are duals but regard them as “mathematical” and so presumably not basic, which I find unsatisfying. 21 of the 36 Ls presented in KP&PK present a lexical *half*, which, recall is not a first order quantifier. Note too that at the level of NP, *both Mary and Sue* and *either Mary or Sue* are duals (as are *and* and *or* at the level of conjunction).

Interim Conclusion

This concludes our review of section 4. Section 5 is a brief reflection by B&C on more general lessons their work supports. The direct mathematical modeling of semantic objects is a major one. It enables us to observe and study many clear and rigorous generalizations about natural language quantification. This is pursued a little further in the Appendices A, B, and C. Appendix A just extends their formal language to include ET sentences, partitives, and some boolean compounds of NPs. B suggests a few conditions on some “non-logical” Dets (below), e.g. **many** is $\uparrow$, etc. And C covers proofs of some of the theorems given in section 4. Most of these are straightforward, with the exception of the non-first order definability of **more than half**. Appendix C exhibits one of the signal merits of B&C's approach B once we have provided a mathematical model we can study the model, see what follows from it (i.e. prove theorems about it) and come to understand better the phenomena under study.

This it seems to me is a huge merit of B&C's work. It also highlights a shift of focus in linguistic semantics, at least within generative grammar. Linguists had tended to focus on syntactic relations between expressions and then comment on their semantic relations: e.g. active and passive Ss (with individual denoting NPs) have the same meaning, or Sentential Extraposition preserves meaning, as in *That Jan left early is strange* and *It is strange that Jan left early* are true in the same conditions. But the fact that *at least four* and *at most four* build NPs of differing monotonicity is (initially) a mere lexical difference, not an obvious syntactic one, just as *Ned can read and write Chinese* and *Ned can read or write Chinese* differ lexically but seem syntactically comparable though their truth conditions and entailments are quite different. This shift in focus was recognized at the time: Van Benthem's "Determiners and Logic" 1983 opens with "A new type of logical semantic enquiry is emerging..." and cites B&C as well as already available work by himself, Keenan & Stavi, Westerståhl, and Zwarts, all with dates 1981 or 1982.

Post B&C

We note first, three topics that grew directly out of the work in the early 80s.

An immediate development was a precise characterization of what constituted "logical" Dets. What distinguishes Dets like *every*, *not every*, *some but not all* from ones like *more male than female*, *John's two or more* B which K&S treated as Dets and which are obviously not logical constants? Their denotations can vary in models with the same universe.

Explicit definitions of logicity were given in van Benthem 1982/1983: 463; Westerståhl 1982/1985 in terms of *permutation invariance* (PI): A Det **d** is PI iff for each permutation π of the universe E , $\mathbf{d}(A)(B) = \mathbf{d}\pi(A)\pi(B)$, all $A, B \subseteq E$. ($\pi(A) =_{\text{def}} \{\pi(a) | a \in A\}$). So $B \in \mathbf{all}(A)$ iff $\pi(B) \in \mathbf{all}(\pi(A))$. That is, it doesn't matter which individuals have which properties. "Logical" Dets denote PI functions. Keenan, 1983 characterized logicity in terms of *automorphism invariance* (AI) which focused on preservation of the boolean structure of denotation sets, including that for S. AI was extensionally the same as PI. Isomorphism invariance generalizes this notion to the case where π above maps the universe of one model bijectively to that of another (see Peters & Westerståhl, 2006: Ch 9).

Interestingly linguists distinguish, without a precise definition, what they call *functional* expressions from others, a distinction that corresponds roughly to the "logical"/"non-logical" distinction just made (see Keenan & Stabler, 2003). So this is a case where mathematical formulation has enabled us to clarify a notion that we have pretheoretical intuitions about. And we may now distinguish "having a constant interpretation" and always interpreted as "logical". Dets like *most* and *several* are plausibly not constants: their denotations may vary (a little) with the model. But it is plausible that all their denotations are PI/AI. E.g. we may interpret *several* as **at least n** ($n > 2$) where n can vary with the model, but all these functions are PI/AI. So *most* and *several* may be PI/AI but not constantly interpreted,

accounting for their mathematical character without requiring them to be logical constants. In contrast, a constantly interpreted expression must denote an AI object (K&S).

A second perspective shift that logicians find natural is abandoning the fixed universe assumption. So Dets D can be thought of as “functionals” that associate with each universe E a function $\mathbf{D}_E$ of type $(et, (et, t))$. Then we can compare denotations of a given Det in different universes E, E' and impose a universe independence condition (called ‘extension’ in van Benthem, 1983):

Universe Independence: For $A, B \subseteq E, E', \mathbf{D}_E AB = \mathbf{D}_{E'} AB$

So a given Det must assign the same value to a given pair of sets regardless of the universe they are subsets of. This notion is commonly used in computer science (Abiteboul et al., 1995). And now conservativity can be given a global formulation, Westerståhl, 1985, which in fact implies Universe Independence:

Global Conservativity: For $A, B \subseteq E, \mathbf{D}_E(A)(B) = \mathbf{D}_A(A)(A \cap B)$

This makes more explicit than our previous local version that the truth of $[[\text{Det} + N] \text{ VP}]$ can be decided by limiting ourselves to the set of things the N denotes. This *global* perspective carries over to other of our earlier notions as well, e.g. Automorphism Invariance becomes Isomorphism Invariance.

Third, Keenan & Westerståhl, 1997 show how to directly interpret NPs when they occur as objects of transitive verbs. E.g. in *John admires all poets* we see that *admires all poets* is interpreted as the set of x which are such that **all(poet)** holds of the set of y that x admires. More generally NPs are arity-1 reducers: they map $n+1$ place relations to n -place ones. The value of a GQ at an $n+1$ -ary relation is determined by its values on the unary ones, the subsets of E , so it can still be given type $((et), t)$. Keenan, 2015 compares this direct interpretation approach with a Quantifier Raising one.

But also many new types of quantifier have been treated. E.g. *more...than...* in *more students than teachers signed the petition* has type $((et, et), (et, t))$, so it maps a pair of properties to a generalized quantifier (see Beghelli, 1994). Also the composition of two ordinary GQs can be treated as a quantifier of type $((et, t), t)$, a function mapping binary relations to truth values. Keenan, 1992 proves that the subject-object pairs in Ss like (15a–c) are inherently of type $((et, t), t)$ and are not the composition of any pair of GQs. See Westerståhl, 1994 for extensive discussion.

- (15) a. Different people like different things
 b. Each student answered the same questions (on the exam)
 c. Ed danced with Sue but *no one else danced with anyone else*

Also Peters & Westerståhl, 2006 discuss inter alia *resumptive* quantification, where we quantify over tuples of objects. E.g. they would treat *Men are usually taller than women* by comparing the set of pairs $\langle x, y \rangle$ with x a man, y a woman and x taller than y , with the full set of pairs **man** $\times$ **woman**. So this would be a case where an A-quantifier was interpreted as a D-one.

We conclude by noting that we have focused on the mathematically original parts of B&C. There is more purely mathematical work in this area, e.g. Krynicki et al., 1995. Also there is significant more formal linguistic work on natural language quantification. Westerståhl, 1994 studies iterated, branching and cumulative quantification. Winter, 2001 treats of collective nouns and plural quantification within a boolean perspective. Link, 1987, 1998 also uses algebraic methods to treat of collectivities and plurals, as well as events and mass terms. B&C and the work we have discussed only focus on individual level not group level quantification. As we have noted, B&C do not distinguish *all* and *each* but, as is well known, *all* combines naturally with collective predicates while *each* does not: *All (of) the students gathered in the courtyard* vs. **Each (of) the students gathered . . .*. B&C treat the two and both identically but Ladusaw, 1985 distinguishes *one of the two books* and **one of both books*. Finally Hackl, 2009 distinguishes semantically between *At least two men shook hands* and *#More than one man shook hands*. We may also cite Reuland & ter Meulen, 1987, with several articles on existential *There* sentences in which analyses richer than that in B&C are proposed.

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Chapter 4

On Carlson’s “A Unified Analysis of the English Bare Plural”



Donka F. Farkas

Abstract This chapter discusses Carlson’s classic paper on the English bare plural, which proposes a unitary account for the quasi-universal and the existential readings of bare plurals in English (*Dogs are easily trained* vs. *Dogs are sitting in my back yard*). The summary points out the empirical basis for this unitary analysis, the structure and force of the argumentation, and the novel semantic tools this the account employs. The last part of the article provides a synopsis of the reactions to this proposal, especially in terms of its limitations when the empirical horizon is widened to include languages other than English, where bare plurals only have the existential reading.

Keywords Bare plurals · Generics · State/individual level predicates

Introduction

It is rare for a paper to make an ‘empirical breakthrough’, i.e., bring upon the linguistic scene rich data whose account reorients a subfield for a long time to come. It is also rare for a paper to make a ‘theoretical breakthrough’, i.e., propose novel analytical tools that become essential in accounting for a large number of empirical generalizations. The paper under discussion here, Carlson (1977a), does both.

This classic article should be read and re-read for both *what* it does, and for *how* it does it. Starting with the second reason, this piece illustrates a model semantic argument constructed with one’s ear close to the facts of language and one’s mind focused on the theories and analytical tools that allow us to carve a path through the empirical jungle. Turning to the first reason, in brief, *what* this paper does is: (i) raise a new question; (ii) marshal a host of empirical arguments for a particular account in the course of which it brings up empirical generalizations that will remain central in

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the field for a long time to come; (iii) give an account of the facts presented clearly and succinctly, an account that employs novel analytical tools whose use goes well beyond the immediate problem at hand.

Below, I will summarize the main claims and main arguments in the paper (Section “[Summary](#)”), and then briefly discuss these claims, and give a necessarily partial overview of the paper’s long and rich ‘afterlife’ (Section “[Discussion](#)”).

Summary

The Puzzle and the Main Claim

The starting point of the paper is the observation that, in English, bare plurals (article-less plural nominals) have both generic uses, exemplified in (1), and ‘indefinite’, or existential, uses, exemplified in (2)¹:

- (1) *The generic reading*
 - a. Horses are widespread. [2a]
 - b. Horses are mammals. [1a]
 - c. Horses are good pets. [1b]
- (2) *The existential reading*²
 Doctors tried to save the dying boy. [3a]

The generic reading itself is heterogeneous. Thus, in (1a), the bare plural is not quantificational, in (1b) it is akin to a universal, while in (1c) it is interpreted as an ‘exception tolerating’ universal. Thus, (1c) is judged true even if, in fact, there are quite a number of horses that are not good pets.

Under the existential reading, on the other hand, the interpretation of the bare plural is close to that of an indefinite plural: (2) can be paraphrased by (3):

- (3) There were some doctors who were trying to save the dying boy.

Such an indefinite paraphrase is unavailable for the generic bare plural.

The initial puzzle, then, is how to capture the interpretive versatility of bare plurals in English.

As its title suggests, the main claim of the paper is that, despite these apparent differences, all uses of bare plurals are ‘merely facets of a syntactically and semantically unified phenomenon’. Accordingly, all the subjects in the examples in (1) and (2) are claimed to have the same syntax and the same semantics; the differences in interpretation, in Carlson’s account, spring entirely from semantic differences across the VPs in these sentences.

¹ Whenever an example is taken from the original paper, its number is given in square brackets.

² This reading is called “indefinite” in Carlson ([1977a](#)).

Empirical Motivation for a Unified Analysis

The argument Carlson produces has the following three sub-claims: (i) the existential bare plural is not the plural equivalent of a singular indefinite; (ii) the existential and the generic uses of bare plurals are not semantically distinct; (iii) the various subtypes of generic uses exemplified in (1) above, are not distinct either. It is important to note, even though Carlson himself does not, that each of these claims is, in principle, independent of the other two. In the rest of this subsection we concentrate on the data Carlson gives in support of each of these claims.

(i) **Bare vs. non-bare existentials**

The strongest empirical argument in support of differentiating existential bare plurals from ordinary singular or plural indefinites comes from their different scopal properties. The generalization is that existential bare plurals are scopally inert, i.e., they can only have narrowest scope relative to intensional predicates, negation, and clause-mate universal nominals. In these contexts, ordinary indefinite singulars as well as non-bare indefinite plurals, i.e., plural nominals preceded by a determiner such as the unstressed *some* or *several*, allow both wide and narrow scope readings. This is exemplified below, with an opacity inducing predicate

- (4) a. Bob wants to see *a/several French movie(s)*.
 b. Bob wants to see *French movies*.

The example in (4a) is scopally ambiguous: the indefinites may have both wide scope relative to *want*, and narrow scope relative to it. Under the wide scope interpretation, the sentence is true iff there is one French movie/a group of them such that Bob wants to see it/them. Under the narrow scope interpretation, Bob's wish is satisfied as long as he sees one/several French movie(s). In contrast, (4b) only has the latter, narrow scope interpretation of the bare plural relative to *want*. Under this interpretation, Bob does not care which French movie(s) he sees.

A related observation that has received less attention than it deserves is that existential bare plurals can only have narrow scope relative to pluractionality inducing expressions, i.e., adverbials whose interpretation makes the VP refer to multiple events. A non-bare indefinite cannot have narrow scope relative to such adverbials, while a bare plural must. Thus, a non-bare indefinite cannot co-vary across the events denoted by the predicate, while a bare plural is not interpreted as constant across them, as exemplified below.

- (5) a. Leon has killed a cow repeatedly since before the depression.
 b. Leon has killed cows repeatedly since before the depression.

Neither example in (5) is ambiguous: (5a) has only a 'wide scope', pragmatically strange reading involving a constantly resurrecting cow, while (5b) has only a natural 'narrow scope' reading, involving different cows killed at different times.

A further difference between non-bare indefinites and existential bare plurals comes from anaphoric processes. Example (6a) shows that non-bare indefinites

introduce referents that stay constant in discourse. This is not so for existential bare plurals, exemplified in (6b), whose behavior is similar to that of mass terms, exemplified in (6c):

- (6) a. Harriet caught *a rabbit* today, and Ozzie caught *it* yesterday. [55]
- b. Harriet caught *rabbits* today, and Ozzie caught *them* yesterday. [54a]
- c. Dad drank *beer* slowly, and I drank *it* fast. [54b]

Thus, in (6a) Harriet and Ozzie must have caught the same rabbit; this is not so in (6b), which parallels the interpretation of (6c): possibly different rabbits were caught by Harriet and Ozzie in (6b), just as different beers were drunk by dad and me in (6c).

These data show conclusively that existential bare plurals differ empirically from non-bare singular or plural indefinites.

(ii) Semantic non-distinctness of existential and generic uses of bare plurals

The next step is a direct argument for a unified analysis of the two uses of bare plurals. The basis of this claim is that the range of interpretation of a bare plural subject is entirely dependent on the interpretation of the VP, and therefore positing a systematic ambiguity across bare plurals is superfluous. To start with, note that in (7a) only the generic reading is possible, while in (7b), we only get the existential reading:

- (7) a. Dogs are easily trained.
- b. Dogs are sitting in my back yard.

In environments in which both readings of the bare plural are possible, Carlson claims, the source of the ambiguity is to be located in the VP rather than in the subject. Thus, note that while (8) is in principle ambiguous, the ambiguity persists when the subject is replaced by a non-bare nominal or a proper name, as shown in (9).

- (8) Dinosaurs ate kelp. [74]
- (9) Maxwell/lots of conductors/the old fireman/a few scientists ate kelp. [75]

A VP like *ate kelp*, Carlson argues, has two interpretations, independently of what nominal is in subject position. First, it can be interpreted as referring directly to kelp eating events, in which case *while Grog watched* would be a natural continuation. Second, it can be interpreted as ‘reporting past characteristics’ of the subject, in which case it can naturally be followed by an adverbial expression such as *in those days*. In the former interpretation, the bare plural subject can only receive an existential reading, while in the latter, it can only be interpreted generically. Once this ambiguity within the VP is taken into account, the two readings of the bare plural turn out to be in complementary distribution.³

³ In support of the unitary interpretation claim, Carlson notes that a generic bare plural can serve as an antecedent to an existentially interpreted pronoun and the other way around. However, this

(iii) Unitary generic reading

Generic uses of bare plurals, as mentioned above, come in several flavors too. In some instances, such as (1b), repeated below, they are synonymous with universal quantifiers:

(10) Horses are mammals.

In others, such as (1c), repeated in (11), they are interpreted as quasi-universal:

(11) Horses are good pets.

Finally, there are contexts such as (12), where they are not quantificational at all:

(12) Plants are widespread/extinct/numerous. [109].

Here too, Carlson argues, the differences in interpretation should be attributed entirely to the nature of the VP. The strongest empirical argument in favor of this position is that replacing the bare plurals in (10)–(12) with *this (kind of) animal* yields the same results, and yet, one would not want to claim that the noun phrase *this kind of animal* is multiply ambiguous as well. The crucial factor at play, this argument suggests, is that in (10)–(12) the subject involves reference to a *kind*.

More generally, Carlson claims, the properties of bare plurals, whether generic or existential, are mirrored by the properties of the noun phrase *this kind of animal*.

This observation is at the heart of the solution the rest of the paper develops. The proposal, in brief, is to treat all instances of bare plurals as referring to kinds, and to explain their special properties by distinguishing various ways in which a VP can attribute properties to kinds.

The empirical breakthrough here consists in bringing onto the scene, via a thorough and insightful discussion of the relevant data, the special properties of bare plurals in English, the mysteries of generic reference more generally, and the related distinctions that need to be drawn across predicates.

The Proposal

The proposed solution to this complex puzzle rests on several new analytical tools, whose introduction is motivated on the basis of data independent of bare plurals. This constitutes the formal breakthrough achieved in the paper.

The first theoretical proposal concerns the ontological classification of entities into two sorts: stages and individuals. Individuals in turn come in two subtypes:

argument loses some of its force when we note that a non-bare plural can serve as an antecedent to a generically interpreted pronoun as well, as illustrated in (i):

(i) *Some raccoons* dug up my plants yesterday and now I hate *them* with a passion.

ordinary, spatio-temporally bounded individuals like you and me, which I call here *objects*, and special individuals, *kinds*, which are not spatio-temporally bounded and are realized by objects. Stages are spatio-temporal realizations of an individual.⁴

A related, entirely novel proposal concerns a parallel distinction across predicates. Thus, predicates such as *widespread* and *extinct* are predicates that apply only to kinds, *intelligent* and *tall* are predicates that apply to individuals, and predicates such as *available*, or *sick* (in the physical sense), as well as predicates expressed by VPs in the progressive, apply to stages. These distinctions are elaborated on in Carlson (1977b).

Turning now to the account of bare plurals, recall that the core issue is to understand the interpretive properties of sentences of the form Bare Pl + VP.⁵

The gist of the proposal is the following. The bare plural itself always denotes a kind, in the same way that a proper name like *Jake* always denotes an object. When these noun phrases combine with an individual-level predicate, as in (13), the result is an expression that attributes intelligence directly to the entity denoted by the subject.⁶

- (13) a. Jake is intelligent.
 Reduced translation: I(j)
 b. Dogs are intelligent.
 Reduced translation: I(d)

In this proposal, then, individual-level predicates such as *intelligent*, translated above as **I**, can apply to both kinds and objects. Consequently, the difference between (13a) and (13b) involves the strategies we employ for determining the truth value of such sentences. In the case of (13a), we have to check whether the person named by *Jake* is intelligent or not. In the case of (13b), we have to check the intelligence of particular individuals realizing the kind *dogs*. The conditions under which (13b) is pronounced true do not require all dogs to be intelligent, though what exactly they require is not discussed here. Finally, though this is not explicitly mentioned, kind-level predicates, such as *widespread* or *extinct*, would be lexically restricted to take only subjects referring to kinds.

The proposed account of existential readings of the bare plural is based on an analogy with sentences like (14), where an individual-denoting subject combines with a stage-level predicate:

⁴ Note that in the philosophical tradition, the existence of kinds as a special type of entity has been recognized since Aristotle. In fact, the semantics of generics has been an area where the linguistic semantic tradition and work in the philosophy of language have interacted in a mutually beneficial way.

⁵ Carlson assumes that the proposal extends naturally to non-subject arguments.

⁶ In Montagovian terms (see Zimmermann, [this volume](#)), the subject in both (13a) and (13b) denotes the set of properties that an individual has (*Jake*, in the first case, and the kind *dogs* in the second). The VP denotes a property of individuals, and the whole sentence is true just in case the property denoted by the VP is in the set denoted by the subject.

(14) Jake is running.

Here, we have a mismatch between the sort referred to by the subject DP (an *object*, in our terminology) and the sort the predicate applies to (a *stage*). A parallel sortal mismatch is to be found, Carlson claims, in cases of existential bare plurals, where the predicate requires a stage-level argument, and the nominal denotes a kind.

The issue, then, is how to resolve the connection between the argument and the predicate in case there is a sortal mismatch between the two. The solution proposed here is based on the introduction of a relation *R* connecting individuals to their stages, without regard to whether the individual in question is of the kind subtype or not.

More concretely, stage-level predicates such as *be sick* or *be running*, are interpreted as in (15):

(15) $\lambda x \exists y [R(y, x) \ \& \ P(y)]$

Such a predicate, then, is a function from entities to truth just in case there is a stage realizing that entity which is in the denotation of the predicate in question.

Assuming the verb *sick* translates as *S*, and the proper name *Jake* translates as the individual constant *j*, while *dogs* translates as the individual constant *d*, we get the translations in (16).

- (16) a. Jake is sick.
 Reduced translation: $\exists y [R(y, j) \ \& \ S(y)]$
 b. Dogs are sick.
 Reduced translation: $\exists y [R(y, d) \ \& \ S(y)]$

The only difference is that the individual referred to by *Jake* is an object, while the individual referred to by *dogs* is a kind.

In this account then, a bare plural such as *dogs* refers to the kind dog in both its generic and in its existential uses, just as the proper name *Jake* refers to the individual who bears this name in both its individual use, exemplified in (13a), and its 'stage' use, exemplified in (16a).

The existential use of the bare plural arises when it is the argument of a stage level predicate. In these cases, the predicate provides an existential quantifier that binds a variable that is related to the subject by the *R*-relation.

In sum, the generic and the existential uses of the bare plural are distinguished in this proposal solely at the level of the predicate. In the existential use, the predicate provides an existential quantifier which binds a variable that realizes the kind denoted to by the subject. This accounts both for the existential nature of these nominals and for the fact that the existential quantifier associated with them can only have narrowest scope. The fact that neither existential nor generic bare plurals are spatio-temporally bound is directly explained by the fact that they refer to a kind. This also explains their special anaphoric properties.

The account solves the empirical issues raised earlier using novel tools whose introduction is independently motivated. This is so because distinguishing between

different sorts of entities at both the argument and the predicate level, as well as having to deal with apparent mismatches between them, is needed to account for data that is independent of bare plurals.

Discussion

This section provides a brief and admittedly incomplete overview of some of the issues – both theoretical and empirical – that Carlson’s paper raises. These issues have been central in the subfield of nominal semantics for the last forty some years.

First, however, let me mention some matters that the field as a whole has accepted based on Carlson’s work. The distinction between two types of individuals, kinds and objects, that Carlson brought onto the linguistic scene is now part of the landscape of the field. The distinction in the realm of predicates, between those that apply to kinds (*widespread*), those that apply to objects (*intelligent*), and those that apply to stages (*available*, *sick*) is also universally accepted. (For an influential article on the individual/stage distinction, see Kratzer, 1995.) Equally uncontroversial is the need for a way to connect entities across sorts, though the formal means employed vary. The fact that existential bare plurals are not merely the plural counterpart of non-bare indefinites is now also taken for granted. Finally, the fact that the bare plural subjects in (10) and (11) should not be distinguished has also been generally accepted.

We discuss next some theoretical and empirical issues that specific aspects of the account have raised. Keeping the focus on English, one should first note that the morpho-syntactic properties of English bare plurals – the fact that they are bare, and that they are plural – do not play a role in Carlson’s formal account. The question that arises is whether they should. Note that, as Carlson (1977a) observes in the conclusion, singular definites, as well as indefinite singulars have generic uses, exemplified below:

- (17) a. The wolf was widespread in this part of the world.
b. A dog gives birth to live young.

Unlike bare plurals or definite singulars, however, indefinite singulars cannot function as subjects of kind-level predicates:

- (18) *A dog is widespread.

Exactly how to account for the role of the absence of the article and the presence of the plural in bare plurals in English, as well as for the connection between bare plurals and generic uses of definite and indefinite singulars have been much debated in the literature since Carlson’s pioneering work. During these debates, various aspects of Carlson’s account have been called into question. (See Geurts (1985), Wilkinson (1991), Cohen (2001), Greenberg (2003) among many others.)

Turning now to analytical matters, the paper does not discuss in detail the issue of how kinds relate to individuals that realize them. This question is relevant to the

analytical choice of giving bare plurals functioning as arguments of individual-level predicates the same treatment as that given to bare plurals functioning as arguments of kind-level predicates (see (13) above). There are two distinct matters that this choice gives rise to.

The first issue is whether sentences where bare plurals combine with individual level predicates, such as (13b), should be treated on a par with those in which such nominals combine with kind-level predicates, such as (13a), as done in Carlson (1977a). An alternative account has been pursued, in which sentences like (13b) are treated on a par with those in which bare plurals combine with stage-level predicates, such as (16b). In such an account, (13b) involves an individual-level predication about individual realizations of the kind *dogs* refers to.

The second issue is how to flesh out the details of the 'exception tolerant' generic statements exemplified by (10) and (11), a question that has been debated in much of the subsequent literature. If, unlike what Carlson proposes here, quantification is involved in one or both statements of this sort, what is the force of the quantifier? What are the limits of the tolerance for exceptions these sentences allow?

When it comes to the treatment of bare plurals combined with stage-level predicates, Carlson's account places all the analytical burden on the VP: the two nominals in (19) receive exactly the same interpretation:

- (19) a. *Dogs* are widespread.
b. *Dogs* were barking in the garden.

The difference, according to Carlson's account, is that the predicate in (19a) denotes a property, while in (19b), the semantics of the predicate involves existential quantification over realizations of the entity denoted by the subject which have the property of barking in the garden. In fact, this unitary account is at the heart of the paper. An immediate issue that arises is how to motivate the assumed complexity of stage-level predicates.

Some of the matters mentioned above were taken up in Carlson's immediately subsequent work, namely Carlson (1977b) and Carlson (1979). In these works, which left a deep mark on the field, Carlson proposes to treat sentences involving a discrepancy between the sort denoted by an argument and the sort required by the predicate by making use of two types of connections.

First, Carlson introduces *realization relations*, which connect stages to the individuals that they realize, and individuals to the kinds they realize. Second, at the level of predicates, he introduces two VP operators: (i) *Gn*, an operator that maps predicates that apply to objects to predicates that apply to kinds, and (ii) *Gn'* which maps stage-level predicates to individual level ones. *Gn* is present in the interpretation of (13b), allowing a predicate like *intelligent* to apply to the kind denoted by the bare plural. This operator accounts for the fact that all individual-level predicates can have generic bare plural subjects. *Gn'* is present in the interpretation of 'characterizing' sentences such as (20) as well as in sentences involving existential bare plurals, such as (16b):

- (20) Jake smokes.

In Carlson (1977b), indefinite generics, exemplified in (17b), are treated as referring directly to kinds, just as bare plurals, with a somewhat *ad hoc* restriction accounting for their inability to combine with kind level predicates.

Two further questions arise at this point, namely (i) what the precise semantics of the Gn operator should be, and (ii) whether Gn and Gn' should be treated as monadic predicate operators, as Carlson proposes, or whether they should be given sentential scope and assumed to have the ability to bind variables introduced by the subject or other arguments of the predicate. This latter approach treats them on a par with adverbs of quantification discussed in Lewis (1975) and exemplified in (21).

(21) Sometimes/often/always/generally Jake reads *a detective story* after dinner.

Here the adverb of quantification binds the variable introduced by the indefinite. Farkas and Sugioka (1983) were the first to argue for the 'adverb of quantification' approach for sentences like (17b), treated as involving an implicit generic adverb of quantification that binds the free variable introduced by the indefinite. The proposal has subsequently become widely, though not unanimously, accepted. Note though that treating indefinite generics along these lines does not necessarily mean that existential bare plurals should be analyzed in a parallel fashion.

Still within English, Carlson (1989) notices further shades of meaning that his original account did not capture, involving the ambiguity of sentences like (22):

(22) Typhoons arise in this part of the Pacific.

This sentence can either be interpreted as a statement about typhoons, in which case the bare plural is generic, or it can be interpreted as a statement about a particular part of the Pacific, in which case the bare plural has an existential interpretation. Carlson's observation connects the problem of bare plural interpretation to the information structure of the sentence, a very fruitful connection that has been pursued in subsequent work. (See, for instance, Cohen, 2002.)

Carlson's work on the semantics of generics in English generated a large amount of literature – articles, dissertations, books. The first milestone in this area after Carlson's work is Carlson and Pelletier (1995), a collection of seminal articles with an excellent introduction that summarizes the relevant literature up to that point.

Above we have concentrated on issues concerning English. Once the horizon is enlarged to consider languages beyond it, further questions arise, related to the ones discussed above. Recall that the main claim in Carlson (1977a) is that generic and existential uses of bare plurals in English should be given a unitary account. This claim is particularly attractive given that in English, the same nominal form – a bare plural – is used in both cases. As noted already in Farkas (1981), this is by no means true across languages. In Romanian, just as in other Romance languages, the equivalent of the English generic bare plural is a definite plural, while the existential use is expressed, just like in English, by a bare plural:

- (23) a. Câinii sunt inteligenți.
 dog.Pl.Def are intelligent.Pl
 'Dogs are intelligent.'
 b. Lătrau câini în grădină.
 bark.Pl dog.Pl in garden
 'Dogs were barking in the garden.'

French also differentiates generic from existential uses: a definite plural is used for the former, and a partitive for the latter:

- (24) a. Les chiens sont intelligents.
 the dog.Pl are intelligent.Pl
 'Dogs are intelligent.'
 b. Des chiens aboyaient dans le jardin.
 Part.pl dog.Pl bark.Imp in the garden
 'Dogs were barking in the garden.'

This systematic differentiation in the morpho-syntactic realization of the two uses of bare plurals in English raises the question of whether their unitary treatment is as appealing as it first seemed. We now have to explain both why some languages are like English, in having one form for both uses, while others are like Romance languages, in having two forms.

Based on the formal differentiation of the generic and existential interpretations in many languages, the view that bare plurals in English are, in fact, ambiguous between a generic and an existential interpretation gained ground. Under a relatively common view in the 1980s, existential bare plurals were treated as introducing free variables bound by a VP-level existential operator, in sentences like (2), or by a generic adverb of quantification, in sentences like (1c).

A further task is to account for the particular forms languages use when differentiating between generic and existential bare plurals. Thus, even if we restrict our attention to the comparison between the Romance language group and English, one has to explain why the generic use is expressed in English by a bare plural while Romance languages use a definite plural. One also has to explain why, in Romance languages, a definite plural is used in examples like (1c) while some form of indefinite occurs in examples like (2).

The issue this multiplicity of forms is relevant to is whether to place the analytical burden wholly within the VP, as Carlson does, or whether to assume a semantic differentiation within the nominal as well.

The problems just mentioned become even more complex once the cross-linguistic coverage is widened to include languages that make use of bare singulars, among which we find some Romance languages such as Brazilian Portuguese, as well as languages which do not normally employ determiners or mark number, such as Chinese. These matters are addressed head on in Chierchia (1998), which is the next milestone in the subfield of 'generic studies', after Carlson et al. (1995). (See Dayal, [this volume](#).) Incorporating Link (1983), a seminal work on the semantics of

number, Chierchia connects generic reference to the semantics of mass nouns. He also takes into account the role of determiners in nominal reference by introducing a cross-linguistic distinction concerning the ability of bare nominals to function as arguments in a language.

While incorporating some proposals from work that came after Carlson (1977a), such as the use of a sentential level adverb of quantification that is able to bind variables in its scope, Chierchia's theory qualifies as Neo-Carlsonian because it stays close to Carlson's original approach. One difference is that instead of R-relations, Chierchia introduces two sort-shifting operators, $\cup$ and $\cap$, that shift nominal reference across objects and kinds. Unlike Carlson's R relation, these operators play a role not only within the semantics of the VP but within the semantics of the nominal argument as well. In the case of existential bare plurals in English for instance, the down operator $\cap$ turns a bare plural from a kind denoting nominal to a nominal denoting an individual instantiating that kind. (Stages do not play a role in Chierchia's ontology – the work they did in Carlson's is taken over entirely by the interpretation of the VP.)

Chierchia's account of examples like (1c), where a kind referring argument combines with an individual-level predicate, is treated as involving a 'derived kind predication' rule. This rule converts a predication of kinds into a statement claiming the existence of objects realizing that kind of which the relevant predicate is true. Chierchia's approach allows a unitary account of generic bare plurals in English and makes clear cross-linguistic predictions that have been discussed and challenged in the vast literature that followed it. The cross-linguistic patterns we find, and those we do not, when it comes to generic reference and its connections to other areas of nominal reference, are far from being fully understood.

This necessarily incomplete survey of the area of 'generic studies' is meant to show the lasting importance and influence of Carlson (1977a), the work that started it all. Reading or re-reading it forty some years later is essential not only in order to understand the work that it inspired so far, but also to take this work further into the future.

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Chapter 5

On Chierchia's "Reference to Kinds Across Languages"



Veneeta Dayal

Abstract This chapter introduces the core aspects of the theory of variation proposed in Chierchia (Nat Lang Semant 6(4):339–405, 1998). Focusing primarily on noun phrases without overt determiners, Chierchia gives a principled account of what is universal and what can vary in the mapping from form to meaning within and across languages. He takes a neo-Carlsonian approach where bare plurals canonically refer to kinds, regardless of whether or not they occur as arguments of kind-level predicates. He not only captures the core readings of bare plurals but also accounts for readings that are not standardly associated with them. Most significantly, he offers a fresh perspective on cross-linguistic variation in terms of a semantic parameter constraining the type of meaning available at the level of NP, the nominal structure below determiners or quantifiers, in a given language. Following a brief sketch of the challenge that bare arguments pose for syntax and semantics in section “[Introduction](#)”, the details of Chierchia’s theory are presented in section “[A Neo-Carlsonian Theory of Variation](#)”. Sections “[On Covert Type-Shifting](#)”, “[The \[-pred, +arg\] Setting](#)”, and “[On Singular and Definite Kind Terms](#)” deal with the very significant responses that different aspects of the theory have generated since Chierchia’s initial proposal.

Keywords Reference to kinds · Cross-linguistic variation · Semantic parameters · Singular-plural-mass nouns · Ranked type-shifts · Blocking effects · Null determiners

Introduction

The introduction of cross-linguistic considerations in the development of theory is among the more significant developments within semantics in recent years. Some of

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it was perhaps inevitable, given the shift in the language expertise of semanticists. But if one had to pick a single piece of work that served to focus the intellectual energy of the field on the issue of what is universal and what is variable in the interpretive module, it is the idea that there are parameters governing the mapping from morpho-syntax to semantics, forwarded in Chierchia (1998).

In order to appreciate the details of Chierchia's theory, it might be useful to understand what is at the heart of the issues he addresses. We can illustrate with a simple example:

- (1) a. [DP Determiner/Quantifier/Numeral [NP (Adj) N]]
 b. [DP $\emptyset$ [NP (Adj) N]]
 c. [NP (Adj) N]

Overt lexical items of the kind we see in the outer layer of (1a) can plausibly be analyzed as head of DP, taking (modified) nouns under NP as their argument. But what happens if there is no overt expression that can be located in D? Is a bare argument still a DP, with a null D, as in (1b), or does it have the simpler NP structure in (1c)?

The choice between (1b) and (1c) is non-trivial, syntactically and semantically. For syntactic theory, the question is whether there are properties of noun phrases that distinguish between smaller and larger structures. For semantic theory, the question is how to map structures like (1b) and (1c) to the types of meaning that are associated with NP and DP in canonical cases like (1a). These issues become even more nuanced when cross-linguistic considerations are brought into the equation. Do bare arguments always have the same structure, be it (1b) or (1c), or can they differ within and across languages? This is the question that Chierchia's theory of variation addresses.

I present the details of Chierchia's theory in section "[A Neo-Carlsonian Theory of Variation](#)", highlighting what I see as its key components. In sections "[On Covert Type-Shifting](#)", "[The \[-pred, +arg\] Setting](#)", and "[On Singular and Definite Kind Terms](#)" I discuss the response to Chierchia's proposals, both empirical and theoretical.

A Neo-Carlsonian Theory of Variation

We start with a brief review of Chierchia (1998), drawing on Dayal (2011a, 2013).

Bare Plurals as Kind Terms

Chierchia's view of bare arguments is rooted in the analysis of English bare plurals in Carlson (1977): bare plurals always refer to kinds (see also Farkas, [this volume](#)). Aspectual specification on the predicate determines the nature of

quantification involved: (quasi) $\forall$ force in generic sentences, $\exists$ force in episodic sentences. However, differently from Carlson's original proposal, the mapping of bare plurals in quantificational structures is not determined by syntax alone. Rather, it is governed by the same principles that operate on regular indefinites in the context of adverbs of quantification (see Krifka et al., 1995 for an overview). This makes it *neo*-Carlsonian.

Chierchia adopts the perspective of flexible types (Partee, 1987; Winter, [this volume](#)) in admitting three basic operations for turning noun phrases with a predicative meaning (type $\langle e, t \rangle$) into arguments (type e or $\langle \langle e, t \rangle, t \rangle$): *iota*, *nom* and $\exists$. The first two map a predicate into an entity, without introducing quantificational complexity. *Iota* picks out the unique maximal individual in the extension of the predicate at the relevant index, if there is one, and is undefined otherwise (Sharvy, 1980; Link, 1983). Sometimes it has a lexical exponent, *the* in English, but in many languages it operates covertly. *Nom* ($\cap$) is the kind forming operation. Chierchia defines it as a partial function that applies to those properties that fit the concept of a kind. It denotes, for any property it is defined for, a function from indices to the unique maximal entity that instantiates the kind at that index.¹ *Nom* operates covertly in English, and is often taken to do so universally. The third, $\exists$, turns predicative expressions into generalized quantifiers of type $\langle \langle e, t \rangle, t \rangle$ (Montague, 1974; Barwise and Cooper, 1981). $\exists$ is a total function and is realized as *a/an* in English but is often claimed to be covert in article-less languages.

The derivation of kind-level statements with bare plurals is straightforward. Since *evolve* is a predicate that applies to the species as a whole, not to individual dogs or wolves, *nom* is the appropriate type-shift for the properties *dogs* and *wolves* in (2a).²

- (2) a. Dogs have evolved from wolves. $\Rightarrow$ evolve-from($\cap$ dogs, $\cap$ wolves)

Predicates that apply to ordinary individuals require some way of accessing the instantiations of the kind when they take bare plural arguments. Chierchia proposes an operation *pred* ($\cup$) which is the inverse of *nom*, a function from kinds to predicates.³ When the tense-aspect specification in the sentence supports a tripartite quantificational structure with a generic operator, we get the quasi-universal reading associated with generic statements. The sentence in (2b) says that

¹ To illustrate, $\cap$ (dogs) and $\cap$ (dogs with bushy tails) are defined at any index but $\cap$ (dogs in this room) is not—the property of being spatially and temporally restricted to a given set of entities does not qualify for kind-hood. The output of $\cap$ (dogs)/ $\cap$ (dogs with bushy tails) is: $\lambda s \iota$ (dogs(s))/ $\lambda s \iota$ (dogs-with-bushy-tails(s)) (Chierchia, 1998: 351). See also Footnote 3.

² Carlson (1977) distinguishes between kind-level individuals and object-level individuals. The latter are ordinary individuals like John, Mary etc. A further ontological distinction between object-level individuals and stage-level individuals is no longer assumed to be part of grammar (see Chierchia, 1982).

³ *Pred* is a function from $\langle s, e \rangle$ to $\langle e, t \rangle$ where the former is a kind term and the latter a set of ordinary individuals: $\cup \cap$ (dogs) = $\lambda x [x \leq \cap$ (dogs)(s)], where $\cap$ (dogs)(s) is the maximal plural individual that is the sum of all the atomic members of the kind at s (Chierchia, 1998: 350).

in most contextually relevant situations that have instantiations of the dog-kind in them, those dogs bark:

- (2) b. Dogs bark. $\Rightarrow \text{Gn } x \text{ s } [\bigcup \text{dogs}(x) \wedge \text{C}(x, s)] [\text{bark}(x, s)]$

Finally, when the aspect supports an episodic interpretation, as in (2c), something further is needed. Chierchia proposes an operation called *Derived Kind Predication* (DKP) that builds on *pred* by introducing an $\exists$ to bind into the predicate created by it.⁴ It says that the statement is true if there exist some instantiations of the dog-kind that are engaged in the activity of barking:

- (2) c. Dogs are barking.
 $\Rightarrow \text{are-barking}(s) (\bigcap \text{dogs})$
 $=_{\text{via DKP}} \Rightarrow \exists x \text{ s } [\bigcup \text{dogs}(x) \wedge \text{are-barking}(s)(x)]$

This completes Chierchia's account of the core readings of English bare plurals.⁵

Before turning to aspects that concern cross-linguistic variation, it is worth commenting on the status of bare plurals as indefinites. We can see that the truth conditions of (2c) are the same as those of a corresponding statement with an overt indefinite *some dogs are barking*. However, there are sound empirical grounds for distinguishing between the properties of bare plurals on their existential reading and those of overt indefinites.

Consider the paradigm in (3) which has a durational adverbial phrase *for 2 months*. We can interpret the adverbial to involve multiple sub-intervals within the larger 2-month interval, such that the relevant eventuality occurs within each sub-interval. Since *die* is a one-time predicate, this leads to an implausible reading when a singular or a plural individual is fed in as an argument in multiple events of dying:

⁴ *Pred* is defined to yield an empty set when there are no instantiations of the kind in the world of evaluation. This makes an episodic statement false, rather than undefined, in such worlds (Chierchia, 1998: 350). For example, $\bigcup \text{dinosaurs}$ yields the empty set in the actual world, which makes *dinosaurs are eating kelp* false: $\exists x \text{ s } [\bigcup \text{dinosaurs}(x) \wedge \text{eating-kelp}(s)(x)]$. There are further issues having to do with the interaction of tense specification and bare plural interpretation that are worth probing, but for present purposes this will suffice.

⁵ As indicated earlier, there is flexibility in the mapping of bare plurals, which can lead to ambiguity:

- i. Typhoons arise in this part of the pacific $\Rightarrow$
 a. $\text{Gn } x \text{ s } [\bigcup \text{typhoons}(x) \wedge \text{C}(x, s)] [\text{arise-in-this-part-of-the-P}(x, s)]$
 b. $\text{Gn } s [\text{in-this-part-of-the-P}(s) \wedge \text{C}(s)] \exists x [\bigcup \text{typhoons}(x) \wedge \text{arise}(x, s)]$

The bare plural is in the restrictor in (ia), which says of typhoons in general that they arise in this part of the Pacific. DKP is needed to interpret the bare plural when it is in the nuclear scope, as in (ib). It says that it is generally true of all contextually relevant situations involving this part of the Pacific that there are typhoons that arise (see Chierchia, 1998: 366–68).

- (3) a. Some rabbits/a rabbit kept dying for 2 months
 $\Rightarrow \exists x [\text{rabbits}(x)/\text{rabbit}(x) \wedge \forall t [t \leq 2\text{-months} \rightarrow \text{die}(x, t)]]$
- b. Rabbits kept dying for 2 months
 $\Rightarrow \forall t [t \leq 2\text{-months} \rightarrow \exists x [\cup \text{rabbits}(x) \wedge \text{die}(x, t)]]$

The fact that (3a) only has the implausible reading tells us that the syntax of English prevents *some rabbits/a rabbit* from scoping under the adverbial. The bare plural in (3b), because it is individual denoting (type $\langle s, e \rangle$), can be the direct argument of the predicate, regardless of where it is in the syntax. It is the wrong sort, however, since kinds can become extinct or die out, but it is only their instantiations that can die. DKP, piggy-backing on *pred*, intervenes to adjust the sort mismatch. Its associated $\exists$ enters the derivation below the adverbial operator and leads to a plausible reading for (3b), supplying distinct instantiations of the kind for each sub-event.

Appealing to reference to kinds for bare plurals and to a generalized quantifier meaning for indefinites can thus yield radically different truth conditions. The contrast between (3a) and (3b) is a dramatic illustration of the neo-Carlsonian position that bare plurals are not just a sub-type of standard indefinites (see Carlson, 1977 for a full list of such differences).

Variation Based on Lexical Determiners

As we saw, Chierchia's basic theory preserves Carlson's original insights but incorporates advances regarding external sources of quantification for indefinites. We now turn to his claims regarding bare nominals across languages. He introduces two principles, *Blocking* and *Ranking* to constrain the interpretive possibilities illustrated in section "Bare Plurals as Kind Terms". While they are both meant to apply universally, we will see that they make room for variation.

Blocking and Ranking

Blocking prevents a covert type shift when a language has an overt exponent for it, Ranking privileges *nom* over *iota* and $\exists$. Blocking explains why bare plurals do not have definite readings in languages with definite determiners, but do so in languages that lack them. Compare English (4a) with Hindi (4b):⁶

⁶ Dayal (2013) discusses putative counterexamples to Blocking, from Condoravdi (1997), that ultimately do not pose a problem for it: *There was a ghost on campus. The students/Students were afraid.*

- (4) a. Some children entered the room. *(The) children seemed happy.
 b. kuch bacce andar aaye. bacce bahut khush the
 Some children inside came children very happy were
 ‘Some children came inside. The children were very happy.’
 (Dayal, 2004a: 403)

Blocking also rules out the covert application of $\exists$ because of the presence of the indefinite determiner in English. This becomes relevant for English bare plurals that do not refer to kinds:

- (5) Parts of this machine are not new.

Carlson (1977) notes that *parts of this machine* is not a kind-term because of the indexical in it and furthermore notes that it can have an $\exists \neg$ reading, as in (5). Since non kind denoting bare plurals (henceforth indexical bare plurals) are not in the domain of *nom*, and since *iota* is lexically blocked by *the*, Chierchia’s Ranking allows them to shift via the low-ranked $\exists$ type shift. That means that just in such cases bare plural indefinites are generalized quantifiers that are scopally active. This is how Chierchia explains Carlson’s observations regarding sentences like (5).

For Chierchia’s explanation to go through, however, *some* *N*_{PLURAL} should not block the $\exists$ type shift. Chierchia provides two diagnostics for identifying a true lexical indefinite: it must allow for generic readings and it should be able to take scope below and above negation. By these tests, *some* is not a lexical exponent of $\exists$, unlike *a*, which is: *A dog barks when it is hungry* is a generic statement about typical dogs but *Some dogs bark when they are hungry* is not.⁷

In sum, Blocking and Ranking apply universally, but variation occurs because of the presence or absence of lexical determiners.

Ranking Recalibrated

Chierchia’s ranking of *nom* above *iota* and $\exists$ is gleaned from two separate discussions. Taking bare NPs in article-less languages to be ambiguous between definite and indefinite, he ranks *iota* and $\exists$ on a par [pp. 360-61]; taking English kind denoting and indexical bare plurals to have distinct scope profiles, he ranks *nom* above $\exists$ [p. 374]. Note though that if *nom* outranks *iota* and $\exists$, kind-denoting bare plurals could never have the possibility of definite or indefinite readings in any language. It is uncontroversial, however, that bare plurals in article-less languages can be kind terms as well as definites, which suggests equal ranking for *nom* and *iota*. The indefinite status of bare nominals is more nuanced. While it is true that they can often be translated with the indefinite determiner *a/an*, they cannot be

⁷ English lacks bare singulars because they are not in the domain of *nom*, and *iota* and $\exists$ are lexically blocked by *the* and *a* respectively.

taken as having been shifted by $\exists$. These indefinite readings typically take obligatory narrow scope, characteristic of kind denoting terms.⁸ Dayal (1999, 2004a) therefore proposes a revision, placing *nom* and *iota* above $\exists$. This revision is more in line with the theoretical motivation for Ranking and preserves Chierchia's explanation for the behavior of bare plurals in English, both kind denoting and indexical. The revised Ranking further predicts that indexical bare plurals in article-less languages will not have such readings because $\exists$ will be outranked by *iota*, a prediction that is borne out in Hindi.⁹

There remains some resistance to the idea that bare plurals in article-less languages are not indefinites due to the widespread perception that they can have both definite and indefinite readings. Claims about ambiguity, however, have never been established with concrete diagnostics.¹⁰ Should a language turn out to have kind, definite and genuine indefinite readings, it would show that covert type shifts are not ranked in that language. While this may go against the theoretical underpinnings of the principle of ranked type shifts, it would still be weakly compatible with it. Clear evidence against ranking would come from an article-less language that privileges $\exists$ over *nom* and/or *iota*, but such a language has not been attested.

The Nominal Mapping Parameter

The aspect of Chierchia's theory that speaks directly to the issue of variation is the *Nominal Mapping Parameter*, which classifies languages along two dimensions, [+/-pred] and [+/-arg], mnemonic for predicative and argumental, respectively. Depending on the value chosen, languages allow or disallow the denotation of NP (as opposed to DP) to be of the relevant type, $\langle e, t \rangle$ for [pred] and $\langle \langle e, t \rangle, t \rangle$ or *e* for [arg]:

- (6) *The Nominal Mapping Parameter (NMP)*: $N \Rightarrow [\pm \text{pred}, \pm \text{arg}]$
 [+pred, +arg] bare arguments allowed: no article $\Rightarrow$ Slavic; articles $\Rightarrow$ Germanic.
 [+pred, -arg] bare arguments disallowed:

⁸ Dayal (2004a, 2011b) emphasizes that there are two sources for indefinite readings of bare nominals. The indefinite readings of bare plurals are due to DKP operating on kinds. Bare singulars, to be discussed later, do not have recourse to DKP. Their indefinite reading is limited to particular positions/constructions, such as pseudo-incorporation (see Dayal, 2011b on differences between incorporation and reference to kinds).

⁹ Dayal (2013) gives arguments in favor of removing $\exists$ from the set of covert type-shifts. A total function like $\exists$ does not predict the infelicity of bare plurals in examples like *#Stones in this monument are precious*.

¹⁰ Dayal (2018) presents an extensive investigation of Hindi bare nominals, singular and plural, that shows the claim of ambiguity to be unfounded. In ongoing work, the investigation is extended to a wider range of languages (Dayal, forthcoming).

null D $\Rightarrow$ Italian, Spanish; no null D $\Rightarrow$ French.
 [-pred, +arg] every (lexical) noun is mass $\Rightarrow$ Chinese
 (Chierchia, 1998: 400)

A [+pred, +arg] language has bare plural NPs that start out as predicative type $\langle e, t \rangle$, but covert type shifts can apply, converting them to argumental types. This is consistent with the [+arg] setting and is sensitive to considerations of economy in grammar, encoded in *Avoid Structure: Apply SHIFT at the earliest possible level* (Chierchia, 1998: 393). SHIFT refers to the set *nom*, *iota* and $\exists$, modulo Blocking and Ranking.

Bare plurals in Germanic languages, which are [+pred, +arg], can occur in any syntactic position because they are NPs with no null element in need of licensing. Languages that have the same setting but lack lexical determiners, like Russian or Hindi, behave syntactically like English in allowing bare plurals in all positions but differ from English semantically in allowing additional definite readings for them.

Languages that are [+pred, -arg] necessarily project DP in argument positions but bare plurals may be possible in languages that have a null determiner. This distinguishes French, which has no bare plurals, from Italian and Spanish, which do. Chierchia, following Contreras (1986), argues that Romance bare plurals are restricted to well-governed positions because the null D has to be licensed. He further argues that the Romance null D encodes SHIFT. Italian bare plurals can denote kinds and have narrow scope indefinite readings but not definite readings, just like English bare plurals.¹¹

The languages discussed so far have a [+pred] setting, which allows expressions in D, demonstratives, quantifiers, numerals and the like, to combine directly with NP. NPs in languages with a negative setting [-pred, +arg] must project structure above NP in order to denote at the level that can combine with expressions in D. For example, Mandarin and Japanese bare nominals can serve freely as arguments but need classifiers to combine with numerals: Mandarin *yi *(ben) shu* ‘one CL book’.¹² Avoid Structure is not violated because the [-pred] setting for NPs makes it impossible for them to denote $\langle e, t \rangle$ type meanings.

Chierchia’s categorization of [-pred, +arg] languages as those in which every (lexical) noun is mass, in contrast to mass/count [+pred, +/-arg] languages, has to be understood against the background of his general view of how number morphology interacts with the mass-count distinction:

¹¹ Typically, a modifier is needed to facilitate the kind-level reading of Italian bare plurals and therefore the claim of kind reference remains controversial (see Dayal, 2004b on the role of modification in such cases).

¹² This is analogous to the requirement for measure phrases with mass nouns, as in English *two *(pieces of) furniture/two *(cups) milk*.

- (7) a.
- | | | | | | |
|-----------------|---|----|-----|----|-------------------------------------|
| Count noun | | | | | Mass noun |
| <i>chairs</i> = | [| | abc | | = <i>milk</i>
= <i>furniture</i> |
| | | ab | ac | bc | |
| PL |] | | | | |
| <i>chair</i> = | | a | b | c | |

b. $PL(F) = \lambda x [-F(x) \wedge \forall y [y \leq x \wedge \text{Atom}(y) \rightarrow F(y)]]$

(Chierchia 1998: 346–347)

A count noun denotes a set of atomic individuals. Pluralization, as defined in (7b), yields the set of pluralities built out of atoms, using sum formation. Mass nouns, instead, denote the set of singularities and pluralities in an undifferentiated manner. The atoms for nouns like *milk* are not accessible to speakers without contextual support, while the atoms for nouns like *furniture* are conceptually on par with atoms for count nouns like *chair*, though these atoms can range over different subtypes of furniture. The first seems to be mass in all languages, while the second is subject to cross-linguistic variation (see also Champollion and Brasoveanu, [this volume](#), on Link, 1983).

If counting requires a differentiated level of atomic individuals, numerals are predicted to combine directly with count nouns but not mass nouns, unless measure phrases are there to make the atoms salient. It is in this sense that Chierchia considers all nouns in [-pred, +arg] languages to be mass: “saying that all members of category NP are mass-like does not mean saying that something resembling the mass/count distinction cannot be found in such languages (see, e.g., Cheng and Sybesma, 1996)” [p. 355]. This fits in with his view of [-pred, +arg] languages as exclusively having mass-like nouns: if NPs denote kinds at the basic $\langle s, e \rangle$ type, turning them via *pred* to type $\langle e, t \rangle$ delivers sets that have the same structure as mass nouns. That is, conceptually count nouns in such languages are predicted to be like *furniture* in manifesting mass-like syntax. Chierchia takes this setting of the NMP to lead to generalized bare arguments, the extension of all nouns being mass[-like], no pluralization, and the existence of generalized classifier systems.

A [-pred, -arg] setting would characterize languages that lack the category NP altogether and, unsurprisingly, remains unattested.

Definite and Singular Kind Terms

In languages with determiners, the singular definite can refer to a unique contextually salient entity as well as to a kind. In some, but not all such languages, the definite plural can do the same. English, for example, can only use the definite singular to refer to kinds, while Italian can use both the definite singular and the definite plural for that purpose:

- (8) a. The dinosaur is extinct />(*The) dinosaurs are extinct.
 b. Il dinosauro è estinto / *(I) dinosaur sono estinti.

Chierchia draws on the idea that the uniqueness imposed by number morphology on kind formation clashes with the notion of kinds, and so rules out the application of *nom* to singular terms (see also Dayal, 1992, 2004a). This explains the absence of bare singulars in English and Italian. He takes definite singulars, however, to undergo an operation MASS: $\text{MASS}(\text{dinosaur}) = \text{dinosaur} \cup \text{PL}(\text{dinosaur})$ (Chierchia, 1998: 381), the output of which includes both singularities and pluralities, signalling that they are conceptually plural even if they are not morphologically so. Furthermore, *iota* picks out the largest member of the set, followed by a group forming operation. When this term is intensionalized, we get a singular kind term.

Plural definites have a simpler derivation as kind terms, being the intensional version of standard plural definites. The near equivalence with bare plurals follows. English does not use plural definites as kind terms because of Avoid Structure. The [+arg] setting privileges NP for the purpose.

Episodic statements present a challenge to accounts of definite kind terms, singular or plural. These kind terms typically do not refer to some instantiation(s) of the kind in the way that bare plurals do. Kind-based indefinite readings are only possible if the instantiations can be seen as representative of the species. On Chierchia's view, the kind level reading of definites comes into play when there is pressure from a kind-denoting predicate or a generic operator. Otherwise, its basic meaning, referring to the unique maximal entity, suffices.

With these essentials in place, we now turn to the responses that the theory has generated since its publication twenty-some years ago.

On Covert Type-Shifting

This section discusses the choice between covert type-shifts and null determiners in interpreting bare nominals.

Interpreting Null Determiners

Recall that Chierchia defines the principles of Blocking and Ranking in terms relevant to covert type shifts but the core account would not be affected by shifting to a DP analysis of bare nominals: $[_{DP} D_{NULL} [_{NP} N]]$. What needs to be determined is which of the type-shifts, *nom*, *iota* or $\exists$, is the meaning of D_{NULL} . We know that bare plurals in languages with determiners do not have definite readings, while those in determiner-less languages do. The null D would therefore have to be sensitive to Blocking. Furthermore, kind denoting bare plurals only have narrowest scope readings, while indexical bare plurals can take wide scope over negation. Chierchia

demonstrates this for English but presumably this would be so in any language that has determiners and bare plurals. Null Ds in all such languages would have to obey Ranking. Positing null determiners, then, does not provide any substantive advantage as far as interpretation is concerned, a point already underscored for English in Carlson (1977). While there is no inherent obstacle to positing null determiners for bare nominals universally, there is also no particular advantage from the semantic perspective. Something like Blocking and Ranking would be needed in order to derive variations in interpretation, whether they are incorporated as constraints on covert type shifts or as constraints on the type-shifts encoded by null determiners.

Structural (In)sensitivity

Covert type-shifts are sometimes thought to be too powerful to provide a restrictive theory of grammar but does a DP analysis do any better? To address this properly we need to understand how issues of interpretation play out in this approach. We can take the following options from Longobardi (1994) to guide the discussion:

- (9) a. [DP D_{NULL} [NP N]]
 b. [DP N_i [NP t_i]]
 c. [DP D_{EXPL} [NP N]]

A null D, on this view, encodes an $\exists$ that can bind a predicative NP, resulting in indefinite readings of bare plurals. Referential expressions can move from N to D or get coindexed with an expletive D, resulting in definite readings. Note, though, that the proposed semantic operation associated with the null D cannot capture the narrowest scope property of bare plurals, which we have seen, is distinct from the scope properties of regular indefinites. Furthermore, covert type-shifting cannot be avoided for definite readings simply by moving a predicate-denoting expression to D or by coindexing it with an expletive D. *Nom* or *iota* would have to convert the predicative expression into an argumental type.

Finally, the fact that some languages allow bare plurals in subject position but not others proves to be a serious empirical hurdle for the DP-only approach. Longobardi, following Contreras (1986), requires syntactic licensing for the Romance null D. He blocks generic readings for them, on the view that they are VP internal and must map into the nuclear scope (Diesing, 1992).¹³ English bare plurals, however, can occur in subject position and allow generic as well as existential readings. Longobardi proposes that the empty D is there to ensure existential interpretation, but English also has covert N to D movement which allows it to

¹³ When syntactic conditions allow bare plurals to occur above VP, generic readings become possible (Longobardi, 2000).

occur in the ungoverned subject position. Existential interpretation is read off prior to LF movement, but licensing is checked at LF (in English as opposed to Italian). We can see that the moves required to make the DP account work semantically are not straightforward.

The distinction between a null D and an N to D structure has also been invoked for Mandarin to explain why definite readings of bare nominals are syntactically unrestricted while indefinite readings are only possible in governed positions. Li (1997) suggests that indefinite readings involve a null D in need of licensing, while definite readings obviate this need by N to D movement. Yang (2001) shows that it is, in fact, possible for Mandarin subject bare nominals to have indefinite readings, as long as they are not clause-initial. Mandarin being an obligatory topic language, the subject moves into topic position when there is no other expression there. The perceived absence of indefinite readings is due to the nature of topics, not an inherent issue with indefinite readings of bare nominals (see also Jiang, 2012, 2018).

In sum, it seems problematic to derive existential readings without regard to position in languages like English and Mandarin but restrict them to governed positions in Italian and Spanish by parameterizing the levels at which licensing and interpretation operate. Chierchia's theory, instead, capitalizes on the difference between covert type-shifts and null determiners with respect to structural sensitivity. The former, being a purely semantic type repair operation, is not expected to be restricted to specific positions, while it is theoretically plausible to require syntactic licensing for the latter. Facts such as these suggest that natural language grammar must include both options.

Alternative Approaches

There are other theories that are compatible with the view that bare nominals need not be DPs, some of which I mention briefly here. Bošković (2005) claims that languages without articles project NPs, while languages with articles project DPs. While this parameterization brings into focus a sensitivity to syntactic processes that claims about structure can be expected to involve, it does not address issues such as differences between kind denoting and indexical bare plurals within a single language, and distributional differences between bare plurals in different languages. Another approach treats bare plurals in object-level predications, as opposed to kind-level predications, as weak indefinites but this approach fails to capture the crucial narrowest scope property of bare plurals (Gerstner-Link and Krifka, 1993; Wilkinson, 1991; Diesing, 1992; Krifka, 2003, among others). Van Geenhoven (1998) treats bare plurals as semantic incorporation, conflating two distinct phenomena. That they behave differently with respect to discourse anaphora, that incorporation typically targets direct objects while $\exists$ readings for bare plurals are generally possible, and the fact that incorporation, unlike bare plurals, shows quirky lexical restrictions remain unaddressed on this view (see Dayal, 2011b for more discussion).

The [-pred, +arg] Setting

The [-pred, +arg] setting of the NMP is perhaps the aspect of Chierchia's theory that has generated the greatest response. We frame our discussion here around the cluster of properties Chierchia identifies as arising from this particular setting.

Generalized Bare Arguments

For Chierchia, [-pred, +arg] languages, by hypothesis, allow argumental meanings of type $\langle\langle e, t \rangle, t\rangle$ or e at the level of NP. Since there is no call for null determiners, it is predicted that bare nominals will be possible in any argument position. A corollary to this is that such languages should not have definite determiners. At the time when Chierchia was developing his theory no such language had been documented but the empirical picture has changed since then.

Jiang (2012, 2018) shows that the classifier language Yi has a definite determiner:

- (10) si-hni ma sini sse-vo ma i-go nyi, si-hni / si-hni ma su jjy nra.
 girl CL and boy CL room sit, girl girl CL the very beautiful
 'A girl and a boy are sitting in the room, the girl is very pretty.'

Does the fact that both the bare nominal and the definite can be used anaphorically in (10) violate Blocking? Jiang argues that, appearances notwithstanding, the two options do not compete. She follows Dayal (2011a) and Trinh (2011) in taking the definite reading of kind terms to be the extension of the kind at an index $e^K(s)$. This makes the bare nominal a number neutral definite, which can be used with singular or plural antecedents. *Iota*, instead, is a type-shift from $\langle e, t \rangle$ to e , requiring the presence of CL. It is marked for number and therefore sensitive to the singular-plural specification on its antecedent.

The existence of the definite determiner in Yi also bears on the status of Avoid Structure, the principle responsible for ruling out definite determiners in classifier languages. As Jiang argues, economy of structure is preserved if a classifier language that develops definite determiners uses them only for structures where the denotation can be predicative. Determiners are thus predicted not to combine directly with nouns in such languages. The larger structure provided by classifiers is needed.

Jiang (2012, 2018) and Chierchia (2018) introduce modifications to the theory prompted by the discovery of [-pred, +arg] languages with D-based definiteness.¹⁴

¹⁴ Bangla, a South Asian classifier language, provides a different type of evidence for D-based definiteness (see Dayal, 2012, 2014; see also Bhattacharya, 1999; Biswas, 2016).

No Pluralization

Chierchia (1998) rules out plural morphology in [-pred, +arg] languages because the pluralization operation he defines in (7b), when applied to a noun that denotes a set of singularities and pluralities, results in an empty set.¹⁵ Since all nouns in such languages denote kinds, the property that *pred* extracts from them includes both types of entities. As such, pluralization results in the empty set. Count nouns in [+pred] languages, on the other hand, do not meet the same fate because pluralization applies to a set of atoms. The plural term can denote the set of plural individuals, eliminating singular individuals from its denotation. However, Chierchia's view of pluralization has been challenged and he has revised it to include atoms: *for any P, *P = $\lambda x \exists Q [Q \subseteq P \wedge x = \cup Q]$* (Chierchia, 2010: 114). This is in line with the standard view of plural nouns where the perceived restriction to sums is treated as an implicature. However, it also means that the original explanation for the generalization *No PL in [-pred, +arg] languages* is lost, though pluralization could still be ruled out as semantically redundant.

The empirical generalization that classifier languages lack pluralization has also been called into question (Li, 1999; Borer, 2005). Candidates for the plural morpheme include Mandarin *men*, Japanese *tachi*, Bangla *-gulo* and *-ra*. Notably, however, none of them is acceptable in counting constructions. One might argue that this would be similar to languages like Finnish or Turkish where only singular forms combine with numerals, a position consistent with the claim in Ionin and Matushansky (2006) that numerals operate on atomic sets. These supposed plural forms, however, have properties that are not typically associated with pluralization: *men*, *tachi* and *-ra* are restricted to animate nouns, *men*, *tachi* and *-gulo* are obligatorily definite, and *-gulo* also applies to mass nouns (see Yang, 1998; Kurafuji, 2004; Dayal, 2014 and references there). So these markers, though interesting in their own right, are not counterexamples to the generalization that [-pred, +arg] languages lack pluralization.

Mass, Kinds, Classifiers

By far the most debated issues have centered around the last two generalizations: the extension of all nouns is mass in [-pred, +arg] languages, and such languages all have generalized classifier systems. Instead of trying to summarize this vast literature, in the interests of space we will restrict ourselves to some recent developments in Chierchia's thinking on these issues. Chierchia (2010) argues for a notion of stable vs. unstable atomicity for count and mass nouns respectively. Briefly put, what counts as an atom for a count noun remains constant across contexts,

¹⁵ This is analogous to the reasoning behind the ungrammaticality of pluralization with mass nouns.

but what counts as an atom for a mass noun is variable. The relevance of this for counting, which looks for a set of stable atoms, is that a measure phrase is needed to deliver the stable set that is not inherent to mass terms. Note that this measure phrase may still need to be contextually specified (and may be subject to vagueness) but once that is done, the stability needed for counting is satisfied.

Recall that Chierchia (1998) does not actually conflate mass nouns and kinds in [-pred, +arg] languages, though he has sometimes been taken to do so. What he claims there is that all nouns start out as kinds in such languages (see Krifka, 1995 for a related position). Classifiers are needed because the shift from kinds to properties does not deliver the set of (stable) atoms that counting calls for.

Chierchia (2018) clarifies why nouns in classifier languages denote kinds rather than number neutral properties, sets of singular and plural individuals. There should be, in principle, two languages of this type if they did denote properties. One which would be [+pred, +arg], like Hindi, and allow argument shift via a covert application of *nom/iota*. Bare arguments would be freely available in such languages. Another type would be [+pred, -arg], like French, which would still require the obligatory presence of classifiers for counting, but it would also require obligatory overt/null determiners. Obligatory determiner classifier languages have not been documented so far. Neither have null determiner classifier languages, where the presence of null determiners would be signaled by subject-object asymmetries. This is the empirical argument for reference to kinds at the NP level, that is for a [-pred] specification.

Chierchia (2018) also presents a typology which represents the changed empirical landscape. Type I: numerals combine directly with all nouns (Indonesian, Yudja, Nez Perce), Type II: numerals cannot combine directly with any noun (Mandarin, Japanese, Korean, Bangla), Type III: count nouns (mostly) can combine directly with numerals, mass nouns cannot (English, Hindi). Interestingly, a language with no visible singular-plural morphology can belong to any of the three types. Dëne Sųliné, for example, requires a measure phrase for mass nouns while allowing direct counting for count nouns (Type III).¹⁶ We note that the system is predictive in the other direction: types I and II could not have robust singular-plural marking systems.

On Singular and Definite Kind Terms

We now consider alternative accounts of singular and/or definite generics, a class that is less well-studied than bare plural kind terms.

¹⁶ See Chung (2000) and Dalrymple and Mofu (2012) on Indonesian, Lima (2014) on Yudja, Deal (2017) on Nez Perce and Wilhelm (2008) on Dëne Sųliné.

Singularity and Kinds

English bare plural and definite singular kind terms raise the question of whether it is definiteness or number that is crucial in distinguishing between them. Dayal (1992, 2004a) argues that kind formation by *nom* is blocked by singularity, since the notion of a kind is antithetical to being necessarily instantiated by a singleton.¹⁷ Singular predicates shift to the taxonomic domain when *nom* cannot apply. If the uniqueness presupposition of the definite determiner is satisfied, we get a definite singular kind term. Other determiners behave in the expected way, resulting in recognizably taxonomic readings. On this view, it is the common noun in a definite singular that is ambiguous, not the determiner. Since an account is independently needed for taxonomic readings of determiners, nothing special need be said about singular definite generics.¹⁸

This approach predicts that if the singular term denotes a novel kind, it will have an indefinite determiner: *this morning Fred invented a pumpkin crusher* (from Guerts, 2001). It also predicts the possibility of bare singular kind terms in article-less languages, through the covert application of *iota*.¹⁹ Chierchia's account is tied too closely to the definite determiner to provide a simple account of these facts.

Another difference between Chierchia's and Dayal's accounts involves the observation that singular definite generics are restricted to well-established kinds: **The green bottle has a long neck* vs. *Green bottles have long necks*. However, the restriction is not absolute: *We manufacture three types of bottles at this plant, green, blue and clear. The green bottle is our particular specialty. It has a long neck* (Dayal, 1992: 56). Kay (1971) posits that in order for an entity to qualify as an element in a taxonomic hierarchy, it must have a contrast set. Dayal (2004a) leverages this to argue that context is needed to make contrast sets possible when the term itself does not refer to a well-established kind.

Other restrictions are inviolable. The definite singular cannot be associated with a reciprocal: **The dog bites each other* vs. *Dogs bite each other*. It is also incompatible with non-normative inductive generalizations: **The Rutgers Professor seems to be born on weekdays* vs. *Rutgers Professors seem to be born on weekdays* (due to E. Williams, Dayal, 2004a). Such facts motivate Chierchia's group forming operation. In Dayal's approach, it is taken as a primitive about taxonomic kinds that they are group-like rather than sum-like. The analogy, on both accounts, is with group denoting nouns like *committee*, as opposed to sum denoting nouns like

¹⁷ More broadly, the restriction may be against quantization, as suggested in Dayal (2013).

¹⁸ Tying the shift in domains to the inapplicability of *nom*, Dayal (2004a) addresses the fact that kind-level mass nouns are not definite: *(*the) sugar is an unhealthy product*. They may, of course, be definite if they are anaphoric, but that is also true of bare plurals: *This machine packages milk/apples. The milk/the apples is/are then shipped out*.

¹⁹ Chierchia (1998: 361) notes that Russian bare singulars cannot be arguments of kind-level predicates but this has been disputed (Dayal, 2004a: 402).

committee members.²⁰ As Jespersen (1927) put it, the singular definite "denotes the species itself" while the plural kind term denotes "members of the species".

That these effects are tied to number morphology rather than definiteness marking is supported by the fact that Romance definite plurals align with English bare plural kind terms and Romance definite singulars align with English definite singular kind terms.

Definiteness and Kinds

Any language that has definite determiners uses it to form singular kind terms, but not necessarily plural kind terms. The Romance languages do, English does not and German does so optionally. The definite plural generic has been seen as a counterpart of bare plurals because both terms show no preference for well-established kinds and both are acceptable in statements that express accidental generalizations. Chierchia treats the Romance definite plural generic as the intensionalized version of *iota* but formally distinct from *nom* in that it does not block the null determiner inside Romance bare plurals from denoting kinds. Dayal (2004a) sees the Romance definite determiner as a lexicalization of *nom*, but then there has to be some other explanation for bare plural kind terms, since a covert application of *nom* would be blocked in Romance. A third approach treats the definite determiner in these cases as semantically vacuous (Krifka et al., 1995). If expletive elements are deleted at LF, definite and bare plural kind terms should be semantically identical but they are not.

The definite plural aligns itself with the definite singular instead of the bare plural in episodic statements. Neither of them can be used to make a statement about some instantiation(s) of the kind, unless they are somehow representative of the species as a whole. While current accounts point towards an explanation, how exactly an overt definite determiner interacts with kind formation bears further investigation.

Conclusion

Chierchia's neo-Carlsonian approach, which draws on the deep connection between kinds and their instantiations and proposes a constrained system of type-shifts between the predicative and argumental domains, has proven resilient in the face of new discoveries and remains the most comprehensive cross-linguistic theory of noun phrase interpretation.

²⁰ Groups and sums differ in how they relate to their parts (Barker, 1992; Schwarzschild, 1996):
**the committee lives in different towns/the committee members live in different towns.*

As we have seen, the empirical landscape has shifted in significant ways since Chierchia's original proposal. We can conclude by asking if we would have known as much as we now do about the semantics of noun phrases if Chierchia's claim of semantic parameterization had not directed our attention to this issue. There is no way of answering this counterfactual with certainty, but whether we agree with the notion of a semantic parameter or not, we can all agree that it provoked us into action, the believers as well as the skeptics. It redefined the syntax-semantics interface and encouraged cross-fertilization of ideas across (sub)-disciplines and, in doing so, raised the bar for the empirical adequacy of explanations.

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Chapter 6

On Davidson's "The Logical Form of Action Sentences"



Paul Pietroski

Abstract This brief essay reviews and elaborates what I take to be Davidson's main arguments for his celebrated "event analyses" for action reports like (1–9), beginning with the interesting pattern of implications and non-implications exhibited by such sentences.

- (1) Scarlet poked Mustard with a pencil in the library.
- (2) Scarlet poked Mustard with a pencil.
- (3) Scarlet poked Mustard in the library.
- (4) Scarlet poked Mustard.
- (5) Scarlet poked Mustard in the kitchen.
- (6) Scarlet poked Mustard with a spoon.
- (7) Scarlet poked Mustard in the kitchen with a spoon.
- (8) Scarlet poked Mustard with a pencil in the kitchen.
- (9) Scarlet poked Mustard in the library with a spoon.

The second part of the essay addresses some potential sources of skepticism about Davidson's proposal, along with some subsequent sources of support, including perceptual reports like 'Plum saw Scarlet poke Mustard' and 'Plum heard Mustard shriek'. Given the spirit of this volume, the presentation of Davidson's leading ideas is more formally explicit – and more focused on issues of interest to contemporary semanticists – than in his seminal and wide-ranging paper, which was intended primarily for audiences in philosophy in the 1960s.

Keywords Actions · Adverbs · Compositionality · Events · Implication · Logical form · Meaning · Truth

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Suppose that yesterday, Miss Scarlet poked Colonel Mustard with a pencil in the library at noon and then at dusk, poked him again in the kitchen with a spoon. Given this scenario, any of the seven sentences listed below can be used to describe something that happened yesterday.

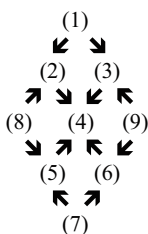
- (1) Scarlet poked Mustard with a pencil in the library.
- (2) Scarlet poked Mustard with a pencil.
- (3) Scarlet poked Mustard in the library.
- (4) Scarlet poked Mustard.
- (5) Scarlet poked Mustard in the kitchen.
- (6) Scarlet poked Mustard with a spoon.
- (7) Scarlet poked Mustard in the kitchen with a spoon.

Patterns of Implication

Intuitively, (1) implies both (2) and (3), each of which imply (4). Likewise, (7) implies (6) and (5), both of which imply (4). But the conjunction of (2) and (5) does not imply (8), which implies both (2) and (5). Nor does the conjunction of (3) and (6) imply (9), which implies (6) and (3).

- (8) Scarlet poked Mustard with a pencil in the kitchen.
- (9) Scarlet poked Mustard in the library with a spoon.

Similarly, (1) is logically stronger than the mere conjunction of (2) and (3). Sentence (1) cannot be used to correctly describe a scenario in which Scarlet poked Mustard with a pencil in the kitchen and again in the library with a spoon, but never with a pencil in the library. Likewise, (7) is stronger than the mere conjunction of (5) and (6). This interesting pattern of implications and non-implications can be indicated with diagrams like the one below.



This same pattern is exhibited by endlessly many sets of sentences like (1–9), in which a “basic” clause like (4) is extended with certain adverbial modifiers. It is also exhibited by sentences (1’–9’), each of which might be used in talking about a certain dinner party.

- (1') There was a banker from Dallas who wore a hat.
- (2') There was a banker from Dallas.
- (3') There was a banker who wore a hat.
- (4') There was a banker.
- (5') There was a banker who wore suspenders.
- (6') There was a banker from Manhattan.
- (7') There was a banker from Manhattan who wore suspenders.
- (8') There was a banker from Dallas who wore suspenders.
- (9') There was a banker from Manhattan who wore a hat.

The systematicity of this pattern presumably reflects some aspect of how speakers of English understand such sentences, whose constituents include expressions like 'with a pencil' and 'from Dallas' that appear as modifiers of other constituents like 'poke Mustard' and 'banker'.¹

For (1'–9'), there is an obvious initial hypothesis. Suppose that if **N** is a noun or noun phrase – like 'banker' or 'banker from Dallas who wore a hat' – and **S** is the sentence formed by adding 'There was a' before **N**, then **S** is understood as an existential generalization according to which (the relevant portion of) the world included an *entity* of the sort specified by **N**. And suppose that modification with a relative clause or a prepositional phrase signifies a kind of conjunctive restriction, as indicated with the simplified "logical forms" below.

- (1'L) $\exists e[\text{Banker}(e) \ \& \ \text{From-Dallas}(e) \ \& \ \text{Wore-a-hat}(e)]$
- (2'L) $\exists e[\text{Banker}(e) \ \& \ \text{From-Dallas}(e)]$
- (3'L) $\exists e[\text{Banker}(e) \ \& \ \text{Wore-a-hat}(e)]$
- (4'L) $\exists e[\text{Banker}(e)]$
- (4'L) $\exists e[\text{Banker}(e) \ \& \ \text{Wore-suspenders}(e)]$
- (6'L) $\exists e[\text{Banker}(e) \ \& \ \text{From-Manhattan}(e)]$
- (7'L) $\exists e[\text{Banker}(e) \ \& \ \text{From-Manhattan}(e) \ \& \ \text{Wore-suspenders}(e)]$
- (8'L) $\exists e[\text{Banker}(e) \ \& \ \text{From-Dallas}(e) \ \& \ \text{Wore-suspenders}(e)]$
- (9'L) $\exists e[\text{Banker}(e) \ \& \ \text{From-Manhattan}(e) \ \& \ \text{Wore-a-hat}(e)]$

This would explain the implications, since inferences of the form ' $\exists e[\Phi(e) \ \& \ \Psi(e)]$, so $\exists e[\Phi(e)]$ ' are obviously valid. If something satisfies two conditions, something satisfies the first condition.

¹ The linear order of modifiers is often irrelevant, but not always. In 'banker who wore suspenders from Manhattan', 'from Manhattan' presumably modifies 'suspenders'. Though we would typically ignore the possibility of using 'with a pencil in the library' to describe an action of manipulating a pencil that was in the library by remote control (e.g., from the conservatory). Likewise 'in the kitchen with a spoon' has a reading – not relevant here – on which 'with a spoon' modifies 'kitchen' instead of 'poked Mustard in the kitchen'.

This invites Davidson's (1967a) corresponding hypothesis about sentences like (1).

- (1) Scarlet poked Mustard with a pencil in the library.

Perhaps this "action report" is understood as an existential generalization, with the basic clause (i.e., the verb and its arguments) corresponding to a predicate – 'Poked(Scarlet, Mustard, e)' – that applies to something if and only if it was an *event* of Mustard being poked by Scarlet. Then the adverbial modifiers in (1–9) correspond to conjunctive restrictions as in (1L–9L) below.²

- (1L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{With-a-pencil}(e) \ \& \ \text{In-the-library}(e)]$
 (2L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{With-a-pencil}(e)]$
 (3L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{In-the-library}(e)]$
 (4L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e)]$
 (5L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{In-the-kitchen}(e)]$
 (6L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{With-a-spoon}(e)]$
 (7L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{In-the-kitchen}(e) \ \& \ \text{With-a-spoon}(e)]$
 (8L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{With-a-pencil}(e) \ \& \ \text{In-the-kitchen}(e)]$
 (9L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{In-the-library}(e) \ \& \ \text{With-a-spoon}(e)]$

Initially, it might seem odd to posit variables corresponding to verbs, in addition to any variables corresponding to nouns like 'Scarlet' or 'banker'. But consider (4) and (10).

- (4) Scarlet poked Mustard.
 (10) Scarlet will poke Mustard.

Such examples already suggest that at least often, tensed sentences are understood as existential generalizations over things that are temporally ordered. Both (4) and (10) seem to imply that some time *t* is such that Mustard gets poked by Scarlet at *t*, with each sentence imposing a further restriction on *t*. And as Davidson notes, examples like (11–13) point in the same direction.

- (11) The clock chimed after Scarlet poked Mustard.
 (12) Scarlet poked Mustard and then the clock chimed.
 (13) Scarlet poked Mustard before the clock chimed.

If 'after' indicates a temporal relation, clauses can presumably specify the relation as in (11L).

- (11L) $\exists f[\text{Chimed}(f, \text{the clock}) \ \& \ \exists e[\text{After}(f, e) \ \& \ \text{Poked}(\text{Scarlet}, \text{Mustard}, e)]]$

² Perhaps every such event was done with something, like a pencil or a finger, in some place at some time. But generalizations concerning pokings (or bankers) can be true without following from the meaning of 'poke' (or 'banker').

It does not follow that (4L) must be elaborated as (4L*) rather than (4L**);

(4L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e)]$

(4L*) $\exists e[\text{Poke-of-by}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{Past}(e)]$

(4L**) $\exists t[\text{Pokes-at}(\text{Scarlet}, \text{Mustard}, t) \ \& \ \text{Past}(t)]$

where 'Poke-of-by(Scarlet, Mustard, e)' applies to any *poke* of Mustard done by Scarlet, and 'Pokes-at(Scarlet, Mustard, t)' applies to any *time* at which Scarlet pokes Mustard. But phrases like 'with a pencil' and 'in the library' are not predicates of times; they seem to be understood as predicates of events that occur at times.³ And as Higginbotham (1983) noted, the untensed clause 'Scarlet poke Mustard' can be used to describe witnessed events, as in (14); see section "[Implication and composition](#)".

- (14) Prof. Plum saw Scarlet poke Mustard with a pencil, and
Mrs. White saw Scarlet poke Mustard with a spoon.

As our initial example highlights, (4) is not a name for any event. Scarlet may have poked Mustard several times. But temporal order is not crucial to the example. Suppose that in the kitchen at dusk, Scarlet poked Mustard gently with a blue spoon and simultaneously poked him sharply with a red spoon. Then (15–20) are all correct partial descriptions of what happened.

- (15) Scarlet poked Mustard gently with a blue spoon.
(16) Scarlet poked Mustard gently.
(17) Scarlet poked Mustard with a blue spoon.
(18) Scarlet poked Mustard with a red spoon.
(19) Scarlet poked Mustard sharply.
(20) Scarlet poked Mustard sharply with a red spoon.

But (21) and (22) are not implied. Scarlet may have poked Mustard with a spoon exactly twice.

- (21) Scarlet poked Mustard gently with a red spoon.
(22) Scarlet poked Mustard sharply with a blue spoon.

No two pokings have exactly the same spatiotemporal properties, down to the details of where they began and ended. But two such events can occur in the same kitchen at the same time.⁴

³ According to Davidson (p. 64), "We may think of 'before' and 'after' as relating events as easily as times. For most purposes, if not all, times are like lengths – convenient abstractions with which we can dispense in favour of the concreta that have them." We can also introduce notions of *reference* time and *speech* time to accommodate complex tenses as in 'By noon tomorrow, Mustard will have departed'; see Reichenbach (1947), Hornstein (1990).

⁴ Likewise, 'a banker from Dallas who wore a hat' is not a name, even if only one person fits the indefinite description; cp. Davidson's discussion of von Wright (pp. 43–4) and reference to Ramsey

Implication and Composition

Let's grant that action reports like (3) and (4) are interestingly similar to (3L) and (4L).

- (3) Scarlet poked Mustard in the library.
- (3L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{In-the-library}(e)]$
- (4) Scarlet poked Mustard.
- (4L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e)]$

This does not yet show that sentences of a naturally acquirable language, like spoken English or ASL, *have* logical forms – much less that (3) and (4) have logical forms that are correctly depicted with the invented formalism of (3L) and (4L). We would not say that (3L) has the grammatical form of (3). So one wants to hear more about Davidson's conception of what it is for a natural sentence, composed of pronounceable words, to have a logical form.

For Davidson, a crucial point is that the same verb appears in (3) and (4). By way of contrast, note that unlike (23), (24) implies (4).

- (23) Professor Plum thinks that Scarlet poked Mustard.
- (24) Professor Plum knows that Scarlet poked Mustard.

This implication is due to a feature of 'knows' that is not shared by verbs like 'thinks', 'hopes', and 'fears'. Similarly, while (25) implies (4), replacing 'and' with 'or' spoils the inference.

- (25) Scarlet poked Mustard and Mustard shrieked.

And while (4) implies (26), (4) does not imply (27).

- (26) Someone poked Mustard.
- (27) Everyone poked Mustard.

These contrasts reflect distinct word meanings – 'knows' vs. 'thinks', 'and' vs. 'or', 'someone' vs. 'everyone'. But with regard to sentences like (3) and (4), Davidson wanted to "give an account of the logical or grammatical role of the *parts or words*" in such sentences that is "consistent with the entailment relations between such sentences and with what is known of the role of *those same parts or words* in other (non-action) sentences." He identifies this project with that of "showing how the meanings of action sentences depend on their *structure*," and goes on to say that he is "not concerned with the meaning analysis of logically simple expressions in so far as this goes beyond the question of logical form" (p. 37, my emphases).

(p. 61). While Davidson's own examples do not involve simultaneous actions that fit a common description, he clearly assumes that such cases are possible. Taylor (1985) offers examples like (15–22), citing Gareth Evans (via Christopher Arnold).

Davidson held that characterizing the logical relations exhibited by natural sentences, including those used to report actions, is part and parcel of characterizing the sentence meanings in a way that reveals how they are determined by word meanings and grammatical structure; see his "Reply to Cargile" (pp. 62–65). From this perspective, the goal is to specify the contributions of expressions like 'poked' and 'in the library' to the meanings of the endlessly many sentences in which such expressions appear. If these sentences exhibit the logical relations that Davidson describes in terms of conjunct reduction, then 'poked' and 'in the library' are like the conjuncts in 'Poke-of-by(x, y, e) & Past-Event(e) & In-the-library(e)' – as opposed to, say, the conjuncts in 'Pokes-in-at(x, y, z, t) & Past-Time(t) & The-library(z)'.

It can be tempting to describe the meaning of a verb as "variably polyadic:" in some sentences, 'poke' expresses a relation exhibited (at times) by pairs of individuals; in other sentences, 'poke' expresses a relation exhibited by triples consisting of two individuals and a place, or two individuals and an instrument; in other sentences, the same verb expresses a relation exhibited by quadruples consisting of two individuals, a place, and an instrument; etc. But *prima facie*, this abandons the idea that 'poke' has a sentence-invariant meaning that it contributes to these many sentences; cp. Kenny (1963), Davidson (1967b, 1968, 1984).

Absent an account of how a single variably-polyadic meaning for 'poke' could accommodate so many different cases of adverbial modification, it seems reasonable to explore the model suggested by (1'–4'), whose constituents seem to have sentence-invariant meanings.

- (1') There was a banker from Dallas who wore a hat.
- (2') There was a banker from Dallas.
- (3') There was a banker who wore a hat.
- (4') There was a banker.

Still, one might doubt that grammatical modification is *logically* significant in a way that is like the explicit conjunctions in (28) and (29).

- (28) Someone was both a banker and from Dallas.
- (29) Something was both a poking of Mustard by Scarlet and done in the library.

A skeptic might challenge Davidson's claim that our pronounceable sentences – as opposed to certain thoughts that we can somehow indicate in contexts – exhibit logical relations that are not indicated with special lexical meanings; cp. (23–27). For not all cases of grammatical modification are cases of conjunctive restriction. A fake diamond is not a thing that is both a diamond and fake. So if 'fake diamond' has a logical form, it is not 'Fake(e) & Diamond(e)'. Put another way, argument (30) is not an instance of the valid form (31).

- | | |
|---|---|
| (30) <u>There is a fake diamond in the library.</u>
There is a diamond in the library. | (31) $\frac{\exists e[\Delta(e) \ \& \ \Phi(e) \ \& \ \Psi(e)]}{\exists e[\Delta(e) \ \& \ \Psi(e)]}$ |
|---|---|

So if ‘fake diamond’ and ‘blue diamond’ are instances of the same grammatical/logical form, then ‘blue diamond’ is not a simple conjunction, and (32) is not an instance of (31).

- (32) There is a blue diamond in the library.
There is a diamond in the library.

Perhaps *pace* Davidson, (32) seems valid because of what we know about ‘blue’ as opposed to ‘fake’; cp. ‘and’ versus ‘or’. Similarly, (33) does not imply (4).

- (33) Scarlet allegedly poked Mustard.
(4) Scarlet poked Mustard.

So if (33) and (34) have the same grammatical/logical form, (34) does not imply (4),

- (34) Scarlet deliberately poked Mustard.

at least not by virtue of its form; cp. ‘knows’ versus ‘thinks’. One can develop this response to Davidson, in a way that hints at variable adicity, via the following conjecture: an adjective can *be* a predicate that combines with tense as in ‘was blue’, in which the logical form of the adjective is ‘Blue-at(x, t)’, with ‘x’ and ‘t’ ranging over entities and times; or it can *complicate* a predicate as in ‘blue diamond’, in which the logical form of the adjective is ‘BLUE[Δ]’, with ‘Δ’ ranging over predicates like ‘Diamond-at(x, t)’. Then one can say that ‘BLUE[Diamond-at(x, t)]’ is like ‘Diamond-at(x, t) & Blue-at(x, t)’ – or ‘Diamond-at(x, t) & Blue-for-a-diamond-at(x, t)’ – while ‘FAKE[Diamond-at(x, t)]’ is more like ‘~Diamond-at(x, t) & Made-to-look-diamondish(x)’. The idea here is that unlike ‘fake’, ‘blue’ complicates ‘diamond’ in a relatively simple and restricting way, but ‘blue diamond’ and ‘fake diamond’ do not differ in *form*. On this view, the contrast reflects the different meanings of ‘blue’ and ‘fake’; see Parsons (1970), Kamp (1975).

Davidsonians will reply that ‘fake diamond’ and ‘blue diamond’ *are not* grammatically type-identical, and that the logical complexity masked by ‘FAKE[Diamond-At(x, t)]’ mirrors the complexity of ‘fake diamond’; cp. ‘faked to look like a diamond’. Perhaps this reply is wrong. But it still seems that endlessly many cases of modification are understood conjunctively, even if some cases are not; and see Higginbotham (1985). If the logical form of ‘banker from Dallas’ is ‘FROM-DALLAS[Banker-At(x, t)]’, one wants to know why it seems so clear that the modifying phrases in (1’) and (7’) can be validly deleted, but not recombined as in (8’) or (9’).

- (1’) There was a banker from Dallas who wore a hat.
(7’) There was a banker from Manhattan who wore suspenders.
(8’) There was a banker from Dallas who wore suspenders.
(9’) There was a banker from Manhattan who wore a hat.

It does not help to say that the logical form of ‘banker from Dallas who wore a hat’ is ‘WHO-WORE-A-HAT[FROM-DALLAS[Banker-At(x, t)]]’, and likewise

for 'banker from Manhattan who wore suspenders', unless this formalism is cashed out in a way that *explains* the apparent implications and non-implications. Similarly, if the logical form of 'poked Mustard with a pencil in the library' is 'IN-THE-LIBRARY[WITH-A-PENCIL[Poked(x, Mustard)]]', why do (1) and (7) seem to have Davidsonian implications *without* jointly implying (8) or (9)?

- (1) Scarlet poked Mustard with a pencil in the library.
- (7) Scarlet poked Mustard in the kitchen with a spoon.
- (8) Scarlet poked Mustard with a pencil in the kitchen.
- (9) Scarlet poked Mustard in the library with a spoon.

In these examples, the modifiers do not merely *complicate* simpler predicates, they seem to *restrict* simpler predicates in a specific systematic way. This suggests that 'fake' and 'alleged' are special cases, which carry their own implications of fakings and allegations.

As Davidson notes, he can accommodate the obvious fact that while a big ant is an ant, nothing is big simpliciter – and while a gentle deliberate poking is a poking, nothing is gentle or deliberate simpliciter. For he can still say that (35) is an instance of the logical form (36),

- (35) There is a big ant in the library.
- (36) $\exists x[\text{Ant}(x) \ \& \ \Phi(x) \ \& \ \text{In-the-library}(x)]$

leaving room for various hypotheses about the conjunct corresponding to 'big', which is somehow comparative. Perhaps the logical form of 'is a big ant' is complex in a way hinted at with 'is an ant and a big one'. One can likewise say that (37) has the logical form shown in (38),

- (37) Scarlet poked Mustard gently.
- (38) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \Phi(e)]$

but grant that 'gently' is comparative, as hinted at with 'a poking of Mustard and a gentle one'.

Evidently, 'deliberately' is more complex, since (34) and (39) do not jointly imply (40);

- (34) Scarlet deliberately poked Mustard.
- (39) Mustard is Batman.
- (40) Scarlet deliberately poked Batman.

cp. 'wants to visit Hesperus' versus 'wants to visit Phosphorus'. Yet deleting 'deliberately' from (34) still seems valid, as if (34) has the logical form shown in (41).

- (41) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \Phi(e)]$

So perhaps 'deliberately' is comparative *and intentional* in a way hinted at with 'a poking of Mustard *and* one done by an agent who *represented it* as such'; cp. (42).

(42) Scarlet poked Mustard because she wanted to do so.

The complexity of ‘because she wanted to do so’ does not show that (42) does not have the logical form shown in (41). The question is whether ‘deliberately’, ‘gently’, and ‘allegedly’ differ grammatically in ways that plausibly correlate with how these words differ logically. *If* these modifiers are grammatically type-identical (cp. ‘red’, ‘green’, and ‘blue’), then deleting the adverb from (34) or (37) is not licensed by virtue of logical form in Davidson’s sense; cp. (33). But if modifiers vary typologically, it seems that many are understood as Davidsonian conjuncts.

This highlights questions about how events are individuated; see, e.g., Davidson (1969, 1980), Goldman (1970), Kim (1976). Suppose that Mustard would not have shrieked at dusk had he not been poked *sharply*. If it follows that the event of *poking him sharply at dusk* differs from the event of *poking him at dusk*, then analogies to adjectival modification are misleading. Each blue diamond is identical with some diamond. And if the event of Scarlet poking Mustard sharply at dusk is not identical with an event of her poking him at dusk, the validity of (31) is irrelevant.⁵

$$(31) \frac{\exists e[\Delta(e) \ \& \ \Phi(e) \ \& \ \Psi(e)]}{\exists e[\Delta(e) \ \& \ \Psi(e)]}$$

These points have ramifications for truth-theoretic conceptions of meaning; see Pietroski (2018). But here, let me simply note that Davidson (1967a) wanted a theory that preserves several desiderata: (i) verbs have sentence-invariant meanings; (ii) action reports are existential generalizations, not names for actions; (iii) adverbial modifiers should be accommodated without resorting to variable polyadicity; (iv) an action can be described in endlessly many ways; so (v) actions should not be individuated as finely as action descriptions, or in any way that would invalidate valid inferences; but (vi) actions should not be individuated so coarsely that Scarlet’s gentle poking with a red spoon and her ungente poking with a blue spoon guarantees a gentle poking with a blue spoon or an ungente poking with a red spoon.

Some Further Evidence and Developments

Davidson’s paper prompted a search for more evidence in support of event analyses, along with a literature that included objections, replies, and emendations of the initial proposal. This is not the place for a review; see, e.g., Pietroski (2003,

⁵ But as Davidson notes (p. 41), in signing a check, he may be writing his name on a piece of paper and paying a debt. He adds that it is “hard to imagine how we can have a coherent theory of action” if corresponding sentential reports are not “made true by the same action.” This invites further discussion of ‘by’, since one can pay by signing, yet not sign by paying; see Hornsby (1980) for discussion of intentional vs. causal ordering.

2018) and references there. But perceptual reports like (43–46) illustrate several Davidsonian points; see Higginbotham (1983), Vlach (1983).

- (43) Plum saw Scarlet poke Mustard.
- (44) Plum heard Mustard shriek.
- (45) Plum heard a shriek, and White heard a shriek.
- (46) White heard Mustard shriek.

First, 'Mustard shriek' is not a name for a particular event. Suppose that Prof. Plum heard Mustard shriek at noon, while Mrs. White heard him shriek at dusk. Second, 'Mustard shriek' seems to be referentially transparent in (44). Given (39), (44) implies (47).

- (39) Mustard is Batman.
- (47) Plum heard Batman shriek.

By contrast, (39) and (48) do not imply (49).

- (48) Plum heard Mustard shrieked.
- (49) Plum heard Batman shrieked.

Sentence (48) has the meaning of (50); cp. (51).

- (50) Plum heard that Mustard shrieked.
- (51) Plum was told that Mustard shrieked.

But (44) is more like (52). Given (39), (52) implies (53).

- (52) Plum heard a shriek by Mustard.
- (53) Plum heard a shriek by Batman.

The grammatical distinction between (44) and (48) is subtle. Nonetheless, it is correlated with an important logical distinction, as if (44L) is logical form of (44).

- (44L) $\exists e \exists f [\text{Heard}(\text{Plum}, e, f) \ \& \ \text{Shriek}(f, \text{Mustard})]$

This suggests a simple diagnosis for the ambiguity of (54).

- (54) Plum heard Mustard shriek in the library.
- (54a) $\exists e \exists f [\text{Heard}(\text{Plum}, e, f) \ \& \ \text{Shriek}(f, \text{Mustard}) \ \& \ \text{In-the-library}(f)]$
- (54b) $\exists e \exists f [\text{Heard}(\text{Plum}, e, f) \ \& \ \text{Shriek}(f, \text{Mustard}) \ \& \ \text{In-the-library}(e)]$

According to (54a), Plum heard a *shriek* that occurred in the library; but he may have been in hall. According to (54b), the event of Plum *hearing* a shriek occurred in the library; but the shriek may have been in hall. This argument for treating 'in the library' as a predicate of events also interacts with the others discussed above. We want some explanation for the apparent network of implications and non-implications for each reading of (55).

- (55) Plum heard Mustard's loud shriek in the library
and then heard Scarlet laugh softly in the hall.

With these points in mind, let's return to the idea of elaborating (4L) as in (4L*).

- (4L) $\exists e[\text{Poked}(\text{Scarlet}, \text{Mustard}, e)]$
(4L*) $\exists e[\text{Poke-of-by}(\text{Scarlet}, \text{Mustard}, e) \ \& \ \text{Past}(e)]$

Initially, one might not see how 'poked' could be logically *triadic*, given that combining 'poked' with *two* names can yield a complete sentence. But if 'Scarlet poke Mustard' is a logically monadic predicate like 'Poke-of-by(Scarlet, Mustard, e)', it can be conjoined with a tense restriction and existential closure as in (4L*). Indeed, (4L*) can be viewed as the result of spelling out 'Poked(Scarlet, Mustard)' in accord with a principle like the following: $\forall x\forall y\{\text{Poked}(x, y) \equiv \exists e[\text{Poke-of-by}(x, y, e) \ \& \ \text{Past}(e)]\}$. So perhaps 'Poke-of-by(x, y, e)' can be further analyzed in accord with (56); cp. Castañeda (1967), Carlson (1984), Parsons (1990).

- (56) $\forall x\forall y\forall e\{\text{Poke-of-by}(x, y, e) \equiv \text{Poke}(e) \ \& \ \text{Patient-of}(e, y) \ \& \ \text{Agent-of}(e, x)\}$

Davidson (1985) adopted this view, according to which (4) implies that Scarlet did something, and that something happened to Mustard – and more strongly, that the poking was both something Scarlet did and something that befell Mustard, as if Scarlet acted *and thereby* Mustard was stabbed.⁶ This preserves a sense in which 'poke' is semantically binary: its meaning is specified in terms of *two* thematic relations, corresponding to the poker and pkee; cp. (57).

- (57) There was a poking of Mustard by Scarlet.

By contrast, the meanings of 'shriek' and 'give' might be specified as thematically unary or ternary. But the details remain matters of debate, in part because of examples like (58–61), which raise questions about how thematically *uniform* verbs are across sentences.

- (58) Scarlet kicked the box.
(59) The box was kicked.
(60) The baby kicked.
(61) Scarlet kicked Mustard the box.

Given (58), one might expect the logical form of 'kick' to be 'Kick-of-by(x, y, e)'. But this predicts covert constituents in (59) and (60), corresponding to 'x' and 'y' respectively; and these claims about grammar are tendentious. In this regard, examples like (61) are potentially instructive. The indirect object, 'Mustard', is not a grammatically essential argument of the verb 'kick'; cp. (62). So it is plausible that the logical form of both (61) and (62) is (62L*);

⁶ Or as Leibniz might have put it, Scarlet poked and *eo ipso* Mustard was poked; cp. Kratzer (1996).

(62) Scarlet kicked the box to Mustard.

(62L*) $\exists e[\text{Kick-of-by}(\text{Scarlet, the box, } e) \ \& \ \text{Past}(e) \ \& \ \text{To-Mustard}(e)]$

where Mustard is represented as a recipient (of the box), and not as the thing Scarlet kicked. So given (59), perhaps the subject of (58) is not an argument of the lexical verb 'kick', as opposed to a covert verbal element that is akin to 'made' in (63), which implies (64); cp. (65) and (66).

(63) Scarlet made the box move.

(64) The box moved.

(65) Scarlet moved the box.

(66) Scarlet saw the box move.

Maybe the logical form of 'kick' is 'Kick(e)' or 'Kick-of(y, e)' – Schein (1993) suggests the former, Kratzer (1996) suggests the latter – with 'Agent-of(e, x)' perhaps reflecting a covert verb akin to 'made'. Then the logical form of (4) is the "thematically separated" (4S*) or (4S**), as opposed to any alternative that includes a ternary predicate like 'Poke(Scarlet, Mustard, e)'.

(4S*) $\exists e[\text{Past}(e) \ \& \ \text{Agent}(e, \text{Scarlet}) \ \& \ \text{Poke-of}(e, \text{Mustard})]$

(4S**) $\exists e[\text{Past}(e) \ \& \ \text{Agent}(e, \text{Scarlet}) \ \& \ \text{Poke}(e) \ \& \ \text{Patient}(e, \text{Mustard})]$

This raises questions about how often verbs are associated with *grammatical configurations* that have thematic significance, in ways that can make logically monadic or dyadic *verbs* seem more polyadic. This in turn highlights questions about plurality.

Consider the collective reading of (67) –

(67) Three musicians pushed five pianos yesterday.

according to which three musicians did the work of getting five pianos pushed, with no implication regarding any degree of cooperation – as opposed to the distributive reading which implies that *each of* three musicians pushed five pianos. One might hope to capture the first reading as follows: $\exists e\exists s\exists s^*[\text{Collection-of-three-musicians}(s) \ \& \ \text{Collection-of-five-pianos}(s^*) \ \& \ \text{Push-of-by}(s, s^*, e) \ \& \ \text{Yesterday}(e)]$.⁷ But Schein (1993, 2017) argues that we need to posit thematically separated logical forms to accommodate variants like (68).

(68) Three artists gave four museums five sculptures.

Note that (68) has a reading according to which three artists were (together) the agents of some events in which each of four museums was given five sculptures, for a total of twenty donations.

⁷ Let's bracket the worry that (67) does not logically imply that there are any *collections* of musicians or pianos, much less collections that push or get pushed; cp. Boolos (1998).

This leads us back to questions about what logical forms *are*, and how they are related to sentences of any invented language, especially given Davidson's preference for first-order quantification and Tarski-style semantics. These issues remain unsettled, though illuminated by Davidson's seminal proposal regarding sentences like (1).⁸

(1) Scarlet poked Mustard with a pencil in the library.

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⁸ Thanks to Zoltán Gendler Szabó for helpful comments.

Chapter 7

On Dowty's "Thematic Proto-Roles and Argument Selection"



Beth Levin

Abstract David Dowty's influential paper "Thematic Proto-roles and Argument Selection" (1991) delineates the problems inherent in the traditional notion of semantic role and then proposes that these challenges can be overcome by adopting a prototype conception of this notion. Dowty introduces two 'proto-roles', each associated with a set of lexical entailments chosen for their relevance to argument realization. He complements the proto-roles with an Argument Selection Principle, which determines how the arguments of a transitive verb map onto subject and object on the basis of their associated entailments. After summarizing the highlights of Dowty's paper, this commentary critically reviews its contributions and limitations. First, the commentary addresses critiques and refinements of the sets of entailments that compose Dowty's proto-roles. Then, it discusses the pluses and minuses of the Argument Selection Principle since Dowty claims that the proto-roles are valuable precisely because they allow a range of argument realization phenomena to be captured. Finally, the commentary considers the broader impact of Dowty's paper and reviews work that develops its ideas further.

Keywords Argument realization · Proto-roles · Semantic roles · Subject and object

Semantic roles appear to offer immense promise in the explanation of a wide range of linguistic phenomena, yet they also seem to consistently fall short of fulfilling this promise. The lure of semantic roles is reflected in the repeated attempts to formulate sets of semantic roles dating back to the Sanskrit grammarian Pāṇini and continuing to current automatic semantic role labelling efforts. Dowty's first goal in 'Thematic Proto-roles and Argument Selection' is to underscore the severity of these shortcomings, pointing to problems inherent in the traditional discrete concept

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of semantic role. In the bulk of the paper, Dowty's aim is constructive: to show a way forward using a revised, prototype conception of semantic role in a domain which has seemed particularly amenable to explanation in terms of traditional semantic roles, but has nevertheless proved recalcitrant: predicting the syntactic realization of a verb's arguments. Dowty (1991: 560–562) argues that success is most likely in a limited domain such as argument realization or, as he calls it, argument selection. Furthermore, if this domain can be conquered, it should provide a stepping stone to other domains of interest.

This commentary presents highlights of Dowty's paper (section “[Proto-Roles and the Argument Selection Principle: An Overview](#)”), critically reviews its contributions and limitations (section “[Critical Comments](#)”), and considers its impact and further developments (section “[Impact of the Paper and Subsequent Developments](#)”).¹ Given space considerations, I focus on the central contributions; facets of the paper that I just touch on include the critique of semantic roles (1991: 553–559), the discussion of the formal foundations of semantic roles, which reviews (Dowty, 1989), and the discussion of psycholinguistic implications (1991: 600–605).

Proto-Roles and the Argument Selection Principle: An Overview

Traditionally, semantic roles (also known as thematic relations or participant roles) are labels associated with the arguments of a verb by virtue of their part in the event denoted by the verb (Fillmore, 1968, 2003; Gruber, 1965; Levin and Rappaport Hovav, 2005: Chapter 2). Each traditional semantic role is taken to be a discrete notion with a definition which picks out a natural class of arguments with respect to linguistic generalizations. The set of roles chosen should be small enough to allow for the statement of strong generalizations, yet large enough that every argument of every verb can be assigned a role from the set. However, it has proved difficult to find a set that meets these desiderata, as Dowty (1991: 553–559) forcefully reviews.

Against this backdrop, Dowty introduces a revised conception of semantic roles by focusing on a key problem in argument realization: given a transitive verb, what determines which argument is its subject and which its object? To answer this question, Dowty re-envision the notions of agent and patient, the two traditional semantic roles most closely tied to subject and object, as collections of semantic properties which contribute to subjecthood and objecthood, respectively. Dowty

¹ There are many critical discussions of Dowty's paper; they include Croft (1998: 36–38), Davis (2001: 61–73), Davis and Koenig (2000: 74–76), Grimm (2011: 519–522), Koenig and Davis (2001: 80–84), Levin and Rappaport Hovav (2005: 51–65, 112, 150), Newmeyer (2002: 67–71), Primus (1999: 33–60), and Tenny (1994: 101–105). For comparisons to other ‘generalized semantic role’ approaches (see section “[Impact of the Paper and Subsequent Developments](#)”), see Van Valin (1999: 385–388). For further references to proto-roles—and semantic roles in general—see Levin (2013).

calls each property cluster a proto-role, a name purposefully evoking the notion of prototype. Section "[The Proto-Role Properties](#)" introduces the two clusters of proto-role properties, and section "[Proto-Role Entailments in Argument Realization](#)" describes how they figure in argument realization.

The Proto-Role Properties

Dowty takes a truth-conditional semantic perspective on the argument realization problem, building on earlier work (Dowty, 1989) which appraises the notion of semantic role from such a perspective. A verb imposes entailments on each of its arguments by virtue of the part that argument plays in the event that it describes. These are 'lexical entailments' in that they follow from the meaning of the verb alone (1991:552); that is, they hold of all its uses and are not influenced by context. Dowty is interested in those lexical entailments which recur across many verbs and, among these, those that figure in subject and object selection. This goal leads him to identify the two clusters of properties in (1) and (2), which he claims are comprised of precisely such lexical entailments.² Dowty refers to the clusters as Agent Proto-Role and Patient Proto-Role properties—or, for short, Proto-Agent and Proto-Patient properties—since each cluster includes properties that commonly figure in definitions of the traditional agent and patient roles, respectively.

- (1) Contributing properties for the Agent Proto-Role (Dowty 1991: 572, (27)):
 - volitional involvement in the event or state
 - sentience (and/or perception)
 - causing an event or change of state in another participant
 - movement (relative to the position of another participant)
 - (exists independently of the event named by the verb)
- (2) Contributing properties for the Patient Proto-Role (Dowty 1991: 572, (28)):
 - undergoes change of state
 - incremental theme
 - causally affected by another participant
 - stationary relative to movement of another participant
 - (does not exist independently of the event, or not at all)

Given Dowty's goals, the proto-roles include only recurring lexical entailments that bear on subject and object selection, whether or not they are pertinent to any traditional semantic role (1991: 562). Indeed, the majority of the proto-role entailments overlap with notions referenced in definitions of semantic roles or mentioned in the literature as semantic determinants of argument realization (e.g.,

² Dowty (1991: 552) notes some of the relevant properties might be presuppositions rather than entailments, but does not find it important to distinguish between the two. Following him, I simply refer to 'entailments'.

Fillmore 1977: 102, Hopper and Thompson 1980, Lakoff 1977: 245, Mohanan 1994: 28). An exception is ‘incremental theme’.

Several properties deserve comment, although see Primus (1999: 36–43) for extensive elaboration on all the proto-role properties. First, movement, since it involves a change, might have been expected to be a Proto-Patient entailment, yet ‘movement relative to the position of another participant’ is listed as a Proto-Agent entailment, with its counterpart ‘stationary relative to the movement of another participant’ listed as a Proto-Patient entailment.³ Dowty includes it to deal with transitive verbs like *cross*, *enter*, and *pass*, as in *The runner crossed the finish line*; he comments that ‘movement usually only counts as a relevant change of state when described as to or from a specific location’ (1991: 574). Second, the notion of sentience involves more than being a sentient entity (1991: 573); Dowty’s intent is perhaps well-captured by Wechsler’s (1995: 35–40) ‘having a notion of’. Third, traditional semantic roles are not aspectual in nature, yet the incremental theme Proto-Patient entailment is. It holds of an event participant whose spatial extent defines a homomorphism with the progress of the event; it ‘measures out’ the event, to quote (Tenny, 1992, 1994). Its inclusion reflects the increased interest at the time that Dowty’s paper was published in grounding semantic determinants of argument realization aspectually, perhaps in part because aspectual notions seemed more amenable to rigorous, formal definition than traditional semantic roles.⁴ Fourth, both proto-role clusters include an entailment referencing independent existence, listed in parentheses. Dowty (1991: 572), who adopts this entailment from Keenan (1976), uses this typographical device because he is unsure whether this entailment might bear instead on the discourse functions of subject.⁵

As the labels suggest, Dowty takes each proto-role to be a concept with a prototype structure (Rosch, 1973; Rosch and Mervis, 1975). When a verb entails most, if not all, of the properties of either the Proto-Agent or the Proto-Patient of one of its arguments, then that argument corresponds to a ‘good’ example of the corresponding traditional role. For instance, the verb *assassinate* entails all the Proto-Agent entailments of its subject, and its subject is indeed considered an uncontroversial agent. An argument with only some entailments of a given proto-role would qualify as a less good example of the relevant role and might even be assigned another traditional role. For example, the perceiver argument of *see* or other perception verbs has the sentience Proto-Agent entailment, but lacks the

³ As Dowty notes, in assessing whether the movement-related properties hold, it may be that a second object is only considered to ‘be ‘stationary’ with respect to the faster first object’s perspective’ (1991: 573). That is, both event participants could be in motion (e.g., the verbs *overtake* and *pass*).

⁴ For discussion of aspectual vs. causal conceptualizations of events and their contribution to argument realization see Levin and Rappaport Hovav (2005: Chapter 4).

⁵ The notion of existence does not figure in most proposed semantic role lists. An exception is Fillmore, who recognizes a ‘factive’ (1968: 25) or ‘result’ (1971: 376) role assigned to ‘effected objects’, i.e. created entities. For discussion of whether effected objects are implicated in argument realization see Hopper (1985) and Martínez Vázquez (1998).

volitionality and causer entailments (Dowty, 1991: 573); concomitantly, despite its realization as a subject, it is often said to be an experiencer rather than an agent. In fact, an argument of a verb need not be associated with any of the proto-role entailments at all (although presumably it is still associated with some entailment by its verb). Thus, an argument of a verb may have all, some, one, or none of the entailments from a given cluster. Further, it might even have entailments from both clusters. For example, the object of the experiencer-object psych-verb *frighten* has the Proto-Agent entailment of sentience and the Proto-Patient entailment of undergoing a change of state in its cause-a-change-of-state use (Dowty, 1991: 579–580). Arguments, then, can show ‘degrees of membership’ in a proto-role; thus, the problems of unclear boundaries and role fragmentation that plague traditional semantic roles are sidestepped.

Although the names ‘Proto-Agent’ and ‘Proto-Patient’ evoke traditional semantic roles, no argument is actually assigned either label; further, the labels themselves do not figure in any grammatical generalizations. Nevertheless, some subsequent work (Alsina, 1996; Davis and Koenig, 2000) does ‘reify’ the proto-roles, to use Davis and Koenig’s (2000: 74, 75) term, treating certain arguments as if they were either a Proto-Agent or a Proto-Patient and stating generalizations that make reference to these labels.

Dowty’s approach offers insight into why traditional semantic roles emerge: these roles simply label certain recurring subsets of proto-role entailments (Dowty, 1991: 577). For instance, causation and movement without sentience correspond to the notion of instrument, while sentience without volition and causation defines an experiencer. Relatedly, the thematic hierarchy, which is often appealed to in the explanation of argument realization and other linguistic phenomena, also emerges from the proto-role clusters (Dowty, 1991: 578). Nested sets of Agent (or Patient) proto-role entailments can be ranked according to the decreasing number of such entailments they encompass. That is, agent outranks experiencer because it has volition and causation entailments that an experiencer lacks, although both share sentience and independent existence. Further, the proto-role approach easily allows for the statement of generalizations of different grain-size: a generalization that makes reference to just one or two entailments will apply more broadly than one that makes reference to a larger number of entailments. However, this option is not available with a set of discrete semantic roles.

Proto-Role Entailments in Argument Realization

Dowty takes a nonderivational approach to argument selection: argument selection reflects constraints on possible verb meanings. In particular, a verb must lexicalize a meaning consistent with his Argument Selection Principle. This principle states that for a given verb, the argument with the most Proto-Agent entailments is lexicalized as its subject, and the one with the most Proto-Patient entailments as its object (1991: 576, (31)). For example, the verb *build* conforms to this principle since

its subject has all the Proto-Agent entailments, but no Proto-Patient entailments, while its object has all the Proto-Patient entailments, but no Proto-Agent entailments (1991: 577).

Since the Proto-Agent and Proto-Patient entailments figure in subject and object selection, respectively, the contributing properties of each qualify as semantic determinants of subjecthood and objecthood. The Argument Selection Principle, taken in conjunction with the entailments subsumed by the Proto-Agent and Proto-Patient, explains why arguments which meet the traditional criteria for agent and patient are always subjects and objects, respectively, in active sentences, and, more generally, why verbs that describe an agent acting on and changing the state of a patient are invariably transitive across languages (Andrews, 1985: 68, DeLancey, 1984; Dixon, 1979: 103, 1994: 114–118, Hopper and Thompson, 1980). However, on Dowty's approach there is no single determinant of subjecthood or objecthood. Arguments qualify as subjects precisely because they have Proto-Agent entailments, but the subject of one verb and the subject of a second could fail to share certain Proto-Agent entailments (and similarly for objects and Proto-Patient entailments). Independent existence aside, the subject of *fear* has the sentence Proto-Agent entailment, while the subject of *enter* has the entailment of movement with respect to another participant.

Dowty introduces two corollaries to the Argument Selection Principle. First, if the two arguments of a verb have roughly equal numbers of Proto-Agent and Proto-Patient properties entailed of them, then either may be the subject or alternatively the object (1991: 576, (32)); that is, the verb exhibits 'argument selection indeterminacy' (1991: 579–581). As support for this corollary, Dowty points to psych-verb doublets such as *fear* and stative *frighten*, which have the same semantic content but differ as to whether the stimulus is the subject and the experiencer is the object as with *frighten* or vice versa as with *fear*. The other corollary, which applies to three-argument verbs, states that the nonsubject argument with the greater number of Proto-Patient entailments is the direct object and the one with fewer Proto-Patient entailments is an oblique or a prepositional object (1991: 576, (33)). Dowty supports this corollary through case studies of the verbs *break* and *hit*, as well as verbs manifesting the locative alternation and the *with/against* alternation.

Given the formulation of the Argument Selection Principle, the number of Proto-Agent properties entailed of an argument matters to subject selection, but not their identity (and similarly for Proto-Patient properties and object selection). Support for this prediction is provided by 'partially symmetric interactive predicates' (Dowty, 1991: 583–586). Such predicates show two argument realization options in English and many other languages. In the first, illustrated by (3a) and (4a), the verb has a collective or conjoined subject, while in the second, illustrated by (3b) and (4b), the verb takes a subject and either an object or oblique complement.

- (3) a. Smith and Jones debated.
- b. Smith debated Jones.
- (4) a. Smith and Jones talked.
- b. Smith talked with Jones.

Although the alternate argument realizations are truth-conditionally equivalent for *debate* and *talk*, they are not for verbs such as *hug* and *collide*. In the (b) sentences in (5) and (6) *Kim* has a Proto-Agent entailment that *Sandy* lacks, while in the (a) sentences, which have a conjoined subject, *Sandy* as well as *Kim* has this entailment. The exact entailment depends on the verb: volitionality in (5) and movement with respect to another participant in (6).

- (5) a. Kim and Sandy hugged.
- b. Kim hugged Sandy.
- (6) a. Kim and Sandy collided.
- b. Kim collided with Sandy.

This alternation, then, conforms to the Argument Realization Principle, but would be difficult to capture using a traditional all-or-nothing definition of agent.

Dowty (1991: 605–613) further suggests that proto-role entailments provide insight into some of the challenges around the phenomenon of unaccusativity (Burzio, 1986, Levin and Rappaport Hovav, 1995, Perlmutter, 1978, 1989, Sorace, 2000). Dowty proposes that the core unaccusative verbs are those intransitive verbs that entail the incremental theme Proto-Patient entailment (or, more generally, telicity) of their subject, while the core unergative verbs are those intransitive verbs that entail the volition Proto-Agent entailment of their subject. Such verbs are consistently unaccusative or unergative across languages. The proto-role entailments offer insight into the crosslinguistic variation observed in the class membership of certain verbs (Rosen, 1984; Sorace, 2000). Many such verbs, which include *sneeze* and *snore*, describe eventualities which may or may not be construed as having a volitional argument; the verb's classification, then, depends on whether it lexicalizes volitionality. Further, Dowty proposes that proto-roles offer insight into why unaccusative diagnostics differ as to the sets of verbs they pick out (Levin and Rappaport Hovav, 1995, L. Levin, 1986): such 'unaccusative mismatches' arise because different phenomena may be sensitive to distinct contributing properties of a particular proto-role.

Critical Comments

Most critical comments about Dowty's paper address either the proto-role entailments or the Argument Selection Principle. I consider each in turn.

The Proto-Role Entailments

The success of Dowty's theory of argument selection depends to a large extent on the entailments it is built on. Dowty argues that a plus of his entailments is that they involve 'distinctions in kinds of events found 'out there' in the real world'

(1991: 575) that are ‘significant to human life’ (1991: 575); they are notions that are salient beyond language, independently identified in work in cognitive psychology and language acquisition. Dowty argues this property means ‘there is no reason to believe that all such classes must have discrete boundaries’ (1991: 575). For instance, he suggests that sentience is a matter of degree, pointing to computers and intelligent animals (1991: 574). For this reason, he (1991: 575) argues that the proto-role entailments are not open to criticism in the way that unclear boundaries have made traditional semantic roles vulnerable to criticism.

Optimally, the entailments in each proto-role cluster would be expected to be independent of each other (cf. Wilks, 1987: 759 on primitive predicates). Indeed, Dowty (1991: 572) cites verbs intended to show that his entailments meet this desideratum. For example, the object of *need* only has the lack of independent existence Proto-Patient entailment (1991: 573–574, (30e)). Nevertheless, the entailments in each proto-role cluster are not all fully independent of each other. Dowty (1991: 573) himself notes that the independent existence Proto-Agent entailment is presupposed by the other Proto-Agent entailments. Further, the volitionality and sentience Proto-Agent entailments overlap in that volitionality presupposes sentience: only sentient entities can be volitional (1991: 607). Among the Proto-Patient entailments, ‘causally affected by another participant’ and ‘undergoes a change of state’ might overlap, depending on exactly how the notion of affectedness is understood. Does affectedness require a change of state in the relevant argument or is it a weaker notion of impingement, corresponding perhaps to the notion of force recipient (Rappaport Hovav and Levin, 2001: 786), inspired by causal chain models of events (Croft, 1991)? As this example shows, it is not always clear precisely what Dowty intends by some of the entailments.

There are suggestions that certain entailments might be refined or that the sets of entailments might be amended. Davis and Koenig (2000: 72) and Primus (1999: 54–55) argue for the addition of paired Proto-Agent and Proto-Patient entailments related to possession to deal with *give* and comparable verbs. Ackerman and Moore (2001: 89–127) argue for an additional Proto-Patient entailment, bounding entity, to handle accusative–partitive case alternations in Finnish and Estonian. Recent work on argument realization suggests that ‘undergoes change of state’ might be reformulated as ‘undergoes scalar change’, a notion which Rappaport Hovav (2008) and Rappaport Hovav and Levin (2010) argue is critical for objecthood. Grimm (2011: 522) unifies the existence and change of state entailments, taking them to be two forms of a property he calls ‘persistence’. Further, some entailments, such as affectedness and causation, involve complex notions that can be broken down further (Grimm 2011: 520).

Dowty’s inclusion of affectedness and incremental theme among the Proto-Patient entailments has inspired considerable discussion of these notions and the relation between them. Tenny (1994) argues that affectedness could be reduced to ‘measuring out’, a ‘cousin’ of incremental theme and that more generally, the Proto-Patient entailments follow from her Aspectual Interface Hypothesis (Tenny, 1992: 21–22). Beavers (2011) argues against Tenny’s position. He identifies several different degrees of affectedness which are characterized by ‘monotonically

weakening truth conditions' (2011: 359), which he defines formally over scales of change.

As Dowty (1991: 574) and others (Croft, 1998: 37, Koenig and Davis, 2001: 83, Primus, 1999: 52–53) note, three of the Proto-Agent properties are necessarily paired with Proto-Patient properties in that they identify event participants in an asymmetric semantic relation. For example, the Proto-Agent entailment 'causing an event or change of state in another participant' necessarily co-occurs with the Proto-Patient entailment 'undergoes change of state'. In each pair the Proto-Patient entailment is dependent on the Proto-Agent entailment. Davis and Koenig (2000: 73) and Primus (1999: 352) see the relations instantiated in the pairs as instances of a more general 'control' relation. That is, as Davis and Koenig (2000: 73) propose, the Proto-Agent entailments 'might then reduce to a general entailment roughly paraphrasable as 'has control over the unfolding of the situation''. Instead, Primus (2006: 56–59) proposes that the dependencies can all be viewed as causal in nature, so the causation-related proto-role entailments are unnecessary. Recognizing dependence between entailments vitiates another criticism levelled at proto-roles: why the entailments cluster as they do rather than in some other way (Croft, 1998: 37, Tenny, 1994: 103). Grimm (2011: 521–522) takes a different approach to these pairings. Rather than reducing them to a single more general relation, he reformulates Dowty's entailments as privative oppositions in which properties of agents are taken as basic.

Some researchers recast additional traditional semantic roles as proto-roles. Primus (1999: 54–55, 137–138) introduces a Recipient Proto-role to deal with phenomena involving ditransitive verbs, benefactives, and addressees. This proto-role combines Proto-Agent and Proto-Patient properties. Rissman and Rawlins (2017: 532) propose that the notion of instrument is best understood as a proto-role rather than a traditional role. They identify two characteristic properties of instruments, intentionality on the part of the agent using the instrument and being 'a direct extension of the force initiated by the agent' (2017: 532), but argue neither is necessary for instrumenthood. Their conclusions are interesting given that (Schlesinger, 1989, 1995) introduces an even broader notion, the C-role, a notion like a proto-role which subsumes the traditional instrument and comitative roles. Additionally, Barker and Dowty (1993) extend the notion of proto-role to the nominal domain; they propose there are two sets of lexical entailments that nouns may impose on their arguments, which they label Proto-Part and Proto-Whole. Going even further, Winter (2018) proposes an account of the partially symmetric interactive predicates discussed in section "[Proto-Role Entailments in Argument Realization](#)" which builds on a notion of proto-predicate.

One advantage of proto-roles is that they involve 'smaller' meaning components than the traditional agent and patient roles. In fact, certain approaches to traditional semantic roles 'unpack' their contents using a small set of meaning components—or features—which together define the set of possible roles (Reinhart, 2001, 2002, 2016; Rozwadowska, 1988, 1989). Interestingly, the best known of these approaches, Reinhart's, uses two features, [c] ('cause change') and [m] ('mental state'), which are reminiscent of Dowty's causation and sentience Proto-

Agent entailments. However, existing feature-based approaches still assume that each semantic role is defined by a specific, fixed combination of features; that is, they are not prototype-based. Thus, from Dowty's (1991: 574, 598–599) perspective they suffer from the same drawbacks as typical semantic role approaches.

Finally, as Dowty's goal is simply to identify the facets of meaning that matter to argument realization, he does not concern himself with a particular theory of lexical semantic representation. Rather, he assumes that his approach is compatible with various approaches to such representations, and, in fact, some subsequent work grounds the proto-role entailments in predicate decompositions (Davis and Koenig, 2000; Primus, 1999). Since these decompositions often involve two-argument predicates, they provide an explanation for the existence of paired proto-role entailments.

Argument Selection

Dowty does not deal with changes in valence or voice that come about through morphosyntactic operations, yet a full theory of argument realization should. Even setting these aside, Dowty's Argument Selection Principle has a major limitation: it presupposes that a verb's transitivity is known before its application (Croft, 1998: 37, Davis, 2001: 64, Davis and Koenig, 2000: 74–75, Grimm, 2011: 520, Levin and Rappaport Hovav, 2005: 60, Primus, 1999: 47). That is, it simply determines which argument of a transitive verb is its subject and which its object. Yet, a theory of argument realization and its semantic determinants should make predictions about whether a two-argument verb is transitive in the first place.⁶ As mentioned in section “Proto-Role Entailments in Argument Realization”, agent-act-on-patient verbs are stably transitive across languages; however, although crosslinguistic variability is observed outside this semantic class as to whether a two-argument verb is transitive, verbs in certain semantic classes are more likely to be transitive across languages than those in others (Malchukov, 2005; Tsunoda, 1981, 1985, 2015). Tsunoda (1985: 388–389) captures the systematic, though differential likelihood that the members of semantic classes of two-argument verbs figure among the transitive verbs of a language with an implicational hierarchy. In simplified form, it is: causative change of state verbs > surface contact verbs > perception/cognition/emotion verbs. Tsunoda suggests that the hierarchy is organized in terms of a decrease in affectedness (in the sense of Hopper and Thompson, 1980). Malchukov (2005: 81) splits Tsunoda's hierarchy into two hierarchies, one according to degree of agenthood and the other according to degree of patienthood. Relatedly, among non-transitive two-argument verbs, the appropriate argument must

⁶ Dowty uses the term ‘transitive’ in its strict, syntactic sense to refer to a verb that takes a subject and an object (in contrast to an intransitive verb, which simply takes a subject). Most verbs that semantically select for two arguments are transitive verbs, but not all are (e.g., look (at), rely (on)).

be chosen as subject, and as Davis (2001: 65–66) and Davis and Koenig (2000: 74–75) discuss, the argument with the most Proto-Agent entailments qualifies as subject.

Dowty's Argument Selection Principle embodies the assumption that there are no priorities among the entailments in either proto-role cluster with respect to subject or object selection; only number of entailments matters. Yet Dowty (1991: 574, 581, 607) acknowledges that perhaps not all proto-role entailments contribute equally. He cites causation as the Proto-Agent entailment with the most weight for subject selection, and suggests that incremental theme is often the most significant for object selection (1991: 607). Ackerman and Moore (2001: 52), Davis (2001: 66–72), and Davis and Koenig (2000: 74–76) argue that the causation entailment outranks the other Proto-Agent entailments in subject selection: an argument with the cause entailment is always subject (cf. the transitivity of agent-act-on-patient verbs. Dowty does not offer insight into why causes have the highest priority, but approaches using predicate decompositions or other structured event structures do: causes are structurally more prominent than entities that change state. Koenig and Davis (2001: 82–83) further propose that sentience and volition matter for non-causative verbs, and, if these entailments are not present, then motion matters. Given these priorities, they argue that number of entailments is not relevant for determining subjecthood, contra the Argument Selection Principle.

Moving beyond causative verbs, Dowty's entailments offer insight into why the Proto-Agent entailments bear on subject selection and the Proto-Patient entailments bear on object selection rather than vice versa. The dependence of the paired Proto-Patient entailments on the corresponding Proto-Agent entailments can be interpreted as the Proto-Agent entailments having greater semantic prominence than the Proto-Patient entailments. Assuming semantic prominence is reflected in syntactic prominence (Jackendoff, 1992: 22, Levin and Rappaport Hovav, 2005: 140–145), then the Proto-Agent entailments should bear on subject selection and the Proto-Patient entailments on object selection.

Impact of the Paper and Subsequent Developments

Dowty's proto-roles represent perhaps the best-known 'generalized semantic roles' (Levin and Rappaport Hovav, 2005: 51–53, Van Valin, 1999). Such approaches to semantic roles are often prompted by a desire to overcome the challenges facing attempts to posit unified semantic determinants of subjecthood or objecthood, and reject semantic roles as defined by sets of necessary and sufficient conditions. Dowty's proto-roles, unlike other generalized semantic roles, are not reified. For instance, Role and Reference Grammar (Foley and Van Valin, 1984; Van Valin and LaPolla, 1997), another well-known generalized semantic role approach, reifies its Actor and Undergoer 'macroroles'; see Van Valin (1999: 386–387) for discussion.

Dowty's paper continues to provide one of the most compelling and comprehensive expositions of the shortcomings of traditional semantic roles. Nevertheless,

semantic roles continue to be used in certain areas, such as NLP applications (Gildea and Jurafsky, 2002; Márquez et al., 2008; Palmer et al., 2005). Although semantic roles have not been completely discarded, they are now treated with more care, and many researchers have embraced Dowty's proto-roles. Dowty's work has also prompted researchers to think deeply about the semantic determinants of argument realization, including re-evaluating some commonly used determinants and recognizing the need for finer-grained determinants than traditional semantic role approaches provide.

Proto-role entailments have been integrated into various syntactic frameworks. Taking a Principles-and-Parameters perspective, Baker (1997) uses proto-role-like notions of agent, patient, and location defined over syntacticized predicate decompositions to show that a wide range of argument realization phenomena are consistent with his Uniformity of Theta Assignment Hypothesis, but see Newmeyer (2002) for a critique. Davis and Koenig (2000: 72) fine-tune the proto-role entailments and use the revised entailments in an HPSG-inspired theory of argument realization, which uses reified proto-roles. To address unaccusative mismatches, Zaenen (1993) integrates the entailments into Lexical-Functional Grammar's Lexical Mapping Theory (Bresnan and Kanerva, 1989). She uses them in the assignment of the intrinsic classification features that in turn determine grammatical function assignment. Also working in LFG, Alsina (1996: 38–43) adopts a reified version of proto-roles to explain morphosyntactic phenomena in Catalan and Spanish.

Proto-roles have been used in the analysis of argument realization phenomena that go beyond the English phenomena that Dowty considers. I mention some representative examples. Dowty's paper has provided a foundation for work on typological issues, with proto-role entailments invoked in the explanation of properties of case-marking systems. Particularly ambitious in scope is Grimm's (2011) study, which uses a set of privative features inspired by the Proto-Agent entailments to define a lattice. He shows that attested morphological case systems are defined over contiguous regions in the lattice, and proposes that the lattice's form constrains available case systems. Aissen (1999) uses reified proto-roles in an Optimality-Theoretic account of the constraints on crosslinguistically available person- and animacy-based case marking splits identified by Silverstein (1976). Primus (1999, 2006) proposes that number of proto-role entailments matters for case-marking and agreement patterns, such as the assignment of nominative vs. accusative or ergative vs. absolutive case to the arguments of core transitive verbs, but that the dependence relations defined between Proto-Agent and Proto-Patient entailments matter for word order.

Other work focuses on specific languages. Ackerman and Moore (2001:89–127) show how proto-role entailments can be used to understand alternate realizations of the arguments of a single verb, such as the accusative–partitive case alternations of Estonian and Finnish. Blume (1998) uses Proto-Agent entailments to capture the properties of interaction verbs, verbs that in many languages take an animate oblique second argument. García-García (2007: 74, 2018: 229–231) invokes Proto-Agent entailments to explain the distribution of differential object marking (Bossong,

1991) with inanimate objects in Spanish. Tollan (2018: 17) builds on Proto-Agent entailments to distinguish core two-argument verbs in Samoan, which take ergative subjects, from unergative verbs and two-argument 'middle' verbs, both of which take absolutive subjects. She proposes that the latter have subjects with fewer Proto-Agent entailments than the former: they always lack volitionality and effortfulness entailments. Brunetti (2009) appeals to Proto-Agent entailments to explain why the dative experiencer argument of psych-verbs is the topic in Spanish and Italian.

Another line of work explores wider implications of proto-roles, and in so doing bears on Dowty's proposal that the proto-role entailments are cognitively salient notions. Kako (2006) presents a series of experiments, some replicated by Reisinger et al. (2015), intended to verify the psychological validity of the proto-role entailments in the argument realization of transitive verbs. The experiments show that subjects across real and nonce transitive verbs have more Proto-Agent than Proto-Patient properties, while the reverse holds of objects. The broad applicability of the proto-role entailments is shown by Reisinger et al. (2015), who report on a large-scale corpus annotation of these entailments. Their work shows that such annotations provide an alternative to semantic role labelling in NLP applications. Further, an examination of the distribution of the entailments across verbs in their annotated data reveals that a small number of combinations of entailments are encountered frequently, but the vast majority occur just a few times. Reisinger et al. (2015: 482–483) take this finding as evidence favoring proto-roles over traditional semantic roles. In particular, this finding provides further evidence that role fragmentation is endemic, supporting Dowty's claim (1991: 553–556) that it poses an obstacle to any effort to posit a fairly small set of traditional semantic roles. However, in order to operationalize the proto-role entailments so they are accessible to non-linguists, both studies omit the incremental theme entailment, which has figured prominently in approaches to argument realization. Finally, taking Dowty's Proto-Agent entailments as inspiration, Mayrhofer and Waldmann (2014) develop empirical criteria for agentivity in an experiment involving video scenarios of (near)-collisions between two balls (i.e. Michotte's (1963) launching events).

Summing Up

At over 50 pages, Dowty's paper 'Thematic Proto-roles and Argument Selection' is not short, but its length is matched by its many contributions, large and small. It argues for a prototype perspective on semantic roles grounded in the entailments that a verb imposes on its arguments, simultaneously rejecting discrete and categorical semantic roles. It proposes an Argument Selection Principle and its corollaries that take advantage of the cluster concept nature of proto-roles. In so doing, the paper confronts the challenge of argument selection, providing a novel solution to the continuing debate over the semantic underpinnings of subjects and objects. Simultaneously, it offers insights into prominent phenomena bearing on argument realization. The paper is so jam-packed with insights into the lexical semantics of

verbs that repeated readings prove rewarding. For all these reasons, Dowty's paper will continue to be appreciated in the years to come.

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Chapter 8

On Groenendijk and Stokhof's "Dynamic Predicate Logic"



Anthony S. Gillies

Abstract This paper can serve as a gentle introduction to (but not exhaustive overview of) Groenendijk and Stokhof's classic paper "Dynamic predicate logic". If you are not new to DPL and so not in need of an introduction to DPL, this paper can alternatively be read as an invitation to re-read it.

Keywords Dynamic semantics · Dynamic logic · Update semantics · Anaphora

Introduction

Sometimes it happens that commentaries on or textbook treatments of important theories or ideas are better to read than the originals. On these occasions the commentary or textbook version is able to turn the benefits of time, hindsight, and intellectual division of labor into a cleaner or more elegant presentation of the essential insights of the original. This is not such an occasion.

Instead this paper is an introduction to, or alternatively an invitation to re-read, Groenendijk and Stokhof's (1991) "Dynamic predicate logic" (DPL): a useful warm-up exercise containing some background and sign-posts, gestures at some nearby connections, and (hopefully) something that will enrich the DPL experience.¹

Here is what you can expect. We will begin (section "Farmers, Donkeys, Files, Boxes") with stage-setting: an argument that no quantificational analysis of indefinites can be right and how this set the stage for DPL. Section "Dynamic Worldview" is an introduction to the dynamic worldview, got at by a gentle introduction to (propositional) dynamic logic. Sections "DPL Basics: $\wedge$ and $\exists$ "

¹ I will use 'DPL' to pick out both the system of dynamic predicate logic and the article itself, hoping that context will disambiguate.

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and “DPL Basics: $\neg$, $\rightarrow$, and $\forall$ (and $\vee$, Too)” illustrate the mechanics of the core of canonical DPL (what you will find in the paper itself) and section “[Truth, Entailment, Equivalence](#)” is an overview of the theoretical properties of the system (truth, entailment, equivalence). Sections “[Updating Information States](#), [DPL Updates](#), and [Dynamic Differences](#)” shift gears a bit, recasting DPL as an update system and looking a little into the ways in which such systems can depart from classical frameworks. Section “[Extensions, Amendments, Revisions](#)” concludes by offering a pointer to one main area of subsequent work in the spirit of DPL. The appendix contains some official definitions.

Farmers, Donkeys, Files, Boxes

Suppose we focus on a small fragment of natural language, one where we can talk about a collection of farmers, livestock, and some properties they have. This kind of fragment is one where the tools of first-order logic can be put to good use. So assume that we have a language L_{FO} of first order logic with individual variables $x, y, \dots$, a stock of predicates, negation ($\neg$), conjunction ($\wedge$), and the existential quantifier ($\exists$).²

Some things we might want to report:

- (1) a. A farmer walked in the field and he whistled.
- b. A farmer walked in the field. He whistled.

There is not much mystery what (1a) and (1b) express: both are true if and only if there is at least one farmer who walked in the field and whistled. But thinking about how they manage to express this (surprisingly) raises some hard issues.³

The anaphoric pronoun *he* in both examples seems to pick out a farmer that walked in the field, adding that that farmer whistled. This seems to suggest that the indefinite noun phrase *a farmer* that is the antecedent for this *he* also picks out that field-walking farmer and thus that the two devices are coreferential. But indefinite noun phrases don’t in general seem to be devices for referring to individuals, much less co-referring. You can see this by embedding them in various ways, for instance under a wide-scope negation:

- (2) Pedro doesn’t own a donkey.

² See Definition 16 for the official definition. Throughout, I will assume a working knowledge of what it takes for sentences of L_{FO} to be true in a model $M = \langle D, I \rangle$ with respect to an assignment function g .

³ Numbering conventions: numbers-(plus-letters) (as in (1a) and (2)) are for example sentences; number. Number labels (as in (8.1) and (8.10)) are for equations and formulas, with the first number picking out the section of the paper in which the equation/formula makes its first appearance.

Here *a donkey* doesn't refer to any donkey. So understanding the relationship between indefinites and pronouns as coreference doesn't in general seem right.

If not referring, then what? The other salient option is to treat *a farmer* as a quantificational expression. This makes a lot of sense. And given the humble fragment we are investigating, we can think of them as regular first-order existential quantificational expressions. After all, the examples seem to say that some farmer walked and whistled. This works well for (1a) but runs into problems for (1b).

$$\exists x(\text{farmer } x \wedge \text{walked-in-field } x \wedge \text{whistled } x) \quad (8.1)$$

$$\exists x(\text{farmer } x \wedge \text{walked-in-field } x) \wedge \text{whistled } x \quad (8.2)$$

With a little imagination and practice, you can see a natural path from (1a) to the representation in (8.1). On this way of thinking, the relationship between the indefinite *a farmer* and *he* is a familiar one: the occurrences of the variable x in $\text{farmer } x \wedge \text{walked-in-field } x$ and the occurrence of x in $\text{whistled } x$ all lie in the scope of the quantifier $\exists x$. This quantifier thus binds these variables. Interpret these first-order representations in the usual way and you therefore get the prediction that (8.1)—and so by extension (1a)—is true if and only if something is a farmer that walked in the field and is also something that whistled.

Things are trickier with (1b). Assume that our representations should go sentence-by-sentence and that a sequence of sentences can be usefully modeled as conjunction: however you represent *A farmer walked in the field* should be conjoined with how you represent *he whistled*. Then we arrive at (8.2). But now the x in $\text{whistled } x$ is not in the scope of $\exists x$ and so we don't predict the right relationship between it and something that is a farmer that walked in the field. As a result we get the wrong truth conditions: (8.2) is true in a model with respect to an assignment g if one farmer walked in the field, no farmers whistled, and $g(x)$ is a non-farmer who whistled.

The strict sentence-by-sentence approach prevented us from getting the right relationship between the quantificational apparatus and the downstream pronouns. Can we insist that the indefinite *a farmer* must have scope over *he whistled*? This gets us back to the representation in (1a) and does deliver the right truth conditions. But it is dodgy. How can the syntactic scope of indefinites cross sentence boundaries? Probably, it can't. This strategy also moves from taking the individual sentences in (1b) as the principal thing semantics assigns meanings to and shifts focus to the whole discourse.⁴ With that comes a whiff of non-compositionality: what looked like a meaningful sentence in natural language (the first sentence of (1b)) isn't being interpreted on its own and the whole discourse therefore has a meaning not directly figure-out-able from the meanings of its parts.

⁴ This is the strategy suggested in Geach (1962/1980). This discourse-first motto is taken up, but for a non-quantificational approach, in DRT (Kamp, 1981).

Whether or not this or some other approach can be defended in this case, things get worse when we again shift to other embedding environments. Notably, in so-called *donkey conditionals*:

- (3) a. If a farmer owns a donkey, he feeds it hay.
 b. Every farmer feeds hay to every donkey he owns.

The conditional in (3a) has a reading where it means what (3b) means. The hypothesis that *a farmer* and *a donkey* are existential quantifiers again runs into trouble. Two attempts:

$$\exists x(\text{farmer } x \wedge \exists y(\text{donkey } y \wedge \text{owns } xy)) \rightarrow \text{feeds-hay-to } xy \quad (8.3)$$

$$\exists x(\text{farmer } x \wedge \exists y((\text{donkey } y \wedge \text{owns } xy) \rightarrow \text{feeds-hay-to } xy)) \quad (8.4)$$

The representation in (8.3) is pretty good: it is a conditional and *a farmer owns a donkey* shows up as the conditional's antecedent, which seems right. But *feeds-hay-to* xy is not in the scope of $\exists x$ or $\exists y$ and so the truth conditions are wrong: the bound reading of the pronouns in *he feeds it hay* are not predicted. Meanwhile the representation in (8.4) ensures that the existential quantifiers both reach all the way to the consequent. Insisting that indefinites always take wide scope is surprising—deep down (3a) isn't a conditional!—and not worth it since the truth conditions are wrong: (8.4) is true if there is a farmer and a donkey he doesn't own but (3a) seems to be true if and only if every farmer feeds every donkey he owns hay.

So: the relationship between indefinites and anaphoric pronouns doesn't seem to be the relationship of coreference and it doesn't seem to be the relationship of quantifier and a variable it scopes over. This is a rough version of arguments in both Kamp (1981) and Heim (1982) that aim to motivate non-quantificational (and non-referential) theories of indefinites and both Discourse Representation Theory (DRT) and File Change Semantics (FCS) deliver just that.⁵

Two features of DRT to keep in mind as a backdrop for reading DRT. First: like the Geach-strategy, it is discourses not sentences that are the primary bearers of semantic values. It is not in general true in DPL that the truth conditions of whole discourses can be decomposed into the truth conditions of the individual sentences that make them up. Second: the analysis of indefinites is thoroughly non-quantificational and representational (making heavy use of discourse referents and an intermediate and distinctive level of representation).⁶ DRT can be seen as a reaction to both of these features.

⁵ DRT was first developed in Kamp (1981); see Geurts (2019) for a reader's guide to it. The canonical version of DRT is developed in Kamp and Reyle (1993); see Geurts et al. (2016) for an introduction. FCS is developed in Heim (1982) and refined somewhat in Heim (1983); see Yalcin (2012) for a recent introduction to FCS.

⁶ Discourse referents originate in Karttunen (1976).

But it is not a baby/bathwater situation: for the target fragment—roughly, anaphoric relationships that can be sensibly represented using the resources of L_{FO} —DPL makes exactly the same empirical predictions as DRT.⁷

Dynamic Worldview

Famously, DPL embraces the dynamic perspective on meaning. Groenendijk and Stokhof put it this way:

The general starting point of the kind of semantics that DPL is an instance of, is that the meaning of a sentence does not lie in its truth conditions, but rather in the way it changes (the representation of) the information of the interpreter. The utterance of a sentence brings us from a certain state of information to another one. The meaning of a sentence lies in the way it brings about such a transition. (DPL, p. 43)

This is powerful imagery and makes dynamic semantics seem in equal parts exotic and revolutionary.⁸

Before getting to DPL proper, and seeing how it makes this imagery concrete, let's look at a nearby idea: dynamic logics.⁹ These are (modal) logics for reasoning about programming languages. They aren't the same thing as dynamic semantics, and that is useful. Like regular modal logics, sentences express propositions (sets of satisfying points) but they also interpret program-denoting parts of the language. Thus you can view the semantics for these logics as an intermediate stepping stone on the road to dynamic semantics.

Here we will just sketch the simplest such logic, propositional dynamic logic (PDL).¹⁰

In classical propositional modal logic, sentences are assigned meanings with respect to a model that includes a non-empty set of worlds and a binary accessibility relation over that set. All sentences have sets of worlds as meanings: $\llbracket \phi \rrbracket$ is the set of worlds at which ϕ is true, a proposition. For modal claims like $\Diamond \phi$ the propositions depend on what's accessible to what: $\llbracket \Diamond \phi \rrbracket$ is the set of w 's such that there is an accessible world from w at which ϕ is true. So far, so familiar.¹¹

⁷ We won't go into the details, but Section 4.2 of DPL (pp. 76–83) is devoted to showing this.

⁸ Dynamics really got a foothold in the 1980s and 1990s: dynamic semantics, belief dynamics, dynamic epistemic logic were all in boom cycles (see, for instance, Muskens et al. 1997). Thanks to the trio of David Beaver, Paul Dekker, and Willem Groeneveld (who, unbelievably in hindsight, taught an online course in dynamic semantics way back then) I got swept up by all of it. Heady times.

⁹ A good textbook: Harel et al. (2000).

¹⁰ This is a way to ease into the dynamic worldview and serves as preparation for tackling Section 4.3 of DPL, where you can see how DPL can be embedded into a fragment of Quantified Dynamic Logic.

¹¹ If not, see Gamut (1991).

PDL is similar in both language and interpretation but is expressively richer. It also has room to express things like `set the value of register b to 3 and do this and then do that`. The language of PDL has both atomic *sentences* and atomic *programs*. Atomic sentences are sentences that are basic and have no interesting parts. So too for atomic programs: they represent the simple, non-decomposable actions. Sentences of PDL can be negated and conjoined to create more complex sentences, but there are also ways of making more complex programs out of simpler ones. For instance, you can *compose* them: $\pi_1; \pi_2$ is a program whenever π_1 and π_2 are programs. Declarative sentences and programs have two points of contact. First, given a sentence ϕ , you can form a *test*: $\phi?$ is a program whenever ϕ is a sentence. Second, the modalities are generated by programs: $\langle \pi \rangle \phi$ and $[\pi] \phi$ are sentences if π is a program and ϕ is a sentence. These take programs and sentences and create possibility and necessity claims in terms of them.

The semantics for PDL is built on two ideas, one familiar and the other new. The familiar: declarative sentences express propositions, the sets of points or worlds or (as they're called in this context) *states* at which they are true. The new: programs express what they do. What a program does depends on the state in which it is executed. Executing the program take 3 steps to the right when I am standing here results in a different state of affairs than it does when it is executed when I am standing over there. Same goes for all programs in whatever state. They link (input) states to (output) states. Which is to say: programs express relations between states.¹²

So, following the set-up for modal logic, a PDL model $M = \langle W, V \rangle$ is a non-empty set of states W and a function V that assigns atomic *sentences* propositions and atomic *programs* relations on W . Where p is an atomic sentence and a is an atomic program:

$$V(p) \subseteq W \text{ and } V(a) \subseteq W \times W \quad (8.5)$$

Just as $V(p)$ is the set of states s such that (atomic) p is true at s , $V(a)$ is the set of pairs of states $\langle s, t \rangle$ such that executing (atomic) a in s possibly lands you in state t .¹³ From here we have to say how more complex meanings are assigned. Focus on three aspects of that assignment, leaving the rest to the official version in Definition 19.

First: possibility modal claims $\langle \pi \rangle \phi$. In PDL possibility is restricted twice over: $\langle \pi \rangle \phi$ at a state s says that ϕ is true at some state reachable by *executing* π in s .

$$s \in \llbracket \langle \pi \rangle \phi \rrbracket \text{ iff there is a } t \text{ s.t. } \langle s, t \rangle \in \llbracket \pi \rrbracket \text{ and } t \in \llbracket \phi \rrbracket \quad (8.6)$$

¹² Sentences and programs differ in semantic type, but still fit together. That is important because even though program constructs are not themselves formulas in the language, they do appear as parts of declarative formulas. The modalities are a case in point.

¹³ Since program meanings are relations, not necessarily functions, we are allowing that programs are not deterministic. Hence the “possibly” in the gloss.

We have lots of modalities in PDL: two for each program we can construct. But where are the accessibility relations? They are the meanings of the programs! The compositional way that complex programs express relations on W as a function of the relations expressed by the simpler programs that make them up is key here.

Second: program composition. Whenever π_1 is a program and π_2 is a program we know $\pi_1; \pi_2$ is, too. And we know that $\llbracket \pi_1; \pi_2 \rrbracket$ will serve as the accessibility relation for the modals $\langle \pi_1; \pi_2 \rangle$ and $[\pi_1; \pi_2]$. As the name suggests such composed programs express the composition of the relations expressed by the programs they are composed of.

$$\llbracket \pi_1; \pi_2 \rrbracket = \llbracket \pi_1 \rrbracket \circ \llbracket \pi_2 \rrbracket \quad (8.7)$$

where $\circ$ is, literally, relational composition. Thus

$$\langle s, t \rangle \in \llbracket \pi_1; \pi_2 \rrbracket \text{ iff, for some } r, \langle s, r \rangle \in \llbracket \pi_1 \rrbracket \text{ and } \langle r, t \rangle \in \llbracket \pi_2 \rrbracket \quad (8.8)$$

The linking here through the intermediate state r is key.

And third: test programs. These show the other way that things fit together by turning a sentence ϕ into a program $\phi?$. The point of such programs is to *check whether* ϕ is true. They don't themselves change the state you're in.

$$\langle s, t \rangle \in \llbracket \phi? \rrbracket \text{ iff } s = t \text{ and } s \in \llbracket \phi \rrbracket \quad (8.9)$$

Test programs are sometimes described as those programs only defined on the diagonal. Draw an array of ordered pairs of states: test programs denote subsets of the diagonal relation $\langle s, s \rangle, \langle t, t \rangle, \langle u, u \rangle, \dots$

PDL models can get represented graphically, as in Fig. 8.1: states are nodes, and each atomic program is represented by a labeled arrow (labeled by the program). Double-circled states represent *fixed-point* states: these are states where there is a successful test program pictured (i.e., a labeled self-loop).¹⁴

DPL Basics: $\wedge$ and $\exists$

Indefinites are shape-shifters, sometimes seeming to contribute existential truth conditions (as is in (1a) where *a farmer* scopes under *and*) and sometimes seeming to contribute universal force (as in (3a) where *a farmer* scopes under *if*). A quantificational analysis of indefinites thus seems hopeless. There is a loophole, though, and DPL exploits it.

¹⁴ In the model in Fig. 8.1 suppose $\llbracket p \rrbracket = \{s, t\}$ and $\llbracket q \rrbracket = \{s, u\}$. It might be useful to put pencil to paper and figure out $\llbracket a; b \rrbracket$, $\llbracket (a; b); a \rrbracket$, $\llbracket q? \rrbracket$, and $\llbracket \langle c \rangle p \rrbracket$.

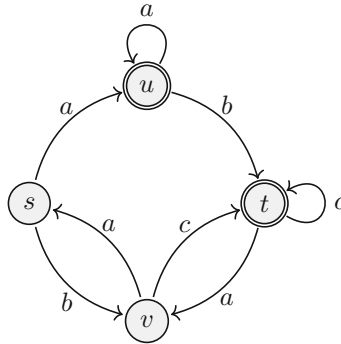


Fig. 8.1 PDL model

DPL offers a new way of interpreting a language of first order logic. It looks at programs in PDL for inspiration and says *every* sentence is a program. Thus meanings will all be relations: sets of ordered pairs of “states”. This will provide a way of treating indefinites as existential quantifiers (albeit, dynamic ones) that can account for their shape-shifting behavior.

What counts as a state (of information) depends on what we are modeling. Given the limited focus here (extensional first order stuff) this means that we are tracking, with respect to a given first order model M , what we are talking about. Thus, information is *discourse information*: which variables are hooked up with which objects in D . That is information usefully encoded by assignment functions.

Definition 1 (Information States in (Relational) DPL) Let V be the set of variables of L_{FO} and $M = \langle D, I \rangle$ be a FO model. The set S of DPL information states is the set:

$$g \in S \text{ iff } g \in D^V$$

where D^V is the set of assignment functions from V to D .

Putting the pieces together: sentences are programs or instructions for changing information. Information is information about variables. So sentences denote sets of input-output pairs of assignment functions.¹⁵

Definition 2 (Meanings in (Relational) DPL) For any ϕ :

$$\llbracket \phi \rrbracket^M \subseteq S \times S$$

¹⁵ The general idea that interpreting things might both depend on and affect assignment functions can also be found in Barwise (1987).

This doesn't tell us just which relation is expressed by ϕ . We'll get to that. Intuitively: $\langle g, h \rangle \in \llbracket \phi \rrbracket$ iff executing (the program expressed by) ϕ in state g (possibly) terminates in state h . Since $\llbracket \phi \rrbracket$ is a relation I'll opt for infix notation, $g \llbracket \phi \rrbracket h$ being a somewhat snappier thing to read and write than $\langle g, h \rangle \in \llbracket \phi \rrbracket$.

What remains is to look at the specific ways different sorts of sentences say to change an information state g .¹⁶ Begin with (1b) and its sentence-by-sentence representation (8.2). We will work up to the DPL treatment of this by talking about simple cases showing how the clauses in the official version of relational DPL work (Definition 20). By the end, we will see why this is true:

Fact 1 $g \llbracket \exists x Fx \wedge Gx \rrbracket h$ only if $h(x) \in I(F)$ and $h(x) \in I(G)$.

Start with the DPL meaning assigned to atomic formulas like Fx . This simply says that whatever we have been using x to talk about is one of the F 's.¹⁷ So Fx expresses a test program to see if that's the case.

$$g \llbracket Fx \rrbracket h \text{ iff : } h = g \text{ and } g(x) \in I(F) \quad (8.10)$$

Thus $\llbracket Fx \rrbracket$ contains those pairs $\langle g, g \rangle, \langle h, h \rangle, \langle f, f \rangle, \dots$ of assignments that map x to a thing in D that is in $I(F)$.

The instruction that $\exists x Fx$ expresses is different. It is a two step procedure. It tells us, first, to move from g to a state that differs at most from g in what it associates with x : the state $g[x]k$.¹⁸ From this state, does interpreting Fx move you to h ? The original g is linked to h by way of $\llbracket \exists x Fx \rrbracket$ if and only if the answer is yes.

$$g \llbracket \exists x Fx \rrbracket h \text{ iff : there is a } k \text{ s.t. } g[x]k \text{ and } k \llbracket Fx \rrbracket h \quad (8.11)$$

Given (8.10), the right hand side simplifies:

$$g \llbracket \exists x Fx \rrbracket h \text{ iff : } \begin{cases} \text{there is a } k \text{ s.t. } g[x]k \text{ and} \\ k = h \text{ and} \\ h(x) \in I(F) \end{cases} \quad (8.12)$$

That is:

¹⁶ It is important that all of this happens within a given model M . We'll typically suppress mention of this, though.

¹⁷ This isn't too far away from the idea in FCS that definites are used to update existing files.

¹⁸ Officially, define:

$$g[x]k \text{ iff } k = g[x/d] \text{ for some } d \in D$$

This is an unspecific x -alternative to g . It is also an information state: one just like g except (possibly) in what it says about x . Note that we are writing this ' $g[x]k$ ' while Groenendijk and Stokhof write ' $k[x]g$ '. The left-to-right way seems more natural to me, but your mileage may vary.

$$g \llbracket \exists x Fx \rrbracket h \text{ iff } :g[x]h \text{ and } h(x) \in I(F) \quad (8.13)$$

So when you interpret $\exists x Fx$ you wind up in a state in which x is associated with a thing that is an F .

In DPL $\wedge$ expresses program composition. So to interpret $\exists x Fx \wedge Gx$ we find a linking information state between $\llbracket \exists x Fx \rrbracket$ and $\llbracket Gx \rrbracket$.

$$g \llbracket \exists x Fx \wedge Gx \rrbracket h \text{ iff } : \begin{cases} \text{there is a } k \text{ s.t. } g \llbracket \exists x Fx \rrbracket k \text{ and} \\ k \llbracket Gx \rrbracket h \end{cases} \quad (8.14)$$

Using (8.13) we can substitute in the right-hand side:

$$g \llbracket \exists x Fx \rrbracket k \text{ iff } g[x]k \text{ and } k(x) \in I(F) \quad (8.15)$$

So the denotation of our conjunction is:

$$g \llbracket \exists x Fx \wedge Gx \rrbracket h \text{ iff } : \begin{cases} \text{there is a } k \text{ s.t. } g[x]k \text{ and} \\ k(x) \in I(F) \text{ and} \\ k \llbracket Gx \rrbracket h \end{cases} \quad (8.16)$$

Similarly, for $\llbracket Gx \rrbracket$:

$$k \llbracket Gx \rrbracket h \text{ iff } h = k \text{ and } h(x) \in I(G) \quad (8.17)$$

Putting those together and skipping a step:

$$g \llbracket \exists x Fx \wedge Gx \rrbracket h \text{ iff } : \begin{cases} g[x]h \text{ and} \\ h(x) \in I(F) \text{ and} \\ h(x) \in I(G) \end{cases} \quad (8.18)$$

The x in Gx is outside the syntactic scope of $\exists x$ but still gets bound by it. The reason is that, as we saw, $\llbracket \exists x Fx \rrbracket$ relates g to h only if $h(x)$ is in $I(F)$. Since conjunction is program sequencing, such h 's are the input states for executing $\llbracket Gx \rrbracket$. Existentially quantified formulas change downstream assignment functions. Conjunction, a.k.a. relational composition, is primed to be sensitive to those changes. You can see this in Fig. 8.2. That completes the demonstration of Fact 1.

This works because there is an interplay between *external* and *internal* dynamics in DPL. External: $\exists x$ can have effects on things outside its syntactic scope. Internal: $\wedge$ has dynamic effects that carry from first conjunct to second.¹⁹ By the time we get to the x in Gx we are therefore in an information state that maps x to a thing

¹⁹ Conjunction is externally dynamic, too.

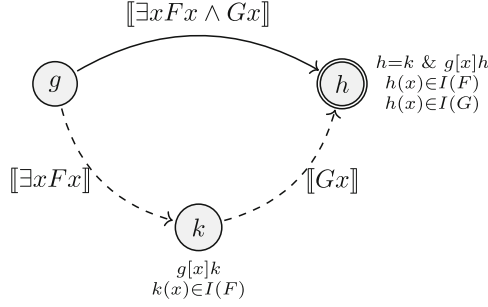


Fig. 8.2 $\llbracket \exists x Fx \rrbracket \circ \llbracket Gx \rrbracket$

in $I(F)$ and the Gx conjunct thus tacks on the information that that thing is also in $I(G)$. We will return to this idea a few more times.

This particular way that external and internal dynamics play together has the upshot that scope and binding come apart in DPL. Binding outstrips syntactic scope.²⁰ So taking on board the information carried by $\exists x Fx$ has the downstream consequence that occurrences of x outside the scope of this $\exists x$ can end up being obligatorily associated with things in $I(F)$.

DPL Basics: $\neg$, $\rightarrow$, and $\forall$ (and $\vee$, Too)

Our FO language takes $\neg$, $\wedge$, and $\exists$ as basic and offers $\vee$, $\rightarrow$, and $\forall$ as abbreviations. This is another distinctive feature of DPL: classically, it doesn't matter what you take as basic and what you take as derived, but it matters in DPL. We'll see why.

In this section we will work up to the DPL treatment of example (3a) by talking about simple instances of the DPL clauses for $\neg$ and $\rightarrow$. Once again the simple, sentence-by-sentence representation in (8.3) will do in DPL. The main thing is that in DPL an indefinite in a conditional antecedent can bind a pronoun in a conditional consequent and bind it with universal force. Working out the meaning of $\exists x Fx \rightarrow Gx$ is enough to make the point.²¹

Fact 2 $g \llbracket \exists x Fx \rightarrow Gx \rrbracket h$ only if $h[x]k$ and $k(x) \in I(F)$ imply $k(x) \in I(G)$.

All of our derived connectives appeal to $\neg$ so let's start with that. Negation in DPL tests for *failure*. An information state g is $\llbracket \neg\phi \rrbracket$ -related to h iff $g = h$ and it's not possible to execute $\llbracket \phi \rrbracket$ in g at all.

²⁰ Section 3.3 of DPL probes how scope and binding come apart (DPL, pp. 58–61).

²¹ This is code for: you should work out the DPL meaning for (8.3) as a homework exercise.

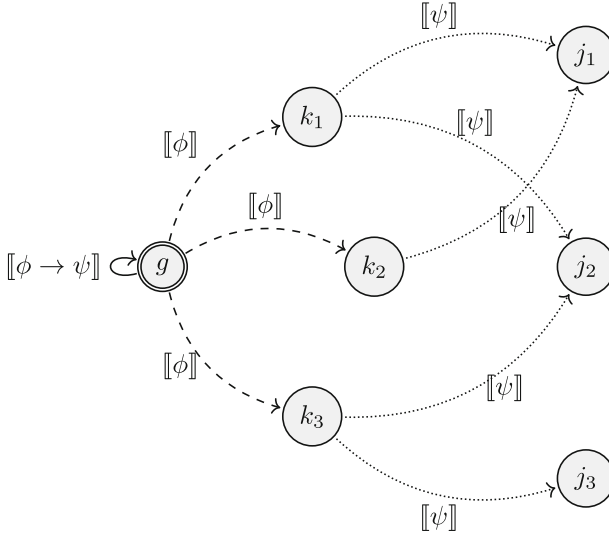


Fig. 8.3 $\llbracket \phi \rightarrow \psi \rrbracket$

$$g \llbracket \neg\phi \rrbracket h \text{ iff : } \begin{cases} h = g \text{ and} \\ \text{there is no } k \text{ s.t. } g \llbracket \phi \rrbracket k \end{cases} \quad (8.19)$$

This makes negation a test-maker: $\neg\phi$ is a test no matter what and so is not externally dynamic (a.k.a., static).

Conditionals in DPL are negated conjunctions: $\phi \rightarrow \psi = \neg(\phi \wedge \neg\psi)$. This has two upshots. The first is that the derived meaning for $\phi \rightarrow \psi$ is a test with a wrinkle:

$$g \llbracket \phi \rightarrow \psi \rrbracket h \text{ iff : } \begin{cases} h = g \text{ and} \\ h \llbracket \phi \rrbracket k \text{ implies } k \llbracket \psi \rrbracket j \text{ for some } j \end{cases} \quad (8.20)$$

The first clause ensures that conditionals are tests. The second clause contains the wrinkle. It implicates an intermediate state k that passes values of variables from antecedent to consequent. So, even though they are tests, conditionals are internally dynamic.

The second upshot: conditionals in DPL are *strict*. You can see this in the second clause: in saying “ $h \llbracket \phi \rrbracket k$ implies ...” we are universally quantifying over all k ’s that are reachable from h (i.e., g) by executing $\llbracket \phi \rrbracket$. Figure 8.3 graphs this relationship: $\phi \rightarrow \psi$ relates g to itself iff every way of executing $\llbracket \phi \rrbracket$ in g lands you in an information state (those are the intermediate k ’s) where it is possible to execute $\llbracket \psi \rrbracket$.

The strictness of conditionals conspires with the wrinkle of internal dynamics in donkey conditionals. Again, the point is already present in the simpler version we’re

dealing with. Applying the clauses above what you'll see is this:

$$g \llbracket \exists x Fx \rightarrow Gx \rrbracket h \text{ iff : } \begin{cases} h = g \text{ and} \\ h[x]k \text{ and } k(x) \in I(F) \text{ implies} \\ k(x) \in I(G) \end{cases} \quad (8.21)$$

That is: the x in Gx is bound by the $\exists x$ in the antecedent, and bound with universal force. The result is that $\exists x Fx \rightarrow Gx$ says all F s are G s. That's Fact 2.²²

The universal quantifier in DPL is similar and for the similar reason that it is derived from the interaction of $\exists$ and $\neg$. The derived meaning for $\forall x \phi = \neg \exists x \neg \phi$:

$$g \llbracket \forall x \phi \rrbracket h \text{ iff : } \begin{cases} h = g \text{ and} \\ h[x]k \text{ implies } k \llbracket \phi \rrbracket j \text{ for some } j \end{cases} \quad (8.22)$$

Working out a simple example, you'll see that $g \llbracket \forall x Fx \rrbracket h$, iff $g = h$ and $h[x]k$ implies $k(x) \in I(F)$. Like $\rightarrow$, the universal quantifier is internally dynamic (a linking state k shows up) but externally static (because: negation). We'll return to what this predicts for bound readings of pronouns (and thus why $\forall$ is introduced as an abbreviation).

But first, let's turn to $\vee$: we are taking $\phi \vee \psi$ as shorthand for the official $\neg(\neg\phi \wedge \neg\psi)$. A useful exercise is to verify this:

$$g \llbracket \phi \vee \psi \rrbracket h \text{ iff : } \begin{cases} g = h \text{ and} \\ \text{there is a } j \text{ s.t. } g \llbracket \phi \rrbracket j \text{ or} \\ \text{there is a } j \text{ s.t. } g \llbracket \psi \rrbracket j \end{cases} \quad (8.23)$$

Like all of these negation-derived operators, DPL disjunction is externally static. But it is also internally static: this is because the first disjunct in $\phi \vee \psi$ is officially embedded under a negation that has scope only over it.

²² We are taking it for granted that $\exists x Fx \rightarrow Gx$ does, invariably, say that all F s are G s. This is a simplification. Consider weak *donkey*- or *parking meter*-conditionals (the examples are from Schubert and Pelletier 1988):

- (i) a. If Pedro owns a donkey, he will ride it to town tomorrow.
- b. If I have a quarter in my pocket, I will put it in the parking meter.

There are readings of these that don't require Pedro to ride all of his donkeys to town tomorrow and don't require that I pump the meter with every quarter in my pocket. The hard-wired universal force that DPL delivers isn't as hard-wired as it seems, though: weak-conditional variants are available as amendments. This is taken up (in somewhat different ways) in Section 4.2 and Section 4.3 of DPL (pp. 81–2, 89). (Hard-wiring of either sort seems incorrect since, as pointed out in Dekker (1996), simply negating the consequent of a weak-donkey conditional brings the strong reading back to center stage.)

The test-making behavior of negation has empirical predictions for the derived operators. We will briefly touch on them.

Since conditionals are tests, and so externally static, we get an explanation for why pronouns beyond the scope of the conditional aren't bound by the indefinite in the conditional's antecedent.

- (4) a. If a farmer owns a donkey, he feeds it hay. #It is happy with that meal.
 b. A farmer walked in the field or he whistled.

While the *he* and the *it* in the consequent of the conditional in (4a) get bound by *a farmer* and *a donkey*, that is where it ends. And in (4b) it is hard to hear *he* as bound by *a farmer*.²³

Similarly:

- (5) a. It's not the case that a farmer walks in the field. #He whistles.
 b. No farmer owns a donkey. #He's happy about it.
 c. Every farmer walks in the field. #He whistles.

There aren't bound readings of *he* in these sorts of discourses.²⁴ This is explained naturally in DPL: the first sentence of (5a) is a test and so doesn't change the values of variables. Hence, it doesn't constrain the *he* in the second sentence. Similarly for (5b) and, for exactly the same reason, the universal (5c).

There are (at least) two related things to keep in mind. First, this isn't meant to be the last word. Second, this is a different means of generating these predictions from the way the very same data gets explained in DRT and FCS. In DRT, for example, the behavior is explained by the rules that govern when discourse referents (officially, DRSS that contain them) are and aren't *accessible* from a given DRS. Here the explanation instead follows from the DPL behavior of the basic connectives ($\exists$, $\wedge$, and $\neg$) and the completely intro-logic-textbook definitions that introduce $\rightarrow$, $\vee$, and $\forall$.

²³ Here, too, things are trickier. Groenendijk and Stokhof discuss discourses like this (DPL, p. 91):

- (i) a. If a client turns up, you treat him politely. You offer him a cup of coffee and ask him to wait.
 b. Either there's no bathroom here or it is in a funny place.

Here *him* seems to talk about the client who hypothetically turned up (and *it* the bathroom). The lesson they suggest is that static, non-dynamic meanings might be recoverable on-demand from the default dynamic meanings and that certain contexts or situations trigger that recovery. For another approach (especially to discourses like (ia)) see Poesio and Zucchi (1992).

²⁴ But see footnote 23.

Truth, Entailment, Equivalence

Meaning in DPL is richer than truth conditions. So when it comes to thinking about logical concepts (truth, entailment, equivalence) we have options: seemingly different ways of thinking that collapse or coincide in a classical universe may well mark out different regions in a dynamic universe.²⁵

In DPL truth in g is a matter of whether it is *possible to execute* the program $\llbracket \phi \rrbracket$ in g .²⁶

Definition 3 (DPL Truth) For any M , g and ϕ :

1. ϕ is true (in M w.r.t. g) iff $g \llbracket \phi \rrbracket h$ for some h .
Notation: $g \models_M \phi$
2. ϕ 's satisfaction set is the set of assignments at which it is true.
Notation: $\llbracket \phi \rrbracket_s = \{g : g \models_M \phi\}$

Thus ϕ is true w.r.t. g iff g is in ϕ 's satisfaction set. For the simplest tests, the classical set-up comes out as a special case. Suppose $g \in \llbracket Fx \rrbracket_s$. So $g \llbracket Fx \rrbracket h$ for some h . Since Fx is a test, $g = h$ and so $g(x) \in I(F)$. This is what it takes for g to classically satisfy Fx in M .

But DPL truth conditions go beyond classical truth conditions. Take our earlier example $\exists x Fx \wedge Gx$. For this to be true w.r.t. an information state g there must be something that is both an F and a G . These are the same as the truth conditions for the wide-scope existential $\exists x(Fx \wedge Gx)$. Summing up:

Fact 3 For any M : $\llbracket \exists x Fx \wedge Gx \rrbracket_s = \llbracket \exists x(Fx \wedge Gx) \rrbracket_s$

Similarly, the donkey-like conditional $\exists x Fx \rightarrow Gx$ is true w.r.t. g iff every x -alternative k to g such that $k(x)$ is an F is such that $k(x)$ is a G , too. These are the same as the truth conditions for $\forall x(Fx \rightarrow Gx)$. Hence:

Fact 4 For any M : $\llbracket \exists x Fx \rightarrow Gx \rrbracket_s = \llbracket \forall x(Fx \rightarrow Gx) \rrbracket_s$

Something strictly stronger is in fact true: each of these pairs is equivalent.

Classically satisfaction, entailment, and equivalence are all tightly constrained: entailment is preservation of satisfaction, sameness of satisfaction sets is equiva-

²⁵ We will only look at things from a semantic perspective. This raises the question: what does proof theory (say, a sequent system or a natural deduction system) for DPL look like? The answer is: interesting but complicated! The only completeness proofs I am aware of are in Veltman (2000) and (in a more general setting) (Dekker, 2016).

²⁶ Compare this to truth conditions for atomic DRSS in DRT. A DRS like

$$[x : \text{farmer}(x); \text{walked-in-field}(x)]$$

is true in M iff *there is* a (partial) assignment g such that $g(x)$ is a farmer in M and $g(x)$ walked in a field in M . The existential force of indefinites comes from the existential truth conditions, not from quantificational meaning of indefinites.

lence, equivalence is mutual entailment. In DPL there is more room. DPL meanings are fine grained, satisfaction is derivative and doesn't determine entailment, and two-way entailment isn't enough for equivalence.

Definition 4 (DPL Entailment, Equivalence) For any $\phi, \phi_1, \dots, \phi_n$ and ψ :

1. ϕ and ψ are equivalent ($\phi \simeq \psi$) iff: for any M

$$\llbracket \phi \rrbracket^M = \llbracket \psi \rrbracket^M$$

2. $\phi_1, \dots, \phi_n$ entail ψ ($\phi_1, \dots, \phi_n \models_M \psi$) iff: for any M

$$g(\llbracket \phi_1 \rrbracket^M \circ \dots \circ \llbracket \phi_n \rrbracket^M)h \text{ implies } h \models_M \psi$$

Equivalence asymmetrically implies sameness of satisfaction sets.

Fact 5 If $\phi \simeq \psi$ then $\|\phi\|_s = \|\psi\|_s$ but $\|\phi\|_s = \|\psi\|_s$ does not imply $\phi \simeq \psi$.

In computing the semantic value of $\exists x Fx \wedge Gx$ we actually showed this:

$$\exists x Fx \wedge Gx \simeq \exists x (Fx \wedge Gx) \quad (8.24)$$

This equivalence is sometimes called *Egli's theorem*.²⁷ Similarly, we did enough to show this:

$$\exists x Fx \rightarrow Gx \simeq \forall x (Fx \rightarrow Gx) \quad (8.25)$$

This equivalence is sometimes called *Egli's corollary*.

To see that sameness of satisfaction sets is not generally sufficient for sameness of meaning, consider $\exists x Fx$ and $\exists y Fy$. Note that $g \models \exists x Fx$ iff something in D is an F . Same for $\exists y Fy$. Thus:

$$\|\exists x Fx\|_s = S = \|\exists y Fy\|_s \quad (8.26)$$

These sentences prime different variables for downstream binding: $\llbracket \exists x Fx \rrbracket$ takes a g to an h_1 that maps x to an F while $\llbracket \exists y Fy \rrbracket$ takes g to an h_2 that maps y to an F . Those are different outputs reflecting their different meanings. Thus:

$$\exists x Fx \not\simeq \exists y Fy \quad (8.27)$$

In canonical DPL the terminology is that $\exists x Fx$ and $\exists y Fy$ differ in their *production sets* (DPL, p. 56).²⁸

²⁷ See Egli 1979.

²⁸ Define

$\|\phi\|_p = \{h : g \llbracket \phi \rrbracket h \text{ for some } g\}$

When it comes to entailment in DPL, it is sequences (not sets) of premises that matter, so you might expect the order and repetition of premises to matter. And since entailment involves composing the programs denoted by the premises, you might expect that the interesting dynamics associated with interpreting indefinites and conjunctions matter for entailment. Putting those together we'll see that there are entailments of classical logic that are not entailments in DPL and the other way around, too.

DPL entailment has many good-making features. Here is one: it validates the deduction theorem.

Fact 6 $\phi_1, \dots, \phi_n \models \psi$ iff $\models (\phi_1 \wedge \dots \wedge \phi_n) \rightarrow \psi$.

DPL entailment also goes beyond its classical counterpart. Whether ψ is entailed by $\phi_1, \dots, \phi_n$ depends on what is true in a state after executing the composite program $\llbracket \phi_1 \rrbracket \circ \dots \circ \llbracket \phi_n \rrbracket$. This means that discourse information can get passed from the premises to a conclusion.

- (6) a. A woman came in and ordered a sandwich. So: she ordered a sandwich.
- b. A woman came to the department. So: she came to campus.

Like the examples of donkey anaphora, these mini-arguments can get straightforward representations in L_{FO} :

$$\exists x(\text{woman } x \wedge \text{order-sandwich } x) \therefore \text{order-sandwich } x \quad (8.28)$$

$$\exists x(\text{student } x \wedge \text{came-to-department } x) \therefore \text{came-to-campus } x \quad (8.29)$$

The common element is that an indefinite seems to reach from premises to an anaphoric pronoun in the conclusion.

Fact 7 $\exists x(Fx \wedge Gx) \models Gx$ and $\forall x(Gx \rightarrow Hx), \exists x(Fx \wedge Gx) \models Hx$

This is all due to the way discourse information gets introduced and passed through externally dynamic operators.

DPL entailment is about how a *sequence* of premises changes an information state. Neither repetition nor order matters when it comes to classical entailment, but both can be relevant here.

Fact 8 $\phi_1, \dots, \phi_{n-1}, \phi_n \models \psi$ does not imply $\phi_1, \dots, \phi_n, \phi_{n-1} \models \psi$.

Similar facts hold about production sets: if $\phi \simeq \psi$ then $\|\phi\|_p = \|\psi\|_p$ but not in general the other way around. Satisfaction sets pool the states where you can get somewhere from whereas production sets pool states gotten to from somewhere. Still, it is not in general true that $\|\phi\|_s = \|\psi\|_s$ and $\|\phi\|_p = \|\psi\|_p$ together entail that $\phi \simeq \psi$. Counterexamples are things like $Fx \vee \neg Fx$ and $\exists x(Fx \vee \neg Fx)$. Exactly one is a test!

Counterexamples involve existential quantifiers with matching variables competing over a free variable in the conclusion.

$$\exists x Fx, \exists x Gx \models Gx \text{ but } \exists x Gx, \exists x Fx \not\models Gx \quad (8.30)$$

Since existential quantifiers can bind across entailment, interloping existential quantifiers bearing the same variable can interfere with this. This is because $\exists x$ resets the value we have associated with x .²⁹

This is also why right monotonicity fails for DPL entailment.

Fact 9 $\phi \models \psi$ implies $\chi, \phi \models \psi$ but $\phi \models \psi$ does not imply $\phi, \chi \models \psi$.

Counterexamples involve things like

$$\exists x Fx \models Fx \text{ but } \exists x Fx, \exists x Gx \not\models Fx \quad (8.31)$$

A somewhat natural instance of this pattern (using singular *they*):

- (7) a. Someone is at home. So: they are at home.
 b. Someone is at home. Someone is at work. ?So: they are at home.

Again we have quantifiers competing over the downstream pronoun.

This is also why idempotence and repetition are not structural properties of DPL entailment.

Fact 10 $\phi \not\models \phi$ and $\phi, \psi, \chi \not\models \psi$.

By now you will see how to concoct the right kind of counterexample.

So the effects of dynamic entailment show up in lots of places, but a common thread is simply that $\exists x Fx \wedge Gx$ doesn't mean the same thing as $Gx \wedge \exists x Fx$.

Fact 11 $\phi \wedge \psi \not\models \psi \wedge \phi$

DPL entailment is not unruly because it lacks some classical property or other. There are systematic, precise and well-understood boundaries to when instances of classical patterns are DPL-valid and when they aren't.³⁰ That said, it may be pushing luck somewhat to mention that DPL entailment isn't transitive, either.³¹

Fact 12 $\phi \models \psi$ and $\psi \models \chi$ do not imply that $\phi \models \chi$.

²⁹ This is sometimes called “destructive update” or the “downdating problem” in DPL. I see it as more feature than bug, an interesting property that arises because of the way existential quantifiers reset values to variables. We will see another reflection of this feature later. That said, this is a place where FCS and DPL differ (see, e.g., Dekker 1996).

³⁰ See in particular the discussion in Section 3.5 (DPL, pp. 68–70).

³¹ Also, no cut: $\exists x Fx, \exists x \neg Fx \models \exists x Fx$ and $\exists x Fx, \exists x \neg Fx, \exists x Fx \models Fx$ but $\exists x Fx, \exists x \neg Fx \not\models Fx$.

Here the counterexamples still have to do with reaching down to conclusions, but the blocking is achieved not by a competing quantifier but by negation.

$$\neg\neg\exists x Fx \models \exists x Fx \text{ and } \exists x Fx \models Fx, \text{ but } \neg\neg\exists x Fx \not\models Fx \quad (8.32)$$

Note that $\exists x Fx \models\!\!\models \neg\neg\exists x Fx$. But they are not equivalent.³²

Fact 13 $\neg\neg\phi \models\!\!\models \phi$ but $\neg\neg\phi \not\models \phi$.

We have been stressing the way in which the richer universe of meanings that DPL opens up interacts with how we think about things like truth and entailment and equivalence and how those things relate to each other. This is different from the more restricted classical set-up where different ways of thinking collapse so quickly we barely notice. To finish off this section, we'll see one more example of this richness.

Return to the choice of basic versus derived connectives. Classically, this is largely a matter of taste and pragmatic considerations. Not so in DPL. For instance, it matters that $\exists$ is basic and $\forall$ is introduced in terms of it. One way to see that:

Fact 14 While $\neg\exists x\phi \simeq \forall x\neg\phi$ it is not in general true that $\exists x\phi \simeq \neg\forall x\neg\phi$. However, $\neg\neg\exists x\phi \simeq \neg\forall x\neg\phi$

Thus if you instead start with $\forall$ and introduce $\exists$ in terms of it, what you'll get will be a test and hence not the same as the DPL meaning for $\exists$ (and not something that would fit together with $\wedge$ very well). For exactly the same reasons it is important that $\wedge$ is basic in DPL and $\rightarrow$ and $\vee$ are introduced in terms of it. These facts percolate to familiar logical patterns: for example, while contraposition and DeMorgan's equivalences don't in general hold, restricted forms with well-placed closure operators do.

Updating Information States

We have focused on canonical, relational DPL. This has pluses and minuses. Among the pluses: the connection to dynamic logic is clearer. Among the minuses: there is some mismatch between the mechanics of DPL and the idea that the meaning of a sentence is "the way it changes (the representation of) the information of the interpreter" (DPL, p. 43).

³² Add the one-place closure operator $\blacktriangleright$ and interpret it this way:

$$\llbracket \blacktriangleright\phi \rrbracket = \{ \langle g, h \rangle : g = h \text{ and } \langle h, k \rangle \in \llbracket \phi \rrbracket \text{ for some } k \}$$

This takes a sentence and (only) strips it of its dynamic potential (this is sometimes symbolized by $\diamond$ (DPL, p. 63)). You should show that $\blacktriangleright\phi \simeq \neg\neg\phi$.

Here's the mismatch. DPL meanings are relations between information states: what links a prior information state to a posterior state is the meaning of the sentence whose information is being taken on board. But hearers do not in general have complete information and so the "information of the interpreter" should be partial. In relational DPL information states are total assignment functions. Thus the picture that relational DPL leaves us with is one in which interpreters lurch from one complete state of information to another.

Fortunately, this is an artifact. We will explore an implementation of DPL that that doesn't involve such lurching. It will also bring out different ways that meanings can be "dynamic" and it will be good preparation for tackling the issues that arise when the dynamics of discourse information and world information mix (in particular, for tackling the system explored in Groenendijk et al. (1996)).

Let's start with a change in subject: update systems. Both the general set-up and the specific system we'll look at are from Veltman (1996).

Definition 5 (Update Systems) Given a language L , an update system for L is a pair $\langle \mathfrak{S}, [\cdot] \rangle$ where $\mathfrak{S}$ is the set of information states (ordered by $\leq$) with minimal state $\mathbf{1}$ and absurd state $\mathbf{0}$, and $[\cdot]: \mathfrak{S} \rightarrow \mathfrak{S}$ is an interpretation function for L .

Intuitively, $s \leq t$ iff t contains at least as much information as s does.

Fact 15 Define $s + t = t$ iff $s \leq t$. Then $+$ is a join in the lattice $\langle \mathfrak{S}, \mathbf{1}, + \rangle$.³³

We will look at a simple example of an update system for a simple propositional modal language $L_\Diamond$ (Definition 17). The resulting dynamic interpretation of it is Update Semantics (US).³⁴

Definition 6 (Information States in US) The set I of information states is the set:

$$s \in I \text{ iff } s \subseteq W$$

where $W = 2^A$ (and A is the set of atomic sentences in $L_\Diamond$).

This takes an operationalist view on possible worlds: they are what sorts out which atomic sentences are true and which are false. Two limiting cases of information states: W plays the $\mathbf{1}$ role, $\emptyset$ plays the $\mathbf{0}$ role. Note that $\subseteq$ is the natural ordering of states and $\cap$ is the join. These information states, unlike DPL information states, represent genuine uncertainty: it is not in general true that every information state is opinionated about every issue.

As in DPL, sentences express programs. Here, though, they are deterministic: meanings in US are functions from information states to information states.

³³ So: (i) $\mathbf{1} + s = s$, (ii) $s + s = s$, (iii) $s + t = t + s$, and (iv) $(s + t) + r = s + (t + r)$.

³⁴ This simple modal system is given as a warm-up exercise in Veltman (1996) before more sophisticated systems are developed. (The version here is different in small ways; e.g., we are allowing the modals to be out-scoped by $\neg$ and $\wedge$.) The disclaimer at the beginning of this paper holds for US, too: what we'll do isn't — couldn't be — an improvement on the original and you are hereby invited to spend quality time with it.

Definition 7 (Meanings in US) For any ϕ :

$$[\phi]_{\text{US}} : I \rightarrow I$$

The convention is to write the function post-fix (and omit the subscript when unambiguous): the result of applying $[\phi]$ to state s is $s[\phi]$.

This doesn't say which function $[\phi]$ denotes. We'll look at some characteristic examples; the official version is in the appendix (Definition 21).

Often, updating is a matter of eliminating worlds from contention. Consider a simple negated sentence like

- (8) It's not raining.
 $\neg p$

Adding the information carried by this to an information state s goes by two steps; the clauses for $\neg$ and atomic sentences conspire to achieve the intuitively right result. Negation works by set complementation:

$$s[\neg p] = s \setminus s[p] \quad (8.33)$$

So, first, we need to hypothetically update s with p :

$$s[p] = \{w \in s : w(p) = 1\} \quad (8.34)$$

Thus $s \setminus s[p]$ is the state t which contains all and only the worlds in s in which p is false. That is: $\neg p$ expresses the program or instruction to eliminate p -worlds from consideration.

Conjunction in US is *functional* composition. So updating s first with $\neg p$ and then with q further refines things: $s[\neg p]$ is the state in which $[q]$ is executed.

$$s[\neg p \wedge q] = s[\neg p][q] = (s \setminus s[p])[q] \quad (8.35)$$

The resulting state is the state t that contains all and only the worlds from s in which p is false and in which q is true.

The existential modal claim $\Diamond\phi$ is a different sort of program. It is a test, checking whether ϕ is compatible with s .³⁵

$$s[\Diamond p] = \{w \in s : s[p] \neq \emptyset\} \quad (8.36)$$

Notice that this test behavior means that either $s[\Diamond\phi] = s$ or $s[\Diamond\phi] = \emptyset$ for any state s and any prejacent ϕ (Fig. 8.4).

³⁵ There's a surprising ambiguity in "compatibility" (Gillies, 2020).

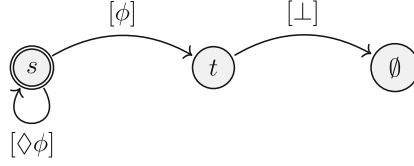


Fig. 8.4 Compatibility test

Definition 8 (US Truth, Entailment, Equivalence) For any $\phi, \phi_1, \dots, \phi_n, \psi$ and any state s :

1. ϕ is true in s iff $s[\phi] = s$
Notation: $s \Vdash_{\text{US}} \phi$
2. $\phi_1, \dots, \phi_n$ entails ψ iff for any s : $s[\phi_1] \dots [\phi_n] \Vdash \psi$
Notation: $\phi_1, \dots, \phi_n \Vdash_{\text{US}} \psi$
3. ϕ and ψ are equivalent iff $[\phi] = [\psi]$
Notation: $\phi \simeq_{\text{US}} \psi$

We are, again, in a rich universe: truth is what you might expect, but entailment is more than preservation of truth at a state, and equivalence is stronger than co-entailingness.³⁶

For the descriptive, non-modal fragment things aren't really that different from the classical interpretation of propositional logic.

Fact 16 For non-modal ϕ, ψ and state s : $W[\phi]$ behaves like a (classical) proposition, updating s is intersection with a proposition, and truth is propositional containment. That is:

1. $W[\neg\phi] = W \setminus W[\phi]$ and $W[\phi \wedge \psi] = W[\phi] \cap W[\psi]$;
2. $s[\phi] = s \cap W[\phi]$; and
3. $s \Vdash \phi$ iff $s \subseteq W[\phi]$.

Things are different with $\Diamond$ and this leads to some similar-to-DPL-looking logical facts: non-commutativity, non-idempotence, non-permutation, etc.

Fact 17 $\phi \wedge \psi \not\approx \psi \wedge \phi$; $\phi \not\Vdash \phi$; and $\phi_1, \dots, \phi_{n-1}, \phi_n \Vdash \psi$ does not imply $\phi_1, \dots, \phi_n, \phi_{n-1} \Vdash \psi$.

The relevant counterexamples all involve $\Diamond$.³⁷

³⁶ There are other sensible choices for entailment: see van Benthem 1996; Veltman 1996 for menus of options. The interesting thing is that various ways of thinking about entailment that tend to collapse in a classical set-up can come apart in a dynamic one.

³⁷ For instance: $\Diamond p \wedge \neg p \not\approx \neg p \wedge \Diamond p$. *Proof:* let $s = \{w, v\}$ where $w(p) = 1$ and $v(p) = 0$ and note that $s[\Diamond p \wedge \neg p] = \{v\}$ while $s[\neg p \wedge \Diamond p] = \emptyset$. Using this information, you can construct counterexamples for idempotence and permutation.

The point for now is to see US as implementing a natural way of updating an information state where those states are partial pictures of the way things are.

DPL Updates

We want to implement the core of DPL in an update system that traffics in partial information states. In DPL the only information is discourse information so we don't want every information state to take a stand on the value of every variable. To achieve this, take sets of variable assignments as our states.

Definition 9 (Information States in Functional DPL) Let $M = \langle D, I \rangle$ be a first-order model and $S = \wp(D^V)$ the set of variable assignments. The set $\mathbf{S}$ of (functional) DPL information states is the set:

$$s \in \mathbf{S} \text{ iff } s \subseteq S$$

That is: information states are *sets* of relational DPL information states. Note that $\mathbf{S}$ plays the **1** role, $\emptyset$ the **0** role, and s contains at least as much information as t iff $s \subseteq t$.

Definition 10 (Meanings in Functional DPL) For any ϕ :

$$[\phi]_{\text{DPL}}^M : \mathbf{S} \rightarrow \mathbf{S}$$

The conventions from US carry-over here. We'll look at characteristic examples of the interesting clauses, saving the official version of $[\cdot]_{\text{DPL}}$ for Definition 22.

The move from $\llbracket \cdot \rrbracket_{\text{DPL}}^M$ to $[\cdot]_{\text{DPL}}^M$ is mostly straightforward: atomic formulas and negations operate on the assignments that make up a state and conjunction becomes functional composition.

So, for instance, applying the meaning $[Fx]$ to a state s :

$$s[Fx] = \{g \in s : g(x) \in I(F)\} \quad (8.37)$$

The instruction: figure out which individual assignments in s map x to something in $I(F)$, and keep those (and only those). Likewise, in applying $[\neg\phi]$ to s , test the individual assignments in s to see if they permit updating by $[\phi]$. Extending our example:

$$\begin{aligned} s[\neg Fx] &= s \setminus \{g : \{g\}[Fx] \neq \emptyset\} \\ &= \{g \in s : g(x) \notin I(F)\} \end{aligned} \quad (8.38)$$

The singleton sets $\{g\}$ on the right-hand side are a giveaway that DPL negation, unlike its US cousin, is entirely point-wise: $s[\neg Fx]$ rides on $\{g\}[Fx]$ for each $g \in s$.

Updating a state s with $[\exists x Fx]$ should output a state that only contains assignments mapping x to an F . There are multiple ways to achieve this.

Definition 11 Fix a model $M = \langle D, I \rangle$ and let s be any state, x any variable, and d any object in D .

$$s[x/d] = \{h : h = g[x/d] \text{ for some } g \in s\}$$

This is the set of assignments that differ from an assignment in s at most in that they map x to d . This achieves a global and specific rejiggering of the information so that all assignments have x pointing to this particular d .³⁸

The official, full-generality, clause for updating with an existentially quantified formula is this:

$$s[\exists x \phi] = \bigcup_{d \in D} (s[x/d][\phi]) \quad (8.39)$$

The intermediate, rejiggered, state $s[x/d]$ is key. What we are doing is taking an object d from D and re-assigning x to map to it, and then updating the resulting $s[x/d]$ with ϕ , and then repeating this process for every object in D .

An example may help. First, given a $d \in D$, we take the state $s[x/d]$ and update it with Fx :

$$g \in s[x/d][Fx] \text{ iff } g \in s[x/d] \text{ and } g(x) \in I(F) \quad (8.40)$$

Next, collect up the results from doing this for all the ways of choosing such a d :

$$g \in \bigcup_{d \in D} (s[x/d][Fx]) \text{ iff for some } d : g \in s[x/d] \text{ and } g(x) \in I(F) \quad (8.41)$$

Thus $[\exists x Fx]$ instructs us to randomly reassign a value to x and then insists that whatever x points to is an F .

Fact 18 Let $s[x] = \{g[x/d] : g \in s \text{ and } d \in D\}$. Then $s[\exists x \phi] = s[x][Fx]$.

So we could have defined $s[\exists x \phi]$ in terms of $s[x]$ and $[\phi]$.³⁹

Truth, entailment, and equivalence undergo similar changes: we just take everything up a level to sets of assignments.

³⁸ Note that it is not in general true that $s[x/d] \subseteq s$. This is not surprising but important: it will come up again in the next section.

³⁹ Notice that this amounts to treating $\exists x$ as categorically expressing the function $[x]$ and thus $\exists x \phi$ (abusing our object language formation rules) as the conjunction $\exists x \wedge \phi$. These two formulations agree but can come apart once modals are added to the mix. Some (canonical) discussions: van Eijck and Cepparello (1994); Groenendijk et al. (1996); Beaver (2001).

Definition 12 (DPL Truth, Entailment and Equivalence, Again) For any $\phi, \phi_1, \dots, \phi_n$, and ψ :

1. ϕ is true in s w.r.t M ($s \models \phi$) iff:

$$\text{for every } g \in s: \{g\} [\phi] \neq \emptyset$$

2. $\phi_1, \dots, \phi_n$ entail ψ ($\phi_1, \dots, \phi_n \models \psi$) iff:

$$\text{for every } M, s: s[\phi_1] \dots [\phi_n] \models \psi$$

3. ϕ and ψ are equivalent ($\phi \simeq \psi$) iff:

$$\text{for every } M: [\phi]^M = [\psi]^M$$

Truth at an information state (set of assignments) is just truth with respect to each point (assignment) in that state. Similarly for entailment. Equivalence is sameness of semantic value.

This really is DPL in an update system's clothing. Here are two ways to see that.

Fact 19 $g \models \phi$ iff $h \in \{g\} [\phi]$.

Fact 20 $s \models \phi$ iff for every $g \in s: g \models \phi$. Similarly, $\phi_1, \dots, \phi_n \models \psi$ iff $\phi_1, \dots, \phi_n \models \psi$.

Thus all of the characteristic predictions and patterns of (non-)entailment and (non-)equivalence of relational DPL hold for the functional implementation. From that perspective, there is no difference.

Dynamic Differences

Why bother with the update reformulation then? A few reasons. First: it shows that the essential insights of DPL are independent of the relational implementation. Second: it sits better with the idea that contexts are characterized by states of uncertainty and that meanings are ways of updating that uncertainty. Third: it makes for a better comparison with the other dynamic system on the table (US). We turn to this comparison now.

DPL and US have different targets: one cares only about discourse information (level of granularity: which pronouns pick out which objects) and one cares only about world information (level of granularity: which worlds are compatible with the information gathered so far). Even so there are similarities. In both set-ups semantic values are functions linking information states to information states and key concepts like truth and entailment are relativized to information states not to

individual unanalyzed points of evaluation. Both say that equivalence is more than mutual entailment and both exhibit the similar non-classical entailment behavior.

For all this similarity, there are deep differences. We will touch on two of them. The first difference is that in US information always accumulates but DPL permits information “loss”.⁴⁰ The second difference: everything in US semantics happens *set-wise*: the information states are key players *as* states. In DPL at key points the call to information states gets flattened out to quantification over what happens at the assignments that make them up. The result is a system in which, in a precise sense, there is no essentially set-wise behavior.

The updates in US are aptly named: posterior states contain at least as much information as their prior states do.

Definition 13 (Eliminativity) An update system $\langle \mathcal{S}, [\cdot] \rangle$ for L with ordering $\leq$ is eliminative iff: for every $s \in \mathcal{S}$ and $\phi \in L$:

$$s \leq s[\phi]$$

Eliminative update systems are ones in which information accumulates: $s[\phi]$ contains at least the information that s does.

Fact 21 $\langle I, [\cdot]_{\text{US}} \rangle$ is eliminative and $\langle \mathcal{S}, [\cdot]_{\text{DPL}} \rangle$ is not eliminative.

You can see why $s[\phi]_{\text{US}}$ is always included in s : each characteristic clause either eliminates possibilities or leaves the prior state unchanged. Not so in DPL: updating a set of assignments s with $\exists x$ globally resets the value associated with x , forgetting the prior value. The way it achieves such resetting is by admitting into $s[x]$ assignments not in s . Hence $s[\exists x \phi]$ will not in general be included in s .

We have already seen how the resetting behavior makes for empirical and structural differences in DPL: it is involved in why conjunctions don’t commute and in failures of right monotonicity, idempotence, and repetition. There is another place where it comes up and where it is alleged to be problematic.

(9) A farmer walked in the field. A farmer whistled.

$$\exists x(\text{farmer } x \wedge \text{walked-in-field } x) \wedge \exists x(\text{farmer } x \wedge \text{whistled } x)$$

Suppose we update s so that x is associated with a farmer who walked and then update the resulting state so that x is associated with a farmer who whistled. The result will be the same as had we just changed s so that x is associated with a farmer who whistled. The information carried by the first sentence is gone. This is the destructive update or downdate problem of DPL, again.

It is tricky to say exactly what the problem is, though. It isn’t that re-using a variable gets us incorrect predicted truth conditions or entailments. You can check that (9) is true in DPL iff there is a farmer who walked and there is a farmer who

⁴⁰ The scare quotes indicate that there is some interpretive controversy about the nature of the loss of information. When we get to it, we’ll flag the controversy.

whistled.⁴¹ Instead it must be more conceptual: there is a nagging feeling that re-using variables shouldn't destroy the information that was associated with an upstream use of that variable and, more generally, that information updates should be bona fide updates. But if we are only tracking discourse information, re-setting on re-use is what we should expect.⁴²

Now to the second deep difference. In US it is whole information states that count: they get implicated in the recursive definition of $[\cdot]_{\text{US}}$, truth at a state isn't defined as truth at the points that make that state up, and entailment is about whether a whole state is a fixed-point of a sequence of updates. In contrast, DPL meanings live and die on the behavior of updates to "states" that are just the individual assignments that make states up.

Definition 14 (Distributivity) An update system $\langle \mathfrak{S}, [\cdot] \rangle$ for L is distributive (a.k.a. continuous) iff: for every $s \in \mathfrak{S}$ and $\phi \in L$

$$s[\phi] = \bigcup_{x \in s} \{x\}[\phi]$$

In a distributive system $s[\phi]$ is always the same thing as collecting up the results from computing $\{x\}[\phi]$ for each $x \in s$. In such systems there is nothing essentially global or set-wise as far as $[\cdot]$ is concerned.

Fact 22 $\langle I, [\cdot]_{\text{US}} \rangle$ is not distributive and $\langle \mathfrak{S}, [\cdot]_{\text{DPL}} \rangle$ is distributive.

It is the modal $\Diamond$ that is responsible for non-distributivity in US.⁴³ Here is a simple example: let $s = \{w, v\}$ where $w(p) = 1$ and $v(p) = 0$.

$$\begin{aligned} \bigcup_{x \in s} \{x\}[\Diamond p] &= \{w\}[\Diamond p] \cup \{v\}[\Diamond p] \\ &= \{w\} \cup \emptyset \\ &\neq s[\Diamond p] = s \end{aligned} \tag{8.42}$$

On the other hand, you can see tell-tale signs of DPL's distributivity: for example, DPL negation doesn't care about the incoming state *as* a state; $s[\neg Fx]$ is determined by what $\{g\}[Fx]$ is for each assignment $g \in s$.

Again, an example may help. Let $M = \langle D, I \rangle$ be a model with just three objects a, b , and c where $I(F) = \{a, b\}$ and let $s = \{h, k\}$. We will look at $s[\exists x \neg Fx]$ and compare this to $\{h\}[\exists x \neg Fx] \cup \{k\}[\exists x \neg Fx]$.

⁴¹ This same point is made in Charlow (2019).

⁴² In a more expressive fragment, we may need a more subtle way of tracking different dimensions of information. For instance, the referent systems in Groenendijk et al. (1996) are designed explicitly to disassociate a variable from the world information associated with it. This suggests there is no downdating problem as such for the tracking of discourse information.

⁴³ That's why Fact 16 only holds for descriptive (non-modal) sentences.

There are two steps to look at. First, randomly reassign x throughout s as a whole:

$$\begin{aligned} s[\exists x] &= s[x] \\ &= \{h[x/a], h[x/b], h[x/c], k[x/a], k[x/b], k[x/c]\} \end{aligned} \quad (8.43)$$

Compare this to applying it to $\{h\}$ and to $\{k\}$:

$$\begin{aligned} \{h\}[\exists x] &= \{h\}[x] \\ &= \{h[x/a], h[x/b], h[x/c]\} \end{aligned} \quad (8.44)$$

$$\begin{aligned} \{k\}[\exists x] &= \{k\}[x] \\ &= \{k[x/a], k[x/b], k[x/c]\} \end{aligned} \quad (8.45)$$

You can probably already see why in general $s[x] = \bigcup_{g \in s} \{g\}[x]$.

The other step that is instructive to look at is negation.

$$\begin{aligned} s[x][\neg Fx] &= s[x] \setminus \{g : \{g\}[Fx] \neq \emptyset\} \\ &= \{h[x/c], k[x/c]\} \end{aligned} \quad (8.46)$$

Compare to the point-wise updates:

$$\begin{aligned} \{h\}[x][\neg Fx] &= \{h\}[x] \setminus \{g : \{g\}[Fx] \neq \emptyset\} \\ &= \{h[x/c]\} \end{aligned} \quad (8.47)$$

$$\begin{aligned} \{k\}[x][\neg Fx] &= \{k\}[x] \setminus \{g : \{g\}[Fx] \neq \emptyset\} \\ &= \{k[x/c]\} \end{aligned} \quad (8.48)$$

Here you can see why in general $s[\neg\phi] = \bigcup_{g \in s} \{g\}[\neg\phi]$. A full, but entirely routine, inductive proof establishes things aren't trickier than this example suggests: $s[\phi]$ is always the same as taking each $g \in s$ one at a time, figuring out $\{g\}[\phi]$, and collecting the results.

Eliminativity and distributivity aren't two randomly chosen properties. Together they characterize the update systems that are *additive*.⁴⁴

Definition 15 An update system $\langle \mathfrak{S}, [\cdot] \rangle$ for L with join $+$ is additive iff for every $s \in \mathfrak{S}$ and $\phi \in L$:

$$s[\phi] = s + \mathbf{1}[\phi]$$

⁴⁴ This basic idea goes back to van Benthem (1986) but its modern formulation can be found in both Groenendijk and Stokhof (1990) and Veltman (1996). See also Rothschild and Yalcin (2016).

Such systems are commutative ($s[\phi][\psi] = s[\psi][\phi]$) and idempotent ($s[\phi][\phi] = s[\phi]$). So there's not much dynamics: $[\phi]$ is less an instruction for how to move from a given state s to a posterior state t and more a description of a state that shares information with both s and $\mathbf{1}[\phi]$.

Fact 23 *An update system $\langle \mathfrak{S}, [\cdot] \rangle$ for L with join $+$ is additive iff it is both eliminative and distributive.*

So US and DPL are both non-additive but in exactly complementary ways.

Extensions, Amendments, Revisions

We have long since run out of space. So covering all the many ways in which DPL (in spirit if not always in letter) has been subsequently put to use is out of the question. But to close, I want to mention one prominent extension.

The fact that US and DPL depart from additive systems in different directions is part of what makes it interesting, and hard, to think about systems that track the dynamics of both discourse and world information. This is taken up in Groenendijk et al. (1996).⁴⁵

One thing that needs grappling with is how $\exists x$ re-assigns values. Above we breezily slid from talk of distributive re-assignment to global random assignment; the difference didn't matter. To highlight the issue, consider these two options:

$$s[\exists x\phi] = \bigcup_{d \in D} (s[x/d][\phi]) \quad (8.49)$$

$$s[\exists x\phi] = (\bigcup_{d \in D} s[x/d])[\phi] \quad (8.50)$$

In the presence of a consistency-testing modality, there is a difference between these.

- (10) a. Someone is hiding in the closet. He might be guilty.
 $\exists x Hx \wedge \Diamond Gx$
- b. Someone who is hiding in the closet might be guilty.
 $\exists x (Hx \wedge \Diamond Gx)$

⁴⁵ There are similar issues when the dynamics of quantification and the dynamics of presupposition mix. A dynamic presupposition operator ∂ naturally has a test profile: $s[\partial\phi] = s$ if $s \models \phi$ and is undefined otherwise. This leads to a non-additive system for reasons of non-distributivity. So modeling the dynamics of quantification together with the dynamics of presupposition poses a lot of the same types of issues and requires a lot of the same kinds of innovations. On this score, see Beaver 2001.

These sentences (arguably) don't say the same thing and the distributive, but not the global, way of re-assigning values to variables can predict this.⁴⁶

This, of course, is not the last word on how the dynamics of discourse and world information fit together, but it will in fact be the last word here.⁴⁷

Acknowledgments This paper's main overarching debt is recounted in Footnote 8. More proximately, this paper is based on handouts and lectures I have used to teach DPL over the years; thanks are due to students and colleagues who tolerated and corrected them along the way, especially David Boylan, Jingyi Chen, Sam Carter, Nico Kirk-Giannini, Simon Goldstein, Jeff King, Zack Kofi, Sara Packalan, Paul Pietroski, Lauren Richardson, and Andrew Rubner. Thanks also to Zoltán Gendler Szabó, Louise McNally, and an anonymous referee for comments and suggestions.

Appendix: Some Official Definitions

Definition 16 (First Order Languages) L_{FO} is a first order language iff L_{FO} is built from a set V of variables, a set of predicate symbols, and the operators $\neg$, $\wedge$, and $\exists$ in the usual way.⁴⁸ Other operators are introduced as abbreviations: $(\phi \rightarrow \psi) = \neg(\phi \wedge \neg\psi)$, $(\phi \vee \psi) = \neg(\neg\phi \wedge \neg\psi)$, and $\forall x\phi = \neg\exists x\neg\phi$.

Definition 17 (Propositional Modal Languages) A propositional modal language $L_{\Diamond}$ is built on a set $A = \{p, q, \dots\}$ of atomic sentences, and operators $\neg$, $\wedge$, and $\Diamond$ in the usual way. $\Box$ abbreviates $\neg\Diamond\neg$.

Definition 18 (PDL w/tests) Fix a set $\mathbf{A}$ of atomic programs. The language L_{PDL} is the smallest set of sentences including L_0 (propositional logic) and is closed as follows:

1. L_{PDL} is closed under $\rightarrow$ and $\neg$;
2. Π is the smallest set including $\mathbf{A}$ and is such that: (a) if $\pi_1, \pi_2 \in \Pi$ then so are $\pi_1; \pi_2$ (composition), $\pi_1 \cup \pi_2$ (choice), and π_1^* (iteration); and (b) if $\phi \in L_{PDL}$ then $\phi? \in \Pi$;
3. If $\phi \in L_{PDL}$ and $\pi \in \Pi$ then $\langle\pi\rangle\phi \in L_{PDL}$.

Definition 19 (Semantics for PDL) A PDL (propositional) model is a pair $M = \langle W, V \rangle$ where W is a non-empty set of states and V is a function taking atomic formulas to subsets of W and atomic programs to subsets of $W \times W$. $\llbracket \cdot \rrbracket^M$ is the interpretation function based on M iff for any p, a, ϕ, ψ, π :

⁴⁶ Suppose you don't know who did it and you don't know who is in the closet. Then (10a) is true. But if you also know that certain people are *not* guilty but you can't rule out that one of them is who is in the closet, then arguably (10b) isn't.

⁴⁷ Recommended reading: van Eijck and Cepparello (1994); Groenendijk et al. (1996); Aloni (2005); Dekker (2012).

⁴⁸ We focus on languages with variables and no constants. When there is no risk of confusion omit the subscript on L_{FO} (this goes for $L_{\Diamond}$ and L_{PDL} , too).

1. Atomic:

- (a) $\llbracket p \rrbracket = V(p)$ for atomic formulas
- (b) $\llbracket a \rrbracket = V(a)$ for atomic programs

2. Boolean:

- (a) $\llbracket \neg\phi \rrbracket = W \setminus \llbracket \phi \rrbracket$
- (b) $\llbracket \phi \rightarrow \psi \rrbracket = (W \setminus \llbracket \phi \rrbracket) \cup \llbracket \psi \rrbracket$

3. Programs:

- (a) $\llbracket \pi_1; \pi_2 \rrbracket = \llbracket \pi_1 \rrbracket \circ \llbracket \pi_2 \rrbracket$
- (b) $\llbracket \pi_1 \cup \pi_2 \rrbracket = \llbracket \pi_1 \rrbracket \cup \llbracket \pi_2 \rrbracket$
- (c) $\llbracket \pi^* \rrbracket = (\llbracket \pi \rrbracket)^*$
- (d) $\llbracket \phi? \rrbracket = \{ \langle w, w \rangle : w \in \llbracket \phi \rrbracket \}$

4. Modalities:

- (a) $\llbracket \langle \pi \rangle \phi \rrbracket = \{ w : \text{there is a } v \text{ s.t. } \langle w, v \rangle \in \llbracket \pi \rrbracket \text{ and } v \in \llbracket \phi \rrbracket \}$
- (b) $\llbracket [\pi] \phi \rrbracket = \{ w : \text{for all } v \text{ if } \langle w, v \rangle \in \llbracket \pi \rrbracket \text{ then } v \in \llbracket \phi \rrbracket \}$

In this definition, $*$ has a dual role. Attached to a program as in π^* this is the program you get by executing π zero or more times; attached to a program's *meaning* (i.e., a relation on states) as in $(\llbracket \pi \rrbracket)^*$ this is the reflexive and transitive closure of the relation $\llbracket \pi \rrbracket$.

Definition 20 (Relational DPL) Let $M = \langle D, I \rangle$ be a first-order model and g an assignment function. Let $g[x]h$ iff for some $d \in D$: $h = g[x/d]$. Then $\llbracket \cdot \rrbracket_{\text{DPL}}^M$ is defined as follows:

1. $g \llbracket Pt_1, \dots, t_n \rrbracket h$ iff $h = g$ and $\langle [t_1]_g, \dots, [t_n]_g \rangle \in I(P)$
2. $g \llbracket \neg\phi \rrbracket h$ iff $h = g$ and for no k : $g \llbracket \phi \rrbracket k$
3. $g \llbracket \phi \wedge \psi \rrbracket h$ iff there is a k s.t. $g \llbracket \phi \rrbracket k$ and $k \llbracket \psi \rrbracket h$
4. $g \llbracket \exists x\phi \rrbracket h$ iff there is a k s.t. $g[x]k$ and $k \llbracket \phi \rrbracket h$

Definition 21 (Update Semantics) $[\cdot]_{\text{US}} : I \rightarrow I$ is defined as follows (where p is any atomic sentence and ϕ, ψ are any sentences of $L_\Diamond$):

1. $s[p] = \{w \in s : w(p) = 1\}$
2. $s[\neg\phi] = s \setminus s[\phi]$
3. $s[\phi \wedge \psi] = s[\phi][\psi]$
4. $s[\Diamond\phi] = \{w \in s : s[\phi] \neq \emptyset\}$

Definition 22 (Functional DPL) Let $M = \langle D, I \rangle$ be a first-order model. Define the DPL interpretation function $[\cdot]_{\text{DPL}}^M : \wp(D^V) \rightarrow \wp(D^V)$ as follows (super-/subscripts omitted when unambiguous):

1. $s[Px_1 \dots x_n] = \{g \in s : \langle g(x_1), \dots, g(x_n) \rangle \in I(P)\}$
2. $s[\neg\phi] = s \setminus \{g : \{g\}[\phi] \neq \emptyset\}$
3. $s[\phi \wedge \psi] = s[\phi][\psi]$
4. $s[\exists x\phi] = \bigcup_{d \in D} (s[x/d][\phi])$

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Chapter 9

On Heim's "On the Projection Problem for Presuppositions"



Mandy Simons

Abstract Irene Heim's 1983 paper, "On the Projection Problem for Presuppositions" is, first, a paper about presupposition projection. But perhaps its major significance lies in its role in launching the dynamic turn in formal semantics, whose central idea is that the conventional meaning of an expression is given by a description of how that expression updates a context. The fundamental ideas of context and context change that Heim presented in this brief paper are now part of the basic toolkit of semantics. At the same time, the paper established presupposition and presupposition projection as a topic of central concern for the emerging dynamic approach. In this commentary, I briefly describe the paper's most direct antecedents, review its central theoretical innovations, and describe some alternative approaches to the formal characterization of contexts and to the analysis of presupposition and presupposition projection.

Keywords Presupposition · Projection · Dynamic semantics · Context change

Introduction: Heim's Antecedents

In a sequence of papers published in the 1970s, Robert Stalnaker (1970, 1973, 1974, 1978, see also 1999, 2002) developed a deeply influential picture of the relation between context and content. On this picture, discourse takes place relative to a context, a background of "given" information. Context has a dual function. It is the body of information to be updated with asserted content, but at the same time has a content-fixing role, for example, by fixing the referents of indexicals. And, as a representation of what the interlocutors currently take as given, it places constraints on what can felicitously be asserted.

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Stalnaker's account of context and context change is entirely pragmatic: context change has the character it has by virtue of the uses to which people put linguistic forms. Heim (1983a), while not explicitly presented as a reformulation of Stalnaker's work, carries out a semanticization of context change, presenting a version of Stalnaker's program in which the conventional meanings of expressions are characterized by the "update effects" that they have on a context. At the same time, Heim's work cements a semantic view of presupposition as a property of linguistic forms.

Heim (1983a) brings together and extends two earlier lines of work. The first is Heim's prior work on noun phrase semantics (Heim, 1982, 1983b), developed in a more restricted dynamic framework. There, Heim introduces the idea of a File Change Potential, the antecedent of the Context Change Potentials of the current work.

The second source is Karttunen's work on presupposition projection, in particular Karttunen (1974). Karttunen there adopts Stalnaker's general picture of context update, but detaches it from its epistemological foundations. Karttunen defines a context as "a set of logical forms that describe the set of background assumptions, that is, whatever the speaker chooses to regard as being shared by him and his intended audience" (p. 182). Each sentence is associated with a set of *admittance conditions*: conditions which must be satisfied by a context in order for the context to admit update with the content of that sentence. For a simple sentence *S*, the admittance condition is that the context entails the presuppositions of *S*. For compound sentences, "we define [admittance] recursively by associating each part of the sentence with a different context" (p. 184; see also Stalnaker, 1973). Karttunen continues: "In case a sentence occurs as part of a larger compound, its presuppositions need not always be satisfied by the actual conversational context, as long as they are satisfied by a certain local extension of it" (p. 185). On his account, a presupposition of a sentential constituent *projects* – becomes a presupposition of the sentence as a whole – when the admittance conditions of that constituent impose a requirement on the conversational, or *global*, context. But when admittance conditions can be met locally, without imposing any requirements on the global context, non-projecting readings ensue. In the case of a sentence of form *A and B*, the local context for *B* is the global context incremented with the content of *A*. Hence, if (the logical form of) *A* entails the presupposition of *B*, that presupposition will be satisfied in the local context for *B* regardless of the properties of the global context. In this case, the presuppositions of *B* are not judged to be shared by *A and B* as a whole.

This is the core of Karttunen's account. Heim adopts Karttunen's treatment of presuppositions as admittance conditions attached to clauses, as well as his account of projection. Heim, however, embeds this account within a general, formal theory of context update, to which we now turn.

Formal Theory of Context Update

Context Change Potentials

Karttunen's projection rules stipulate the local contexts for constituents of compound sentences in order to derive the correct observations about presupposition projection and filtering. Stalnaker (1974) critiqued the explanatoriness of this model, arguing that context change is to be explained entirely in terms of the communicative effects of utterances. He argues that Karttunen's rules are an unnecessary complication of the semantics, because the determination of the local contexts for embedded clauses follows from the pragmatics of context change. For example, he argues that the observations regarding conjunction simply follow from the pragmatic fact that after a speaker has uttered the first conjunct of a conjunction, she is entitled to assume that its content is part of the common ground relative to which the second conjunct will be interpreted.

Heim similarly aims for a non-stipulative theory of context change. However, she does not aim to ground the theory in pragmatic principles, but instead in the truth conditions of atomic and complex expressions. Like Karttunen's, her theory seeks to specify, for any sentence form, its Context Change Potential (CCP): the update that utterance of that sentence will effect on its context. In Heim's approach, a context is no longer identified with a set of logical forms, but (initially) with a set of worlds. (In the next section, we'll see a modification of this treatment.) The CCP of any atomic sentence requires intersecting its content (the set of worlds compatible with it) with the current context; this adds the content of the atomic sentence to the context. The CCPs of complex sentences are, as in Karttunen's model, to be defined recursively.

In her paper, Heim argues that the CCPs for complex sentences are uniquely determined by their truth conditions, and thus that the projection behavior of complex sentences is predictable from their truth conditional content. However, this position was subsequently refuted by Soames (1989, pp. 597–8).¹ Consider again the case of conjunction. In order to reflect the truth conditions of conjunction, update with *A* and *B* must eliminate from the context *c* all non-*A* worlds and all non-*B* worlds. According to Heim, this determines that conjunction must have the CCP in (1), where "+" indicates update:

$$(1) \quad c + A \text{ and } B = (c + A) + B$$

This CCP has the desired output, while also characterizing the local contexts for each of the conjuncts: as per Karttunen's rule, the local context for *A* is the global context, while the local context for *B* is $(c + A)$, i.e., the global context updated with

¹ Rothschild, 2011 reports that the same observation was made independently by Mats Rooth in private correspondence with Heim, acknowledged by Heim (1990). Rothschild also gives a careful articulation of how to best understand the position of Heim (1983a) on the predictive nature of her framework.

the content of *A*. So, for example, updating a context with the sentence *Malia has a dog and her dog is loyal* involves first updating the context with *Malia has a dog*, and then updating the result with *Her dog is loyal*. As the first update guarantees that the presupposition of the second clause (that *Malia has a dog*) is entailed by its local context, the sentence as a whole imposes no presuppositional requirements on the starting context.

But as pointed out by Soames, the procedures in (2) below have the same output, but would each make different predictions about local contexts and hence about presupposition projection.

- (2) a. $(c + B) + A$
 b. $(c + A) \cap (c + B)$

A CCP, then, does two things. It specifies the ultimate effect of a sentence on a context; but it also specifies a procedure for deriving that effect. The specification of the procedure does the same work as Karttunen's stipulations: it specifies the local context for each sentence constituent. Soames' observation shows that the procedures which Heim specifies are no less stipulative than Karttunen's original definitions of local contexts. In some simple cases – the CCPs for conjunction and negation, perhaps – it is plausible to think that CCPs could be motivated in terms of pragmatics or even language processing, in line with Stalnaker's approach. But the CCPs for conditional sentences and for the quantificational sentences we'll consider below can really only be conceptualized as part of a formal semantic system, which is indeed the spirit in which Heim proposes them.

In later work, Heim (1992) extends the context change framework to the intensional domain, offering an account of the presuppositional behavior of attitude sentences. In this extension, local contexts are given intensional interpretations (e.g. as the belief states of agents). For other work extending and modifying Heim's approach, see Beaver (1992, 2001) and Zeevat (1992). In a rather different vein, Rothschild (2011) reformulates Heim's system with the goal of formalizing a systematic relation between the truth conditional properties of operators and their projection properties, as Heim originally hoped to do.

Extension to Quantification: Context as Assignment-World Pairs

The second formal innovation in Heim's work is the introduction of a more fine-grained model of context. Initially, as noted, Heim follows Gazdar (1979) and Stalnaker in modeling contexts as sets of possible worlds. In the refined model (originally developed in full detail in Heim, 1982, 1983b), contexts become sets of assignment-world pairs. An assignment (also called a *sequence*) is a mapping

from the set of natural numbers (corresponding in practice to the indices assigned to noun phrases and variables at the level of logical form) to entities.²

This extension of the model of context allows us, in effect, to treat variables as temporarily denoting entities, and hence to assign context update effects to open sentences as if they were proposition denoting. For example, consider the open sentence x_3 barked. This can be assigned the CCP in (3):

$$(3) \quad c + [x_3 \text{ barked}] = \{ \langle g, w \rangle \in c : g(3) \text{ barked in } w \}$$

Thus an open sentence, just like a complete sentence, can effect a proposition-like update to the context.

The importance of this move for Heim is that it enables her to provide a dynamic semantics for quantified sentences. Following Barwise and Cooper (1981), Heim assumes a generalized quantifier analysis of quantification, and so assumes tripartite logical forms for quantified sentences. For example, sentence (4) has the logical form shown in (5), where the restrictor (*dog*) and scope (*barks*) of the quantifier are each rendered as the predicate of an open sentence.

(4) Every dog barks

(5) Every _{x_2} [x_2 is a dog][x_2 barks]

Now, the machinery which allows her to specify CCPs for open sentences allows Heim to specify a CCP for the *every* sentence. These CCPs specify distinct local contexts for the restrictor and scope of the quantifier, and hence allow Heim to use the admittance conditions of each of these clauses to predict the presuppositional properties of quantified sentences containing presuppositional expressions.

Heim was the first to offer such predictions, and even to suggest that the presuppositional behavior of quantified sentences is governed by the same basic mechanism that governs projection in compound sentences. However, the accuracy of her predictions remains unclear. Experimental work exploring speakers' judgments about the presuppositions of quantified sentences suggests that presuppositions triggered by expressions which occur in the scope of universal quantifiers can project both existentially and universally. What this means is illustrated by the interpretations of sentence (6). When read with a universal presupposition, the sentence presupposes that everything satisfying the CN *students* satisfies the presupposition (reading 7a); but the sentence can also be understood with a weak existential presupposition, requiring only that at least one thing satisfying the CN also satisfies the presupposition (reading 7b).

(6) None of the students stopped eating donuts.

(7) a. Every student had been a donut-eater.

(universal presupposition)

b. At least one student had been a donut-eater

(existential presupposition)

² The use of sequences is standard in mathematical logic, going back to Tarski (1933).^o

In addition, the sentence seems to allow for non-presuppositional readings with relative ease. The experimental work also suggests that presuppositions project differently under different quantifiers. For details, see Chemla (2009), Sudo et al. (2012), Geurts and van Tiel (2015), Zehr et al. (2016), Cremers et al. (2018). Similarly, one significant point of difference between the various theories of projection subsequent to Heim (see references in Section “[Alternative approaches to context](#)” below) is in their predictions with respect to quantified sentences.

Accommodation, Global and Local

Accommodation Defined

On Heim’s account, the presuppositions of a clause S must be satisfied (entailed) by any context c which admits S . The process of *accommodation* (Lewis, 1979) provides an escape clause. Accommodation is a process whereby a context is updated with a required presupposition of S prior to update with the ordinary content of S , ensuring definedness of the update. But which context? In the case of complex sentences, each constituent clause updates its own local context; and this provides multiple options for accommodation. For illustration, consider the case of the negated sentence (8), with the CCP in (9):

(8) The king of France does not have a son.

(9) $c + (8) = c \setminus c + \text{The KoF has a son}$ ³

Suppose that c does not satisfy the presupposition of the definite, but is compatible with it. This context does not admit (8). Accommodation allows us to first update c with the required presupposition to create a revised context, c' , and then to update the revised context with the sentence content. This sequence is illustrated in (10).

- (10) i. $c' = c + \text{there is a KoF}$
 ii. $c + (8) = c' \setminus c' + \text{The KoF has a son}$

The result will be a context which entails that there is a king of France, and that this king has no son.

However, the CCP for negation allows for a second possibility. We can accommodate into the local context to which the presupposing clause is added, leaving untouched the global context which is to be intersected with the complement of that operation. In other words, we proceed as follows:

(11) $c + (8) = c \setminus [c + \text{there is a KoF}] + \text{The KoF has a son}$

³ $A \setminus B$ is the intersection of A with the complement of B .

Now the local context which is required to satisfy the presupposition does so, but is different from the context being updated with the overall content of (8). The result will be the set of worlds from *c* where either there is no king of France, or there is a king of France who has no son. This is a reading on which (8) may be true either because there is a king of France who has no son, or because there is no king of France to have a son in the first place. Note that here, as in all cases of local satisfaction, the apparent absence of a presupposition is a consequence of local satisfaction, not of cancellation or suspension of the presupposition of the definite.

The first strategy illustrated here is called *global accommodation*; the latter is called *local accommodation*. As Heim says, "[Global accommodation] is more like pretending that *c* & *p* obtained all along...; [local accommodation] is rather like adjusting the context only for the immediate purpose of evaluating the constituent sentence *S*." (p. 120). The more complex the CCP, the more options there are for local accommodation, and hence the more possible interpretations are generated (although sometimes distinct accommodation possibilities may have the same interpretational effect).

For a thorough discussion of accommodation in dynamic semantics, see Beaver and Zeevat (2006).

Justifying Accommodation

Heim's notion of accommodation builds on prior work by Stalnaker and Lewis. From his earliest work on presupposition, Stalnaker has noted that the (speaker) presuppositions required by an utterance need not in fact be common ground at the time an utterance is made. In many circumstances, it suffices that the addressee can recognize that the speaker is *acting as if* the common ground is a particular way. By allowing the addressee to recognize this, the speaker brings about a change in the common ground, as long as the addressee does not object to what is manifestly being presupposed. For Stalnaker, this process of context change does not involve pretense on the part of the speaker, or "context fixing" by the addressee, but is rather the normal kind of belief change that "must take place as events take place, and people become aware that they have" (Stalnaker, 2002, p. 711) and is motivated by general considerations of cooperativity.

It is from related work by David Lewis (Lewis, 1979) that we derive the notion of accommodation as context fixing, which Heim utilizes in her account. Lewis identifies a range of linguistic phenomena, including presupposition as understood in Stalnaker's sense, which impose requirements on the context of utterance, but notes that "it's not as easy as you might think to say something that will be unacceptable for lack of required presuppositions" or other contextual features. He proposes a (schematic) "rule of accommodation" for "fixing" the context so as to meet the requirements of utterances made. In contrast to Stalnaker, Lewis does not offer any broad pragmatic motivation for accommodation, and assumes that

accommodation is driven by potential violations of linguistic rules associated with particular words or constructions.⁴

Both Stalnaker and Lewis take themselves to be modeling, at some level, actual conversational practice. When Heim introduces her version of accommodation, she similarly invokes what happens “in practice” (p. 119). Her global accommodation strategy straightforwardly models the process of accommodation that Lewis envisions. But what about local accommodation? If this is also supposed to model something that people do, it seems to commit Heim to claiming that the update procedures encapsulated in CCPs have some degree of psychological reality; but, beyond the most basic cases, this seems rather implausible. An alternative interpretation is that the local accommodation case is *not* intended as a model of what people actually do, but simply provides a formal model of the variety of readings of complex sentences containing presupposition triggers. This leaves us with some open questions: for example, Heim’s use of local contexts allows her to treat presuppositions as inviolable constraints on contexts; but if local contexts – at least, the local contexts required for her theory – have no cognitive correlates, then in actual interpretation it may be that presupposition triggers have a probabilistic interpretation, sometimes understood as imposing contextual constraints and sometimes not. In other words, this interpretation means that we cannot look to the context change model for an understanding of how presuppositional triggers are interpreted in practice.

Alternative Approaches to Context

Context in Discourse Representation Theory

Discourse Representation Theory (DRT; Kamp, 1981; Kamp & Reyle, 1993; see Geurts et al., 2016 for an overview, and Geurts’s contribution to this volume) is, like Heim’s Context Change Semantics, a dynamic semantic theory. DRT differs from Heim’s theory in that it is a representationalist account, positing a level of semantic representation called Discourse Representation Structures. These structures are constructed from the linguistic input via Construction Rules, and are the input to interpretation rules. A DRS consists of two components: a universe (a set of discourse referents: roughly, representations of the entities under discussion in the discourse) and a set of conditions. The conditions contain information about the properties of, and relations between, discourse referents. Crucially, conditions can themselves contain (subordinate) DRSs, resulting in a hierarchical structure.

⁴ See Simons, 2003 for detailed discussion of Stalnaker’s notion of accommodation and how it differs from Lewis’s, and von Fintel, 2008 for an extended defense of accommodation in a broadly Stalnakerian framework.

Van der Sandt (1992) (for elaboration, see Geurts, 1999) develops an account of presupposition in DRT which is in many ways the DRT analogue of Heim's. In DRT, contexts are DRS's: the context for any clause is the DRS which has been constructed from the discourse so far. As in the Karttunen/Heim model, presuppositions are taken to impose constraints on the prior context. Specifically, presuppositions are taken to be *anaphors* which require an isomorphic antecedent in an accessible position within the DRS. This analysis is motivated by clear parallels between the availability of local satisfiers for presuppositions, and the availability of antecedents for anaphors, as illustrated for the case of conditionals in (12):

- (12) a. If France has a king, then Jane met the king of France.
 b. If Frank was at the party, then Jane met him.

Just as in the Karttunen/Heim model, presuppositions can be satisfied either locally or globally (or in an intermediate position). Specifically, a presupposition can find its antecedent either in the global DRS or in a subordinate DRS. If a presupposition finds an antecedent in a subordinate position, the presuppositional information is not entered into the main DRS, and so does not become part of the content of the DRS as a whole. In (12)a. just above, the presupposition of the consequent finds an antecedent in the DRS for the antecedent of the conditional, and therefore fails to project.

In Heim's account, the CCP of the conditional guarantees that when the antecedent satisfies the presupposition of the consequent, that presupposition does not project. In van der Sandt's account, there is no analogue of the CCP. In principle, a potential antecedent anywhere along the accessibility path of the presupposition (between the sub-DRS in which it is introduced and the global DRS) is a possible resolution site of the presupposition. However, van der Sandt proposes that there is a preference for binding of presuppositions as close as possible to where they are introduced. This preference predicts that the non-presupposing reading of sentence (12)a. is preferred. But what if the presupposition does not find an antecedent anywhere? Like Heim, van der Sandt posits that presuppositions can be accommodated. Accommodation simply involves adding the required DRS condition into an accessible location. As with binding, there may be multiple locations in which the presupposition could be accommodated, resulting in multiple possible interpretations; but there is a presumed preference to accommodate in the highest possible location within the DRS (typically, the global DRS). This assumption echoes Heim's observation that global accommodation is preferred over local accommodation.

While there are obvious parallels between Heim's account and van der Sandt's, the two theories do not make identical predictions for all cases. For discussion of differences, see van der Sandt (1992, p. 349ff.), Geurts (1999), and Beaver (2002).

Schlenker on Local Contexts

Schlenker (2009, 2010) rejects the claim that an account of presupposition requires a dynamic framework. He develops an alternative notion of local contexts within a static semantic theory, in order to utilize the notion of local satisfaction of presuppositions without the dynamic commitments. Like Karttunen/Heim, Schlenker assumes that (i) presuppositions are compositionally associated with sentence subparts and (ii) the presuppositions of any sentence constituent must be entailed by its local context. Hence, Schlenker derives projection facts in a way analagous to Heim. (For a closely related view, see George, 2008).

For Schlenker, the local context for the evaluation of any sentence *S* is the smallest (most restricted) set of worlds relevant for establishing the interpretation of *S*, that is, assigning to *S* a function from worlds to truth values. The function assigned, Schlenker suggests, should be sensitive in a particular way to the context in which *S* occurs: specifically, it should be indifferent to worlds which are already determined to be “not the way things are.” So, suppose that a speaker utters *S*, where the established context is *C* (a set of possible worlds). Whatever function from worlds to truth values we assign to *S*, it can be one which maps all non-*C* worlds to false, because we have already excluded non-*C* worlds as not true.⁵ This idea corresponds to the non-controversial claim that a sentence *S* uttered in context *C* is informationally equivalent to (*C* & *S*).

Now let’s consider a more substantive case: a conditional of form *If A, B*. Like Heim, Schlenker (2010) assumes at least for purposes of illustration a material implication semantics for conditional sentences. To calculate the function corresponding to the sentence as a whole, the interpreter must calculate the functions corresponding to each constituent sentence. Let’s assume that the interpreter has already interpreted the antecedent *A*, and now must interpret the consequent. The question is: which worlds must the interpreter treat as “undecided” parts of the domain of the corresponding function? As before, the interpreter can safely predetermine that the function should map all non-*C* worlds to false. In addition, she can predetermine that the function should map to true all worlds at which the antecedent of the conditional is false. That’s because the truth conditions of the conditional as a whole guarantee that the conditional is true at all such worlds, regardless of the truth value of the consequent at those worlds. But for all other worlds – all *C* worlds where *A* is true – the interpreter must compute a value based on the actual meaning of *B*. Hence, the local context for *B* (the consequent) consists of *C* & *A*, just as in the Karttunen/Heim account.

Schlenker (2009) demonstrates that his account largely makes the same predictions as Heim’s. (Schlenker accounts for the facts pertaining to quantified sentences by allowing the local context for a VP to be a set of properties.) The two central assumptions required for predicting local contexts are (i) the

⁵ A natural alternative would be to restrict the domain of the function to *C* worlds. Schlenker eschews this option in order to maintain a bivalent semantics.

truth conditional requirements of complex sentences and (ii) the assumption that sentences are interpreted incrementally, left-to-right. This latter assumption means that the computation of the local context for a constituent K in a complex sentence can take into account the information already provided by constituents earlier in the sentence, as illustrated in the case of the conditional. Schlenker also, though, considers a symmetric variant of the account, one which also allows material which follows a constituent to be informative with respect to the local context. This approach is motivated by the observation that in some cases, presuppositions can be filtered by later material, as illustrated by disjunctions like (13)a. and conditionals with postposed antecedents, like (13)b.

- (13) a. Either John has stopped smoking, or he never smoked.
 b. John has stopped smoking, if in fact he ever smoked.

Schlenker bills his account as pragmatic and static. It is static, in that it assumes straightforward static truth conditional contents for sentences. It is pragmatic in the sense that it makes no new claims about semantic content. It is perhaps also intended as a pragmatic account in the sense that local contexts are motivated by claims about what is easy for interpreters to do (Schlenker, 2010, p. 381); but as Schlenker's model cannot possibly be taken as a processing model of any type, it is unclear what might ground such claims.

We noted earlier (Section "Context change potentials") that the local contexts which play such a central role in the Karttunen/Heim account are motivated by the very observations which they are used to explain, opening these accounts to charges of being stipulative. Schlenker, while defining the notion of local context in a very different way, arrives (largely) at the same identification of local contexts; similarly, as we saw above, van der Sandt offers parallels with anaphora as further motivation for an analysis which has many commonalities with Karttunen/Heim. This convergence of results gives us good reason to believe that local contexts really are an important part of an explanatory semantic theory.

Projection Without Lexical Triggering

Contextual Presupposition

All of the literature on projection that we have discussed so far, beginning with Karttunen (1974), shares a basic assumption: that presuppositions are lexically triggered, hence are associated with any constituent that contains the trigger. In contrast, Stalnaker (1974) conjectures "that one can explain many presupposition constraints in terms of general conversational rules without building anything about presuppositions into the meanings of particular words or constructions" (212).

Are there cases of presuppositions clearly not associated with a lexical trigger? Kadmon (2001, pp. 213–217), and Simons (2004, 2013) have argued that there are,

and that these presuppositions (which Simons calls *contextual presuppositions*) can show standard projection behavior. For example, consider the following:

- (14) Ann: Shall we go on a picnic?
 Bud: It's raining.

Bud implicates an answer to Ann's question (no picnic); in order for Ann to recognize this implicature, she must assume that Bud assumes some relation between going on picnics, and rain – presumably, that rain excludes picnics. Bud's utterance signals that he takes this assumption to be part of the common ground, i.e. to be presupposed. Now, suppose that Bud had replied with any of the following:

- (15) a. It's not raining.
 b. It might be raining.
 c. Is it raining?

To establish any of these responses as relevant requires the same presupposition, that rain excludes picnics. So, in the context given, utterance of the atomic sentence *it's raining* generates a presupposition that it shares with utterances of the standard projection variants. Yet this presupposition clearly is not lexically triggered. It is generated by the need to make sense of the speaker's raising of the issue of rain in the context of the discussion of going on a picnic.

Kadmon is optimistic that the projection behavior of presuppositions of this type can be adequately handled within Heim's system. In particular, she suggests (p. 216) that the familiar relevance constraints could be taken to operate at the level of local contexts. Working out this suggestion in full detail would require coming to grips with the questions raised earlier about the status of Heimian local contexts; for conversational principles cannot be applied to purely formal constructs.⁶ More generally, the question is whether it is plausible to assume that contextual presuppositions are appropriately analyzed as properties of sentence subparts, or should instead be taken to be properties of the utterance. This remains an open question.

Projection and At-Issue-ness

Kadmon, while entertaining the possibility that at least some presuppositions have a pragmatic source, remains committed to the view that projection is governed by (dynamic) semantic mechanisms. Roberts et al. (2009) and Simons et al. (2010) argue instead that projection is a discourse based, pragmatic phenomenon, while agreeing that both conventional and pragmatic factors play a role in determining

⁶ In a similar vein, Abrusan (2011a, b) proposes a pragmatic account of verbal presuppositions, but assumes that her pragmatic principles will generate presuppositions that associate with atomic clauses, whose projection behavior can be accounted for by an independent projection mechanism.

what components of meaning have the capacity to project. In their approach to projection, these authors argue that accounts of projection should take seriously the observation that many kinds of contents can project. (See e.g. Potts, 2005 on projection of Conventional Implicatures.) Rather than developing an account of projection properly applicable only to (purported) presuppositional contents, and looking for alternative accounts of "special cases," Roberts and Simons (together with their collaborators David Beaver and Judith Tonhauser) argue that projection should be recognized as a broad linguistic phenomenon which cross-cuts the standard distinctions among content types. They argue further that there is one unifying property of projective content: roughly, the property of being non-main-point, or *not-at-issue*, content (Potts, 2005; Amaral et al., 2007; see Abbott, 2000 for a related view). In this work, at-issueness is characterized in terms drawn from Roberts' (2012) model of discourse: not-at-issue content is content which is not presented as intended to address the Question Under Discussion. Some content is signalled as not-at-issue by virtue of linguistic form; for example, the use of a non-restrictive relative clause may have this function. Some content is signalled as not-at-issue by virtue of intonation (see Beaver, 2010; Simons et al., 2017; Stevens et al., 2017; Tonhauser, 2016). And sometimes, content is recognized as not-at-issue through conversational inference (again, see Simons et al., 2017). Typically presuppositional contents project when not-at-issue; in special cases where those contents become at-issue, they fail to project, and instead fall under the scope of the relevant sentence embedder. For experimental investigation of the correlation between projection and at-issueness, see Xue and Onea (2011), Destruel et al. (2015), Tonhauser et al. (2018).

Beaver, Roberts, Simons and Tonhauser have not, as yet, provided a fully general and predictive account of projection and filtering. This is at least partly due to the very different approach of this group to the phenomenon. On the standard view, theories of projection are theories of properties of sentence types. Sentences are reasonably well behaved objects, and it is relatively easy to identify some well defined class of them to theorize about. On the BRST view, a theory of projection is a theory of properties of utterances and discourses. Utterances and discourses are difficult to individuate or to organize into clearly defined classes. Nonetheless, work on the ways in which context and discourse contribute to projection behavior is ongoing.

Final Remarks

Heim (1983a) is, of course, a paper about presupposition projection. But perhaps its major significance lies in its role in launching the dynamic turn in formal semantics. The fundamental ideas of context and context change that Heim presented in this brief paper are now part of the basic toolkit of semantics. At the same time, the paper established presupposition and presupposition projection as a topic of central concern for dynamic semantics.

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Chapter 10

On Jacobson's "Towards a Variable-Free Semantics"



Simon Charlow

Abstract Jacobson (*Linguist Philos* 22(2):117–184, 1999) proposes an account of anaphora that eschews variables and variable assignments, instead treating pronouns as identity functions and extending functional application with operations that pass up and close off anaphoric dependencies. I review the central aspects of Jacobson's variable-free semantics, counterpoising it with standard accounts of anaphora. I discuss conceptual and empirical virtues of Jacobson's theory, and some shortcomings. I sketch an alternative compositional implementation of variable-free semantics that draws on certain design features of the standard account, connect this approach to the computer science concept of 'applicative functors' (and thereby to frameworks as varied as alternative semantics and continuations), and clarify which of the variable-free theory's properties should be regarded as proprietary, and which can be easily repurposed into a theory with variables.

Keywords Variable-free semantics · Variables · Pronouns · Binding · Functional gaps · Paycheck pronouns · Applicative functors

Reviewing the Standard Account

Formal semantic theories characterize the kinds of meanings expressions can have and the procedure that compositionally assembles meanings for complex constituents from the meanings of their parts. A simple model in the Fregean vein is given below: (1) says that meanings can either be entities (type e), propositions (type t), or functions from meanings to meanings; (2) says that the meaning of a binary branching node is gotten by doing functional application on the meanings of its daughters (throughout this paper, all binary compositional rules should be

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assumed to have ‘backwards’ variants, in which the function is on the right, and the argument on the left; I’ll leave such rules implicit).¹

(1) $\tau ::= e \mid \tau \mid \tau \rightarrow \tau$

(2) If $\llbracket L \rrbracket = f : a \rightarrow b$, and $\llbracket R \rrbracket = x : a$, $\llbracket L R \rrbracket := \llbracket L \rrbracket \llbracket R \rrbracket : b$.

Figure 10.1 represents this grammar as a simple deductive system. Rule (2) is characterized by the inference scheme on the left. Reading bottom-up: given (an expression denoting) a function f of type $a \rightarrow b$ and (an expression denoting) an argument x of type a , the semantic value of the two together is fx , of type b .

Two notational points are in order. First, to encourage the reader to reason using types rather than values, I set the latter in *gray*. Second, the reason I present derivations bottom-up is that the resulting ‘upside-down’ proof trees echo the syntactic structures that undergird them. A sample derivation of *Amy saw Bob* in these terms is given on the right of Fig. 10.1.

$$\frac{f x : b}{f : a \rightarrow b \quad x : a} \qquad \frac{\frac{\text{gray} \text{ saw } b a : \tau}{a : e \quad \text{gray} \text{ saw } b : e \rightarrow \tau} \quad \text{gray} \text{ saw} : e \rightarrow e \rightarrow \tau \quad b : e}{\text{gray} \text{ saw } b a : \tau}$$

Fig. 10.1 Orienting a grammar around functional application, and a derivation of *Amy saw Bob*

This theory can be used to theorize about a large fragment of natural language. But there’s a lot it doesn’t handle, including pronouns. No single object is a good candidate for the meaning of a free pronoun, whose reference shifts with the context of utterance as in (3), or a bound pronoun, whose meaning is controlled by a higher operator as in (4).

(3) Amy saw **him**.

(4) Every philosopher_{*i*} thinks **they**_{*i*}’re a genius.

In the standard treatment of pronouns (e.g., Heim & Kratzer, 1998), meanings are uniformly determined relative to some way of valuing pronouns, generally an *assignment function*, with $\llbracket \text{pro}_n \rrbracket^g := g_n$ (that is, whatever the assignment g maps the index n to) and binary-branching nodes interpreted via assignment-relative functional application:

(5) $\llbracket \alpha \beta \rrbracket^g := \llbracket \alpha \rrbracket^g \llbracket \beta \rrbracket^g$

A rule like (5) requires semantic composition to be done with a specific assignment g in view. Less committally, we can see meanings *simpliciter* as functions from

¹ Here is a summary of the notational conventions used throughout this paper. First, types: ‘ $a ::= b$ ’ means that type a is being defined as b ; ‘ $a \rightarrow b$ ’ names the type of functions from type a to type b ; and ‘ $x : a$ ’ means that x has type a . Second, parentheses are dropped whenever possible, under the following conventions: types associate to the right, such that $a \rightarrow b \rightarrow c$ is equivalent to $a \rightarrow (b \rightarrow c)$; dually, application associates to the left, such that ‘ $fx y$ ’ is equivalent to ‘ $(f(x))(y)$ ’. Finally, ‘ $\lambda_x \Delta$ ’ names the function f such that for all type-appropriate d , $fd = \Delta^{x:=d}$ (that is, Δ with d substituted for x).

assignments to values (cf. Lewis, 1980). Taking this perspective in lieu of (5), the rule in Fig. 10.1 (left) is replaced with Fig. 10.2 (left), where ‘ τ^g ’ abbreviates the type $g \rightarrow \tau$ (and ‘ g ’ is the type of assignments).² This lets us derive a type- τ^g meaning for *Amy saw him*₀ in Fig. 10.2 (right). (Notice in particular how the types “under the g ” compose exactly as in Fig. 10.1.) To extract propositional content from the derived meaning $\lambda_g \text{ saw } g_0 a : \tau^g$, this meaning can be applied to a (contextually furnished) assignment supplying a value for the index 0.

$$\frac{\lambda_g f g (x g) : b^g}{f : (a \rightarrow b)^g \quad x : a^g} \qquad \frac{\lambda_g \text{ saw } g_0 a : \tau^g}{\lambda_g a : e^g \quad \lambda_g \text{ saw } g_0 : (e \rightarrow \tau)^g} \quad \frac{\lambda_g \text{ saw } : (e \rightarrow e \rightarrow \tau)^g \quad \lambda_g g_0 : e^g}{\lambda_g \text{ saw } g_0 a : \tau^g}$$

Fig. 10.2 Assignment-passing functional application, and a derivation of *Amy saw him*₀

We round out this setup with the rule for binding in Fig. 10.3 (left). When this rule applies, m is evaluated at a modified assignment mapping n to x , with x anchored to the functional abstract λ_x , over which $\mathcal{F}$ scopes. Figure 10.3 (right) shows how this rule allows pronouns to be bound, with a derivation of *everyone t₀ likes their₀ mom*.³

$$\frac{\lambda_g \mathcal{F} g (\lambda_x m g^{n=x}) : \tau^g}{\mathcal{F} : ((e \rightarrow \tau) \rightarrow \tau)^g \quad m : \tau^g} \triangleright_n \qquad \frac{\lambda_g \text{ eo } (\lambda_x \text{ likes } (\text{mom } x) x) : \tau^g}{\lambda_g \text{ eo } : ((e \rightarrow \tau) \rightarrow \tau)^g \quad \lambda_g \text{ likes } (\text{mom } g_0) g_0 : \tau^g} \triangleright_0$$

$\vdots \quad \vdots \quad \vdots$

Fig. 10.3 A rule for binding, and an abbreviated derivation of *everyone t₀ likes their₀ mom*

Variable-Free Semantics

Pronouns

Jacobson (1999) proposes a variable-free account of pronouns and binding that eschews indices and assignments, instead treating pronouns as identity functions on entities:⁴

$$(6) \llbracket \text{pro} \rrbracket := \lambda_x x : e^e$$

² This rule corresponds closely to von Stechow and Heim’s (2011) intensional interpretation function $\llbracket \cdot \rrbracket_{\mathcal{C}}$. Related approaches to assignments include (Montague, 1970; Rooth, 1985; Poesio, 1996; Sternefeld, 1998, 2001).

³ This structure might be generated by covert movement (i.e., Quantifier Raising) of *everyone* or, more likely, overt movement out of a vP-internal subject position (e.g., Heim & Kratzer, 1998: 218ff.).

⁴ Hepple (1990) was the first author to treat pronouns as identity functions. Other variable-free treatments of anaphora exist which I regrettably lack the space to discuss here. The reader is referred to works by Szabolcsi (1983), Szabolcsi (1989), Szabolcsi (1992), Szabolcsi (2003), Steedman (1987), Steedman (2000), Dowty (1992), Dowty (2007), Dekker (1994), van Eijck (2001), Shan (2001), Shan (2004), Jäger (2005), and de Groote (2006).

The definition repurposes the superscripting convention for types introduced in section “[Reviewing the Standard Account](#)”: e^e is inhabited by functions from individuals to individuals. Importantly, though, we will follow Jacobson in grammatically distinguishing $a \rightarrow b$ and b^a , even though these types are potentially inhabited by the same functions (in other words, types can be used to make finer-grained distinctions than set-theoretic meanings alone). More on this soon.

Composition

There is an immediate compositional challenge: although pronouns are type e^e , they typically occur in places where something of type e is expected. This is analogous to variable-full theories, where pronouns are type e^g . But whereas theories with variables deal with pronouns by ‘generalizing to the worst case’, making every lexical meaning assignment-relative and *replacing* functional application wholesale with the assignment-passing enrichment in Fig. 10.2, Jacobson retains conservative lexical entries for non-pronominals, *supplementing* functional application with additional compositional rules.

Jacobson’s central rules are **G** and **Z**, defined in Fig. 10.4. **G** lets a function $f : a \rightarrow b$ apply to an ‘incomplete’ value $m : a^c$ by function-composing f with m , passing up the anaphoric dependency. **Z** similarly performs a kind of application to an ‘incomplete’ value $m : a^c$, but it does so by identifying the missing c with the second argument of $f : a \rightarrow c \rightarrow b$, closing off the anaphoric dependency. Note: I’ve given **G** and **Z** as binary compositional rules since I believe this presentation is more intuitive for beginners, but readers should keep in mind that on Jacobson’s official proposal, **G** and **Z** are unary rules (‘type-shifters’) which apply directly to the function f (e.g., Z_{unary} maps $f : a \rightarrow c \rightarrow b$ to $Z_{unary} f : a^c \rightarrow c \rightarrow b$, which then may combine with $m : a^c$ by functional application). We will occasionally appeal to appropriately flagged unary versions of **G** and **Z** as we go.

$$\frac{\lambda_x f(m x) : b^c}{f : a \rightarrow b \quad m : a^c} \text{ G} \qquad \frac{\lambda_x f(m x) x : c \rightarrow b}{f : a \rightarrow c \rightarrow b \quad m : a^c} \text{ Z}$$

Fig. 10.4 The **G** and **Z** rules, which supplement functional application, Fig. 10.1 (left)

A derivation of *every boy_i thinks Amy saw him_i* illustrating the basic workings of Jacobson’s system appears in Fig. 10.5. Focus first on the embedded clause, *Amy saw him*. **G** allows **saw** to compose with the pronominal object, followed by the subject, resulting in a meaning of type t^e . (Since **G** requires the constituent hosting the pronoun to play the role of argument to some function, we invoke **Lift** on the embedded subject to turn it into such a function. See Partee (1986), Winter this volume.) Next, an application of **Z** is used to compose this clause with **think**, yielding a garden-variety predicate which identifies the values of the matrix subject and the embedded pronoun: $\lambda_x \text{thinks}(\text{saw } x \text{ } a) x : e \rightarrow t$. The quantified subject

merges via regular functional application, and we are done. The result represents the binding reading: for every boy x , x thinks Amy saw x .

$$\begin{array}{c}
 \text{eb } (\lambda_x \text{ thinks } (\text{saw } x \text{ } a) \text{ } x) : \mathfrak{t} \\
 \hline
 \text{eb} : (\mathfrak{e} \rightarrow \mathfrak{t}) \rightarrow \mathfrak{t} \quad \lambda_x \text{ thinks } (\text{saw } x \text{ } a) \text{ } x : \mathfrak{e} \rightarrow \mathfrak{t} \\
 \hline
 \text{thinks} : \mathfrak{t} \rightarrow \mathfrak{e} \rightarrow \mathfrak{t} \quad \lambda_x \text{ saw } x \text{ } a : \mathfrak{t}^{\mathfrak{e}} \quad \mathbf{Z} \\
 \hline
 \lambda_K \text{ } K \text{ } a : (\mathfrak{e} \rightarrow \mathfrak{t}) \rightarrow \mathfrak{t} \quad \lambda_x \text{ saw } x : (\mathfrak{e} \rightarrow \mathfrak{t})^{\mathfrak{e}} \quad \mathbf{G} \\
 \hline
 a : \mathfrak{e} \quad \text{saw} : \mathfrak{e} \rightarrow \mathfrak{e} \rightarrow \mathfrak{t} \quad \lambda_x x : \mathfrak{e}^{\mathfrak{e}} \quad \mathbf{G} \\
 \hline
 \text{Lift}
 \end{array}$$

Fig. 10.5 A typical variable-free derivation: *every boy_i thinks Amy saw him_i*

Of course, pronouns do not *need* to be bound. In such cases, the anaphoric dependency is simply passed up all the way: **G** replaces **Z** in Fig. 10.5, and another instance of **G** is used to fold in the matrix subject. The result, $\lambda_x \text{ eb}(\text{thinks}(\text{saw } x \text{ } a)) : \mathfrak{t}^{\mathfrak{e}}$, is a function from individuals x to the proposition that every boy thinks Amy saw x , and can be applied by the speaker/hearer to a contextually salient individual. Though readers may find it odd to think of a sentence with an unbound pronoun as denoting a property of individuals, Jacobson argues that the standard account is not really in better shape, since in that theory *all* sentences denote properties, of assignments (as in Fig. 10.2).

Now’s a good time to see why we can’t conflate $a \rightarrow b$ and b^a . If we did, the embedded VP’s meaning, $\lambda_x \text{ saw } x : (\mathfrak{e} \rightarrow \mathfrak{t})^{\mathfrak{e}}$, could apply directly to the embedded subject $a : \mathfrak{e}$, yielding $\text{saw } a : \mathfrak{e} \rightarrow \mathfrak{t}$ and disastrously scrambling subject and object. Distinguishing $a \rightarrow b$ and b^a prevents this undesirable outcome, guaranteeing that a superscripted type can only be passed up by **G** or discharged (bound) by **Z**.

Stepping Back

The central features of Jacobson’s theory are as follows. Pronouns lack indices, whence the ‘variable-free’ moniker. (Don’t be misled by the presence of variables and variable binding in our metalanguage. ‘Variable-free’ is a claim about the *object language*.) Given the absence of indices, there can be no co-indexing relationships between pronouns and anything else. Instead, the ‘co-construal’ relationship between a pronoun and its binder is established *combinatorially*: **G** passes up anaphoric dependencies, and **Z** closes them off just before the binder is merged. Likewise, it is impossible to formulate rules that refer to, let alone constrain, the distribution of indices—e.g., Conditions A, B, and C of the Binding Theory, prohibitions on Weak Crossover, etc. Insofar as such phenomena are grammatical, they will need to be captured some other way. (There is reason for optimism, though. I discuss Weak Crossover in sections “[Binding Between Non-coarguments](#)” and “[Binding](#)”.)

Jacobson’s account is also *flexible* in a way the variable-full account is not. Whereas the latter theory treats all expressions as assignment-sensitive (often trivially so) and therefore lacks any non-assignment-passing modes of composition, Jacobson supplements functional application with additional modes of composition. We will see in section “[New Foundations](#)” that theories with variables and assignments can avail themselves of a similar flexibility in lieu of the usual uniformity-based approach. This points the way to improvements in both variable-full and variable-free systems and lets us clearly see where their fundamental differences (and similarities) lie.

Some Empirical Results

This section fleshes out the bare-bones system of section “[Variable-Free Semantics](#)” with a basic account of relativization. It does this by extending our baseline variable-free theory with simple hypotheses about the meanings of gaps and relative pronouns. I use this slightly modified substrate to showcase some marquee empirical results of variable-free semantics.

Gaps, Pied-Piping, and i-within-i

We’ll begin by implementing a simple account of extraction in relative clauses. Some basic examples are given in (7)–(9), where ‘[]’ marks an extraction gap. Summarizing, (7) is a basic object relativization construction; (8) is subject relativization with binding of a pronoun; and (9) involves pied-piping of -’s *mom* by the relative pronoun *who*.

(7) The woman who John saw [] left.

(8) The woman who []_i married her_i childhood sweetheart left.

(9) The woman whose mom John saw [] left.

What’s the semantics of gaps and relative pronouns? We make the typical assumption that the semantics of gaps is *pronominal*, and (somewhat less typically) extend this treatment to relative pronouns such as *who* (Jacobson, 1998: 81). In the present setting, this means gap and relative pronoun denotations are identity functions, type e^e .⁵

⁵ Since pronouns can’t be replaced with gaps while preserving well-formedness (though gaps can be replaced with *resumptive* pronouns), gaps and pronouns should likely be distinguished at the type level. For example, we might take pronouns to be type- e^e identity functions and gaps to be type- e_e identity functions, generalizing the type of G/Z in order to interact appropriately with both. (The fact that pronouns can replace gaps, but not vice versa, may additionally suggest that e^e is a proper subtype of e_e , cf. Bernardi & Szabolcsi, 2008.)

This theory of gapped structures differs from Jacobson's in that it assumes (with, e.g., Shan & Barker, 2006) that gaps are null pronominal elements, while Jacobson does not actually countenance gaps! Instead, in the tradition of Combinatory Categorical Grammar, Jacobson uses syntactic function composition to assemble gapped clauses, without gaps (cf., e.g., Lambek, 1958; Ades & Steedman, 1982; Szabolcsi, 1989, 1992; Steedman, 1987, 2000). Why admit gaps? First, it streamlines the presentation. Second, and more importantly, I wish to highlight that the key ideas and techniques of Jacobson (1999) are independent of many of the syntactic assumptions therein.

With our meanings for gaps and relative pronouns we have essentially all we need to account for (7)–(9). Figure 10.6 shows this in one fell swoop with a derivation of *whose mom* [i saw herself] $_i$, a relative clause with binding and pied-piping. The relative pronoun, gap, and reflexive pronoun all denote identity functions. The application of **Z** guarantees that the subject and object of the relative clause are covalued, and the applications of **G** pass up pronominal dependencies, as usual.⁶ I additionally appeal to a **Front** rule which recognizes a complete relative clause and adjusts its type to that of a property, which allows the fronted phrase *whose mom* to be folded in (cf. Shan & Barker, 2006: 114).

$$\begin{array}{c}
 \frac{\lambda_x \text{ saw } (\text{mom } x) (\text{mom } x) : \mathbf{t}^e}{\lambda_x \text{ mom } x : \mathbf{e}^e} \text{ G} \quad \frac{\lambda_y \text{ saw } y' y' : \mathbf{e} \rightarrow \mathbf{t}}{\lambda_y \text{ saw } y' y' : \mathbf{t}^e} \text{ Front} \\
 \frac{\lambda_x x : \mathbf{e}^e \quad \text{mom} : \mathbf{e} \rightarrow \mathbf{e} \quad \lambda_y y' : \mathbf{e}^e \quad \lambda_y \text{ saw } y' y' : \mathbf{e} \rightarrow \mathbf{t}}{\text{saw} : \mathbf{e} \rightarrow \mathbf{e} \rightarrow \mathbf{t} \quad \lambda_y y' : \mathbf{e}^e} \text{ G} \quad \text{Z}
 \end{array}$$

Fig. 10.6 Deriving the pied-piped relative clause *whose mom* [i saw herself] $_i$

The most notable feature of this derivation is that the meanings of gaps and relative pronouns, together with the **G** rule, generate an immediate account of pied-piping (as pointed out in Jacobson, 1998). Thus, though the relative pronoun's movement brings along additional material, that material is in the end interpreted as if it were in situ (that is, in the gap). In other words, a form of semantic reconstruction in pied-piping is automatic, in contrast with many standard theories (see, e.g., Engdahl, 1986; Nishigauchi, 1990; Cresti, 1995; von Stechow, 1996; Sharvit, 1997; Heck, 2008; Cable, 2010; Dayal, 2016).

A final empirical point discussed by Jacobson (1994b, 2014) is so-called *i*-within-*i* effects in relational noun phrases. Examples (10) and (11) are ungrammatical, though the meaning that they are trying to express, that of (8), is coherent.

(10) *The wife $_i$ of her $_i$ childhood sweetheart left.

(11) *Her $_i$ childhood sweetheart's wife $_i$ left.

⁶ Nothing I've said explains why a reflexive's antecedent must be a higher argument of the same predicate (but see Szabolcsi, 1989, 1992 for a variable-free treatment of reflexives). I've used the reflexive here because Jacobson's theory does not actually generate a binding reading of *whose $_i$ mom* [i saw her $_i$]. See section "Binding Between Non-coarguments".

As Jacobson notes, Z provides a ready explanation of these facts. Z insists that its input be a two-place function $f : a \rightarrow c \rightarrow b$ (Fig. 10.4). But relational nouns like *wife*, though semantically two-place, are syntactically one-place. (Jacobson points out that relational nouns are impossible in small clauses such as **with Calista wife of Bill*.) Thus, relational nouns should in fact be of type $e \rightarrow n$ (with ‘ n ’ the type of nouns and noun phrases), rather than $e \rightarrow e \rightarrow t$. Z cannot apply, and so binding is correctly ruled out. Parallel explanations appealing to the intransitivity of relational nouns are available in standard theories, but it is remarkable that *i*-within-*i* facts fall out of the variable-free system.

Functional Gaps and Pronouns

Functional readings, exemplified by (12) and (13), suggest that *wh* and nominal quantification sometimes ranges over functions (e.g., Engdahl, 1980, 1986; Groenendijk & Stokhof, 1983; Sharvit, 1997, 1999; Winter, 2004). Standard analyses of these examples have two features. First, the gapped constituents (underlined) have functional meanings. The question in (12) requests an $f : e \rightarrow e$ such that every Englishman x loves fx ; and the subject of (13) denotes the (woman-valued) $f : e \rightarrow e$ such that every Englishman x loves fx . Second, the meaning of *his_i mom* is $\lambda_x \text{mom } x : e \rightarrow e$. In (12) this function is the answer to the question; in (13) it is equated with the meaning of the copular subject.

(12) Who does every Englishman_i love []? His_i mom.

(13) The woman every Englishman_i loves [] is his_i mom.

What explains functional gaps? Standard accounts following Engdahl (1980) posit that functional gaps are *complex variables*. In existing compositional implementations this requires stipulations and/or interpretive apparatus beyond what is strictly necessary to account for pronouns and binding in simple cases. The issue is that one expression, the gap, contributes a complex meaning fx to the functional gapped clause, even though in non-functional readings gaps contribute a simplex meaning, with a single variable. It is unclear how a single meaning for gaps can do all this work (but cf. Charlow, 2019).

A significant virtue of variable-free semantics, one emphasized by Jacobson (1999), is its ready compatibility with functional readings. To show this, I’ll sketch a variable-free treatment of the functional readings of (12) and (13)’s gapped clauses.

So far we have treated variable-free pronoun meanings as identity functions with type e^e . We now generalize this treatment a bit. After all, an identity function is naturally polymorphic: its type is $a \rightarrow a$, for any type a (cf. Barker, 2018; Pierce, 2002 is a useful reference on polymorphism). A generalized pronominal type, then, would be a^a , again for any type a . This is in fact a tad *too* general, since pronouns should eventually deliver a type- e value to the semantics (rather than something ridiculous like, say, a relation). In other words, only a restricted sub-space of identity

function types is suitable for pronouns. I therefore propose the following hierarchy of pronominal types:

- e^e is a pronominal type.
- If b^a is a pronominal type, $(b^e)^{(a^e)}$ is a pronominal type.
- Nothing else is a pronominal type.

It should be emphasized that the semantics of pronouns/gaps is unitary and unchanged: a pronoun's meaning is $\lambda_x x$. But x may be of type e , e^e , $(e^e)^e$, and so on.

The generalization of pronominal types is all that is required to derive functional meanings in gapped structures; **G** and **Z** do the rest. A schematic derivation of Q V $[\]$ (e.g., *every Englishman loves* $[\]$) is given in Fig. 10.7. The gap meaning is $\lambda_f f : (e^e)^{(e^e)}$, an identity function over type- e^e meanings, and the derivation is a mundane series of applications of **G** and **Z**. One notable feature, foreshadowed in section "Composition", is that **Z** is applied here as a *unary* rule (as per Jacobson's official proposal), turning V 's type from $e \rightarrow e \rightarrow t$ to the out-to-bind $e^e \rightarrow e \rightarrow t$, which then composes via **G** with the functional gap. The result is a property of functions f such that Q -many x 's stand in the V relation to $f x$; the complex gap meaning $f x$ has been conjured combinatorially.

$$\begin{array}{c}
 \frac{\lambda_f Q (\lambda_x V (f x) x) : t^{(e^e)}}{Q : (e \rightarrow t) \rightarrow t} \quad \frac{\lambda_f \lambda_x V (f x) x : (e \rightarrow t)^{(e^e)}}{\lambda_f \lambda_x V (f x) x : e^e \rightarrow e \rightarrow t} \quad \frac{\lambda_f f : (e^e)^{(e^e)}}{V : e \rightarrow e \rightarrow t} \quad \text{Z}_{unary} \\
 \hline
 \frac{\lambda_f Q (\lambda_x V (f x) x) : t^{(e^e)}}{Q : (e \rightarrow t) \rightarrow t} \quad \frac{\lambda_f \lambda_x V (f x) x : e^e \rightarrow e \rightarrow t \quad \lambda_f f : (e^e)^{(e^e)}}{\lambda_f \lambda_x V (f x) x : e^e \rightarrow e \rightarrow t} \quad \text{G} \\
 \hline
 \frac{\lambda_f Q (\lambda_x V (f x) x) : t^{(e^e)}}{Q : (e \rightarrow t) \rightarrow t} \quad \frac{\lambda_f \lambda_x V (f x) x : e^e \rightarrow e \rightarrow t \quad \lambda_f f : (e^e)^{(e^e)}}{\lambda_f \lambda_x V (f x) x : e^e \rightarrow e \rightarrow t} \quad \text{G}
 \end{array}$$

Fig. 10.7 A schematic analysis of functional readings

Finally, as noted, (12) and (13) suggest a symmetry between their functional gapped structures and *his_i mom*: the former denotes a property of functions (Fig. 10.7), and the latter denotes such a function. On standard theories, the second half of this symmetry requires us to bind the free variable *his_i*. While that's feasible, it's noteworthy that the variable-free account requires no such shift. If *'s mom* denotes an $e \rightarrow e$ function from people to their mothers, the meaning of *his mom* (i.e., *he's mom*) is $\lambda_x \text{mom} x : e^e$ (via **G**).

This variable-free analysis of functional gaps extends immediately to examples of functional pronouns, more commonly known as *paycheck pronouns* and exemplified in (14) (e.g., Karttunen, 1969; Cooper, 1979; Engdahl, 1986; Jacobson, 2000a). Here the second sentence means that every philosopher x spent $f x$, with f a function from x to x 's paycheck. As in (12) and (13), the function in question corresponds to other material in the example, here *their paycheck*, which the functional pronoun is anaphoric to.

(14) Every linguist_{*i*} deposited [their_{*i*} paycheck]_{*j*}. Every philosopher_{*k*} spent it_{*j*}.

Figure 10.7 doubles as an analysis of paycheck pronouns (Jacobson, 2000a). Set Q to the meaning of *every philosopher* and V to the meaning of *spent*. **G**, **Z**, and the

pronoun's general type yield a type- $\tau^{(e^e)}$ property of functions f such that every philosopher x spent $f x$. This property may be applied to a salient e^e function to yield a proposition. A natural candidate in (14) is $\lambda_x \text{paycheck } x : e^e$, the meaning of *their paycheck*!⁷

Pointers to Other Work

There is a great deal more work on variable-free semantics than I can hope to cover in this contribution. Jacobson herself develops and defends variable-free semantics in a substantial body of work (1992, 1994a, 1994b, 1996, 1998, 2000a, 2000b, 2004, 2014, 2016, *inter alia*). While the empirical focus of these contributions is pronominal binding and adjacent phenomena, Jacobson has also argued that the variable-free architecture offers (or is consistent with) competitive accounts of verb phrase ellipsis and antecedent-contained deletion (1992, 1998, 2004, 2009; see also Szabolcsi, 1992, 2013; Charlow, 2008). Subsequent to Jacobson's main developments, continuations-based theories of anaphora and binding (e.g., Shan & Barker, 2006; Barker & Shan, 2008, 2014; Barker, 2018) have been developed which extend variable-free semantics with general semantic mechanisms for scope-taking and fine-grained control of evaluation order. The reader is additionally referred to the contributions listed in footnote 4, some of which approach variable-free semantics from a very different set of foundations than Jacobson.

Some Conceptual Points

Direct Compositionality

Variable-free semantics is commonly associated with *Direct Compositionality* ('DC'). The essence of DC is that surface structures directly receive interpretations, without abstract mediating levels of representation such as LF—indeed, without any explanatorily important notion of logical form whatsoever. In DC architectures, the output of any operation in syntax is *immediately* paired with its interpretation; the syntax and the semantics operate in tandem (e.g., Barker & Jacobson, 2007). According to Jacobson, DC is, other things being equal, a simpler, more austere, and ultimately more explanatory view of the syntax-semantics interface than its alternatives.

⁷ Jacobson assumes pronouns are type e^e and derive further instances of our pronominal type by generalizing G to apply to functions of type b^a in addition to functions of type $a \rightarrow b$ (Jacobson, 2014: 329). While this is elegant, allowing pronouns to be polymorphic incurs little cost and streamlines the presentation.

That said, there is no *necessary* connection between variable-free semantics and DC: theories with variables may be DC, and theories without them can fail to be (Jacobson, 2014: xviii–xix). What does seem true is that variable-free semantics reduces the appeal of certain non-DC architectures. As noted in section “[Stepping Back](#)”, rules that regulate the distribution of variables in LF cannot be formulated if there are no variables to begin with. And the operation of Quantifier Raising, in some respects the central motivation for LF, is incompatible in its standard formulation with variable-free semantics (since variables are used to link the raised expression and its ‘trace’). Overall it is probably fair to say that variable-free semantics makes Direct Compositionality more appealing at the margins.

Issues with Variables

Compositionality aside, one might simply be skeptical that variables have an explanatory role to play; after all, natural language syntax certainly *looks* variable-free.⁸ With Harris (2019), we might add that it can seem absurd to think language users to bear referential intentions towards assignments. The commitments of the variable-free theory appear lighter. ‘Open’ propositions require mere referential intentions towards individuals—or towards sequences of individuals, in the general case (sections “[Generalized Rules](#)” and “[On Applicatives](#)”).⁹

Variables can also create unintended technical issues. I might wish for assignments that can value variables of arbitrary types (for cross-categorical movement, semantic reconstruction, and so on; see Heim & Kratzer, 1998: 213; Charlow, 2019, and citations therein). However, Muskens (1995: 179ff.) cautions that a type theory with assignments powerful enough to value variables of any type requires additional stipulations to avoid inconsistency. Again, the variable-free theory is on solid ground, since all it requires to value ‘variables’ of arbitrary types is the identity-functional type $a \rightarrow a$.

Finally, variables cause problems for processes like ellipsis that are licensed (as is standardly supposed) in virtue of syntactic or semantic identity. ‘Rebinding’ configurations like (15) are especially vexing (Evans, 1988; Rooth, 1992b; Takahashi & Fox, 2005). The sentence allows a ‘sloppy’ reading, i.e., with $\Delta = \text{likes him}_j$, and so Δ must be identical (either in form or meaning) with the antecedent VP *likes him_i*. But, then again, it *can’t* be: the overt *him_i* and the elided *him_j* are distinct

⁸ Recently, Kuhn (2016) has motivated a variable-tree treatment of spatial loci in American Sign Language (which had been argued to be overt manifestations of variables, cf. Schlenker, 2011).

⁹ However, sequences of individuals are isomorphic to partial assignments, and so it is appropriate to take this argument with a grain of salt. See section “[On Applicatives](#)” for discussion of a related point.

variables.¹⁰ (Additionally, since the sloppy pronoun him_j in (15) isn't bound by the subject, the antecedent and elided VPs cannot be treated as 'alphabetic variants', cf. Keenan, 1971; Sag, 1976.)

(15) John_i 's mom likes him_i . Bill_j 's mom does Δ too.

The solution proposed by Rooth (1992b) (and widely adopted in subsequent literature) is that the syntactic identity requirement on ellipsis licensing *allows variables to differ*. But this is, of course, suspicious: variables exist, and variables matter—until they don't.

Some Problems

Generalized Rules

The appealingly simple picture summarized in Fig. 10.4 under-generates. Constructions like (16) are impossible to analyze: the subject is type e^e , and the VP is readily assigned type $(e \rightarrow t)^e$ (i.e., via **G**). But that is as far as we can go: we have no ability to combine these two values to derive the expected final value $\lambda_y \lambda_x \text{saw } yx : (t^e)^e$.

(16) He_i saw her_j .

(17) Every professor _{i} gave their _{i} opinion to the dean.

Example (17) highlights an issue for **Z**. The formulation in Fig. 10.4 allows a pronoun inside the initial argument of some function $f : a \rightarrow c \rightarrow b$ to be bound by the *very next* argument of f (i.e., the c). In (17), however, the pronoun inside the direct object is bound by the matrix subject, skipping over the indirect object (assuming the verb combines first with the direct object).

Jacobson (1999: 138ff.) is aware of both issues and proposes generalizations of **G** and **Z** allowing these rules to 'skip over' arbitrary amounts of irrelevant material. I do not give the fully general versions of **G** and **Z** here, but Fig. 10.8 provides the relevant (unary) instances of the general schemas (additional instances simply allow for more and more intervening material in place of the type d).

$$\frac{\lambda_y \lambda_m \lambda_x f \gamma (m x) : (a^c \rightarrow b^c)^d}{f : (a \rightarrow b)^d} \text{G}_{\text{unary}} \qquad \frac{\lambda_m \lambda_y \lambda_x f (m x) \gamma x : a^c \rightarrow d \rightarrow c \rightarrow b}{f : a \rightarrow d \rightarrow c \rightarrow b} \text{Z}_{\text{unary}}$$

Fig. 10.8 Instances of generalized **G** and **Z** rules

¹⁰ Allowing the overt and elided pronouns to be 'accidentally' identical for the purpose of licensing the ellipsis creates enormous problems of its own, as discussed by Heim (1997: 217ff.).

In lieu of presenting detailed derivations of (16) and (17), I simply note that generalized G_{unary} allows *saw her* to shift from type $(e \rightarrow t)^e$ to type $(e^e \rightarrow t^e)^e$, making room for the pronominal subject in the inner e^e . Meanwhile, generalized Z_{unary} allows *gave* to shift from type $e \rightarrow e \rightarrow e \rightarrow t$ to type $e^e \rightarrow e \rightarrow e \rightarrow t$, with the superscripted and right-most e 's identified. So the generalized formulations of G and Z solve our issues.

There is a sense in which G need not actually be generalized, so long as we admit *nullary* occurrences of G/Z , i.e., we allow them to be instantiated as *inputs* to further applications of G/Z . As the formulation of Jacobson (1999: 138) suggests, 'generalized' G is just G applied to itself (and further instances can be derived by iteratively G -ing G).¹¹

The required generalization of Z is more costly, and gives rise to an infinite family of distinct Z operations (since in principle there is no upper bound on the number of arguments between a^c and c). More technically, the generalization of Z is a species of 'ad hoc' polymorphism (in which an operation behaves *differently* depending on the types of its inputs) and contrasts with 'parametric' polymorphism (in which an operation behaves *uniformly* on inputs of different types; e.g., $\lambda_x x : a \rightarrow a$ is a parametric-polymorphic identity function). Other things being equal, theories without ad hoc polymorphism may be preferable to, and potentially more explanatory than, theories requiring it.

Binding Between Non-coarguments

Even with the generalization of Z , other under-generation issues remain. Importantly, (generalized) Z only allows binding relationships to be established between two coarguments of the same predicate (with the binder the 'higher' or later argument of that predicate, and the bind-ee contained in the 'lower' or earlier argument). Effectively, this means that binding needs surface c-command, much as proposed by Reinhart (1983).

This is a two-edged sword. On the one hand, as Jacobson (1999: 135ff.) points out, it correctly rules out instances of Weak Crossover like (18). This is especially notable in view of the fact that traditional accounts of Weak Crossover (such as Reinhart's) are representational in nature. As mentioned back in section "Stepping Back", this is one reason to be optimistic that the variable-free program is consistent with grammatical explanations of phenomena that have previously received only representational accounts.

(18)*His_i mom likes every boy_i.

¹¹ That is, given that G is itself of type $(a \rightarrow b) \rightarrow a^c \rightarrow b^c$, GG has type $(a \rightarrow b)^d \rightarrow (a^c \rightarrow b^c)^d$. Thanks to Jeremy Kuhn for pointing this out.

A challenge for Jacobson's non-representational account of Weak Crossover and Reinhart's representational theory alike is that binding doesn't require surface c-command (Safir, 2004; Barker, 2005, 2012), and arguably doesn't require 'LF c-command' (i.e., scope) either (Kamp, 1981; Heim, 1982). Against surface c-command, we have possessor binding (19), inverse linking (20), and binding into adjuncts (21, 22). Against 'LF c-command', we have cross-sentential anaphora (23) and donkey anaphora (24).

- (19) Everyone_i's mom likes them_i.
- (20) Somebody from every city_i likes it_i.
- (21) We will sell no wine_i before its_i time.
- (22) Our critic eats in each restaurant she reviews_i before assigning it_i a score.
- (23) A linguist_i walked in the park. She_i whistled.
- (24) If there's [a train or a bus]_i leaving Dallas, I hope you're on it_i.

It would arguably be too much to demand an account of (23) and (24) from a 'static' theory of binding such as Jacobson's (see the Conclusion for more on this point). But the inconsistency of *Z* with cases such as (19), (20), and (21) suggests that the explanation of Weak Crossover and the sole reliance on *Z* to effect binding are untenable.¹²

On Becoming Variable-Free

Variable-free semantics is both conceptually and empirically attractive. However, it isn't embedded in a general theory of which kinds of supplements to functional application ('type-shifts') are available in grammar, or related to existing modes of theorizing about other kinds of phenomena. Importantly, general techniques for *integrating* variable-free theorizing with extant semantic architectures have not been specified, such that it is unclear what tradeoffs or new possibilities might be associated with going variable-free.

Jacobson herself has considered the interaction of variable-free semantics with 'alternative semantics', a framework proposed Hamblin (1973) for questions and extended to focus by Rooth (1985, 1992a, 1996). In Roothian alternative semantics, constituents are associated with sets of alternative meanings, their 'focus values'. Jacobson (2000b, 2004) argues that focus values of constituents with pronouns may

¹² A caveat on (21): Jacobson independently assumes a 'Wrap' analysis of direct objects, such that the complex transitive *sell _ before its time* is circumfixed around the direct object *no wine*. In effect, Wrap means that direct objects generally c-command more oblique post-verbal material (much as in the cascading VP shells of Larson, 1988; see Dowty, 2007 and Larson, 2014 for useful overviews of Wrap and its motivations, and pointers to relevant literature). If this approach is on the right track, (21) is not a counter-example to surface c-command, after all. That said, the general point is not in doubt: pronoun binding does not require surface c-command, as binding out of DP (19, 20) establishes clearly. Thanks to Pauline Jacobson for discussion of these points.

in some circumstances be sets of functions, type $a^e \rightarrow \tau$, and in others functions into sets, type $(a \rightarrow \tau)^e$. However, she does not give general mechanisms deriving these results.

New Foundations

The Essence of the Standard Theory

Our strategy for improving variable-free semantics will be to crib some tools from the standard account based on assignments, while retaining the flexibility and modularity of Jacobson's variable-free architecture (and, of course, her lexical semantics for pronouns!). Recall that the standard account is built on *uniformity*: lexical meanings are all assignment-relative, and meaning composition always passes around assignments.

Another approach is to abstract out and modularize the core features of the standard account. Instead of treating the lexical-semantic values of non-pronominals as uniformly, trivially dependent on an assignment, we invoke a function η which turns any x into a constant function from assignments into x (Fig. 10.9, left). Instead of making semantic composition uniformly relative to an assignment, we'll help ourselves to a function $\otimes$ which performs assignment-friendly function application on demand (Fig. 10.9, right).

$$\frac{\lambda_y. x : a^r}{x : a} \eta \qquad \frac{\lambda_y. m \ y \ (n \ y) : b^r}{m : (a \rightarrow b)^r \quad n : a^r} \otimes$$

Fig. 10.9 Abstracting out the two operations underlying environment-sensitivity

The definitions replace the superscripted g with r , which can stand in for any type (for example, e !). The reason to (parametrically) generalize η and $\otimes$ in this way is that their form has exactly nothing to do with assignment functions per se. Rather, these operations together capture the logic of *environment-sensitive* composition, a feature shared equally by variable-full and variable-free architectures: in theories with variables, the environment relative to which values are determined is an assignment; in variable-free theories, the environment relative to which values are determined is an individual.

Thus, the operations in Fig. 10.10 offer a completely general way to do environment-sensitive composition, with or without variables. Variable-full and variable-free derivations of *she left* using η and $\otimes$ are provided in Fig. 10.10. We use η to 'lift' environment-insensitive values, and $\otimes$ to combine environment-sensitive functions and arguments.

$$\begin{array}{c}
\frac{\lambda_g \text{left } g_0 : \mathbf{t}^g}{\lambda_g g_0 : \mathbf{e}^g} \quad \frac{\lambda_g \text{left} : (\mathbf{e} \rightarrow \mathbf{t})^g}{\text{left} : \mathbf{e} \rightarrow \mathbf{t}} \eta \quad \circledast \quad \frac{\lambda_x \text{left } x : \mathbf{t}^e}{\lambda_x x : \mathbf{e}^e} \quad \frac{\lambda_x \text{left} : (\mathbf{e} \rightarrow \mathbf{t})^e}{\text{left} : \mathbf{e} \rightarrow \mathbf{t}} \eta \quad \circledast
\end{array}$$

Fig. 10.10 Variable-full and variable-free composition of *she left* using η and $\circledast$

These derivations highlight that η and $\circledast$ are a *decomposition* of $\mathbf{G}$: instead of directly using $\mathbf{G}$ to compose $\text{left} : \mathbf{e} \rightarrow \mathbf{t}$ and $\lambda_x x : \mathbf{e}^e$, we first use η and then $\circledast$. Additionally, we observe that the lexical and compositional uniformity of the standard account turn out to be negotiable: a semantics oriented around variables can be compositionally grounded using precisely the same flexibly applying operations as the variable-free system.¹³

Binding

The re-engineered variable-free semantics needs just one more piece: a way to effect binding. Whereas Jacobson’s appeal to $\mathbf{Z}$ under-generates by linking binding too closely to the semantic structure of the verb (and thereby to coargument-hood and surface c-command), the standard account does better. For example, the rule in Fig. 10.3 easily generates possessor binding for LFs like *everyone* t_0 ’s *mom* likes *them*₀.

In effect, the standard account improves on $\mathbf{Z}$ here because it links binding to scope rather than coargument-hood (as appropriate for a static semantics). This feature can be imported into the variable-free setting. The rule in Fig. 10.11 gives a scopal expression $\mathcal{F} : (\mathbf{e} \rightarrow \mathbf{t}) \rightarrow \mathbf{t}$ an anaphoric charge, turning it into something which expects to scope over and plug two $\mathbf{e}$ -sized holes, one of them pronominal.¹⁴

$$\frac{\lambda_\kappa \mathcal{F} (\lambda_x \kappa x x) : (\mathbf{e} \rightarrow \mathbf{t}^e) \rightarrow \mathbf{t}}{\mathcal{F} : (\mathbf{e} \rightarrow \mathbf{t}) \rightarrow \mathbf{t}} \triangleright$$

Fig. 10.11 A rule for binding that gives the binder an anaphoric charge

Variable-free treatments of scope (e.g., Hendriks, 1993; Shan & Barker, 2006; Szabolcsi, 2011) go well beyond what I can cover here (but see Appendix). Instead I take a shortcut, defining a descriptive ‘rule’ for scope in Fig. 10.12, in which

¹³ This flexibility allows us to readily generate functional readings as well. See Charlow, 2019 for discussion.

¹⁴ The rule is identical to Shan and Barker’s (2006) rule for binding, and like it draws on Szabolcsi’s (1989, 1992) use of the duplicator combinator $\mathbf{W} := \lambda_\kappa \lambda_x \kappa x x$. Unlike Jacobson’s variable-free semantics, Shan and Barker adopt a scopal treatment of *pronouns* as well, which requires a pronoun to take scope immediately under its binder (see also Dowty, 2007). This may be problematic since binding is not subject to locality restrictions.

operations instantiate a general, widely used abstraction in computer science, with properties particularly well suited to the compositional behavior of pronouns and their interactions with other facets of meaning.

Of course, we should be sure we haven't lost anything in the bargain. We haven't. For one, η and $\otimes$ together *entail* $\mathbf{G}$, and $\triangleright$ generates strictly more binding configurations than $\mathbf{Z}$ (even so, *i*-within-*i* is ruled out, for basically the same reason as before: relational nouns lack subjects, and so there is nothing for $\triangleright$ to target). Examples that may raise concerns are pied-piping with binding (cf. Fig. 10.6) and functional readings (cf. Fig. 10.7), since the toy 'rule' for scope does not seem to allow gaps and $\triangleright$ to interact in the right ways. In reality there are no problems here. See Appendix for a brief discussion.

On Applicatives

Abstracting out η and $\otimes$ puts us in the presence of something known to computer scientists and functional programmers as an **APPLICATIVE (FUNCTOR)** (McBride & Paterson, 2008). The essence of an applicative functor is an enriched type-space with η and $\otimes$ operations that support some correspondingly enriched notion of functional application.

More formally, an applicative functor is a type constructor F associated with mappings $\eta : a \rightarrow Fa$ and $\otimes : F(a \rightarrow b) \rightarrow Fa \rightarrow Fb$.¹⁵ η and $\otimes$ must satisfy several laws which guarantee that $\otimes$ embodies an enriched notion of functional application, and that η does nothing more than trivially inject values into the enriched type-space characterized by F . We pass over these here, but see (McBride & Paterson, 2008; Charlow, 2019).

Applicative functors can be factored out of a great deal of existing semantic theory. I'll mention just two examples. Alternative semantics can be built from an applicative functor for sets, such that $Sa ::= a \rightarrow \mathbf{t}$, and with η and $\otimes$ as defined as follows:

$$\begin{aligned} (25) \quad \eta x &::= \{x\} & \eta : a \rightarrow Sa \\ (26) \quad m \otimes n &::= \{f x \mid f \in m, x \in n\} & \otimes : S(a \rightarrow b) \rightarrow Sa \rightarrow Sb \end{aligned}$$

Likewise, continuations-based theories of scope (Shan & Barker, 2006; Barker & Shan, 2014) are built on two combinators (**Lift** and **Scope**) that directly instantiate the applicative functor for continuations, such that $C_r a ::= (a \rightarrow r) \rightarrow r$, and with η and $\otimes$ as follows:¹⁶

$$\begin{aligned} (26) \quad \eta x &::= \lambda_\kappa \kappa x & \eta : a \rightarrow C_r a \\ (27) \quad m \otimes n &::= \lambda_\kappa m (\lambda_f n (\lambda_x \kappa (f x))) & \otimes : C_r(a \rightarrow b) \rightarrow C_r a \rightarrow C_r b \end{aligned}$$

¹⁵ Type constructors are mappings from types to types. You can think of them as ways of abbreviating complex types. For example, our superscripted-e notation is a type constructor: if a is a type, then so is a^e .

¹⁶ In fact, linguistic uses of continuations rely on *indexed* or *parametrized* applicatives, a somewhat more general notion. See Wadler, 1994 and Kobele, 2018, as well as Appendix, for discussions of related constructs.

Applicative functors enjoy an important property: they’re closed under composition (e.g., McBride & Paterson, 2008; Kiselyov, 2015): if F and G are applicative, then $F \circ G$ is too. The recipe for assembling a composite $\otimes$ operation for $F \circ G$ is given in Fig. 10.14 (to assemble a composite η , just apply G and F ’s η ’s in succession).¹⁷

$$\frac{\frac{\frac{F(G(a \rightarrow Gb))}{F(G(a \rightarrow Gb))} \quad \frac{F(G(a \rightarrow Gb))}{F(G(a \rightarrow Gb))} \quad \otimes_F \quad F(Ga)}{F(G(a \rightarrow Gb)) \rightarrow Ga \rightarrow Gb} \quad \eta_F \quad F(G(a \rightarrow Gb))}{\otimes_G : G(a \rightarrow b) \rightarrow Ga \rightarrow Gb} \quad \otimes_F$$

Fig. 10.14 Composing two applicative functors F and G

For example, using ‘ E ’ to stand in for the variable-free semanticist’s type constructor (i.e., $Ea ::= a^e$), $E \circ S$ yields environment-sensitive alternative sets, with $\eta x = \lambda_y \{x\}$ and $m \otimes n = \lambda_y \{f x \mid f \in m y, x \in n y\}$ (Kratzer & Shimoyama, 2002; Charlow, 2014, 2020). Conversely, $S \circ E$ gives alternative environment-sensitive meanings, with $\eta x = \{\lambda_y x\}$ and $m \otimes n = \{\lambda_y f y (x y) \mid f \in m, x \in n\}$ (Poesio, 1996; Romero & Novel, 2013). And composing E with *itself*—taking $E \circ E$ —yields an applicative for double-sensitivity $\eta x = \lambda_y \lambda_z x$ and $m \otimes n = \lambda_y \lambda_z m y z (n y z)$.

The ‘compositionality’ of applicative functors is significant in part because it guarantees that applicative pieces of grammar can be theorized about separately and modularly. Any set of potentially disparate analyses relying on applicatives automatically generates a ‘composite’ analysis in terms of composed applicatives. For example, the existence of applicative E and S within a grammar immediately generates the compositional resources needed to handle both environment-sensitive alternative sets ($E \circ S$), and sets of alternative environment-sensitive meanings ($S \circ E$) (cf. section “On Becoming Variable-Free”).

Moreover, composing the variable-free applicative functor with *itself* ($E \circ E$) provides an easy solution to the problem of multiple pronouns (cf. section “Generalized Rules”). This composite applicative functor embodies sensitivity to two

$$\frac{\frac{\lambda_x \lambda_y \text{ saw } y' x : (t^e)^e}{\lambda_x x : e^e} \quad \otimes_\eta \quad \frac{\lambda_x \lambda_y \text{ saw } y' : ((e \rightarrow t)^e)^e}{\lambda_y \text{ saw } y' : (e \rightarrow t)^e} \quad \otimes_\eta}{\frac{\lambda_y \text{ saw } : (e \rightarrow e \rightarrow t)^e}{\text{saw} : e \rightarrow e \rightarrow t} \quad \eta \quad \lambda_{y'} y' : e^e} \quad \otimes$$

Fig. 10.15 Deriving *he saw her* by composing the variable-free applicative functor with itself

¹⁷ It may appear from Fig. 10.14 that we require a *nullary* instance of $\otimes_G$ (i.e., we allow our rules to operate on *themselves*). See Barker and Shan (2014: 118) and White et al. (2017) for alternative ways to get the same result.

type- e environments. A derivation of *he saw her* is provided in Fig. 10.15. Here, I use ‘ $\circledast$ ’ for the composite $\circledast$ operation, and ‘ $\circledast\eta$ ’ as shorthand for applying η ‘under the e ’ (i.e., in precisely the way **G** had allowed).¹⁸

Consider now the familiar Curry-Uncurry isomorphisms in Fig. 10.16: Curry converts a function from tuples into a higher-order function, and Uncurry turns a higher-order function back into a function on tuples. Uncurrying the two-place function derived by the variable-free theory for *he saw her* in Fig. 10.15 gives $\lambda_{(x,y)} \text{ saw } yx = \lambda_p \text{ saw } p_2 p_1$. Something very much like dependence on an assignment (dependence on a sequence) springs organically into being, albeit with a complete absence of object-language variables. In other words, not only can variable-free and variable-full approaches be treated with an identical set of applicative combinatory tools, and cognate rules for binding—the semantic values thereby generated turn out to be equivalent up to isomorphism.

$$\frac{\lambda_x \lambda_y. f(x, y) : (c^b)^a}{f : c^{(a \times b)}} \text{ Curry} \qquad \frac{\lambda_{(x,y)}. f x y : c^{(a \times b)}}{f : (c^b)^a} \text{ Uncurry}$$

Fig. 10.16 The Curry-Uncurry isomorphisms

Again, this does not imply any sort of ultimate equivalence between variable-free and -full theories. Having object-language variables allows the variable-full theorist to impose representational constraints that the variable-free theorist cannot even contemplate.

Conclusion

Our time’s up. I hope you’ve gotten a glimpse of the elegance and viability of variable-free theorizing, along with a sense of which of its differences from standard theories are negotiable, and which aren’t. For example, both sorts of theories can readily exploit flexible compositional architectures (though variable-full theories typically do not), and both sorts of theories can have a rule for binding oriented around scope (though Jacobson does not). The status of object-language variables, by contrast, is definitional and, therefore, non-negotiable. Nevertheless, the two kinds of theories can be stated in ways that make them seem much more alike than we might have initially supposed. This suggests in turn that one may not need to be a variable-free semanticist in order to reap the empirical insights of variable-free semantics (see Charlow, 2019).

Where do we go from here? One area ripe for investigation is variable-free *dynamic* semantics, in order to explain data like (23) and (24). While there are

¹⁸ Intriguingly, the continuations-based analysis of inverse scope (e.g., Shan & Barker, 2006) involves composing C_r with itself, in a way that is remarkably parallel to the strategy for dealing with multiple pronouns.

existing ‘variable-free’ (or nearly so) dynamic systems (e.g., Dekker, 1994; van Eijck, 2001; Bittner, 2001; de Groote, 2006; Murray, 2014; Charlow, 2014), these theories are generally quite different in their aims and form from Jacobson’s static variable-free system, and cannot be properly regarded as dynamic extensions of it. Preliminary work has been initiated by Shan (2001) and Szabolcsi (2003), with Shan’s decomposition of canonical dynamic systems into ‘adjoint’ functors plus nondeterminism especially close to the outlook of section “[New Foundations](#)”.

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Appendix: Scope Without Variables

This appendix presents a pared-down version of Shan and Barker’s (2006) variable-free account of scope based on the continuations (indexed) applicative functor, shows how composing this applicative functor with the variable-free applicative of section “[New Foundations](#)” allows scope to feed binding, and gives hints about how to derive pied-piping with binding (cf. Fig. 10.6) and functional readings (cf. Fig. 10.7).

Continuations-based theories of scope are built on two combinators (Lift and Scope) that instantiate the *indexed* applicative functor for continuations, with $\mathbb{C}_i^o a ::= (a \rightarrow i) \rightarrow o$, and with η and $\otimes$ as follows (note that the λ -terms are unchanged from (26) and (27)):

$$\begin{aligned} (29) \quad \eta x &:= \lambda_{\kappa} \kappa x & \eta : a \rightarrow \mathbb{C}_o^o a \\ (30) \quad m \otimes n &:= \lambda_{\kappa} m (\lambda_f n (\lambda_x \kappa (f x))) & \otimes : \mathbb{C}_i^o (a \rightarrow b) \rightarrow \mathbb{C}_j^i a \rightarrow \mathbb{C}_j^o b \end{aligned}$$

Indexed applicatives have more general types than garden-variety applicatives, but they still obey the four applicative laws and compose freely with themselves and with other applicatives, following the general recipe in Fig. 10.14. For example, the indexed continuations applicative may be composed with the variable-free applicative:

$$\begin{aligned} (31) \quad \eta x &:= \lambda_{\kappa} \kappa (\lambda_y x) & \eta : a \rightarrow \mathbb{C}_o^o a^e \\ (32) \quad m \otimes n &:= \lambda_{\kappa} m (\lambda_f n (\lambda_x \kappa (\lambda_y f y (x y)))) & \otimes : \mathbb{C}_i^o (a \rightarrow b)^e \rightarrow \mathbb{C}_j^i a^e \rightarrow \mathbb{C}_j^o b^e \end{aligned}$$

Our two applicatives allow us to give a binder scope over its bind-ee, as in Fig. 10.17. Similar to Fig. 10.15, ‘ $\otimes \eta$ ’ means we use the continuations $\otimes$ to apply the variable-free η under the $\mathbb{C}$ (and ‘ $\otimes \otimes$ ’ is the composite $\otimes$). As in other analyses based on continuations, a **Lower** rule ends a derivation by applying the resulting

$$\begin{array}{c}
\frac{\text{eo}(\lambda_x(\lambda_y \text{ likes } y(\text{mom } x))x) : \mathbf{t}}{\lambda_x \text{ eo}(\lambda_x \kappa(\lambda_y \text{ likes } y(\text{mom } x))x) : \mathbf{C}_t^e \mathbf{t}^e} \text{Lower} \\
\frac{\lambda_x \text{ eo}(\lambda_x \kappa(\lambda_y \text{ mom } x)x) : \mathbf{C}_t^e \mathbf{e}^e}{\lambda_x \text{ eo}(\lambda_x \kappa(\text{mom } x)x) : \mathbf{C}_t^e \mathbf{e}} \odot \eta \quad \frac{\lambda_x \kappa(\lambda_y \text{ likes } y) : \mathbf{C}_t^e (\mathbf{e} \rightarrow \mathbf{t})^e}{\lambda_y \text{ likes } y : (\mathbf{e} \rightarrow \mathbf{t})^e} \eta \\
\frac{\lambda_x \text{ eo}(\lambda_x \kappa x x) : \mathbf{C}_t^e \mathbf{e}}{\text{eo} : \mathbf{C}_t^e \mathbf{e}} \triangleright \quad \frac{\lambda_x \kappa \text{ mom} : \mathbf{C}_t^e (\mathbf{e} \rightarrow \mathbf{e})}{\text{mom} : \mathbf{e} \rightarrow \mathbf{e}} \odot \eta \quad \frac{\lambda_y \text{ likes} : (\mathbf{e} \rightarrow \mathbf{e} \rightarrow \mathbf{t})^e}{\text{likes} : \mathbf{e} \rightarrow \mathbf{e} \rightarrow \mathbf{t}} \eta \quad \frac{\lambda_y y : \mathbf{e}^e}{\lambda_y y : \mathbf{e}^e} \odot
\end{array}$$

Fig. 10.17 Deriving *everyone_i's mom likes them_i* using two applicatives

term to a trivial continuation, here an identity function of type $\mathbf{t}^e \rightarrow \mathbf{t}^e$. The result after **Lower** is equivalent via a β -reduction to $\text{eo}(\lambda_x \text{ likes } x(\text{mom } x))$, as desired.¹⁹

Using these applicatives, we can derive pied-piping with binding (cf. Fig. 10.6) and functional readings (cf. Fig. 10.7). Details are omitted, but breadcrumbs are provided. There are two kinds of derivations of pied-piping with binding; one identifies the gap and pronoun directly via the variable-free $\otimes$, and another is closer in form to Fig. 10.17, but juggles the gap's anaphoric dependency as well.²⁰ Derivations of functional readings will likewise be similar in form to Fig. 10.17, but additionally require us to project up the anaphoric dependency on an $\mathbf{e}^e$ function using (what else?) another applicative layer.

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¹⁹ The form of **Lower** we rely on is actually inconsistent with Shan and Barker's (2006) account of Weak Crossover! If we wish to help ourselves to their account, we can follow them in only allowing **Lower** to apply to types of the form $\mathbf{C}_t^o \mathbf{t}$. This will in turn necessitate an auxiliary mapping $\lambda_m \lambda_\kappa \lambda_x m x \kappa : (\mathbf{C}_t^o a)^e \rightarrow \mathbf{C}_t^o a$. Applied to constituents containing an unbound pronoun, this mapping allows binder and bind-ee to meet via C's $\otimes$ operation, rather than via **Lower**. See Shan and Barker (2006) for in-depth discussion of related matters.

²⁰ The latter of these is probably to be preferred since, as per the considerations in fn. 5, gaps and pronouns should be analyzed as having different types (and so cannot be directly identified by $\otimes$).

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Chapter 11

On Kamp's "A Theory of Truth and Semantic Representation"



Bart Geurts

Abstract First published in 1981, Hans Kamp's paper, "A theory of truth and semantic representation", introduced Discourse Representation Theory, or DRT for short, which soon became one of the main frameworks in semantics and pragmatics, and in the meantime has been applied to an exceptionally wide range of phenomena, including anaphora, definites and proper names, quantification, tense and aspect, presupposition, implicatures, and attitude reports. DRT's key innovations were two. First, it introduced a level of representation which was new to semantics. Kamp defined a formal language, with a model-theoretic interpretation, which served to represent the contents not only of individual sentences, but of entire discourses as well. Secondly, this formal language featured a new species of variables, called *discourse referents*, which were the keystone of DRT's unified treatment of anaphoric pronouns. Discourse representations and discourse referents are the main topics of this commentary.

Keywords Discourse representation theory · Discourse referents · Anaphora · Quantification · Conditionals · Compositionality

Introduction

Hans Kamp's paper, "A theory of truth and semantic representation" (TSR), was presented at the third Amsterdam Colloquium in 1980, printed in the conference proceedings (Kamp, 1981), reprinted as Kamp (1984) and several times since (in the following, all page references are to the 1984 edition). Only after the paper had been published did Kamp's theory come to be known as "Discourse Representation Theory", or DRT for short.

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Soon after its inception, DRT became one of the main frameworks in semantics and pragmatics, and in the meantime has been applied to an exceptionally wide range of phenomena, including anaphora, definites and proper names, quantification, tense and aspect, presupposition, implicatures, and attitude reports. Significant book-length extensions of DRT may be found in Asher and Lascarides (2003), Kamp and Reyle (1993), and Kamp et al. (2011). Geurts et al. (2017) provide a synoptic survey.

As TSR's title indicates and Kamp discusses in his introduction, DRT brings together two ways of looking at meaning. On the one hand, the theory stands in a long tradition that views meaning as that which determines the truth conditions of a declarative sentence. On the other hand, there is an even longer tradition which views meaning as "that which a language user grasps when he understands the words he hears or reads." (p. 1) DRT brings these two lines of thought together, and this is one of the reasons why the theory has been so influential. Another reason is that DRT was the first theory to offer a unified account of anaphora within and between sentences. For example, although (1) and (2) have quite different types of meanings, and it used to be thought that the neuter pronouns they contain must function differently, Kamp's paper showed for the first time how they can be interpreted uniformly:

- (1) Pedro owns a donkey. He beats it.
- (2) Every farmer who owns a donkey beats it.

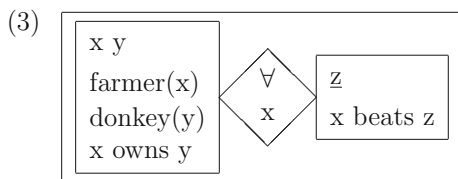
Technically, DRT's key innovations were two. First, it introduced a level of representation which was new to semantics (if that is the right term to use; see below). Kamp defined a formal language, with a model-theoretic interpretation, which served to represent the contents not only of individual sentences, but of entire discourses as well. Secondly, this formal language featured a new species of variables, called "discourse referents", which were the keystone of DRT's unified treatment of anaphoric pronouns. Discourse representations and discourse referents will be the main topics in the following.

Preliminaries

The first formulation of a theory is rarely the best, and TSR is no exception. Following the publication of that paper, DRT quickly caught on, and as the theory was being reformulated and expanded, its notation became more perspicuous, and while most of the key ideas were retained, some substantial revisions were made too. I will explain the most important changes with the help of example (2).

In the original version of the theory, syntactically analysed sentences were mapped onto discourse representation structures (DRSs) in one go. It soon became clear that it was simpler and more illuminating to divide this mapping into two stages, with the first stage consisting of something like a compositional translation, which left unresolved various types of indeterminacies, including anaphora. These

indeterminacies were marked in the output of the first stage (here I use underlining for that purpose), to be resolved in the second stage:

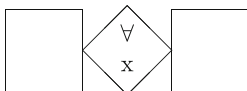


Apart from the fact that this DRS is as yet incomplete, because it is not settled yet whether z is to be equated with x , y , or neither, it differs from its TSR-counterpart in two main respects. First, the notation is cleaner, chiefly because it renders explicit structure that was merely implicit in the original notation, and it comes with a linear variant, too:

(4) $[\vdash [x, y: \text{farmer}(x), \text{donkey}(y), x\ \text{owns}\ y] (\forall x) [\underline{z}: x\ \text{beats}\ z]]$

Secondly, while TSR treated conditionals and universal quantifiers on a par (which made some sense, because (3) seems equivalent to “If a farmer owns a donkey, he beats it”), it was soon observed that this will not generalise to other quantifiers, notably “most”, and therefore a more orthodox analysis of quantification was adopted instead (Kamp & Reyle, 1993).

There are two ways of spelling out the interpretation of so-called “duplex conditions” of the form:



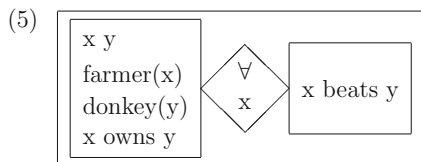
According to one way, an embedding function f verifies the duplex condition in (3) iff the following holds for every individual i :

If any g extends f in such a way that $g(x) = i$ and $g(x)$ is a farmer who owns a donkey, $g(y)$, then $g(x)$ beats $g(y)$.

Thus construed, (3) is true iff every donkey-owning farmer beats all his donkeys. The second interpretation is weaker: it says that every donkey-owning farmer beats at least one of his donkeys. While speakers' intuitions are not always clear, it does seem to be the case that both readings occur, and that in some sense donkey sentences are ambiguous between a strong and a weak reading. However, the nature of this ambiguity has been debated for some time now, and with increasing sophistication; see Champollion et al. (2018) for a recent contribution.

Whether we adopt the strong reading or the weak one, duplex conditions are dynamic generalised quantifiers, which is to say that the restrictor DRS on the left

is accessible to the scopal DRS on the right, so that the discourse referent y in (3) is accessible to z . Therefore, the latter may be equated to the former, which after a minor simplification results in:



On the strong reading of “every”, the truth conditions associated with (5) are the same as in TSR, but this style of analysis is cleaner and more general.

Of the changes discussed here, the shift to a two-stage architecture was probably the most significant. Since it seems rather natural to label these stages “semantics” and “pragmatics”, respectively, this architecture could have played a useful role in the long-standing debate over the semantics/pragmatics distinction (see, e.g., Szabó ed. 2005). Regrettably, it has failed to do so thus far.

Discourse Representations and Compositionality

To a large extent DRT owes its explanatory power to the fact that it adopts a form of representation that other theories lack. It is widely agreed in the humanities and the social sciences that we mentally represent our social interactions in general and our communicative exchanges in particular. Against this background, Kamp’s discourse representations may not seem very innovative. However, in formal semantics this idea met with reserve, because it was thought to threaten the compositionality principle. According to that principle, the meaning of a compound expression is fully determined by two factors: the meanings of its parts and its syntax. The new form of representation proposed by Kamp introduces a third factor, and it is essentially involved in the interpretation process, too. Take, for example, the neuter pronoun in (6), which is represented in (3) by $\underline{z}$.

(6) Every farmer who owns a donkey beats it.

This discourse referent occurs in a part of the DRS associated with the sentence’s verb phrase, and is equated to a discourse referent in a part associated with the subject; which would be impossible in a strictly compositional system.

While DRT was being conceived in the late 1970s, the paradigm framework in formal semantics was Montague’s (1974). The Montagovian framework was strictly compositional, and many people working in the field considered this a signal virtue. At the same time, Kamp’s treatment of anaphora held considerable appeal, and thus began the search for a compositional but non-representational alternative to DRT. A raft of proposals were made over the years, starting with

Barwise (1987), Zeevat (1989), and Groenendijk and Stokhof (1990, 1991). These theories covered the original TSR fragment and more, and employed essentially the same sort of semantics while eschewing discourse representations, thus disproving Kamp's conjecture that strict compositionality is "ultimately untenable" (p. 20) for the TSR fragment. However, if this was progress of a sort, it was hardly a leap forward, because the TSR fragment was rather small, and many of the extensions that have been proposed since seem to be irredeemably non-compositional. This holds, for example, for the DRT account of presupposition that was first proposed by van der Sandt (1992), and extended and modified by Geurts (1999) and Kamp (2001), for which a compositional equivalent hasn't been reported for three decades.

Besides, it is less than self-evident that we should want to have a fully compositional account of the TSR fragment, in the first place. Even if there is a way of linking "it" in (7) to "a donkey", what good will it do, especially when one considers that, in principle, it is always possible to recode a semantic theory as a compositional one, provided we are sufficiently flexible about the kinds of objects we are willing to call meanings (see Zadrozny, 1994 for proof).

(7) Pedro owns a donkey. He beats it.

It doesn't seem to have occurred to Kamp that compositionality might be feasible or desirable for discourses, for although TSR discusses the role that syntax has to play in a theory of interpretation (pp. 20–22), compositionality is not even mentioned. But Groenendijk and Stokhof maintain that there is such a thing as "discourse semantics", and that anaphora is a *semantic* phenomenon within and across sentences: "Like most authors, we start from the assumption that co-reference and anaphora are, by and large, semantic phenomena." (Groenendijk and Stokhof, 1991: 94; the claim that there is, or was, broad support for this assumption seems rather overstated.) Apparently, Groenendijk and Stokhof take the compositionality principle to apply in discourse semantics just as it applies in sentential and sub-sentential semantics. However, this line of thinking presupposes that discourses are compound expressions which are assembled in ways that must be substantially similar to the ways in which syntactic constituents are assembled, and have the same kind of meanings, too boot.

To see why these presuppositions are problematic, consider the following discourse fragment from Sacks (1974), simplified by Clark (1996):

- Ken: You wanna hear—My sister told me a story last night.
 Roger: I don't wanna hear it. But if you must.
 Al: What's purple and an island. Grape, Britain. That's what his sister told him ...
 Ken: No. To stun me she says uh, there was these three girls and they just got married?
 ...

How plausible is it that discourses like this are compound expressions sufficiently similar to "pink elephants" or "Chiquita snores" that one semantic treatment will fit all? Not very, it seems to me. So, while it may be instructive to learn that the pronouns in (7) can be treated "compositionally", it is not at all clear what that might imply.

Compositionality aside, Kamp's discourse representations raise deeper issues, as well. To explain, let me quote TSR at some length:

Discourse representations can be regarded as the mental representations which speakers form in response to the verbal inputs they receive. I should point out that the specific theory that is presented below does not render such identification essential. Even if the representations it posits are thought of as purely theoretical devices whose *raison d'être* is to be found solely in the contribution they make to an effective account of certain semantic properties of sentences and sentence complexes, the theory may merit comparison with other schemes of linguistic description which have been applied to the same phenomena. But this is not how I would like to see the proposal of this paper myself. (pp. 5–6)

Since DRT is usually taken to be a mentalist theory, it is of some importance to note that Kamp insists that such a construal is not mandatory. It may be natural to view DRT this way, and it is the way Kamp prefers, but his formal theory doesn't require that we do so. What other ways are there? Kamp himself points to the somewhat anaemic possibility that DRSs could be thought of as "purely theoretic devices", but we might also want to consider a more dashing approach along the following lines.

Communication is a form of social interaction in which speakers build and rely on common ground (Clark, 1996; Stalnaker, 2002). On a mentalist construal of DRT, common ground exists merely as an elaborate structure of correspondences between representations in the minds of its participants. By contrast, on a socialist construal, common ground exists in a more robustly social sense. Whereas the mentalist views discourse referents as existing primarily in the minds of speakers, the socialist regards them as denizens of the social space between speakers. Intuitively, the idea is that discourse presupposes and interacts with a relational structure which is defined by social relations between participants rather than what goes on in their minds (see Brandom, 1994; Geurts et al., 2017, to appear). In short, whereas common ground may be seen as a correspondence structure between mental representations, it may also be seen as a social structure simpliciter. These two views are not incompatible, and in fact it is natural to suppose that they complement one another. If so, DRSs would be mental and social entities at the same time (see Geurts, 2017 for further discussion).

Discourse Referents

The key elements of DRT's newfangled discourse representations were discourse referents. (The concept of discourse referent as well as the name hail from Karttunen, 1971.) Names and indefinites introduce discourse referents into the current DRS; pronouns pick them up. Formally, discourse referents are so-called dynamic variables, which differ from the static variables of classical logic in that they are able to retain their values in certain configurations where a classical variable becomes unbound. To illustrate, let's compare two attempts at representing sentence (8): one in predicate logic (9a), the other in DRT (9b).

- (8) Every farmer who owns a donkey beats it.
 (9) a. $\forall x[[\text{farmer}(x) \wedge \exists y[\text{donkey}(y) \wedge x \text{ owns } y]] \rightarrow x \text{ beats } y]$
 b. $[: [x, y: \text{farmer}(x), \text{donkey}(y), x \text{ owns } y](\forall x)[x \text{ beats } y]]$

In (9a), the rightmost *y*-token is not bound by the existential quantifier, and therefore no semantic connection exists between this token and the *y*-tokens to its left. By contrast, in (9b), the interpretation of the last *y*-token covaries with the interpretation of *y*-tokens in the restrictor of the quantifier. This difference has nothing to do with the notation: as was shown by Groenendijk and Stokhof (1991), a dynamic semantics for standard predicate logic can render (9a) synonymous with (9b).

If discourse referents are variables, then why call them "referents"? I believe that this curious terminology betrays a tension between the official theory and the intuitive picture on which it is based, a tension which also shines through in the more discursive parts of TSR, where the discussion of this topic seems not entirely sure-footed. For example, Kamp claims that indefinites are referential terms rather than existential quantifiers (p. 5), which may still sound odd even to the converted (like myself), but then he also puts "referent" in scare quotes when discussing the relation between an indefinite and the discourse referent it introduces (p. 16).

Let us try to locate, if not resolve, this tension by going back to our running examples, (8) and (10).

- (10) Pedro owns a donkey. He beats it.

In both cases, it is tempting to say that the neuter pronoun "refers back" to the individual introduced by the indefinite. But which individual? If (8) is true then several donkeys are likely to be involved, and the discourse in (10) may be true even if Pedro owns more than one donkey. Then perhaps our intuitions may be given their due by saying that the neuter pronoun refers to the discourse referent associated with the indefinite. That is also what Kamp says and it may seem plausible at first, until one realises with a shudder that this discourse referent is really a variable. But then again it is no ordinary variable, for in some contexts it behaves *almost* as if it were a constant.

Many years ago, Henk Zeevat told me that discourse referents are "very mysterious things". At the time, I failed to see the profundity of his remark, but now I think I know what he meant, and I believe he was right.

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Chapter 12

On Karttunen's "The Syntax and Semantics of Questions"



Elena Guerzoni

Abstract Karttunen's article on the syntax and semantics of questions is a milestone in the truth-conditional compositional semantics of interrogatives and of verbs that embed them. It is the first comprehensive study of the mapping between the syntax and the interpretation of the three different types of questions (polar, constituent and alternative questions) and presents the first semantic analysis of question-embedding verbs (QEVs henceforth) that assumes the same intensions for matrix and embedded interrogatives. This analysis continues to vastly inspire the ongoing research on the properties of questions and QEVs. This chapter illustrates Karttunen's theory focusing on those formal details that have been the most influential in subsequent literature. In doing so, however, I will take the liberty to suggest a less than literal rendition of these details, in an attempt to make the discussion more accessible to today's reader. The main departure that I make here from Karttunen's 1977 is in the formal framework. Whereas Karttunen adopts Montague's PTQ, here I will expose his ideas in Heim and Kratzer's type driven semantics.

Keywords Interrogatives · Questions · Syntax-semantics interface · Compositionality · *Wh*-movement · Question-embedding predicates · *De re/De dicto* ambiguities · Strong exhaustivity

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Introduction: Questions as Sets of (Instantial) Answers

In the early days of truth-conditional semantics, interrogative sentences represented a puzzle for the obvious reason that they are neither true nor false. At the same time a truth-conditional and compositional account of their interpretation was desirable, not merely for theoretical reasons, but also, and more importantly, for empirical ones. This is so because all questions evidently include a declarative-like component whose interpretation must be obtained via the same mechanisms at work in simple declaratives.

The first challenge for such an account was to establish what kind of objects questions denote. In two articles from 1958 and 1973, Hamblin offered a simple and elegant answer to this challenge. He suggested that every root question denotes the collection of all its possible (semantically congruent)¹ answers. This idea finds its plausibility in what Hamblin took to be the discourse function of matrix questions; that is the presentation of alternative propositions, among which the addressee is expected to select the true one(s).²

Since the options presented are *propositions*, this picture explicitly connected the notion of truth-conditions (of the answers) and the meaning of questions.

Karttunen's article adopts Hamblin's insight but claims that the denotation of a question in a given world of evaluation is restricted to the set of answers that are true in that world, rather than the set of all its possible answers. The examples below illustrate this difference. (Following common practice, I use *that*-clauses to refer to propositions and @ to refer to the actual world.)

- (1) a. Is Paris in France?

Polar Question

Karttunen's denotation in @:

- b. {that Paris is in France}

Hamblin's denotation in @:

- c. {that Paris is in France, that Paris is not in France}

¹ The difference between a semantically congruent answer and a mere response to a question is crucial in this view and it will become clearer in the coming sections.

² Hamblin (1958) suggests that the denotation of a question is the set of its possible complete answers, whereas the more formally explicit theory in Hamblin (1973) derives as their denotations the sets of their possible "instantial" answers, a notion that is defined later in this chapter. Karttunen (1977) follows Hamblin (1973) in this respect.

- (2) a. Which semifinalists played in the 2018 World Cup final match? **Wh-question**

Abbreviations

B = That Belgium played in the '2018 final match

C = That Croatia played in the '2018 final match

UK = That UK played in the '2018 final match

F = That France played in the '2018 final match

b. {C, F} (Karttunen's denotation in @)

c. {B, C, UK, F} (Hamblin's denotation in @)

Before delving into the details of Karttunen's analysis, it is worth evaluating this claim. According to Karttunen, what justifies the restriction to true answers is the existence of QEVs that 'refer' to the true answers of the questions they embed. One such predicate is *depend on*, which expresses a dependency between the true answers to the questions it combines with. His example is repeated in (3).

- (3) Who is elected depends on who is running.

More recent work by Sigrid Beck and Hotze Rullmann challenged this view on both empirical and technical grounds. The empirical challenge Beck and Rullmann (1999) presents is the existence of embedding predicates expressing attitudes towards merely possible answers as well. For instance they discuss the example of *agree on*, as in (4a), which is true in the scenario described in (4b).

- (4) a. John and Mary agree on who was elected.
b. *Scenario*: Trump was elected, but both Mary and John believe that Hillary was.

Mary and John may both be wrong about the elected president, but may still agree on who was elected.

In addition to this, from a technical perspective, the set of true answers in a given world (i.e. Karttunen's set) can be systematically obtained from the set of possible answers (i.e. Hamblin's set) via the following function that Rullmann and Beck (1998) define. (' Q_H ' is a function from possible worlds to sets of propositions.)

- (5) For any arbitrary possible world w :
 $\text{Kartt}(Q_H)(w) = \{p: p \in Q_H(w) \wedge w \in p\}$

The function *Kartt* maps Hamblin's intensions (Q_H) to the corresponding Karttunen denotations in any given world, thus making the restriction to true answers unnecessary for an account of *depend on* (and of other predicates 'referring' to true answers), as illustrated (6) below.

- (6) $\llbracket \text{depends on} \rrbracket^w(Q_H)(Q_{1H}) = \text{liff}$
 $\cap \text{Kartt}(Q_H)(w) \text{ depends on } \cap \text{Kartt}(Q_{1H})(w)$

Instantial vs. Complete Answers

The considerations presented in the previous session allow us to forgo the restriction to true answers, at least for the time being. With this in mind, let me return to example (2a), whose denotation is repeated below in (7).

- (7) a. $\llbracket \text{Which semifinalists played in the 2018 World Cup final match (?)} \rrbracket^@ =$
 b. $\{B, C, UK, F\}$

This case is useful to understand the notion of ‘instantial answer’ in Hamblin’s as well as Karttunen’s semantics of *wh*-questions. The elements of the set in (7b) are ‘instantial’ answers to the question in that each proposition corresponds to one *instance* of a semifinalist national team. Notice that, in this case, no true instancial answer in the actual world (C or F) constitutes by itself the complete true answer to the question, only their conjunction does. Heim (1994) formalizes this notion as the following function that derives the complete true answer to any question in any world of evaluation, from the set of its possible instancial answers.

- (8) For any world w :

$$\text{ans}_1(Q_H)(w) = \cap \{p: w \in p \text{ and } p \in Q_H(w)\}$$

$$(= \cap \text{Kartt}(Q_H)(w))$$

Sets of Answers and Answerhood Conditions

The example in (2a) helped me to concretely illustrate some core distinctions, such as instancial vs. complete and true vs. possible answers, because we all share the relevant world knowledge that is necessary to list and talk about all these types of answers to that question. The reason is twofold: the number of semifinalist soccer teams in the 2018 World Cup is finite and we do know (or can easily find out) which ones they were.

Importantly, this much factual knowledge is not generally necessary in order to understand what a question means, just as we do not need to know if a sentence is true in order to understand its meaning. This is why an English speaker in 2019 has the ability to interpret questions like *which semifinalists will play in the 2022 World Cup final match?* or *which integers are prime numbers?*

So, what is it that a speaker knows when s/he understands a question? Hamblin and Karttunen’s insight is that understanding the meaning of a question boils down to the understanding what conditions a proposition must satisfy in order to qualify as a semantically congruent (instantial) answer to that question, i.e. its *answerhood conditions*. *Answerhood conditions* are formally represented as the membership conditions in the abstract definitions of the denotation of questions. Given this, an abstract definition as in (9) of sets like (7b) is a more suitable description of this semantic ability.

- (9) a. $\llbracket \text{Which semifinalists played in the 2018 World Cup final match} \rrbracket^@ =$
 b. $\{p: \exists x [\llbracket (2018 \text{ WC}) \text{ semifinalist} \rrbracket^@ (x) = 1 \wedge p = \text{that } x \text{ played the 2018 World Cup semifinals}]\}$

Since the actual four 2018 semifinalists were Belgium, Croatia, UK and France, (9b) can be rewritten as (9b'):

- (9) b'. $\{p: \exists x [x \in \{\text{Belgium, Croatia, UK, France}\} \wedge p = \text{that } x \text{ played the 2018 World Cup semifinals}]\}$

Although it might take some work to show this, the abstract definition in (9b') and the list definition in (7b) pick out exactly the same set.

From LFs to Sets of Answers

Whereas establishing what questions denote is a fundamental step towards a compositional truth-conditional approach to their meaning, the main task for such an approach is uncovering the mechanisms that can systematically derive the desired set of propositions from each question's structure and the meanings of its parts. This section illustrates a type driven rendition of Karttunen's model of these mechanisms.³

From Declarative to Question Semantics: Proto-Questions

Karttunen assumes that the Logical Form (LF) of all types of interrogatives contain declarative constituents (see (10)), with declarative denotations (open or closed propositions) obtained compositionally in a customary fashion. (10), below, illustrates this point; each example shows a constituent of a question and its corresponding intension, that is a proposition. $(\llbracket \alpha \rrbracket_c^g$ is an abbreviation for $\lambda w. \llbracket \alpha \rrbracket^{g,w}$ I adopt from Montague Grammar, therefore it indicates the intension of α relative to the assignment function g ; cf. Dowty et al., 1981, p. 147). Since all the intensions in (10) are propositions, I have abbreviated them as *that*-clauses. IPs are Inflectional Phrases, these phrases are syntactic complements of Complementizer Phrases (CPs).⁴

³ For this rendition of Karttunen's semantics I am immensely indebted to Irene Heim's lectures and lecture notes on the semantics of interrogatives.

⁴ Unlike Hamblin, whose *wh*-phrases are interpreted in their base position, Karttunen adopts the syntactic hypothesis that all *wh*-phrases occupy a position above C at LF. Given this, (10b) results from the covert movement of the object *wh*-phrase. In (10c), the first trace is left by the overtly moved *who* and the second trace corresponds to the covert movement of *whom*. All traces are

- (10) a. [_{IP} *Mary smiled*] is an LF constituent of *did Mary smile?*
 $\llbracket \text{Mary smiled} \rrbracket_{\epsilon}^g = \text{That Mary smiled}$
 b. [_{IP} *John met t_i*] is an LF constituent of *which student did J. meet?*
 $\llbracket \text{John met } t_i \rrbracket_{\epsilon}^g = \text{That John met } g(i)$
 c. [_{IP} *t_i met t_h*] is an LF constituent of *who met whom?*
 $\llbracket t_i \text{ met } t_h \rrbracket_{\epsilon}^g = \text{That } g(i) \text{ met } g(h)$
 d. [_{IP} *Mary cooks*] and [_{IP} *John eats out*] are LF constituents of
Does Mary cook or John eats out?
 $\llbracket \text{Mary cooks} \rrbracket_{\epsilon}^g = \text{That Mary cooks and}$
 $\llbracket \text{John eats out} \rrbracket_{\epsilon}^g = \text{That John eats out}$

What distinguishes questions from declaratives is a silent question complementizer “?”, whose only semantic function is to lift the semantic type from propositions (type $\langle s, t \rangle$) to sets thereof (type $\langle st, t \rangle$). Once applied to a proposition p , the output is the singleton containing p . The proto-questions in the questions in (10) are shown in (11). Notice that, Karttunen proposes that there are as many proto-questions in alternative questions as there are disjuncts in them (see (11d)).

- (11) a. [_{C'} ? [_{IP} *Mary smiled*]]
 b. [_{C'} ? [_{IP} *John met t_i*]]
 c. [_{C'} ? [_{IP} *t_i met t_h*]]
 d. [_{C'} ? [_{IP} *Mary cooks*]]
 [_{C'} ? [_{IP} *John eats out*]]

- (12) $\llbracket ? \rrbracket_{\epsilon}^g = \lambda p. \{p\}$ (In full function talk: $\llbracket ? \rrbracket_{\epsilon}^g = \lambda p. \lambda q. q=p$)⁵

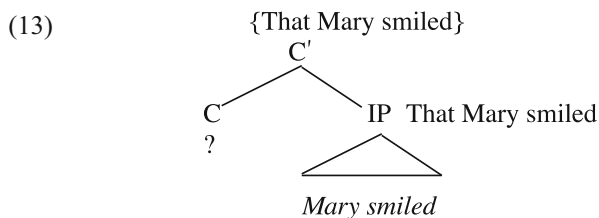
Karttunen names constituents headed by “?” ‘proto-questions’ and takes them to be the ‘embryonic structures’ of questions, both semantically and syntactically.

interpreted via the assignment function g , to be bound higher in the structure. This generates in both cases a constituent denoting an open sentence.

⁵ Karttunen’s original denotation of “?” also filters out possible but false answers from the final denotation of the question.

- (i) For any possible world w , $\llbracket ? \rrbracket^w = \lambda p. \{q : w \in q \text{ and } q = p\}$

This question morpheme generates the singleton containing p if p is true, and the empty set otherwise. As mentioned above, there are good reasons to maintain a semantics for “?” that does not exclude possible but false answers.



While proto-questions already have the same semantic type as complete questions, they don't yet present the alternative possible (or true) answers to polar, constituent or alternative questions.⁶ This second semantic step is achieved by the semantic contribution of *whether (or not)*, *wh*-quantifiers and disjunctions, respectively.

Wh-Questions

Since Karttunen's semantics of multiple *wh*-questions is one of the most noteworthy contributions of his article, I will turn to this case first. The core of the proposal is in the above mentioned division of labor between the semantics of the *?*-morpheme and that of *wh*-phrases. Specifically, the desired set of instantial answers denoted by a *wh*-question is obtained from an open proposition in two main steps: i) type lifting from $\langle s, t \rangle$ to $\langle st, t \rangle$ which occurs only once in each *wh*-question, regardless of the number of *wh*-phrases in it and ii) quantifying in as many existential quantifiers as there are *wh*-phrases in the question via a special **WH-quantification** rule. This strategy allows Karttunen to maintain an unambiguous semantics for *wh*-words as well as a unified account of the semantics of proto-questions across all type of questions.

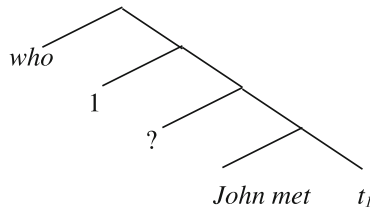
Examples of Karttunen-style LFs for single and multiple *wh*-questions are in (14a) and in (14b) respectively.⁷

⁶ The proto-question in (13) also fails to represent the 'original' Karttunen denotation, because it fails to contain the true answer in all the worlds in which Mary doesn't smile.

⁷ Following Heim and Kratzer (1998)'s analysis of binding, I assume an LF where the index of each moved DP is a separate head immediately inserted under the DP's landing site and that constituents headed by indices denote predicate abstracts.

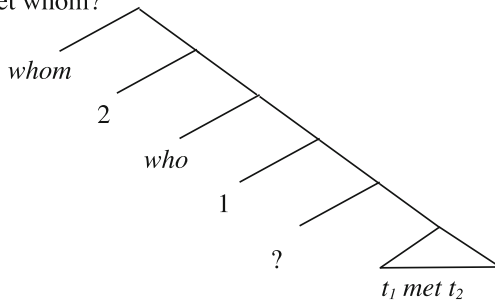
- (14) a. Who did John meet?

LF:



- b. Who met whom?

LF:



Let us first consider (14a). For any arbitrary assignment function g , the declarative component in it denotes the open proposition given in (15a), the proto-question denotes the singleton containing it in (15b), and the constituent headed by the index 1 denotes the abstract in (15c).

- (15) a. $\llbracket \text{John met } t_i \rrbracket_{\mathcal{C}}^g = \text{That John met } g(i)$
 b. $\llbracket ? \text{ John met } t_I \rrbracket_{\mathcal{C}}^g = \{\text{That John met } g(1)\}$
 c. $\llbracket 1 ? \text{ John met } t_I \rrbracket_{\mathcal{C}}^g = \lambda x. \{\text{That John met } x\}$
 (proto-question abstract)

Karttunen stipulates that *wh*-words are equivalent to existential determiners and *wh*-phrases to existential quantifiers (see 16), which combine with proto-question abstracts via the construction-specific compositional rule in (17).

- (16) a. $\llbracket \text{which} \rrbracket^{g,w} = \llbracket \text{some} \rrbracket^{g,w}$
 $= \lambda Q_{\langle e, \triangleright \rangle}. \lambda P_{\langle e, \triangleright \rangle}. \exists x_e [P(x)=1 \wedge Q(x)=1]$
 b. $\llbracket \text{who} \rrbracket^{g,w} = \llbracket \text{someone} \rrbracket^{g,w}$
 $= \lambda Q_{\langle e, \triangleright \rangle}. \exists x_e [x \text{ is a person in } w \wedge Q(x)=1]$

(17) WH-quantification (WHQ) Rule (Modified)⁸

For any world w and assignment function g ,
if α 's immediate daughters are β and γ , $\llbracket \beta \rrbracket^{g,w} \in D_{\langle et, t \rangle}$, and $\llbracket \gamma \rrbracket^{g,w} \in D_{\langle e, \langle st, t \rangle \rangle}$,
then:

$$\llbracket \alpha \rrbracket^{g,w} = \{p : \llbracket \beta \rrbracket^{g,w} (\lambda x. p \in \llbracket \gamma \rrbracket^{g,w} (x)) = 1\}.$$

This generates the desired set of propositions.

$$\begin{aligned}
 (18) \quad & \llbracket \text{who } 1 \text{ ? John met } t_1 \rrbracket^{g,w} \\
 &= \{p : \llbracket \text{who} \rrbracket^{g,w} (\lambda x. p \in \llbracket 1 \text{ ? John met } t_1 \rrbracket^{g,w} (x)) = 1\} \\
 &= \{p : \llbracket \text{who} \rrbracket^{g,w} (\lambda x. p \in [\lambda y. \{\text{That John met } y\}](x)) = 1\} \\
 &= \{p : \exists z [z \text{ is a person in } w \wedge [\lambda x. p \in [\lambda y. \{\text{That J. met } y\}](z)](z) = 1\} \\
 &= \{p : \exists z [z \text{ is a person in } w \wedge [\lambda x. p \in \{\text{That John met } x\}](z) = 1\} \\
 &= \{p : \exists z [z \text{ is a person in } w \wedge p \in \{\text{That John met } z\}]\} \\
 &= \{p : \exists z [z \text{ is a person in } w \wedge p = \text{That John met } z]\}
 \end{aligned}$$

Since the output of both the denotation of the ?-morpheme and of the WHQ rule are sets of propositions, abstraction and then the WHQ rule can be simply reiterated as many times as there are *wh*-phrases in the question. This provides a straightforward account of multiple *wh*-questions.

$$\begin{aligned}
 (19) \quad & a. \llbracket ? t_1 \text{ met } t_2 \rrbracket^{g,w} = \{\text{That } g(1) \text{ met } g(2)\} \\
 & b. \llbracket 1 \text{ ? } t_1 \text{ met } t_2 \rrbracket^{g,w} = \lambda x. \{\text{That } x \text{ met } g(2)\} && \lambda\text{-abstraction} \\
 & c. \llbracket \text{who } 1 \text{ ? } t_1 \text{ met } t_2 \rrbracket^{g,w} = \\
 & \quad \{p : \exists x [x \text{ is a person in } w \wedge p = \text{That } x \text{ met } g(2)] && \text{WHQ} \\
 & d. \llbracket 2 \text{ who } 1 \text{ ? } t_1 \text{ met } t_2 \rrbracket^{g,w} = \\
 & \quad \lambda y. \{p : \exists x [x \text{ is a person in } w \wedge p = \text{That } x \text{ met } y] && \lambda\text{-abstraction} \\
 & e. \llbracket \text{whom } 2 \text{ who } 1 \text{ ? } t_1 \text{ met } t_2 \rrbracket^{g,w} = \\
 & \quad \{p : \exists x \exists y [x \text{ is a person in } w \text{ and } y \text{ is a person in } w \wedge p = \\
 & \quad \quad \text{That } x \text{ met } y] && \text{WHQ}
 \end{aligned}$$

⁸ Karttunen's rule also contains a syntactic component, which makes it construction specific by limiting its applicability to cases where β is a *wh*-phrase and either the head of γ is the question morpheme ?, or its specifier is another *wh*-phrase.

Moreover, in Montague's system, which Karttunen adopts, index binding and quantification are obtained by one single compositional rule, similar to (i) below:

$$(i) \quad \text{If } \alpha \text{'s immediate daughters are } \beta_i \text{ and } \gamma \text{ and } \llbracket \beta_i \rrbracket^{g,w} \in D_{\langle et, t \rangle} \text{ and } \llbracket \gamma \rrbracket^{g,w} \in D_t, \text{ then} \\ \llbracket \alpha \rrbracket^{g,w} = \llbracket \beta_i \rrbracket^{g,w} (\alpha x. [\llbracket \gamma \rrbracket^{g,w}]^{g[x/i], w})$$

Given this, the following is a more faithful rendition of the semantic component of Karttunen's rule (assuming that the index of the moved *wh*-phrase is a subscript rather than heading its sister node):

$$(ii) \quad \text{If } \alpha \text{'s immediate daughters are } \beta_i \text{ and } \gamma \text{ and } \llbracket \beta_i \rrbracket^{g,w} \in D_{\langle et, t \rangle} \text{ and } \llbracket \gamma \rrbracket^{g,w} \in D_{st, t}, \text{ then} \\ \llbracket \alpha \rrbracket^{g,w} = \{p : \llbracket \beta_i \rrbracket^{g,w} (\alpha x. p \in [\llbracket \gamma \rrbracket^{g,w}]^{g[x/i], w})\}$$

This kind of recursion is possible because of the division of labor between the ?-morpheme and *wh*-phrases. If *wh*-words or WHQ were responsible for the type shifting from propositions to sets of propositions the structure would be un-interpretable due to a type-mismatch between the first proto-abstract and the *wh*-phrase above it.⁹

From Existential to Wh-Quantifiers: Dispensing with WHQ

Current type-driven developments of Karttunen's semantics have abandoned his assumption that *wh*-phrases are regular existential quantifiers and, instead, have analyzed them as question quantifiers.¹⁰

- (20) a. $\llbracket who \rrbracket^{g,w} =$
 $\lambda Q_{\langle e, \langle st, t \rangle \rangle} . \{p: \exists x [x \text{ is a person in } w \wedge p \in Q(x)] \}$
 b. $\llbracket which \rrbracket^{g,w} =$
 $\lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, \langle st, t \rangle \rangle} . \{p: \exists x [P(x) = 1 \wedge p \in Q(x)] \}$

One immediate benefit of question quantifiers is that Karttunen's construction specific WHQ rule can be dispensed with. The semantic component of this rule is an ad hoc mechanism of "quantifying-in" crafted to combine existential quantifiers with proto-questions. With question quantifiers, simple function application suffices. The "construction-specificity" of the rule (see footnote 8) prevents its application to regular existential quantifiers in order to block a reading of (21)a equivalent to (21)b. If *wh*-phrases denote question quantifiers, the absence of equivalent interpretations of (21)a and b follows straightforwardly.¹¹

- (21) a. Mary saw someone.
 b. Who did Mary see?

⁹ This mismatch can be avoided under the undesirable additional stipulation that *wh*-words are lexically ambiguous.

¹⁰ Abbreviating $\langle st, t \rangle$ as τ_q reveals the similarities and differences between the two types of determiners:

$$\begin{aligned} \llbracket some \rrbracket^{g,w} &= \lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, t \rangle} . \exists x [P(x) = 1 \wedge Q(x) = 1] \\ \llbracket which \rrbracket^{g,w} &= \lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, \tau_q \rangle} . \{p: \exists x [P(x) = 1 \wedge Q(x)(p) = 1]\} \end{aligned}$$

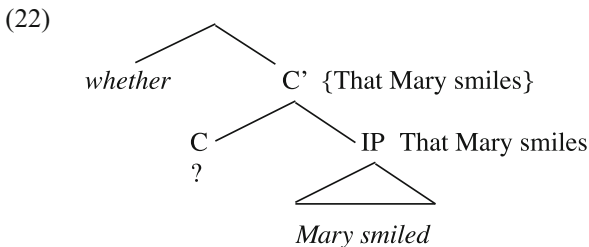
¹¹ This is so because both the LFs below lead to type mismatches and are therefore un-interpretable.

- (i) $[_{CP} \text{ someone } [_1 [_{C'} ? [_{IP} \text{ Mary saw } t_1]]]]$
 (ii) $[_{IP'} \text{ who } [_1 [_{IP} \text{ Mary saw } t_1]]]]$

Notice that Karttunen also needs to stipulate that the movement of *wh*-phrases above C is compulsory for mere syntactic reasons, in order to block (ii).

Polar Questions

According to Karttunen, *whether* (or *not*) occurs (overtly or covertly) in both embedded and root polar questions. Given this, the LF of *did Mary smile* is the structure below.



The denotation of *whether* is given in (23)¹² ($\sim p$ below stands for $\{w: w \notin p\}$ and Q is a set of propositions).

$$(23) \quad \llbracket \textit{whether} \rrbracket^{g,w} = \lambda Q. \{p: p \in Q \text{ or } \sim p \in Q\}$$

This function combines with the denotation of the proto-question in (22) and returns the set containing the propositions corresponding to the positive and the negative answer.

$$\begin{aligned}
 (24) \quad & \llbracket \textit{whether} \text{ ? } \textit{Mary smiled} \rrbracket^{g,w} = \\
 & \llbracket \textit{whether} \rrbracket^{g,w} (\{\textit{That Mary smiled}\}) \\
 & = \{p: p \in \{\textit{That Mary smiled}\} \text{ or } \sim p \in \{\textit{That Mary smiled}\}\} \\
 & \textit{That Mary smiled and } \sim(\sim\textit{That Mary didn't smile}) \text{ are} \\
 & \text{elements of } \{\textit{That Mary smiled}\} \text{ therefore} \\
 & \{p: p \in \{\textit{That Mary smiled}\} \text{ or } \sim p \in \{\textit{That Mary smiled}\}\} \\
 & = \{\textit{That Mary smiled}, \sim\textit{That Mary smiled}\}
 \end{aligned}$$

Since the denotation of *whether* can combine with sets of propositions and returns sets of propositions, nothing in this semantics prevents it from occurring inside *wh*-questions. Given this, Karttunen stipulates a grammatical constraint

¹² Due to the restriction to true answers, Karttunen's denotation of the proto-question (i.e. $Q(w)$) might be empty. Given this, semantics of *whether* needs to be more complex:

$$\llbracket \textit{whether} \rrbracket^w = \lambda Q_{\langle s, \langle st, t \rangle \rangle}. \{p: p \in Q(w) \text{ or } [Q(w) = \emptyset \rightarrow p = \{w: Q(w) = \emptyset\}]\}$$

If Mary smiled in w , $Q(w) = \{\textit{that Mary smiled}\} = \llbracket \textit{whether Mary smiled} \rrbracket^w$. If Mary didn't smile in w , $Q(w) = \emptyset$ and $\llbracket \textit{whether Mary smiled} \rrbracket^w = \{p: p = \{w: Q(w) = \emptyset\}\} = \{p: p = \{w: w \notin \textit{that Mary smiled}\} = \{\textit{That Mary didn't smile}\}.$

against this possibility. However, more recent research on *wh*-questions foregoes this constraint and proposes that strongly exhaustive interpretations and the licensing of negative polarity items (NPIs) in questions follow from a silent *whether* in *wh*-questions (see Guerzoni, 2003, 2004; Guerzoni & Sharvit, 2014). For a discussion of Strong Exhaustivity see the Section “*Exhaustivity and De Dicto*” below.

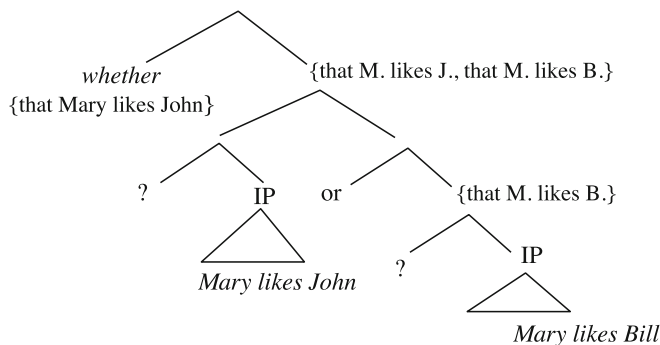
Alternative Questions

According to Karttunen, questions like (25) below are ambiguous between a polar interpretation where a *yes* or a *no* answer is expected, and an alternative interpretation where the addressee is expected to indicate which of the two propositions conjoined by *or* is true.

(25)whether Mary likes John or Mary likes Bill

Karttunen proposes that the LF generating the alternative reading contains as many proto-questions as there are disjuncts in the question. Each disjunction of two proto-questions is the union of the sets they denote. He also assumes that both root and embedded alternative readings contain *whether*, although this *whether*, unlike the *whether* in polar questions, doesn’t do any semantic work.

(26) {that Mary likes John, that Mary likes Bill}



The analysis works if the alternatives combined by *or* denote sets of propositions. Give this, and since Montague’s PTQ cannot generate structures like *whether John or Bill arrived late* from *whether John arrived late or Bill arrived late*, Karttunen considered the availability of an alternative reading for these questions problematic for his theory (see footnote 10 in Karttunen’s article). Current theories of ellipsis under identity, however, do allow such derivations. On an ellipsis analysis, *whether*

John or Bill arrived late may result from the deletion of the VP *arrived late*, which is legitimated by the presence of an identical VP in the same question.

(27) a. ... whether John ~~arrived late~~ or Bill arrived late.

Although other analyses of these cases are feasible, Han and Romero (2004) provide convincing arguments in favor of this account.

Whether and Presuppositions

As Karttunen himself points out, his semantics of alternative questions fails to derive the presupposition that one and only one of the alternatives in them is actually true. For instance, under its alternative interpretation, *did John play chess or checkers* presupposes that John played exactly one of the two games. However, nothing in Karttunen's semantics precludes the possibility that John played both games in the world of evaluation or the possibility that he played neither of them.

A second weakness in Karttunen's analysis of alternative readings is that *whether* in them is essentially semantically vacuous. Under this view, the fact that polar and alternative questions are introduced by this same *wh*-word falls out as a mere coincidence.

In Guerzoni and Sharvit (2014), we suggest that these two problems are related to one another. Specifically, we argue that a proper unified semantics of *whether* in polar and alternative questions can also predict the presuppositions of the latter. Our proposal is that polar questions are derived from structures of alternative questions whose disjuncts are sentences denoting the positive and negative answer. Following Larson (1985), we claim that *whether* in alternative questions makes roughly the same contribution as *which* in *which one of either p or q*, namely, that of a *wh*-determiner whose restrictor consist of a disjunction of propositions. Given this, the existence and uniqueness presuppositions of alternative questions are the same as those of singular *which*-questions (see Dayal, 1996 for an account of the latter).¹³

Embedding Questions

Karttunen's analysis of question embedding is one of the most influential contributions of his paper. All three types of questions discussed in the previous section can occur in embedded positions (see (28)).

¹³ In polar questions, the presupposition that at least and at most one answer is true is trivially satisfied, since their answers are mutually exclusive and jointly exhaustive.

- (28) Tom knows/wonders/predicted ... who was at the party.
 ... whether Susan was at the party.
 ... whether Sue or Bill cooked dinner.

The theory has it that the meaning of each embedded question, derived in the previous section, functions as the argument of the denotations of embedding verbs like those in (28).¹⁴ Karttunen claims that all QEVs denote functions that take intensions of questions and that *know*, *tell*, *predict*, and many other verbs which can embed questions (as seen in (28)) as well as propositions (as shown in (29)), are actually lexically ambiguous between a QEV and PEV (proposition embedding verb).

- (29) Tom knows/ told me/predicted ... that Susan was at the party.

The hypothesis of a lexical ambiguity between QEVs and their proposition embedding counterparts, when there are any, correctly predicts the possibility of QEVs lacking a proposition embedding homophonous counterpart (this happens with *ask* and *wonder*) and vice-versa.

Despite the differences in their combinatorial properties, Karttunen also identifies a systematic relation between QEVs and PEVs in his discussion of *know*. Question embedding *know* ($know_Q$ henceforth) expresses roughly the same attitude towards the true answers of the question it embeds as proposition embedding *know* ($know_P$) does towards the proposition it embeds. To see this let us consider Karttunen's Semantics of $know_Q$ defined in terms of ans_1 .¹⁵

- (30) $\llbracket know_Q \rrbracket^w (\lambda Q_{<s, \tau_q>}) (x)$ is true iff
 (i) x knows $ans_1(Q)(w)$ in w and
 (ii) $[Kartt(Q)(w) = \emptyset \rightarrow x \text{ knows } [\lambda w'. Kartt(Q)(w') = \emptyset]]$ in w

This semantics predicts that x knows Q in w if x knows all the true answers to Q in w . This is so under the following conditions. Condition (i) in (30) is self-explanatory: x must know the true proposition in the extension of Q in w , assuming that there is at least one, in order to know all the true answers to Q in w . However, there might be no true propositions in the extension of Q in w at all. For instance, the extension of *which students got an A?* has no true members in worlds where no student got an A. Without condition (ii), we would incorrectly predict that any

¹⁴ Karttunen's theory is built on the assumption that questions denote the same sets of propositions whether they are embedded under a QEV or are uttered in isolation. In fact, Karttunen claims that the root questions too are occurrences of embedded questions under silent performative verbs. This kind of 'performative hypothesis' is problematic and it is also un-necessary within current semantic theorizing, where free and embedded occurrences of questions can be analyzed as sharing the same LF structure and therefore the same meaning. I will return to this point in my conclusions.

¹⁵ Recall that for any world w :

$$ans_1(Q_H)(w) = \cap \{p : p \in w \wedge p \in Q_H(w)\} = \{w : \forall p [p \in Kartt(Q_H)(w) \rightarrow w \in p]\}$$

sentence of the form *x knows which student got an A* should be trivially true in these worlds.¹⁶ Condition (ii) guarantees that also in these cases *x* knows *Q* only if *x* knows its true answer. To see this, consider again the worlds where no student got an *A*. In this case, according to condition (ii), *x* knows which student got an *A* only if *x* knows that indeed no students got an *A*, because the set of worlds in which no students received an *A* coincides with the set of worlds in which the extension of the question contains no true instantial answers.

Exhaustivity and *De Dicto*

There are two problems in Karttunen's analysis that have been at the center of a long-standing debate in the literature on questions. The first problem, which Karttunen himself noticed, is that it fails to predict the validity of inferences such as (31).

- (31) Mary knows who was at the party last night
 Bill wasn't at the party last night

 Mary knows that Bill wasn't at the party last night

This is so because Karttunen's truth conditions of the first premise are 'weakly exhaustive', rather than 'strongly exhaustive':

- (32) *Mary knows who was at the party last night* is true iff for every *x* in the domain of quantification:
- a. If *x* was at the party Mary knows that *x* was at the party
weakly exhaustive
 - b. Mary knows **whether** *x* was at the party
strongly exhaustive

Weakly exhaustive TCs are compatible with situations where Mary mistakenly believes of some absentees that they were at the party or she simply has no opinion on whether they were or not at the party. (c.f. Groenendijk & Stokhof 1982, 1992; Heim, 1994).

Groenendijk and Stokhof were the first to notice the second problem. Although sentences like (33) are ambiguous, Karttunen's analysis derives only one of their interpretations.

- (33) Mary knows which students are at the party.

¹⁶ This is so because in each one of these worlds $\{w: \forall p[p \in \text{Kartt}(Q_H)(w) \rightarrow w \in p]\}$ is *W*, which is a tautology.

Under Karttunen's interpretations, (33) follows from (34).

(34) Mary knows who is at the party.

Karttunen's true complete answer to *who is at the party* entails the true complete answer to *which students are at the party*¹⁷; therefore, knowledge of the former entails knowledge of the latter.

However, there is a second, very natural, interpretation of (33) that doesn't follow from (34). This interpretation is false in situations where (34) is true, for example, if Susan and Bill are students who were at the party and, though Mary believes that they were at the party, she is unaware that they are students.¹⁸

Groenendijk and Stokhof call the first interpretation *de re* and the second *de dicto*. The truth conditions of these two readings are given below.

(35) *De re*: For every x that is a student at the party, M . believes that x is at the party.

De dicto: For every x that is a student at the party, M . believes that x is a **student** at the party.

Strong Exhaustivity and the distinction between *de re* and *de dicto* readings are thoroughly argued in Groenendijk and Stokhof's work to be among the empirical grounds for rejecting Karttunen's semantics altogether in favor of the partition semantics of questions they propose.

Heim (1994), however, shows that it is possible to maintain Karttunen's original semantics of questions and derive *de dicto* interpretations and Strong Exhaustivity, by allowing the lexical semantics of QEVs to make reference to strongly exhaustive answers which can be systematically obtained from weakly exhaustive ones via the following function 'ans₂'.

¹⁷ Here is a formal proof of this claim: For every world w ,

(i) $\llbracket \textit{student} \rrbracket^w \subseteq \llbracket \textit{person} \rrbracket^w$
 (ii) $\{p: \exists x [\llbracket \textit{student} \rrbracket^w(x)=1 \wedge p = \text{that } x \text{ came to the party}]\}$
 $\subseteq \{p: \exists x [\llbracket \textit{person} \rrbracket^w(x)=1 \wedge p = \text{that } x \text{ came to the party}]\}$
 (iii) $\llbracket \textit{which students were at the party} \rrbracket^w \subseteq \llbracket \textit{who was at the party} \rrbracket^w$
 (iv) $\text{Kartt}(\lambda w'. \llbracket \textit{which student was at the party} \rrbracket^{w'}(w)) \subseteq \text{Kartt}(\lambda w'. \llbracket \textit{who was at the party} \rrbracket^{w'}(w))$
 (v) $\cap \text{Kartt}(\lambda w'. \llbracket \textit{who was at the party} \rrbracket^{w'}(w))(w) \subseteq$
 $\cap (\text{Kartt}(\lambda w'. \llbracket \textit{which students were at the party} \rrbracket^{w'}(w))(w))$

¹⁸ Heim (1994) observes that Karttunen's analysis actually makes the correct prediction in the special case where some people were at the party but none of them are students. In this special case, due to the second condition in Karttunen's semantics of *know_Q*, Mary should know that of none of those who were at the party were students for (33) to be true. This amount of knowledge does not follow from (34).

- (36) For any world w ,
 $\text{ans}_2(Q_H)(w) = \{w' : \text{ans}_1(Q_H)(w') = \text{ans}_1(Q_H)(w)\}$
 For any world w :
 $\text{ans}_1(Q_H)(w) = \cap \{p : p \in w \text{ and } p \in Q_H(w)\}$

$\text{Ans}_2(Q)$ in w is the proposition 'that ans_1 is the answer to Q in w '. The lexical entry for strongly exhaustive *know*_Q that Heim proposes is given in (37).

- (37) $[[\text{know}_Q]]^w = \lambda Q. \lambda x. x \text{ believes } \text{ans}_2(Q)(w)$

Heim demonstrates that the ans_2 of *which students are at the party* in w is equivalent to the proposition in (38).

- (38) $\lambda w'. \forall x[x \text{ is a student and at the party in } w$
 $\leftrightarrow x \text{ is a student and at the party in } w']$

It follows from (37) that *Mary knows which students are at the party* is true in w iff Mary believes the proposition in (38).

This semantics of *know*_Q predicts the strongly exhaustive inference in (39) to be valid.

- (39) Mary knows which students are at the party
 Bill is a student who isn't at the party

 Mary knows that Bill isn't at the party

If the consequent is false, then one of the premises has to be false too. If Mary doesn't believe that Bill isn't at the party, but he in fact isn't, then there is at least one world compatible with her beliefs in which (38) is false. It would then follow that she doesn't know who was at the party. If Mary doesn't believe that Bill isn't at the party and she knows (38), then it cannot be that Bill isn't at the party.

This proposal also derives Groenendijk and Stokhof's *de dicto* interpretation of *Mary knows which students are at the party*. Suppose that Mary knows that John is at the party but doesn't know that he is a student. This means that there is some world w compatible with her beliefs where John is not a student. Also in these worlds, (38) is predicted to be false.

Heim points out that her amendment of Karttunen's analysis has an advantage over Groenendijk and Stokhof's semantics, in that it can also derive weakly exhaustive interpretations independently of strongly exhaustive ones, via ans_1 .¹⁹ This kind of flexibility predicts that there can be QEVs which only support weakly exhaustive inferences. The verb *surprise* appears to be one such predicate. Consider the sentence in (40)a. This sentence is intuitively false in the scenario described in (40)b.

¹⁹ In a theory like Groenendijk and Stokhof's where questions are inherently strongly exhaustive, ans_1 cannot be retrieved.

- (40) a. It surprised Ann who called.
 b. *Scenario*: The , individuals , in , the , domain , of quantification are Susan and Bill and Tim. Susan and Bill called, but Tim didn't. Ann expected Susan and Bill to call, but she also expected Tim to call. Mary later discovered that Tim called too.
 c. ans₁ *to who called*:
 That Susan and Bill called *weakly exhaustive*
 d. ans₂ *to who called*:
 That Susan and Bill called, and Tim didn't. *strongly exhaustive*

Surprise indicates that the subject's past expectations relative to the answer of the question it embeds had been false and presupposes that, by the time of evaluation, the subject realizes that they were false. Heim notices that a strong exhaustive semantics for *surprise* would incorrectly make (40)a true in (40)b, while a weakly exhaustive semantics would make it false in this scenario (cf. also Bermann 1991). On the one hand, (40)b meets all the conditions to make (40)a defined and true under a strongly exhaustive semantics. Ann had had false expectations relative to the strongly exhaustive answer to *who called*, she had believed (40)d to be false because she expected Tim to call, but she later learned that Tim called too. On the other hand, (40)b meets the conditions to make (40)a defined but not the conditions to make it true under a weakly exhaustive semantics, because Ann's past expectations relative to the proposition in (40)c, the weakly exhaustive answer, had not been false. She had expected (40)c.

Given this, the intuition that (40)a is false in the scenario described in (40)b can straightforwardly be captured if the semantics of *surprise* expresses false expectations regarding ans₁.²⁰

Since in Heim's analysis, reference to ans₂ is necessary to derive *de dicto* interpretations, this analysis predicts that these interpretations should always be strongly exhaustive as well. Beck and Rullmann (1999) and Sharvit (2002) offer a detailed discussion of this prediction.

Final Remarks

This chapter offered a simplified type driven rendition of the basic results and of some subsequent developments of Karttunen's semantics of interrogatives and question embedding verbs. My rendition is simplified because it excludes some of Karttunen's assumptions, which have subsequently proven to be questionable and/or unnecessary. One of these assumptions is Karttunen's performative view

²⁰ George (2013) argues that the above facts about *surprise* are not sufficient to prove the existence of a weakly exhaustive semantics of questions, suggesting instead that Heim's weakly exhaustive judgments with *surprise* can be derived from a notion of 'mention some' answers; see George (2011) for additional discussion. The topic remains a matter of debate.

of root questions, according to which questions like *who called?* are equivalent to sentences like *I ask you to tell me who called* (see footnote 14). A second one is the restriction to true answers, which I discussed to some extent in Section "Introduction: Questions as sets of (instantial) answers".

In addition, due to space limitations, and in the interest of clarity and formal explicitness, I could not include Karttunen's discussion of 'pair list' interpretations of multiple *wh*-questions and of *wh*-questions containing universal quantifiers.

Finally, I merely touched upon some of the basic issues in the current debates on *de dicto/de re* interpretations, Strong and Weak Exhaustivity and presuppositions of questions.

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Chapter 13

On Kratzer's "The Notional Category of Modality"



Cleo Condoravdi

Abstract A. Kratzer's "The Notional Category of Modality" (NCM) develops a unified analysis of modals and conditionals, which has come to be the classic approach to modality in linguistic semantics and beyond. This chapter presents an overview of NCM's main contributions and clarifies the motivations behind particular analytical choices, concentrating on the role of conversational backgrounds in the theory. It also highlights new research that has enriched the framework or raised outstanding issues.

Keywords Modality · Conditionals · Premise semantics · Context dependence · Conversational background · Modal base · Ordering source

Modality in the Kratzer Framework

In a series of seminal works, A. Kratzer has developed an approach to modals and conditionals that forms the basis of work on modality in linguistic semantics and is a standard of reference for work on modality more generally (Kratzer, 1977, 1979, 1981a, b, 1989, 1991a, b, 2002, 2012).¹ "The Notional Category of Modality" (NCM) develops what has become the standard version of the theory and puts empirical meat on it by explicating the behavior of modals in German. Kratzer's approach to modality is based on the following two tenets.

¹ For overviews and explications of the approach see Frank (1997), von Stechow (2006), Kaufmann et al. (2006), Swanson (2008), Nauze (2008), Portner (2009), von Stechow & Heim (2011), Hacquard (2011), von Stechow (2011), Kaufmann & Kaufmann (2015).

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Tenet 1

The meaning of all modal expressions is to be characterized in terms of necessity and possibility.

Tenet 2

Modal statements express propositions.

Standard and unquestioned in their time, they remain the mainstay of current work although they have more recently been challenged.

Premise Semantics for Modality

Kratzer (1977) lays the groundwork for Kratzer's approach to modality with two major analytical moves. The first move is to dissociate from the lexical meaning of modal expressions the flavor of modality in terms of which we usually describe their interpretation. Such major flavors include among others epistemic, deontic, dispositional (aka dynamic), bouletic (aka preferential), exemplified below (examples from Kratzer (1977)).

- | | | |
|-----|--|-----------------|
| (1) | The ancestors of the Maoris must have arrived from Tahiti | [epistemic] |
| (2) | All Maori children must learn the names of their ancestors | [deontic] |
| (3) | If you must sneeze, at least use your handkerchief | [dispositional] |
| (4) | Rakaipaka must become our chief | [bouletic] |

The claim is that despite the distinct interpretations of (1)–(4), the meaning of *must* remain constant. Epistemic, deontic, dispositional, bouletic are different uses of *must*, determined by the conversational context.

The second move is to connect the semantics of necessity modals and possibility modals to entailment and consistency of their scope – what is nowadays referred to as 'prejacent' – relative to some set of premises, hence the term 'premise semantics'. Premises are propositions, just like the semantic value of the prejacent. Premise sets track in some way the different flavors of modality and they vary across different uses of modals in a way that is both context-dependent and world-dependent.

Kratzer (1977) argues for the first move by showing that there is no limit to the number of distinct interpretations of a modal expression. As Kratzer explains in the following passage, the categories of epistemic, deontic, dispositional, preferential modality are individuated in more fine-grained ways in different uses, which are sensitive to different relativizations of knowledge, evidence, duties, dispositions, or preferences to situations, agents, times, etc.

How many kinds of 'must' do we have to distinguish? How many deontic ones? How many epistemic ones? How many dispositional ones? And how many preferential ones?

Obviously many in each group. We do not just refer to duties. We refer to duties of different kinds; to different duties different persons have towards different persons at different times.

We do not simply refer to a bit of knowledge or information — once and for ever the same. We refer to different kinds of knowledge or information in different situations.

We do not simply consider dispositions. Dispositions change. My dispositions now are not the same as my dispositions two minutes ago.

We do not always refer to the same wishes or preferences when we use a preferential 'must'. Sometimes it is the wish of the Queen, sometimes it is the wish of our tribe, sometimes we consider even our own wish. (Kratzer, 1977, p. 339)

Once you start making fine-grained classifications, you realize that these distinctions do not come from the invariant meaning of the modal expressions themselves. Rather, Kratzer argues, they should be factored out as the contribution of an independent parameter that the interpretation of modal expressions is relativized to.²

Kratzer calls this parameter 'conversational background' and construes conversational backgrounds as functions from worlds to premise sets. They can be supplied overtly by linguistic expressions like *what we know*, *what the University regulations provide*, *what Emma wants*,³ or covertly by the context of use. Relativizing modals to conversational backgrounds and factoring out this parameter as a contextual contribution allows for a uniform semantics across different types of modality.⁴

Different contexts supply different conversational backgrounds. These correspond to the well-recognized types of modality, such as epistemic, deontic and bouletic, as they pertain to the relevant agents, times, etc. The unlimited variety of interpretations comes from the unlimited way in which the high-level categories can be parameterized, e.g., *x's duties to y at time t*. A fixed conversational background delivers different premise sets across different worlds – what Emma wants obviously varies from world to world – which captures the contingency of modal statements.

As a first approximation, a necessity modal relative to a given premise set states that its prejacent is entailed by the premise set, a possibility modal that its prejacent is consistent with the premise set. This cannot be quite right though, because premise sets, by virtue of what they are supposed to model, are not guaranteed to be consistent – rules, laws, desires can certainly be inconsistent – and a semantics based on entailment and consistency with inconsistent premise sets is trivialized. A

² A recent extension to this argument comes from Knobe & Szabó (2013). It is based on their claim that modals can have impure flavors, where normative values are implicated in the interpretation of non-deontic modals. "If modals were ambiguous, it would be natural to posit a list of possible flavors and to assume that any given modal had to fit neatly into one of them. As soon as one gives up the idea that modals are ambiguous and shifts instead to a theory based on context, the assumption of modal purity begins to look suspect. [W]hen one shifts from a theory based on ambiguity to a theory based on context, it is only natural to suppose that there can be impure modals." (p. 24–25).

³ Note that these free relatives are semantically functions from worlds to sets of propositions, so of the same type as conversational backgrounds.

⁴ Groenendijk & Stokhof's (1975) in-depth study of epistemic modals at roughly the same time is a useful point of comparison, which puts the importance of Kratzer's contribution in historical perspective. They also relativize the semantics of epistemic modals to a contextually determined premise set, what they call 'conversational information', but they do not conceive of such a relativization as enabling a unified account of the diverse interpretations of modal expressions.

major contribution of Kratzer's premise semantics is the proposal it incorporates to resolve inconsistencies: it deals with inconsistent premise sets by quantifying over the domain of consistent subsets of the premise set. Necessity modals require of every set in the domain to have a superset in the domain that entails the prejacent. Possibility modals require there to be some set in the domain all of whose supersets in the domain are consistent with the prejacent. Kratzer (1979) extends this type of analysis to conditionals.

The analysis proposed by Kratzer (1977) thus highlights the relativity and context dependence of modal expressions, while accounting for the contingency of modal statements and the duality between necessity and possibility modals.

Relativity The interpretation of modal expressions is always relative: necessary/obligatory/possible/permissible etc. in view of . . .

Context Dependence The parameter that modal expressions are relativized to, the conversational background, is context-dependent.

Contingency The truth of a modal statement can vary from world to world, because the value of the conversational background varies from world to world.

Duality *Can* p iff $\neg Must \neg p$ when *Can* and *Must* are interpreted relative to the same conversational background.

Modal Bases and Ordering Sources

In NCM, Kratzer develops and enriches the analysis and brings under the fold a wider range of modal expressions, as well as conditionals. The main innovation of NCM is to make the interpretation of modals relative to two types of conversational backgrounds with distinct roles. One type is the modal base, the other type is the ordering source. The motivations for making modals doubly relative and distinguishing between modal base and ordering source are many-fold. What they have in common is the need for one conversational background, the modal base, to be given priority over the other, the ordering source. Perhaps the easiest way to appreciate this need is by considering cases of factual constraints on realizing an agent's desires.

Suppose that all that Emma wants is to become mayor and to not hang out at the pub and that, as it happens, she will become mayor only if she hangs out at the pub. This is "the horrible story of someone who wants something but rejects the necessary means leading to the fulfillment of her desires." (NCM, p. 66) Intuitively, in any world in accord with this scenario, (5) is false.

- (5) Considering the relevant circumstances and what Emma wants, she could become mayor and not hang out at the pub.

What we have here is two conversational backgrounds, let's call them f and g , f delivering, for any w , the relevant facts of w , and g what Emma wants in w .

Emma's preferences are not logically contradictory; they are contradictory in view of the facts. $f(w)$ and $g(w)$ can each be consistent but are jointly inconsistent. The proposition $\neg(\text{Mayor} \wedge \neg\text{Pub})$, as a relevant fact of w , is in $f(w)$. Let's assume for simplicity that it is the only relevant fact. The premise set $g(w)$ consists of Emma's two desires, **Mayor** and $\neg\text{Pub}$. If the modal in (5) is relativized to one premise set, the premise set in this case would correspond to the relevant facts together with what Emma wants, $f(w) \cup g(w)$. $f(w) \cup g(w)$ comprises a set of inconsistent propositions, but $g(w)$ is one of its maximally consistent subsets,⁵ which means that (5) comes out true in w . The problem here is that by not separating out the premise set corresponding to the relevant facts from the premise set corresponding to Emma's desires when considering consistent subsets of $f(w) \cup g(w)$, we end up ignoring the factual constraints on realizing the desires. The facts need to be given priority over the desires. We need to ensure, in effect, that the domain of consistent subsets of $f(w) \cup g(w)$ referred to in the semantics of the modal is restricted so as to consist only of supersets of $f(w)$. NCM gets at this indirectly by keeping $f(w)$ and $g(w)$ separate, not relativizing to their union, and having them play distinct roles.⁶

Modal bases map each world to premise sets consisting of propositions true in that world. Therefore, the premise sets delivered by a modal base are consistent. Any consistent premise set determines a non-empty set of possible worlds: the set of worlds verifying all the propositions in the premise set. Ordering sources map each world to premise sets that characterize 'ideals' in that world, including similarity to the world itself, normalcy/stereotypicality, or normative ideals of various kinds. The premise sets delivered by an ordering source are used to define an ordering of comparative possibility between worlds, such as comparative similarity, relative likelihood, relative goodness, relative desirability, etc.

In NCM the semantics of modals is not defined directly in terms of the premise sets delivered by conversational backgrounds. Rather, in line with work in philosophical logic, especially Kripke (1963) and Lewis (1973), it is defined in terms of a domain of accessible worlds and an ordering imposed on these worlds. The role of the modal base is to derive the accessibility relation. For any conversational background cb , there is a relation R_{cb} which pairs up every world w with all and only the worlds in $\cap cb(w)$. The role of the ordering source is to derive the ordering on possible worlds. If $\mathcal{P}$ is any set of propositions, we can define an order on worlds that depends on the propositions in $\mathcal{P}$ they each verify, as in (6). For any $\mathcal{P}$, $\leq_{\mathcal{P}}$ is a pre-order (transitive and reflexive).

$$(6) \quad u \leq_{\mathcal{P}} v \text{ iff } \{p \in \mathcal{P} \mid v \in p\} \subseteq \{p \in \mathcal{P} \mid u \in p\}$$

⁵ If we drop the simplifying assumption that $f(w)$ contains only a proposition that is inconsistent with $g(w)$, then $g(w)$ would not be a maximally consistent subset but any other propositions from $f(w)$ that we add to it would still be consistent with the prejacent of the modal in (5).

⁶ Frank (1997) develops an analysis of "multiply relative modality" more in line with Kratzer (1977). She defines 'compatibility-restricted union': $f(w) \cup !g(w)$ is the set comprised of premise sets, each being the union of $f(w)$ together with a subset of $g(w)$ that is maximally consistent within $f(w) \cup g(w)$.

If $u \leq_{\mathcal{P}} v$, then u verifies all the propositions in $\mathcal{P}$ that v verifies, and possibly more. At each world w , a conversational background cb induces a preorder $\leq_{cb(w)}$ on the set of possible worlds as in (6). It is important to keep in mind that this order does not give us “degrees of closeness” to the relevant ideal; it only gives us a way of comparing worlds and determining if some world is closer than, not as close as, equally close, or incomparable to another one. In the interpretation of modals, we are interested in the ordering $\leq_{cb(w)}$ not throughout the set of worlds, but restricted to the subset of worlds as determined by the conversational background that is the modal base.

The intuitive flavor of a doubly relativized modal depends sometimes on the modal base, sometimes on the ordering source. Kratzer distinguishes between two basic types of modal bases: epistemic and circumstantial. Epistemic modal bases correspond to what is known by some agent(s), or to the available evidence. Circumstantial modal bases are supposed to deliver the ‘relevant facts’. Epistemic uses of modals arise from epistemic modal bases. A deontic, bouletic, or teleological reading arises when the ordering source is, respectively, deontic, bouletic, or teleological. Portner (2009) groups together these readings under the category of ‘priority’ interpretations.⁷ Priority modals are construed with a circumstantial modal base. In ability and dispositional readings the modal base is circumstantial, but, as we will see in section “[Dispositional modality as pure circumstantial modality](#)”, there is to date no characterization of the types of circumstantial modal bases that are associated with such readings. In special cases, a conversational background delivers no premises, that is, its value is the empty set.

The analysis of particular types of modal expressions makes reference to the following formal properties of conversational backgrounds, among others.

Consistency A conversational background cb is consistent if for any world w , $cb(w)$ constitutes a set of consistent propositions, i.e., $\bigcap cb(w) \neq \emptyset$.

Realism A conversational background cb is realistic if for any world w , $cb(w)$ constitutes a set of propositions true at w , i.e., $w \in \bigcap cb(w)$.

Total Realism A conversational background cb is totally realistic if for any world w , $cb(w)$ constitutes a set of propositions fully characterizing w , i.e., $\bigcap cb(w) = \{w\}$.

Emptiness A conversational background cb is empty if for any world w , $cb(w) = \emptyset$, in which case $\bigcap cb(w) = W$ and $\leq_{cb(w)}$ is such that for any v , $u \in W$, $v \leq_{g(w)} u$.

The distinction between modal bases and ordering sources is well suited to account for the compatibility between primary obligations and contrary-to-duty obligations. For example, suppose the traffic regulations in w both prohibit speeding and require that anyone caught speeding pays a fine. Let g be a deontic background encoding traffic regulations at the relevant jurisdiction. Then the propositions that no one speeds, S , and the material conditional that if someone speeds, that person

⁷ “The idea behind the term “priority” is that such things as rules, desires, and goals all serve to identify some possibility as better than, or as having higher priority than, others.” (p. 135)

pays a fine, $S \supset F$, are in $g(w)$. Suppose further that John was caught speeding in w . Let f, f' be circumstantial backgrounds such that $f(w)$ contains the proposition that John was caught speeding and $f'(w) = \emptyset$. Then, as indicated in (9) and (10), the truth of (7) and (8) depends on which of the two circumstantial backgrounds is given as the modal base for the modal *ought to*. (7) is a primary obligation, independent from the facts. (8) is a contrary-to-duty obligation, dependent on the violation of a primary obligation. Once an obligation is violated and this fact is taken into account by the contextually supplied circumstantial modal base, then optimal worlds relative to $g(w)$, where both $\neg S$ and $S \supset F$ are verified (the latter guaranteed true in any $\neg S$ -world), are not in the set of worlds determined by the modal base.

- (7) There ought to be no speeding.
- (8) John ought to pay a fine.
- (9) a. $\text{OUGHT}_{f,g}(\text{No one speeds})$ [False in w]
 b. $\text{OUGHT}_{f',g}(\text{No one speeds})$ [True in w]
- (10) a. $\text{OUGHT}_{f,g}(\text{John pays a fine})$ [True in w]
 b. $\text{OUGHT}_{f',g}(\text{John pays a fine})$ [False in w]

More crucially, the separation of modal bases and ordering sources allows for stronger and weaker notions of necessity and possibility and, more importantly, for graded modality. To begin with, necessity and possibility relative to one premise set, as in Kratzer (1977), discussed in section "[Premise semantics for modality](#)", now becomes a special case of necessity and possibility relative to some modal base and an empty ordering source. NCM refers to these as 'simple necessity' and 'simple possibility'. Relative to a non-empty ordering source we get what NCM refers to as 'human necessity' and 'human possibility'.

- (11) A proposition p is a human necessity in a world w with respect to a modal base f and an ordering source g iff
 for all $u \in \bigcap f(w)$, there is a $v \in \bigcap f(w)$ such that
 a. $v \leq_{g(w)} u$ and
 b. for all $z \in \bigcap f(w)$: if $z \leq_{g(w)} v$, then $z \in p$
- (12) A proposition p is a human possibility in a world w with respect to a modal base f and an ordering source g iff $\neg p$ is not a human necessity in w with respect to f and g .

The reason for the complexity of (11) is the need to allow for infinite chains of worlds closer and closer to the ideal. The idea about the type of necessity that (11) is meant to capture is that as we move across worlds in each linearly ordered chain as determined by the ordering $\leq_{g(w)}$, we will eventually reach a world such that all the worlds beyond it on that chain (i.e., closer to the ideal) are p -worlds. A non-empty ordering source, in effect, restricts the domain of quantification to a subset of the set of worlds determined by the modal base. Human necessity is thus weaker than simple necessity and human possibility is stronger than simple possibility.

The ordering on possible worlds induced by ordering sources also serves to define relations of comparative possibility between propositions, and graded

notions of modality on the basis of them. The semantics of gradable modal expressions (abstractly, as sentential operators) is then identified with these graded notions of necessity and possibility, making them quantificational expressions. *Probably*, *likely*, *more likely than* are analyzed as epistemic modals relying on a qualitative notion of likelihood. *Good* and *better than* with sentential complements are analyzed as priority modals relying on a qualitative notion of desirability or goodness of some kind (e.g., moral/ethical, goal achievement, rule adherence). (13), from Kratzer (1991b),⁸ lists different graded types of modality along with modal expressions from the epistemic domain exemplifying them.

- | | | |
|------|--------------------|---|
| (13) | human necessity | <i>must</i> |
| | weak necessity | <i>probably</i> |
| | human possibility | <i>there is a good possibility that</i> |
| | possibility | <i>might</i> |
| | slight possibility | <i>there is a slight possibility that</i> |
| | better possibility | <i>is more likely than</i> |

Analyzing gradable modal expressions as quantificational expressions based on qualitative comparative notions is now one of the most controversial aspects of the Kratzer framework, see especially Yalcin (2010), Lassiter (2011, 2015, 2017), Klecha (2014). Kratzer motivates analyzing expressions of probability and gradable possibility as belonging to the category of epistemic modals on the basis of the observation that the two types of expression are semantically related by entailment (e.g., *must p* entails *probably p*, *there is a good/slight possibility that p* entail *might p*), consistency (e.g., between *might p* and *there is a slight possibility that ¬p*), or inconsistency (e.g., between *must p* and *there is a slight possibility that p*). Another motivation for this analytical grouping is that when conditionalized, they can receive the same analysis as conditionalized modals.

Gradable epistemic modals must be construed with a non-empty ordering source, but can regular epistemic modals like *must*, *might* and *can* (which allows for an epistemic reading under negation) express simple necessity and possibility? An epistemic *must* construed with an empty ordering source ought to entail its prejacent, since its modal base is realistic. Karttunen (1972) originally observed two key properties of epistemic *must*: weakness and indirectness. In uttering (14a), a speaker appears to be making a weaker claim compared to uttering (14b), and indicates lack of direct evidence about the phone's whereabouts. (14a) would be infelicitous, for instance, in a context where the speaker is in direct visual contact with the phone in the kitchen and fully aware of this fact.

- (14) a. The phone must be in the kitchen.
 b. The phone is in the kitchen.

⁸ NCM's 'human necessity' and 'human possibility' correspond to Kratzer's (1991b) 'necessity' and 'good possibility'.

NCM takes Karttunen's observation about the relative strength of (14a) and (14b) as an argument for ordering sources and assumes that epistemic *must* requires a non-empty ordering source. On this view, the weakness of *must* is of semantic origin, while its indirectness is a pragmatic implicature. The question of the source of the apparent weakness of *must* is now under active debate, especially since von Fintel & Gillies (2010) turned the tables around by defending a truth-conditional content for *must* as simple necessity and by encoding indirectness into its meaning (Lassiter, 2016; Goodhue, 2017; Mandelkern, 2019; Pinal & Waldon, 2019; Roberts, 2019; von Fintel & Gillies, 2021).

In order for the various strengths of necessity and possibility to not collapse, the set of worlds determined by the modal base must contain infinite chains of worlds as determined by the ordering induced by the ordering source. This is a reason why Kratzer rejects a strong version of the limit assumption, which requires that every linearly ordered chain terminates with worlds that are closest to the ideal.⁹

NCM, Kratzer (1991b), and Kratzer (2012) define different relations of comparative possibility (see also Katz et al. 2012). Kratzer (2012) argues that different applications might call for different types of relations and suggests that the range of relations of comparative possibility goes well beyond what has been proposed already. As an example of the first point, she contends that the relation of comparative possibility given in NCM, meant for epistemic comparative possibility and criticized by Yalcin (2010) as yielding problematic inferential properties for probability expressions, might be better suited for moral reasoning.

Finally, NCM provides a uniform account of conditionalized modals. Following Lewis (1975) and building on Kratzer (1979), Kratzer argues that there is no dyadic conditional operator corresponding to *if ...*, *then*. A crucial part of the argument is that such an operator cannot be made to interact properly with the modal in the consequent (see also Kratzer 1991a) for further arguments, as well as Gillies (2010) and Kaufmann & Kaufmann (2015) for proposals on how a modal conditional operator can interact with a modal in the consequent without unwelcome consequences). *If*-clauses do not introduce an operator of their own, but rather, they restrict some operator that is independently present, like a modal in the consequent. This is known as the 'restrictor' approach to *if*-clauses. Semantically, an *if*-clause is a modifier of a modal, operating on its modal base parameter: the proposition that is the semantic value of the antecedent is added to the premise set delivered by the modal base for each world.

The restrictor approach has two features that distinguish it from a conditional operator analysis. One feature is that the same modal can be restricted by more than one *if*-clause. This guarantees the equivalence between the nested conditional in (15a) and the conditional with a conjunctive antecedent in (15b).¹⁰

⁹ A weaker version of the limit assumption requires only that there are closest worlds, which implies that some chain terminates. See Kaufmann (2017) for discussion.

¹⁰ This equivalence is known as the Import-Export Principle. See Kaufmann & Kaufmann (2015) for discussion of the principle under different assumptions about the interpretation of

- (15) a. If Mary got home early, then if it was not raining, they all must have gone out for a walk.
 b. If Mary got home early and it was not raining, then they all must have gone out for a walk.

The other feature is that one *if*-clause can restrict multiple modals and at a distance, as pointed out by Kratzer (2012, ch. 4) with examples of modal subordination as in (16).¹¹

- (16) If a wolf entered the house, he must have eaten grandma, since she was bedridden. He might have eaten the girl with the red cap, too. In fact, that's rather likely. The poor little thing wouldn't have been able to defend herself.

For conditional sentences that contain no modal or other quantificational expression, like bare indicative conditionals, Kratzer assumes that there is a covert epistemic necessity operator. Special construals of conditionals can then be characterized in terms of the types of modal base and ordering source the modal operator is relativized to. A material implication is characterized by a totally realistic modal base and an empty ordering source. A strict implication is characterized by an empty modal base and an empty ordering source. A counterfactual is characterized by an empty modal base and a totally realistic ordering source (uniquely characterizing any given world).

A class of conditionals with priority modals in the consequent pose a special problem, because the hypothesized proposition affects in some way the premise set delivered by the ordering source of the priority modal (what Condoravdi & Lauer (2016) call 'the problem of conditioning on norms'). This problem was originally pointed out by Frank (1997) for conditionals which hypothesize something about the law, and independently by Sæbø (1985, 2001) for anankastic conditionals. Such cases are illustrated in (17).

- (17) a. If the new laws for opening hours of shops go through, salespeople will have to work longer. (Frank, 1997)
 b. If jaywalking is illegal in this town, that guy over there has to be punished. (von Stechow & Iatridou, 2005)
 c. If you want to go to Harlem, you have to take the A train. (Sæbø, 2001)

Frank (1997) proposed that modalized conditionals can have a nested modal structure: the outer modal is a covert modal operator, like the one that Kratzer

the conditional and an illuminating comparison between the restrictor approach and a modal conditional operator approach.

¹¹ "In [(16)], the *if*-clause restricts four modal expressions in a row: *must* in its own clause, *might* in the second clause, *rather likely* in the third clause, and *would* in the fourth clause. This suggests that natural languages have overt or non-overt anaphoric mechanisms that allow modal expressions to pick up possibilities introduced in preceding discourse." Kratzer (2012, p. 94)

assumes for conditional sentences without an overt modal, and it is the one whose modal base is restricted by the *if*-clause; worlds in which the antecedent is true then provide the world of evaluation for the inner, overt modal. Importantly, in the case of conditioning on norms, as in the examples in (17), the truth of the antecedent at a world *w* affects the value at *w* of the ordering source for the overt priority modal. For instance, in (17b) the outer modal operator quantifies over (a subset of) worlds in which *jaywalking is illegal* is true, and the value of any relevant deontic conversational background at those worlds would include the proposition that no jaywalking occurs and propositions about what ensues if it occurs. The analysis of such conditionals relies on an alignment between certain facts of a world *w* and the premise sets for the relevant ordering sources (deontic, preferential or teleological) at *w*. Such an alignment is present independently, and in principle, it can result in a dependence between the premises in a modal base premise set and those in an ordering source premise set. As we will see in section “[Adverbial phrases and modal flavor](#)”, it can also make priority-oriented adverbial phrases capable of targeting either the modal base or the ordering source parameter of a modal.

On the analysis of modality proposed by Kratzer and outlined above, the quantificational force of a modal is determined by its lexical meaning. Its flavor is determined by context. However, not all modals have the entire range of flavors.¹² The range of acceptable flavors for a modal also has to be determined by its lexical meaning, via constraints on the types of conversational backgrounds it can be construed with. For instance, the lexical meaning of *might* and of *probably* specifies their force as possibility and weak necessity, respectively, and requires the modal bases they are construed with to be epistemic and the ordering sources stereotypical.¹³ On the strongest version of this bare-bones semantics for modals, then, we would expect that any reference to the flavor of a modal in its meaning specification would be confined to the constraints on the types of conversational backgrounds it can be construed with, and that the flavor of a modal on a particular use will be determined by the type of conversational backgrounds the context assigns to it. As discussed earlier, sometimes it is the modal base that determines the type of modality and sometimes the ordering source: an epistemic modal base makes the modal epistemic, a priority ordering source makes the modal deontic, bouletic, or teleological.

¹² Hacquard (2013) points out that lexical modals generally have a more limited range of interpretations than modals that are function words, like modal auxiliaries, but even the latter exhibit restrictions.

¹³ A non-empty ordering source is essential for *probably*, since it is a gradable modal. Whether *might* allows for an empty ordering source depends on one's view of the domain of epistemic modality.

Adverbial Phrases and Modal Flavor

Adverbial phrases like *in view of / given / considering X*, minimally, give clues as to what kind of conversational backgrounds the speaker has in mind in using a modal and can help get a fix on the modal base or the ordering source parameter. For example, in (18a) the adverbial pertains to the modal base of the modal *should*, while in (18a) it seems to fix the ordering source.

- (18) a. Given/In view of the accident he caused, he should be arrested.
 b. Given/In view of the regulations, he should be arrested.

Missing is an explicit semantic treatment of such adverbial phrases, their effect on a parameter of a modal, and, more generally, how they resolve the context-dependency of the modal. Do they simply reset the value of a parameter, or do they modify the value that is fixed by context? Two works that are more explicit about this question are Portner (2009) and Knobe & Szabó (2013), but their discussion is mostly for illustrative purposes. Portner treats them as resetting the contextually supplied value. Knobe & Szabó treat them as restricting the modal base. These proposals still leave open the internal compositional semantics of the adverbials – for instance, how they provide something of propositional type when their complement phrase is not clausal, as in (18) – and whether their uses as modifiers of modalized as well as unmodalized clauses, shown in (19), can be semantically unified.

- (19) a. Given/In view of the accident he caused, we did not renew his contract.
 b. Given/In view of the regulations, we did not renew his contract.

As part of an argument that the type of modality is not determined by conversational backgrounds, Nauze (2008) shows that an adverbial phrase making reference to a deontic conversational background is not sufficient to determine the flavor of the modal as deontic. He provides the example in (20a), which in the given context leads to an epistemic reading for the modal *may*. Nauze's observation is further supported by the examples in (20b) and (20c). *Might*, which has no deontic reading, is fine in the variant (20b), and so is *must* in (20c) in a context in which the verdict is assumed to have already been reached (in *must have* the modal only has an epistemic reading; see Ninan (2005) for discussion and a potential explanation). The examples in (20) are about potential outcomes of the trial, and thus about how the world turns out, not about what ought to be the case.

- (20) [We are discussing potential outcomes of the trial of the suspected criminal Jockl]
 a. In view of what the law provides, Jockl may be executed.
 b. In view of what the law provides, Jockl might be convicted.
 c. In view of what the law provides, Jockl must have been convicted.

Nauze concludes that "contextually given conversational backgrounds fix the content against which a modal proposition will be evaluated but do not fix the type of modality" (p. 157). Yalcin (2016) makes similar observations, but comes to a more cautious conclusion that "the connection between the interpretation of modals and *in view of*-phrases is loose and indirect at best" (p. 236).

As undeniable as Yalcin's conclusion is, I think we can say something more systematic about what is going on in cases like (20). Analogously to the cases of conditioning on norms, a premise set that is the value at world w of a conversational background about the law can consist of the content of the law at w , or it can consist of facts of w about what the law prohibits and what punishments it provides for in cases of violation. In the former case, it would include propositions of the form that no activity of type X occurs, and that if activity of type X occurs, there is punishment of type Y ; in the latter case, it would include propositions of the form that the law prohibits activities of type X , and that the law provides for punishment of type Y for activities of type X . The adverbial phrase in (20) is of the latter type, and as such it helps determine the modal base, not the ordering source, of the modals. The fact that the law has the provisions that it has is part of the relevant evidence, just as facts about Jockl and the crime he is accused of are part of the relevant evidence. On this view then, the modal base determines the flavor of the modals in (20) as epistemic, contrary to Nauze's conclusion.

Conversational Backgrounds

As we saw in section "[Premise semantics for modality](#)", the contextually supplied values of conversational back- grounds are temporally relativized. Temporal relativization is not just a matter of context, it is also dependent on the compositional make-up of a modal or conditional statement. A lot of work has focused on figuring out how the temporal parameter of conversational backgrounds is determined by the interaction of modals with tense and aspect, and also on tying the interpretation of different types of conditionals, especially indicative vs. subjunctive conditionals, more directly to their compositional make-up (Crouch 1993, Enç, 1996; Abusch, 1997; Condoravdi, 2002; Ippolito, 2004; Kaufmann, 2005; Ippolito, 2006; Hacquard, 2006; Arregui, 2007; Schulz, 2008; Arregui, 2009; Hacquard, 2010; Romero, 2014; Khoo, 2015; Klecha, 2016; Rullmann & Matthewson, 2018; among many others).

As we saw in section "[Modal bases and ordering sources](#)", premise sets are used to define accessibility relations and orderings between worlds, which the truth conditions of modal expressions make reference to. In going from a set of propositions to an accessibility relation or an ordering we lose information, since the mapping is not one-to-one, but rather many-to-one. The extra structure that premise sets provide has been critical to Kratzer's work on counterfactuals, which centers around characterizing the type of totally realistic ordering sources invoked in the interpretation of counterfactuals (Kratzer, 1981b, 1989, 2002, 2005, 2012: ch. 5).

Otherwise, it remains a mostly untapped resource of the framework. Exceptions include von Fintel & Gillies (2010, 2021) on (non-gradable) epistemic modals; von Fintel & Iatridou (2005, 2008) on the difference between (non-gradable) strong and weak necessity modals (*must*, *have to* vs. *should*, *ought to*), appealing, in effect, to ranked premise sets; Katz et al. (2012) on combining ordering sources for a variety of applications, including cases that are on the face of it serious challenges for the Kratzer framework; Kaufmann (2013) on developing causal premise semantics for counterfactuals; Khoo (2015) on reformulating Condoravdi's (2002) version of the diversity condition (a constraint on the assignment of modal bases to modals) in terms of premise sets.

For example, von Fintel & Gillies (2010) reconcile their claim that non-gradable epistemic modals express simple necessity and possibility with Karttunen's observation about the weakness of *must* by exploiting the difference between premise sets not closed under logical consequence and the set of worlds they determine.¹⁴ They attribute to epistemic modals a presupposition requiring that the premise set delivered by an epistemic modal base does not settle whether the prejacent is true, i.e., no proposition in the premise set either entails the prejacent or is incompatible with it.¹⁵ Epistemic *must* is truth-conditionally strong, thus entailing its prejacent, but its presupposition, generally, makes it infelicitous in contexts in which the bare prejacent is assertable, and felicitous in contexts in which the bare prejacent is unassertable.¹⁶

NCM distinguishes between two kinds of realistic conversational backgrounds – epistemic and circumstantial – and underscores the difference between epistemic and circumstantial premise sets.

There is a clear intuitive difference between the two kinds of occurrences of modals which I grouped under the two headings [epistemic and circumstantial]. It is a difference in the kind of premises from which we reason. If we use an epistemic modal, we are interested in what else may or must be the case in our world, *given everything we know already*. And if we use a circumstantial modal, we are interested in what can or must happen, *given circumstances of a certain kind*. Circumstances of a certain kind are facts of a certain kind. Facts concerning the outside world, our bodies or our mind, for example. Usually, circumstances permit or exclude that certain things happen. Only sometimes do they necessitate an event or an action: We have to die, to cough, to vomit, to laugh, to cry or to realize that we are lost.

Epistemic modality and circumstantial modality involve a different categorization of the facts. The problem is now to find out some more details about this partition. (NCM, p. 52)

¹⁴ It is worth pointing out that the same kind of analysis was proposed by Groenendijk & Stokhof (1975) to account for a host of observations about epistemic modals in Karttunen (1972).

¹⁵ This is one of two non-equivalent characterizations of a premise set settling a proposition that von Fintel & Gillies give.

¹⁶ The presupposition is stated as a definedness condition for truth value in von Fintel & Gillies's analysis and is specific to epistemic modals. It is thus in conflict with the strong hypothesis that the meaning specification of modals makes no reference to their interpretive flavor, except for the restrictions on the conversational backgrounds they can be construed with. Khoo (2015) points out that it is a consequence of the diversity condition, and hence there is no need for a presupposition specific to epistemic modals.

A category of modality which, like epistemic modality, is determined by the modal base is dispositional modality. The distinction between epistemic and dispositional interpretations is, of course, very clear, but characterizing the distinction analytically within the Kratzer framework depends on an appropriate characterization of the relevant circumstantial backgrounds and that has proven to be elusive.

Dispositional Modality as Pure Circumstantial Modality

NCM reduces dispositional modality to pure circumstantial modality, an interpretation that arises when a modal is construed with a circumstantial modal base and an empty ordering source.¹⁷ An individual's dispositions, abilities, or skills constitute inherent properties of that individual and that an individual has a particular property in a world is a fact of that world. Semantically, there is no sharp dividing line between the *can* of ability and the *can* of opportunity. Their distinction is a matter of whether the premises delivered by the modal base are only about an agent's skills or whether they are also about external circumstances.

The continuum between ability and opportunity readings, along with the role of inherent properties and external circumstances in determining the particular flavor of *can*, are well demonstrated by (21). (21a) can convey that I have learned how to swim, or that my physical condition is good enough to allow me to swim, or that some free time and being near a body of water will give me the opportunity to swim. (21b) can be a testament to her special driving abilities, the power of her car, or the thinness of the layer of mud.

- (21) a. I can swim this summer.
b. She can drive through this mud.

Similarly, (22) can be used to convey that he knows or doesn't know how to play the trombone; that his physical condition, such as the condition of his lungs, makes it possible or impossible for him to play; that the trombone is within or out of reach.

- (22) He can/cannot play the trombone.

What shifts from context to context — what NCM refers to as the "slippery matter" of "the kinds of facts we take into account for circumstantial modality" (p. 53) — is how inclusive the premise sets returned by related circumstantial backgrounds are. In the first case, the circumstantial background takes into account only the agent's know-how, ignoring his physical condition and the instrument's availability; in the second case, it takes into account both the agent's know-how and

¹⁷ The issues regarding dispositional modality discussed in this section are orthogonal to whether the ordering source must be assumed to be empty. Rather they have to do with how to circumscribe the type of modal bases involved in dispositional/ability readings.

physical condition but ignores the instrument's availability; in the third case, it takes into account all three.

Hydrangeas and Circumstantial vs. Epistemic Modality

In NCM, Kratzer sought to characterize the difference between epistemic and circumstantial backgrounds, but was ultimately unable to provide a positive characterization of circumstantial backgrounds. A hallmark of circumstantial backgrounds that Kratzer's discussion focusses on is that they deliver premise sets that may include just a subset of the known facts.

In using a circumstantial modal, we neglect certain kinds of facts.

...

Again, we have to neglect certain facts for [(23)], although we might be aware of them.
(NCM, p. 53)

This particular attribute of circumstantial modal bases is motivated by a contrast between dispositional *can* and epistemic *might*, famously illustrated by (23) and (24).¹⁸ Suppose we know that no hydrangea seeds ever made their way here but soil and climate are favorable to hydrangeas. (24) is false in any world w that accords with this scenario, while (23) can be true.

- | | |
|--|-----------------|
| (23) Hydrangeas can grow here. | [True in w] |
| (24) There might be hydrangeas growing here. | [False in w] |

The epistemic premise set and the circumstantial premise set share the facts about the type of soil and the type of climate in w . On the other hand, the fact $\neg$ Seeds of w enters the epistemic premise set, but it stays out of the circumstantial one. This allows for (23) to be true, while (24) is false. Given the way the scenario is presented, in this case the epistemic background delivers a superset of the set of propositions that are delivered by the circumstantial background. Hence, the set of worlds corresponding to the epistemic modal base is more restricted than the set of worlds corresponding to the circumstantial modal base. When $cb^{circ}(w) \subset cb^{epi}(w)$, as in this scenario, it will be the case that $\bigcap cb^{epi}(w) \subset \bigcap cb^{circ}(w)$ and this pattern of truth and falsity can emerge.

Do epistemic premise sets always bound circumstantial ones? NCM does not explicitly address this question, but it is important to get clear on the answer, since it is consequential for the relation between epistemic and circumstantial possibility.

¹⁸ The hydrangea set appears in Kratzer (1991b). NCM makes the point with German examples. (23) is a generic sentence, while (24) is not. As is often done in discussions of the contrast, I put aside this difference here.

Nauze (2008) construes the Kratzer framework as being committed to a positive answer,¹⁹ which has the undesirable consequence that epistemic possibility entails circumstantial possibility.

This argument [about how the pattern of truth and falsity in (23) and (24) arises] is, as far as I can see, making the right diagnosis: when concerned with circumstantial ability, only certain facts (from our knowledge) are relevant. However, even though the diagnostic is essentially right, it has unwarranted consequences in the framework.

... [W]ith the same kind of reasoning we can conclude that *might* entails *can*: if something is epistemically possible, it is a circumstantial possibility too. If I think that hydrangeas might be growing here, it would seem that I would better think that they can grow here too. Unfortunately, this line of reasoning is not valid and it poses a very serious problem for the standard [i.e., Kratzer's] framework. (p. 162)

Nauze considers a scenario where we know less than in the scenario above. Suppose it is known that hydrangeas need a temperate climate and an alkaline soil to grow, the climate is known to be temperate, earlier inhabitants of the place are known to have had hydrangea seeds (which they might or might not have planted), but where there is uncertainty about the soil – it could be alkaline or acid. He concludes:

In this situation, sentence [(24)] is true. It is possible that the soil is alkaline, that the pilgrims planted some hydrangeas and thus that they still grow in the region. However, sentence [(23)] is not true in the present situation (as would follow in the standard [i.e., Kratzer's] framework); it is not false either but it is undecided awaiting for further knowledge about the soil's pH. . . .

The problem is that any circumstantial modal base based on an epistemic modal base compatible with a proposition *p* will also make this proposition true. (p. 163)

This would, of course, be a serious problem. Nauze takes it to be an argument in favor of a non-uniform analysis of epistemic and ability modals. But the assumption that circumstantial premise sets contain no more information than epistemic ones is not a necessary one. The lesson to draw from Nauze's critique is instead that there are facts that **must** enter a circumstantial premise set – not any fact can be ignored – and that such facts are not restricted to known facts.

In the setup of the two scenarios, the status of the soil's pH is different from the status of the availability of the seeds. In the original scenario, the unavailability of seeds, $\neg$ Seeds, is a fact that is not in the circumstantial premise set, thus allowing for the possibility that Seeds, and hence for the truth of (23). In Nauze's scenario, in a world *w* in which we do not know the soil's pH, $cb^{circ}(w)$ still includes the fact about the soil's pH. If the soil is alkaline in *w*, $cb^{circ}(w)$ includes the proposition $\neg$ Alkaline and (23) is true in *w*. If the soil is acidic in *w*, $cb^{circ}(w)$ includes the proposition Alkaline and (23) is false in *w*. Neither Alkaline nor $\neg$ Alkaline are

¹⁹ Frank (1997) appears to have the same understanding, as suggested by the following: "So while in the epistemic reading the conversational background specifying our knowledge about facts in the world is taken into account as a whole, the circumstantial conversational background neglects some of these facts." (p. 35).

included in $cb^{epi}(w)$. On this view, (23) has a determinate, albeit unknown, truth value in w .

What Nauze takes to be a matter of indeterminacy of truth value is a matter of epistemic indeterminacy.

If there are facts that must enter a circumstantial premise set, this suggests that there is ultimately a way of characterizing backgrounds for pure circumstantial modality.

It is worth pointing out that in her 2012 book, Kratzer goes in the opposite direction and concludes that looking for such a characterization was misguided:

It now seems to me a hopeless enterprise to try to characterize formal objects like conversational backgrounds as “circumstantial” versus “epistemic.” Both types of backgrounds are functions that map possible worlds to sets of factual premises. What is it that would allow us to single out some of those functions as epistemic, but not circumstantial, or the other way round? There don’t seem to be any characteristic properties that could produce such a distinction (see Nauze (2008) for an insightful objection along those lines). (p. 24)

Kratzer frames this discussion not in terms of the interpretive distinction between pure circumstantial vs. epistemic modality, but in terms of the related grammatical distinction between root and epistemic modals. She then suggests that we can make sense of the interpretive difference along the lines that Hacquard (2006, 2010) has proposed to link conversational backgrounds to grammar: the difference between epistemic and circumstantial modal bases is due to the different kinds of events which provide the “anchors” from which modal bases are projected.²⁰ But use of events on its own does not guarantee the interpretive distinction that we need for Nauze’s case. Suppose we take a possible event of hydrangeas growing around here and fix how things stand in the actual world with respect to that event’s participants. The question remains, which participants should be taken into account and which ones can be ignored? The seed is a participant in any hydrangea growing event. But (23) can be true if no hydrangea seed ever made it into the ground, or even to our present location. Just like Kratzer’s original analysis, the view of modal bases as projected from anchors needs to be supplemented with an analysis of the type of content encoded in circumstantial premise sets and the role of context in selecting the backgrounds that supply them.

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²⁰ Hacquard’s project was to explain the interdependence between a modal’s structural position and its possible flavors. She proposed that conversational backgrounds take events, rather than worlds, as arguments and that those events are determined by the local structural configuration in which the modal appears.

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Chapter 14

On Krifka's "Nominal Reference, Temporal Constitution and Quantification in Event Semantics"



Hana Filip

Abstract Krifka, in his paper “Nominal reference, temporal constitution and quantification in event semantics”, provides the first formal mereological (algebraic) analysis of the relation between nominal reference and temporal constitution (also based on his 1986 PhD thesis). The focus is on two manifestations of this relation in the grammar of natural languages. First, as many observed, there are direct structural analogies between the following two sets of distinction: namely, mass/count and atelic/telic. They are clearly reflected in their parallel cooccurrence patterns with quantifiers, numerical and measure expressions. Second, nominal reference and temporal constitution interact and mutually constraint each other in the derivation of meaning of complex verbal predicates. One key example is *aspectual composition(ality)* e.g., *eat soup* (atelic) versus *eat two apples* (telic). In order to provide an adequate analysis of the relevant data Krifka's principal innovation is to assume a single join semi-lattice structure, undetermined with respect to atomicity, relative to which he defines two higher-order, cross-categorical predicates for reference types of natural language predicates: namely, quantized and cumulative. Specifically in the case of aspectual composition, the interactions and mutual constraints between the structure of objects and eventualities stem from the systematic mappings (homomorphisms) whose source is the lexical semantics of verbs. Such mappings are also independently motivated by other phenomena exhibiting systematic interactions objects and eventualities.

Keywords Mereology · Grammatical aspect · lexical aspect · Nominal reference · Aspectual composition · Measure function · Event semantics

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An Overview

Main Issues and Challenges

The term ‘temporal constitution’¹ subsumes the telic/atelic distinction (Garey, 1957). Intuitively, a verbal expression (a lexical V, a complex verbal predicate or a sentence) is telic if its denotation includes a terminal point (e.g., *recover (from an illness)*, *solve the puzzle*, *walk a mile*, *enter the room*), and atelic if it does not (e.g., *walk*, *rain*, *stand*). Given that Krifka (1989, p. 75) aligns the telic/atelic distinction with Vendler’s (1957) accomplishment/activity distinction, it is plausible to view *temporal constitution* as implicitly corresponding to *aspectual class* (Dowty, 1979, building on Johnson, 1977) or *situation aspect* (Smith, 1991/1997), which are commonly taken to concern the classification of verbal predicates based on the internal temporal structure, or ‘temporal contours’ (Rappaport Hovav, 2008, and references therein), of their denotata.

The relation between nominal reference and temporal constitution is manifested in two ways in the grammar of natural languages, according to Krifka (1989). First, there are parallels based on the intuition that singular count nouns (or NPs) ((a) *cat*) and telic predicates have discrete countable entities, or atoms (Bach, 1986), in their denotation, while prototypical mass nouns (or NPs) (*beer*, *air*, *mud*) and atelic predicates do not (they take their meaning from the non-atomic domain, Bach, 1986).² One of Krifka’s key contributions is to define this parallelism in formal (mereological) terms. Observations regarding this parallelism have a long tradition across virtually all schools of thought in linguistics and philosophy (Krifka, 1989 cites Leisi, 1953; Taylor, 1977; Bach, 1986, for instance). In general terms, and assuming that telic predicates denote (sets of) EVENTS and atelic ones PROCESSES (Bach, 1981; Parsons, 1990, and references therein), we have the following structural analogy: EVENT: PROCESS = OBJECT: STUFF (Bach, 1986)

¹ The term ‘temporal constitution’ is a translation of the German ‘Zeitkonstitution’, coined by François (François, 1985).

² The use “count nouns (or NPs)” and “mass nouns (or NPs)” here reflects the fact that there is no general agreement concerning the level of linguistic description on which these categories are grammatically relevant. Starting with Verkuyl (1971, 1972) at least, there have been debates whether the telic/atelic distinction, and subsequently also the mass/count distinction (Pelletier & Schubert, 2002), are distinctions that are relevant at the level of lexical items at all, and whether they should instead be viewed as distinctions at the level of phrasal constructions, as constructional properties of NPs, VPs and sentences.

(see Rothstein's commentary on Bach, 1986, this volume).³ (States are set aside, given their 'atemporal', or non-temporal, character, see e.g., Bach, 1981, 1986).

Nominal and verbal predicates also exhibit parallels in their co-occurrence patterns with quantifiers, numerical and measure expressions:

- (1) a. There was a lot of beer in the barrel. $\approx$ The baby slept a lot today.
- b. There were three books on the table. $\approx$ He knocked three times on the door.

This led to proposals that verbal predicates, just like nominal predicates, have the grammatical feature 'count' or 'mass' (see e.g., Allen, 1966; Leech, 1969; Verkuyl, 1971, 1972; Gabbay & Moravcsik, 1973; Bolinger, 1975; Mourelatos, 1978/81), although there is no general agreement on whether these are lexical or constructional/phrasal features (See fn.2.)

Krifka (1989) gives new impetus to this research tradition by arguing that the basic distinction between singular count nouns (denoting sets of singular OBJECTS) and mass nouns (denoting STUFF) in the nominal domain is insufficient to account for all the relevant semantic and distributional analogies with the verbal domain. Among others, telic predicates parallel not only singular count predicates (*(an/one/the) cat*), but also counting constructions (*three cats*) and measuring constructions (*a glass of beer, three pounds of apples*), while atelic predicates are similar not only to mass predicates (*beer*) but also to bare plurals (*cats*). The basic structural analogy EVENT: PROCESS = OBJECT: STUFF is insufficient to account for such parallels, along with its grounding in the basic atomic/non-atomic distinction that is presupposed by the lattice theory of Link (1983, 1987) (see also Champollion and Brasoveanu, this volume). It is one of Krifka's main theses that such parallels are best analyzed in terms of two new lattice-theoretic properties of quantization and cumulativity (see below), which leads him to revisions and extensions of Link's and Bach's theories.

Second, the relation between nominal reference and temporal constitution (telicity) is manifested in their interactions and mutual constraints, a phenomenon widely-known as *aspectual composition(ality)* (coined by Verkuyl, 1971, 1972).⁴

³ For Krifka (1989, 1998), these two sets of distinctions are taken to be semantic properties of nominal and verbal predicates, rather than being inherent in entities in the domain, in the external world. There are debates and numerous misunderstandings concerning the status of these categories. The key question is whether these are ontological categories (Bach, 1986; Parsons, 1990, i.a.) or whether these are categories that are properties of verbal and nominal predicates (Krifka, 1986, 1989; Filip Filip, 1993a, b; Partee, 1999, i.a.) (See Filip, 2011, 2012 for summaries of these debates.)

⁴ There is a long-standing tradition of observations related to aspectual composition(ality), which can be traced to the nineteenth century philology (e.g., Streitberg, 1891). Some notable precursors are Poutsma (1926) and Jacobsohn (1933) (cited by Verkuyl, 1971, 1972, also 2005 and elsewhere, who credits them as the major sources of inspiration for his theory of aspectual compositionality), Garey (1957) (inspiration for Filip, 1985 and Krifka, 1986) and Leech (1969, p.137) who speaks of 'semantic concord' between nominal arguments and complex verbal predicates.

Krifka's examples are given below, along with his judgments marked as '*' (see (2a,b) in Krifka, 1989):

- (2) a. Mary drank beer (for 10 min)/(*in 10 min). ATELIC
 b. Mary drank a glass of beer (*for 10 min)/(in 10 min). TELIC

The compatibility with durative adverbials (e.g. *for 10 min*) and time-span adverbials (e.g. *in 10 min*) provides the fundamental diagnostic test for the telic/atelic distinction. The domain of application of durative adverbials is restricted to atelic predicates, while that of time-span adverbials to telic ones.⁵ For Krifka, the key observation tied to the above pair of sentences is that the telicity difference between them must stem from the difference in the referential type of their direct object NPs, given that there is no other difference between them: so, the VP *drink beer* is atelic due to its mass direct object *beer*, while the VP *drink a glass of beer* is telic due to the singular measure pseudo-partitive NP *a glass of beer*.

What any theory of aspectual composition must explain is not only when the referential type of an argument NP influences the telicity of a complex verbal predicate, as in (2a,b), but also when it does not, as the minimal pair of sentences below illustrates:

- (3) a. Mary swirled beer in her glass (for 10 min)/(? in 10 min). ATELIC
 b. Mary swirled a glass of beer (for 10 min)/(? in 10 min). ATELIC

(3a,b) are both atelic, despite the difference in the referential type of their direct object NPs, *beer* and *a glass of beer*. Given that (2) and (3) only differ in their main lexical verbs, *drink* versus *swirl*, Krifka (1989) argues that the source of telicity in (2) lies in the lexical semantics of *drink* and the way it composes with its direct object argument, which systematically leads to a telic or an atelic interpretation of a complex verbal predicate. This phenomenon is referred to as aspectual composition(ality). Notice that *drink* and *swirl* are both episodic (dynamic) and their direct object is an 'affected object', or a Patient/Theme argument, put in traditional terms. Therefore, such semantic notions will not do, as Krifka also argues, to motivate why direct objects of *drink*, but not of *swirl*, participate in aspectual composition. Instead, to this goal, he defines a novel notion of a thematic

⁵ The interpretations that are relevant for this test concern the temporal extent of singular eventualities denoted by predicates in the scope of these temporal modifiers. The interpretations that are irrelevant are iterative and generic interpretations. Moreover, for time-span modifiers, we need disregard the shifted inchoative interpretation of atelic predicates under which the time-span modifier denotes the measure of time until the onset of denoted eventualities from 'now' or some other reference point (see also Vendler, 1957, p. 147) (e.g., *The children ran in an hour* understood as meaning they started running after an hour from some understood reference point), and for durative modifiers, the irrelevant interpretations regard the duration of the result state that follows the set terminal point in the denotation of telic verbal expressions (e.g., *John put the wine into the fridge for half an hour*).

role in terms of incremental relations between the parts of an object subjected to an eventuality and the parts of that eventuality (see further below for details).

By arguing that the source of aspectual composition lies in the lexical semantics of verbs, Krifka (1986, 1989 and elsewhere) improves on the previous analyses of this phenomenon, which were feature-based and relied on rules projecting features from nominal arguments of verbs to VPs and sentences, but left what is their fundamentally semantic grounding at a pre-theoretical level (see also the same criticism of Dowty, 1979, p. 76). As Krifka (2001) observes, this holds for Verkuyl's (1972) first pioneering analysis in the spirit of Generative Semantics which uses the feature $[\pm \text{ SPECIFIED QUANTITY}]$, for Platzack's (1979) proposal couched in the Extended Standard Theory (Chomsky, 1970, 1975) and based on the feature $[\pm \text{ DIVID}]$, but also for later proposals, such as Tenny's (1987, 1992) within the Government and Binding Theory (Chomsky, 1981).

Theoretical Prerequisites: Lattices and Measure Functions

Krifka's (1989) lattice-theoretic (or mereological) event semantics builds on Link (1983, 1987) who was the first to use the algebraic device of a complete join semi-lattice (without bottom element) for the semantic analysis of mass and plural terms. Another key source of inspiration is Bach's (1981) recasting of the aspectual classes (STATE, PROCESS, EVENT) in mereological terms, and Bach's (1986) extension of Link's (1983) lattice-theoretic analysis of mass and plural nouns to the verbal domain. Both Link and Bach assume a model with two separate, but related, domains, structured by complete join semi-lattices: one atomic and the other non-atomic. Atomic semi-lattices consist of minimal, discrete elements, or atoms. Non-atomic semi-lattices are 'not known to be atomic' (Link, 1983) or 'not-necessarily-atomic' (Partee, 1999). On Bach's view, EVENT (telic) predicates and count nouns take their denotation from the ATOMIC semi-lattice domain, while PROCESS (atelic) predicates and mass nouns from the NON-ATOMIC one.

The binary split between the atomic and non-atomic conception of the world, as many noticed, raises inconsistencies and problems in the semantic analyses of nominal and verbal predicates. For instance, take a measure (pseudo-partitive) construction like *a pound of pudding*. As Krifka (2001) observes, it is singular count, and so on Link's and Bach's theory, its denotation would have to be atomic, but what it denotes is some pudding-stuff (to the amount of one pound), which corresponds to a non-atomic denotation. This problem carries over to the meaning of complex predicates derived by means of aspectual composition. It is unclear how, on the original proposals of Link and Bach, we can compositionally derive an EVENT (telic) interpretation of *eat a pound of pudding*, if *a pound of pudding* were to take its denotation from a non-atomic (mass) lattice. To take another example, *eat apples* is PROCESS-denoting (atelic), and hence takes its denotation from a non-atomic lattice, but *apples* denotes atoms (OBJECTS), and hence takes its denotation from an atomic lattice (ibid.).

Such observations, among others, lead Krifka (1986, 1989, p. 81, 2001 and elsewhere) to assume a single join semi-lattice structure, undetermined with respect to atomicity.⁶ While atomic semi-lattices have smallest discrete elements (atoms), non-atomic ones may not, they are “not necessarily atomic” (Partee, 1999, p. 94). Moreover, Krifka assumes three domains, each having the structure of a non-atomic semi-lattice: namely, objects, eventualities⁷ and times. If, for instance, a predicate *O* is used to denote objects⁸ (whereby the domain of entities of the sort *O* includes discrete things like apples, animate beings like people, and also stuff like water), as opposed to eventualities *E* and times *T*, its denotation will be structured by a semi-lattice that is defined by means of a two-place *join* operation $\cup_O$, and the two-place relations *part* $\subseteq_O$, *proper part* $\subset_O$ and *overlap* $\circ_O$, all of which are indexed by *O*.

Based on such algebraic structures, Krifka defines higher-order, cross-categorical predicates and relations over first order predicates that allow him to analyze the meanings of natural language predicates. Of key importance are two predicates for reference types: namely, cumulative and quantized. Some examples are given below:

- | | | |
|-----|------------------------------------|--|
| (4) | cumulative predicates | quantized predicates |
| | nominal: <i>beer, cats</i> | nominal: <i>a cat, two cats, three glasses of beer</i> |
| | verbal (atelic): <i>rain, walk</i> | verbal (telic): <i>recover, build a house</i> |

The notion of cumulative reference is due to Van Orman Quine (1960). For predicates of objects, Krifka’s mereological definition is given below (see Krifka, 1989, D12; also 1998, (3)), where *P* is a variable of the type of first order predicates applying to entities of the sort *O* (object). (The formulas given below, and also other formulas in this paper, are taken from different published papers by Krifka, and also from some of his handouts. Given that they span 20 years or so, there are occasional slight variations in the treatment of specific logical components.)

- (5) $\text{CUM}_O(P) \leftrightarrow \forall x, y [P(x) \wedge P(y) \rightarrow P(x \cup_O y)] \wedge \exists x, y [P(x) \wedge \neg x = y]$
 Example: If *beer* applies to *x* and *y*, then it also applies to *x* and *y* taken together.

⁶ This, among others, obviates the minimal part problem posed by the putative divisive reference of mass nouns and process-denoting (atelic) predicates (Taylor, 1977; Bach, 1981, i.a.), and also unintuitive results such that there is a sharp sortal difference between what *walk* (atelic, non-atomic domain) and *walk a mile* (telic, atomic domain) describe, even though it arguably is the same eventuality in the world under two different descriptions (Krifka, 2001).

⁷ In compliance with later developments in event semantics, here the term ‘eventuality’ (coined by Bach, 1981) is used instead of ‘event’, given that ‘event’ is now restricted to mean an entity in the denotation of telic (accomplishment, and also achievement, according to some at least) predicates, and given that by ‘events’ Krifka (1989) intends to cover the domain from which both telic and atelic predicates draw their denotation.

⁸ Krifka (1989) uses *S*, rather than *O*, for the relevant predicate variable.

Cumulative predicates of eventualities (*rain*, *walk*, *sleep*) can be defined in an analogous way, where P is a variable of the type of first order predicates applying to entities of the sort E (eventualities):

- (6) $\text{CUM}_E(P) \leftrightarrow \forall e, e' [P(e) \wedge P(e') \rightarrow P(e \cup_E e')] \wedge \exists e, e' [P(e) \wedge P(e') \wedge \neg e = e']$

Example: If an eventuality e constitutes (some) activity that satisfies the predicate *walk*, and e' does, as well, then the sum of e and e' constitute (some) activity that also satisfies the predicate *walk*.

All quantized predicates are derived from cumulative ones by means of EXTENSIVE MEASURE FUNCTIONS. The property of QUANTIZED reference is defined for predicates of objects (Krifka, 1989, D14, 1998, (4)) and for predicates of eventualities below:

- (7) $\text{QUANT}_O(P) \leftrightarrow \forall x, y [P(x) \wedge P(y) \rightarrow \neg y \subset_O x]$

Example: If a *cat* applies to each of the two (different) entities x and y separately, then one cannot be a proper part of the other. A predicate P is quantized if and only if whenever it holds of x , it does not hold of any of its proper parts.

- (8) $\text{QUANT}_E(P) \leftrightarrow \forall e, e' [P(e) \wedge P(e') \rightarrow \neg e' \subset_E e]$

Example: If an eventuality e falls under the predicate of *recover* and e' as well, then e' cannot be a proper part of e .

The intuition behind the definitions of quantized predicates, both nominal and verbal, is that they determine what is 'one' discrete and indivisible entity in their denotation, which is modeled by an extensive measure function in their logical structure. The notion of an extensive measure function grounds the analysis of counting and measure constructions as well as classifier constructions in classifier languages like Chinese.

Extensive measure functions are lexicalized by relational nouns that refer to standard measures, such as *pound*, *hour* or *liter*, and they are also derived from sortal nouns that denote container nouns, such as *cup*, *basket*, *bag*, *bottle*, and may denote either standard(ized) measures like *cup*, or non-standard measures like *bottle* (the notion of a 'non-standard measure' is here used approximately in the sense of Cartwright, 1975; Lønning, 1987). An extensive measure function is a function (possibly partial) from individuals to numbers which preserve certain structures in the object domain (Krifka, 1989, p. 78, following Suppes & Zinnes, 1963). Its defining property is additivity^{9,10} For instance, adding 2 pounds of apples to 3 pounds of apples yields 5 pounds of apples which weigh more in pounds than the quantity of any of the proper parts of the total quantity of 5 pound of apples.

⁹ For a formal definition of an extensive measure function see Champollion and Krifka, 2016, §13.21.

¹⁰ Schwarzschild (2002, 2006) relies on a closely related property of monotonicity in his analyses of closely related phenomena.

Nouns that denote standard or non-standard extensive measure functions are used to form a measure (or pseudo-partitive) construction, as illustrated in the following example:

$$(9) \quad \llbracket \text{three meters of snow} \rrbracket = \lambda x [\text{SNOW}(x) \wedge \text{METER}(x) = 3]$$

A measure (pseudo-partitive) construction consists of a MEASURE PHRASE (*three meters (of)*) which is an operator on the denotation of a CUMULATIVE predicate that denotes what is measured, either some stuff (*snow*) or plural entities (such as *books*, as in *three meters of books*).^{11,12} According to Krifka, measure phrases apply to “predicates that are not quantized yet” (Krifka, 1998, p. 202) and yield quantized predicates. Empirical support comes from examples like the following one:

$$(10) \quad ? \text{ fifty pounds of three meters of snow}$$

Here, the measure phrase *fifty pounds (of)* is applied to the predicate denoted by the measure (pseudo-partitive) construction *three meters of snow* which leads to an ungrammatical result, which would not be expected if *three meters of snow* were not quantized. Krifka (1989, 1998, and elsewhere) uses such ungrammatical examples in support of his argument that generally measure (pseudo-partitive) constructions denote quantized predicates.

However, there are uses of measure (pseudo-partitive) constructions which are not quantized: e.g., *Two glasses of wine equals one glazed doughnut* (comparative construction) or *As much as 2 feet of snow is expected/ possible /forecast over the next 3 days*. Were *2 feet of snow* here quantized, then it would have to govern a plural verb agreement, as in *More than two feet of snow were measured on Christmas Day*, for instance, and be compatible with the quantifier *many*, as in *There are still as many as five feet of snow at higher elevations* (see Rothstein, 2011; Landman, 2016; Filip & Sutton, 2017 for a discussion of the quantized property of the measure (pseudo-partitive) construction).

The notion of an extensive measure function also grounds the meaning of count nouns. Although mass and count nouns are interpreted with respect to the same non-atomic domain, they are typically different, because the meanings of count nouns are derived by means of a noun-specific measure function incorporated in the lexical structure of count nouns, namely, the ‘natural unit’ NU function. Consequently,

¹¹ Measure phrases, such as *five meters*, are of type [N/N] and analyzed by means of the NUMBER (*n*) expressed by the numerical word (*five*) and a measure function (μ), expressed by some measure word (*meter*). Syntactically speaking, numerals (*five*) belong to a basic category NM (numerical number), and consequently measure words (*meter*) have the category [N/N, NM] (see (4), Krifka (1989, p.83).

¹² Krifka (1989) analyzes extensive measure phrases (such as *five ounces(of)*) as ‘quantizing modifiers’ that derive quantized predicates from non-quantized ones, namely, denoted by mass terms (*beer, gold*) and plural terms (*apples*): $\forall P \forall P[\text{QMOD}(P,P) \leftrightarrow \neg \text{QUA}(P) \wedge \text{QUA}(P(P))]$ (Krifka, 1989, D28, p.82). In later works, he specifies the input of measure phrases in terms of the property of cumulative reference.

count nouns denote two-place relations between entities x and numbers n (Krifka, 1989, p. 85), as illustrated by a simplified lexical entry for *cat*:

$$(11) \quad \llbracket cat \rrbracket = \lambda x [\text{CAT}(x) \wedge \text{NU}(\text{CAT})(x) = 1]$$

NU operates on entities in the non-atomic domain, or more precisely on the intension of the mass noun (Krifka, 1989, pp. 84–85). It is NU that makes singular count nouns semantically quantized by capturing what is intuitively ‘one’ discrete and indivisible object in their denotation, or their QUANTITATIVE criterion of application (Krifka, 1989, p. 84). In contrast, mass nouns like *water* are cumulative, because they lexically specify only a QUANTITATIVE criterion of application (ibid.), which makes them one-place predicates:

$$(12) \quad \llbracket water \rrbracket = \lambda x [\text{WATER}(x)]$$

One of Krifka's main innovations is to use measure functions to capture a variety of quantization phenomena in natural languages, including the similarities between COUNT nominal expressions and TELIC verbal expressions. Both denote QUANTIZED predicates, which means that they have only indivisible discrete entities of the appropriate sort in their denotation, and just what they are is determined by the requisite measure function in their logical structure. Such discrete entities need not be the absolutely smallest elements (atoms), unlike in Bach (1986) and Link (1983, 1987). They may be specified relative to either one unit of measurement (as in *a lone/the cat* or *eat an apple*) or multiples thereof (as in *two cats* or *eat two apples*).

Krifka's analysis of quantization/cumulativity phenomena in the verbal domain rests on two fundamental assumptions about the nature of eventualities. Inspired by Davidson (1967, 1969), Krifka takes eventualities to be entities in their own right, apart from objects (O) and times (T). Second, unlike entities of the sort of objects, i.e., things like apples or stuff like water, eventualities per se are assumed to be measurable via their run times or via various measurable dimensions of objects which bear a relation to eventualities (where ‘object’ is widely construed as subsuming concrete objects like apples, but also paths or property scales).¹³ Krifka's original proposal is that measure functions (or relations) on eventualities are *derived* from measure functions on times or objects. This idea capitalizes on the overarching hypothesis that there are systematic parallels between their domains, each structured by an algebraic join semi-lattice.

Take, for instance, *Kim walked for an hour*. As many semanticists agree, durative adverbials like *for an hour* are adverbial counterparts to adnominal measure

¹³ This is reminiscent of Strawson's (1959) view that the possibilities for identifying eventualities without reference to objects are limited, because eventualities fail to provide “a single, comprehensive and continuously usable framework” of reference of the kind provided by physical objects (Strawson, 1959, p. 46ff.).

phrases like *a pound (of)* (Vlach, 1981; Bach, 1981, i.a.).¹⁴ Durative adverbials and adnominal measure phrases are quantizing modifiers. When applied to non-quantized predicates they derive quantized predicates, and they cannot be applied to quantized predicates: ?*a pound of a watermelon*, ?*solve the puzzle for an hour*. While a measure phrase like *a pound (of)* can directly apply to non-quantized predicates of objects, as in *a pound of cherries*, a durative adverbial like *for an hour* cannot directly apply to non-quantized predicates of eventualities, because its constituent temporal measure function HOUR applies to time-intervals, not to eventualities. However, a measure function on times like HOUR can indirectly measure eventualities by measuring their ‘run times’, or temporal traces, which are mapped from eventualities by the temporal trace function τ (see Appendix for a definition). τ is a homomorphism (a structure-preserving mapping) relative to the join operations for eventualities $\cup_E$ and times $\cup_T$ (Krifka, 1989, p. 97, (D40); Link, 1987), and it is used to analyze temporal adverbials like *for an hour*, as in *walk for an hour*:

$$(13) \quad \llbracket \textit{walk for an hour} \rrbracket = \lambda x, e [\text{WALK}(e) \wedge \text{AGENT}(e, x) \wedge \text{HOUR}(\tau(e)) = 1]$$

Measure functions can be ‘transferred’ from one domain to another with ease (Krifka, 1990, p. 519), which affords the derivation of quantized verbal predicates that may be delimited not just based on their run times, but also relative to a variety of dimensions along which properties of objects can be measured. For instance, extensive measure functions on distances in space like MILE can yield derived measure functions on eventualities based on a path trace function π (a homomorphism) and used to analyze locative adverbials like *(for) two miles*, as in *walk (for) two miles* (see Appendix for details). This idea can also be extended to the analysis of locative and directional adverbials like *in/to/towards the station* in the derivation of the meanings of quantized (telic) and cumulative (atelic) complex verbal predicates: cp. *walk to the station* (telic) versus *walk towards the station* (atelic).

Temporal Constitution of Complex Verbal Predicates: Aspectual Composition

Another ‘transfer’ mechanism, which also relies on the notion of a homomorphism, mediates the transfer of reference properties, quantized and cumulative, between nominal arguments and the temporal constitution of verbal predicates. It is in a nutshell, Krifka’s view of the phenomenon known as aspectual composition(ality)

¹⁴ “An hour full of running is naturally assumed to be without gaps, like a bathtub full of water. A year full of winning (iterative) has got to have gaps, like a street full of policemen” (Vlach, 1981, p. 282, fn. 17).

(see also above). Krifka's examples (along with his judgments marked as '*'), which were introduced at the outset of the paper, are here repeated for convenience:

- (14) a. Mary drank beer (for 10 min)/(*in 10 min). ATELIC
 b. Mary drank a glass of beer (*for 10 min)/(in 10 min). TELIC

Intuitively, drinking of some quantity of beer is such that every proper part of that beer corresponds to a proper part of the eventuality of drinking of that beer, and vice versa. A change of state that the beer undergoes proceeds in distinguishable separate stages, each tracking a successive stage of the relevant drinking eventuality. It is Krifka's key claim that it is precisely verbs like *drink* which entail this type of 'synchronized' object-eventuality change that induce aspectual composition.¹⁵

The argument with respect to which this entailment holds (*beer* and *a glass of beer* in the examples above) is an Incremental Theme argument. As a historical aside, Krifka's (1986, 1989, 1992) original term is 'Gradual Patient'. The term 'Incremental Theme' was coined by Dowty (1988, 1991), and subsequently adopted by Krifka (e.g., Krifka, 1998). Dowty (ibid.) integrates the Incremental Theme relation (Krifka's Gradual Patient relation), along with its lattice-theoretic underpinnings, into his theory of thematic Proto-Roles and argument selection, where it is treated as one of the properties of the Patient Proto-Role.¹⁶

Formally speaking, a verb like *drink* entails a homomorphism between the part structure (semi-lattice structure) associated with an Incremental Theme argument and the part structure (semi-lattice structure) associated with the eventuality argument. The homomorphism is defined in terms of five two-place relations for thematic roles (see Appendix). They define incremental relations between the eventuality argument and one other semantic argument, which is paradigmatically instantiated by an Incremental Theme argument.

Assuming, as Krifka does, that the source of aspectual composition lies in incremental relations like *drink*, and assuming the cross-categorical quantization/cumulativity reference properties (defined in (5)–(8)), aspectual composition directly follows the standard semantic composition of a sentence: namely, the way in which verbs normally combine with their arguments by applying compositional semantic rules to independently motivated syntactic structures (Krifka, 1989, p. 91ff.):

¹⁵ As Krifka (1998, and elsewhere) also observes, similar notions and relations mediating between participants and eventualities were proposed by others: e.g., [+ADD-TO] V property (Verkuyl, 1972, 1993), 'measuring out' tied to the internal direct object DP (Tenny, 1987, 1994), 'structure-preserving binding relations' (Jackendoff, 1996).

¹⁶ Dowty's (1987, 1989, 1991) treatment of Incremental Theme as one of the lexical determinants of argument selection is not entirely uncontroversial. For instance, Jackendoff (1996), Rappaport Hovav and Levin (2002, 2005, pp. 284–285) argue that Incremental Theme is not a factor in argument selection, while agreeing that the intuition behind it, which concerns structure-preserving mappings between eventualities and some suitable objects, plays an important role in a variety of aspectual phenomena.

- (15) $\phi = \lambda e \exists x [\alpha(e) \wedge \delta(x) \wedge \Theta(e, x)]$ Krifka (1989, p. 93, (12))
 If Θ is an Incremental Theme relation, then (i) when δ (nominal predicate) is cumulative, ϕ is cumulative (atelic, activity); (ii) when δ (nominal predicate) is quantized, ϕ is quantized (telic, accomplishment), provided ϕ is a singular predication, i.e., the existential closure over the event argument e is not a plural quantification. Example:
- a. $\llbracket \text{eat two apples} \rrbracket = \lambda x, e \exists y [2\text{APPLES}(y) \wedge \text{EAT}(x, y, e)]^{17}$ QUANTIZED
- b. $\llbracket \text{eat soup} \rrbracket = \lambda x, e \exists y [\text{SOUP}(x) \wedge \text{EAT}(x, y, e)]^{18}$ CUMULATIVE

This also correctly predicts that the predicate *push two carts* is cumulative, even if its Theme argument *two carts* is quantized, because *push* is a non-incremental relation (for the term a ‘non-incremental relation’ see e.g., Krifka, 1998):

- (16) a. $\llbracket \text{push two carts} \rrbracket = \lambda e. \exists x [\text{PUSH}(e) \wedge \text{CARTS}(x) = 2 \wedge \text{THEME}(e, x)]$ CUMULATIVE
- b. $\llbracket \text{push carts} \rrbracket = \lambda e. \exists x. \exists n [\text{PUSH}(e) \wedge \text{CARTS}(x) = n \wedge \text{THEME}(e, x)]$ CUMULATIVE

A striking feature of Krifka’s mereological approach to aspectual classes is that it relies on non-temporal cross-categorical mereological properties, in departure from Vendler (1957) and others, who following Vendler, focus on defining their ‘temporal contours’ in terms of properties of time intervals and instants (Levin & Rappaport Hovav, 2005, and references therein).

There are two important (and often misunderstood) corollaries of Krifka’s mereological approach to aspectual composition:

- Incrementality does not guarantee quantization (telicity).
- Neither incrementality nor quantization (telicity) are exclusively linked to the (internal) direct object argument.

Let us examine them in turn. Incrementality and quantization (telicity) are independent of each other. There are quantized (telic) predicates that are non-incremental. For example, *make a dot* (Krifka, 1998) is quantized by virtue of having atomic events in its denotation, but it is non-incremental, because its denotation has no proper temporal parts, and a dot itself has no spatial proper parts. There are also complex incremental predicates that are cumulative, because their Incremental Theme argument is cumulative: e.g., *eat soup*, *burn fall leaves*, *polish silverware* (?in an hour/for an hour). Being a direct object argument is neither a sufficient nor a necessary condition for it to function as an Incremental Theme and/or to serve as the single most important ingredient for determining the quantization (telicity)

¹⁷ The simplified logical representation of the complex predicate *eat two apples* is taken from Krifka (1998, example (53b)).

¹⁸ The simplified logical representation of the complex predicate *eat soup* follows the style of (15a), which in turn is taken from Krifka (1998, example (53b)).

of complex verbal predicates. It is not sufficient, because many verbs that take a direct object argument are non-incremental (*push a cart*, *see a photograph*). It is not necessary, because, for instance, the Incremental Theme can also be realized as the subject argument of transitive verbs like *cross*, *penetrate*, *permeate*, *pass*, *skirt* (Verkuyl, 1972; Declerck, 1979; Filip, 1990; Dowty, 1991; Jackendoff, 1996; Levin & Rappaport Hovav, 2005):

- (17) a. John entered the icy water (very slowly). Dowty (1991)
 b. At the turtle race, the winning turtle crossed the finish line in 42 seconds.

It is also worth mentioning that such data provide a compelling argument against syntactic theories of aspect that assume that *only* the (internal) direct object argument participates in defining the telicity of complex verbal predicates (e.g., Verkuyl, 1972/73; 1993; Tenny, 1987, 1994; Kratzer, 2004; Borer, 2005, and references therein). This argument is spelled out in Dowty (1991, p. 571, fn. 15).

The object-eventuality homomorphism motivates a variety of interactions between nominal arguments and verbal predicates in English, as we have just seen. In English, aspectual composition turns on the referential properties of the Incremental Theme DP, where the main operators are quantifier and measure phrases with which it may be formed. In different languages, such interactions are subject to additional constraints stemming from the realization of the Incremental Theme argument, and in languages with grammatical aspect also from the semantics of aspectual forms by which verbal predicates are expressed. For instance, in German, the Incremental Theme can be syntactically realized either as an accusative-marked direct object or as a 'partitive' ('conative') *an*-PP. The latter has an effect on a complex verbal predicate that is taken to be close to the progressive aspect in English: e.g., '*Das Kind aß an einem Fisch* 'The child was eating a fish' versus *Das Kind aß einen Fisch* 'The child ate a fish' (see also Filip, 1989, 1993a, b; 1999; Kratzer, 2004). Krifka (1989) also mentions Finnish, where the accusative/partitive case alternation on certain Theme arguments, not just Incremental Themes, results in an effect comparable to the perfective/imperfective aspect (see also Filip, 1993a, b, 1999; Kiparsky, 1998).

Given that the object-eventuality homomorphism 'works in both directions' (Krifka, 1992, p. 49), we also have a converse case where a verbal predicate operator has effects on the interpretation of the Incremental Theme argument. Some of the clearest examples come from languages that have a well-developed grammatical category of aspect and that lack a (fully developed) system of (in)definite articles. Krifka (1989) mentions Chinese and Slavic languages. Regarding Slavic languages, Krifka's theory (see Krifka, 1986, 1992) sheds a new light on the often cited, but ill-understood, 'definiteness' effect of perfective verbs on their arguments. Building on Wierzbicka (1967) for Polish and Filip (1985) for Czech, the object-eventuality homomorphism makes the correct generalization that the 'definiteness' effect will be restricted to bare mass and plural Incremental Theme arguments of perfective verbs, while singular count ones are exempt from it, as illustrated by the following examples:

- (18) a. Petr snědl^{PFV} kaši / olivy.
 Peter ate porridge.SG.ACC / olives.PL.ACC
 ‘Peter ate (up) (all) the porridge/(all) the olives.’
- b. Petr snědl^{PFV} hrušku.
 Peter ate pear.SG.ACC
 ‘Peter ate (up) a pear/the pear.’

Krifka’s analysis relies on the common (though not entirely uncontroversial) assumption that nouns in languages lacking articles are lexically ambiguous between a definite and an indefinite interpretation. Specifically, in Slavic languages, as Krifka also assumes, mass nouns (*wine*) and plural nouns (*pears*) are quantized on the definite reading and cumulative on the indefinite reading; singular count nouns (*pear*) are quantized in both interpretations. Perfective verbs, which Krifka takes to be uniformly quantized, enforce the quantized interpretation of their Incremental Theme, which for mass or bare plural nouns must also be definite.

The advantage of Krifka’s analysis of Slavic data like those above is that it improves on analyses that assume a uniform ‘definiteness’ effect of perfective verbs on all and only arguments that fill the (internal) direct object slot (e.g., Borer, 2005; Schoorlemmer, 1995; Verkuyl, 1993, 1999, and references therein). Such accounts massively overgeneralize. For instance, they wrongly predict that the direct object in (18b) is required to have a definite interpretation only. Instead, Krifka’s analysis is more nuanced, as the ‘definiteness’ effect related to perfectivity depends on the thematic role, countability properties and grammatical number of arguments of perfective verbs.

Critical Comments

The notion of quantization has borne the main brunt of criticism. To illustrate the nature of contentious issues, four central problematic areas will be addressed below.

Quantization and the Grammatical Count Property

In the nominal domain, quantization, as defined in (7), is not a necessary condition for a noun to be grammatically count, contrary to Krifka’s (1989) proposal. Krifka (1989, p. 87, due to Partee, p.c.) himself draws attention to this problem, and gives two examples of count nouns that fail to be quantized, namely *twig* and *sequence*. Other examples like *twig*, a sortal count noun, are *wall*, *fence*, *hedge*, *bush*, *stick*, *ribbon*, *bouquet*, among many others. (A) *sequence* (of) exemplifies a relational count noun that fails to be quantized, and among other examples are *portion*, *quantity*, *piece*, *distance*, for instance, some of which can also be taken as denoting a

non-standard measure (a 'non-standard measure' roughly in the sense of Cartwright, 1975; Lønning, 1987). (The distinction between sortal and relational nouns is not always easy to draw, but the above examples, however, should suffice to illustrate the intuition behind it.) We can best show what it means for a noun to be count, and yet fail to denote a quantized predicate with *sequence*, as in *a sequence of numbers* (also discussed in Zucchi & White, 1996). For instance, take a sequence of numbers like 1, 2, 3, 4. It has proper parts, such as 1, 2, 3 or 2, 3, 4. The predicate *a sequence (of numbers)* felicitously applies both to the main sequence 1, 2, 3, 4 and also to 1, 2, 3 or 2, 3, 4, which are its proper parts. But this means that the predicate *a sequence (of numbers)* fails to be quantized, according to (7).

Various solutions to this problem converge on the idea that the meaning of such singular count nouns is partially specified by context. That is, what we view as one entity in their denotation, for the purposes of grammatical counting operations, depends on context, and can be rather arbitrary. However, once the context is fixed, such nouns denote a quantized set (see Champollion & Krifka, 2016; Chierchia, 2010; Rothstein, 2010). Such observations also lead to the conclusion that no context-independent notion, be it quantization as defined in (7) which relies on the extensive measure function NU, or an atom in a Boolean algebra (see Link, 1983, 1987; Bach, 1986, i.a.), will do to analyze the meaning of singular count nouns as a whole class (see e.g., Zucchi & White, 1996, 2001; Rothstein, 2010; Filip & Sutton, 2017, i.a.). Instead, the notion that is needed to ground the count property of nouns is a notion that allows for what counts as 'one' in their denotation to be fixed relative to a particular context.

Non-quantized Incremental Themes in Quantized Predications

The context-dependency of count nouns like *twig* and *sequence* also carries over to aspectual composition. Despite failing quantization as defined in (7), they still pattern with prototypical count nouns like *cat* or *letter*, which satisfy quantization (7), in so far as they enforce quantized (telic) predicates:

- (19) a. Sally wrote [NON-QUANT a sequence of numbers] in 10 seconds/ ?for 10 seconds.
 b. Sally wrote [QUANT a letter to her friend] in 10 seconds/ ?for 10 seconds.

A similar quantization problem also arises with DPs that fail to be quantized, according to (7), due to their determiners, rather than due to the lexical properties of their head nouns.¹⁹ Salient examples include DPs with vague determiner quantifiers like *many*, *a lot*, *(a) few*, *some* and *most*, and cardinal numerals combined with approximative modifiers like *at least/at most*. Nonetheless, such non-quantized DPs

¹⁹ Related observations were also made by L. Carlson (1981), Mittwoch (1988), Dahl (1991) and Moltmann (1991).

still enforce a quantized (telic) interpretation on complex verbal predicates, when they function as Incremental Themes:

- (20) Sally ate [NON-QUANT at least three apples] in an hour / ?for an hour.

In order to solve this puzzle, Krifka (1998, p. 221) proposes that quantificational indefinite DPs (*subject to existential closure*), such as *at least three apples*, have a wide-scope reading with respect to time-span adverbials like *in an hour*, because only this satisfies the application conditions of such adverbials. The LF structure of the above sentence then approximately amounts to: [at least three apples]₁ [Sally [[ate x_1] in an hour]]. However, this wide-scope solution predicts odd or anomalous readings for many sentences with quantificational indefinites. For instance, *Every guest ate some muffins* would mean that every guest ate the same muffins (Zucchi & White, 1996, p. 241ff.).

Quantized Incremental Themes in Cumulative Predications

There are also Incremental Themes that are quantized in compliance with (7), and which, however, contrary to Krifka's aspectual composition (15), do not force a quantized (telic) interpretation on complex verbal predicates, but rather also allow for them to have a non-quantized (atelic) interpretation in an appropriate context (see also Kratzer, 2004; Partee, 1999):

- (21) a. Bill ate the apple bit by bit for 10 min
(and still didn't finish it). Jackendoff (1996)
b. She ate the sandwich in 5 min/ for 5 min. Hay et al. (1999)
c. We both ate the soup for a couple of minutes
while the rest of the dishes cooled down.
- (22) a. He read a book in/for an hour. Fillmore (1971) (in Dowty, 1979, p. 61)
b. Bill washed the car
(i) ... (clean) in an hour.
(ii) ... for an hour, but only got its hood clean.
... for an hour, but none of its parts got completely clean.

Given that an indisputably quantized Incremental Theme argument does not necessarily force a quantized interpretation on a verbal predicate, as the above examples show, obligatory quantization (telicity) by aspectual composition (15) is a more limited phenomenon than commonly assumed (Filip, 2004, i.a.).

Notice also that examples like those in (21a-c) clearly invalidate a common claim about the link between the definiteness of a direct object DP (or an internal direct object DP) and telicity of verbal predicates (e.g., Borer, 2005, and

references therein).²⁰ Definite argument DPs, even if they are uncontroversially quantized, according to (7), such as *the apple*, are not sufficient to enforce a telic interpretation on a complex verbal predicate (Filip, 1996, 2004; Jackendoff, 1996). Worth mentioning is also that the definite article is best not viewed as a quantifier or an expression of quantity (*pace* Borer, 2005, i.a.), because it "asserts neither universality nor distributivity nor any particular cardinality" (see Partee, 1995, p. 581).

Perfective Semantics Reduced to Quantization

For languages that have a grammatical perfective/imperfective aspect, it is commonly assumed that all perfective verbs are uniformly telic, and so uniformly quantized (in the sense of (8)), according to Krifka (1986, 1989, 1992). However, what is less known is that a typical feature of such languages is a sizeable class of verb forms (verbs or constructions) that are perfective, according to the relevant language-specific distributional and semantic tests, and yet fail quantization (8), due to the meaning of the lexical material making them up.

In Slavic languages, as Filip (1992, 2000, 2005a, b) argues, the main 'culprits' are prefixes which are used to form perfective verbs from (im)perfective ones. Given that prefixation is the most common process by which perfective verbs are derived, perfectives which fail quantization are also common. A case in point are prefixes that have meanings/uses resembling those of measure phrases or vague quantifiers in the nominal domain, such as *a small/large quantity (of)*, *many*, *much*, *a lot (of)*, *(a) little (of)* (ibid.; also Kiparsky, 1998). For instance, the attenuative (glossed as 'ATN') use of the prefix *po-* (illustrated in the Czech example in (23) below) contributes the meaning of a relatively low degree on some measurement scale (which would be most likely related to time in (23)) and relative to some contextually provided standard.

- (23) po.povídat^{PERF} si
 ATN.chat REFL
 'to chat for a (short) while'

While nominal measure (pseudo-partitive) constructions like *a small quantity of wine* are non-quantized due to their vague measure phrases (here *a small quantity (of)*), perfective verbs like *popovídat si* 'to chat for a (short) while' are non-quantized due to prefixes which contribute the meanings of vague quantity or measurement to them. The perfective verb *popovídat si* 'to chat for a (short) while' fails to be quantized, as defined in (8), given that an eventuality of chatting for a while has proper parts that also qualify as chatting for a while. But this means that

²⁰ Borer (2005) argues that a telic interpretation must be licensed by a quantity DP, with a definite DP being one subtype, including definite mass nouns and definite plurals.

such Slavic perfective verbs are problematic for a uniform semantic characterization of perfective verbs in terms of quantization, as defined in (8).

Moreover, Slavic perfective verbs that are formed with prefixes that denote some vague measure tend to enforce a non-specific indefinite and non-quantized interpretation of their mass and bare plural Incremental Themes (Filip, 1997, 2005a, b), which is unexpected on any account of Slavic aspectual composition:

- (24) Na.vařili jsme hodně jídla na sváteční večeři.
 CUM.cooked AUX.IPL a.lot.of food.SG.GEN for holiday dinner
 ‘We cooked up a lot of food for the holiday dinner.’

Impact and Most Important Subsequent Developments

Krifka’s (1986, 2007, and elsewhere) proposal to base the analysis of predicates in natural languages on a single non-atomic domain, as opposed to Link’s (1983, 1987) original two-domain approach, has been steadily gaining more proponents, especially in the last 10 years or so (Chierchia, 2010; Landman, 2016; Filip & Sutton, 2017, i.a.).

Many discussions have also concerned the meaning of the formally marked plural form, whether *Ns* in English, for instance, means *one or more N*, as Krifka (1986 and elsewhere) suggests (Sauerland, 2003; Sauerland et al., 2005; Chierchia, 2010; Yatsushiro et al., 2017, i.a.) or *two or more Ns*, as Link (1983) originally proposed (Chierchia, 1998, and references therein); another possibility is to treat it as ambiguous between *one or more N* and *two or more Ns* (Farkas & de Swart, 2010).

Krifka’s (1989) observations and formal representations of classifier constructions like *five head of cattle* and their similarities to classifier constructions in classifier languages like Chinese energized debates concerning formal analyses of nominal constructions in classifier languages. Krifka (1995) argued that classifiers combine with numerals first, and then the result combines with nouns (see also Bale & Coon, 2014 on Chol (Mayan) and Sudo, 2017 on Japanese), which is in contrast to a more common view that classifiers combine with nouns to yield countable predicates which then combine with numerals and quantifiers (Chierchia, 1998, 2010, 2015, and references therein).

Many recent algebraic (mereological) theories of the mass/count distinction agree on the key role of context, in contrast to Krifka’s (1989 and elsewhere) algebraic (mereological) theory which relies on context-independent notions of cumulativity and quantization. For instance, Rothstein (2010, 2017) treats all count noun denotations in a uniform way of type $\langle e \times k, t \rangle$, i.e., functions from an individual and a counting context, in which that individual counts as one, to truth values. Chierchia (2010, 2015) argues that count concepts have ‘stable atoms’ in their denotation, i.e., minimal entities that are the same ones in every context, while mass concepts are vague in so far as they can be construed as atomic or not, depending on context.

Broadening the empirical scope beyond prototypical mass nouns (*water*) and count nouns (*cat*) led to further challenges for a uniform analysis of nominal count predicates in terms of quantization (7). Chierchia (1998, 2010) and Landman (2011, 2016) focus on object mass nouns like *furniture*, *kitchenware*, *silverware* (also known as fake mass nouns), which are grammatically mass, but their denotations consist of discrete objects, rather than undifferentiated stuff. This is problematic for Van Orman Quine's (1960) view that count nouns divide their reference, while mass nouns do not, adopted in Link (1983). Landman (2011, 2016) argues that all mass nouns denote overlapping sets, which leads to overspecification (Landman, 2011, p. 17) with respect to how many entities we can distinguish as counting as 'one' simultaneously in the same context, and prevents (grammatical) counting (cp. *#three kitchenware(s)* versus *three cups*). Given this individuation overspecification, it is unsurprising that noun concepts of collective artefacts like FURNITURE exhibit a variation in their encoding as mass or count in a particular language and cross-linguistically (Sutton & Filip, 2016, 2021).

Granular aggregates like rice or sand and collective aggregates like insects or small fruit conceptually exclude overlaps among what we view as the smallest entities in their denotation, and yet allow either mass or count encoding (Sutton & Filip, 2016, 2021). Based mainly on English, Welsh, Turkana (Nilo-Saharan), Maltese (Afroasiatic), and Dagaare (Niger-Congo), Grimm (2012) introduces the notion of *cluster* (of externally connected individuals) which allows him to differentiate a mereological sum counting as a clustered entity (e.g., *rice*, *sand*) from a mereological sum viewed as one individual entity. Predicates of discrete individuals like *cat* or *grain of rice* involve the notion of a *maximally strongly self connected* (MSSC) individual, i.e., an entity for which every part internally overlaps with the whole.

Turning to temporal constitution, Krifka (1998) develops a comprehensive framework of "telicity by precedence and adjacency", in order to provide an explicit formal analysis of telic predicates built with locative and directional adverbials like (*for*) *two miles* and *in/to/towards the station* (see Section "Theoretical prerequisites: lattices and measure functions"). Krifka (1998) models both temporal and path traces as one-dimensional directed PATH structures. In the Figure/Path constellations with directed motion predicates, the PATH is implied by the measure construction (*500 miles*) or by the Source- and/or Goal-PP, as we see below:

- (25) a. Sam drove 500 miles/(from Boston) to New York.
 b. We flew over the lake in an hour/over water for hours. Talmy (1985)

The requisite incremental relations hold between the parts of some eventuality and the spatially adjacent segments of a PATH structure. A quantized PATH yields a quantized (telic) verbal predicate, while a cumulative PATH, a cumulative (atelic) predicate. Among others, directed motion predicates like *walk to the post office* are problematic for this straightforward correlation, because they fail to be quantized (8), and yet are telic. This forces Krifka to abandon a uniform analysis of all telic predicates in terms of quantization, and to define telicity, a weaker notion than quantization as follows: "An event description *R* is telic if and only if it applies

to events e such that all parts of e that fall under R are initial and final parts of e " (Krifka, 1998, p. 207).

The analysis of predicates describing motions in a physical space provides a basis for the modeling of changes in other domains.²¹ This also allows us to analyze changes of various qualitative properties of objects, including those denoted by Themes of so-called 'degree-achievement' verbs (in the sense of Dowty, 1979) or scalar verbs (Hay et al., 1999; Rappaport Hovav, 2008, and references therein), as the examples below illustrate:

- (26) a. The temperature rose in/for 6 days.
 b. We cooled the metal (from 90 °C) to 20 °C.
 c. The room darkened (in 10 min/?for 10 min).

Krifka's unified treatment of various kinds of changes presupposes that both Goal PPs and Result XPs are represented as endpoints on a generalized PATH. The notion of result, completion or culmination, which many consider a key semantic property of telic (accomplishment) predicates (see e.g., Tenny & Pustejovsky, 2000), also in the spirit of Vendler (1957), has no separate place in this theory. This means that differences between Goal-based and Result-based telicity (Levin & Rappaport Hovav, 2005; Rappaport Hovav, 2008) cannot be accounted for.

Krifka's (1998) development of a comprehensive framework of "telicity by precedence and adjacency" coincided with the emergence of a SCALAR turn in the theory of telicity (see e.g., Hay et al., 1999, also Filip, 1993a, 1997, 1999). The first purely scalar analysis is Hay et al. (1999) who focus on the variable telicity of so-called 'degree achievements' (Dowty, 1979), which are lexically associated with a scale, as in (26a-c) above. Kennedy and Levin (2008) provide a detailed analysis of this phenomenon, also building on the insights of Kennedy and McNally (2005) on gradable adjectives.

Beavers (2013) extends Krifka's (1998) "telicity by precedence and adjacency" to include a class of ternary FIGURE/PATH theta-relations that allow for double, interdependent Incremental Themes that are implicated in the derivation of (a)telic interpretations of predication that involve complex Figure/Path constellations. For a predication of this type to be telic, the FIGURE must have quantized reference and the PATH must be bounded. We may illustrate this with examples taken from Filip (1993a, b, 1999):

- (27) a. The earthquake shook a book off the shelf in/#for a few seconds. Filip (1993a, b, 1999)
 b. The earthquake shook books off the shelf for/#in a few seconds.

Krifka's (1986, 1989, 1992) classic mereological approach to aspectual composition, as outlined above (see Section "Temporal constitution of complex verbal

²¹ This idea is inspired by the Localist Theory (Gruber, 1965), which inspired the framework of Conceptual Semantics of Jackendoff (1991, 1996).

predicates: Aspectual composition") is characterized in Krifka (1998) as a static theory of telicity or "telicity by sums and parts", and extended by means of a dynamic theory of "telicity by precedence and adjacency", which is closely related to scalar theories of telicity. In effect, this amounts to two types of telicity, albeit both of which crucially rely on incremental relations.

This led to various attempts at combining Krifka's classic mereological approach to aspectual composition (aka Krifka-Dowty framework) with scalar (or alternately path or degree) semantics (Filip & Rothstein, 2005; Filip, 1997, 2008, 2017; Piñón, 2008; Beavers, 2008; Rappaport Hovav, 2008; Kennedy, 2012, i.a.). One of the challenges for such an integration is that the class of incremental verbs and that of scalar verbs (including degree achievements) cannot be (straightforwardly) reduced one to the other, because incremental verbs are not lexically associated with a scale, while scalar verbs do not take an Incremental Theme.

One attempt at such an integration is Filip (Filip, 1993a, b, 1999) who extends Krifka's classic mereological approach to aspectual composition with the notion of a generalized scale in order to account for a variety of telicity phenomena that are not of compositional nature. The requisite incremental relations (or structure-preserving mappings) are treated as having their source in the lexical semantics of verbs, as in the original proposals of Krifka (1986, 1989, 1992) and Dowty (1991), but in addition can also be a property of grammatical constructions. Among others, the latter allows for the analysis of (a)telic interpretations of verbal predicates which depend on the lexical material of a sentence interacting with context and general world knowledge (about how various states of affairs take place). For instance, in the following sentences the 'degrees' on the relevant scale are ordered steps of script-like scenarios. The requisite incremental relations then generally obtain between the parts of some eventuality and the 'degrees' (or segments) of the scale, which allow us to track its progress.

- | | | | |
|------|----|---|------------------------|
| (28) | a. | The doctor examined the patient (in an hour). | Filip (1993a, b, 1999) |
| | b. | John was becoming an architect but was interrupted. | Dowty (1991) |

In this connection with such examples, we may also observe that the varied sources of incremental relations (or structure-preserving mappings), which motivate telic and atelic interpretations of complex verbal predicates, also led to questioning their status with respect to the grammar of natural languages. Rather than having their source in the lexical semantics of verbs, as Krifka (1986, 1989, 1992) and Dowty (1991) originally proposed, they are taken to be an emergent property of the lexical structure of verbs interacting with pragmatics (Jackendoff, 1996), a property of the [telic] inflectional head above VP (Kratzer, 2004), and even not part of the grammar of natural languages at all and instead entirely determined by world knowledge and pragmatic factors (Borer, 2005).

Another approach that attempts at combining Krifka's classic mereological approach to aspectual composition with scalar semantics is proposed by Filip and Rothstein (2005) and Filip (2008, 2017). Their main goal is to solve the

quantization puzzle posed by predicates like *write a sequence of numbers* or *eat at least three apples* (Section “Non-quantized incremental themes in quantized predications” above). On their account, such predicates denote sets of maximal events with respect to an induced SCALE, and relative to context. The scale is induced by the Incremental Theme argument, provided it contains some scale-inducing expression, such as a numerical expression, quantifier (and so giving rise to scalar implicatures), or a count noun denoting an object with an inherent internal structure (whose parts can be ordered relative to a scale). The requisite incremental relations hold between the parts of some eventuality and the segments of the relevant scale. This maximization approach to quantization (telicity) is also extended to apparently non-quantized perfective verbs in Slavic languages, as exemplified by (23) and (24) above. Here, the requisite scale is induced by verbal prefixes, as they carry a quantitative, measurement or change of state meaning. Crucially, verbal prefixes in Slavic languages do not contribute telicity (quantization) to the meaning of perfective verbs, contrary to what is commonly assumed, but rather they introduce an ordering criterion on eventuality stages, or alternately they directly introduce a scale,²² which is a prerequisite for the application of a context-relativized maximization operator that derives the quantized (telic) interpretation of perfective verbs in which they occur.

The idea of maximization that grounds the quantized (telic) interpretation of verbal predicates in Filip’s and Rothstein’s work is inspired by the notion of a *Maximal Participant* introduced by Zucchi and White (1996, 2001). They propose that indefinite DPs like *a sequence of numbers* denote *Maximal Participants* in sentences like *John wrote a sequence of numbers* in order to solve the quantization puzzle that such indefinite DPs raise (Section “Non-quantized incremental themes in quantized predications” above). That is, while *a sequence of numbers* on its own fails to be quantized, in *John wrote a sequence of numbers* it denotes the maximal (participant) sequence written by John at some contextually determined reference time, which then motivates the telic (quantized) interpretation of the whole sentence. However, this amounts to having the maximization built into an indefinite DP, which leads to wrong interpretations for non-quantized (atelic) sentences (Rothstein, 2004, p. 153). For instance, it wrongly predicts that *The emperor has ruled fewer than 5 countries for the last 10 years* requires that the same maximal set of fewer than 5 countries be ruled by the emperor at all times in the last 10 years. Therefore, to avoid such wrong predictions, Filip and Rothstein (2005) and Filip (2008, 2017) treat the requisite maximization operator as a property of VPs (as in English) or perfective verbs (as in Slavic languages).

An important, though less noticed, refinement of Krifka’s theory concerns the introduction of a Strictly Incremental Theme (Krifka, 1998). Its single most important characteristic is the UNIQUENESS OF EVENTS relation, which an Incremental Theme lacks (see Appendix for more details). It holds between the

²² Kagan (2013) applies this idea to a full-fledged scalar approach to the semantics of Russian prefixes.

eventuality argument and one other argument of verbs that entail a permanent and non-reversible change for that argument, which means that for a given entity token denoted by that argument there can be at most one (unique) eventuality related to it. A good example is *eat an apple*: you can eat one and the same apple at most once. Other examples are: *drink, build, burn, destroy, compose, write*. In contrast, incremental relations like *read* do not entail UNIQUENESS OF EVENTS. For instance, you can reread the same book or the same passages from a given book more than once. The class of *read*-type verbs is very large, some examples are: *examine, analyze, barbecue, roast, iron, bathe, massage, wash, comb, brush, fry, polish, explain, confuse, pollute, control, cover, insulate, test, decorate, describe, drain, mop, survey, check ...* (Kratzer, 2004, p. 396, (10)).²³ The distinction between an Incremental Theme and a Strictly Incremental Theme is, among others, relevant for specifying the conditions under which a verbal predicate will have an obligatory quantized (telic) interpretation consistently in any context (Section "Quantized incremental themes in cumulative predications" above), although Krifka (1998) does not exploit it to this purpose. The conditions are sensitive to two main parameters, as Filip (2008) suggests: A verbal predicate will have a quantized (telic) interpretation consistently in all its occurrences if its Strictly Incremental Theme argument contains some expression of quantification, cardinality or measure, as we see below:

- (29) Mary ate all the sandwiches/three sandwiches
 (i) ... ? for an hour/in one hour.
 (ii) ... ?? but only finished two.

In contrast to (29), and as we have seen in examples in (21a-c), mere definiteness of the Strictly Incremental Theme argument does not enforce the quantization (telicity) of complex verbal predicates. Moreover, Incremental Theme arguments (which are not Strictly Incremental Theme arguments) may not enforce the quantization (telicity) of complex verbal predicates, even if they contain some expression of quantification, cardinality or measure: e.g., *Kim baked three pizzas for 20 min/ in 20 min*.

The impact of Krifka's (1989) paper "Nominal Reference, Temporal Constitution and Quantification in Event Semantics" goes well beyond the above observations. They represent merely a selection of some of the themes that have been recurrent in the theory on aspect since its publication. They have to suffice to illustrate the richness of this paper, and its indelible impact on how semanticists and philosophers conceive of the parallels between nominal and verbal domains.

²³ The line between incremental and non-incremental verbs is not always easy to draw, given that nearly all episodic verbs may be interpreted as incremental in a suitable context, cp. 'latent incremental theme verbs' (Rappaport Hovav & Levin, 2005).

Appendix

- (1) Temporal trace function $\tau: E \rightarrow T$ Krifka (1989), p. 97 (D 40)
 $\forall e \forall e' [\tau(e \cup_E e') = \tau(e) \cup_T \tau(e')]$
 The run time of the sum of two events e and e' is the sum of the run time of e and the run time of e' .
- (2) Locative trace function: $\pi: E \rightarrow L$ Lasersohn (1995), Krifka (1998)
 $\forall e \forall e' [\pi(e \cup_E e') = \pi(e) \cup_L \pi(e')]$
 The path trace of the sum of two events e and e' is the sum of the path trace of e and the path trace of e' .²⁴
 Example: $\llbracket \text{walk two miles} \rrbracket = \lambda x, e [\text{WALK}(x, e) \wedge \text{AGENT}(e, x) \wedge \text{MILE}(\pi(e)) = 2]$
 A set of sums of walking eventualities, each to the amount of two miles
- (3) A DERIVED MEASURE FUNCTION μ' . Intuitively, it describes the transfer of a measure function from one domain to another, on the assumption that there is a homomorphism h from one domain to the other, i.e., a function that preserves some structural relation defined on its domain in a similar relation defined on the range (Krifka, 1989, p.80). For example, a measure function μ for times like HOUR, WEEK or YEAR can be used as a derived measure function μ' on temporal traces of eventualities:
- (a) $\forall e [\mu'(e) = \mu(\tau(e))]$ Krifka (1989), p. 97 (D41)
 where $\tau(e) = t$, the temporal trace of e
- (b) $\text{HOUR}'(e) = \text{HOUR}(\tau(e))$
- (c) $\llbracket \text{sing for an hour} \rrbracket = \lambda x, e [\text{SING}(e) \wedge \text{AGENT}(e, x) \wedge \text{HOUR}'(e) = 1]$

²⁴ As Krifka (1998, Section 4.4) observes, this does not work in cases of backtracking, walking in curves or circles. In order to treat such a case, we can construct a measure function for motion eventualities from a measure function for distances which makes use of the mapping of motion eventualities to distances. This measure function can be standardized by linear moving events (for them, we have only to measure the distance between the start and the end), and it can be generalized by claiming additivity for any moving eventualities (that is, if e is a moving event of 6 kilometers and e' is a moving event of 4 kilometers, and e and e' do not overlap, then $e \cup_E e'$ is a moving event of 10 kilometers, even if the start and the end of $e \cup_E e'$ is less than 10 km apart).

- (4) Two-place predicates that capture the structure-preserving 'transfer' properties of thematic relations that mediate between objects and eventualities (Krifka, 1998, D 29-D 33, p. 92, and 2001)

- *Summativity (cumulativity)*

$$\forall R[\text{SUM}(R) \leftrightarrow \forall e, e', x, x' [R(e, x) \wedge R(e', x') \rightarrow R(e \cup_E e', x \cup_O x')]]$$

A general condition for the relation between thematic relations and the join operations. For example, two events of drinking a glass of wine yield an event of drinking two glasses of wine (Krifka, 1989, D29, p.92).

- *Uniqueness for Objects*

$$\forall R[\text{UNI-O}(R) \leftrightarrow \forall e, x, x' [R(e, x) \wedge R(e, x') \rightarrow x = x']]$$

There can be no two distinct objects which bear the thematic relation R to the same event (Krifka, 1989, D30).

- *Uniqueness for Eventualities*

$$\forall R[\text{UNI-E}(R) \leftrightarrow \forall e, e', x [R(e, x) \wedge R(e', x) \rightarrow e = e']]$$

There can be no two distinct events which bear R to the same object, that is, an event is related to a specific object. E.g., a drinking of a glass of wine is related only to this glass of wine as a theme/patient and to nothing else (Krifka, 1989, D31).

(o.k.: *eat, write*; not o.k.: *read, see, push, ride*)

- *Mapping to Subobjects*

$$\forall R[\text{MAP-O}(R) \leftrightarrow \forall e, e', x [R(e, x) \wedge e' \subset_E e \rightarrow \exists x' [x' \subset_O x \wedge R(e', x')]]^{25}$$

If an event bears R to an object, any subpart of the event bears R to some subpart of the object. E.g. every proper subpart of an event e of drinking a glass of wine corresponds to a proper subpart of the glass of wine (Krifka, 1989, D32).

(o.k.: *eat, write*; not o.k.: *read, see, push, ride*)

- *Mapping to Subeventualities*

$$\forall R[\text{MAP-E}(R) \leftrightarrow \forall e, x, x' [R(e, x) \wedge x' \subset_O x \rightarrow \exists e' [e' \subset_E e \wedge R(e', x')]].^{26}$$

If an event bears R to an object, any subpart of the object bears R to some subpart of the event (Krifka, 1989, D32).

(o.k.: *eat, write, read*; not o.k.: *see, push, ride*)

²⁵ Following Krifka (1998, 2001), the *Mapping to Subobjects* is defined in terms of ' $\subset$ ', instead of ' $\subseteq$ ' used in Krifka (1989).

²⁶ Following Krifka (1998, 2001), the *Mapping to Subevents* is defined in terms of ' $\subset$ ', instead of ' $\subseteq$ ' used in Krifka (1989).

(5) Classification of thematic relations (Krifka, 1989, p. 96 (14), 1998, 2001)

Example	SUM	UNI-O	MAP-E	MAP-O	UNI-E	
<i>eat an apple, write a letter</i>	+	+	+	+	+	Strictly Incremental Theme
<i>read a book</i>	+	+	+	—	—	Incremental Theme
<i>push a cart, see a movie</i>	+	+	—	—	—	Theme/Stimulus

(6) a. Strictly Incremental Theme: $\forall R[\text{SINC}(R) \leftrightarrow \text{UNI-O}(R) \wedge \text{MAP-O}(R) \wedge \text{MAP-E}(R) \wedge \text{UNI-E}(R)]$

b. Incremental Theme: $\forall R[\text{INC}(R) \leftrightarrow \text{UNI-O}(R) \wedge \text{MAP-O}(R) \wedge \text{MAP-E}(R)]$

(7) Maximal Participant:

$\forall x[\text{MAX}(P, x) \leftrightarrow P(x) \wedge \neg \exists y[P(y) \wedge x < y]]$ Zucchi and White (1996, 2001)

An individual is a maximal P iff it is P and it is not a proper part of another P.

(8) $\llbracket \text{write a sequence} \rrbracket = \lambda y \lambda e \exists x[\text{WRITE}'(e) \wedge \text{AG}(y, e) \wedge \text{PAT}(x, e) \wedge \text{MAX}(\lambda z \exists e'[\text{WRITE}'(e') \wedge \text{AG}(y, e') \wedge \text{PAT}(z, e') \wedge \text{SEQUENCE}'(z) \wedge \tau(e') \leq tR], x)]$

Zucchi and White (2001), p. 261)

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Chapter 15

On Ladusaw's "On the Notion *Affective* in the Analysis of Negative-Polarity Items"



Jon Gajewski

Abstract Ladusaw's paper "On the notion *affective* in the analysis of negative polarity items" presents a theory about the class of expressions that license negative polarity items in English. The result was first presented in chapter 6 of Ladusaw's 1979 PhD dissertation and would go on to have a profound influence on the field of linguistics. Ladusaw's account of the distribution of negative polarity items helped to bring about the maturity of the field of formal semantics, built on Montague's foundations, and its arrival as a core part of the study of the grammar of natural languages. Ladusaw shows that an insightful analysis of the distribution of negative polarity items can be given in simple, elegant model-theoretic terms within a denotational semantics.

Keywords Affective · "Any" · Downward entailing · Negative polarity items

Introduction

Ladusaw (1980) presents a theory about the class of expressions that license negative polarity items in English. The result was first presented in chapter 6 of Ladusaw (1979) and would go on to have a profound influence on the field of linguistics. Ladusaw's account of the distribution of negative polarity items helped to bring about the maturity of the field of formal semantics, built on Montague's foundations, and its arrival as a core part of the study of the grammar of natural languages. Ladusaw shows that an insightful analysis of the distribution of negative polarity items can be given in simple, elegant model-theoretic terms within a denotational semantics. The distribution of NPIs had been the object of intense inquiry in (at least) the 15 years prior to the publication of Ladusaw (1979). Those prior studies had attempted to analyze the distribution through syntactic mechanisms

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or otherwise through reference to aspects of meaning that were not explicitly formulated in a comprehensive theory of semantics. Ladusaw (1979) applied the latest updates of Montague's views (primarily Cooper (1975)) to assign precise functional meanings to licensing expressions and applied advances in the analysis of quantifiers (such as Barwise, 1978) to classify them by their formal properties. Ladusaw (1979) embeds the theory of licensing expressions within a finely detailed approach to scope within a framework of Cooper storage (Cooper, 1975). The great virtue of Ladusaw (1980) is that it eschews the finer points of Montagovian interpretation to present the fundamental result in a way that is instantly accessible to any linguist with a basic familiarity with set theory and logic. Formal semantics was now essential to any linguist who was paying attention.

In this brief paper, I will focus on Ladusaw's ideas and their direct developments. Regrettably, there will be little space for the alternatives that have developed in response. Most prominent among these is the approach advocated by Giannakidou (1998) et seq., which proposes that the distribution of polarity sensitive items can be more profitably analyzed in terms of (non-) veridicality.

The Scene

Ladusaw's theory of NPI licensing did not come out of nowhere. The intense study of NPIs and their distribution that commenced in earnest with Klima (1964) had produced a great number of insights that culminated in Ladusaw's achievement. Klima (1964) argues that NPIs must be c-commanded by a negative expression. The challenges for Klima were to explain why c-command was relevant and to precisely characterize the expressions that were negative in the relevant sense. The difficulty of the latter problem drove Klima to propose the feature [Affective] that triggered the presence of expression like *any* by transformation. This feature was simply stipulated to be associated with a list of licensing expressions. Jackendoff (1969) argued against Klima's analysis in terms of syntactic rules and in favor of the relevance of semantic scope. Jackendoff's theory of scope, however, differed greatly from those being developed in the newly forged Montague tradition. Baker (1970) was the first to argue that entailments should play a central role in the distribution of negative (and positive) polarity items. Fauconnier (1975) carried this idea a step further and argued that the ability of an operator to reverse the direction of entailment within a scale was relevant to the distribution of polarity sensitive items.

The Main Results

Ladusaw observes that one challenge for syntactic theories like Klima's is that there is apparently no syntactic unity among the expressions that license the presence of

NPIs in English. They include sentential negation, determiners, noun phrases, verbs, adjectives, prepositions, adverbs and adverbial conjunctions. See the examples below illustrated with the negative-polarity adverb *ever*.

- (1) a. *Bill will ever leave.
b. Bill **won't** ever leave
- (2) a. *Some professor will ever leave.
b. **No** professor will ever leave.
- (3) a. *Hillary always ever ate healthy foods.
b. Hillary **never** ever ate healthy foods.
- (4) It's *(**un**)likely that Pat will ever leave.

Examples (1)–(4) display the diversity of the categories of licensing expressions. However, these four examples include licensers that one might argue contain morphological instances of negation. A further problem for Klima is that within these categories it is also possible to find NPI licensers that bear no surface resemblance to negation and appear to differ substantially in meaning.

- (5) a. **Only** professors will ever leave.
b. Hillary **rarely** ever ate healthy food.
c. It's **surprising** that Hillary ever left.

Furthermore, Ladusaw observes that even if you could justify the assignment of the [+affective] feature to a certain set of lexical items, a theory that license NPIs based on their syntactic position relative to those lexical items will still fail. This is demonstrated by the case of the quantificational determiner *every*. Compare the licensing abilities of *every* with those of the determiner *no*.

- (6) a. **No** professor will ever leave.
b. ***Every** professor will ever leave.
- (7) a. **No** professor who ever got a raise will teach that class.
b. **Every** professor who ever got a raise will teach that class.

The determiner *no* evidently would bear the [+affective] feature as it licenses NPIs in its noun phrase restrictor and its verb phrase scope. To explain the licensing in (7)b, one would want to assign the feature to *every*, as well. But then one cannot explain the failure of licensing in (6)b: *every* and *no* bear the same syntactic relation to *ever* in (6) and by hypothesis both bear the affective feature.

Building on Fauconnier's ideas, Ladusaw proposes that what unites the licensing expressions is a pattern of entailments that derives from their semantic denotations. Specifically, Ladusaw proposes that NPI licensers validate inferences from weaker expressions to stronger ones. For example, consider the inference pattern of negation with respect to the verb phrase. Dancing implies moving, but not the other way around. The verb *dance* is stronger than the verb *move* in the sense that ((8)b) entails ((8)a).

- (8) a. The toy dog moves.
- b. The toy dog dances.
- (9) a. The toy dog doesn't move.
- b. The toy dog doesn't dance.

When negation is added, as in (9), the direction of entailment is reversed. Now (9)a entails (9)b. In other words, the sentence with the weaker verb entails the sentence with the stronger verb. Another way to look at this is in terms of the extensions of the verbs. Each verb determines a set of individuals that perform the action with respect to a point of reference (world and time). The set of dancers is necessarily a subset of the set of movers. So, in (8), we see an entailment from a subset to a superset. In (9), we see that negation supports entailment from supersets to subsets. Ladusaw suggests that affectivity can be defined in terms of such inferences.

- (10) An expression is affective iff it licenses inferences from supersets to subsets.
(Ladusaw, 1980, 22)

This theory applies well to all the cases of licensing in (1)–(5). Most strikingly this notion of affectiveness is able to explain the variety of licensing abilities of quantificational determiners. The determiner *no* licenses inferences from supersets to subsets both in its restrictor and in its scope: (the set of children is necessarily a subset of the set of people.)

- (11) a. No person moved. $\models$
- b. No child moved.
- (12) a. No person moved. $\models$
- b. No person danced.

This contrasts with the case of the universal quantifier. While *every* licenses superset to subset inferences in its nominal restrictor, it does not license such inference in its verb phrase scope.

- (13) a. Every person moved. $\models$
- b. Every child moved.
- (14) a. Every person moved. $\not\models$
- b. Every person danced.

This disparate pattern of inferences correlates with the ability to license NPIs in the restrictor but not in the scope. The move from inference patterns to licensing happens in two steps. First, licensors are assigned functional denotations. The formal properties of these functions underwrite the observed inference patterns. Second, a licensing principle must be stated that references the class of functions with the relevant properties.

Following Montague (1973) and subsequent work, Ladusaw proposes that quantificational determiners denote functions from sets to functions from sets to truth values. Ladusaw provides such denotations for *some*, *every* and *no*. I reproduce the denotations below in their intensional logic translations.

- (15) a. $\lambda X.\lambda Y.\exists x[X(x) \wedge Y(x)]$ *some*
 b. $\lambda X.\lambda Y.\forall x[X(x) \rightarrow Y(x)]$ *every*
 c. $\lambda X.\lambda Y.\neg\exists x[X(x) \wedge Y(x)]$ *no*

These functions can now be classified in terms of their entaillingness properties. The crucial definition is in (16). The definition is stated to apply when the range of δ' is either sets ($\subseteq$) or truth values ($\rightarrow$).

- (16) An expression δ is *downward entailing* (affective) iff its denotation δ' is a monotone decreasing function.

δ' is monotone decreasing iff $\forall X\forall Y \square [[X \subseteq Y] \rightarrow [\delta'(Y) \subseteq/\rightarrow \delta'(X)]]$

- (17) An expression δ is *upward entailing* (affective) iff its denotation δ' is a monotone increasing function.

δ' is monotone increasing iff $\forall X\forall Y \square [[X \subseteq Y] \rightarrow [\delta'(X) \subseteq/\rightarrow \delta'(Y)]]$

By these definitions, the determiners *every* and *no* turn out to be downward entailing, and *some* does not. Instead, *some* is upward entailing, as defined in (17). This explains why the former license NPIs within the nominal restrictor as in (7), but *some* does not:

- (18) *Some professor who ever got a raise will teach that class.

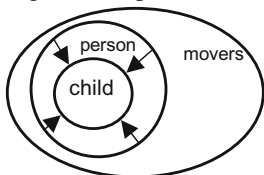
The important thing to notice now is that downward entaillingness is not only a property of lexical items. It is defined for all expressions that denote the appropriate kind of function. So, in addition to being able to diagnose whether or not a determiner is downward entailing, we can also ask whether a complete noun phrase is downward entailing. That is once a determiner applies to a common noun, we can inquire as to whether the denotation of the output is a monotone decreasing function.

- (19) a. $\lambda Y.\exists x[N'(x) \wedge Y(x)]$ *some N*
 b. $\lambda Y.\forall x[N'(x) \rightarrow Y(x)]$ *every N*
 c. $\lambda Y.\neg\exists x[N'(x) \wedge Y(x)]$ *no N*

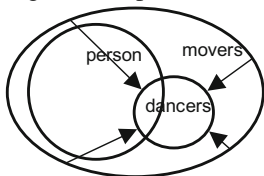
It turns out that (19)a is not monotone decreasing, just like (15)a; and (19)c is monotone decreasing like (15)c. On the other hand, while (15)b is monotone decreasing, (19)b is not. This tracks the pattern of entailments that we observed in (11)–(14).

It can be helpful to visualize some of these inferences with Euler diagrams, as Ladusaw does. The denotation for *every* indicates that the extension of its first argument (the nominal restrictor) is a subset of the extension of its second argument (the scope). It is clear that if A is a subset of B and C is a subset of A, then C will be a subset of B as well.

- (20) Figure 1. Superset to subset inferences succeed in the restrictor with *every*.



- (21) Figure 2. Superset to subset inferences fail in the scope of *every person*.



To fully grasp the significance of this difference between *every* and *every N* for licensing it is necessary to see how Ladusaw states the licensing principle for NPIs. Klima (1964) had suggested that an NPI must be commanded by an affective expression. Ladusaw (1979) argues in detail that the relevant notion is the relative scope of the NPI and the affective expression.¹

- (22) For any two expressions α and β , constituents of a sentence ϕ , α is in the scope of β
with respect to an interpretation of ϕ , ϕ' , iff the interpretation of α is used in the formulation of the argument of β 's interpretation in ϕ' .
- (23) A negative-polarity item is acceptable only if it is interpreted in the scope of a downward entailing expression.

In (7)b, repeated below as (24), the interpretation of the NPI *ever* is used in the formulation of the argument of the interpretation of the determiner *every*, which as we saw denotes a monotone decreasing function. In (6)b, on the other hand, repeated below as (25), *ever* does not contribute to the argument of *every*. Instead, *ever*'s interpretation is used in the formulation of the argument of the interpretation of the full noun phrase *every professor*.

- (24) Every professor who ever got a raise will teach that class.
(25) *Every professor will ever leave.

As we have just shown, full noun phrases like *every professor* are not downward entailing. This concludes the main, quite compelling point of Ladusaw (1980).

¹ These notions collapse in frameworks where scope derives from c-command (at LF).

Questions and New Directions

Much of the discussion that followed Ladusaw's results centered on the extent to which the generalization was correct. Though the introduction of downward entailingness represented a great step forward in the theory of negative polarity items, there were legitimate questions about the extent of its application. Ladusaw acknowledges that there are expressions that are neither upward nor downward entailing but he does not investigate their licensing properties in any depth. Furthermore, among the cases that Ladusaw identified as downward entailing, there are several expressions for which the actual entailment pattern is far from clear. Finally, there are cases in which NPIs can be licensed but there is no obvious way to apply the definition of downward entailingness as given in Ladusaw (1979, 1980). We will look at each case in turn and track some of the developments in each area.

The Licensing Properties of Non-Monotone Quantifiers

Ladusaw (1979) observes that quantificational determiners such as *exactly three* are neither upward nor downward entailing and that *exactly three* projects noun phrases that are also neither upward nor downward entailing. Without producing any examples, Ladusaw concludes that such non-monotone expressions display licensing behavior similar to upward entailing expressions. That is, he proposes that they do not license NPIs. This conclusion was questioned very quickly by Linebarger (1980). Linebarger produces several convincing examples that raise even deeper questions about Ladusaw's approach.

- (26) Exactly three students said anything in yesterday's discussion section.
- (27) Exactly thirty students said anything in yesterday's discussion section.

The acceptability of (26) contrasts with the relative unacceptability of (27) in a context where discussion sections typically have around thirty students. Linebarger proposes that NPIs may be licensed by Negative Implicatures, which may derive from defeasible contextual information. For example, in the appropriate context, (26) could be taken to carry the Negative Implicature below.

- (28) Not many students said anything in yesterday's discussion section.

According to Linebarger the presence of negation in the Negative Implicature is enough to license the NPI. For Ladusaw, these examples raise the question of how licensing could vary with context. A noun phrase is or is not downward entailing. Ladusaw's theory leaves no room for contextual variation.

On the other hand, Linebarger's (1980, 1987) theory was not adequately constrained. No systematic method of generating Negative Implicatures was produced and it was unclear how one could know where negation may or may not be appearing in the unspoken implicatures.

Examples like (26) have led some researchers to propose explicitly that Ladusaw's notion of affective is not correct. A weaker condition has been proposed in several iterations (Progovac, 1993; Rothschild, 2006). The suggestion is that an NPI licenser must only be *not* upward entailing. This allows downward entailing and non-monotone expressions to license NPIs. The problem for such accounts is that they tend to be too permissive. First, they face the same difficulty Ladusaw does in explaining how (27) could have a different judgment from (26). *Exactly thirty students* is not upward entailing, so on this view it should license NPIs. Second, there are clearly many non-monotone quantifiers that do not license NPIs.

- (29) *An odd number of students did any work. (Schlenker p.c. to Rothschild, 2006)

A promising line of inquiry has recently been opened by Crnič (2014). Crnič analyzes *any* as containing a covert *even* as part of its meaning. He proposes that the felicity conditions on the use of *even* determine *any*'s distribution. *Even* is assumed to carry a presupposition that it's focus is the least likely among a set of alternatives. There is an important connection between likelihood and entailment: If α entails β , then β cannot be less likely than α . Several results follow. If *any* is in an upward entailing context, it will be infelicitous. If it is in a downward entailing context, it will be felicitous. If it is in a non-monotone context, the felicity of *any* will depend on context, as desired. This theory weakens the licensing restriction on NPIs like *any* in a principled way without opening the floodgates to any non-monotone environment. For details, I refer the reader to Crnič's (2014) paper.

Expressions That May Not Be Downward Entailing as Claimed

Ladusaw (1980) focuses on two test cases for the downward entailing theory: the quantificational determiners and conditionals. Ladusaw shows that typically NPIs are licensed in the antecedent of conditionals and are not licensed in the consequent of conditionals.

- (30) a. If *anyone ever* discovers that the money missing, then John will return it.
 b. *If John doesn't return it, then anyone will ever discover the money missing. (Ladusaw, 1980, 34)

Ladusaw assumes that the conditional connective can be represented by material implication. He then shows that material implication is downward entailing with respect to its first argument, the antecedent, and upward entailing with respect to its second argument, the consequent.

- (31) a. $\forall p \forall q \forall r \Box [[p \rightarrow q] \rightarrow [[q \rightarrow r] \rightarrow [p \rightarrow r]]]$
 b. $\forall p \forall q \forall r \Box [[p \rightarrow q] \rightarrow [[r \rightarrow p] \rightarrow [r \rightarrow q]]]$

Ladusaw bolstered this conclusion with the intuition about actual inferences involving sets and supersets. Assuming that the set of brussels sprouts is necessarily a subset of the set of green vegetables, Ladusaw argues that the following entailments are valid.

- (32) a. If John eats a green vegetable for dinner, then Mary will reward him. $\models$
 If John eats brussels sprouts for dinner, then Mary will reward him.
 b. If Mary rewards him, then John will eat brussels sprouts for dinner. $\models$
 If Mary rewards him, then John will eat a green vegetable for dinner.

Even at the time it was known that material implication was not an adequate analysis of the conditional in English, cf. Stalnaker (1975). Furthermore, it was immediately questioned whether conditionals actually support superset to subset inferences even in their antecedents. Linebarger (1980) and Heim (1984) produce examples like the following, which are related to the classic examples from Lewis (1973) and Stalnaker (1975).

- (33) a. If you read a newspaper, you are well informed.
 b. If you read a newspaper and cannot retain information, you are well informed.
 c. If you read any newspaper at all, you are well informed.
 (34) a. If the USA throws its nuclear weapons in the ocean tomorrow, there will be war.
 b. If the USA and every other nation throw their nuclear weapons in the ocean tomorrow, there will be peace.
 (35) a. If you strike a match, it will light.
 b. If you strike a wet match, it will light.

In this case, (33)a clearly does not entail (33)b. Nevertheless, we see that an NPI can appear in the corresponding conditional in (33)c. Heim argues that the NPI is allowed here because there is a form of "Limited Downward Entailingness" that applies here. Heim argues that contextual information can play a crucial role in establishing DE-ness. In her opinion, the acceptability of (33)c, requires the presence of a contextual assumption like this:

- (36) If using a set of information A makes you well informed and B is a superset of A, then using B will also make you well informed.

Heim proposes the principle below where c represents the contextual set of background assumption and $X[A/B]$ is just like X except that all instance of A have been replaced by B.

- (37) c; $X[A/B] \rightarrow X$; If X, then Y
 $\models$ if $X[A/B]$, then Y

How principles like this fit into a general theory of Limited Downward Entailingness proved difficult to state formally. Von Fintel (1999) took up the challenge of NPIs in conditional antecedents again, arguing that licensors need only meet a weaker condition than DE-ness. Von Fintel (1999) calls the new weaker property of licensors Strawson Downward Entailingness. This notion of DE-ness holds when the presuppositions of the conclusion of the inference are assumed to hold. See (38) presented in the style of Ladusaw's (1980) definition in (16) with the additional condition that X belongs in the domain of the possibly partial function δ' .

- (38) An expression δ is *Strawson downward entailing* iff
 it denotes a (partial) function δ' such that
 $\forall X \forall Y \square [[X \subseteq Y] \rightarrow [[\delta'(Y) \wedge X \in \text{dom}(\delta')] \rightarrow \delta'(X)]]$

In the case of conditionals, von Fintel argues that a conditional are interpreted relative to a set of background assumptions represented in a modal horizon. According to most approaches to the failure of strengthening of the antecedent, as shown in (33) and (34), conditionals quantify over a set of worlds that are most similar to our world in which the antecedent is true. The closest worlds to ours in which we strike a match are worlds where the match is dry. So, in (35), a completely different set of worlds is quantified over. The closest worlds to ours in which you strike a match and the match is wet are not worlds in which the match is dry. Von Fintel (1999) argues that the modal horizon is a contextual parameter and the conditionals carry a presupposition that the modal horizon contains worlds in which the antecedent is true. When assessing Strawson Downward Entailingness, contextual parameters such as the modal horizon must be held constant in the superset to subset inference. Strawson Downward Entailingness prescribes that the presupposition of the conclusion should be added as a premise of the inference.

- (39) a. If you strike a match, it lights.
 For every world $w' \in H_w$ such that you strike a match in w' , it lights in w' .
 b. If you strike a wet match, it lights.
 For every world $w' \in H_w$ such that you strike a wet match in w' , it lights in w' .
 c. Presupposition of (39)b: H_w includes worlds in which you strike a wet match.

Thus, (39)b does follow from (39)a together with (39)c. This is a great simplification of von Fintel's account of conditionals and the reader is referred to von Fintel (1999, 2001) for details.

Only

Strawson Downward Entailment can also be useful in the analysis of several other constructions that license NPIs but appear not to license inferences from supersets to subsets. A classic example is the case of *only*. *Only NP* clearly licenses NPIs in its scope as in (40). However, most speakers feel that inference from (41)a to (41)b is not valid.

- (40) Only Pat ate anything.
 (41) a. Only Pat ate a green vegetable.
 b. Only Pat ate brussels sprouts.

The reason the inference fails appears to be that (41)b requires that Pat actually ate brussels sprouts, but that cannot be inferred from (41)a. However, if we follow Horn (1969) in proposing that *Only NP* presupposes that the VP is true of the NP and asserts that it is true of no other individual, then we can view this as a valid Strawson inference.

- (42) Only Pat ate a green vegetable.
 Pat ate brussels sprouts.
 $\models$ Only Pat ate brussels sprouts.

In fact, Ladusaw (1979) already offers an analysis of this case. He proposes to view Horn's presupposition as a conventional implicature in the sense of Karttunen and Peters (1979). Karttunen and Peters (1979) sort truth conditions and conventional implicatures into two separate dimensions. Ladusaw proposes that the second dimension, in which conventional implicatures are expressed, is irrelevant to the calculation of DE-ness.

Definite Descriptions

The analysis of apparently non-monotone licensors in terms of Strawson entailment raises some interesting questions about NPI licensing in definite descriptions. In (43), we give a supremum-based semantics for definites based on Link (1983).

- (43) a. $\llbracket \text{the } \rrbracket(P) = \sigma(P)$ (Cf. Link, 1983)
 b. $\sigma(Q)$ is defined only if $\sqcup Q \in Q$
 When defined, $\sigma(Q) = \sqcup Q$

When P is the denotation of a singular predicate, the presupposition in (43)b amounts to a uniqueness predicate that there is one and only one atomic individual in P . When P is the denotation of a plural predicate, (43)b amounts to an existence presupposition. If we follow the scheme of Strawson DE-ness, then both singular and plural definite descriptions turn out to be Strawson DE in their nominal restrictions.

- (44) The student arrived.
 There is a unique salient French student. (Presupposition of conclusion)
 $\therefore$ The French student arrived.
- (45) The students arrived.
 There are French students. (Presupposition of conclusion)
 $\therefore$ The French students arrived.

The consensus appears to be, however, that singular definite descriptions typically do not license NPIs in their restrictions, but that plural definite descriptions may.

- (46) a. *The student who had ever studied NPIs dropped my class immediately.
 b. The students who had ever studied NPIs dropped my class immediately.

Lahiri (1998) observes that this is the case and proposes a solution. He notes that in addition to being Strawson Downward Entailing, singular definite descriptions also turn out to be – somewhat surprisingly – also Strawson Upward Entailing:

- (47) The French student arrived.
 There is a unique student.
 $\therefore$ The student arrived.

This is not true of plural definite descriptions. The attempted Strawson inference in (48) fails. The arrival of the French students and the existence of students does not guarantee the arrival of any non-French students.

- (48) The French students arrived.
 There are students.
 $\# \therefore$ The students arrived.

Consequently, Lahiri (1998) proposes that Strawson DE-ness is not sufficient for licensing. He proposes instead that an NPI must be contained in an environment that is Strawson DE and not Strawson UE. We can then rewrite our licensing principle as follows:

- (16) An NPI α must occur in the scope of an expression δ that is
 Strawson DE and not Strawson UE.

For further developments of and possible alternatives to Lahiri's account see Guerzoni and Sharvit (2007) and Gajewski and Hsieh (2014).

NPIs in Questions

NPIs occur quite comfortably in many different kinds of questions.

- (49) a. Did you eat any soup?
 b. Who would like anything to eat?

As stated, it is unclear how to apply Ladusaw's definition of Downward Entailment to questions. There have been attempts to generalize the definition of entailment to the denotations of questions, for example as sets of propositions, cf. Groenendijk and Stokhof (1997). But these efforts have failed to draw the correct lines between licensors and non-licensors. See for example, the critiques in Guerzoni (2003).

The most productive strategy in this domain has been to follow Karttunen (1977) in focusing on the semantics of embedded interrogatives and their NPI licensing abilities. Guerzoni and Sharvit (2007) is a very detailed study of NPIs in embedded interrogatives. Interestingly, Guerzoni and Sharvit conclude that (Strawson) Downward Entailingness does not provide an adequate account of the distribution of NPIs in questions. They propose that the property that best predicts whether or not NPIs are licensed in embedded interrogatives is whether the predicate embedding the interrogative requires a strongly exhaustive interpretation of the interrogative. For exhaustivity, see Groenendijk and Stokhof (1984) and especially Heim (1994).

Nicolae (2013) builds on Guerzoni and Sharvit's observation and proposes that strong exhaustivity can be analyzed in terms of a covert *only* in the denotation of the question. As we have seen above, *only* is an NPI licensor and can be shown to be Strawson DE. For an alternative in terms of non-veridicality see Giannakidou (1998).

Why Downward Entailment?

Accurately specifying the formal properties of contexts that license NPIs is an important step in developing a theory of negative polarity and polarity sensitive items. However, it has long been acknowledged that it is necessary to move beyond this stage to greater explanatory depth. Ultimately, we need to understand the semantics and pragmatics of polarity sensitive items and how that gives rise to their limited distribution. It has long been observed that there are tendencies in which items show sensitivity to negative polarity. Expressions of minimal quantity are especially frequent as NPIs. It has been proposed that there are essentially pragmatic reasons why minimal quantities prefer to be in negative environments. The reason is that it is in these environments that expressions of minimal quantity are most informative.

This kind of intuition has been developed in many different but related strands in the literature. Kadmon and Landman (1993) propose that *any* is subject to two conditions. First, *any* as an indefinite must widen its domain relative to the domain of quantification in the context. Widening the domain of an existential makes it a weaker quantifier. Second, Kadmon and Landman propose that *any* is subject to a pragmatic requirement to produce a stronger proposition. To satisfy

this requirement, the weakened, widened existential quantifier must be placed in the right environment. An entailment reversing environment, as in the scope of a DE quantifier, will do the trick. Kadmon and Landman show how their proposal of widening and strengthening is related to and distinct from Ladusaw's (1979) proposal.

Lahiri (1998) gives a finely worked out theory of the distribution of polarity items in Hindi. These items contain morphologically an instance of an *even*-like particle combined with an indefinite. *Even* requires that the phrase that it associates with is the least likely among a set of alternatives. An indefinite, however, is weak relative to other quantifiers, i.e., it is entailed by other quantifiers such as the universal. Consequently, one way to satisfy the requirements of *even* is for the indefinite to be embedded in a polarity-reversing context. Though an indefinite is the most likely among the set of alternative quantifiers, the negation of an indefinite entails the negation of other stronger quantifiers and is thus the least likely among the alternatives. This derives the polarity sensitivity of these *even* + indefinite items in Hindi.

Chierchia (2013) combines properties of Kadmon and Landman (1993) and Lahiri (1998) in an analysis of English *any* and a great variety of other polarity sensitive items across languages. Chierchia's analysis also takes inspiration from Krifka's (1995) approach to polarity sensitivity in terms of alternatives and their interactions with different varieties of assertion operators. Chierchia (2013) proposes that *any* invokes a particular set of alternatives relative to its quantification domain. Specifically, he proposes that *any* invokes *subsets* of its domain as alternatives. Because *any* is analyzed as an existential, the subdomain alternatives represent stronger quantifier alternatives. These alternatives feed the interpretation of covert focus particles which may correspond to *even* or *only*. Chierchia shows that in unembedded contexts the requirements of the particles combined with the evoked alternatives lead to an unassertable proposition. However, when embedded in the right environments, including in the scope of a DE operator, the requirements of the particle can be met.

A quite recent paper, Barker (2018), proposes that the distribution of negative polarity items can be derived from the idea that NPIs exist to mark unambiguous narrow scope. In particular, Barker claims that NPIs are only licensed in environments in which that wide scope reading of an indefinite would fail to entail the narrow scope reading. This derives a distribution for NPIs that corresponds most closely to the non-monotonic theories discussed above in section "The licensing properties of non-monotone quantifiers".

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Chapter 16

On Lewis's "Adverbs of Quantification"



Mats Rooth

Abstract David Lewis's "Adverbs of Quantification" provided an account of words such as *always* and *usually* in examples where they occur in construction with restrictive clauses and indefinite descriptions. According to Lewis, adverbs of quantification quantify "cases", which are tuples of witnesses for indefinite descriptions. This paper is an elementary introduction to Lewis's approach. In a formalization, clauses contribute properties of cases, and adverbs of quantification contribute relations between properties of cases.

Keywords Adverb of quantification · Dynamic semantics · Indefinite description · Anaphora · Compositional semantics · Case quantification

Introduction

Adverbs of quantification in David Lewis's sense are words that semantically have a quantifying role, and morpho-syntactically are adverbs that can occur in the pre-verbal auxiliary position of English sentences (Lewis, 1975). (1) is a list of common adverbs of quantification, and (2) are some sentences where they are used. (3) lists some complex items. Lewis's paper called attention to the interest and complexity of establishing a compositional structure and semantics for adverbs of quantification, and made a specific proposal about it.

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- (1) always invariably universally
 usually mostly generally
 often frequently commonly
 rarely infrequently seldom
 sometimes occasionally
 never
- (2) a. The fog usually lifts before noon here.
 b. If a dart hits a unit square, it almost never hits it on the diagonal.
 c. People who commit mass murders generally suffer from an extraordinary lack of impulse control.
- (3) almost always nearly always
 almost never nearly never

Calling the adverbs quantificational is justified by the fact that sentences with adverbs of quantification (henceforth AQs) sometimes have paraphrases or near-paraphrases involving quantificational determiners.¹ For instance sentence (4a) is paraphrased by (4b) with a quantificational determiner *most* and nominal generalized quantifier *most times*. In sentence (5b), the paraphrase has a nominal quantifier *every quadratic equation* in subject position.²

- (4) a. This clock is usually off by at least half an hour.
 b. At most times, the time showing on this clock differs from the actual time by at least half an hour.
- (5) a. Quadratic equations always have two real solutions.
 b. Every quadratic equation has two real solutions.

In (4a), *usually* presumably quantifies times, and AQs are sometimes morphologically related to the root *time* (for instance *some-times*, or Swedish *al-tid*, morphologically ‘all time’). However in (5) time is irrelevant, and on the face of it equations are being quantified, just as in (5b) with a nominal quantifier.³ Similarly, (6a) with the AQ *sometimes* is talking about adverbs of quantification in English and other languages, not about how the nature of adverbs of quantification changes over time.

- (6) a. Adverbs of quantification are sometimes morphologically related to the root *time*.
 b. Some adverbs of quantification are morphologically related to the root *time*.

¹ As pointed out by Zoltán Gendler Szabó, similar paraphrases are found in Russell (1905), where they go in the other direction. Russell says that *C(everything)* means “*C(x)* is always true”.

² This leaves open why the determiners in the paraphrases are to be considered quantificational, and Lewis does not try to give an account of this. As worked out in sections “[Semantics of the LF](#)” and “[Dynamic Case Quantification](#)”, the terminology is justified in Lewis’s theory, where the semantics of AQs is isomorphic to the semantics of determiners, as analyzed in generalized quantifier theory.

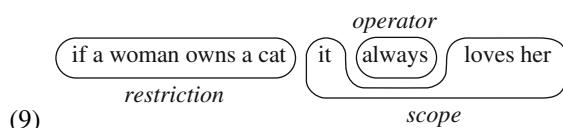
³ This point is made by Lewis. I’m not going to say this every time I could.

An important group of examples are ones where the AQ occurs together with an adverbial *if*-clause, see (7).⁴ A feature of these examples is that what is quantified is introduced by some phrases or combination of phrases in the *if*-clause. In (7c) one can reasonably say that headlines are quantified, because of the near paraphrase (8). The phrase *a headline* is part of the *if*-clause in (7c).

- (7) a. If you drop an unbreakable object, it always lands on something more valuable.
 b. If two cars are driving in opposite directions on a long road with a one-way bridge, they always meet at the bridge.
 c. If a headline ends in a question mark, the answer is usually "no".
- (8) Most headlines that end with a question mark are answered negatively.

Logical Forms for AQs

A *logical form* or *compositional structure* in natural language semantics is a structure that encodes in its syntax how the semantics of parts of a sentence are composed to give a semantics for the whole, so that each structure has an unique semantic interpretation. Usually, a logical form (or LF) is a complex of functions or operators, and the arguments of those functions or operators. Accordingly, an important issue in establishing the logical form for a sentence is identifying the main function or operator, and establishing the number and identity of the arguments of that operator. For sentences with an AQ and an *if*-clause, Lewis made the hypothesis that the adverb of quantification is the main operator, and that it has *two* arguments. The *if*-clause supplies one argument (called the *restriction* of the AQ), and the main clause of the sentence (minus the AQ itself) supplies the other argument, termed the *scope*. This is shown graphically in (9).



Beyond articulating sentences with an AQ into an operator, a restriction, and scope, Lewis hypothesized that any number of variables can be quantified by the AQ. This is the analysis of AQs as *unselective* quantifiers. The variables that are to be quantified are those corresponding to indefinite descriptions. For expository purposes, here we begin by writing the quantified indices as numerical subscripts on the AQ.⁵ In a notation with the operator on the left, the logical form of a sentence

⁴ (7a) and (7b) are cases of Murphy's law.

⁵ Then in section "[Dynamic Case Quantification](#)", the subscripts are eliminated, by encoding the variables that are quantified in the semantics.

with an AQ is as exemplified in (10), where indices appear on the operator [always], and also on the phrases [a woman]₁, [a cat]₂, [it]₂, and [her]₁ in the restriction and scope. Two arguments follow the AQ.⁶

(10) [[always]_{1,2} [a woman₁ owns a cat₂] [it₂ loves her₁]]

Lewis proposed that the semantics of the structure is quantification over *cases*, which are tuples of individuals. With two quantifying indices, the quantified cases have two slots, and the semantics of (10) is as stated in (11).

(11) Semantics of (10)

For every case $\langle x_1, x_2 \rangle$ such that x_1 is a woman, x_2 is a cat, and x_1 owns x_2 :
 x_2 loves x_1 .

How can the structure (10) be derived systematically, and what is the source of the clauses such as “ x_1 is a woman” seen in (11)? The second chapter of Irene Heim’s dissertation addressed this with a sequence of syntactic operations (Heim, 1982). We start with a syntactic structure (12a), that already has indices on indefinite descriptions and pronouns. First, indefinite descriptions are adjoined to a dominating clause, introducing an index on the moved constituent, and leaving behind an indexed empty category that functions as a variable.⁷ See (12b), where the two indefinite descriptions have been moved to the level of the restrictive clause, introducing the empty categories e_1 and e_2 . Second, the indices of indefinite descriptions are copied to the minimal c-commanding quantificational adverb, which in this case is *always*.⁸ Finally, an indexed indefinite description [a α]_i is rewritten as [$e_i \alpha$]. See (12c), where the indices 1 and 2 have been copied as subscripts onto [always]_{1,2}, and where [a cat]₂ has been rewritten as [e_2 cat].⁹ In our example, these steps transform the restriction into a sequence of atomic formulas, consisting of relation symbols with indexed arguments (traces or pronouns) that have the status of bound variables. Indices coming from indefinite descriptions appear as subscripts on the AQ.

(12) a. [[always] [a woman₁ owns a cat₂] [it₂ loves her₁]]
 b. [[always][[a woman]₁ [[a cat]₂ [e_1 owns e_2]]] [it₂ loves her₁]]
 c. [[always]_{1,2}[[e_1 woman] [[e_2 cat] [e_1 owns e_2]]] [it₂ loves her₁]]

Heim pointed out that additional treatment is required for sentences such as (13a) where there is an indefinite description in the scope. Somehow, *a shed* should be interpreted existentially in the scope, rather than being quantified by *usually*. For this, Heim proposed that there is an existential operator adjoined to the scope, and

⁶ Lewis encodes the same information by introducing variables such as x and y , with the understanding that all such variables in the restriction are quantified. This is (10) in the notation used by Lewis: always if x is a woman and y is a cat and x owns y , y loves x .

⁷ Adjunction is the tree transformation that maps [_A ... X ...] to [_AX_i [_A ... e_i ...]].

⁸ A node A c-commands a node B if and only if the node dominating A dominates B, and A does not dominate B.

⁹ This last step, which is not found in Heim, is included here in order to produce something that has the syntactic shape of an atomic formula.

that the indefinite $[a \text{ shed}]_3$ is indexed with it by the procedure that copies the index of an indefinite description to the minimal c-commanding adverbial operator. The result in this case is (13b), where the index originally on $[a \text{ shed}]_3$ ends up as a subscript on the existential adverb of quantification $\exists_3$, and as a variable in the atomic formula $[e_3 \text{ shed}]$.

- (13) a. If a woman₁ owns a cat₂, she₁ usually keeps it₂ in a shed₃.
 b. $[\text{usually}_{1,2} [[a \text{ woman}_1][[a \text{ cat}_2] [e_1 \text{ owns } e_2]]] [\exists_3 [[e_3 \text{ shed}] [she_1 \text{ keeps } it_2 \text{ in } e_3]]]$

Revising the syntactic procedure stated earlier, first an existential operator $\exists$ is adjoined to the scope of each adverb of quantification, and then we proceed as before, with $\exists$ together with AQs treated as a target of indexing.¹⁰

Copying indices from indefinite descriptions and nothing else has another good consequence. When as in (14) there is an indexed pronoun in the *if*-clause, the index does not get copied, because it is an index on a pronoun rather than an indefinite description. This explains why the pronoun $[it_3]$ is interpreted in the context (perhaps it refers to a certain famous beautiful and friendly cat), rather than being quantified by the AQ.

- (14) If a woman₁ owns $[a \text{ cat}]_2$ that resembles it₃, it₂ always loves her₁.

Semantics of the LF

The case-quantifying semantics for *always* needs to be generalized to other adverbs of quantification. A neat way of achieving this is to assume that AQs *always*, *usually*, *never*, (and so forth) have the same root meanings as nominal determiners *every*, *most*, *no*, (and so forth), except that the AQs quantify cases rather than individuals. (15) gives the semantics for some quantificational determiners, as they are defined in generalized quantifier theory (Barwise and Cooper, 1981).¹¹ A and B are sets of individuals.

- (15) Let A and B be sets of individuals. Two-place relations corresponding to determiners have the following truth conditions.

every (A, B)	true iff $A \subseteq B$
most (A, B)	true iff the cardinality of $A \cap B$ is greater than the cardinality of $A - B$
no (A, B)	true iff $A \cap B = \emptyset$

¹⁰ $\exists$ is not an adverb of quantification, because it has one argument rather than two.

¹¹ See Chap. 3 in this volume.

(16) uses the same definitions for the root meanings of some adverbs of quantification. The switch from (15) is that A and B are sets of cases, rather than sets of individuals.

(16) Let A and B be sets of cases. Two-place relations corresponding to adverbs of quantification have the following truth conditions.

always (A, B)	true iff $A \subseteq B$
usually (A, B)	true iff the cardinality of $A \cap B$ is greater than the cardinality of $A - B$
never (A, B)	true iff $A \cap B = \emptyset$

The relation between AQs and quantificational determiners is supported by the fact that the phrases of the form *in D case* or *in D cases*, where D is a determiner, can function as AQs. See the examples in (17).¹²

- (17) a. If a woman owns a cat, $\left\{ \begin{array}{l} \text{in every case} \\ \text{in most cases} \\ \text{in some cases} \end{array} \right\}$ it loves her.
- b. If a woman owns a cat, it $\left\{ \begin{array}{l} \text{in all cases} \\ \text{in most cases} \\ \text{in some cases} \end{array} \right\}$ loves her.

To link up the root meanings with the LF, we use the indices on the AQ to extract a set of cases from the restriction, and another set of cases from the scope. The root meaning of the AQ takes those two sets of cases as arguments. In (18), the lambda binder $\lambda n_1, \dots, n_k$ is understood to be of an abstractor over cases, so that $\lambda n_1, \dots, n_k \phi$ and $\lambda n_1, \dots, n_k \psi$ are sets of cases of length k . Here $n_1, \dots, n_k$ are the numerical indices seen in the LF.¹³

- (18) Case quantification interpretation for the logical forms of AQs
 A clause of the form $[\text{AQ}_{n_1, \dots, n_k} \phi \psi]$ is semantically equivalent to
 $\text{AQ}(\lambda n_1, \dots, n_k \phi, \lambda n_1, \dots, n_k \psi)$.

Let us work through an example that exercises everything in the analysis. Sentence (19a) has indefinite descriptions in both the restriction and scope, and also

¹² While Lewis theorized about quantifying cases, the noun *case* hardly figured in his database of examples. Moltmann (2017) is a study of uses and interpretations of this noun.

¹³ So on Lewis's analysis, AQs quantify cases, and cases are tuples of individuals. Or if the analysis is generalized to pronouns of other types, cases are tuples of model-theoretic objects of various types. The principal competing account of cases is that they are situations (Berman, 1987; Von Stechow, 1994). This results in a really different theory. Notably, the account of pronouns has to be different, because situations do not provide antecedents for pronouns in the same direct way as tuples. An approach to this is that pronouns in AQ constructions are elliptical definite descriptions (Elbourne, 2001). A paraphrase of the semantics of (2b) is roughly "in almosted situation of a dart hitting a unit square, does the dart in that situation hit the unit square in that situation on the diagonal".

has pronouns in both the restriction and scope. The sentence is transformed into an LF that is interpretable by the semantics by a sequence of syntactic transformations, as follows. In (19b), an existential operator is added to the scope of *always*. In (19c), indefinite descriptions are adjoined to dominating clauses, either the *if*-clause or the main clause, leaving behind indexed traces. In (19d), the indices of indefinite descriptions are copied to the minimal c-commanding adverbial operator, which is *always* in the case of $[a \text{ woman}]_1$ and $[a \text{ cat}]_2$ in the restriction, and $\exists$ in the case of $[a \text{ shed}]_4$ in the scope.

- (19) a. If a woman₁ owns a cat₂ that resembles it₃, she₁ always keeps it₂ in $[a \text{ shed}]_4$.
 b. $\exists$ *adjunction*
 $[always$
 $[a \text{ woman}]_1 \text{ owns } [a \text{ cat}]_2 \text{ that resembles it}_3]$
 $[\exists [she_1 \text{ always keeps it}_2 \text{ in } [a \text{ shed}]_4]]]$
 c. *Movement of indefinite descriptions*
 $[always$
 $[[a \text{ woman}]_1 [[a \text{ cat}]_2 \text{ that } [e_2 \text{ resembles it}_3]] [e_1 \text{ owns } e_2]]$
 $[\exists [[a \text{ shed}]_4 [e_1 \text{ keep } e_2 \text{ in } e_4]]]]]$
 d. *Indexing of indefinite descriptions to AQ and $\exists$*
 $[always_{1,2}$
 $[a \text{ woman}]_1 [[a \text{ cat}]_2 \text{ that } [e_2 \text{ resembles it}_3]] [e_1 \text{ owns } e_2]]$
 $[\exists_4 [[a \text{ shed}]_4 [e_1 \text{ keep } e_2 \text{ in } e_4]]]]]$

The semantics stated in (18) maps the restriction to a set of cases, by binding the indices on the AQ. In this case, there is a free pronoun $[it_3]$, which intuitively has to take its value from the context. Writing the value for this pronoun as $g(3)$, where g is an assignment function used to give values to free variables, we obtain the set of cases defined in (20).¹⁴

- (20) *Set of cases derived from the restriction (= A)*
 $[[\lambda 1 2] [[a \text{ woman}]_1 [[a \text{ cat}]_2 \text{ that } [e_2 \text{ resembles it}_3]] [e_1 \text{ owns } e_2]]]$
 The set of all pairs $\langle x_1, x_2 \rangle$ such that
 x_1 is a woman and
 x_2 is a cat and
 x_2 resembles $g(3)$ and
 x_1 owns x_2 .

Similarly, a set of cases is derived from the scope, by binding the same indices 1 and 2. In this case, there is an existential quantifier on the scope, which quantifies the index 4. The set of cases described in (21) is the result.

- (21) *Set of cases derived from the scope (= B)*
 $[[\lambda 1 2] [\exists_4 [[a \text{ shed}]_4 [e_1 \text{ keep } e_2 \text{ in } e_4]]]]]$

¹⁴ The notation " $\lambda 1 2$ " indicates that the indices on the AQ are used to form cases by lambda-binding of variables. This notation is not part of the LF.

The set of all pairs $\langle x_1, x_2 \rangle$ such that there is an x_4 such that
 x_4 is a shed and
 x_1 keeps x_2 in x_4 .

According to (18), these two sets of cases become arguments of the root meaning for the AQ, which in this case is the subset relation on sets of cases that is defined in (16). The resulting truth condition is worked out in (22).

- (22) Let **always** be the relation between sets of cases defined in (16). Let A and B be the sets of cases defined in (20) and (21). The structure

$$\begin{aligned} & [\text{always}_{1,2} \\ & \quad [[\text{a woman}]_1][[\text{a cat}]_2 \text{ that } [\text{e}_2 \text{ resembles it}_3]][[\text{e}_1 \text{ owns e}_2]] \\ & \quad [\exists_4[[\text{a shed}]_4[\text{e}_1 \text{ keep e}_2 \text{ in e}_4]]]] \end{aligned}$$

is true with respect to an assignment g if and only if **always**(A, B). That is, it is true iff the set of cases

$$\left\{ \langle x_1, x_2 \rangle \left| \begin{array}{l} x_1 \text{ is a woman and} \\ x_2 \text{ is a cat and} \\ x_2 \text{ resembles } g(3) \text{ and} \\ x_1 \text{ owns } x_2 \end{array} \right. \right\}$$

is a subset of the set of cases

$$\left\{ \langle x_1, x_2 \rangle \left| \exists x_4 \left[\begin{array}{l} x_4 \text{ is a shed and} \\ x_1 \text{ keeps } x_2 \text{ in } x_4 \end{array} \right] \right. \right\}.$$

The truth condition in this case is equivalent to the intuitively correct condition that is stated in (23) using universal and existential quantifiers over individuals.

- (23) The structure is true with respect to assignment g iff for every x_1 and every x_2 such that x_1 is a woman and x_2 is a cat and x_2 resembles $g(3)$ and x_1 owns x_2 , there is an x_4 such that x_4 is a shed and x_1 keeps x_2 in x_4 .

Dynamic Case Quantification

In the analysis from sections “[Logical Forms for AQs](#)” and “[Semantics of the LF](#)”, variables in the *if*-clause get quantified by an adverb of quantification, as expressed by indexing in the notation with subscripts on the AQ. The movement and indexing steps ensure that the indices that are quantified are exactly the maximal-scope indefinite descriptions in the restriction of the adverb. This account of the meaning of AQs and the interaction between AQs and indefinite descriptions is mediated by the syntax of LFs, and by the syntactic transformations that construct the LFs. While the theory is workable, the syntactic transformations are not motivated by any observable aspects of English syntax. A response is to try to treat the interaction

between AQs and indefinite descriptions semantically, instead of syntactically. In particular, we would like for it to be a *semantic* property of the restriction that certain variables get quantified by an AQ.¹⁵

Some terminology from Karttunen (1969) hints at a way of conceptualizing the problem. The sentence (24) introduces two *discourse referents* (one for a woman, and one for a cat). The semantics for this clause should encode possible values for these discourse referents that witness the truth of the sentence, in the sense that that they are values for the discourse referents that make the sentence true. We will look at an account that develops this idea, while staying as close as possible to Lewis's formalization in terms of cases.

(24) A woman owns a cat.

A significant fact is that indefinite descriptions allow for *discourse anaphora*, where one sentence contains an indefinite description, and a following sentence contains a pronoun that refers back to the indefinite description. This is illustrated in (25) for various types of indefinite descriptions. Examples (26a,b) indicate that nominals with cardinal determiners, and nominals with the plural determiner *some* can also be used in the AQ construction.

- (25) a. A woman owns a crocodile. She keeps it in her swimming pool.
 b. Two people in the department specialize in historical linguistics. They have different interests, though.
 c. Some people in the department are interested in deep learning. They have organized a reading group.
- (26) a. If two people have the same last name, they are usually related.
 b. If some people have the same last name, they always know each other.
 c. If some people are jointly interested in an arcane subject, they usually organize a reading group on it.

A clue about a way to proceed technically comes from the interpretation rule (18) in the last section, where a set of cases $\lambda n_1, \dots, n_k \phi$ is obtained from the restriction, and used as the first argument of the AQ. If the semantics could be modified so that this set of cases is directly the semantic value of the restriction, this could simply be fed in as the first argument of the AQ. Notice by the way that an element of $\lambda n_1, \dots, n_k \phi$ can be viewed as a case that records choices for the discourse referents corresponding to the top-level indefinite descriptions in ϕ . Anyway, to summarize the basic idea, the semantic value of sentence (27) should be the set of cases (27b), or equivalently (27c).

¹⁵ This follows the plot of Heim (1982), where first an analysis using indexing and existential closure is stated, and then the analysis is re-formulated in dynamic terms. Rather than following Heim's file change semantics, the second part as stated here is close to Paul Dekker's formulation of dynamic semantics, see Dekker (1994, 2012).

(27) Illustration of case semantics for clauses (Version 1)

- a. A woman owns a cat.
- b. The set of cases $\langle x, y \rangle$ such that x is a woman, y is a cat, and x owns y .
- c. $\lambda xy. x$ is a woman and y is a cat and x owns y

This design does not take into account the fact that a clause can contain pronouns that refer back to something earlier in the sentence or discourse, as we saw in section “[Logical Forms for AQs](#)”. What set of cases is contributed by clause (28a), which has a pronoun $[it_i]$ that intuitively has to pick up its referent from the context? An answer is to say that it denotes a set of cases *relative to a case*. In (28b), c is an arbitrary case, and the semantics takes the value of the indexed pronoun $[it_i]$ to be $c[1]$, which is the first element of c .¹⁶ An additional idea is that *cases are cumulative*. In (28), relative to an arbitrary case c , a case in the semantics of (28a) as described in (28b) is formed by incrementing c with a witness y for the cat, and a witness x for the woman.

(28) Illustration of case semantics for clauses (Version 2)

- a. A woman owns a cat that resembles it_i . (Sentence A)
- b. Relative to a case c , sentence A denotes the set of cases of the form xyz , where x is a woman, y is a cat, x owns y , and y resembles $c[1]$.

To explain the increment notation, where c is a tuple $\langle z_1, \dots, z_k \rangle$, xc designates the tuple $\langle x, z_1, \dots, z_k \rangle$. Where $d = xc$, the new value x is added at the beginning in the tuple, and can be accessed as $d[1]$. Pronouns index into cases. Relative to case c , the value of the pronoun $[it_i]$ is $c[i]$.

(29) introduces some terminology that will be used in defining the semantics of AQs. Where A is a clause and c is a case, cA is the set of cases corresponding to clause A , relative to case c . The semantics will be phrased as a recursive definition of cA , for arbitrary c and various choices of clause A . The *domain* of A , written $Do(A)$, is the set of all cases c such that cA is non-empty.¹⁷ This projects away witnesses, and is conceived of as the set of cases relative to which A is true.

(29) Let c and d be cases, and A be a clause.

- cA Semantics of A as a set of cases, relative to case c .
- $Do(A)$ The set of all c such that cA is non-empty.

For a clause headed by *always*, we assume the logical form $[always A B]$, where the restriction A and the scope B are clauses that may contain indefinite descriptions and pronouns. Unlike in sections “[Logical Forms for AQs](#)” and “[Semantics of the LF](#)”, there is no indexing on the AQ, and no existential AQ adjoined to the scope. Given the format (29), we need to define the set of cases $c[always A B]$ for arbitrary

¹⁶ The choice of i in $[it_i]$ depends on details of the sub-clausal semantics. (34) switches to $[it_3]$, on the assumption that 1 and 2 are discourse referents for the woman and the cat.

¹⁷ To explain the terminology, $Do(A)$ is the domain of the relation that holds between cases d and e iff e is an element of dA .

c , A , and B . In the term $c[\text{always } A \ B]$, the part following c is a clause, just like A in the term cA . Before looking at a definition, let us check empirically whether a clause of the form $c[\text{always } A \ B]$ sets up any discourse referents. (30) checks whether indefinite descriptions in A or B sets up discourse referents that can be referred to with a pronoun in later discourse. In (30a), [a woman] in the restriction can not be picked up with the pronoun *she* in the second sentence. In (30b), [a shed] in the scope can not be picked up with the pronoun *it* in the second sentence. This is partial evidence that clauses headed by AQs do not set up discourse referents.

- (30) a. If a woman owns a cat, she always keeps it in a shed. #She is cruel.
 b. If a woman owns a cat, she always keeps it in a shed. #It is resentful.

Clauses that do not set up discourse referents are known as *tests*. A test A has the property that cA is either the unit set of c , or the empty set. The value $\{c\}$ corresponds to A being true relative to c , and the value $\emptyset$ corresponds to A being false relative to c . This is illustrated in (31) for a transitive clause with two pronouns.

- (31) $c[\text{he}_1 \text{ admires her}_2]$ = $\{c\}$ if $c[1]$ admires $c[2]$
 = $\emptyset$ otherwise.

(32) is a definition of the semantics of *always*. It has the same superstructure as (31), with a description of conditions for the value being the unit set $\{c\}$, corresponding to truth relative to c . The condition requires that B be true relative to each case in cA . Since cases in cA record witnesses for indefinite descriptions in A , this allows indefinite descriptions in A to set up antecedents for pronouns in B . This is a core result of dynamic semantics for adverbs of quantification.

- (32) Semantics for *always*
 $c[\text{always } A \ B]$ = $\{c\}$ if $cA \subseteq Do(B)$
 = $\emptyset$ otherwise.

To illustrate, we assume the semantic values described in (34) and (35) for the restriction A and the scope B of the sentence that is repeated in (33). The semantics (32) refers to cA and $Do(B)$. cA is already described in (34), and it can be rewritten as in (36). Given the definition in (29), $Do(B)$ is (37). The semantics (32) performs a subset check between the two sets of cases, and we obtain the semantics (38). The value $\{c\}$ corresponds to truth, and truth conditionally, (38) is equivalent to the truth condition for the same sentence at the end of section "Semantics of the LF". An interesting change in the mechanics is that while before cases of length two were quantified, now longer cases beginning with all of c are quantified. This does not matter, because since c is constant, the short cases are in one-to-one correspondence with the long cases.

- (33) If a woman owns a cat that resembles it, she always keeps it in a shed.
 (34) Where A is [a woman owns a cat that resembles it]₃ and c is a case, cA is the set of all cases x_1x_2c such that

x_1 is a woman and
 x_2 is a cat and
 x_2 resembles $c[1]$ and
 x_1 owns x_2 .

- (35) Where B is [she_1 keeps it_2 in a shed] and d is a case, dB is the set of all cases yd such that
 $d[1]$ keeps $d[2]$ in y and
 y is a shed.

$$(36) \left\{ d \left[\begin{array}{l} d[1] \text{ is a woman and} \\ d[2] \text{ is a cat and} \\ d[1] \text{ resembles } d[3] \text{ and} \\ d[1] \text{ owns } d[2] \end{array} \right] \right\}$$

$$(37) \left\{ d | \exists y \left[\begin{array}{l} d[1] \text{ keeps } d[2] \text{ in } y \text{ and} \\ y \text{ is a shed} \end{array} \right] \right\}$$

$$\begin{aligned}
 (38) \quad & c[\text{always [a woman owns a cat that resembles it] [she keeps it in a shed]}] \\
 &= \{c\} \text{ if} \\
 &\quad \left\{ d \left[\begin{array}{l} d[1] \text{ is a woman and} \\ d[2] \text{ is a cat and} \\ d[1] \text{ resembles } d[3] \text{ and} \\ d[1] \text{ owns } d[2] \end{array} \right] \right\} \subseteq \left\{ d | \exists y \left[\begin{array}{l} d[1] \text{ keeps } d[2] \text{ in } y \text{ and} \\ y \text{ is a shed} \end{array} \right] \right\} \\
 &= \emptyset \text{ otherwise.}
 \end{aligned}$$

In (39), the analysis is generalized to arbitrary adverbs of quantification that have root meanings that are relations between sets of cases.

- (39) Dynamic case-quantifying semantics for adverbs of quantification
- $$\begin{aligned}
 c[Q \ A \ B] &= \{c\} && \text{if } Q(cA, Do(B)) \\
 &= \emptyset && \text{otherwise.}
 \end{aligned}$$

There is work to do in completing the sub-sentential parts of the semantic analysis, which is only illustrated by example here. (40) gives the primitive semantics of an indefinite description, and of some atomic formulas with indexed arguments. (41) defines a dynamic conjunction. With these one can work out the semantics of the discourse anaphora in (42), assuming that dynamic conjunction is used for combining [a woman] with [e_1 rests], and combining [[a woman][e_1 rests]] with [she_1 is tired].

- (40) $c[\text{a woman}] \quad \{xc | x \text{ is a woman}\}$
 $c[e_1 \text{ rests}] \quad \{d | d = c \wedge c[1] \text{ rests}\}$
 $c[she_1 \text{ is tired}] \quad \{d | d = c \wedge c[1] \text{ is tired}\}$

- (41) Dynamic conjunction
- $$b[A \ B] \quad \{d | \exists c [c \in [bA] \wedge d \in [cB]]\}$$

- (42) $c[[[a \text{ woman}]_i[e_i \text{ rests}]]][she_i \text{ is tired}]]$
 $= \{yc | y \text{ is a woman and } y \text{ rests and } y \text{ is tired}\}$

Summing up the results of this section, by switching to a dynamic semantics where clauses directly contribute sets of cases (relative to a case), it is possible to eliminate indexing from the formulation in sections "[Logical Forms for AQs](#)" and "[Semantics of the LF](#)", and to build Lewis's hypothesis of unselective case quantification quite directly into the compositional semantics of adverbs of quantification.

Syntactic Splitting

In the simple picture from section "[Logical Forms for AQs](#)", the main clause supplies the scope of an AQ, and the *if*-clause supplies the restriction. But components of the main clause can also contribute to the restriction. This is illustrated by the sentences in (43), where the paraphrases in (44) indicate that the subject is contributing to the restriction.

- (43) a. A man who owns a dog usually resembles it.
 b. A dog is usually happy if it has a happy owner.
- (44) a. Most men who own a dog resemble it.
 b. Most dogs that have a happy owner are happy.

Focusing on bare plural subjects, Diesing (1992) pointed out that with certain predicates, the subject can ambiguously contribute to the restriction or to the scope. This is shown by the competing paraphrases in (45b,c) for sentence (45a). In the reading (45b), the subject *ghosts* has contributed to the restriction, and is quantified by *usually*, or by *most* in the paraphrase. In the reading (45c), the subject has contributed to the scope, taking on an existential reading, as indicated by the determiner *some* in the paraphrase, or by the existential *there*-construction in (45d).

- (45) a. Ghosts are usually visible.
 b. Most ghosts are visible.
 c. On most occasions, some ghosts are visible.
 d. On most occasions, there are ghosts visible.

With other predicates, a bare plural subject contributes unambiguously to the restriction. Sentence (46a) has the reading (46b), and can not have the reading (46c).

- (46) a. Ghosts are usually honest.
 b. Most ghosts are honest.
 c. #On most occasions, there are some honest ghosts.

Diesing proposed that these data follow from a syntactic splitting procedure that maps the verb phrase to the scope of an AQ, and the rest of the sentence to the restriction. Example (46a) without the adverb has the structure (47). The subject is outside the VP, and so is mapped to the restriction. Example (45) has a more

complicated syntactic structure, where the subject has been raised out of a base position inside the VP, as in (48.) Semantically, the subject *ghosts* can be either in its raised position, resulting in splitting into the restriction of the AQ, or in its base position, resulting in splitting into the restriction. This accounts for the two readings of (45a).

(47) [_s ghosts [_{vp} are honest]]

(48) [_s ghosts_i [_{vp} e_i are visible]]

The predicates where the subject is raised out of VP on Diesing's analysis are the *stage level predicates* in the sense of Carlson (1977).¹⁸ Predicates such as 'being honest' are his individual-level predicates. Carlson had proposed an ontological distinction between stages or time-slices of individuals, and individuals in the ordinary sense. Some predicates are primitively predicates of stages, and others are primitively predicates of individuals. In Diesing's development, the distinction is argument-structural, with only stage-level predicates having VP-internal subjects.

Support for the VP-splitting analysis comes from German and Dutch, where the possibilities for restriction and scope readings correlate with surface word order. (49) is a German word-final embedded clause, where the bare plural object *books about wombats* follows the adverb of quantification *immer* 'always'. This sentence has a scope reading for the bare plural, as glossed at the bottom in (49). In (50), the object is to the left of the AQ, and gets a restriction reading. These data (which are from Diesing 1992) follow from the hypothesis that the AQ is at the left edge of the verb phrase, and that arguments in the VP can be optionally moved leftward out of the VP and past the adverb, by the operation of scrambling. When the bare plural object is not scrambled, it is in the VP and is mapped to the scope. When it is scrambled, it is outside VP and is mapped to the restriction. Thus we see evidence in German surface phrase order for the hypothesis that material inside the VP is mapped to the scope of an adverb of quantification.

(49) ... daß Otto immer Bücher über Wombats liest.
... that Otto always books about wombats reads

that Otto is always reading some books about Wombats.
that Otto, whenever he reads, reads books about Wombats

(50) ... daß Otto Bücher über Wombats immer liest.
... that Otto books about wombats always reads

that whenever there are books about wombats, Otto reads them.

The issue of splitting is largely independent of the approach to semantic interpretation. Diesing (1992) assumed the indexing analysis from sections "Logical Forms for AQs" and "Semantics of the LF". We can say that a clause headed by an AQ is first split by mapping the VP to the scope and the rest of the sentence to

¹⁸ See Chap. 4 in this volume.

the restriction. Then indexing proceeds as before, with adjunction of an existential adverb to the scope, and indexing of each indefinite to the minimal c-commanding AQ. Or we can say that splitting proceeds syntactically, with the interpretation of the resulting configuration by the dynamic procedure in section "[Dynamic Case Quantification](#)".

More on Restrictive *if*-clauses

A dramatic aspect of Lewis's analysis of adverbs of quantification used in combination with *if*-clauses is that *if* has no meaning of its own. It is simply a syntactic marker for the role of being a restriction of an operator. This comes up in some other places. Kratzer (1978) proposed that in the examples (51) and (52), the modals are the main operators, while the *if*-clauses supply arguments of the modals.¹⁹

- (51) a. If Jack eats the panna cotta, he will become sick.
 b. If Jack eats the panna cotta, he might become sick.
- (52) a. If Jack had eaten the panna cotta, he would have become sick.
 b. If Jack had eaten the panna cotta, he might have become sick.

Lewis (1976) looked at contexts with a probabilistic pragmatics, and at the phenomenon of *if*-clauses expressing conditional probability in such contexts. (53) exemplifies some syntactic constructions where such interpretations arise. (53a) can be understood to describe the probabilistic judgment that the probabilistic measure of the set of options where John ate the panna cotta and is sick is significantly greater than the probabilistic measure of the set of options where John ate the panna cotta and is not sick. Or equivalently, the probability that Jack is sick, conditioned on his eating the panna cotta, is significantly greater than 0.5. Similarly, suppose we stipulate that belief states are characterized by probability measures, so that (53b) is true if and only if Keisha's epistemic probability measure assigns high probabilistic measure to the proposition that Jack ate the panna cotta. With this theoretical and empirical mindset, (53c) can be understood to describe a constraint on Keisha's epistemic probability measure that is stated in terms of conditional probability—it seems to be true if and only if Keisha's epistemic probability measure is such that the measure of the set of options where John ate the panna cotta and is sick, divided by the measure of the set of options where Jack ate the panna cotta is high (close to 1.0). Or equivalently, Keisha's epistemic probability measure is such that the probability of Jack being sick, conditioned on his eating the panna cotta, is high.²⁰

- (53) a. If Jack ate the panna cotta, he is probably sick.
 b. Keisha believes that Jack ate the panna cotta.
 c. Keisha believes that if Jack ate the panna cotta, he is sick.

¹⁹ See Chap. 13 in this volume.

²⁰ Khoo (2016) is a recent discussion of this empirical picture and counterarguments to it.

Lewis (1976) proved a technical result that there is no semantics that defines a proposition contributed by *if p then q* in terms of propositions *p* and *q*, and also allows the conditional-probability readings seen in (53) to be derived compositionally. In other words, the logical form of (53a) could not be as in (54), with [if Jack ate the panna cotta, he is sick] contributing a proposition that is the argument of *probably*. Instead (going beyond what Lewis said explicitly) the *if*-clause supplies an argument of a two-place conditional probability operator.

(54) [probably [if Jack ate the panna cotta, he is sick]]

Kratzer (1986) generalized the cases of adverbs of quantification, modals, and probabilistic readings to the hypothesis that *if* is never meaningful, and summarized the situation like this: “The history of the conditional is the story of a syntactic mistake. There is no two-place *if* . . . *then* connective in the logical forms for natural languages. *If*-clauses are devices for restricting the domains of various operators.”

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Chapter 17

On Link's "The Logical Analysis of Plurals and Mass Terms: A Lattice-theoretical Approach"



Lucas Champollion and Adrian Brasoveanu

Abstract Few works in formal semantics have been as influential as Godehard Link's 1983 paper *The logical analysis of plurals and mass terms: A lattice-theoretical approach*. Here we give an overview of the paper and highlight various aspects of its legacy.

Keywords Count/mass distinction · Distributivity · Plurals · Group nouns · Collectivity · Conjunction · Hydras · Cross-domain generalizations · Plural discourse reference · Experimental semantics

Introduction

Few works in formal semantics have been as influential as Godehard Link's paper "The logical analysis of plurals and mass terms: A lattice-theoretical approach" (Link, 1983, hereafter L83). Here we give an overview of the paper (section "Overview of the Paper") and highlight various aspects of its legacy. In section "Subsequent Work on Issues Discussed by Link", we comment on subsequent developments concerning topics discussed in the paper: the count/mass distinction, distributivity, plurals, group nouns and collectivity, conjunction, and hydras. We then turn to topics not directly discussed, but nevertheless influenced, by the paper. Section "Extensions to Cross-Domain Generalizations" focuses on generalizations across the nominal and verbal domain, and section "Plural Discourse Reference" on the role of mereological sums in dynamic treatments

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of quantificational dependencies. Section “[Experimental Work on Distributivity](#)” highlights some relevant experimental work, and section “[Conclusion](#)” concludes.

Overview of the Paper

Early Montague grammar (Montague, 1973b) laid the foundations for formal semantics as practiced today and established its core methodology by focusing on logic and compositionality, but did not account for a variety of natural language phenomena including plurals, mass terms, and collectives. These phenomena were of great interest to philosophers of language as well as to semanticists already then and are linked to one another in various ways. For example, plural nouns and mass nouns both exhibit the cumulative reference property:

- (1) a. If the animals in this camp are horses and the animals in that camp are horses, then the animals in both camps are horses.
- b. If *a* is water and *b* is water, then *a* and *b* taken together is water.

Singular and plural count nouns license inferences that parallel those of distributive verbs:

- (2) John and Bill are boys. $\Leftrightarrow$ John is a boy and Bill is a boy.
- (3) John and Bill smiled. $\Leftrightarrow$ John smiled and Bill smiled.

These inferences contrast with group terms and collective verbs:

- (4) John and Bill are a team. $\nLeftrightarrow$ #John is a team and Bill is a team.
- (5) John and Bill met. $\nLeftrightarrow$ #John met and Bill met.

Mass nouns and collective nouns alike combine with collective verbs:

- (6) a. The students gathered around their teacher.
- b. The water gathered in big pools.

Material constitution and collective nouns both exhibit a similar behavior:

- (6) a. The ring is new but the gold it consists of is old.
- b. The committee is old but the people it consists of are young.

To capture such facts within Montague grammar, it is necessary to determine what plurals, mass terms, group terms, as well as distributive and collective verbs denote. Early treatments of plurals in formal linguistics used formal devices from set theory (Bennett, 1974; Hausser, 1974; Kroch, 1974; Scha, 1981). However, set theory has sometimes been seen as an imperfect match for the modeling of mass terms because it does not straightforwardly accommodate nonatomic structures (we return to this point below). L83 develops a unified logic of plural and mass expressions by building on ideas from the philosophical literature and on notions from lattice theory. A large part of L83 is devoted to presenting a detailed formal fragment in

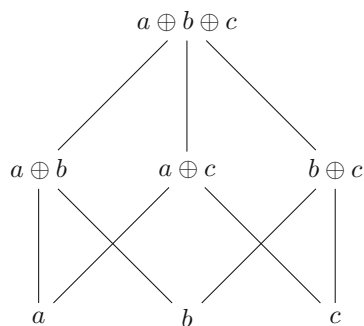
Montague grammar; here we focus on the central aspects of his model of plurality and the count-mass opposition.

Since plural nouns are derived from singular nouns, their denotations must be considered in relation to each other. Following Massey (1976), L83 extends the domain of entities E to include not only ordinary (singular) individuals but also pluralities, which are modeled as sums or fusions in the sense of classical extensional mereology. Intuitively, a sum of some entities is another entity that combines or fuses these entities together. A sum is taken to be of the same type as its parts.

Mereology can be axiomatized in a way that gives rise to semilattice structures like the one in Fig. 17.1. The nodes in this figure stand for the individuals a , b , and c , and for the sums that are built up from them. The lines stand for the parthood relations that hold between the various individuals and sums. In mereology, the parthood relation $\leq$ is reflexive, transitive, and antisymmetric; that is, it is a partial order. The behavior of parthood and sum is constrained so that $x \leq y$ if and only if $x \oplus y = y$. The binary sum operation $\oplus$ is the join of the underlying Boolean algebra. It is associative, commutative and (unlike arithmetic sum) idempotent. For an overview of linguistic applications of mereology, see Champollion and Krifka (2016). Structures like the one in Fig. 17.1 are isomorphic to complete Boolean algebras, or alternatively to complete Boolean lattices, with the bottom element removed. L83 uses an actual complete Boolean algebra with bottom element $\mathbf{0}$ but then proceeds to use $\mathbf{0}$ only as a dummy object and excludes it in most subsequent definitions.

To incorporate mereology into Montague Grammar, L83 follows the strategy adopted in Montague (1973b) and translates English into an intermediate language, an extension of first-order predicate logic which has been enriched with symbols such as $\oplus$ (sum) and Π (parthood). These symbols, in turn, are interpreted in a structured domain. This is a model that contains, among other things, a mereological sum operation $\sqcup_i$ over E and a parthood relation $\leq_i$; this relation is atomic, and the sum operation is complete over E . The subscript i on $\leq_i$ and $\sqcup_i$ stands for *individual*; it distinguishes these symbols from a different pair, $\leq$ and $\sqcup$, which will be discussed shortly.

Fig. 17.1 An algebraic structure



Much work in formal semantics, starting with Montague (1970), dispenses with the kind of intermediate language used in L83, in favor of giving English syntactic rules a direct model-theoretic interpretation. Distinctions such as the one between $\oplus$ and $\sqcup_i$ are then no longer relevant. From here on, we use the symbol $\oplus$ to refer both to the binary connective $\oplus$ and to its model-theoretic operation $\sqcup_i$.

The fact that the sum operation is complete means that every subset of E has a sum that is also in E ; the fact that the parthood relation $\leq_i$ is atomic means that it is not possible to find infinitely descending chains $\dots \leq_i e_3 \leq_i e_2 \leq_i e_1$, even though they are in principle consistent with the axioms of mereology (in contrast to ordinary, i.e. well-founded, set theory, which disallows such chains). Rather, everything in E is ultimately built up from $\leq_i$ -atoms, or individuals that have no $\leq_i$ -parts other than themselves. Among the $\leq_i$ -atoms, A , L83 includes all entities in the denotation of singular nouns; their sums are the pluralities that are in the denotation of plural nouns. Since these sums are in E , their type is the same as that of singular entities.

The lack of inherent criteria of individuation in mass terms has been taken to imply that mass nouns cannot be modeled as denoting sets the way count nouns do in Montague grammar, since membership in a set might qualify as a criterion of individuation (Bunt, 1979). But in fact there is nothing in Montague grammar that prevents us from modeling mass terms as sets, and Montague himself suggested doing so (Montague, 1973a). Similarly, Link assumes that mass nouns denote sets of portions of matter. He equips the set D of portions of matter with its own sum operation $\sqcup$ and parthood relation $\leq$. To distinguish these concepts from the ones used for plural terms, $\sqcup$ and $\leq$ are referred to as material fusion and material part while $\sqcup_i$ and $\leq_i$ are referred to as individual sum and individual part. The material fusion operation is complete over D , meaning that the material fusion of any number of portions of matter is again a portion of matter. But the material parthood relation is not necessarily atomic. For any given mass term, there may not be such a thing as a smallest portion of matter to which the term applies; thus, we do not exclude infinitely descending chains $\dots \leq m_3 \leq m_2 \leq m_1$.

In summary, the semantic values of both plurals and mass terms have a lattice structure, with the only difference being that the lattice for plurals is atomic while the lattice for mass terms is not. Link describes this as one of the virtues of his system, as it makes it possible to give parallel accounts of cumulative reference for plurals and mass terms. To do this, he takes mass terms to denote sets of portions of matter that are closed under fusion. As for plurals, he assumes that they are derived from singular terms semantically as well as morphologically. He takes the meaning of a singular term P to be a set of $\leq_i$ -atoms; the meaning of the corresponding plural term, *P , is the closure of that set under mereological sum minus the atoms themselves. For example, in Fig. 17.1, a given singular term P_1 might apply to a , b , and c ; the plural term *P_1 would then apply to $a \oplus b$, $a \oplus c$, $b \oplus c$ and $a \oplus b \oplus c$. A different singular term P_2 that applies just to a and b would correspond to a plural term *P_2 that applies just to $a \oplus b$. In general, pluralized terms have cumulative reference. Based on this, Link gives a compositional treatment of the plural morpheme as denoting the * operator.

Link also gives a treatment of the distinction between distributive and collective verbs. In L83, unlike in later work by Link and others, this distinction is treated as a lexical property and comes down to whether the verb applies only to plural individuals or also to the atoms they consist of. Distributive verbs apply to $\leq_i$ -atoms, in the same way as count nouns. In a sentence like *John and Bill smiled*, where the conjoined noun phrase is taken to refer to the sum $j \oplus b$ of the $\leq_i$ -atoms denoted by John and by Bill, the distributive verb *smiled* is assumed to undergo closure under sum, written $*$. This operation is identical to the $\oplus$ operation that singular count nouns undergo when they are pluralized, except that it does not eliminate atoms. As a result, the derived predicate $*\textit{smiled}$ applies to the sum $j \oplus b$ so long as *smiled* applies to j and to b . Collective verbs like *gathered*, by contrast, apply directly to plural individuals such as $j \oplus b$, without $*$ being involved, and to portions of matter, but never to atomic individuals. This accounts for the contrast between *The students gathered*, *The water gathered*, and *#John gathered*. Link stresses that one advantage of this account is that all predicates have the same type (they all apply to entities in E); therefore mixed collective-distributive predicates like *carry the piano* can be represented without postulating an ambiguity. For a more detailed overview, see Champollion (2019).

The relationship between count and mass terms is related to the problem of constitution, that is, the question whether things are identical to the stuff that constitutes them or makes them up. What is the relationship between the ring and the gold of which it is made? If they are identical, how can the ring be new while the gold is old? If not, what are the properties of this constitution relation? For an overview of the problem and the standard answers in philosophy, see Wasserman (2018). Following a popular philosophical view, Link takes the ring and the gold to be distinct but related by a constitution relation, which he models as a function h from individuals to those portions of matter that constitute them. This function is constrained so that $x \leq_i y$ only if $h(x) \leq h(y)$. For example, the ace of clubs is an individual part of the deck of cards only if the matter making up the ace is a material part of the matter making up the deck of cards. Furthermore, for any x and y , it is required that $h(x \oplus y) = h(x) \oplus h(y)$. This means the individual sum of two entities x and y is made up of the material fusion of the matter that constitutes x and the matter that constitutes y . The use of the constitution relation allows Link to treat pairs like *the gold* vs. *the ring* as equivalent in certain contexts, and likewise for plural terms like *the cards* vs. collective terms like *the deck of cards*. At the same time, by distinguishing between individual sums and portions of matter, he is able to avoid treating these pairs of expressions as coreferential.

Subsequent Work on Issues Discussed by Link

Perhaps the most influential effect of L83 on formal semantics is that structured domains are now widely used, with their mereological sums occurring as possible referents of conjoined terms and plural definite descriptions, in the denotations

of predicates and quantifiers, and so on. While Link was not the first to use mereological sums (see e.g. Massey, 1976), and many of his individual moves have precursors (such as Sharvy, 1980 on a unified treatment of definite descriptions across the count and mass domains), his paper marks the first time a large compositional semantic fragment in the style of Montague (1973b) incorporates mereological concepts and structures. Aside from this model-theoretic innovation, L83 also made substantive contributions and claims concerning the account of a variety of phenomena, including the count/mass distinction, plurals, distributivity, group nouns, conjunctions and hydras. In this section, we highlight subsequent developments in formal semantics concerning these issues. We largely focus on work by scholars other than Link himself, whose subsequent work on these and related issues has been reprinted as Link (1998a) and is summarized in the introduction of that collection. For reasons of space, we omit many topics on which L83 has proved influential, such as the interaction of plurals with generalized quantifier theory (van der Does, 1993b), extensions to individual concepts (Krifka, 2009), investigations of the lexical semantics of distributivity across multiple verbs (Glass, 2021), or the connections between the logic of plurals and second-order logic (Boolos, 1984; Lewis, 1991; Schein, 1993).

The Count/Mass Distinction

L83 treats the ring and the gold as distinct, and more generally posits two separate sorts, or domains, for count and mass denotations (see also Quine, 1960). This explains why they have different properties. By taking the meaning of singular count nouns, but not mass nouns, to be sets of $\leq_i$ -atoms, he also explains why only the former are compatible with numerals (*three rings* vs. **three gold*).

Subsequent research on the count/mass distinction has followed L83 in adopting classical extensional mereology (with some exceptions, see e.g. Moltmann, 1997, 1998; Grimm, 2012) but has not upheld Link's two-sortal approach. Thus, Krifka (1989a) models the difference between count and mass nouns by assuming that they have different semantic types: the former but not the latter expect a number to be supplied (for example, by a numeral). Rather than relying on atomicity for counting, Krifka assumes that count nouns involve the application of a built-in extensive function called "natural unit". For more on Krifka (1989a), see also Filip (2022).

The primary challenge for the two-sortal approach comes from mismatches between the count-mass distinction and the object-substance distinction (for an overview of the empirical landscape across languages, see Grimm, 2018). The challenge can be illustrated by two kinds of nouns not discussed by L83: count nouns that are semantically similar to mass nouns, and mass nouns that are semantically similar to count nouns. The former include nouns like *fence*, *twig*, and *rock*; some examples of the latter are *jewelry*, *furniture*, and *cattle*. These are also

called object mass nouns or fake mass nouns, and they pattern with count nouns in experimental quantity comparison judgment tests (Barner and Snedeker, 2005).

Nouns like *fence* pose a problem for Link's atomicity assumption because what counts as one fence for certain purposes can have several parts that each count as one fence for other purposes (Zucchi and White, 2001). Rothstein (2010) proposes that nouns denote sets of pairs of an entity and a counting context; these contexts in effect let the same noun involve reference to different sets of individuals on different occasions, so that when the context is fixed, no part of a fence is a fence.

Object mass nouns pose a converse problem for the atomicity assumption in the sense that they intuitively involve reference to the same individuals as count nouns, as suggested for example by pairs like *jewel/jewelry*, *spaghetti/noodles*, *letter/mail*, or English *hair* vs. the French plural count noun *cheveux*. Maintaining that these near-minimal pairs involve reference to distinct sorts of individuals would be problematic. Krifka (1989a) proposes that the "natural unit" function is also defined on object mass nouns and can be supplied by nouns such as *piece* or *item* in constructions like *a piece of furniture* or *ten items of jewelry*. Chierchia (1998, 2010) proposes that count nouns and mass nouns have the same referents (the ring is the same as the gold) but that they differ in the way they refer: the criteria of identity of mass nouns, including object mass nouns, are vague in a way that those of count nouns are not. Numerals are assumed to test for atomicity and absence of vagueness. Landman (2011, 2016) proposes that for each noun, a distinguished set of entities in its denotation is identified; when these entities do not overlap, the noun is count; when they do overlap, the noun is mass; when they overlap but consist of non-overlapping parts, the noun is object mass. On this approach, atomicity is no longer required. See also Sutton and Filip (2016, 2021) for a synthesis of Landman's and Rothstein's views.

Distributivity

Link's translation rule for distributive predicates involves the star operator *. As he points out, the rule predicts that distributivity is a lexical property. Subsequent work has revealed two problems with this approach. The first problem stems from mixed predicates like *win* or *build a raft*, which can be read both distributively and nondistributively. This is problematic because lexical properties are not typically thought of as optional. The second problem arises when a complex (phrasal) predicate is understood distributively and contains a scope-taking element that can covary. In these cases, we observe what Winter (2000, 2001) terms *Q-distributivity* (see also Kratzer, 2008; Champollion, 2016a; de Vries, 2015, 2017). Predicates that involve a scope-taking element, such as an indefinite (*have an espresso*) or a disjunction (*walk or cycle*), can involve covariation when they are interpreted distributively.

(8) a. The women had an espresso.

$\forall > \exists$ possible

- b. The women were walking or cycling. $\forall > \vee$ possible

This is problematic because the translation rule in L83 is conceived of as a lexical property, but complex predicates are typically not stored in the lexicon as a unit.

Link (1991[1984], 1987b) treats mixed predicates like *build a sand castle* through a different mechanism: a VP-level operator that shifts nondistributive predicates into distributive predicates. This *D* operator, as it is called, is often mistakenly attributed to L83. It takes a predicate *P* over individuals and returns a predicate that applies to any individual whose atomic parts each satisfy *P*. Its intuitive meaning and function corresponds to that of the English adverb *each* (Choe, 1987) or German *je* (Link, 1998b): when the operator is inserted into one of the sentences in (8), it takes scope over the existential or disjunction. Distributive markers like *each* and *je* have been analyzed in terms of the *D* operator; their meaning varies across languages along interesting dimensions (e.g. Zimmermann, 2002; Champollion, 2016b).

There are important differences between the *** and *D* operators having to do with whether the predicate quantifies over each atomic part of the plural individual denoted by the subject, or over a cover of such parts, the cells of which need not each be atomic. (A cover of an entity is understood here as a set of entities whose sum is that entity.) As Schwarzschild (1996) notes, the *** operator amounts to existential quantification over covers, while the *D* operator amounts to universal quantification over atoms. This distinction is the topic of Landman (1989b), who tries to use groups (see Section “Group Nouns and Collectivity” below) as a way to collapse the difference between *D* and *** in order to unify distributivity and plurality.

The *D* operator has been adopted by many subsequent authors (e.g. Roberts, 1987; Lasersohn, 1995; Winter, 2001). Various generalizations and modifications have been proposed that make it compatible with event semantics (e.g. Lasersohn, 1998; Champollion, 2016a), nonatomic distributivity (e.g. Schwarzschild, 1996; Moltmann, 1997), or dynamic semantics (see section “Plural Discourse Reference”). For a detailed discussion of these issues, see Champollion (2019).

Plurals

A central idea in L83 is that the plural morpheme denotes an operator that compositionally derives plural count nouns from singular ones. L83’s proposal for the denotation of the plural morpheme *-s* is couched in terms of the operator $\oplus$, which denotes closure under mereological sum with all the atoms removed. In other words, for L83, *doctors* means *two or more doctors*. While this assumption rules out sentences like **John is doctors*, it poses problems in downward-entailing contexts such as *No doctors are in the room* (for this argument and its sources, see Schwarzschild, 1996, 5). This sentence is false in scenarios where exactly one doctor is in the room. To account for this and related facts, modern theories of the plural tend to assume that it simply denotes closure under sum, and that the atoms

are not removed; that is, $-s$ denotes $*$ rather than $\otimes$. Sentences like **John is doctors* are assumed to be ruled out for pragmatic reasons (Krifka, 1989b; Sauerland et al., 2005; Spector, 2007; Zweig, 2009). A combination of these two views is found in Farkas and de Swart (2010) and Grimm (2013), in which both the "one or more" and the "two or more" interpretations are assumed to be available, with hearers selecting among them based on factors such as monotonicity, world knowledge, and Questions under Discussion.

Group Nouns and Collectivity

L83 considers group nouns like *committee* or *deck of cards* (German *Kartenspiel*) to denote sets of atoms, analogously to singular ordinary count nouns such as *person* or *card* (German *Karte*). In doing so, he departs from previous analyses like Bennett (1974), where group nouns are treated analogously to plural count nouns such as *cards*. On Link's view, the relationship between an atom x representing a given set of cards and an atom y representing the ace of spades in that set is indirect: both are atoms and thus neither is an individual part of the other, but their underlying portions of matter stand in the material parthood relation. Most subsequent researchers have followed Link's approach; explicit discussions and proposals are found in Barker (1992), Schwarzschild (1996), and Winter (2001).

Link's treatment of collective predicates as applying directly to mereological sums, which we mentioned earlier, has not been universally accepted. Landman (1989a, 1996, 2000), building on later work by Link (1984, 1991[1984]) and in analogy to Link's treatment of group nouns just described, argues that collective predicates apply to a special kind of atoms he calls groups rather than directly to sums. Sums of individuals are taken to be sent to group atoms by an operation $\uparrow$, which is similar to the inverse of the operation h (recall that h takes individuals to their underlying material parts). The $\uparrow$ operation is not one-to-one; for example, two committees may consist of the same individuals. Likewise, h is not one-to-one; for example, a golden ring melted and fashioned into two earrings has the same material parts as these earrings, but is distinct from them. For a review of empirical phenomena related to collectivity and distributivity, see Champollion (2021).

Conjunction and Hydras

At the end of his paper, Link considers relative clauses with more than one head noun, which he calls *hydras*, like *boy and girl who dated each other*. He analyzes *boy and girl* as denoting the set of all sums of a boy and a girl, with the conjunction *and* introducing the sum operation $\oplus$. This analysis is subsequently developed and generalized in Link (1984) and Krifka (1990). It is critiqued by Winter (2001) and Champollion (2016c), who argue for an approach in the spirit of generalized

quantifier theory in which *and* is always analyzed as involving the intersection of two sets (e.g. von Stechow, 1974; Partee and Rooth, 1983); various covert operators are then used to derive the sum-like behavior of hydras. Schein (2017) also rejects a sum-based analysis of *and* and argues that it always denotes a sentential connective. The question whether a sum-based interpretation of conjunction is available at the predicate level is not resolved: Schmitt (2013, 2019), for example, argues for a sum-based approach to conjunction more in the style of L83, also found in truthmaker semantics (Fine, 2017).

Extensions to Cross-Domain Generalizations

Bach (1986) extends Link's approach to the verbal domain to account for well-known parallels between the mass-count distinction and the aspectual classification of verbal expressions. The two-sorted approach is extended to verbs in the following way: telic predicates (accomplishments and achievements) denote sets of atomic events, like count nouns do; atelic predicates (activities and states) denote sets of "bits of process" like mass nouns do. A function *h* mediates between the two domains much in the same way as in the case of nouns. Bach uses this setup to link various puzzles in the nominal and verbal domain and sketch parallel accounts, such as the imperfective paradox (why is it that *John was crossing the street* does not entail that John crossed the street?) and the partitive puzzle (why is it that *This is part of a Roman aqueduct* does not entail that said aqueduct ever existed in its complete form?). Link himself extends his approach to events as well (Link, 1987a). For more on Bach (1986), see also Rothstein (2022).

Krifka (1989a, 1989b, 1992, 1998) builds a theory of the count-mass and telic-atelic distinctions on top of a mereological framework inspired by Link's. The approach is broadly similar to Bach (1986) in that numerous parallels are drawn between the nominal and verbal domains, but there are many extensions. Importantly, the sortal distinction between count and mass nouns, so central in Link's account, plays virtually no role. One of the main results of Krifka's system is a principled explanation of aspectual composition: why is it that *see apples*, *see two apples* and *eat apples* are atelic, while *eat two apples* is telic (Verkuyl, 1972)? Moreover, Krifka provides a detailed semantic analysis of pseudopartitives like *three liters of water* that explains, among other things, why they are incompatible with intensive measure functions like temperature (**three degrees Celsius of water*). Within a mereological framework, Krifka formalizes notions such as quantization (which sets *two apples* apart from *apples*) or incrementality (which sets *eat* apart from *see*). To do so, he defines or reuses higher-order properties of predicates such as cumulative, divisive, and quantized reference, and relies on properties of thematic roles such as strict incrementality.

Cross-domain generalizations are also taken up in Champollion (2010, 2015, 2017), which aims to subsume previous results concerning aspect and measurement (Krifka, 1998) and distributivity (Link, 1987b; Schwarzschild, 1996), as well as

certain distinctions within the class of collective predicates (Dowty, 1987; Winter, 2001). Champollion's unified account is based on the idea that singular, telic, and collective predicates are delimited or bounded in ways that plural, mass, atelic, and distributive predicates are not. The difference between boundedness and unboundedness is formalized through a parametrized higher-order property called stratified reference, which is used to characterize atelicity, distributivity, plural and mass reference, as well as differences within the class of collective predicates. When a predicate that has stratified reference holds of an individual or event, it also holds of subparts of that individual or event along a certain dimension and down to a certain level of granularity.

Plural Discourse Reference

Link's seminal contributions have also fruitfully interacted with dynamic semantics and its relatives (Kamp, 1981; Heim, 1982; Groenendijk and Stokhof, 1991). Within the dynamic tradition, a notion of plural discourse reference was developed that is distinct from reference to mereological pluralities in Link's sense. In this line of research, plural discourse reference is used to interpret quantificational dependencies involving anaphoric expressions such as pronouns, definites, partitives, or reciprocals (Rooth, 1987; Kamp and Reyle, 1993; Dekker, 1994; van den Berg, 1996; Asher and Wang, 2003; Nouwen, 2003, 2006 a.o.). These anaphoric expressions can be morphologically singular or plural, so plural discourse reference is not the same as morphological plurality.

By plural discourse reference, we mean reference to a quantificational dependency between sets of entities that is established at one point in a discourse and can be elaborated upon at later points. Take the sentence in (9), where antecedents are superscripted with the discourse referent they introduce, while anaphors are subscripted with the discourse referent they retrieve.

- (9) Kim bought a^{x₁} gift for every^{x₂} close friend in her class and asked their_{x₂} deskmates to hide them_{x₁} under their_{x₂} desks.

The first conjunct in (9) introduces a quantificational dependency between the set x_1 of gifts Kim bought and the set x_2 of close friends for which she bought the gifts. That dependency links each friend in x_2 to the corresponding gift in x_1 . The second conjunct elaborates on this dependency: for each friend in x_2 , Kim asked that friend's deskmate to hide the corresponding gift in x_1 (and no other gift) under the friend's desk.

In the dynamic semantic tradition, plural discourse reference is modeled through sets of variable assignments rather than through mereological sums. Dynamic Plural Logic (van den Berg, 1996) evaluates a natural language expression relative to a set of assignments, and not relative to a single assignment as in ordinary (dynamic) predicate logic. We write $G, H, \dots$ for sets of assignments and $g_1, g_2, \dots, h_1, h_2, \dots$ for assignments. A set of assignments is called a plural

Table 17.1 An example of a plural information state

Plural information state G	...	x_1	x_2	x_3	...
g_1	...	α_1	β_1	γ_1	...
g_2	...	α_2	β_2	γ_2	...
g_3	...	α_3	β_3	γ_3	...
...	...	...	...	...	...

information state, and can be visually represented as a matrix in which each row is an assignment and each column is a variable—see Table 17.1.

Plural information states encode the values of discourse referents as well as the quantificational dependencies that hold between them. In Table 17.1, the discourse referent x_1 stores the individuals $\alpha_1, \alpha_2, \dots$, the discourse referent x_2 stores the individuals $\beta_1, \beta_2, \dots$, and so on. The rows of the matrix induce n -ary relations between objects, which encode quantificational dependencies. Thus, in each row in G , the individual assigned to the discourse referent x_1 corresponds to the individual assigned to the discourse referent x_2 . As a whole, the plural information state G induces the binary relation $\{\langle \alpha_1, \beta_1 \rangle, \langle \alpha_2, \beta_2 \rangle, \dots\}$, which captures the dependency between gifts in x_1 and friends in x_2 .

While van den Berg (1996) assumes that α_1, β_1 etc. are ordinary atomic individuals, Brasoveanu (2008) proposes a generalized version of Dynamic Plural Logic in which each cell in the table can contain either an atom or a mereological sum. This allows Brasoveanu to extend van den Berg’s account to plural indefinites, such as a variant of (9) that replaces *a gift* by *two gifts* (*Kim bought two gifts for every close friend in her class and asked their deskmates to hide them under their desks*). In addition, the generalization makes it possible to account for the interaction of collectivity with dynamic phenomena such as donkey sentences. For example, by countenancing domain-level plurality, Brasoveanu (2008) captures the observation that morphologically singular, but not plural, donkey pronouns are incompatible with collective predication (Kanazawa, 2001):

- (10)
- a. # Every ^{x_1} farmer who owns a ^{x_2} donkey gathers it _{x_2} around the fire at night.

b. Every ^{x_1} farmer who owns three ^{x_2} donkeys gathers them _{x_2} around the fire at night.

The notion of plural discourse reference raises several—still open—questions. First, how much can event plurality (e.g., as used in Schein, 1993) and plural information states trade against each other? Second, how can discourse-level and domain-level plurality trade off across domains? On the latter question see for example Dotlačil and Nouwen (2016) in the degree domain. Finally, having taken on board Link’s notion of domain-level plurality, do we still need van den Berg’s notion of evaluation-level plurality? Brasoveanu (2008) argues that plural information states are needed to account for the fact that different donkey pronouns can be interpreted existentially or universally, sometimes even within the same sentence (van der Does, 1993a). For example, (11) has a reading that requires buyers of multiple books to

pay for each book they bought, but doesn't require owners of multiple credit cards to use each card they own:

- (11) Every^{x₁} person who buys a^{x₂} book on amazon.com and has a^{x₃} credit card uses it_{x₃} to pay for it_{x₂}.

The gist of Brasoveanu's account is that indefinites like a^{x₂} *book* simultaneously access every possible value of their discourse referent that satisfies the surrounding context, while those like a^{x₃} *credit card* only access some possible values. In other words, a^{x₂} *book* introduces a maximally large discourse-level plurality, while a^{x₃} *credit card* does not.

On the other hand, Champollion et al. (2019) argue that donkey sentences do not motivate a move to evaluation-level pluralities. Their account of donkey sentences forgoes plural information states in favor of a simple dynamic system that evaluates expressions with respect to single assignments. To capture existential and universal readings of donkey pronouns such as those in (11), they adapt the theory of homogeneity in Križ (2016) that combines trivalence with sensitivity to Questions under Discussion.

To be sure, Brasoveanu's argument for discourse-level pluralities is not based exclusively on donkey sentences. A whole range of phenomena, including quantificational and modal subordination (12), suggest that the grammar keeps track of quantificational dependencies that can be modeled in similar ways by using discourse-level pluralities (van den Berg, 1996; Krifka, 1996; Nouwen, 2003; Asher and Wang, 2003; Brasoveanu, 2007, 2010). The argument in Champollion et al. (2019) does not extend to these phenomena.

- (12) a. Every^{x₁} student wrote an^{x₂} article. They_{x₁} sent it_{x₂} to L&P.
 b. A^{x₁} wolf might^{w₁} come in. It_{x₁} would_{w₁} eat you first.

Experimental Work on Distributivity

Link's paper and the work it influenced opened up several thriving avenues of research in formal semantics, which in turn sparked a rise in experimental work on distributivity and plurality in natural languages. For example, distributive and collective readings of predicates have been studied in connection with disambiguators like *each*, *every* and *together*. We briefly summarize some of this research here and refer the readers to the overview in Syrett (2019) for more discussion and references.

One line of research examines the pattern of acquisition for *each* and contrasts it with *all*, *every* or *together* (Brooks and Braine, 1996; Ferenz and Prasada, 2001; Brooks and Sekerina, 2006; Musolino, 2009; Pagliarini et al., 2012; Syrett and Musolino, 2013; Syrett, 2015a). Until about the age of eight, children do not take *each* to be a strong cue for distributivity, and will accept sentences like *Each boy is building a snowman* (from Pagliarini et al., 2012) in a collective situation where the boys are working on the same snowman. However, they understand that *together*

is a cue for collective readings, and that *all* contributes some form of collectivity and exhaustivity. Moreover, as Syrett (2015b) shows, children are sensitive to the distributivity of predicates and correctly infer that distributive adjectives like *round* or *big* ('stubbornly distributive predicates,' Schwarzschild, 2011) apply to the atomic individuals that are part of pluralities.

Another line of research investigates the processing of distributivity by adult speakers. In a landmark study, Frazier et al. (1999) study the interpretive interaction between conjunctions of proper names in subject position and two disambiguators *each* and *together* that come early in the sentence (immediately after the conjoined NP) vs. late (after the VP), for example: *Edna and Milton [each/together] mailed one care-package [each/together] after hearing about the severe droughts in Africa*. Frazier et al. find that distributive readings are harder to process, as evidenced by higher first-pass and total times, as well as more regressions, but only when the disambiguator *each* occurs in a late position. This suggests that adults initially read sentences with a conjoined noun-phrase subject as collective, and upon encountering *each* late in the sentence, they backtrack to assign a distributive reading to the misinterpreted noun-phrase conjunction. Dotlačil and Brasoveanu (2021) replicate the Frazier et al. result using a different methodology (self-paced reading), but also show that the preference for collective readings is only observed when the distributive reading arises via phrasal distributivity, e.g., it is triggered by distributive quantifiers. The preference disappears in the case of lexical distributivity, e.g., the distributive interpretation of *win*. These findings provide evidence for different mental representations of the two types of distributivity, consistent with theoretical work such as Winter (2001), Kratzer (2008) and Champollion (2016a).

Building on the seminal work on quantifier processing in Tunstall (1998) (see also Anderson, 2004 and references therein), Brasoveanu and Dotlačil (2015b) compare *each* and *every* in terms of their ability to contribute inverse (wide) scope distributive readings in a forced-choice task and a self-paced reading follow-up. They find that *each* does indeed seem to contribute a distributive event-differentiation condition, as hypothesized by Tunstall (1998). In a separate investigation, Brasoveanu and Dotlačil (2015a) identify a processing cost for *each* in sentences with sentence-internal *same* (e.g., *Each student saw the same movie*), which provides further evidence for the specific kind of event-differentiation distributivity that *each* contributes.

Conclusion

In the wake of Link's seminal paper, structured domains are now widely used in formal semantics, whether formalized in terms of lattices or other model-theoretic structures. The investigation of plurality, distributivity, and mass reference across domains and languages, using both introspective and experimental methodologies, is a thriving direction of current semantic research. After 35 years, *The Logical*

Analysis of Plurals and Mass Terms remains a regularly assigned reading in graduate courses and a classic paper in the field.

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Chapter 18

On Montague's "The Proper Treatment of Quantification in Ordinary English"



Thomas Ede Zimmermann

Abstract In a series of papers published between 1970 and 1973, the late Richard Montague developed a mathematical theory of semantics and its interface with syntax. The theory, later to be called *Montague Grammar* (or *Montague Semantics*), was not only intended as an alternative to generative grammar but also designed so broadly as to cover both natural language and systems of formal logic. An outline of the theory was contained in the short and somewhat hermetic essay 'Universal Grammar' (1970). However, it was the slightly more accessible descriptive application in Montague's last publication that was to define the vantage ground of semantic research for at least the next decade: 'The Proper Treatment of Quantification in Ordinary English' (1973), aka '*PTQ*'. *PTQ* was presented both as an illustration of the power of Montague's general theory and as a contribution to the semantic description of English, thereby extending the range of phenomena addressed in his earlier papers. Avoiding technicalities, this survey offers a glimpse of Montague's general theoretical contributions and the more specific descriptive aspects of *PTQ* from a modern perspective. It also addresses some later developments in formal semantics that were arguably provoked by inadequacies in the theoretical apparatus and descriptive details of *PTQ*.

Keywords Compositionality · Intensional logic · Intensional transitive verbs · Montague grammar · *PTQ* · Temperature paradox · Types

In a series of papers published between 1970 and 1973, the late Richard Montague developed a mathematical theory of semantics and its interface with syntax. The theory, later to be called *Montague Grammar* (or *Montague Semantics*), was brought to the attention of a larger linguistic community only after its originator's death, and chiefly through Barbara Partee's (1975). Montague Grammar was not only

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intended as an alternative to contemporary generative grammar but also designed so broadly as to cover both natural language and systems of formal logic. An outline of the theory was contained in the short and somewhat hermetic essay ‘Universal Grammar’ (= Montague, 1970a). However, it was the slightly more accessible descriptive application in Montague’s last publication that was to define the vantage ground of semantic research for at least the next decade: ‘The Proper Treatment of Quantification in Ordinary English’ (= Montague, 1973), *vulgo* (and henceforth) *PTQ*.¹

The following survey can only provide a rather superficial glimpse of *PTQ* and the theory behind it without doing full justice to its linguistic, philosophical, and logical depth and breadth. For more on the topic readers should consult a decent Montague Grammar text (e.g., Link, 1979; Dowty et al., 1981; Partee, 1997) or, in case they are not easily scared off by mathematical jargon and density, Montague’s original papers.

PTQ was presented both as an illustration of the power of the general theory expounded in Montague (1970a) and as a contribution to the semantic description of English, thereby extending the range of phenomena addressed in Montague’s earlier papers, Montague (1970a, b). So there is both a general theoretical and a more specific descriptive aspect to *PTQ*. For ease of presentation, the following two sections will tease these two aspects apart – sometimes artificially, and not always quite in line with Montague’s original views. As we go along, later developments in formal semantics will be addressed that were arguably provoked by inadequacies in the theoretical apparatus and descriptive details of *PTQ*.

Theoretical Aspects: The Framework

In a nutshell, Montague Grammar may be characterized as an (v) *indirect*, (iii) *model-theoretic* approach to the (i) *compositional* behavior of (ii) *extensions and intensions* in a hierarchy of (iv) *functional types*. The following five sub-sections will address these theoretical aspects in the order indicated.

Compositionality: The Algebra of Constructions

Montague’s work is generally seen as marking the beginning of modern *compositional* semantics.² In fact, his algebraic characterization of the interface between

¹ See Partee (2011) and Caponigro (to appear) for the historical and biographical background of Montague Grammar.

² The term ‘compositionality’ goes back to the generative tradition (cf. Katz & Fodor, 1963: 171f.) and does not appear in Montague’s writings, though.

syntax and semantics basically boils down to the tenet that the meanings of non-lexical expressions can be obtained by combining the meanings of their immediate parts. In order to achieve this, Montague assumed that, underlying any constituent, there is a *disambiguated expression* from which its immediate parts and the way they have been combined can be determined; and it is this underlying disambiguated expression that eventually receives a compositional interpretation. In this respect, Montague's disambiguated expressions are like May's (1977) *LFs*, especially as featured in Heim and Kratzer (1998).

In *PTQ* certain tree graphs were introduced that specify both the (semantically relevant) part-whole structure and the ways in which the (immediate) parts are combined – roughly corresponding to (semantically relevant) grammatical constructions. Thus, the trees in (1), redrawn from *PTQ* (p. 228), are disambiguated expressions underlying the surface string **John tries to find a unicorn**:

- (1) a. **John tries to find a unicorn**, 4
-
- b. **John tries to find a unicorn**, 10,0
-

In (1), the numbers n following the boldface surface strings of complex (disambiguated) constituents refer to the pertinent syntactic operation (or construction), called F_n . Thus, F_2 is the operation of pre-posing the appropriate form of the indefinite article – i.e., **a** or **an** – to (the surface string of) a disambiguated expression and it is regulated by a rule S2 to the effect that applying F_2 to a (not necessarily lexical) noun results in a determiner phrase. F_2 is unary: the article does not count as a lexical expression but as 'logical material' that marks the operation of forming an indefinite from a noun. Analogous constructions F_0 and F_1 are used to form DPs of the forms '**every** N ' and '**the** N '. We will get to the semantic effect of

these constructions in due course, but already at this point one difference between the Montagovian approach and type-driven frameworks along the lines of Klein and Sag (1985) or Heim and Kratzer (1998) becomes apparent: in disambiguated expressions, unary branching does not always represent a meaning-preserving operation.

The other constructions in (1) are all binary. F_4 brings together subject and predicate to form a sentence. F_5 combines a transitive verb with (the accusative form of) its direct object to construct a verb phrase. F_6 attaches an infinitival object to a control verb, again to produce a verb phrase. Finally, $F_{10,n}$ is the predecessor of May's (1977) Quantifier Raising, where n is the subscript of the trace (which in *PTQ* surfaces as an indexed pronoun); by $F_{10,n}$, a DP is woven into a sentence to form another sentence, in which the DP takes widest scope. The morpho-syntactic details of these binary operations are formulated in a somewhat *ad hoc* manner – which is why they are not worth going into here.³

Semantic interpretation in Montague Grammar proceeds by moving up trees like those in (1) that represent derivations by the operations mentioned at their nodes: the way the meanings of the immediate constituents of an expression are combined is determined by the syntactic operation that combines the (disambiguated) constituents. In particular, the same expressions may stand in different grammatical constructions that differ in their meanings. As a case in point, in addition to the simple subject-predicate construction F_4 , *PTQ* has five operations $F_{11} - F_{15}$, which introduce negation, non-present tense (future or past), or both.

Generally speaking, to any syntactic operation F there corresponds some semantic operation G on meanings so that the result of combining any disambiguated expressions $\alpha_1, \dots, \alpha_n$ by F gets assigned the meaning that results from combining the meanings of $\alpha_1, \dots, \alpha_n$ by G :

$$(2) \quad \llbracket F(\alpha_1, \dots, \alpha_n) \rrbracket = G(\llbracket \alpha_1 \rrbracket, \dots, \llbracket \alpha_n \rrbracket)$$

In mathematical parlance, (2) says that meaning (designated by the 'semantic brackets') is a *homomorphism*. So Montague's (by now standard) criterion of compositionality is that (2) holds for all constructions F operating on underlying disambiguated expressions. These remarks should suffice to give an impression of the general features of the compositional meaning assignment. The following subsections will illustrate how the algebraic formulation (2) of the principle of compositionality is specified in *PTQ* in order to give an account of the meanings of disambiguated English expressions.⁴

³ Montague (1970a: 388ff.) offers a more principled approach to subject-verb agreement by parallel recursion. See von Stechow (1979) for a general account of feature matching in German along those lines.

⁴ Actually, *PTQ* deviates from the general framework of Montague (1970a) to some extent. In particular, the construction-driven character of compositionality is at times traded for a *rule-driven* account, which comes close to Janssen's (1983: 41ff.) approach in terms of sorted algebras; the interpretation of quantifying determiner phrases and the scope-taking mechanisms are cases in point. We will gloss over these details here.

Frege's Functionality, California Style: Extensions and Intensions

The general approach to meaning pursued in *PTQ* can be traced back to the core piece of Montague's *Universal Grammar*: the 'Theory of Reference' (Montague, 1970a: 378ff.). Although presented as a development of Frege's (1892) logical semantics, the roles of sense and reference are played by (generalized versions of) Carnap's (1947) *extensions* and *intensions*. In particular, the intension of an expression α is a function (in the set-theoretic sense) that assigns to any point of evaluation α 's extension at that point, where a point of evaluation is an ordered pair of a possible world and a time.⁵ As a consequence, Montague's intensions suffer from a certain lack of fine-grainedness vis-a-vis Frege's realm of senses. However, the specific way in which extensions and intensions interact does give the approach a distinctly Fregean flavor: not the meanings, but the extensions of complex expressions are the outputs of compositional operations; and as a rule, the extensions of their parts form their inputs, though at times their intensions may do so. As a case in point, the conjunctive coordination S_1 and S_2 of two declarative sentences can be described by a truth-table which combines the extensions, i.e. the truth values, of the two immediate parts, S_1 and S_2 – or, as Frege (1892) would have put it: their ordinary referents ('gewöhnliche Bedeutungen'). On the other hand, in order to determine the truth value of a causal connection S_1 because S_2 , the informational contents of the two parts need to be taken into account, and these contents are (arguably) provided by their intensions – which thus play the role of indirect referents ('ungerade Bedeutungen') that Frege assigned to their senses. Hence, compositionality is, first and foremost, a matter of extensions, but intensions come to the rescue when everything else fails. In general, then, two fundamentally different ways of compositionally obtaining extensions need to be distinguished: in *extensional* (or direct) environments, extensions are combined, in *intensional* (or indirect) constructions, intensions are employed to obtain the extensions of larger expressions. It is this interplay of the two basic semantic values that Montague took to be central to his semantic theory and that he called 'Frege's functionality principle'.⁶

Although extensions are the results of the compositional operations used in semantic analysis, they are not themselves the meanings that the principle of

⁵ There is no specific term for such pairs in *PTQ*; I am indebted to Zoltán Gendler Szabó for suggesting *point of evaluation* to me.

⁶ See Montague (1970c: 75f.), where the principle is characterized with reference to certain formal languages: 'the extension of a formula is a function of the extensions [...] of those of its parts not standing within indirect contexts [...], together with the intensions [...] of those parts that do stand within indirect contexts.' A few paragraphs earlier (*ibid.*, p. 75), Montague makes the connection between extension and Frege's [*gewöhnliche*] *Bedeutung* (which he translates as [*ordinary*] *extension*), and also mentions that Frege did not intend his functionality principle to apply to variable binding.

compositionality (2) is about: after all, the extension of a sentence is its truth value, and while speakers ideally know the meanings of all sentences, they can hardly be expected to know of all of them whether they are true, and certainly not solely on the basis of their linguistic competence. Fortunately, though, there is a direct route from extensional compositionality to the compositionality of intensions: if the extension of a complex expression can be determined from the extensions of its parts, then so can its intension from the intensions of its parts. Again, conjunction is a case in point: the fact that the extension of S_1 and S_2 can be calculated by a truth table implies that its intension can be obtained by a Boolean operation on propositions (construed as sets of points of evaluation): intersection in the case at hand (or, rather, the corresponding operation on characteristic functions). More generally, writing ‘ $\llbracket \alpha \rrbracket^{w,t}$ ’ for the extension of the expression α at point $\langle w, t \rangle$, the extensional compositionality (3) of a construction F (at arbitrary points of evaluation) in terms of some operation H , is easily seen to imply the existence of a corresponding meaning combination G satisfying (2) above – provided that meanings are identified with intensions:

$$(3) \quad \llbracket F(\alpha_1, \dots, \alpha_n) \rrbracket^{w,t} = H(\llbracket \alpha_1 \rrbracket^{w,t}, \dots, \llbracket \alpha_n \rrbracket^{w,t})$$

For on the basis of (3), G can be defined in a pointwise fashion: the intension that G assigns to the intensions $\llbracket \alpha_1 \rrbracket, \dots, \llbracket \alpha_n \rrbracket$ is the function that takes each point $\langle w, t \rangle$ to the corresponding H -value:

$$(4) \quad G(\llbracket \alpha_1 \rrbracket, \dots, \llbracket \alpha_n \rrbracket)(\langle w, t \rangle) = H(\llbracket \alpha_1 \rrbracket^{w,t}, \dots, \llbracket \alpha_n \rrbracket^{w,t})$$

Since each extension $\llbracket \alpha_i \rrbracket^{w,t}$ is merely the value of $\llbracket \alpha_i \rrbracket$ at $\langle w, t \rangle$, i.e., $\llbracket \alpha_i \rrbracket(\langle w, t \rangle)$, (4) is indeed a pointwise definition of G in terms of the intensions of the parts $\alpha_1, \dots, \alpha_n$. And *a fortiori*, a similar recipe can be given for intensional constructions, where the extensions already depend on the intensions of (some of) the parts. Hence, the (allegedly Fregean) strategy of obtaining extensions from the extensions or intensions of the parts of expressions guarantees the compositionality of meanings – at least to the extent that the meanings of expressions coincide with their intensions.

But do meanings and intensions coincide? In *PTQ*, Montague seems to have given a positive answer to this question, since his semantic account of English largely consists in a recursive characterization of the intensions of expressions. Hence, the *PTQ* framework has rightly been seen as being in the spirit of what Putnam (1975: 133ff.) called ‘California semantics’: the meanings speakers grasp as part of their mastery of a language are identified with intensions, which in turn enable them to identify corresponding extensions on the basis of further factual knowledge. However, two qualifications to the identification of meaning and intension should be made, even according to Montague:

- The intensions of expressions containing deictic material depend on the utterance context.
- The intensions of formulae with free variables depend on a variable assignment.

In view of these two exceptions, Montague (1970a) actually identified meanings with functions from utterance contexts and (or, rather: including) variable assign-

ments to intensions. However, since *PTQ* features no deictic expressions, the context dependence of intensions has been suppressed throughout. On the other hand, variables do play a role in the interpretation of quantifier scope constructions like $F_{10,0}$ in (1) above; and these variable-binding operations are known to not conform to extensional or Fregean compositionality. For ease of presentation, however, we will follow semantic mainstream and ignore the compositionality complications arising from variable binding, for which the interested reader is referred to Chap. 10 of Zimmermann and Sternefeld (2013).

In *PTQ*, then, even though intensional constructions are in focus (and variable binding occurs), it is extensions that play first fiddle: meaning composition is (largely) described in terms of operations deriving the extensions of complex expressions from the extensions, and sometimes the intensions, of their parts. The format in which these extensions and intensions are defined will be addressed in "[Translating into IL: indirect interpretation](#)".

A Touch of Structuralism: Model Theory

Whether or not they are identified with intensions, Montagovian meanings ultimately serve the purpose of determining referents and truth values, thereby connecting words and worlds. However, the general theory behind *PTQ* is set up in such a way that any semantic account of any (formal or natural) language is bound to leave out the details of this connection: not the meanings of the expressions under scrutiny but only some of their structural features and relations are specified. This surprising omission is a design feature of the model-theoretic framework, according to which the interpretation of a language is not a mapping from expressions to their meanings but a family of mappings from expressions to objects that share (part of) their set-theoretic structure but may hugely disagree as to what the objects and points of evaluation are. These families of mappings to meaning-like objects are called *models*, and in *PTQ* [p. 230] they come in the form of quintuples $\langle A, I, J, \leq, F \rangle$ whose components are sets representing objects that play the roles of: potential referents of individual terms (A); possible worlds (I); moments of time (J); the temporal order ($\leq$); and the lexical meaning assignment (F), whose extrapolation to complex expressions will be addressed in "[Translating into IL: indirect interpretation](#)". All of these components are arbitrary sets, which means that they do not have to contain any physical objects, worlds, or times. Yet while it may happen that none of the A -components in a model M of the above form contains any wolves or sheep, the roles of these animals (at certain points of evaluation) will be played by the elements of the extensions determined by $F(\mathbf{wolf})$ and $F(\mathbf{sheep})$; yet in another model of the same structure at least some of the wolves according to F play the roles of sheep. And both of these models count as equally legitimate representations of the form-meaning mapping in English. In terms of classic structuralism, it is only (set-theoretic) *structure* that counts in model-theoretic interpretation, not the

(set-theoretic) *material* that displays it.⁷ On the other hand, the model-theoretic interpretation of English in *PTQ* does not fully specify the structure of the meaning assignment. In fact, it comes with a huge class of models that not only vary as to their substance but are also structurally divergent. Thus, e.g., apart from demanding non-emptiness of the domains *A*, *I*, and *J*, nothing is said about their sizes. As a consequence, there are models with only one (object playing the role of an) individual in their domain as well as infinite sets of (representatives of) individuals of any cardinality, and similarly for worlds and times. Moreover, given a selection of domains *A*, *I*, and *J*, a lexical item α may receive any meaning constructed from them, as long as $F(\alpha)$ is of the type matching α 's category (about which more in the next subsection) and certain 'reasonable' *meaning postulates* [*PTQ*, p. 235] are respected to take care of semantic effects like⁸:

- *Rigidity*: the intensions of proper names should be constant across all points of evaluation;
- *Transparency*: the complements of certain verbs and prepositions should take scope over them;
- *Lexical decomposition*: **seek** can be paraphrased in terms of **try** and **find**.

The *PTQ* meaning postulates are formulated in terms of higher-order intensional logic and exert their effect on the expressions of English by way of the indirect interpretation mechanism to be addressed in "[Translating into IL: Indirect Interpretation](#)"; in particular, they are not expressed in the object language itself, nor can they be without loss of substance (cf. Zimmermann, 1985).

Even given the above meaning postulates (and others to be addressed in "[Descriptive Aspects: The Fragment](#)"), the type assignment and the nine schematic postulates leave a lot of wiggle room for potential meaning assignments. As a case in point, according to some models with a singleton domain of individuals, the sentence **Some man is not Bill** comes out as false at every point of evaluation; the same effect will arise in a model according to which the extension of **man** coincides with the (characteristic function of) the singleton of the (rigid) extension of **Bill** at every point of evaluation. Clearly, such 'degenerate' models (Rooth, 1985: fn. 13) can be eliminated by *ad hoc* postulates but there seems to be no general

⁷ Cf. Saussure's (1916: 122) famous dictum that 'language is a form, not a substance' [*la langue est une forme et non une substance*].

⁸ To be sure, the labels are anachronistic. The rigidity schema can already be found in Montague (1970a: 393); hence Montague's treatment of proper names as "logically determinate" [*PTQ*, p. 235, in scare-quotes] foreshadows Kripke's (1972) rigid designators. Transparency blocks both non-specific and intensional readings in verbs like **find** (and unlike **seek**) and in prepositions like **in** (and unlike **about**); it should be noted, though, that non-specificity and intensionality are both theoretically and empirically independent, as pointed out, respectively, in Zimmermann (1983: 73, fn. 9) and Mador-Heim and Winter (2015). The lexical decomposition of **seek** goes back to Quine (1956, 1960: §32), where it is presented as a reductive analysis rather than an accidental observation; see Montague (1969: 174ff.) for criticism and den Dikken et al. (2018) for a recent revival of the Quinean spirit in syntactic terms.

systematic way for doing so short of assuming a fixed *intended* model whose meaning assignment would be fully isomorphic to the real thing – thereby in effect giving up the model-theoretic approach (cf. Zimmermann, 2011: 799).

Though rarely explicitly questioned, the value of the model-theoretic approach to natural language semantics is moot. In particular, in some of the most influential works in the (post-) Montagovian tradition, model classes are traded for material truth conditions and meaning assignments. As a case in point, the meaning assignments in Heim and Kratzer (1998) are given in terms of reference to persons and other concrete objects, as opposed to arbitrary set-theoretic constructions. In the remainder of this brief survey, reference to model-space will be largely suppressed too.

The Hierarchy of Extensions: Functional Types

According to Montague Grammar, the category of an expression not only determines its syntactic behavior but also the *kind* of meaning it has. The rationale for carving up the realm of meanings into different types goes back to Frege's (1891) strategy of distinguishing *saturated* ['gesättigt'] expressions which, like names and sentences, have truth values or (individual) referents, from those whose extensions compositionally contribute to the referents or truth values of the expressions in which they occur, where these contributions are construed as functions assigning extensions of mother constituents to extensions of sister constituents. Thus, e.g., the extensions of verb phrases *V* come out as characteristic functions assigning truth values (of sentences with *V* as a predicate) to individuals (as the extensions of referential subjects of said sentences). By iteration, this strategy, which has become part of many textbook accounts of modern semantics, eventually leads to a whole hierarchy of ever more complex functional extensions, including: functions from individuals to truth values (VP-extensions); functions from individuals to VP-extensions (extensions of transitive verbs); functions from VP-extensions to truth values (quantifier extensions); etc. For notational convenience, Montague followed Church (1951) and used mnemonic labels for the layers of this hierarchy – his system of *intensional (functional) types*, starting with the extension types *e* [*entities*] and *t* [*truth values*] for referential expressions and sentences, and labelling the layer of functions from layer *a* to layer *b* by the ordered pair of *a* and *b*: $\langle a, b \rangle$.⁹ Hence, just for the record, the by now ubiquitous angular brackets *per se* do not have anything to do with types but merely serve to indicate ordered pairs; in particular, Montague did not use them to mark the ground types *e* and *t* themselves. In order

⁹ Montague's hierarchy was clearly inspired by Church's (1951) formalization of Frege (1892). In fact, both systems of types are derived by adding senses to Church's (1940) hierarchy of extensional functional types. However, they differ in that Church restricted the hierarchy of senses to the two ground types *t* and *o* (which correspond to Montague's *e* and *t*). See Kaplan (1975: 721) for the relation between the two hierarchies.

to cope with intensional constructions by way of Fregean compositionality, the type hierarchy also needs to contain intensions, which serve as the compositional contributions of those constituents that give rise to substitution problems. Since, according to Montague (and presumably Frege, too), expressions of any category might in principle contribute their intensions, the functional type hierarchy contains intensions corresponding to any type of extension, i.e., functions from points of evaluation to any layer a of extensions, which, paying reverence to Frege (1892), Montague (1970a: 379) dubbed $\langle s, a \rangle$; again for the record: the ‘ s ’ in the intension types stands for *sense* (or *Sinn*), not for *situation*, even though arguably this is what the points of evaluation that constitute their domains come down to.

The set *Type*, then, contains the labels e and t as well as all pairs $\langle a, b \rangle$ and $\langle s, b \rangle$, where a and b are members of *Type*, and nothing else [PTQ, p. 229]; and the (model-dependent) domains D_a of objects of any type – the *possible denotations* of that type – are then defined by the following recursion [PTQ, p. 230]: D_e is the set of individuals provided by the model; D_t is the set $\{0, 1\}$ of truth values; $D_{\langle a, b \rangle}$ is the set of total functions from D_a to D_b , i.e., those functions that assign to each argument of type a a value of type b ; and $D_{\langle s, b \rangle}$ is the set of total functions from the (model-dependent) set of evaluation points to D_b , i.e., those functions that assign to each pair $\langle i, j \rangle$, where i is a world and j is a time, a value of type b .

It should be noted that Montague’s type hierarchy differs from later versions in that it does not treat s as a type of its own: due to their special function of determining extensions relative to logical space, points of evaluation do not count as objects of reference. By the same token, there are no functional extensions that would assign points of evaluation to extensions of potential sister nodes, and as a consequence, there appears to be no need for any types of the form $\langle a, s \rangle$. It would certainly have been possible (for Montague or anyone) to include s in the type hierarchy as well as all pairs formed from the so-extended types – thus arriving at the more inclusive set *Type2* of *two-sorted types* containing e , t , and s , as well as all pairs of members of *Type2*, and nothing else.¹⁰ In this extended hierarchy, intensions are merely special cases of functional extensions. One reason why Montague did not take this somewhat more straightforward route lies in his motivation to build the hierarchy of extensions as part of a reconstruction of Frege’s functionality principle in terms of Carnapian intensions. A further potential reason for this restriction on *Types* will be brought up in the next subsection.

Given the motivation of types in terms of compositional contributions to extensions, it should not come as a surprise that expressions that behave syntactically alike also have extensions of the same type. In Montague Grammar, this connection between syntax and semantics is made by a *type assignment* called ‘ f ’, which (as the term indicates) associates syntactic categories with corresponding functional types. One obvious condition on this association is that the category of sentences

¹⁰ The term *two-sorted* was introduced by Gallin (1975: 58) in this connection to reflect the fact that in *Type2* and its applications to type logic (more about which below), the sets D_e and D_s play a role analogous to that of sorts of individuals in many-sorted predicate logic.

is matched with the type t of truth values. The theoretical framework of Montague (1970a) imposes no further restrictions on type assignments in general. However, in *PTQ*, Montague borrowed some semantically motivated notation from categorial grammar (Ajdukiewicz, 1935; Bar-Hillel, 1953; Lambek, 1958) to replace the arbitrary category labels: sentence predicates (or verb phrases) fall under the category t/e , where (non-italicized) t stands for the category of sentences and e for the category of individual-denoting expressions. In classical categorial grammar, this categorization would have the consequence that putting a verb next to an expression of category e results in a sentence. But in *PTQ*, the notation merely indicates the type of extensions of expressions of a category. On the basis of numerous descriptive applications and extensions of Montague Grammar as well as several later developments, it is natural to assume that this type should be the type $\langle e, t \rangle$ of (characteristic functions of) sets of individuals.¹¹ The reasons why, in *PTQ*, Montague decided otherwise, lies in his attempt to achieve a maximum of semantic *uniformity* in syntactic categories, to which we will now turn.

Since in general, expressions with functional extensions may give rise to substitution problems, their sister constituents may have to contribute their intensions (rather than their extensions) to the extension of the mother constituent. Sentence adverbs are a case in point. In *PTQ*, they naturally fall under the category t/t , because they combine with a sentence to produce another sentence, with **necessarily** (construed in the logicians' sense of *everywhere in Logical Space*) serving as the sole example in *PTQ*. Clearly, in this case the corresponding type needs to be $\langle \langle s, t \rangle, t \rangle$: the truth value of a sentence of the form '**Necessarily** S ' cannot be determined on the basis of the truth value of S alone (not if S is true, that is). So the type assigned to sentence adverbs is that of functions taking *intensions* of the sister type to *extensions* of the mother type. However, this does not make them *per se* intensional: even negation (if it were a sentence adverb) could be regarded as an operator of type $\langle \langle s, t \rangle, t \rangle$: given a point of evaluation, its extension would assign the opposite truth value of whatever its sister assigns to that point.¹² Of course, there is something hybrid to this construal of negation as operating on propositions, since it would do the very same job as the familiar truth table of type $\langle t, t \rangle$. But it would unify the types of intensional and extensional sentence adverbs. And it is readily seen that, quite generally, at any point $\langle i, j \rangle$, any function $g \in D_{\langle a, b \rangle}$ could be 'simulated' by a hybrid function $g^+ \in D_{\langle \langle s, a \rangle, b \rangle}$ that does the same job: $g^+(x) = g(x(\langle i, j \rangle))$, whenever $x \in D_{\langle s, a \rangle}$. This elementary observation gives rise to *PTQ*'s [p. 232] unified treatment of the types $f(A/B)$ of expressions of a category A/B that combine with category B expressions to produce expressions of category A : whether or not they create intensional contexts, they are treated as operating on the intensions of their sisters (in LF, as it were) and thus receive the extension type $\langle \langle s, f(B) \rangle, f(A) \rangle$. Hence verb phrases, whose category is t/e , come out as having extensions of type $\langle \langle s, e \rangle, t \rangle$. Thus, at a given

¹¹ See, e.g., Dowty et al. (1981) or Heim and Kratzer (1998).

¹² Cf. Thomason and Stalnaker (1973: 203), where **actually** is given as another example of an extensional sentence adverb besides negation.

point of evaluation $\langle i, j \rangle$, the extension of **walk** contains all functions of type $\langle s, e \rangle$ whose value for that point is an individual that walks at that point. If one thinks of such functions as intensions of descriptions or modes of presentation of objects, then the extension of **walk** may be said to contain each walking individual in any way this individual can be described or presented. This is certainly an overkill vis-à-vis the simpler extension that would merely collect the walking individuals themselves. But then the type $\langle e, t \rangle$ of that simpler extension would not work for intensional predicates. And, according to Montague, there are such predicates; we will turn to them in “[The Temperature Paradox](#)”.

The unification of the semantic types of extensional and intensional verb phrases is part of a general strategy that has aptly been described as ‘generalizing to the worst case’ (Partee, 1997: 75): if a class K of expressions is syntactically homogenous enough to form a category, K is assigned the most complex type observed among the compositional contributions of its members, all of whose extensions are adapted to that type. The adaptations from the originally observed (or stipulated) types – $\langle e, t \rangle$ in the case at hand – to the most complex (or ‘worst’) one are known as *type shifts*. The map from sets of individuals to a corresponding set of individual concepts (of type $\langle s, e \rangle$) is a case in point. Another type shift is the so-called *Montague Lift* that maps the referents of proper names to corresponding quantifiers. Here the relevant syntactic category is that of a Term, i.e., a nominal expression that may combine with a predicate (of category IV) to form a sentence (t), which is reflected in the label of this category: $t/(t/e)$, abbreviated as T. Given the above intensionality-sensitive pattern of mapping categories to types, $f(T)$ comes out as $f(t/(t/e)) = \langle \langle s, f(IV) \rangle, t \rangle$, which spells out as: $\langle \langle s, \langle \langle s, e \rangle, t \rangle \rangle, t \rangle$. As a consequence, the extensions of names according to *PTQ*, are (characteristic functions of) sets of properties of individual concepts. More will be said about the exact nature of these sets later; for the time being we only note that they need to be defined in such a way that they fit the quantificational subject construal F_4 .

While differences in the semantic behavior of syntactically similar expressions can be evened out by generalizing to the worst case, the semantic similarity among expressions does not imply syntactic uniformity. In particular, two distinct categories may be assigned the same extension type. In *PTQ* the category of nouns, whose extensions come out as sets of functions of type $\langle s, e \rangle$, is a case in point. Given their different syntactic behavior, their category thus needs to be distinguished from the category of verb phrases while still wearing its extension type on its notational sleeves. Since the categorial label t/e has already been used up, Montague came up with the *ad hoc* device of adding another slash to distinguish the category of nouns from that of Intransitive Verbs and defined it as $t//e$. The same hack is used to distinguish adverbs (IV/IV) from control verbs (IV//IV), and again it is quite arbitrary which is which: one or two slashes, the effect on the extension type is the same: $f(A/B) = f(A//B) = \langle \langle s, f(B) \rangle, f(A) \rangle$, for any categories A and B [*PTQ*, p. 232].

Translating into IL: Indirect Interpretation

Montague Grammar owes much of its flavor to the particular format used to specify extensions (and intensions), viz. by formulae of the language of intensional type logic – *IL*, for short.¹³ In the *PTQ*-grammar, as already in Montague (1970a), an algorithm is specified that translates the expressions of English into *IL*-formulae, whose (model-dependent) denotations are specified independently and carry over to the English sources. Crucially, the algorithm proceeds in a compositional fashion so as to guarantee the compositional behavior of the English expressions it translates. This means that the translation into *IL* constitutes a detour on the way from natural language to meaning – and hence an *indirect interpretation*. Yet despite its eliminability, the procedure offers a number of advantages, not the least of which is that *IL* (or whichever language is chosen as a medium) may serve as a *lingua franca* for the semantic community. However, indirect interpretation must not be misconstrued as representationalism: the meanings of English expressions are the model-theoretic interpretations of their *IL*-translations, not the *IL*-formulae themselves.

The algebraic details of indirect interpretation, and particularly its compositionality and eliminability, are beyond this short survey, in which the translation algorithm will only be presented by way of examples.¹⁴ The same goes for the language *IL* itself. The following remarks are mainly meant as a help to decipher the formulae used in the *PTQ*-translation.

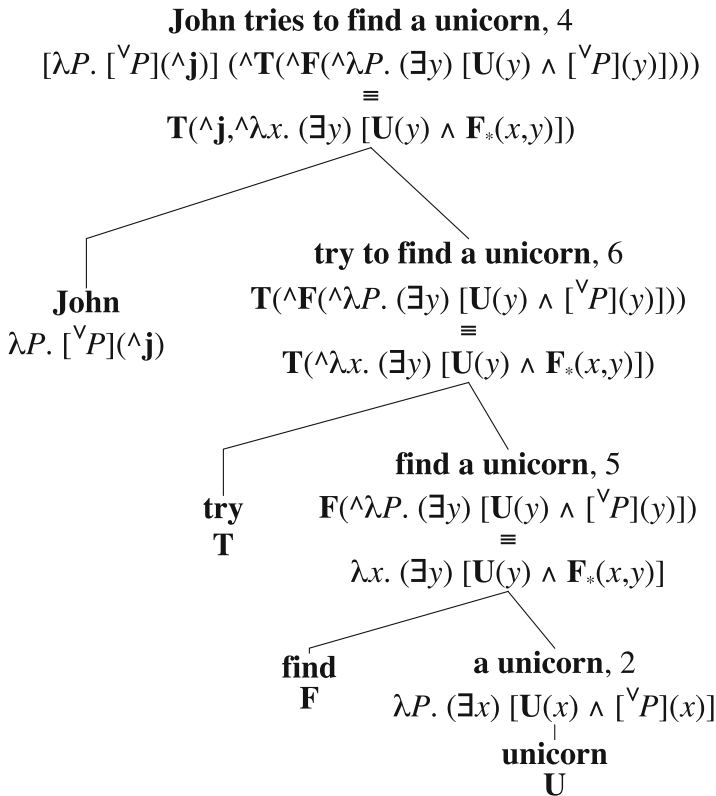
The formulae of *IL* are stratified by their *Types*; the idea is that the translation of an English expression of any category *A* will be an *IL*-formula of type $f(A)$ denoting an object of that type. Thus, sentences translate as formulae of type t ; verb phrases as formulae of type $\langle\langle s, e \rangle, t \rangle$; etc. As a rule of thumb, lexical expressions get translated by corresponding constants, whereas the translations of complex expressions are obtained by combining the translations of their immediate parts in accordance with the syntactic constructions by which they have been built. Thus, e.g., the verb **try** and the verb **sleep** translate as certain constants of types $\langle\langle s, \langle\langle s, e \rangle, t \rangle \rangle, \langle\langle s, e \rangle, t \rangle \rangle$ ($= f(IV//IV)$) and $\langle\langle s, e \rangle, t \rangle$ ($= f(IV)$), which in *PTQ* are referred to as **try'** and **sleep'**, respectively. As a consequence, their extensions may vary wildly across the class of all models (though not necessarily across the worlds within one model; cf. Zimmermann, 2017). While such wide ranges of extensions appear to reflect the inferential behavior of non-logical lexical items, it would be inappropriate in worst case generalizations. Thus, e.g., a type-shifted

¹³ This by now common name appears to be due to Gallin (1971). In Montague (1970a), the system was called L_0 , and no shorthand name was used for the slightly extended version in *PTQ*. We will also follow semantic practice and use the term (*IL*-) *formula* for all terms or 'meaningful expressions' [*PTQ*: *passim*] of that language, not just for truth-valued ones (of type t : see below), as logicians would prefer.

¹⁴ The algebraic aspects of indirect interpretation are expounded in Montague (1970a: 383f.). An interesting generalization (in terms of 'safe derivers') can be found in Chap. II of Janssen (1983).

name should not be able to denote an arbitrary quantifier: if, say, the extension of **John** (at some point of evaluation in some model) were the same as that of **no entity**, the inference from **John walks** to **Some entity walks** would not go through. To capture their inferential behavior, then, proper names like **John** are translated by type-theoretic formulae of the form $\mathbf{c}^*$, where $\mathbf{c}$ is a constant of type e and the $*$ -operator expresses Montague Lifting in *IL*. Before getting into the specifics of this type shift, it will be instructive to take a brief look at how complex expressions of English are translated into *IL*. As a first step in this direction, let us adorn the tree in (1a) with type-logical translations:

(5) *IL*-translation of (1a)

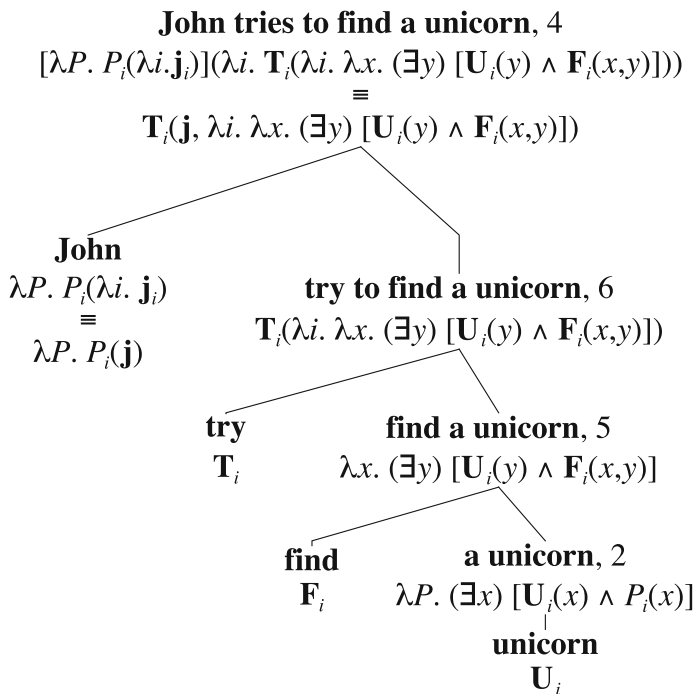


Apart from a few minor notational variations,¹⁵ (5) displays the stepwise *IL*-translation of (1a) according to the *PTQ*-algorithm, plus some logical reductions (following the triple bars). Readers unfamiliar with the *IL*-formalism may prefer the following equivalent representation in terms of a more transparent notation that

¹⁵ Thus, Montague uses capitalized versions of the connectives ‘ $\wedge$ ’ and ‘ $\vee$ ’ to denote universal and existential quantification, where we use the more common inverted ‘A’s and ‘E’s. Moreover, we have given mnemonic names to the constants and omitted predictable brackets around λ -terms.

makes explicit reference to points of evaluation by way of variables, a notation known as *Ty2* (for *two-sorted type theory*):

(6) *Ty2*-translation of (1a)



The ubiquitous index i is a variable ranging over points of evaluation; and subscripting is merely a (common) notational variant for functional application, $\mathbf{U}_i$ being shorthand for $\mathbf{U}(i)$, etc. The logical formulae in (6) are built up using bold-face constants of the appropriate *intension* types. Thus, given that the extension of the noun **unicorn** is of type $f(t/e)$, $\mathbf{U}$ is of type $\langle s, \langle \langle s, e \rangle, t \rangle \rangle$. Slightly deviating from the *PTQ* type assignment (see below), the constant $\mathbf{F}$ in (6) denotes the intension of a binary relation between individual concepts and is thus of type $\langle s, \langle \langle s, e \rangle, \langle \langle s, e \rangle, t \rangle \rangle \rangle$. And given that the verb **try** is of category (IV//IV), the constant $\mathbf{T}$ is of type $\langle \langle s, f(\text{IV}) \rangle, f(\text{IV}) \rangle$ ($= \langle \langle s, \langle \langle s, e \rangle, t \rangle \rangle, \langle \langle s, e \rangle, t \rangle \rangle \rangle$); as a consequence (to which we will also return), the constant $\mathbf{j}$ that fills the 'subject' argument of $\mathbf{T}$, is of type $\langle s, e \rangle$. The rest of the notation is hopefully self-explanatory. In particular, the triple bar between the type-logical translations and their reduced forms stands for logical equivalence; the suggested reductions all make use of the familiar laws of λ -conversion.¹⁶

¹⁶ The leftmost equivalence (below **John**) also takes advantage of the fact that $\lambda i. \mathbf{j}_i$ reduces to $\mathbf{j}$, due to the logical law of η -conversion, which reflects the extensionality of set-theoretic functions.

Readers may feel that the formulae in (5) are somewhat simpler than those in (6), particularly because they seem to do without the evaluation-point variable i . As a matter of fact, though, rather than being absent, it is cleverly hidden. Its suppression is the main feature that distinguishes *IL* from the seemingly more complicated notation in (6). Closer inspection of the translations of all parts of (1a) but the embedded verb reveals how the evaluation-point variable can be made to reappear by a systematic procedure:

- *IL*-constants $\mathbf{c}$ of any extension type a correspond to constellations of the form ' $\mathbf{c}_i$ ', where $\mathbf{c}$ is a homographic Ty2-constant of the corresponding intension type $\langle s, a \rangle$ and i is a fixed variable of the (non-Montagovian but) two-sorted type s .
- The *cup* operator preceding the variable ' P ' (of type $\langle s, \langle \langle s, e \rangle, t \rangle \rangle$) corresponds to application of the homographic Ty2-variable ' P ' to the evaluation-point variable i ; in fact, any *IL*-formula of the form ' $[\forall \alpha](\beta)$ ' corresponds to $\alpha^*(i)(\beta^*)$, where α^* and β^* correspond to α and β .¹⁷
- The *cap* operator preceding intensional arguments corresponds to the prefix ' λi '.

Using these three correspondence rules and leaving everything else as is, almost any *IL*-formula can be translated into the more transparent Ty2-notation.¹⁸ The one remaining difference between (5) and (6) then concerns the translation of **find**, which is a constant of the worst case for transitive verbs, which take Term intensions of the dazzlingly complex type $\langle s, f(T) \rangle = \langle s, \langle \langle s, \langle \langle s, e \rangle, t \rangle \rangle, t \rangle \rangle$ as their arguments. The justification for this type assignment will be supplied in '[Intensional Transitive Verbs](#)'. However, like its namesake in (6), the **F** in (5) merely relates individual concepts represented by x and y – except that the type of the latter has been adapted by a Montague Lift. It is this binary relation between x and y that features in the translation tree (5)¹⁹:

$$(7) \quad \mathbf{F}_* := [\lambda y. \lambda x. \mathbf{F}(x, y^*)]$$

where the upper asterisk again indicates a Montague Lift, the definition of which can be gleaned from the translation of the subject in (5):

$$(8) \quad \alpha^* := [\lambda P. [\forall P](\alpha)]$$

In *PTQ*, (7) is implemented as (a consequence of) a meaning postulate – an *IL*-formula that is supposed to be true at points of evaluation of all “reasonable” models. The same effect could have been achieved by taking the right-hand side of (7) as the

¹⁷ In *PTQ* the constellation ' $[\forall \alpha](\beta)$ ' is usually abbreviated as ' $\alpha\{\beta\}$ '.

¹⁸ Tense and modal operators need a special treatment, thence ‘almost’; see, e.g., Köpping & Zimmermann (2021) – The correspondence, which is a special case of the so-called *standard translation* of modal logic (Fine, 1975: 19), was first described in Gallin (1975: 61); given that the system Ty2 is the obvious substratum underlying the very construction of *IL*, Montague was certainly aware of it, though.

¹⁹ In fact, by the meaning postulate (4) given in *PTQ* [p. 235], $\mathbf{F}_*$ even downgrades finding to a relation between *individuals* (instead of individual concepts); however, we refrain from doing so here for expository reasons. Similar remarks apply to (10) below.

translation of **find**, instead of having a constant **F** (which, it should be recalled, is the *IL*-version of the Ty2-formula F_i). Indeed, for the translation of proper names, Montague [*PTQ*, p. 233, T1. (d)] followed this alternative strategy and put:

$$(9) \quad \text{John}' = j^*,$$

where **j** is a constant of type *e* (and thus, again, abbreviates j_i) and the asterisk is defined as in (8).

The type reduction in (7) can be used to express that any quantificational object of **find** will be interpreted so as to take scope over the verb – which is precisely what the transparency postulate demands [*PTQ*, p. 235, (4)]:

$$(10) \quad \mathbf{F} = [\lambda \wp . \lambda x. [\forall y](\wedge \lambda y. \mathbf{F}_*(x, y))]$$

(10) offers an alternative route to what Heim and Kratzer (1998: 178) famously dubbed 'the problem of quantifiers in object position'. The Term is interpreted *in situ* (and by functional application) but still manages to quantify over objects found by the subject. This is possible because the verb (or its *IL*-translation **F**) does not denote the relation of finding between individuals but a type-shifted relation between individual concepts *x* and Term intensions $\wp$, viz. the relation that holds if the property of *being a y that is found by x* is in the extension of the quantifier – where the italicized condition is expressed in terms of $\mathbf{F}_*$, not **F**.

These illustrations must suffice to give readers a glimpse of how the indirect translation process works. For the specifics of the translation algorithm and the target language *IL*, they are referred to *PTQ* itself or the surveys mentioned in the Introduction.

Before we get to the descriptive aspects of the *PTQ* grammar, a word is in order about the very choice of an intensional target language over its obvious – and obviously more transparent – two-sorted rival. For apart from being shorter and at times closer to predicate logic, the *IL*-formulae have little to recommend them. In particular, they are much harder to manipulate: the principle of λ -conversion, which is vital when it comes to reduction for readability, is not valid in its familiar form, the reason being that the suppressed evaluation point variable may lead to a variable clash but cannot be renamed. Pertinent examples can be found in any good reference text.²⁰ Another, less frequently noticed deficiency of *IL* is revealed in connection with the rigidity of proper names, which in Ty2 can be dealt with by type shifting constants from *e* to $\langle s, e \rangle$ – but not in *IL*, because *IL*-constants have intension types only (Zimmermann, 1983: 160, fn.20).

Given these problems with Montague's *IL*, one may wonder whether the simplification is worth the trouble.²¹ One potential reason for still using the language – and possibly Montague's own main motivation – may be seen in its restricted

²⁰ See, e.g., Dowty et al. (1981: 167). The correct, restricted version of λ -conversion in *IL* has first been formulated in Gallin (1975: 19) as an axiom schema.

²¹ The questions in this paragraph are addressed in more detail in Zimmermann (2021), where relevant references can be found too.

expressivity: *IL* avoids apparently unnecessary types like $\langle\langle s, t, \rangle, s \rangle$ and has only one ‘invisible’ evaluation-point variable (*i*) at its disposal. So if the intensional language suffices for indirect interpretation, its limitations might tell us something about the complexity of natural language, viz. that its combinatory power is below full two-sorted type theory. There are, however, two reasons that might speak against such an interpretation. On the one hand, it is quite uncertain that the avoidance of multiple (free) variables of type *s* is even descriptively adequate (cf. Cresswell, 1990). On the other hand, the expressivity gap between *IL* and *Ty2* is much smaller than it may appear: ignoring multiple evaluation-point variables and non-intensional types, anything that can be expressed in *Ty2* can also be expressed in *IL* – but usually in a much more roundabout and much less conspicuous way (cf. Zimmermann, 1989). Hence there is every reason to do away with Montague’s elegant but hard to read and manipulate intensional logic and use the much more transparent two-sorted version in indirect interpretation – as many semanticists have done since the 1980s.

Descriptive Aspects: The Fragment

The semantic phenomena covered by the *PTQ fragment* of English are too varied to all be addressed here. We will therefore concentrate on a few of the most innovative and influential descriptive contributions. It needs to be mentioned though that, apart from Montague’s previously published work, as a precise account of compositional semantics, *PTQ* was without precedent. In that sense, practically everything in that paper was both original and innovative.

Extensional Constructions

Although Frege (1879) had characterized quantification as second-order predication and Russell (1905) had extended Frege’s approach to definite descriptions, thereby rejecting Frege’s (1892) own, referential treatment, Richard Montague is usually credited with the first systematic compositional interpretation of nominal quantification.²² Given this achievement, it is somewhat surprising that his treatment of quantificational Terms does not proceed by a binary composition of determiner and noun, but rather lifts the latter to quantifier status by one unary operation per determiner. We thus have:

²² Essentially the same account of nominal quantification (also covering the determiner **no**) was already given in Montague (1970a). There is evidence (Barbara Partee, p.c.) that it was motivated by, and started with, the compositional analysis of intensional transitives, as announced in Montague (1969: 177).

$$(11) \quad F_0(\mathbf{unicorn}) = \mathbf{every\ unicorn},$$

which translates as:

$$(12) \quad \lambda P. (\forall x)[\mathbf{U}(x) \rightarrow [\forall P](x)],$$

which in turn is an instance of a general rule applying to translations α of nouns lifted by F_0 [PTQ, p. 233, T2]:

$$(13) \quad \lambda P. (\forall x)[\alpha(x) \rightarrow [\forall P](x)]$$

And we do not have anything like:

$$(14) \quad F_{100}(\mathbf{every, unicorn}) = \mathbf{every\ unicorn},$$

which would translate as:

$$(15) \quad \mathbf{every}'(\mathbf{U}),$$

where $\mathbf{every}'$ is the following *IL*-formula of type $\langle\langle\langle s, e \rangle, t \rangle, \langle\langle\langle s, e \rangle, t \rangle, t \rangle\rangle$:

$$(16) \quad \lambda Q. \lambda P. (\forall x)[[\forall Q](x) \rightarrow [\forall P](x)]$$

– and similarly for the determiners $\mathbf{a[n]}$ and $\mathbf{the}$.²³ In other words, instead of forming one lexical category, determiners are treated as each indicating its own unary syntactic operation. Hence Montague's analysis of nominal quantification does not lend itself so easily to the study of determiner denotations as the generalized quantifier theory it inspired, starting with Barwise and Cooper (1981). The reason why Montague preferred (11) over (14) may have to do with the *logicality* of determiners, which shows in their definability in terms of parameter-free type-logical formulae (i.e. formulae that do not contain any non-logical constants or free variables) and suggests that they be treated as part of the grammatical structure of language. However, the characterization of the distinction between lexical and grammatical meaning in terms of logicality is not part of the general theory of Montague (1970a).²⁴

As pointed out by Thomason (1974: 261, fn. 12), the translation (13) (like its variants for the other determiners) leads to undesired results: if α happens to contain a free pronoun with the same index n as the (fixed) variable ' x ', the universal quantifier would accidentally bind it – a case in point being: $\alpha = \mathbf{unicorn\ such}$

²³ A comparison of (12) and (16) nicely brings out the asymmetry between constants and variables in *IL*: in (12) the dependence on the point of evaluation is implicit in the constant $\mathbf{U}$, which corresponds to $\mathbf{U}_i$ in *Ty2*, whereas in (16) it needs to be marked by the cup operator preceding Q .

²⁴ A distinction is still made though: according to the theory of reference in Montague (1970a), abstraction from context, with variable binding as a special case, is confined to semantic composition. – Definability in a suitable logical system is one way of characterizing logicality. Another, more popular approach goes back to Tarski (1986) and uses invariance with respect to permutations on the ground domains (i.e., the individuals and the points of evaluation); this criterion was first proposed for (lexical) determiners in Keenan and Stavi (1986: 310f.).

that it loves him_n, which according to (12), would result in quantification over the set of narcissist unicorns. The following corrected version of (13) takes care of this nuisance²⁵:

$$(17) \quad [\lambda N. \lambda P. (\forall x)[N(x) \rightarrow [\vee P](x)]](\alpha)$$

Using (17), Montague's analysis turns out to be equivalent to the by now standard generalized quantifier account (15) + (16). Moreover, it helps bring out an elementary feature of the unary constructions $F_0 - F_2$: they are *extensional* in that the extensions of the quantifying Terms so constructed are determined solely on the basis of the extensions of their (unique) nominal parts. In (17) this shows in the position of the meta-variable α , which does not stand in the scope of any variable binder (λ , $\forall$, $\exists$) or intensional operator ($\hat{}$, $\Box$, W , H).

Apart from the Term formation operations $F_0 - F_2$, there is only one other family of extensional constructions in the *PTQ* fragment of English, viz. the coordinations of sentences, verb phrases (IVs), and Terms; their *PTQ* treatment formed the starting point for numerous investigations into (generalized) Boolean conjunction and disjunction.²⁶ We will not go into them here except for noticing that Term coordination in *PTQ* is restricted to disjunction (by T13 on p. 234), thereby evading any semantic problems posed by plural formation.²⁷

Interestingly though, the (later) so-called *in situ* interpretations of quantificational subjects and objects, both of which are listed among the rules of functional application [*PTQ*, p. 233], fall prey to Montague's strategy of generalizing to the worst case, ending up as intensional constructions, which is why we will only turn to them in "*Intensionality*". Finally, the reader should bear in mind that, on account of the extension type $\langle\langle s, e \rangle, t \rangle$ assigned to nouns and verbs, the *IL*-variable x in (13) ranges over individual concepts of type $\langle s, e \rangle$. This peculiarity of the *PTQ*-treatment of nominal quantification will also be addressed in "*Intensionality*".

²⁵ Thomason (*ibid.*) proposes to choose the variable x as depending on α , which is however not in line with Montague's (1970b: 383f.) 'polynomial' construal of indirect interpretation.

²⁶ See, in particular, Partee and Rooth (1983) and Hendriks (1990); Geach (1970) is an important predecessor.

²⁷ Barbara Partee (2011: 26) reports that 'there are handwritten pages in [Montague's] files from 1970 when he was working on PTQ that show failed attempts to treat quite a number of phenomena that never made it into PTQ. For example, he had intended to include a much larger class of quantifiers than the three (*a*, *the*, *every*) that ended up being treated in PTQ. But he abandoned the attempt to include a treatment of plural expressions in PTQ, which eliminated most quantifiers and eliminated term phrases conjoined with *and*, leaving only those three singular determiners and term phrases conjoined with *or*.'

Binding and Scoping

Failure of extensional substitution is not always caused by intensionality in the narrow sense. For apart from the point of evaluation, extensions also depend on context; and according to Montague's (1970a) theory of reference, this dependence can be abstracted from by variable binding, where the assignment plays the role of context. In the *PTQ* fragment, there are two kinds of construction that are interpreted as variable-binding operations: (restrictive) relative clause formation and quantifier raising.²⁸ On the syntactic side, both involve removing subscripts on pronouns, thus indicating their bound status; according to Montague's (1970a) theory of reference, any remaining subscripted ('free') pronouns need to be interpreted by a contextually given assignment.²⁹

The *PTQ* fragment does not really cover relative clauses but only circumlocutions like **unicorn such that it speaks**. The reason for this is presumably that, unlike the traces of relative pronouns, the bound anaphors in these circumlocutions do not call for any syntactic restrictions, as witnessed by **unicorn such that it or John speaks**. On the semantic side, Montague implemented Quine's (1960: 110f.) suggestion that restrictive relative clause formation and modification correspond to set abstraction and intersection, respectively, which are rolled into one construction $F_{3,n}$, translated by the following rule [*PTQ*, p. 233, T3]:

$$(18) \quad [\lambda x_n. [\alpha(x_n) \wedge \varphi]]$$

where α and φ are the respective *IL*-translations of the modified noun and the relative clause (bar the **such that**-prefix), and n is the index of the pronoun corresponding to the trace. It may be noted in passing that (18) suffers from similar problems as (13) above, and that analogous cures may be applied.³⁰

As to scope construal, the *PTQ* fragment distinguishes three cases according to the landing site of the quantifying Term, which may take scope over (a) a sentence, (b) a noun, or (c) a predicate. Case (a), which applies in (1b) above, naturally

²⁸ As pointed out by an anonymous reviewer, the references to quantifier raising in the following presentation need to be taken with a grain of salt: the *PTQ* scoping mechanism directly characterizes the relevant immediate constituents of the underlying disambiguation ($\approx$ LF); see von Stechow (1991: 100f.) for the relation between the two approaches.

²⁹ This suggestion has since been taken up by many semanticists – see, e.g., Kaplan (1989: 527) or Heim and Kratzer (1998: 242ff.); still, its coherence is debatable: cf. Wehmeier (2018).

³⁰ More specifically, (18) could either be traded for the more cumbersome (i), in analogy to (17), or the relative clause could be constructed independently, translated as (ii), in analogy to (15), and then get attached by predicate modification (in the sense of Heim & Kratzer, 1998: 65):

- (i) $[\lambda P. \lambda x_n. [P(x_n) \wedge \varphi]](\alpha)$
- (ii) $[\lambda x_n. \varphi]$

A third possibility is offered in Montague (1970a: 393, def. of H_7), where the relative clause gets generalized to the worst case of an intensional modifier.

generalizes the quantifier construal of predicate logic, by binding a variable in a ‘matrix’ sentence and then applying the quantifier to the resulting set:

$$(19) \quad \Theta(\lambda x_n. \varphi),$$

where Θ translates the quantifying Term and φ the sentence from which it has been scope-raised. To be sure, in (19) it is the λ -operator and not the quantifier itself that binds the variable; but then the predicate logic notation (20) may be taken as shorthand for (19) anyway – it’s just that predicate logic lacks λ -abstraction:

$$(20) \quad (\Theta x_n) \varphi$$

Due to the uniform type assignment, (19) is not quite the *PTQ* translation for scope-raised quantifiers of type (a) yet: since Terms are of category t/IV , their extension type comes out as $\langle\langle s, f(IV) \rangle, t \rangle$, and thus their argument needs to be intensionalized. So we have (21) as translating case (a) – even though no intensionality effect is observed here³¹:

$$(21) \quad \Theta(\hat{\lambda} x_n. \varphi)$$

Whenever a Term translation Θ is inserted in (21), it so happens that the cap operator eventually washes out by a variant of λ -reduction. (22), which is (equivalent to) the translation of (1b), is a case in point:

$$(22) \quad [\lambda P. (\exists x)[U(x) \wedge [\hat{\vee} P](x)]]([\hat{\lambda} x_0. \mathbf{T}(\hat{\mathbf{j}}, \hat{\lambda} z. \mathbf{F}_*(z, x_0))])$$

As readers should verify for themselves, using the above *Ty2*-representation of *IL*, (22) can be λ -converted to (23), where the underlined formula has taken the place of the λ -bound variable P :

$$(23) \quad (\exists x) [U(x) \wedge [\hat{\vee} [\hat{\lambda} x_0. \mathbf{T}(\hat{\mathbf{j}}, \hat{\lambda} z. \mathbf{F}_*(z, x_0))]](x)]$$

The underlined part of (23) is in the scope of the cup operator, thus producing a formula of the form: $[\hat{\vee} [\hat{\alpha}]]$, which in *Ty2* spells out as:

$$(24) \quad [\lambda i. \alpha^*](i)$$

which in turn λ -reduces to α^* .³² As a consequence, the cup-cap sequence turns out to be redundant and we get the following logical law³³:

³¹ In *PTQ* notation, (21) gets contracted to $\Theta(\hat{x}_n \varphi)$.

³² β -reduction always applies to constellations of the form $[\lambda x. \alpha](x)$ because x cannot be accidentally bound when it replaces a free occurrence of itself; moreover, the replacement $\alpha[x/x]$ is obviously the same as α .

³³ The law – whose name goes back to (Dowty et al., 1981: 154) – features as an axiom schema in Gallin (1975: 19) but is not mentioned in *PTQ* or Montague (1970a), although it is crucial for many of the reductions given there. – Incidentally, there is no ‘reverse’ law of up-down cancellation: $[\hat{\vee} [\hat{\alpha}]]$ is generally not equivalent to α ; where it is, it boils down to an instance of η -conversion in *Ty2*. Readers are invited to construct their own counterexamples, starting by inspecting corresponding *Ty2*-constellations $[\lambda i. \alpha(i)]$.

(25) *Down-up cancellation*

For any *IL*-formula α : $[\vee [\wedge \alpha]] \equiv \alpha$

With the help of (25), (22) ultimately reduces to:

$$(26) \quad (\exists x)[U(x) \wedge T(\wedge j, \wedge \lambda z. F_*(z, x))]$$

The same kind of spurious intensionality is built into the variants (b) and (c) of the scope-mechanism, which allow Terms to quantify into nouns and IVs, respectively. These possibilities are needed to account for readings that cannot be obtained by quantifying into sentences, as in the following examples:

(27) **John wishes to find a unicorn and eat it**

(28) **Every man such that a woman loves him such that he loves her loves a woman**

(27) is the example adduced in *PTQ* [p. 240] to motivate rule T16 [p. 234] that translates $F_{10,n}(T, P)$ as applying to a Term T and a predicate P , which themselves respectively translate as Θ and α :

$$(29) \quad \lambda y. \Theta(\wedge \lambda x_n. \alpha(y))$$

Using (29), the indefinite **a unicorn** in (27) can receive a non-specific reading and at the same time bind the pronoun **it** by taking scope over the embedded infinitive. – (28) illustrates how quantifier raising to the noun level allows for anaphora across 'stacked' relative clauses.³⁴ We leave it to the reader to figure out the details.

The flexibility concerning the landing sites is but one of a number of major differences between the *PTQ* construal of quantifier scope and later, more sophisticated approaches. Most importantly, it is completely unrestricted and thus allows for numerous unattested scope constellations such as (30), one *PTQ* translation of which reduces to (31):

(30) **A woman such that every man loves her loves him**

$$(31) \quad (\forall x) [\text{man}'(x) \rightarrow (\exists y) [\text{woman}'(y) \wedge \text{love}'_*(x, y) \wedge \text{love}'_*(y, x)]]$$

Moreover, due to restrictions on scope constellations corresponding to what was later going to be called c-command (Reinhart, 1983), *PTQ* has no way of capturing the apparent covariation between indefinites and pronouns in so-called donkey sentences like:

³⁴ No justification for raising Terms above nouns is given in *PTQ*. (28) is inspired by Partee's (1975: 236) **Every man who has lost a pen who does not find it will walk slowly**, which however happens to receive the intended reading by **such that** circumlocution if the second relative clause is attached to the noun **pen**.

(32) Every farmer who owns a donkey beats it

Though the problem had been known since Geach (1962: 117ff., 128f), Montague apparently made no attempts at solving it; indeed, legend has it that he refused to even recognize the relevant reading of (32) [Terence Parsons, p.c.]. As it turns out, its solution seems to require a revision of several basic assumptions of Montague Grammar, including the treatment of indefinites as existential quantifiers, as first pointed out in Lewis (1975) and later systematized in so-called *dynamic* accounts starting with Kamp (1981) and Heim (1982).

Intensionality

The above examples already offered a glimpse of how intensionality – in the usual sense of substitution resistance on the level of extension – is treated in *PTQ*: in an intensional construction, substitution objectors get marked by a cap operator in front of their usual, extensional translation. Thus, in particular, object clauses of the form **that** *S* translate as *IL*-formulae [$\hat{\varphi}$], where φ is of type *t* and translates the embedded sentence *S*. Since the cap operator merely abbreviates λ -abstraction from the invisible variable *i*, intensionality boils down to binding *i*, and substitution resistance turns out to be the result of a variable clash: if the translations $L_i(j, b)$ and $L_i(b, j)$ of **Jane loves Bill** and **Bill loves Jane** happen to have the same truth value under a given assignment, they may replace each other *salva extensione* – though not in the scope of λi , $\forall i$ or $\exists i$.

In a similar vein, tense morphology is translated into *IL*-operators that act on the proposition expressed by the whole sentence. As already mentioned in “[Compositionality: The Algebra of Constructions](#)”, this is achieved by a number of alternative subject-predicate construals that introduce temporal operators and/or negation. However, Frege’s functionality principle does not restrict or reduce intensionality to clausal embedding: any expression has an intension, which may in principle contribute to the extension of the mother constituent when combined with an intensional sister. Indeed, *PTQ* has several intensional constructions that do not operate on propositions, i.e., objects of type $\langle s, t \rangle$, to which we now turn.

Infinitival Complements

As can be gleaned from the above examples, in *PTQ*, control verbs like **try** and **want** take uninflected predicates (as opposed to clauses with PRO-subjects) as arguments and their extensions operate on the intension of the embedded infinitive. Interestingly, the ensuing complex type of control verbs has the semantic potential of differentiating infinitival embedding from propositional attitude reports. In particular, it is consistent with this approach that the relation denoted by **T** – the *IL*-constant of type $\langle \langle s, f(IV) \rangle, f(IV) \rangle$ that translates **try** – cannot be reduced to a

relation $\mathbf{T}_*$ of type $\langle\langle s, t \rangle, f(IV) \rangle$ in the sense that the following equivalence would hold (for any individual concept x and any property Q of individual concepts)³⁵:

$$(33) \quad \mathbf{T}(x, Q) = \mathbf{T}_*(x, [\wedge^{\vee} Q](x))$$

In effect – and ignoring the fact that the 'x' stands for individual concepts – (33) says that for any individual x and property Q , trying to have Q means trying for it to be the case that x has Q – i.e., if Q happens to be the property of performing a certain action A : x tries to do A means x tries to bring it about that x does A . Crucially, *PTQ* is *not* committed to (33). This is good news insofar as it has been argued that, quite generally, the implicit subjects of infinitival objects – aka PRO – receive an irreducibly *de se* reading, according to which (roughly) the subject portrays the attitude holder as his or her own self (Chierchia, 1989; Higginbotham, 2003), which in turn means, following Lewis' (1979) influential account of attitudinal self-identification, that the object of the attitude ought to be a property rather than a proposition – the property denoted by the infinitive, more likely than not.

That the *PTQ* treatment of control verbs is consistent with them irreducibly expressing *de se* attitudes does not mean that it implies it: the bad news about (33) is that it is as consistent with the *PTQ* approach as its negation. And though its absence from the list of meaning postulates might suggest that Montague had his doubts about it or even reasons to reject it, there is no indication that he did so, let alone to account for the *de se* reading of PRO.

Whether or not control verbs are interpreted as denoting propositional attitudes, infinitival embedding may be regarded as a variant of clausal embedding: with only the subject missing, the intension operated on is almost a proposition. In fact, given Lewis' (1979) account, since properties are the proper objects of mental attitudes, they may well be regarded as epistemic propositions [David Kaplan, p.c.]. On that view, then, the *PTQ* analysis of infinitival embedding confirms the *propositionalist* prejudice that intensionality is a matter of propositional attitude.³⁶

Intensional Transitive Verbs

The connection between intensionality and propositional embedding is less clear in the case of intensional transitives, which to the semantically uninitiated seem to denote relations between individuals and objects of desire (aka *intentional* objects). As already pointed out in Montague (1969: 175), though, a compositional treatment of the non-specific (in Quine's parlance: *notional*) reading of objects of intensional

³⁵ Readers can test their *IL* proficiency by verifying that, according to the correspondence established in "Translating into IL: Indirect Interpretation", (33) is just shorthand for the following $\mathcal{T}_2$ -formula: $\mathbf{T}_i(x, Q) = \mathbf{T}_*(i)(x, [\lambda i. Q_i(x)])$.

³⁶ The term 'propositionalism' originates with Forbes (2000), which marks the starting point of a still ongoing debate; cf. the contributions in Grzankowski and Montague (2018). – Lewis' account also raises the question of how to deal with clausal embedding as reporting attitudes towards properties; see Schlenker (2011) for a survey.

transitives can be obtained on the basis of Quine's (1956) proposal to paraphrase them in terms of propositional embedding, and more specifically by: (a) analyzing the paraphrase, (b) isolating the contribution the non-specific object makes to that paraphrase, and (c) identifying the remainder of the paraphrase as the contribution of the intensional verb. As a case in point, (a) the *IL*-translation (35) of Quine's paraphrase (34) is taken to carry over to (36):

(34) **try to find a unicorn**

(35) $\mathbf{T}(\hat{\lambda}x. (\exists y)[\mathbf{U}(y) \wedge \mathbf{F}_*(x, y)])$

(36) **seek a unicorn**

Next, (b) the infinitival complement may be dissected into the contributions of its immediate constituents, rendering (35) equivalent to:

(37) $\mathbf{T}(\hat{\lambda}x. \underline{[\lambda P. (\exists y)[\mathbf{U}(y) \wedge [\forall P](y)]]} (\hat{\lambda}z. \mathbf{F}_*(x, z)))$

The underlined formula marks the contribution of the indefinite **a unicorn** to the infinitival complement in (34), which however does not coincide with its contribution to the entire predicate. This becomes clear if one tries to λ -abstract from it: the following formula is *not* equivalent to (37):

(38) $[\lambda \wp. \mathbf{T}(\hat{\lambda}x. \wp (\hat{\lambda}z. \mathbf{F}_*(x, z)))](\lambda P. (\exists y)[\mathbf{U}(y) \wedge [\forall P](y)])$

As the reader is invited to verify, the main argument in (38) contains two invisible occurrences of *i*, which would be bound by the cap operator in the body of the $\lambda \wp$ -functor. To avoid this kind of variable clash, Montague frequently employed redundant reformulations in terms of Down-Up-Cancellation (25) – which also helps in the case at hand, turning (37) into:

(39) $\mathbf{T}(\hat{\lambda}x. \forall [\underline{\lambda P. (\exists y)[\mathbf{U}(y) \wedge [\forall P](y)]]] (\hat{\lambda}z. \mathbf{F}_*(x, z)))$

Now the underlined functor of type $\langle s, f(T) \rangle$, which refers to the *intension* of **a unicorn**, can be λ -abstracted, thereby isolating its contribution to the extension of the predicate **try to find a unicorn**:

(40) $\underline{[\lambda \wp. \mathbf{T}(\hat{\lambda}x. [\forall \wp](\hat{\lambda}z. \mathbf{F}_*(x, z)))]}(\lambda P. (\exists y)[\mathbf{U}(y) \wedge [\forall P](y)])$

Since, by assumption, (40) also represents the extension of **seek a unicorn**, (c) the (underlined) main functor may be taken to be the contribution made by the intensional verb **seek**; and the construction of feeding it its (non-specific) object is interpreted by (intensional) functional application. Closer inspection of (40) reveals that the type of the underlined functor is $\langle \langle s, f(T) \rangle, f(IV) \rangle$, which thus features as the worst case for transitive verbs. In the end, then, **seek** gets translated by a constant **S** of that type, which is postulated to be equivalent to the functor in (40)³⁷:

³⁷ Actually, (41) merges *PTQ*'s transparency postulate for **find** (p. 235, (4)) and the reduction postulate (*ibid.*, (9)); the latter is formulated as a bi-conditional rather than an equation. –

$$(41) \quad S = \lambda \wp . T(\wedge \lambda x. [\vee \wp](\wedge \lambda z. F_*(x, z)))$$

In a nutshell, then, the *PTQ* treatment of **seek** is a compositional reformulation of Quine's (1956) paraphrase in terms of possible worlds semantics. Montague would not have put it that way, though, pointing out that his account of intensional transitives does not necessitate their reducibility by propositionalist paraphrase. However, as in the above case of control verbs, even in the absence of reduction postulates like (41), Montague's analysis of intensional transitives does not imply their irreducibility: contrary to Montague's (1969: 177) suggestions, an analysis without reduction is not an analysis of irreducibility but merely incomplete. Moreover, as Larson (2002) argued, given Montague's abstract analysis, the fact that clear cases of irreducibly higher-order verbs seem hard to find appears somewhat mysterious.³⁸

Due to the non-specific reading of its object, **seek** represents the worst case of a transitive verb. On the specific reading though, it behaves like an ordinary, extensional verb. In *PTQ* this ambiguity is captured by the general scope mechanism $F_{10,n}$: as in the paraphrase (1b) above, the object **a unicorn** can be raised above the verb (and its referential subject), leaving a trace of type $\langle s, e \rangle$. Hence, using the notation introduced in (7) for ordinary transitive verbs like **find**, the specific reading of (42) comes out as in (43), which in the presence of (41), is indeed equivalent to the reduced translation (44) [$\equiv$ (26)] of (1b):

(42) **John seeks a unicorn.**

(43) $(\exists y)[U(y) \wedge S_*(\mathbf{j}, y)]$

(44) $(\exists y)[U(y) \wedge T(\wedge \mathbf{j}, \wedge \lambda x. F_*(x, y))]$

Readings like (26) and (44), which are derived by raising a quantifying Term out of an intensional position (or, from a top-down perspective, *quantifying it into* that position), raise a vexing problem concerning the truth conditions. Due to the asymmetry of variables and constants in *IL*, the *PTQ* treatment of such constellations invariably predicts that the embedded proposition or property is *about the object* (= Latin *de re*) that the bound variable denotes, independently of how that object is described. When spelt out in possible worlds semantics and using Kripke's (1972) familiar term, this means that the variable acts as a *rigid designator*.³⁹ This

According to Montague, the higher-order type $\langle s, f(T) \rangle$ corresponding to the object position of intensional transitives can also be found in the subject position of raising verbs like **appear**. Generalizing to the worst case, then, all verbs end up as denoting operators acting on the intensions of their quantifying subjects. Apparently for reasons of presentation, this complication of the type assignment was avoided in *PTQ* [cf. p. 222], though not in Montague (1970a).

³⁸ Another problem with the *PTQ* type of intensional transitives was noticed in Zimmermann (1993): given its generality, non-specific readings are expected to also occur with non-existential quantifiers – contrary to fact. See Schwarz (2021) for a survey of alternative analyses.

³⁹ One may have hoped that this descriptive content is taken care of since the variable ranges over individual concepts. However, for one thing, the sole function of that type in *PTQ* is to deal with the data discussed in "The Temperature Paradox"; and for another thing, the flexibility needed to cover *de re* constellations exceeds that provided by locally bound individual concepts.

assumption has been argued to be inadequate for various reasons; see, e.g., Kaplan (1968: 192ff.) or Lewis (1981). As it turns out, more adequate truth conditions for *de re* reports require a number of restrictions on the descriptive content contributed by the bound variable; see, e.g., Kaplan (1968: 197ff.), Lewis (1979: 538ff.), Aloni (2001: ch. 2), and many others. As a result, post-Montagovian semanticists have tried to implement these conditions in the semantics of *de re* reports, which has proven to be quite tricky; see, e.g., Cresswell and von Stechow (1982), Percus and Sauerland (2003), or Maier (2009).

Adverbial Modification

In *PTQ* adverbs like **slowly** are in the syntactic category (IV/IV) of (potentially intensional) modifiers of verb phrases, which they share with PPs like **in London**. The latter are interpreted by having the preposition operate on the intension of its Term argument. Rather than going into the details of this approach to adverbial modification, which was further elaborated by Thomason and Stalnaker (1973), it is worth pointing out some of its shortcomings that eventually led to various proposals to seriously revise the framework of Montague Grammar by invoking event arguments, in the spirit of Davidson (1967) and Parsons (1990).

Since attitude predicates, including intensional transitives and control verbs, are supposed to have extensional subject positions, prepositions happen to be the only expressions in the *PTQ* fragment that may reveal double intensionality, a case in point being **about**. This seems well motivated. For one thing, the predicates in (45) and (46) are likely to differ in extension even though the italicized (non-specifically construed) indefinites do not:

(45) **talk about a unicorn**

(46) **talk about a centaur**

For another thing, even if the talkers and the thinkers are the same persons, those who think about a unicorn do not have to talk about a unicorn (again in the non-specific sense):

(47) **talk about a unicorn**

(48) **think about a unicorn**

As in previous cases, the intensionality of prepositions is only treated by omission: the absence of any transparency postulate for **about** blocks any extensional substitution. And again, neither does this treatment imply intensionality nor does it explain or describe just how the intensions of the arguments (Term and predicate) manage to determine the resulting extensions of the predicates in (45)–(48).

Moreover, the *PTQ* analysis of PPs as predicate modifiers encounters serious problems when it comes to apparently extensional prepositions like **in** (in its local

reading), which, unlike **about**, is subject to a transparency postulate to the effect that its Term argument outscopes the VP argument, in analogy to (7) and (10)⁴⁰:

$$(49) \quad \mathbf{I}_* := [\lambda y. \lambda Q. \mathbf{I}(Q, y^*)]$$

$$(50) \quad \mathbf{I} = [\lambda \wp . \lambda Q. [\lambda y. \mathbf{I}_*(Q, y)]]$$

To see why (50) cannot be the full story about the local preposition **in**, one may consider (51) whose *in situ* construal comes out as (52) (assuming a transparency postulate for **meet**):

(51) **John meets a woman in Paris.**

$$(52) \quad \mathbf{I}_*(\lambda x. (\exists y)[\mathbf{W}(y) \wedge \mathbf{M}_*(x, y)], \mathbf{p}^*)(\mathbf{j})$$

(52) roughly says that John's meeting of some (non-specific) woman took place in Paris, but does not imply that John did meet a (specific) woman in Paris; this is easily repaired by the following, apparently harmless veridicality postulate:

$$(53) \quad \mathbf{I}_*(Q, p)(x) \rightarrow [\lambda Q](x)$$

According to (53), what is done in a specific place (by a specific individual), is done *simpliciter* (by that individual). Given (53), (52) does imply that John meets a woman and that his meeting of some woman takes place in Paris; but this does not mean that there is a woman he meets in Paris. In other words, (52) does not imply (54); in fact, (52) is consistent with the negation of (54) even in the presence of (53):

$$(54) \quad (\exists y)[\mathbf{W}(y) \wedge \mathbf{I}_*(\lambda x. \mathbf{M}_*(x, y), \mathbf{p})(\mathbf{j})]$$

The problem is that the local adverbial **in Paris** only seems to target the lexical verb **meet**, not its (existentially) quantifying object; and, as readers are invited to verify, this observation is independent of the kind of quantifier in object position. Again, it would seem that this gap could be closed, too; in fact, the following scope principle seems to do so:

$$(55) \quad \mathbf{I}_*(\lambda x. [\lambda \wp . (\lambda y. [\lambda R](x, y), p)))(x) = [\lambda \wp . (\lambda y. \mathbf{I}_*(\lambda x. R(x, y), p)(x))]$$

Roughly, (55) says that a quantificational object of a transparent verb takes scope over a referential local PP that modifies it. As it turns out, however, (53) and (55) together imply (56), which says that predicate modifiers like **in Paris** are redundant – clearly an unwelcome result⁴¹:

$$(56) \quad \mathbf{I}_*(Q, p)(x) = [\lambda Q](x)$$

The only known safe way to avoid this embarrassment is to interpret adverbial modifiers as predicates on event arguments introduced by lexical verbs.

⁴⁰ For reasons unknown to the author, the relevant *PTQ* postulate (8) (on p. 235), which is again formulated in terms of a bi-conditional and does not make use of the $_*$ -notation, does not reduce the predicate argument to a *set* but only to a *property*, albeit one of individuals. For simplicity, it has been left untouched in (50).

⁴¹ See Zimmermann (1987), where Engesser (1980) is credited with the basic observations on adverbial modification in *PTQ* and (56) is proved for **slowly** in lieu of **in Paris**.

The Temperature Paradox

The intensional constructions discussed so far have in common that they all operate on *predicative* (or *Boolean*, or *conjoinable*) types, i.e. types $\langle a_1, \dots, \langle a_n, t \rangle \dots \rangle$ that end in a t and may thus be construed as hosting relations between objects of (not necessarily predicative) types $a_1, \dots, a_n$. One may wonder whether intensionality is more general than that and also occurs with arguments of non-predicative types, i.e. those that end in e (given that all Montagovian Types end in either e or t). According to Montague, this is indeed so: in *PTQ*, [p. 239], he describes intensionality in first-order predicates, i.e. those that may combine with referential expressions, as ‘a kind of intensionality not previously observed by philosophers’, crediting Barbara Partee for its discovery.⁴² As already indicated, it is this type of intensionality that is responsible for the worst-case analyses of nouns and predicates as denoting sets of individual concepts rather than merely individuals. The following type of example (taken from *PTQ*, p. 239) motivates this decision:

(57) **The temperature is ninety.**

(58) **The temperature rises.**

(59) **Ninety rises.**

Like all Terms in *PTQ*, the definite descriptions in (57) and (58) are treated as generalized quantifiers and thus give rise to truth conditions à la Russell (1905). Moreover, following Quine (1960: 118), Montague decided to interpret the copula verb **to be** uniformly as denoting identity. If, in addition, the extensions of predicates and nouns were sets of individuals, (57) and (58) would jointly entail (59). To escape this absurdity, Montague proposed to interpret the noun **temperature** as denoting a set of *individual concepts* – functions whose (numerical) values change across points of evaluation, thus in particular reflecting fluctuations over time. By the same token, the verb **rise** would have only certain such functions in its extension⁴³ – but crucially no constant functions like the one denoted by **ninety**. As a consequence, (58) may be true while (59) comes out false. In order for (57) to also be true under the same circumstances, it would not express identity between individual concepts, but rather identity between their values (at a given point). The following *IL*-translation of the copula achieves this [*PTQ*, p. 233: T1. (b)]:

(60) $\lambda \wp . \lambda x. [\forall \wp](\wedge \lambda y. [\forall x] = [\forall y])$

⁴² Partee (2004: 17) reminisces: ‘I first gave this example [i.e., (57)–(59), TEZ] [...] to David Lewis [...] He had stated (Lewis, 1970) that intransitive verbs are never intensional [...] So I [...] came up with examples about the price of milk rising or changing and wondered why they didn’t count as intensional. And David told them to Montague [...]’. The example may have had an influence on Montague’s decision to give up the Russellian approach to types advocated in Montague (1970b) in favor of a Fregean system – which would however have been premature in view of later equivalence proofs: see Kaplan (1975), Muskens (1989), and Liefke (2015).

⁴³ To wit, those whose value at a given point of reference increases at that point. Moreover, the place, which is not part of the point of evaluation, needs to be fixed to make the point about (57)–(59). Such obvious details have been omitted from the *PTQ* analysis.

(60) generalizes the binary relation of value equivalence (between individual concepts) to the worst-case type of **seek**; in this way, the copula can be treated as an extensional transitive verb (for better or for worse). Given this construal and the Russellian approach to definite descriptions, (57)–(59) come out as follows:

$$(61) \quad (\iota y. \mathbf{T}(y)) [\vee y] = \mathbf{n}$$

$$(62) \quad (\iota y. \mathbf{T}(y)) \mathbf{R}(y)$$

$$(63) \quad \mathbf{R}(\wedge \mathbf{n})$$

In the interest of readability, (61) and (62) make use of a common abbreviatory convention not found in *PTQ*:

$$(64) \quad (\iota y : \varphi) \psi := (\exists y)[(\forall x)[[\lambda y. \varphi](x) \leftrightarrow x = y] \wedge \psi]$$

(64) expresses the Russellian truth condition that there is precisely one individual concept y that satisfies φ and that every (or equivalently: some) y satisfying φ also satisfies ψ .

In (61) and (63), $\mathbf{n}$ is a constant of type e , whose extension is the value 90 at any point of evaluation; as a consequence, $[\wedge \mathbf{n}]$ denotes a constant function, thus falsifying (63) throughout any decent model. However, for (61) and (62) to be true at one point, the (unique) temperature curve only needs to take the value 90 at a point and still increase. Such models, then, provide counter-examples to the alleged inference (57)–(59) if construed along the lines of (61)–(63).

As repeatedly pointed out above, in *PTQ* individual concepts are required for the extensions of a few nouns (viz. **temperature** and **price**) as well as the verb **rise** which display the inference behavior observed in (57)–(59). For all 'ordinary' nouns and verbs, sets of individuals would have sufficed as extensions. However, due to Montague's strategy of generalizing to the worst case, the latter receive the extension type $\langle\langle s, e \rangle, t \rangle$, too. Interestingly, though, extensional nouns and verbs do not come out alike in *PTQ*. First of all, a closer look at the interpretation of the definite description **the temperature** reveals that the Russellian uniqueness condition concerns individual concepts, not individuals: the noun needs to have a singleton extension whose member must be in the extension of **rise**, which in turn only contains individual concepts (by the very idea of the *PTQ* approach to the puzzle). For 'ordinary' nouns like **woman** that get lifted from (extension) type $\langle e, t \rangle$ to $\langle\langle s, e \rangle, t \rangle$, this means that each individual in the original extension needs to be represented by precisely one individual concept in the type-lifted version; otherwise the uniqueness condition would not carry over. To achieve this, each individual u gets represented by the constant (or rigid) function that assigns u to any point of evaluation; thence the following schematic meaning postulate (*PTQ*, p. 235, (2)):

$$(65) \quad [\delta(x) \rightarrow (\exists u) x = \wedge u],$$

where δ translates an 'ordinary' noun and x and u are variables of types $\langle s, e \rangle$ and e , respectively. So sets of individuals get represented by corresponding sets of rigid intensions. However, this representation does not work for 'ordinary' intransitive verbs like **sleep**, which also need to be type-adjusted from $\langle e, t \rangle$ to $\langle\langle s, e \rangle, t \rangle$. For

otherwise prices and temperatures could never have mundane properties, like being even or, indeed, identical to 90. As a consequence, the lifted predicate extensions need to contain all individual concepts whose value at a given point is in the original set of individuals. Again, this feature is captured by a postulate schema⁴⁴:

$$(66) \quad [\forall x = \forall y \rightarrow [\delta(x) \leftrightarrow \delta(y)]],$$

where δ translates an ‘ordinary’ verb and x and y are variables of type $\langle s, e \rangle$. In particular, then, in *PTQ* two distinct roads from sets of individuals to sets of individual concepts co-exist: ‘strategies of intensionalization’ (in van Benthem’s, 1988 words) need not be uniform – *pace* de Groote and Kanazawa (2013). Still, as pointed out in *PTQ* (p. 236), the original set δ_* of individuals can be uniformly retrieved from an extension δ , no matter whether it satisfies (65) or (66):

$$(67) \quad \delta_* = \lambda u. \delta(\hat{u})$$

A further strange feature of the *PTQ* solution to Partee’s temperature puzzle had apparently escaped its author: the *double intensionality* of nouns like **temperature** – both their intensions and the elements of their extensions abstract from the Logical Space of evaluation points – leads to the unwelcome consequence that the inference from (68) and (69) to (70) is not predicted to be valid, as argued by Anil Gupta and reported in Dowty et al. (1981: 284f.):

(68) **Necessarily the temperature is the price.**

(69) **The temperature rises.**

(70) **The price rises.**

The source of this gap has been located in the intensional overkill indicated (Hansen, 2016): since the temperatures and price values are already fully recorded within the extension of the nouns, their intension should not vary with the points of evaluation but rather needs to be rigid, thus in effect dislocating the original variation of the extension from the entire set to its individual members. The shift is possible because of the *functionality* of the noun in question: at any given point there is precisely one (relevant) temperature or price; on the other hand, in view of a number of related phenomena, it seems unlikely that it is lexically based, as the *PTQ* analysis would have it (Janssen, 1984; Schwager, 2007).

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⁴⁴ In *PTQ* (p. 235, (3)), a slightly different but equivalent postulate is given: $(\exists M) (\forall x) \Box [\delta(x) \leftrightarrow [\forall M](\forall x)]$.

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Chapter 19

On Partee's "Noun Phrase Interpretation and Type-Shifting Principles"



Yoad Winter

Abstract Montague's classic article "The Proper Treatment of Quantification in Ordinary English" (PTQ, 1973) treated all NP occurrences as quantificational. Partee's article "Noun Phrase Interpretation and Type-Shifting Principles" (1987) reconciles PTQ's uniform quantificational strategy with the older distinction between three NP types: entities, predicates and quantifiers. On top of this distinction, Partee introduces operators that allow shifting the denotation of an NP to a different type than the one it is initially assigned. Using these type-shifters, one and the same NP may receive each of the three interpretations. In addition to this synthesis of previous approaches, Partee's article contains a rather elaborate analysis of predicative NPs, as well as insightful hints about the treatment of definite NPs, nominalization phenomena, plural, mass and generic NPs, and the mathematical principles underlying type-shifting. At a more global level, Partee's article marks a methodological transition in formal semantics, highlighting general principles that are relevant to different languages and to different linguistic frameworks, rather than technicalities of artificial language fragments. This general account and the new ways it opened for semantic theory, together with the paper's lucid and friendly style, have made "Noun Phrase Interpretation and Type-Shifting Principles" one of the modern classics in formal semantics. After some necessary background on NPs in PTQ, this review covers the main innovations in Partee's article, and comments on the work and its influence.

Keywords Type shifting · Noun phrase · Quantification · Predication

Different usages of noun phrases appear to involve different semantic objects. In (1a) below the NP seems to refer to a concrete entity, in (1b) the NP looks like a unary predicate applying to an entity, while in (1c) the NP functions like a quantifier.

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- (1) a. Look at *the trumpeter* over there!
 b. Mary is *a trumpeter*.
 c. *Every trumpeter* likes Schoenberg.

In a notable step towards unifying the three usages of NPs, Montague's classic article (Montague, 1973, henceforth *PTQ*) avoided this traditional distinction, treating all NP occurrences as quantificational.¹ Partee's article *Noun Phrase Interpretation and Type-Shifting Principles* (1987/2004) reconciles PTQ's uniform quantificational strategy with the older distinction between three NP types.² This three way distinction between NPs is encoded using the following semantic types:³

e	entities
et	one-place predicates, or functions from entities to truth-values
$(et)t$	quantifiers, or functions from one-place predicates to truth-values

On top of this distinction, Partee introduces operators that allow shifting the denotation of an NP to a different type than the one it is initially assigned. Using these *type-shifters* (TSs), one and the same NP may receive entity-referring, predicative and quantificational interpretations. In this way, Partee reintroduces the traditional distinction between NPs while retaining Montague's quantificational treatment. In addition to this synthesis of previous approaches, Partee's article contains a rather elaborate analysis of predicative NPs as in (1b), as well as insightful hints about the treatment of definite NPs, nominalization phenomena, plural, mass and generic NPs, and the mathematical principles underlying type-shifting.

At a more general level, Partee's article marks a methodological transition in formal semantics. Following the example of PTQ, many early works in formal semantics focused on language fragments, which demonstrated the application of the theory to selected linguistic facts, usually about English. This focus on small, well-defined fragments of English often obscured the wider linguistic relevance of semantic theory. In contrast, Partee's article concentrates on general principles that are relevant to different languages and to different linguistic frameworks, and it is constructively removed from irrelevant technicalities of specific fragments. This general account and the new ways it opened for semantic theory, together with the paper's lucid and friendly style, have made *Noun Phrase Interpretation and Type-Shifting Principles* one of the modern classics in formal semantics.

This review is structured as follows. Section “Quantificational and Predicative NPs in PTQ” gives necessary background on quantifier denotations and predicative NPs in PTQ. Section “Partee's Type-Shifting Strategy” reviews the main proposals

¹ See Chap. 18. Barwise and Cooper (1981) is another classic article in semantic theory, which follows PTQ in analyzing all NPs as quantifiers (see Chap. 3).

² All page references in the present article are to the reprinted version (Partee, 2004).

³ Here and henceforth I use short notation for extensional types, where e and t are the types for *entities* and *truth-values* respectively, and $(\tau\sigma)$ is the type of functions from objects of type τ to objects of type σ . Suppressing outermost parentheses, we abbreviate this type as $\tau\sigma$. Partee uses the $\langle\tau, \sigma\rangle$ notation for types of this form, hence, for instance, her notation $\langle\langle e, t \rangle, t\rangle$ for the quantifier type that I here denote $(et)t$.

in Partee's article, and sections "Critical Comments" and "Influence on Further Work" comment on the work and its influence.

Quantificational and Predicative NPs in PTQ

PTQ treats all NPs as quantifiers, i.e. functions from predicates to truth-values. This immediately accounts for the interpretation of NPs in subject positions. For example, in an extensional framework, the noun phrase *a trumpeter* denotes the following function:⁴

$$(2) \lambda P_{et}. \exists x_e. \mathbf{trumpeter}(x) \wedge P(x)$$

In words, this is the function mapping a predicate P to *true* if and only if P holds of some trumpeter. For example, in the analysis (3b) of sentence (3a) below, the function (2) applies to the predicate **smile**. By simplifying the analysis, we get the intuitive result in (3c):⁵

- (3) a. A trumpeter smiled.
 b. $(\lambda P_{et}. \exists x_e. \mathbf{trumpeter}(x) \wedge P(x))(\mathbf{smile}_{et})$
 c. $\exists x_e. \mathbf{trumpeter}(x) \wedge \mathbf{smile}(x)$

The quantificational analysis is applicable to all NPs in argument position. Notably, even proper names, which intuitively denote simple entities, can be analyzed using quantificational operators of type $(et)t$. For example, the extensional version of PTQ uses the following quantifier as the denotation for the proper name *Berg*, relying on the entity **berg**:

$$(4) \lambda P_{et}. P(\mathbf{berg})$$

In words, this is the $(et)t$ function that sends a predicate P to *true* if and only if P holds of the entity **berg**. In the sentence *Berg smiled*, this quantifier applies to the predicate **smile**. In formula:

$$(5) (\lambda P_{et}. P(\mathbf{berg}))(\mathbf{smile})$$

After simplification, we get the intuitive analysis **smile(berg)**.

Let us now consider NPs with a predicative use, as in *Mary is a trumpeter* (=1b)). To analyze such sentences, PTQ treats the copula *is* as a function that

⁴ Like Partee, I ignore the intensional aspects of the PTQ analysis.

⁵ Here and henceforth I refer by "simplification" to the β -conversion rule of the Lambda Calculus, which is used to simplify terms with function application. A λ -term $(\lambda x. \varphi)(exp)$ describes a function $\lambda x. \varphi$ applying to an expression exp . The β -reduced term $\varphi[x := exp]$, which is obtained by substituting exp for all free occurrences of x within φ , denotes the same semantic object as the original λ -term. For instance, the expression (3b) has the same denotation as (3c) since the latter is obtained by substituting the predicate **smile** for P in the term $\exists x_e. \mathbf{trumpeter}(x) \wedge P(x)$ within (3b).

takes quantifiers and turns them into predicates. Specifically, in sentence (1b) the quantifier denotation (2) of *a trumpeter* is mapped to a predicate. To see how this mapping works, let us consider that the function in (2) can isomorphically be described as *the set of all sets that contain a trumpeter*.⁶ To illustrate this fact, let us look at the following situation:

- (6) Let S be a situation with three entities: two trumpeters $\mathbf{t}_1$ and $\mathbf{t}_2$ and a non-trumpeter $\mathbf{u}$.

In this situation, the sets of entities that contain a trumpeter are the following:

- (7) $\{\mathbf{t}_1\}$ $\{\mathbf{t}_1, \mathbf{u}\}$ $\{\mathbf{t}_2\}$ $\{\mathbf{t}_2, \mathbf{u}\}$ $\{\mathbf{t}_1, \mathbf{t}_2\}$ $\{\mathbf{t}_1, \mathbf{t}_2, \mathbf{u}\}$

With the description (7) of the quantifier for *a trumpeter*, let us now look at PTQ's analysis of the predicate *be a trumpeter*. This analysis is derived in two steps:

- Step 1** Of the sets that the quantifier function describes, consider the singleton sets. In example (7), these are the singletons $\{\mathbf{t}_1\}$ and $\{\mathbf{t}_2\}$.
Step 2 Construct the predicate that holds of the entities in these singletons. In our example this is the predicate holding of $\mathbf{t}_1$ and $\mathbf{t}_2$ and nothing else.

Consequently, in the situation S , sentence (1b) is analyzed as claiming that Mary is either $\mathbf{t}_1$ or $\mathbf{t}_2$, i.e. she is one of the trumpeters in the situation. This is the desired reading.

In more general terms, the extensional version of PTQ's treatment of *be* is defined as the following function:

- (8) $BE = \lambda\wp_{(et)t}.\lambda y_e.\wp(\lambda z_e.z = y)$

In words, BE maps any quantifier $\wp$ to the predicate holding of an entity y if and only if $\wp$ holds of the function $\lambda z_e.z = y$. The latter function is the predicate holding of y and nothing else, which in set notation is described as the singleton $\{y\}$. Accordingly, BE can be described as the function mapping every quantifier $\wp$ to the predicate holding of y iff $\wp$ holds of $\{y\}$. Thus, for every entity y , if the singleton $\{y\}$ is one of the sets described by the quantifier $\wp$, we have y as one of the entities in the predicate assigned to $\wp$ (cf. Step 2).

There is an intuitive reason for Montague's choice to use the BE function in his analysis of the English copula. Since PTQ only treats singular NPs, the BE operator

⁶ The operator (2) sends a function P of type et to *true* if and only if the proposition $\exists x_e.\mathbf{trumpeter}(x) \wedge P(x)$ holds of P . Of the entities in a given set E , let $S_P \subseteq E$ denote the set of entities $\{x \in E \mid P(x)\}$ of which such a predicate P holds. Since S_P includes at least one trumpeter, we conclude that the set $S_P \cap \{x \in E \mid \mathbf{trumpeter}(x)\}$ is not empty. Formally: the function (2) is isomorphic to the following set of sets:

$$\{S_P \subseteq E \mid S_P \cap \{x \in E \mid \mathbf{trumpeter}(x)\} \neq \emptyset\}$$

This presentation of a quantifier function as a set of sets follows from a general isomorphism between type-theoretical and set-theoretical perspectives on meaning. For more on this isomorphism and its use in formal semantics, see Chap. 3 and 4 in (Winter, 2016).

only applies to denotations of NPs like *Berg*, *a trumpeter*, *every trumpeter*, etc. By only considering the singletons in such quantifiers, the *BE* function does something rather basic: it recovers the predicate denotation of the noun, at least in those cases where the copular sentence is clearly acceptable.⁷

To see how this analysis works in lambda notation, let us consider what happens when we apply the *BE* operator to the quantifier in (2):

(9) *BE* ($\lambda P_{et}.\exists x_e.\mathbf{trumpeter}(x) \wedge P(x)$)

This expression can be simplified as follows:

$$\begin{aligned}
 &= (\lambda\phi.\lambda y.\phi(\lambda z.z=y))(\lambda P.\exists x.\mathbf{trumpeter}(x) \wedge P(x)) &> \text{by definition of } BE \\
 &= \lambda y.(\lambda P.\exists x.\mathbf{trumpeter}(x) \wedge P(x))(\lambda z.z=y) &> \text{simplification} \\
 &= \lambda y.\exists x.\mathbf{trumpeter}(x) \wedge (\lambda z.z=y)(x) &> \text{simplification} \\
 &= \lambda y.\exists x.\mathbf{trumpeter}(x) \wedge x=y &> \text{simplification} \\
 &= \lambda y.\mathbf{trumpeter}(y) &> \text{because for every predicate } P' \text{ and entity } y: \\
 &\quad \exists x.P'(x) \wedge x=y \text{ is equivalent to } P'(y)
 \end{aligned}$$

This sequence of simplifications readily illustrates the complexity of applying the *BE* function to quantifiers. However, in the case of existential quantifiers like the denotation of the noun phrase *a trumpeter*, it leads to a simple result: the expression *be a trumpeter* ends up denoting the same *et* predicate as the noun *trumpeter*.⁸ This happens because, in any situation, similarly to our example of situation *S* above, the singletons $\{y\}$ within the existential quantifier for *a trumpeter* are precisely the singletons $\{y\}$ where *y* is a trumpeter.

Partee's Type-Shifting Strategy

Although Partee (1987) embraces the main foundational assumptions of PTQ, in one respect her proposal is substantially different from Montague's. In PTQ all NPs are treated as quantificational, and quantificational only. Thus, if predicates or entities come in handy for analyzing certain NP occurrences, this can only happen in PTQ by virtue of the meaning of the NP's syntactic environment, and not due to the NP meaning itself. For instance, the PTQ analysis in (9) treats the expression *be a trumpeter* as denoting a predicate. This is due to the analysis of the English copula *be*, but the "predicative" NP receives the same quantificational treatment of NPs in argument positions.

In contrast to PTQ, Partee lets NPs have entity and predicate denotations on top of their quantificational analysis. These different readings are derived using operators

⁷ On the vexing question of whether universal quantifier denotations of NPs like *every trumpeter* can ever appear as arguments of *BE* in predicate constructions, see Sect. "Critical Comments".

⁸ Formally: $\lambda y.\mathbf{trumpeter}(y)$ describes the same function as $\mathbf{trumpeter}$. In the Lambda Calculus this fact is generally captured using the η -conversion.

that Partee refers to as *type-shifting principles*—here, in short, *type-shifters* (TSs). For instance, in agreement with traditional accounts, Partee assumes that proper names initially denote entities. Thus, a sentence like *Berg smiled* immediately receives the intuitive analysis **smile(berg)**, without PTQ’s intermediate step in (5). At the same time, following the proposal in (Partee and Rooth, 1983), Partee allows all entity-denoting NPs to be shifted to quantifiers using the following operator:

$$(10) \text{LIFT} = \lambda x_e. \lambda P_{et}. P(x)$$

This function sends every entity x to the quantifier that holds of all predicates holding of x . For instance, for the entity **berg** we have:

$$(11) \text{LIFT}(\mathbf{berg}) = (\lambda x_e. \lambda P_{et}. P(x))(\mathbf{berg}) = \lambda P_{et}. P(\mathbf{berg})$$

This is the same quantifier that the extensional PTQ analysis in (4) assumes as the lexical denotation of the proper name *Berg*. In this way, all proper names in Partee’s proposal are potentially ambiguous between entities and the quantifiers they denote in PTQ.

Type-shifting is also used in Partee’s analysis of predicative NPs. Following Williams (1983), Partee proposes that in *be* constructions like (1b), quantificational NPs are shifted to predicates. To do that, Partee still uses Montague’s *BE* function, though not as the meaning of the English copula. Following Williams, Partee (2004, p. 212) assumes that the copula *be* requires the expression following it to be of type *et*. This triggers the application of *BE* in copular sentences like (1b). Thus, Partee employs Montague’s *BE* as a TS that applies to the NP’s quantificational meaning and turns it into a predicate. The resulting analysis is identical to the PTQ analysis (9), but it is derived in a different way.

To summarize, we have seen two major elements in Partee’s analysis of NPs:

1. **Non-quantificational denotations:** in addition to quantifiers, NPs can also denote entities and predicates.
2. **Type-shifting:** an entity-denoting NP can be mapped to a quantifier using the type-shifter *LIFT*; a quantifier-denoting NP can be mapped to a predicate using the type-shifter *BE*.

This proposal modifies one of the central principles of Montague’s program: the matching between syntactic categories and semantic types. As a follower of the categorial tradition (Ajdukiewicz, 1935; Bar-Hillel, 1953; Lambek, 1958), Montague assumed a close link between the syntactic category of an expression and its semantic role in the sentence. In the PTQ-style grammar architecture, we are always able to predict the semantic type of an expression from its syntactic category. Officially, we refer to this principle as follows:

- (12) **Category-Type (C-T) Matching Principle:** *If we assign a syntactic category C to an expression α , then the semantic type of α is predictable from C .*

According to this principle, any occurrence of an expression of category *NP* must have the same semantic type. This rigid assignment of types to categories gives compositional semantics a pleasingly restrictive architecture. Once we have decided

on the workings of syntactic categories, we have little room to hesitate about the types of objects they denote. Specifically, according to the C-T Matching Principle, once we have decided to treat some NPs as quantificational, then all the expressions bearing the *NP* category should receive a quantificational treatment. Partee renounces this C-T matching. Instead, she allows the same NP to have denotations of different types when it appears in different syntactic positions.

Partee considers some empirical advantages of her approach over the PTQ analysis. First, as Partee (2004, p. 207) mentions, proper names are among the NPs that license singular anaphoric expressions in discourse, as in (13a) below. By contrast, singular NPs like *every trumpeter* do not easily license such discourse anaphoric relations with singular pronouns (13b).

- (13) a. Berg came in. He looked tired.
 b. Every composer came in. *He looked tired.

Following Kamp (1981) and Heim (1982), Partee assumes that only entity-denoting NPs contribute "discourse referents".⁹ Thus, only entity-denoting NPs appear as antecedents of anaphors such as "he" in (13). Proper names in Partee's system lexically denote entities. She proposes that this accounts for the acceptability of examples like (13a). Definite and indefinite NPs like *thela composer* also appear in discourse anaphora, similarly to proper names. To account for such effects, Partee's system also allows definites and indefinites to denote entities.¹⁰ By contrast, NPs like *every composer* usually fail to denote entities in Partee's system, even after type-shifting.¹¹ Partee sees this as the reason for the unacceptability of examples like (13b).¹²

Another motivation for Partee's proposal comes from predicative NPs. Montague's work only treats predicative uses of NPs in copular sentences like (1b). However, Partee points out that other constructions also require predicative meanings of NPs. For example:

- (14) Berg considers Mary *a trumpeter*.

⁹ On Kamp's article see Chap. 11.

¹⁰ Definites can be interpreted as entities using the *IOTA* type-shifter (see below). As for indefinites, (Partee, 2004, p. 216–7) tentatively proposes to replace PTQ's treatment by free variables over entities as in (Kamp, 1981) and (Heim, 1982).

¹¹ Partee's *LOWER* type-shifter (see below) maps quantifiers to entities, but it is only defined for quantifiers that have single entities as their "basis", i.e. quantifiers that can be described as the result of applying *LIFT* to some entity. This is not generally the case for NPs like *every composer*. Further, shifting *every composer* to a predicate is possible using *BE*, but that predicate cannot always be shifted further to an entity using Partee's TSs, since the available strategy (the *IOTA* type-shifter, see below) requires the predicate to hold of only one entity.

¹² In situations with only one composer Partee's system treats the universal NP *every composer* as possibly referring to that composer. Partee does not fully analyze the status of discourses like (13b) in such situations. On this problem and other limitations of Partee's treatment of universal NPs see Sect. "Critical Comments".

In (14), Berg's attitude is described as taking the noun phrase *a trumpeter* to be a predicate that holds of the entity for *Mary*. In this kind of sentence, the syntactic and the semantic treatment should explain how the verb *consider* combines with the small clause *Mary a trumpeter*.¹³ While PTQ does not analyze such sentences, Partee proposes to treat them by letting the verb *consider* select for a predicate of type *et* as one of its arguments. Partee's *BE* operator shifts the quantificational denotation of *a trumpeter* to the required *et* type in the same way as in copular sentences like (1b). As further evidence for this account, Partee (2004, p. 207) points out that indefinites like *a trumpeter* can appear in coordinations with adjectives and adjective phrases. This is illustrated by the following examples:

- (15) a. John is competent in semantics and *an authority on unicorns*.
 b. Mary considers John competent in semantics and *an authority on unicorns*.

Coordinations as in (15) cannot be accounted for using the PTQ methods. This is because adjectives are a *prima facie* case of predicates, or, as in PTQ, predicate modifiers. Such denotations—of types *et* and *(et)(et)* respectively, cannot be coordinated with quantifiers under the PTQ analysis. PTQ's treatment of coordination (which has been standard in formal semantics ever since) requires the two conjuncts to be of the same type. Thus, cases like (15) are problematic for Montague's uniform quantificational analysis of indefinite NPs. Partee's proposal easily deals with such coordinations as an instance of the predicative reading of indefinites that she derives using the *BE* operator.

Importantly, these motivations for using entities and predicates as NP denotations do not undermine the quantificational treatment. As Partee mentions, Montague's quantificational analysis is still required for NPs in conjunctions like *King John and every peasant* or *a student I know and every teacher*, where a proper name or an indefinite is conjoined with a universal NP. Since the prominent reading of universal NPs is quantificational, standard treatments of conjunction require that the other conjunct is also analyzed as a quantifier. With proper names like *King John*, this quantificational analysis is obtained using the *LIFT* operator from (Partee and Rooth, 1983). As for indefinites like *a student I know*, Partee considers two methods to derive their quantificational reading: either they initially denote quantifiers as in (2), or, as in (Kamp, 1981) and (Heim, 1982), they are treated as free variables ranging over entities, which are shifted to quantifiers (Partee, 2004, p. 216).

In addition to the *LIFT* and *BE* type-shifters, Partee introduces five other extensional operators between entities, predicates and quantifiers. Two of these TSS map predicates to quantifiers:

$$(16) A = \lambda P_{et} . \lambda B_{et} . \exists x_e . P(x) \wedge B(x)$$

$$(17) THE = \lambda P_{et} . \lambda B_{et} . \exists x_e . \forall y_e . (P(y) \leftrightarrow y = x) \wedge B(x)$$

¹³ For a review of small clauses and theoretical proposals for their syntactic treatment see Citko (2011).

The *A* operator is the same function that generates the existential quantifier analysis of indefinites in (2). However, in addition to its use as the lexical meaning of the indefinite articles *a* and *some* in English, the *A* operator is also used as a TS. Similarly, the *THE* operator maps every predicate *P* to Montague's quantificational analysis of definites. For instance, the sentence *the trumpeter smiled* is analyzed as follows:

$$\begin{aligned}
 & (THE(\mathbf{trumpeter}))(\mathbf{smile}) \\
 &= ((\lambda P_{et}.\lambda B_{et}.\exists x_e.\forall y_e.(P(y) \leftrightarrow y=x) \wedge B(x))(\mathbf{trumpeter}))(\mathbf{smile}) \\
 &\quad \triangleright \text{by definition of } THE \\
 &= (\lambda B.\exists x.\forall y.(\mathbf{trumpeter}(y) \leftrightarrow y=x) \wedge B(x))(\mathbf{smile}) \quad \triangleright \text{simplification} \\
 &= \exists x.\forall y.(\mathbf{trumpeter}(y) \leftrightarrow y=x) \wedge \mathbf{smile}(x) \quad \triangleright \text{simplification}
 \end{aligned}$$

In words: there exists a unique trumpeter *x*, and *x* smiled.

The *THE* operator implements the truth-conditional (aka. Russellian) analysis of definites. Partee also uses a TS from predicates to entities, the *IOTA* operator, which models the presuppositional (aka. Strawsonian) analysis of definites. The *IOTA* operator maps a predicate *P* to an entity *c* if *c* is the only entity of which *P* holds, otherwise it is undefined. Formally:

$$(18) \ IOTA = \lambda P_{et}.\begin{cases} c_e & \text{if } P = \lambda y_e.y=c \\ \text{undefined} & \text{otherwise} \end{cases}$$

For instance, the formula *IOTA(trumpeter)* derives an entity as the denotation of *the trumpeter* when the set of trumpeters includes only one member. Otherwise, this Strawsonian treatment leaves the denotation undefined. For a review of Russellian and Strawsonian analyses of definites in philosophy of language and formal semantics, see (Heim, 2011).

Partee recognizes some mathematical properties of the *BE*, *A* and *THE* operators, and claims that the "naturalness" of these properties helps to explain why in different languages, these functions "may be encoded either lexically or grammatically or not explicitly marked at all" (Partee, 2004, p. 204). This is an important part of her proposal, which deserves some elaboration.

Unlike English, in many languages there is no direct expression of (in)definiteness or predication similar to the English use of the words *the*, *a* and *be*.¹⁴ Consider for instance the following sentences in Polish and Hebrew:

- (19) Anna je jabłko
Anna eat apple

¹⁴ Additionally, many languages only optionally express concepts of (in)definiteness and predication, or express them using morphological processes rather than single words. See (Dryer, 2013a, 2013b, Stassen, 2013) for cross-linguistic overviews of these phenomena.

Polish: ‘Anna is eating the/an apple’ (Bielec, 1998, p. 270)

(20) dani ha-more le-matematika

Dani the-teacher to-Math

Hebrew: ‘Dani is the math teacher’ (Doron, 1983, p. 71)

Partee assumes that cross-linguistically, nouns and verbs are of the same predicate type as in English. Sentences as in (19) and (20) then require some additional “semantic glue”. For instance, the Polish noun *jablko* (‘apple’) in (19) must be shifted to an entity (using *IOTA*) or to a quantifier (using *THE* or *A*). In (20), a Russellian quantificational reading of the Hebrew noun phrase *ha-more le-matematika* (‘the math teacher’) can be mapped to a predicate using the *BE* operator (8). A Strawsonian entity denotation of that definite can be mapped to a predicate using Partee’s *IDENT* operator.¹⁵

With respect to *BE*, *A* and *THE*, Partee claims that the mathematical properties of these operators make them natural candidates for providing the necessary “glue”. Thus, she sees these operators as *universal type-shifters*: semantic operators that may be used in any language whenever the meanings of the words (or morphemes) in an NP are not sufficient for interpreting it. This is why, according to Partee, in any language where sentences like (19) are acceptable, they can only be interpreted as definite or indefinite, and not, say, as universal statements like “Anna is eating every apple”. Similarly, the *BE* operator is used to explain why (20) can only be interpreted as claiming that Dani is the math teacher, and not, say, as the negative statement “Dani is not the math teacher”: the *BE* operator is part of Partee’s scheme of “natural” type-shifters, whereas a putative negative TS is not.

We can summarize these ideas as follows:

- (21) **Natural Type-Shifting Principle:** *Some central semantic concepts, including (in) definiteness and predication, involve mathematical definitions that allow them to be conveyed using phonologically silent TSs rather than overt linguistic elements.*

Partee’s article suggests that a principle of this sort can be extended to other areas in semantic theory, including the following:

1. *Nominalization operators*: Following Carlson (1980) and Chierchia (1984), Partee proposes that bare plurals like *dogs* (as in *dogs bark*) and proper names like *blue* (as in *blue is my favorite color*) have entity denotations.¹⁶ These entities are derived by an operator *NOM* applying to the basic predicative denotations of the common noun *dogs* and the adjective *blue* respectively. Importantly, *NOM*, and its inverse *PRED*, are intensional TSs: they operate on property denotations of nouns and adjectives, rather than on extensional functions of type *et*.

¹⁵ The *IDENT* operator maps any entity x to the predicate describing the singleton $\{x\}$. In formula: $IDENT = \lambda x_e. \lambda y_e. y = x$.

¹⁶ On (Carlson, 1977), which covers central aspects of (Carlson, 1980), see Chap. 4.

2. *Plurals*: Following (Link, 1983), Partee assumes that plurals like *the boys* have entity denotations that represent "groups" from an entity domain having a mereological structure.¹⁷ Partee introduces two TSs that are used for plurals—*LINK*, and its inverse *DELINK*. She proposes that they regulate the semantic relations between Link's groups and quantifier denotations of plurals as in (Barwise and Cooper, 1981). In this way, Partee aims to capture ambiguities as in *the three boys ate a pizza*, where the distributive interpretation (one pizza per boy) is derived by the quantificational denotation of the NP, and the collective interpretation (one pizza for all the boys together) is derived using its group reading.
3. *Definites*: As mentioned above, Partee's account takes all definite NPs like *the king* to be ambiguous between the quantificational reading *THE(king)* and the entity reading *IOTA(king)* obtained by the *IOTA* type-shifter. The quantificational reading is responsible for sentences where definites do not give rise to presuppositions: when there is no unique king, the definite quantifier *THE(king)* simply does not hold of any predicate. By contrast, the entity reading of definites is presuppositional: when there is no unique king, the denotation *IOTA(king)* is undefined. The *IOTA* operator is another TS that Partee considers as natural, hence potentially universal.

Although these and other possible implications of the Natural Type-Shifting Principle (21) are not developed in detail in Partee's article, this principle is one of its most influential contributions. As formal semantics has become more ambitious in its cross-linguistic tenets, principles in the spirit of (21) occupy a central place in contemporary work, with the usual controversies that characterize the search for universals in cross-linguistic research (Von Stechow and Matthewson, 2008, pp. 154–156).

Figure 19.1 summarizes the operators in Partee's proposal. Except for the *THE* operator, each "raising" operator—from entities to predicates and quantifiers, and from predicates to quantifiers—has a corresponding "lowering" operator that inverts it. For instance, the *BE* operator "undoes" the effects of the *A* operator due to the following fact:

(22) For any predicate P : $BE(A(P)) = P$.

Similarly, the *IOTA* operator "undoes" *IDENT*:

(23) For any entity x : $IOTA(IDENT(x)) = x$.

Similarly, Partee introduces an invert operator to *LIFT*, which is defined as follows:

(24) $LOWER = \lambda Q_{(et)t}. \begin{cases} c & \text{if } Q = LIFT(c_e) \\ \text{undefined} & \text{otherwise} \end{cases}$

¹⁷ On Link's article, see Chap. 17.

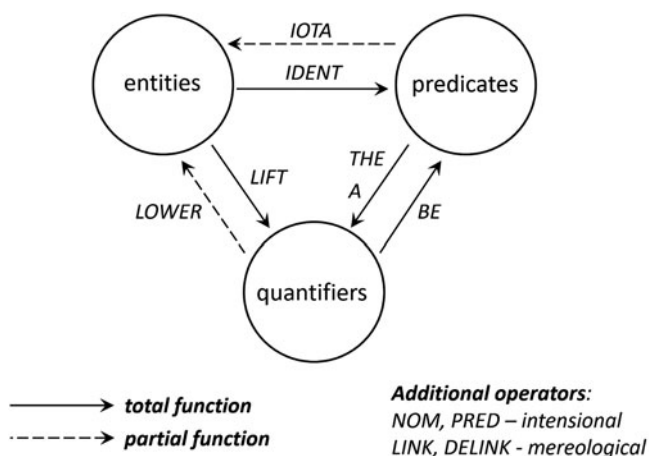


Fig. 19.1 Partee's operators

In words: the *LOWER* operator maps a quantifier Q to an entity c if Q is the lifted version of c , and is undefined otherwise. Thus, the *LOWER* operator trivially “undoes” the effects of *LIFT*.

Partee goes over these and other mathematical properties of her proposal. However, it is important to note that her work is not intended as a minimalist mathematical model of the relevant phenomena. The array of operators in Fig. 19.1 lacks the parsimony and generality that we might expect from such a model.¹⁸ The value of Partee's proposal is in its empirical implications and broader programmatic outlook, to which we now turn.

Critical Comments

Partee's paper modifies the matching between syntactic categories and semantic types, which is one of PTQ's central principles (12). Developing the type-shifting strategy of (Partee and Rooth, 1983), Partee proposes a comprehensive reorganization of semantic theory, which has ramifications for some of the most central issues involving NP interpretation: the semantic distinction between argument NPs and predicate NPs, mass and count, singular and plural, definites and indefinites, genericity, quantification, presupposition, anaphora, and nominalization. This broad ambition is presented using programmatic ideas that are not easy to test, or more

¹⁸ For instance: as mentioned above the *THE* operator cannot be inverted; the *BE* lowering is a total function, whereas other lowering operations are partial; the *NOM/PRED* and *LINK/DELINK* operators require additional ontological assumptions that are not incorporated into the quantificational domain; etc.

concrete proposals that are quickly sketched without full development. Partee is of course aware of this unavoidable limitation of her contribution, and she duly makes caveats about "the wealth of research these suggestions need to be tested against and integrated with" (Partee, 2004, p. 211). Indeed, some of Partee's proposals have come under closer scrutiny.

One of the central ideas in Partee's paper is that all NPs may in principle have predicative readings. The *BE* operator is defined for all quantifiers, and since Partee adopts Montague's idea that all NPs have quantificational readings, we might expect all NPs to also have predicate denotations and appear in predicative positions. This expectation is not borne out. For instance, as illustrated by (25) below, universal NPs are often unacceptable in predicative positions:¹⁹

(25) ?Berg is every composer.

Partee's system interprets (25) using the following predicative denotation of the noun phrase *every composer*:

(26) $BE(\text{every}(\text{composer}))$

$$= \begin{cases} \text{composer} & \text{if } \text{composer} = (\lambda x.x = c) \text{ for some entity } c \\ \lambda x.\perp & \text{otherwise} \end{cases}$$

The predicate in (26) corresponds to the singleton $\{c\}$ if there is a unique composer c , otherwise it corresponds to the empty set. Consequently, Partee's treatment expects sentence (25) to be true if Berg is the unique composer, and false otherwise. However, since sentence (25) is hardly acceptable, it is unclear if it has this interpretation. Addressing this problem, Partee (2004, pp. 214–5) claims that with certain NPs, predicative readings are ruled out for pragmatic reasons. She argues that sentences like (25) are unacceptable because a sentence like *Berg is the (only) composer* would be a more natural alternative that describes the same situations.

This pragmatic reasoning faces some problems. One problem is pointed out by Winter (2001, pp. 143–5), considering cases like the following:

(27) If Beethoven was the only pianist present at the premiere of his 4th Piano Concerto,²⁰ then

- a. *every pianist at that premiere* was among the performers.
- b. ?the soloist was *every pianist at that premiere*.

¹⁹ I am ignoring here the use of universal NPs in case like *this house has been every color*, where the only valid interpretation is "every color x is such that the house has been (colored) x ". Partee (2004, p. 219) refers to this kind of effect as the "Williams puzzle", and treats it straightforwardly by letting the quantifier *every color* take sentential scope. The resulting predication over the *entity-referring* variable "(the color) x " is independent of the issues that I mention above, which involve Partee's general mapping of *quantifiers* to predicates.

²⁰ Beethoven's notable concert of 22 December 1808 featured the public premieres of his 4th Piano Concerto (with the composer as soloist), 5th and 6th Symphonies, and Choral Fantasy.

This example shows an asymmetry between argument positions (27a) and predicate positions (27b). Sentence (27a) is somewhat scholastic but acceptable, and in the given context, it does not have any implication that there was more than one pianist at the premiere. This is not the case for the predicative NP in (27b), which is as unacceptable as (25). If the unacceptability of (25) was only pragmatic, we should have expected this unacceptability to be pragmatically suspended in (27b) in the same way it is in (27a).²¹

The syntactic-semantic restrictions on predicative NPs are another source of problems for Partee's proposal. As shown in Doron's (1983) study of Hebrew, there are important differences in distribution even among the NPs that easily appear in predicative positions. Specifically, unlike Hebrew definites as in (20), proper names cannot appear in predicative positions without a copula.²² Furthermore, Doron points out that English clauses that omit the copula (as in (15b) above), show similar distinctions to those in Hebrew. Consider for instance the following examples:

- (28) a. I considered Mary a champion/the champion/*Sue.
b. I considered Mary to be a champion/the champion/Sue.

The contrast in (28) involves a predicative environment where the *be* copula is required with proper names, but not with definite and indefinite NPs. Partee's account does not distinguish proper names from other descriptions: denotations of all three kinds of NP can be mapped to predicates. The unacceptability of *Sue* in (28a) shows that there are restrictions on the predicative readings of proper names that do not follow from Partee's paradigm. More generally, Partee's paper follows PTQ and analyzes simple English sentences of the form NP_1 is NP_2 and small clauses as in (28) as involving two quantifiers (or entities) that are hammered into the sentence using logical operators (*BE* or *IDENT*). This approach avoids the traditional distinction between "predicative" usages of *be* (*Cicero is an orator*) and "equative" usages of *be* (*Cicero is Tully*). Observations by Doron, Geist (2007), Mikkelsen (2005) and others pose serious challenges for this approach.

Another critical point about Partee's proposal concerns the central role it gives to entity-denoting NPs. Partee motivates the use of entities as NP denotations by linguistic tradition, as well as by the special status of "referential NPs" (proper names, definites and simple indefinites) with discourse anaphora as in (13). However, while we may concede Partee's claim that certain semantic operators on NPs need entities as their input, we should also note that under Partee's other assumptions this is not

²¹ Another potential problem for Partee's shifting of quantifiers to predicates is observed in (Landman, 2004, p. 30), concerning examples with predicative plurals like *the people at the meeting were at most 30 deaf composers*. While such examples may perhaps not be fully acceptable, Landman shows that any theory that would like to treat them using TSs on quantifiers would face serious complications, since *no* function can derive sensible predicative meanings from standard quantificational denotations of NPs like *at most 30 deaf composers*.

²² For instance, (i) below is ungrammatical, and does not mean "Dan is Mr. Blum":

(i) * dan adon blum
Dan Mr. Blum

conclusive motivation for introducing entities as NP denotations. Partee's *LOWER* operator allows us to emulate any necessary operation on entities by adjusting it to PTQ-like quantifiers, leaving the result undefined when the quantifier cannot be associated with a unique entity.²³ This shows a weakness of Partee's argument for entity-denoting NPs.

Lastly, a major controversy surrounds Partee's treatment of uniqueness and existential quantification as stemming from free operators like *THE*, *IOTA*, and *A*. In languages that generally allow bare NPs (cf. (19)) without (in)definite markers, various restrictions have been observed on the availability of readings that convey uniqueness ("definite" readings) or existential quantification ("indefinite" readings). This has led to an on-going debate on the ways in which languages like Polish and Russian encode uniqueness and existentiality, if at all. While proposals differ on many points, authors generally agree that there is a need to posit strong restrictions on Partee's operators in such languages, or maybe to avoid some of them altogether. Thus, some of these operators might be considered less "natural" than what Partee assumed, to the extent that they are cross-linguistically available. For relevant facts, proposals and discussion, see (Dayal, 2004; Geist, 2010; Borik, 2016; Šimík and Demian, 2020).

Influence on Further Work

Partee's paper has inspired much further research, and it would be impossible to summarize here all the important ideas in these contributions and the ways that they relate to Partee's work. Instead, the non-exhaustive list below mentions some of the areas where Partee's paper has been influential, with relevant references.

1. *Bare nominals and reference to kinds across languages*: Chierchia (1998), Cheng and Sybesma (1999), Dayal (2004), Krifka (2004), and see also Chaps. 4 and 5.
2. *Predicative NPs*: in relation to the semantics of definites (Coppock and Beaver, 2015), plurals (Winter, 2001; Landman, 2004; Champollion, 2016), *there* sentences (McNally, 1998, 2016; Landman, 2004), numerals (Kennedy, 2015), syntax-semantics interface (Zamparelli, 1995; Adger and Ramchand, 2003; Julien, 2005; Mikkelsen, 2005; Geist, 2007), or more general perspectives on type-shifting for various phenomena involving predication (Pustejovsky, 1993; de Swart, 2001).

²³ The *LOWER* operator "retrieves" an entity from PTQ's quantificational denotation of proper names and definites, as well as from Partee's quantificational treatment of indefinites (Partee, 2004, pp. 216–7). Thus, if an operation *op* is defined on entities, we can instead use an operation *OP* on quantifiers by composing *op* with *LOWER*. In formula, for any quantifier *Q*:

$$OP(Q) = op(LOWER(Q)) = \begin{cases} op(c_e) & \text{if } Q = \lambda P_{et}.P(c) \\ \text{undefined} & \text{otherwise} \end{cases}$$

3. *Predicative nominals as verb modifiers* (aka. *semantic incorporation*): Van Geenhoven (1998), Chung and Ladusaw (2003).
4. *Type-shifting, indefinites and scope*: Hendriks (1993), Reinhart (1997), Winter (1997, 2007), Kratzer (1998), Landman (2004), Charlow (2020).
5. *Unifying Montague Semantics and Dynamic Semantics*:²⁴ Groenendijk and Stokhof (1990), Chierchia (1995), Muskens (1996), de Groote (2007).

In addition to this direct influence of Partee's article, her work was also one of the first general usages of the idea of type-shifting as introduced by Partee and Rooth (1983). As such, it helped to open the way for new usages of type-shifting in linguistics. For further references and a review of work on type-shifting from a psycholinguistic perspective, see Pykkänen (2008). In relation to type-shifting in general, it should be mentioned that one of Partee's TSs, the *LIFT* operator, adopted from Partee and Rooth (1983), comes out as a straightforward result in one of the most popular frameworks in type-logical categorial grammar: the Lambek Calculus (Lambek, 1958; Moortgat, 2011). Work on *continuations* in natural language (Barker, 2001; de Groote, 2001; Barker and Shan, 2014) draws on this type-logical tradition as well as on the type-shifting analysis in Partee and Rooth (1983) and Partee (1987). Additionally, much work in Combinatory Categorical Grammar (Steedman, 1997; Jacobson, 2014) employs type-shifting principles.

Given this remarkable popularity of type-shifting as a tool in different grammatical frameworks, it is notable that there has not been much research on the theoretical foundations of Partee's conception of type-shifting and its relation with categorial grammar and continuation-based semantics. Anticipating the linguistic importance of this topic, Partee (2004, p. 224) claimed that this is one of the important areas for further research. I believe that, like many of Partee's proposals in her seminal article, this claim still holds, more than 30 years after its first publication.

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Chapter 20

On Stalnaker's "Indicative Conditionals"



Fabrizio Cariani

Abstract Stalnaker's "Indicative conditionals" is not primarily about conditionals. Its main goal is to showcase the power of a general-purpose theoretical framework and not to solve isolated puzzles in conditional semantics. The aim of the present paper is to introduce the central cogs, the main innovations and the general significance of that framework. My focus throughout is on unpacking Stalnaker's views and working through some of the unfinished agenda of "Indicative conditionals". I start out by developing the paper's central puzzle focusing on the direct argument (the inference from *she ate apples or pears* to *if she didn't eat apples, she ate pears*). Stalnaker's solution is to invalidate the direct argument, but recover its plausibility by classifying it as a reasonable inference. As part of my expansion on Stalnaker's discussion, I explore Stalnaker's definition of reasonable inference, the logical properties of the relation, and its relation to the related concept of Strawson entailment. Following this analogy, I highlight the little emphasized fact that reasonable inference is intransitive. I conclude by discussing Stalnaker's application of the apparatus of this paper to a purported argument for fatalism.

Keywords Conditionals · Entailment · Pragmatics · Selection Semantics · Transitivity

Stalnaker's *Indicative Conditionals* (1975, henceforth IC) is not *primarily* about conditionals. I would call this a "secret" if Stalnaker didn't go to some lengths to tell us himself, emphasizing that the essay's main goal is to showcase the power of a general-purpose theoretical framework and not to solve isolated puzzles in conditional semantics (IC, §VI). My aim is to introduce the central cogs, the main innovations and the general significance of that framework. Since IC does lead with a puzzle about indicative conditionals, that's what I'll do as well. Sections "[The Core](#)

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Puzzle” and “Stalnaker’s Solution” reconstruct the puzzle and Stalnaker’s solution; sections “Interlude: Indicative vs. Subjunctive Conditionals”, “On the Intransitivity of Reasonable Inference”, and “On Fatalism” develop three research themes that emerge out of IC. My focus throughout will be on unpacking Stalnaker’s views and working through some of the unfinished agenda of IC.¹

The Core Puzzle

Background

IC stages a competition between two analyses of conditional (*if...then...*) sentences. Some notation first: I use the two-place connective ‘ $\blacktriangleright$ ’ to symbolize our target—the conditional construction. The two connectives we will consider should be thought of as possible candidates for the meaning of ‘ $\blacktriangleright$ ’. I use ‘A’, ‘B’, ‘C’ as variables ranging over sentences.

The material conditional $A \supset C$ is true at world w iff either A is false at w or C is true at w . According to the *material analysis*, the material conditional is the correct semantic value for ‘ $\blacktriangleright$ ’.

The selectional conditional $A > C$ is true at world w iff C is true at the world v that is the closest A-world to w .² According to the *selectional analysis*, the selectional conditional is the correct semantic value for ‘ $\blacktriangleright$ ’ (Stalnaker, 1968; Stalnaker and Thomason, 1970). I call it ‘selectional’ because the model theory for a conditional language including ‘ $>$ ’ is often presented in terms of a “selection function” *sel* that inputs a proposition p and an evaluation-world w and outputs the selected world in p from w ’s perspective. Famously Stalnaker thinks that selection is grounded in facts about similarity between worlds. For that reason, he supplements the selectional analysis with restrictions on closeness, and so on selection functions.

The theoretical task is to incorporate one of these analyses, the material or the selectional, in a theory that accurately predicts which inferences involving conditionals are ordinarily judged valid by speakers of a language like English.

¹ For that reason, I will not survey the whole debate in which it fits. For broader surveys, see Gillies (2012, 2017).

² This demands the existence and uniqueness of a closest world. An important debate thread in conditional semantics has focused on these demands. See Lewis (1973); Stalnaker’s replies are in his (1981) and (1984, ch.7).

The Puzzle

Minimal assumptions about the logic of *if* make it equivalent to the material conditional. "Collapse", as we call it, happens if $A \blacktriangleright C$ and the corresponding material conditional $A \supset C$ are co-entailing. Using ' $\vdash$ ' to denote entailment, we can write this as follows:

COLLAPSE. $A \blacktriangleright C \vdash A \supset C$

The left-to-right direction of COLLAPSE follows from MODUS PONENS (and little else).³

MODUS PONENS. $A \blacktriangleright C, A \vdash C$

Stalnaker's focus is on the right-to-left entailment. All it takes to derive this is the plausible claim that sentences like *either Ada was guilty or Carmen was* entail sentences like *if Ada wasn't guilty, Carmen was* Stalnaker calls this *the direct argument*. In schematic form:⁴

OR-TO-IF. $A \vee C \vdash \neg A \blacktriangleright C$.⁵

To see this, suppose $A \supset C$; by truth-functional equivalence, we get $(\neg A \vee C)$; but if the direct argument is valid, $A \blacktriangleright C$ follows (assuming double-negation elimination).

Perhaps the argument for COLLAPSE is an invitation to start loving the material analysis.⁶ But, Stalnaker argues, that isn't such a great path either. If *if* were the material conditional, then negating *If Ada is not guilty, then Carmen is guilty* should entail *Ada is not guilty*. That seems wrong: one may negate $A \blacktriangleright C$ by way of putting forward some alternative conditional $A \blacktriangleright B$, without settling the status of A . For instance, one may negate *If Ada is not guilty, then Carmen is*, by putting forward *If Ada is not guilty, then Barbara, but not Carmen, is guilty*, while leaving it open whether Ada is guilty. After all, allowing for these kinds of hypothetical disagreements appears to be precisely what conditionals are for.⁷ Formally, we want:

NEGATED CONDITIONALS. $\neg(A \blacktriangleright C) \not\vdash A$

³ The literature features many interesting attempts to invalidate MODUS PONENS (McGee, 1985; Lycan, 1993; Kolodny and MacFarlane, 2010). Moreover, as Khoo (2013) makes clear, Kratzer's semantics for conditionals (Kratzer, 1991, 2012) also invalidates MODUS PONENS. These should not be dismissed, but they aren't ways of avoiding collapse results because attempts to invalidate modus ponens do not usually extend to bare conditionals. That is, they do not extend to conditionals whose antecedents and consequences are not modal sentences, like *if Ada is innocent, Carmen is guilty*. A form of collapse that is restricted to bare conditionals is no better than general collapse.

⁴ The 'or-to-if' nomenclature comes, I believe from Bennett (2003).

⁵ Adams (1975, p. 15) gives an example that purports to show OR-TO-IF to be *probabilistically* invalid. What he means by this is that there are cases in which $A \vee C$ has high probability while $\neg A \blacktriangleright C$ has low probability.

⁶ Possibly supplemented by implicature-based moves in the style of Grice (1989), Lewis (1976), Jackson (1987).

⁷ See Edgington (1995) for a comprehensive array of nails in the material conditional's coffin.

Unfortunately, the material analysis cannot fulfill that wish.

That's the puzzle: if you like MODUS PONENS, and OR-TO-IF, you will quickly get COLLAPSE. But COLLAPSE is bad, among other things because it violates the NEGATED CONDITIONALS desideratum.

A subtler, modern take on this puzzle (Gillies, 2004, 2009) accepts COLLAPSE but rejects the material analysis. According to this view, the conditional might well be co-entailing with the material conditional but co-entailment does not suffice for identity of meaning. In fact, it might not even suffice for logical equivalence—depending on how logical equivalence is understood. This view does not appear to be on IC's radar—and indeed, IC's theoretical development is a key step on the path that leads to it. For this reason, I mention it here out of duty to the fuller dialectic but I will refrain from developing it further.

The only other option is to reject the validity of OR-TO-IF. This is what happens with '>'. Of course, we didn't need the collapse result to know that OR-TO-IF fails for '>'. It is easy to sketch a counterexample. Suppose that in the actual world (w) Ada alone is guilty (A), but in the closest world in which she is innocent (v), Bo is guilty (B) and Carmen is not (C).



Then $\neg A > C$ is false at w even if $A \vee C$ is true there. The additional value of the collapse result is the implication that invalidating OR-TO-IF is required to avoid co-entailment with the material conditional.

Here, then, is IC's local puzzle: how might we invalidate OR-TO-IF while accounting for its plausibility?

Stalnaker's Solution

Stalnaker solution is that OR-TO-IF, though invalid, is pragmatically acceptable. Spelling this out is a two-part task. The first step is to explain how an inference pattern might be acceptable even if invalid. To this end, Stalnaker develops the concept of "reasonable inference". Then, one must check that instances of OR-TO-IF count as reasonable inferences.

The explicit inspiration is Grice's pragmatic defense of the material analysis (Grice, 1989, ch. 4). Like Grice, Stalnaker proposes that much of the inferential profile of the conditional comes from features that are not determined by its truth-conditions. One difference is that, while Grice primarily applies the theory of implicatures to prune off some unwanted entailments of the material conditional, Stalnaker's central concern is that the selectional conditional undergenerates the acceptable inference patterns. Moreover, the patterns it misses are not rescued by the theory of implicatures.

There are important novel elements in Stalnaker's approach. I mention three. *One*: Stalnaker's system relies on constraints tied to the use of specific lexical items in context (see below for illustration). By contrast, the theory of conversational implicatures (at least, in vintage-Grice version) only leverages general principles about rationality and cooperation between conversational participants. *Two*: the acceptability of some reasonable inferences depends on relationships between the information conveyed and the common ground. *Three*: the account appeals to pragmatic principles that are themselves formalizable. Indeed, the appendix to IC goes to some lengths to state the pragmatic theory formally.

Introducing Reasonable Inference

It is worth quoting Stalnaker's characterization of reasonable inference in full. More recent appeals to IC's playbook often rely on simplified notions that have different structural properties from the original.⁸

[A]n inference from a sequence of assertions or suppositions (the premisses) to a conclusion is a reasonable inference just in case in every context in which the premisses can be asserted or supposed, it is impossible for anyone to accept the premisses without committing himself to the conclusion. (IC, p. 271)

Let us break down the main technical concepts in this definition with an eye towards coming up with an algorithm for testing reasonableness.

Context

A Stalnakerian context is a record of those propositions that are commonly accepted by the participants to a conversation. Stalnaker's thinking on the nature of context is complex and evolves over four decades (Stalnaker, 1974, 1978, 2002, 2016). One thing that stays constant is the idea that each context *c* is associated with a set of possible worlds—the *context set* of *c*. In particular, the context set of *c* is the set of

⁸ For example, reasonable inference is sometimes lumped together with the relation of "quasi-validity" (see e.g. Kolodny and MacFarlane, 2010, fn. 39). An argument is *quasi-valid* if, whenever the premisses are known, the conclusion must also be known. Quasi-validity might function in some similar respect like reasonable inference (Nolan, 2003, p. 231), but it is significantly different in concept and mechanics. The closest ancestor to quasi-validity is the concept of "doxastic indefensibility" developed by Hintikka (1962) to account for Moore's paradox. Other nearby notions that are often associated with reasonable inference are Veltman's (1996, p. 224) *validity*₂ (aka *update-to-test*, in the terminology of van Benthem, 1996 and Gillies, 2004) and *validity*₃ (aka *test-to-test*, or also *informational consequence* in the terminology of Yalcin, 2007). Finally, the distinction between preservation of truth and preservation of acceptance also shows up in the debate surrounding McGee's counterexamples to modus ponens (see for example, McGee, 1985, p. 90).

worlds that verify all of the propositions that are common ground in c . Since the context set is the only feature of context we need to attend to, I will deliberately conflate contexts and their context sets.

Commitment and Context Acceptance

It is somewhat surprising to see Stalnaker talk about speakers' *commitments*. However, the technical appendix to IC reveals that talk of speaker's commitment is just short-hand for talk about the shared elements of context. The possible world framework Stalnaker adopts allows definitions that clarify the relevant cluster of concepts.

UNSTRUCTURED CONTENTS. $|A| = \{w \mid A \text{ is true at } w\}$

CONTEXT ACCEPTANCE. c accepts A iff $c \subseteq |A|$.⁹

CONTEXT INCOMPATIBILITY. c is incompatible with A iff $c \cap |A| = \emptyset$

A successful assertive utterance of A in c updates c to $c_A = c \cap |A|$. To say that a speaker who assertively utters A in c is committed to some other proposition p is just to say that $c_A \subseteq p$.

Assertibility

The definition of reasonable inference quantifies over contexts *in which the premises can be asserted (or supposed)*. To understand this restriction, note that in the background of IC is the idea that assertive utterances come with requirements that the context must meet—as we may call them, *presuppositions*. Some (though not all) of these requirements are specifically tied to the use of particular lexical items. One cannot assertively utter *Raz returned to Valencia* in a context that doesn't that Raz ever was in Valencia. Stalnaker's account of OR-TO-IF essentially references the constraints imposed on context by conditionals and disjunctions.

The Reasonable Inference Test

Let's put this all together. To check whether C is a reasonable inference from premises $A_1, \dots, A_n$.

- consider an arbitrary context c in which all the premises are assertible.
- let c' be the context that results from asserting all of the premises in sequence.
- the inference is reasonable if and only if c' accepts C .

⁹ In IC, context acceptance is a relation between a context and a set of worlds. For exposition purposes, I prefer to define it as a relation between contexts and sentences (but one that treats any two sentences that have the same unstructured content in the same way).

Stalnaker's Explanation at a Glance

It is useful to break down Stalnaker's account of OR-TO-IF into two modules—a general module and one for card-carrying Stalnakerians. The general module, which I introduce in this section, does not presuppose the possible-worlds framework. It only presupposes that there is some way of cashing out the relevant concepts of compatibility and acceptance, such that the following principles are satisfied:

D-CONSTRAINT. $A \vee C$ is assertible in c only if c is compatible with $\neg A \ \& \ C$ and c is compatible with $A \ \& \ \neg C$.

D-UPDATE. Successful assertive utterance of $A \vee C$ in c updates c to a context that is:

- incompatible with $\neg A \ \& \ \neg C$.
- compatible with $\neg A \ \& \ C$ if c was.

C-ACCEPTANCE. If c is incompatible with $\neg A \ \& \ \neg C$ and c is compatible with $\neg A \ \& \ C$, then c accepts $\neg A \blacktriangleright C$.

D-CONSTRAINT is justified on familiar Gricean grounds. D-UPDATE is a direct consequence of the account of assertive update I sketched in section “[Introducing Reasonable Inference](#)” according to which $c' = c \cap |A \vee C|$. As for C-ACCEPTANCE, it might be defended in a variety of ways, depending on one's semantics for conditionals. I introduce Stalnaker's defense of C-ACCEPTANCE for section “[Inside Stalnaker's Box](#)”.

For now, let us check that this is enough to classify OR-TO-IF as a reasonable inference.

1. Let c be an arbitrary context in which $A \vee C$ is assertible.
2. By D-CONSTRAINT, c must be compatible with $\neg A \ \& \ C$.
3. Let c' be the result of updating c with the assertion of $A \vee C$.
4. By D-UPDATE and 2., c' is incompatible with $\neg A \ \& \ \neg C$ but compatible with $\neg A \ \& \ C$.
5. By C-ACCEPTANCE and 4., c' must accept $A \blacktriangleright C$.

Inside Stalnaker's Box

I postponed the defense of C-ACCEPTANCE. Here it is, in Stalnaker's words:

PRAGMATIC CONSTRAINT. If the conditional is being evaluated at a world in the context set, then the world selected must, if possible, be within the context set as well. [...] In other words, all worlds within the context set are closer to each other than any worlds outside it. (IC, pp. 275–276)

The formal upshot is that selection functions must satisfy the following constraint (where *sel* denotes the selection function itself):¹⁰

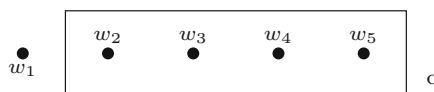
$$\forall w \in c \text{ and } z \notin c, \text{ sel}(w, A) = z \text{ only if } c \cap |A| = \emptyset$$

In turn, this constraint is justified as follows:

Normally a speaker is concerned only with possible worlds within the context set, since this set is defined as the set of possible worlds among which the speaker wishes to distinguish. So it is at least a normal expectation that the selection function should turn first to these worlds before considering counterfactual worlds—those presupposed to be non-actual. (IC, pp. 276)

Some comments in bullet-point format:

- C-ACCEPTANCE only holds by default, or in normal circumstances. This should draw our attention to those cases in which this default presumption is suspended. For instance, Stalnaker remarks that it is suspended by subjunctive conditionals, such as *if I had a house, it would not have a front-yard*. (IC, pp. 276–277).
- The pragmatic constraint breaks the visual analogy between spatial distance and modal similarity. Consider:



In this depiction, w_2 is spatially closer to w_1 than it is to w_5 . But because both belong to the context set c , w_2 and w_5 must be more similar to each other than w_2 is to w_1 . It should be immediately apparent that this is no accident of the depiction: given any depiction of logical space as a convex set, worlds in c can be chosen that are arbitrarily close (spatially) to worlds outside c . Of course, there are ways to depict the structure of logical space that is imposed by the pragmatic constraint, especially if the depiction is allowed to include points that do not correspond to worlds.

Pictorial musings aside, there is a broader theoretical point. If we try to supplement Stalnaker's proposal with a theory of context-dynamics, the pragmatic constraint will require that every (non-empty) informational update to the context must also rearrange the ordering of modal similarity.

¹⁰ Stalnaker gives a different condition—namely: $\forall w \in c, \text{ sel}(A, w) \in c$. This condition does not match the informal presentation of the constraint. However, the condition does follow from a strengthening of the pragmatic constraint which I am about to discuss.

Another thing to consider is whether this stronger condition might be weakened by appealing to accessibility relations. While IC's formalism is a notch looser than Stalnaker (1968) and Stalnaker and Thomason (1970), both these earlier works involve an accessibility relation. In a recent paper, Mandelkern (2018) proposes a role for accessibility relation that bears directly on the formal analysis of the pragmatic constraint.

Moving back to the main argumentative thread, Stalnaker uses the pragmatic constraint to support:

ANTECEDENT COMPATIBILITY. $A \blacktriangleright C$ presupposes that A is compatible with the context set (IC, p. 277).

This idea has been influential and it is sometimes simply quoted in place of the original constraint. However, it is clear that PRAGMATIC CONSTRAINT does not entail ANTECEDENT COMPATIBILITY. The former, but not the latter, is trivially satisfied if there are no antecedent worlds in the context set. IC's text strongly suggests that what fills the gap between PRAGMATIC CONSTRAINT and ANTECEDENT COMPATIBILITY is the availability of the subjunctive conditional as a tool to talk about worlds out of the context set (more on this in the next section). However, it is unclear exactly what principles are relied on in this derivation.

Be all that as it may, the pragmatic constraint is enough for the job at hand.

Fact 1 *Given the framework, PRAGMATIC CONSTRAINT entails C-ACCEPTANCE*

Proof See appendix for proof of this and other facts.

That completes the official path to a pragmatic explanation of OR-TO-IF. There are, of course, other paths that leverage other theories of the conditional. As an illustration, let ' $\rightarrow$ ' denote a strict conditional restricted to the context set. That is, say that $A \rightarrow C$ is true at w and context c iff c is incompatible with $A \ \& \ \neg C$. Then:

Fact 2 *If entailment is defined as preservation of truth at a world and a context,*

$$A \vee C \not\models \neg A \rightarrow C$$

Fact 3 *However, $\rightarrow$ satisfies C-ACCEPTANCE.*

Given Fact 3, OR-TO-IF for $\rightarrow$ can too be rescued as a reasonable inference. I don't mean to take this point very far, but it does show that the main strategy of IC is not exclusively available to defenders of the selectional analysis.

Interlude: Indicative vs. Subjunctive Conditionals

It is a remarkable feature of *Indicative Conditionals* that one of its digressions opened its own research thread. The digression in question concerns the distinction between indicative and subjunctive conditionals. There are many important contrasts between indicatives and subjunctives. The central contrast in IC's dialectic is that indicatives and subjunctives pattern differently with respect to information that is settled by the common ground. Contrast:

- (1) # Ada is guilty. If she isn't, Carmen is.
- (2) Ada is guilty. If she weren't, Carmen would be.

There is an incongruence in (1), as if the speaker is changing their mind or hedging their assertion. No such incongruence affects (2).

One might account for contrasts like this by distinguishing two kinds of conditional *connectives*. According to this view, (1) uses “the indicative conditional” and the (2) uses “the subjunctive conditional”.

Stalnaker would have us avoid this approach. According to him, we can hold on to a single meaning for *if*, while also explaining how indicative and subjunctive conditional *sentences* diverge in meaning. From a pragmatic point of view, Stalnaker thinks that subjunctives are not subject to either version of the pragmatic constraint. Interestingly, this pragmatic suggestion has a syntactic basis. The syntactic proposal is that subjunctive conditionals feature one ingredient that is not present in indicatives: subjunctive mood. Here is how a toy, IC-inspired syntax for (2) might go. Let S-MOOD be a sentential operator that captures the effect of subjunctive mood.

(3) S-MOOD(Ada is not guilty) > WOULD(Carmen is)

Many authors have since questioned the link between “subjunctive mood” and counterfactuality. The alternative view (Dudman, 1983, 1984; Lycan, 2001; Iatridou, 2000; Ippolito, 2013; Starr, 2014; Khoo, 2015) is that “subjunctivity” comes from an extra layer of past tense on the antecedent. What is special about this extra layer is that it does not appear (at least on the surface) to get a past interpretation.¹¹ As Iatridou notes, this is especially salient when “fake past tense” is restricted by future adverbials, as in:

(4) If Odysseus returned tomorrow, his dog would recognize him.

Despite this, the right perspective on the legacy of IC should be that, though it might have gotten the syntactic details wrong, it helped advance the powerful idea that subjunctive meaning ought to be constructed out of the syntactic ingredients of subjunctive conditionals.¹²

The other key component of IC’s account of the contrast is the idea that there are pragmatic differences between indicatives and subjunctives. As I mentioned, the proposal is that subjunctive conditionals are not subject to the pragmatic constraint. But how is that supposed to work, given that we have already rejected the idea that *if* is ambiguous?

Stalnaker sketches an interesting proposal:

I take it that the subjunctive mood in English and some other languages is a conventional device for indicating that presuppositions are being suspended, which means in the case of subjunctive conditional statements, that the selection function is one that may reach outside of the context set. (p. 276)

¹¹ There is a further debate, including among the authors I cited, as to whether the modal force of subjunctive conditionals might be recovered by giving that past tense an ordinary past interpretation.

¹² It should be mentioned that a critical step in this path was the application of Kratzer’s (1991; 2012) framework to conditionals.

This is how I read this: the PRAGMATIC CONSTRAINT is a presupposition introduced by *if*.¹³ However, it can be *lifted* by other devices. Stalnaker's idea seems to be that, while presuppositions are requirements on context, some phrases may *suspend* those requirements (in other words: introduce permissions). I find this idea fascinating, interestingly counter to the current orthodoxy, and deserving of careful development. As far as I know, it hasn't received one.

It is natural to see this lack of development as a significant lacuna. For instance, Starr writes:

Why [...] did Stalnaker not propose a meaning for the subjunctive mood and construe the different propositions expressed by subjunctive and indicative conditionals as a function of this mood's contribution? (Starr, 2014, p. 1031)

I think that Starr is exactly right. It is not enough to point to an extra syntactic constituent, be it mood or tense, and claim that it *might* do the job. Much of the best contemporary work on the indicative/subjunctive contrast (including that of the authors I mentioned in this section) is precisely directed at developing a compositional account. Additionally, Starr (2014, p. 1030) identifies some technical issues with Stalnaker's approach to the indicative/subjunctive contrast which are well worth working through—though space prevents me from doing that here.

On the Intransitivity of Reasonable Inference

Arguably, the most important contribution of IC is the introduction of the concept of reasonable inference. Earlier, I have noted three innovations: (i) that reasonable inference involves formalizable principles, (ii) that it is anchored by non-truth-conditional features of the meaning of particular lexical items, and (iii) that it involves relations between utterances of conditional sentences and the common ground. In light of these innovations, it is natural to view the concept of reasonable inference as a critical step on the path to dynamic notions of consequence (à la Veltman, 1996).

But it would be hasty to assimilate Stalnaker's notion of reasonable inference and the dynamic concepts of consequence. Stalnaker himself has recently emphasized some differences between the dynamic semantics program and his preferred "dynamic pragmatics" (Stalnaker, 2016). Here, I want to emphasize some under-explored structural features of reasonable inference. In particular, I will focus on the little-known-but-obvious-once-you-see-it fact that reasonable inference is intransitive.¹⁴ I argue that the intransitivity of reasonable inference might come with some explanatory benefits. However, I will also question whether Stalnaker is in a

¹³ This might need to be massaged a bit in light of the fact that Stalnaker is skeptical about the idea of semantic presupposition, (Stalnaker, 1974). But I don't see a way of reading the text of IC without linking the presupposition to this lexical item.

¹⁴ I learned this from Simon Goldstein and I have not found the issue discussed elsewhere.

position to reap those benefits, given some of the other commitments he undertakes in IC. More than anything else, my aim is to stress that there are under-explored choice-points in developing the theory of reasonable-inference.

Let us use ‘ $\approx$ ’ to symbolize the reasonable inference relation and restate its definition:

$A_1, \dots, A_n \approx C$ if and only if for all contexts c , if all of the A_i are assertible in c and c' is the result of asserting all of the A_i ’s in sequence, c' accepts C .

Here are some important facts concerning $\approx$. First, every semantically valid argument is a reasonable inference.

Fact 4 *if $\Sigma \models C$ then $\Sigma \approx C$*

Second, the inference from a sentence to its presuppositions is always reasonable.

Fact 5 *if A presupposes C , then $A \approx C$*

Given these facts, it is easy to check that $\approx$ is intransitive in the following sense.

Fact 6 (Intransitivity) *Let Σ be a set of sentences. $\Sigma \approx B$ and $B \approx C$ do not guarantee $\Sigma \approx C$*

In establishing this, it is convenient (but, as I will show shortly, not essential) to add to the formal language a possibility operator $\Diamond$ that tracks what is compatible with the context set. So, $\Diamond A$ is true at w in c iff $c \cap |A| \neq \emptyset$.

By D-CONSTRAINT a disjunction $A \vee C$ is only assertible in those contexts that are compatible with $\neg A$ & C . In particular, $A \vee \neg A$ requires that c be compatible with $\neg A$ & $\neg A$. Though I have defined compatibility as a relation between contexts and sentences, the relation treats intensionally equivalent sentences in the same way. In other words, compatibility with $\neg A$ & $\neg A$ reduces to compatibility with $\neg A$. Using our new possibility operator, we say that $A \vee \neg A$ presupposes $\Diamond \neg A$. Thus, by Fact 5,

$$A \vee \neg A \approx \Diamond \neg A$$

However, $A \vee \neg A$ is a logical truth; given Fact 4, it can be reasonably inferred from the empty set of premises (i.e. every context accepts it). By contrast, $\Diamond \neg A$ cannot be reasonably inferred from the empty set of premises. Putting it all together:

$$\begin{aligned} & \approx A \vee \neg A \\ & A \vee \neg A \approx \Diamond \neg A \\ & \not\approx \Diamond \neg A \end{aligned}$$

□

Though this fact is surprising, there is nothing intrinsically problematic about it. Reasonable inference is a technical concept that is invoked to play a certain role—to characterize the acceptable patterns of inference. There is no *a-priori* reason why that job should be best played by a transitive notion.

Still, we may ask whether intransitivity is indispensable to Stalnaker’s explanation. A useful first experiment in this direction is to compare $\approx$ with its transitive closure. Define $\approx^+$ as the smallest transitive relation that extends $\approx$. Since $\approx^+$

extends $\approx$, it classifies OR-TO-IF as acceptable. More generally, any empirical comparison between $\approx$ and $\approx^+$ must be based on inferences that are sanctioned as acceptable by $\approx^+$ but not by $\approx$.

This comparison reveals some unique advantages of $\approx$. One of the advantages of Stalnaker's theory is that it doesn't validate the paradoxes of material implication. The theory invalidates both these patterns:

FALSE ANTECEDENT. $\neg A \vdash A > C$

TRUE CONSEQUENT. $C \vdash A > C$

Not only are these patterns invalid, they are also not reasonable, in the sense of $\approx$. However, both are sanctioned as acceptable by $\approx^+$. We can see this by noting another violation of transitivity for $\approx$:

Fact 7

- (i) $\neg A : \approx \neg A \vee C$
- (ii) $\neg A \vee C : \approx A > C$ but
- (iii) $\neg A \not\approx A > C$

I give this proof in the main text because it illustrates some of the critical elements I have been drawing attention to.

(i) follows from Fact 4. (ii) is the reasonable-inference analogue of OR-TO-IF for ' $>$ '. As for (iii), let c be a context in which $\neg A$'s presuppositions are satisfied and also such that all the A -worlds are $\neg C$ -worlds. Consider some arbitrary world $w \in c_{\neg A}$. To evaluate $A > C$ at w we must reach out of $c_{\neg A}$. We do know however that all the A -worlds outside of $c_{\neg A}$ are $\neg C$ -worlds, so $sel(A, w) \notin |C|$, which must mean that $A > C$ is false at w . Note that PRAGMATIC CONSTRAINT is vacuously satisfied in $c_{\neg A}$, since there are no A -worlds to select. Furthermore, ANTECEDENT COMPATIBILITY is violated. However, neither of these is incompatible with (iii). $\square$

By contrast, because $\approx^+$ is transitive and extends $\approx$, (i) and (ii) force $\neg A : \approx^+ A > C$.

Many theorists accept the validity of FALSE ANTECEDENT and TRUE CONSEQUENT as the cost of doing business. But the intransitive Stalnakerian package undoubtedly maps onto clear intuitions. Consider:

- (5) Ada is not guilty
- (6) Either Ada is not guilty or Carmen is guilty
- (7) If Ada is guilty, Carmen is guilty

The intransitive package predicts that:

- the inference from (5) to (6) is reasonable
- the inference from (6) to (7) is also reasonable
- the inference from (5) to (7) is not reasonable.

These predictions seem just right and a system relying on intransitive notions is uniquely positioned to capture them.¹⁵ None of this is a conclusive argument for Stalnaker's intransitive package. Any such argument would require a much more systematic analysis than what I can offer here. But it does mean that the intransitive package shows up to the jury of theory-choice criteria with a unique profile.

That is all good, but there is one more twist. The official story of IC is a bit more complicated than the story I just told. We can see this by attending to Stalnaker's discussion of the contraposition pattern.

CONTRAPOSITION. $A > C \vdash \neg C > \neg A$

Like OR-TO-IF, CONTRAPOSITION sounds plausible in the indicative case but not in the subjunctive case (Stalnaker, 1968, p. 107; Lewis, 1973, §1.8). Once again, Stalnaker proposes to invalidate the general pattern, while explaining the plausible cases *via* reasonable inference. Unfortunately, this strategy doesn't work.

Fact 8 *The reasonable-inference analogue of CONTRAPOSITION fails. (i.e. $A > C \not\approx \neg C > \neg A$).*

Stalnaker suggests a patch:

Strictly, [CONTRAPOSITION] is reasonable only relative to the further assumption that the indicative conclusion is not inappropriate" (Stalnaker, 1975, note 15)

What could it mean that an inference pattern is reasonable relative to a further assumption?

One possibility is that there isn't one, monolithic notion of reasonable inference, but a family of them. In addition to $\approx$, we ought to consider what I'll call *Strawsonian Reasonable Inference*.

$A_1, \dots, A_n \approx^S C$ if and only if for all contexts c , if all of the A_i are assertible in c and c' is the result of asserting all of the A_i 's, if C is assertible in c' , then c' accepts the proposition corresponding to C .

This is just like reasonable inference, except for the underlined bit. Moreover, just like reasonable inference $\approx^S$ is intransitive (it is easy to check that the instance of intransitivity I sketched in establishing Fact 6 also works for $\approx^S$). In essence $\approx^S$ is a pragmaticized version of *Strawson Entailment*.¹⁶ Since $\approx^S$ extends $\approx$, we can update the Stalnakerian proposal to the claim the acceptable arguments are exactly the ones that are captured by $\approx^S$.

¹⁵ For some related technical work involving an intransitive notion of entailment, see Cariani and Goldstein (2020).

¹⁶ In a trivalent setting (e.g. in the context of a semantic account of presupposition), Strawson Entailment is equivalent to the claim that valid arguments are exactly those that cannot have true premises and false conclusions. (If this sounds like classical entailment, recall that in the trivalent setting the "standard" notions of consequence are preservation of truth and preservation of non-falsehood.) For discussion: Strawson (1952), Smiley (1967), Fintel von Fintel (1997), Cariani and Goldstein (2020).

So, does this Strawsonized package catch CONTRAPOSITION, as Stalnaker claims? It depends.

Fact 9 *If the presupposition of indicatives is ANTECEDENT COMPATIBILITY, then CONTRAPOSITION, TRUE CONSEQUENT and FALSE ANTECEDENT are all Strawsonian Reasonable Inferences.*

Fact 10 *If the presupposition is just the PRAGMATIC CONSTRAINT, then none of CONTRAPOSITION, TRUE CONSEQUENT and FALSE ANTECEDENT count as Strawsonian Reasonable Inferences.*

Two things are interesting here. *One*: there is a tradeoff between invalidating the paradoxes of material implication and classifying CONTRAPOSITION as acceptable. So it is not clear that Stalnaker can exploit the intransitivity of reasonable inference to avoid the paradoxes of material implication. *Two*: if we prioritize catching the CONTRAPOSITION pattern, we need the full force of ANTECEDENT COMPATIBILITY. This is unlike OR-TO-IF for which PRAGMATIC CONSTRAINT is enough.

On Fatalism

The word limit for this essay is nearing, and I must catch up with Stalnaker's application of IC's apparatus to the "idle" argument for fatalism (see also Dummett, 1964). Extra motivation here is provided by the recent publication of the contemporary gold standard on the idle argument (Bledin, 2020), and by the fact that the idle argument is closely connected with the much-discussed miners paradox.¹⁷ Consider:

- (1) Either this essay will be mistake-ridden or it will not be.
- (2) | Suppose it will be.
- (3) | If I am careful, the essay will be mistake-ridden.
- (4) | Being careful is ineffective.
- (5) | Suppose it will not be.
- (6) | If I am not careful, the essay will not be mistake-ridden.
- (7) | Being careful is unnecessary.
- (8) Being careful is either ineffective or unnecessary.

A satisfactory diagnosis of the idle argument ought to identify where it fails and explain why each step appears compelling. Here are the highlights of Stalnaker's account:

¹⁷ Kolodny and MacFarlane (2010), Charlow (2013), Cariani et al. (2013), Willer (2012), Cariani (2016).

- The steps $(2) \Rightarrow (3)$ and $(5) \Rightarrow (6)$ are invalid, since they are instances of TRUE CONSEQUENT.
- The inferences corresponding to each of these steps might be “saved” pragmatically by appealing to Strawsonized Reasonable Inference.
- The application of constructive dilemma (or “reasoning by cases”) in step (8) is not justified as a reasonable inference. That is:

$$A:\approx C, B:\approx C \text{ does not guarantee } A \vee B:\approx C,$$

- Despite this, the last step appears compelling because constructive dilemma is *valid* when the subarguments are valid.

The aggregated effect is that the premises of the argument do not entail its conclusion, nor is it possible to support the conclusion as a reasonable inference from the premises. However, the conclusion appears plausible because each step is justifiable by at least one of these relations.

This dialectic is subtle and needs some unpacking. First, we need to clarify the claims about the constructive dilemma pattern. Contrast:

- CD. if $A \vdash C$ and $B \vdash C$, $A \vee B \vdash C$,
 RCD. if $A:\approx C$ and $B:\approx C$, then $A \vee B:\approx C$

The key observation is that while CD states a true claim about the logic, RCD does not. Although every entailment is a reasonable inference, it is not the case that, for every true claim about entailment, there is a corresponding true claim about reasonable inference.

Consider this analogy. The relationship between the concept of entailment and the concept of reasonable inference is, in the relevant sense, like the relationship between the concept of *being a Roman* and the concept of *being Italian*: instances of the former are guaranteed to be instances of the latter. That does not mean that for every true claim about Romans there is a true claim about Italians. This could fail in particular if the concept appears in the antecedent of a conditional (much like $\vdash$ and $\approx$ do in CD and RCD respectively). For example, *if Joe is a Roman, Joe lives within driving distance of the beach* does not entail *if Joe is Italian, Joe lives within driving distance of the beach*.

The observation that constructive dilemma behaves in this unusual way for some non-classical consequence relations that extend classical validity is important and it reappears in different guises in much of the contemporary discussion of information-sensitive modals.¹⁸

Another noteworthy item—our final one—is that the dialectic I spelled out at the end of section “[On the Intransitivity of Reasonable Inference](#)” resurfaces here. The steps $(2) \Rightarrow (3)$ and $(5) \Rightarrow (6)$ both have the form of TRUE CONSEQUENT. As we

¹⁸ Some key references for the discussion of constructive dilemma in, and around, dynamic semantics: Veltman (1996); Yalcin (2012); Willer (2012); Bledin (2014, 2015, 2020); Marra (2014); Moss (2015); Charlow (2019); Goldstein (2019).

know, TRUE CONSEQUENT is not a reasonable inference. Stalnaker acknowledges this (fn. 17) and claims that the steps are reasonable in the Strawsonian sense (i.e. according to $\approx^S$). As we know from Facts 9–10 this is only true if we stipulate ANTECEDENT COMPATIBILITY. In fact, IC's account of the idle argument *requires* this particular combination ($\approx^S$ and ANTECEDENT COMPATIBILITY). Given the essential role of that combination in capturing CONTRAPOSITION we likely should characterize it as the "official" version of IC's pragmatics.

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Appendix

Fact 1 *Given the framework, PRAGMATIC CONSTRAINT entails C-ACCEPTANCE*

Proof Suppose that c is incompatible with $\neg A \ \& \ \neg C$ and c is compatible with $\neg A \ \& \ C$. Now consider any world w in c . We must show that the closest $\neg A$ -world to w is a C -world. By the pragmatic constraint, the $\neg A \ \& \ C$ -world v that is in the context set must be closer to w than any $\neg A \ \& \ \neg C$ world. Now, because Stalnaker's selection functions are grounded by an ordering of similarity, the closest $\neg A$ -world is either the closest $\neg A \ \& \ C$ -world or the closest $\neg A \ \& \ \neg C$ -world. But it cannot be the latter, so it must be the closest $\neg A \ \& \ C$ -world. $\square$

Fact 2 *If entailment is defined as preservation of truth at a world and a context, then $A \vee C \not\models \neg A \rightarrow C$.*

Fix c and w so that $A \vee C$ is true at w and c but also so that the context c contains some $\neg A \ \& \ \neg C$ -world. (Note that I assume that disjunction is Boolean in the sense that the truth-conditions of $A \vee C$ at w and c only depend on w .) So c is not incompatible with $\neg A \ \& \ \neg C$, which would make $\neg A \rightarrow C$ false. $\square$

Fact 3 *However, $\rightarrow$ satisfies C-ACCEPTANCE.*

Fix c and w . Suppose (i) $c \cap |\neg A| \cap |\neg C| = \emptyset$ and (ii) $c \cap |\neg A| \cap |C| \neq \emptyset$. Note that (i) alone is enough to guarantee $\neg A \rightarrow C$ at every world within c which is what needed to be shown. $\square$

Fact 4 *if $\Sigma \models C$ then $\Sigma \approx C$*

Fix a c in which every member of Σ is assertible. Let c_Σ be the context resulting from the sequential update of c with every premise in Σ . Let $|\Sigma|$ denote the set $\bigcap \{|A| \mid A \in \Sigma\}$. Because $c_\Sigma \subseteq |\Sigma|$ and $|\Sigma| \subseteq |C|$, $c_\Sigma \subseteq |C|$, which is what we needed. $\square$

Fact 5 *if A presupposes C , then $A \approx C$*

Suppose c is an arbitrary initial context in which A is assertible. Since the assertibility conditions for A include C , we must have $c \subseteq |C|$. Since $c_A \subseteq c$, we must have $c_A \subseteq |C|$. $\square$

Fact 6 (Intransitivity) *Let Σ be a set of sentences. $\Sigma \approx B$ and $B \approx C$ do not guarantee $\Sigma \approx C$.*

[proven in the main text]

Fact 7

- (i) $\neg A \approx \neg A \vee C$
- (ii) $\neg A \vee C \approx A > C$ but
- (iii) $\neg A \not\approx A > C$

[proven in the main text]

Fact 8 *The reasonable-inference analogue of CONTRAPOSITION fails. (That is: $A > C \not\approx \neg C > \neg A$).*

This failure can be illustrated along the same lines as the failure of part (iii) of Fact 7. Let c be a context in which $A > C$'s presuppositions are satisfied. Suppose that c only contains C -worlds, but also that there are some $\neg C$ -worlds in the background model. Suppose however that all such $\neg C$ -worlds are A -worlds. Let c' be the result of updating c with $A > C$. Now, consider the evaluation of $\neg C > \neg A$ in c' : since there are no $\neg C$ -worlds in c' , the selection function must reach outside of it to find some $\neg C$ -worlds. However, such worlds are A -worlds, so c' does not accept $\neg C > \neg A$. $\square$

Fact 9 *If the presupposition of conditionals is ANTECEDENT COMPATIBILITY, then CONTRAPOSITION, TRUE CONSEQUENT and FALSE ANTECEDENT are all Strawsonian Reasonable Inferences.*

Let c be a context in which $A > C$'s presuppositions are satisfied. Let c' be the result of updating c with $A > C$. Because of Stalnaker's commitment to strong centering (if w makes A true, w is the closest A -world to itself), part of the effect of the update from c to c' is to rule out all the $A \& \neg C$ -worlds.

Now suppose c' satisfies the presuppositions of $\neg C > \neg A$. This must mean that c' contains a $\neg C$ -world v . But we just argued that the $A \& \neg C$ -worlds have been excluded in the transition from c to c' . So v must be a $\neg A \& \neg C$ -world, and indeed this must be the case for any $\neg C$ -world in v . Since the worlds in c' are to be considered closer than the worlds out of c' , v (or some world relevantly like it) must be the closest $\neg C$ -world. So the closest $\neg C$ -world is a $\neg A$ -world.

This same argument, however, will go through for TRUE CONSEQUENT and FALSE ANTECEDENT. $\square$

Fact 10 *If the presupposition is just PRAGMATIC CONSTRAINT, then none of CONTRAPOSITION, TRUE CONSEQUENT and FALSE ANTECEDENT count as Strawsonian Reasonable Inferences.*

The countermodel we constructed in establishing Fact 8 vacuously satisfies PRAGMATIC CONSTRAINT in the updated context. (Similarly for analogous countermodels for TRUE CONSEQUENT and FALSE ANTECEDENT.) $\square$

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Chapter 21

On von Stechow's "Comparing Semantic Theories of Comparison"



Cécile Meier

Abstract The syntax and semantics of expressions used to represent comparison is a central topic in linguistic theory. Arnim von Stechow was interested in how these expressions interact with other linguistic expressions and what the entailment relations are between sentences that contain expressions of comparison. In "Comparing Semantic theories of Comparison", Von Stechow evaluates several proposals for a formal analysis of comparison constructions still current in the early 1980s. The basis for his impressive evaluation is a precise translation of the most important existing accounts into a uniform formal language. Only this enables him to compare the descriptive adequacy of the proposals. Even if one of the reviewed approaches is not able to make the correct predictions, von Stechow discusses strategies to repair it. The proposals are tested against a fixed set of phenomena. In fact, von Stechow demonstrates in an exemplary way how puzzling phenomena may play the same role in linguistics as experiments do in physics. The phenomena help us to test linguistic hypotheses and theories. Von Stechow's conclusion is that a Russellian, i.e. a scopal theory, is correct for comparison constructions. Comparative complements are in fact nominals: the *than*-phrase stands in for a definite description of a degree.

Keywords Comparative · Scope · Maximality · Intensionality

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Introduction

Comparative constructions are used in order to describe relations between individuals with respect to attributes of the individuals like length or beauty, or our preferences for individuals. The comparisons establish an order and we may even start assigning numbers to the individuals with shared measures so that the order on the numbers reflects the order of the individuals. That is, we may start constructing scales like the height scale or a beauty scale or a scale for our preferences. Furthermore, we may want to calculate with the numbers, expressing that some individual has twice as much of a property (like length, for example) as another individual. Expressing statements with a comparative is the empirical basis for measurement theory (Krantz et al., 1971).

The syntax and semantics of expressions used to represent comparison is a central topic in linguistic theory. von Stechow (1984a) was interested in how these expressions interact with other linguistic expressions and what the entailment relations are between sentences that contain expressions of comparison. In “Comparing Semantic Theories of Comparison”, von Stechow evaluates several proposals for a formal analysis of comparison constructions still current in the early 1980s. The basis for his impressive evaluation is a precise translation of the most important existing accounts into a uniform formal language. Only this enables him to compare the descriptive adequacy of the proposals. Even if one of the reviewed approaches is not able to make the correct predictions, von Stechow discusses strategies to repair it.

The proposals are tested against a fixed set of phenomena. In fact, von Stechow demonstrates in an exemplary way how puzzling phenomena may play the same role in linguistics as experiments do in physics. The phenomena help us to test linguistic hypotheses and theories.

Von Stechow’s conclusion is that a Russellian, i.e. a scopal theory, is correct for comparison constructions. Comparative complements are in fact nominals: the *than*-phrase stands in for a definite description of a degree. A comparative like (1a) and the respective construction with descriptions in (1b) basically mean the same thing.

- (1) a. Mary is taller than John.
- b. Mary exceeds (in height) the height of John.

The definite description participates in scope interactions with other operators and, in particular, it can take intermediate scope as predicted for definite descriptions in general. Contra Russell (and more in line with Sharvy (1980) and Link (1983) for plurals and mass nouns), von Stechow argues that the definite article in the paraphrase should be enriched by a maximalization operation. The formal account makes use of degrees: The height of John is the description of a value (or number) on a suitable scale, for example. Following Cresswell (1976), von Stechow assumes that degrees have metaphysical reality.

The paper provoked a number of responses both from authors whose papers were criticized and from others, and von Stechow's view is only complete with his reaction to these responses von Stechow (1984b). All those papers were published 1984 in the first two issues of the third volume of the *Journal of Semantics*. Understanding the issues in the papers, and especially von Stechow's in its full range, requires some familiarity with intensional semantics: how attitude reports are interpreted, how counterfactuals and modals work, and some willingness is needed to follow the arguments concerning the application of the method of so-called double-indexing.

We will present the main results of von Stechow's theory in section "The Main Results": (a) his theoretical assumptions in analyzing the comparative and (b) his evaluation of other approaches at the time. Section "Critical Comments" contains a collection of criticism by other scholars of von Stechow's theory. The debate circles around four types of questions: (a) how the comparative construction interacts with binding phenomena, (b) whether movement is involved in the derivation of comparative clauses and what exactly moves, (c) questions involving the compositionality of comparative constructions, i.e., where (and even when) the *than*-phrase attaches, and (d) how the comparative interacts with logical operators. Section "Impact" is concerned with further developments in the field.

The Main Results

The main results are twofold: First, von Stechow proposes a new analysis for capturing comparatives and explaining how they interact with operators in the same sentence; see section "A Theory for Comparatives". Second, he introduces a series of arguments: (i) against the delineation approach (the idea that comparative interpretation is based on the interpretation of positive form as proposed by Lewis and Klein), (ii) against the method of double-indexing (Kamp), (iii) for the use of a maximality operator in the semantics of the comparative, see section "Evaluation". Von Stechow also proposed methods of interpretation for the positive form, equatives, and result clause constructions. These types of constructions are (almost completely) set aside in my overview.

A Theory for Comparatives

As far as the inventory of denotations is concerned, von Stechow assumes that there are degrees in addition to the standard types of denotations in possible-world semantics. The set of semantic types is augmented by the basic type *d*. Degrees can be conceived of in different ways: sets of objects, numbers on a scale or extents, i.e., intervals of numbers on a scale. Let us introduce the proposal by looking at the analysis of an example, as in (2).

(2) Mary is at least two inches taller than John.

The basic lexicon has to specify entries for John and Mary (and other proper names).¹

- (3) a. $\llbracket \text{Mary} \rrbracket^w = \text{Mary}$
 b. $\llbracket \text{John} \rrbracket^w = \text{John}$

Measure phrases that serve as differentials are captured as existential quantifiers over degree predicates (von Stechow, 1984a, p. 56). Von Stechow ignores the internal composition of measure phrases.

- (4) $\llbracket \text{at least 2 inches} \rrbracket^w(P) = 1$ iff there is a degree d at least as long as 2 inches such that that degree d satisfies P in w .

The comparative form *taller* is assumed to consist of two morphemes: one for the adjective *tall* and one for the comparative morpheme *-er*, which are interpreted separately. In fact, comparative forms sometimes occur in two words as in *more beautiful*, for example. The periphrastic form is taken to be the underlying form. Gradable adjectives are relations between individuals and degrees in von Stechow's system. The truth conditions for *tall* are captured as in (5) (See von Stechow, 1984a, 5).

- (5) $\llbracket \text{tall} \rrbracket^w(d)(x) = 1$ iff d is a degree of tallness and x has d in w .

The lexical entry contains information about the kind of order that is involved in assigning a degree to an individual and it states that the individual has that degree. If degrees are just equivalence classes established by ordering individuals by their tallness, then to have a degree means for an individual to belong to a certain equivalence class.

The comparative morpheme *MORE* that may be realized by *-er* or *more* combines with a degree that is specified by a measure phrase, an adjective, and another degree that is specified by the *than*-phrase, in addition to an individual. Von Stechow's 1984 comparative morpheme is a four-place relation as defined in (6).

- (6) $\llbracket \text{MORE} \rrbracket^w(d1)(A)(d2)(x) = 1$ iff $d1 + d2$ is an A -degree and x has $d1 + d2$ in w .

The underlying syntactic structure of our sentence in (2) is more complex than first meets the eye. The *than*-phrase is a reduced version of an underlyingly sentential structure called Complementizer Phrase (CP) more recently.² It contains a *wh*-element in the position of the degree argument of the adjective as represented in (7a), following Chomsky (1977).³ The CP is moved to the left periphery of the comparative complement. The comparative morpheme precedes the adjective (as it

¹ The system of denotations we will be using is the same as the one introduced in chapter 11 in Heim and Kratzer (1998).

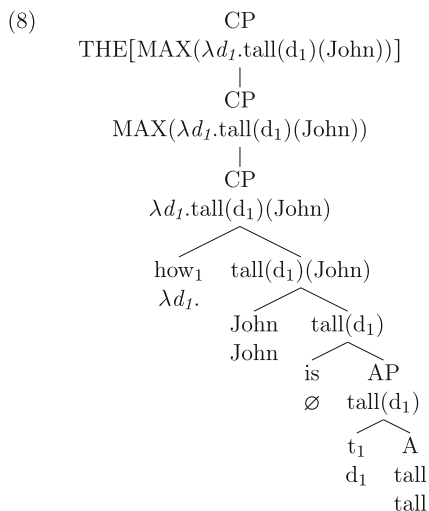
² Compare Heim and Kratzer (1998, p. 88) on relative clauses.

³ The alternative to the clausal analysis would have been a so-called direct analysis (see Lechner, 2019, for the full picture).

does in the periphrastic form) and combines first with the measure phrase, then with the adjective and only then with the *than*-phrase. The comparative morpheme and the *than*-phrase do not form a constituent in this system. The measure term *at least two inches* as well as the comparative complement *than John* move at the level of LF as represented in (7b). Elided material is set in angled brackets. (7b) is the input for interpretation.

- (7) a. [Mary is [_{AP} [_{NP} at least two inches] -er tall] [_{CP} than <how₁> John is <t₁ tall>]]]
 b. [_{CP} than <how₁> John (is) <t₁ tall>]₃ [_{NP} at least two inches]₂ [Mary is [_{AP} t₂ MORE tall] t₃]

The *than*-phrase as such denotes a degree predicate. Its interpretation is analogous to the interpretation of relative clauses (see also Heim and Kratzer, 1998, Chapter 5): the *wh*-element introduces predicate abstraction over the degree trace that is left behind as represented in (8). Before the degree predicate may enter the interpretation, it undergoes an operation of maximalization, i.e., it is reduced to a function that characterizes a singleton, a set that contains only the largest/highest degree. Moreover, the predicate is shifted to a nominal quantifier with *THE* in von Stechow's system.



MAX is in fact a degree modifier, i.e. type $\langle dt, dt \rangle$, as defined in (9) (see von Stechow, 1984a, p. 37).

- (9) $\llbracket \text{MAX} \rrbracket^w(P)(d) = 1$ iff $P(d) = 1$ and there is no other d^* such that $P(d^*) = 1$ and d^* exceeds d .

The definite article *THE* is a quantificational determiner that relates the maximalized degree predicate and another property of degrees in the familiar way proposed by Russell (see von Stechow, 1984a, p. 55).

- (10) $\llbracket \mathbf{THE} \rrbracket^w(P)(Q) = 1$ iff there is an x such that (i) for any y the following holds: if $P(x) = 1$, then x equals y , and vice versa (uniqueness and existence) and (ii) $Q(x) = 1$.

The maximalization operation so to speak ensures that the uniqueness condition imposed by the definite article is met. The definite article could be replaced by an indefinite article with no change in meaning of the whole. This strategy seems unnecessary complicated from today's perspective. Today, many researchers use a maximality operator that selects a single element from a set of degrees like a choice function. But this more recent method would turn the degree expression into an individual term that is scopally inert. The more recent method is more Fregean in style for the interpretation of a definite expression. Von Stechow's *than*-phrase may enter into scope relations with other operators.

With the syntactic analysis and the basic lexicon in place, we may calculate the truth conditions of our example sentence. The quantifier denoted by the *than*-phrase is applied to the denotation of the rest of the sentence as represented in (11). (Following Heim and Kratzer (1998) we assume that a movement index is subscripted to the respective binder. And we follow usual practice in using lambdas for the meta-language as well in order to abbreviate denotations of functions.)

- (11) $\llbracket \mathbf{THE} \rrbracket^w(\llbracket \mathbf{MAX}(\text{than } \langle \text{how}_1 \rangle \text{ John is } \langle \text{t}_1 \text{ tall} \rangle) \rrbracket^w)$
 $(\llbracket \lambda_3 \text{ at least 2 inches } (\lambda_2 \llbracket \text{Mary is } [\text{AP } \text{t}_2 \text{ MORE tall}] \text{ t}_3 \rrbracket^w) \rrbracket^w)$
 $= \llbracket \mathbf{THE} \rrbracket^w(\lambda d. \text{tall}(w)(d)(\text{John}) \ \& \ \forall d^*[\text{tall}(w)(d^*)(\text{John}) \Rightarrow d^* \leq d])$
 $(\llbracket \lambda_3 \text{ at least 2 inches } (\lambda_2 \llbracket \text{Mary is } [\text{AP } \text{t}_2 \text{ MORE tall}] \text{ t}_3 \rrbracket^w) \rrbracket^w)$
 $= \llbracket \mathbf{THE} \rrbracket^w(\lambda d. \text{tall}(w)(d)(\text{John}) \ \& \ \forall d^*[\text{tall}(w)(d^*)(\text{John}) \Rightarrow d^* \leq d])$
 $(\llbracket \lambda_3 \text{ at least 2 inches} \rrbracket^w (\llbracket \lambda_2 \llbracket \text{Mary is } [\text{AP } \text{t}_2 \text{ MORE tall}] \text{ t}_3 \rrbracket^w))$
 $= \llbracket \mathbf{THE} \rrbracket^w(\lambda d. \text{tall}(w)(d)(\text{John}) \ \& \ \forall d^*[\text{tall}(w)(d^*)(\text{John}) \Rightarrow d^* \leq d])$
 $(\lambda d'. \exists d''[d'' \geq 2 \text{ inches} \ \& \ \text{tall}(w)(d' + d'')(\text{Mary})])$
 $= 1$ iff
 $\exists d[\forall d^*[\text{tall}(w)(d^*)(\text{John}) \ \& \ \forall d^{**}[\text{tall}(w)(d^{**})(\text{John}) \Rightarrow d^{**} \leq d^*] \leftrightarrow d = d^*] \ \& \ \exists d' [d' \geq 2 \text{ inches} \ \& \ \text{tall}(w)(d' + d)(\text{Mary})]]$

Evaluation

This strategy of analysis makes the correct predictions for interpreting several phenomena discussed in von Stechow's paper. The phenomena are used as test cases for the other theories. Von Stechow looks at ten phenomena in total that a theory of comparatives has to deal with. Some phenomena concern the interaction of the comparative morpheme with other operators from propositional logic and predicate logic and can be treated in an extensional framework (c, d, e, g, h,

i). Others concern the behavior of the comparative in intensional contexts (a, b, f, j). Our presentation deviates from the order of presentation in von Stechow's paper and highlights three milestones set by the paper: section "[Movement or Double Indexing?](#)" is concerned with the arguments in favor of moving the *than*-phrase (phenomena a, b, j), section "[Uniqueness or Maximality?](#)" motivates maximality as apart of the denotation of the comparative (phenomena c, d, e, f), and section "[What Is Wrong with Delineations?](#)" argues against the analysis of comparison constructions without degrees (phenomena (h,i). Phenomenon (g) is dealt with below in section "[Additional Phenomena](#)".

Movement or Double Indexing?

The first two phenomena (a,b) deal with ambiguities that result from interpreting the comparative complement (i.e., the *than*-phrase) with respect to the actual situation in which the sentence is uttered and not with respect to the modal or attitudinal context.

(a) Russell (1905, p. 489) observed that an attitude report as in (12) has (at least) two readings, a trivial one where the speaker thinks something contradictory and an informative one where the speaker thought that the actual size of the yacht is smaller than what he thought the size is. This kind of ambiguity is named *Russell's Ambiguity* and Russell uses the example in order to illustrate that definite descriptions—as quantifiers—show scope effects in combination with verbs of attitude. Von Stechow accepts this argument.

(12) I thought your yacht was larger than it was.

Von Stechow uses a Hintikka semantics in order to analyze attitudinal expressions. In this framework verbs of attitude like *think* express universal quantifiers over possible worlds. Since the *than*-phrase expresses a definite description of the quantificational type it is subject to quantifier raising and may take narrow scope as in (13a) or wide scope as in (13b) with respect to the attitude.⁴

- 13 a. I thought [_{CP} than <how₁> the yacht was <t₁ large>]₃ DIFF₂[the yacht was [_{AP} t₂ MORE large] t₃]
 b. [_{CP} than <how₁> the yacht was <t₁ large>]₃ DIFF₂ I thought [the yacht was [_{AP} t₂ MORE large] t₃]

Think gets the semantics in (14) where $Dox(x)(w)$ represents the set of worlds that an individual x in a world w admits as possibilities for the worlds he inhabits. If a

⁴ The binder of the trace t_2 is an unpronounced existential quantifier that ensures that there is a difference in degree between the compared items. It is abbreviated *DIFF* here, and it is defined in analogy to the differential *at least* (4) above. $\llbracket \text{DIFF} \rrbracket^w(P) = 1$ iff there is a degree d greater 0 such that that degree d satisfies P in w .

proposition is true in all circumstances that the individual admits as possibilities, it can be said to be believed by that individual.

$$(14) \llbracket \text{think} \rrbracket^w(p)(x) = 1 \text{ iff } \forall w^*[w^* \in \text{Dox}(x)(w) \rightarrow p(w^*) = 1]$$

Russell's ambiguity then amounts to two interpretations: (15a) represents the narrow scope analysis of the *than*-phrase, see (13a), and (15b) its wide scope, see (13b). The two interpretations only differ with respect to the evaluation of the degree predicate contained in the *than*-phrase. In (15a) the world variable of the *than*-phrase is bound by the universal quantifier expressed by *think*; in (15b) it is not. Keep in mind that the ι notation represents a quantifier and not an individual term. In the first reading, the definite description is construed *de dicto*, and in the second, *de re*.

$$(15) \text{ a. } \llbracket (13a) \rrbracket^w = 1 \text{ iff } \begin{aligned} &\forall w^*[w^* \in \text{Dox}(I)(w) \rightarrow \\ &\quad \exists d'[d' > 0 \ \& \ \text{tall}(w^*)(d' + \iota d[\text{tall}(w^*)(d)(\text{the yacht}(w^*))])(\text{the yacht}(w^*))]] \end{aligned}$$

$$\text{ b. } \llbracket (13b) \rrbracket^w = 1 \text{ iff } \begin{aligned} &\forall w^*[w^* \in \text{Dox}(I)(w) \rightarrow \\ &\quad \exists d'[d' > 0 \ \& \ \text{tall}(w^*)(d' + \iota d[\text{tall}(w)(d)(\text{the yacht}(w))])(\text{the yacht}(w^*))]] \end{aligned}$$

(b) Postal (1974, p. 392) claims that counterfactual conditionals containing comparatives are ambiguous very much in the same way as attitude reports are. For the counterfactual in (16), at least four readings are discussed: One with a contradictory antecedent and an informative consequent and vice versa, one where both antecedent and consequent are contradictory, and one where neither of them is contradictory. In fact, only the latter reading is easily available where both comparative complements are construed *de re*. Cresswell (1976, p. 281) discusses similar examples.

(16) If Ede had smoked less (than he did), he would be healthier (than he is).

The ambiguity could be explained by different scopal positions for the interpretation of the *than*-phrase. Being a nominal it may move (see the method above). In this case, the ambiguities are simple scope ambiguities.

However, there is another formal technique to capture these phenomena: An additional operator may be introduced—first described by Kamp (1971)—within the comparative complement that basically evaluates the complement with respect to the actual world of utterance. This technique is called the double-indexing approach. Von Stechow opts for the scope solution on the basis of the phenomena presented in (j) below. But Postal endorses a double-indexing solution. The basic idea of this solution is the introduction of an operator *actually* that shifts the evaluation of the targeted predicate to the world of utterance and out of the responsibility of the intensional operator. The *than*-phrase may remain in its base position and be interpreted as an individual concept (Zimmermann, this volume). (16) may get the syntactic analysis in (17) in this approach on one reading. On the other readings only one *actually* operator modifies the comparative complement or none.

- (17) If [Ede did smoke [_{AP} DIFF LESS much] [_{CP} than <how₁> Ede did **actually** smoke <t₁ much>][Ede would be [_{AP} DIFF MORE healthy] [_{CP} than <how₁> Ede is **actually** <t₁ healthy>]]

At first sight, a double-indexing approach may fare better in these cases. The comparative complement may stay in situ and the *actually* operator shifts the evaluation of the predicate it applies to the actual world c_o . The interpretation function is dependent on two indices, as defined in (18).

- (18) $\llbracket \text{actually } p \rrbracket^{w, c_o} = \llbracket p \rrbracket^{c_o, c_o}$

But von Stechow (1984a) adds the phenomenon (j) that seems to decide on the issue of movement vs. double-indexing. Examples of iterated modality (i.e., constructions with more than one modal) constitute a classical test case for the hypothesis that an expression is a quantifier. Quantifiers may take intermediate scope between two operators. The sentence in (19) has a reading, where the speaker thinks that Plato could have been more boring than the speaker thought Plato was. The comparative complement seems to be evaluated in the scope of the attitude but outside of the scope of the modal verb. This phenomenon is used by von Stechow in order to dismiss the double-indexing approach: It is impossible to derive the intended meaning by introducing an actuality operator because the actuality operator only allows shifts of evaluation to the world of utterance. Therefore, the double-indexing approach cannot handle more than two readings. (19) may have an intermediate reading in addition to the wide scope reading and the narrow scope reading.

- (19) I thought Plato could have been more boring.

Nevertheless, the scopal treatment of the ambiguity has been criticized. See section "[Binding Issues](#)".

Uniqueness or Maximality?

The phenomena (c, d, e), examples with connectives and quantifiers in the comparative complement, and (f), one that involves intensions, are used in order to argue that a maximalization operation is part of the meaning of comparatives. That maximality, as proposed by von Stechow, is part of the meaning of comparatives is widely acknowledged nowadays. Maximalization is added to a theory of comparatives that is rooted in Russell's proposal where unique degrees are compared.

(c) The occurrence of negative polarity items (nominal quantifiers) in the *than*-phrase was first mentioned in Ross (1968, p. 294) in the literature on formal grammar. Consider (20).

- (20) a. Ede is cleverer than *any one* of us.
b. Max is as well as *ever*.

These items were originally thought to be licensed only in contexts that contain a negation or some other element that has a negation as a semantic component (see

also Postal, 1974). One of the accounts criticized by von Stechow, Seuren (1973), explicitly takes this phenomenon as evidence that an analysis for comparatives that contains a negative element is correct. The negative polarity items would be licensed syntactically; see Alrenga and Kennedy (2014) for a more sophisticated version of that idea. Von Stechow, however, adopts the proposal by Ladusaw (1979) that negative polarity items are licensed in downward-entailing contexts, a semantic property; see Gajewski, this volume. Russell's account cannot handle these examples. The maximalization operation in von Stechow's account ensures downward-entailingness of comparatives and therefore explains the occurrence of these items.

(d) Cresswell (1976, p. 270) mentioned the behavior of connectives and quantifiers in the *than*-phrase, in general. In combination with comparatives these elements may trigger interesting entailments. A comparative with a disjunction or a negative polarity item (=an existential quantifier) may trigger a free choice effect.⁵ For the premise in (21), the maximalization operation introduced by von Stechow picks out the maximal degree the nicer one of Düsseldorf or Stuttgart has. The conclusion remains true.

(21) Konstanz is nicer than Düsseldorf or Stuttgart.

Therefore: Konstanz is nicer than Düsseldorf and Stuttgart.

For the premise in (22), the maximalization operation picks out the maximal degree of fatness realized by one of the speaker's group, which ensures the conclusion to be true (provided the universal quantifier is interpreted with wide scope).

(22) Ede is fatter than any one of us [= someone of us].

Therefore: Ede is fatter than everyone of us.

(e) Negative quantifiers in the *than*-clause render comparatives and equatives in English unacceptable; see Cresswell (1976, p. 270) and already Lees (1961, p. 175). These sentences are ruled out on semantic grounds in von Stechow's account: The maximal degree such that nobody has it is undefined (23a). Similarly, the maximal degree such that Bill does not have it on the scale of *intelligent* is not locatable (23b). This argument rests on the assumption that the negative quantifier does not move.

(23) a. *Ede is more intelligent than no one of us.

b. *Ede is as intelligent as Bill is not.

(f) Von Stechow adds phenomena with existential modals in the *than*-clause to the discussion that were argued to have quantificational force as well.⁶ (24) seems to

⁵ The fact seems to have been observed by R. E. Jennings in an unpublished manuscript in 1969 called "Punctuation and the Truth Functionality".

⁶ But it seems that Seuren (1979) had already discovered that there is an ambiguity, as pointed out to me by Hotze Rullmann.

mean that the height of a polar bear could exceed all possible heights of grizzly bears, even the maximal degree that a grizzly bear could have.

(24) A polar bear could be bigger than a grizzly bear could be.

Some scholars questioned the tacit assumptions on the scopal behavior of the connectives, negation and quantifiers in the comparative complement: Negative polarity items and negative quantifiers are interpreted with narrow scope in von Stechow's approach, while all other quantifiers must have wide scope with respect to the comparative relation in order to make correct predictions. The connectives show ambiguities. Schwarzschild and Wilkinson (2002) for example reject the possibility of scoping any quantifier out of the *than*-phrase.

The presence of the maximality operator, however, is widely accepted. Only Pinkal (1989) notes a counterexample to maximality (25). The square root of two is irrational and can be approximated by rational numbers like $\frac{3}{2}$, $\frac{17}{12}$, $\frac{99}{70}$,... by the Babylonian Method, for example, where the approximations slowly decrease in value. All of those fractions are smaller than 2 and represent more and more decimal places of $\sqrt{2}$ correctly. But, it is impossible to represent $\sqrt{2}$ by a crisp rational number. And, evidently there is no maximal lower approximation of $\sqrt{2}$ in the set of rational numbers.

(25) 2 is greater than any rational lower approximation of $\sqrt{2}$. (Pinkal, 1989 ex. 67)

What Is Wrong with Delineations?

The types of constructions (h, i) confirm the view that the objects compared are degrees (more abstract entities) and not just individuals or sets of individuals. Based on these data, von Stechow opts for a degree based account of comparatives.

The alternative delineation approach takes the positive form of the adjective as the basis for constructing lexical entries for the comparative. The main insight is that gradable adjectives in the positive form are vague. An adjectival attribute may vary with respect to the extent to which it is true of different objects. This variation establishes an order on the objects that can be used for the interpretation of the comparative form.

There are different kinds of approaches to vague predicates. It is sometimes assumed that sentences get *degrees of truth values* between perfectly true and perfectly false and these degrees of truth values may be compared in comparatives (as in Fuzzy Logic, Cintula et al., 2017).

Another approach, the delineation approach, assumes that the meaning of a sentence containing a vague positive adjective can be made precise by eliminating borderline cases of application for the adjective step by step. This process of elimination is called precisification. If a sentence remains true under all possible precisifications of the meaning of the adjective, the sentence counts as supertrue (van Fraassen, 1971). And a predication by a gradable adjective may be captured as

the sets of precisifications such that the predication remains true under each of them. Informally, the sentence in (26) may get the paraphrase in (27) in this approach.

(26) Mary is taller than John.

(27) “*Mary is tall* is true under all precisifications of the predicate *tall* under which *John is tall* is true and *Mary is tall* is true under some precisification of the predicate *tall* under which *John is tall* is not true.”

The main arguments against a delineation approach as proposed by Lewis (1970), Kamp (1975), Klein (1980) are constructions with measure phrases, negative polarity items and a certain type of clausal comparative called comparative subdeletion. In order to explain negative polarity occurrence in the comparative complement, proponents of a delineation approach may use a syntactic explanation (see above), where the comparative morpheme is a licensing operator.

(h) Stechow’s examples for comparative subdeletion are usually brought forward in order to argue that the source of comparatives is clausal in nature; see Bresnan (1973) for the first proposal and Kennedy (2002), for example, for discussion. Von Stechow uses it as evidence for a case where degrees (i.e., values of measure functions) are necessary for interpretation. (28) does not mean that an individual is compared to itself. Rather, the measures of the attributes of tallness and breadth of the individual are compared. Sentence (28) should get the informal paraphrase in (29) in the delineation approach. But this paraphrase is absurd and it is evident that von Stechow’s proposal has no problems dealing with it.

(28) Ede is taller than he is broad.

(29) “*Ede is tall* is true under all precisifications of the predicate *tall* under which *Ede is broad* is true and *Ede is tall* is true under some precisification of the predicate *tall* under which *Ede is broad* is not true.”

(i) Stechow’s examples with differentials and factor phrases point in the same direction. Since these expressions can be combined with comparatives and equatives, this shows that measures (or numbers) must play a role sentence internally in the interpretation of comparatives.

(30) a. John is at least 2 inches smaller than Mary.

b. Ede is twice as fat as Angelika.

The two approaches, the delineation approach and the degree based account, still coexist in different versions; see Klein (1991) for a reply to von Stechow. More recently Van Rooij (2011), Cobreros et al. (2012) and Burnett (2015) revived the delineation approach and they proposed a new line of solutions for the comparative subdeletion problem raised by von Stechow by comparing not individuals directly, but rather pairs of individuals and predicates.

Critical Comments

Note that von Stechow's account was much more detailed than anything around at the time (maybe with the exception of Cresswell, 1976). Nevertheless it was criticized.

Binding Issues

The movement account of von Stechow has been criticized. Hoeksema (1983, 1984) first noted that the analysis leads to scope paradoxes in cases where the *than*-phrase gets a true transparent reading (widest scope) and contains a pronoun that is bound by a quantifier that has narrow scope with respect to a verb of attitude (narrower scope). Scoping the *than*-phrase out of the attitudinal scope of *expect* in (31) makes the contained pronoun not bindable for the universal quantifier *every schoolboy*. Note that it is possible that the evaluation of the *than*-phrase may be shifted to the expectations of the speaker's group or to the actual world of utterance.

- (31) We expect that every schoolboy_{*i*} thinks that he_{*i*} is brighter than he_{*i*} is.
[Hoeksema, 1984, ex. 62]

Heim (1985, p. 14) offers a similar sentence to Hoeksema's and leaves the problem unresolved. Consider (32). If the *than*-phrase takes wide scope with respect to the conditional, the pronoun contained in it cannot be bound anymore.

- (32) If at least one member of the team_{*i*} were faster than he_{*i*} is, we could win the game. [Heim, 1985, ex. 20]

Von Stechow cannot capture this sentence. He acknowledges that in von Stechow (1984b) and draws attention to so-called B  uerle sentences (B  uerle, 1983). (33) is an instance of such a sentence. The point of this example is that there are two verbs of attitude and two quantifiers. Both quantifiers may take narrow scope with respect to the embedded attitude (of George) but still the restriction of the universal quantifier may be evaluated with respect to what Max thinks. The sentence may express: "George holds a belief about a set of books which are the books by Chomsky according to Max's belief only" (von Stechow, 1984b, p. 193). This construction constitutes a scope paradox, as well, like Hoeksema's example. The universal quantifier should have wide scope with respect to *believe* and narrow scope at the same time.

- (33) Max thinks that George believes that no student knows every book by Chomsky.

There have been several solutions proposed for B  uerle sentences in the meantime; see Percus (2001), Keshet (2010) and von Fintel and Heim (2011) for an overview. They all involve some kind of theory that manages the binding options for world

variables of the restrictions of quantifiers independently of whether parts of a construction are moved or not.

These examples contradict the movement analysis at first sight. But they do not show in principle that a movement solution (or any other kind of scope solution) explaining the interaction of the comparative and intensional operators is not necessary.

Ellipsis

There are other interesting phenomena that are best analyzed if the comparative construction, in fact, involves some movement. Most notably, it is possible to observe Antecedent Contained Deletion with comparative constructions Heim (2001), and Antecedent Contained Deletion is usually used as a test case for quantifier movement. In elliptical constructions, the elided material should be identical to overt material. But in (34) the elided part is contained in the part that should serve for the resolution of missing material.

(34) John [_{VP} was climbing higher trees than₁ Bill < was climbing t₁ high trees >].

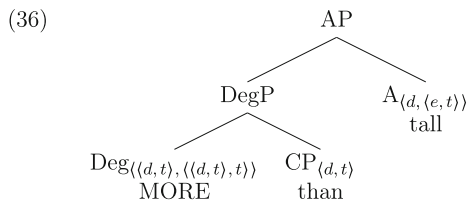
Bhatt and Pancheva (2004) opt for a solution where the Degree operator *MORE* is restricted by the *than*-phrase and is moved to the right periphery. The *than*-phrase is introduced by late merge.

(35) John [_{VP} was climbing t₁ high trees [_{DegP} MORE than₂ Bill < was climbing t₂ high trees >]₁].

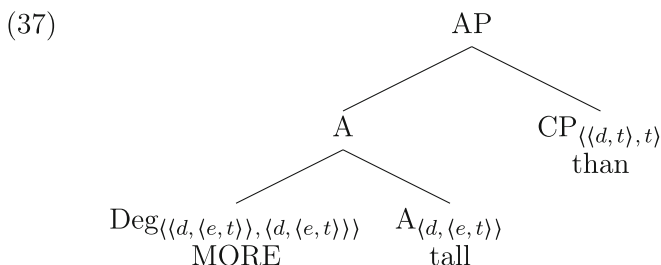
So it seems that a solution for the Bäuerle kind of sentences with comparatives by world variable management combined with a suitable movement analysis is on the right track. And it seems necessary to decompose adjectives in the comparative at a deeper level into the adjectival meaning part and the comparative morpheme *MORE*. The *than*-phrase acts as a complement to the comparative morpheme *MORE*.

Questions of Compositionality

Questions of compositionality concern the external syntax of the AP. Cresswell's and von Stechow's approaches differ with respect to the attachment site of the *than*-phrase. In Cresswell's approach the comparative morpheme is a quantificational determiner relating functions that characterize sets of degrees. The *than*-phrase is conceived as the restriction of the determiner in analogy to nominal quantificational determiners like *every* as illustrated in (36).



Von Stechow, on the other hand, gives quantificational force to the *than*-phrase alone, as illustrated in (37).⁷



Both approaches share the property that comparison constructions are in some sense quantificational and trigger movement or some other method of interpreting the quantifier in-situ. The differences in predictions may be seen with Russell ambiguities and ambiguous counterfactuals as already discussed above (see also Pinkal, 1989). Consider the example in (38).

(38) Mary claimed that Burj Khalifa is higher than it is.

This sentence seems to have three readings: a non-sensical contradictory *de dicto* one and two *de re* readings. Mary may have claimed: "Burj Khalifa is higher than 828 Meters". In this case she uttered a comparative construction. Or she may have claimed: "Burj Khalifa is 882 Meters". In the latter case, she used a measure phrase indicating how high she thinks the tower is. Both claims may be reported faithfully by (38). Von Stechow's analysis predicts that Mary implicitly uttered a comparative construction even in the latter case. But this seems implausible as Pinkal points out.

Therefore the additional *de re* reading is also a further point for movement of the complex of degree determiner (containing the comparison relation MORE) *together* with the *than*-phrase as proposed by Cresswell and Heim, for example, and Bhatt and Pancheva (2004). In their account the *de dicto* reading arises if the degree quantifier is interpreted in the scope of the attitude and the two other readings are compatible with the degree quantifier taking wide scope with respect to the attitude.

⁷ The denotation of more is simplified to a three-place relation here incorporating the contribution of a silent differential: $\llbracket \text{MORE} \rrbracket^w(A)(d2)(x) = 1$ iff there is a $d1 > 0$ and $d1 + d2$ is an A -degree and x has $d1 + d2$ in w .

Even von Stechow used a version where degree operator and *than*-phrase form a unit in later teaching material. The comparative may be read with wide scope with respect to the attitudinal verb.

But keep in mind that counterfactual ambiguities point in exactly the opposite direction, as argued by von Stechow (1984a, p. 13f.), see (16) above. The degree operator *MORE* should always remain in situ in such constructions according to von Stechow and only the *than*-phrase moves out of the scope of the conditional in his variant of the theory. But the judgments about the data are not crystal clear, as Pinkal (1989) already observes.

If-clauses are generally considered islands for movement. Moving the *than*-phrase would violate this restriction on movement. Even if there are syntactic restrictions for moving quantificational expressions out of the conditional clause, it is still possible to bind the world variables contained in the *than*-phrase independently of the conditional (Percus, 2001, investigated this matter first). The *than*-phrase may therefore stay in situ. So, it seems there is no reason to move the *than*-phrase independently from the degree determiner *MORE*, contra von Stechow.

More recent accounts also differ with respect to where the maximality operator is introduced in the compositional analysis. Maximality is an implicit modifier in von Stechow's account, and the definite article is Russellian, i.e., an existential quantifier that checks for uniqueness of its restriction. Many more recent accounts depart from this view. They assume that maximality is part of the denotation of the comparative morpheme. One of the first maximality-based accounts in this sense is Rullmann (1995) (see also Beck, 2011). Or they assume that maximality is built into the meaning of the adjective itself (Kennedy, 1999).

There is some discussion with respect to the internal structure of the *than*-phrase—almost everybody agrees with Chomsky's account for the sentential comparative clauses, which is akin to relative clause constructions. However, Kennedy (2002) proposed a matching analysis and Lechner (2004) argued that the *than*-phrase may better be analyzed as a raising relative clause construction; see Sauerland (2000) for the difference.

Logical and Other Operators

A great deal of attention has been paid to the treatment of nominal and modal quantifiers in comparative constructions and their interaction with the quantificational elements of the comparative. Kennedy (1999) sides with von Stechow that the comparative morpheme *MORE* is always interpreted with lowest scope, but Heim (2001) contributes examples from Stateva (1999) that show that the degree operator *MORE* must be able to move. Consider (39). The sentence is ambiguous between a reading where it is required that the paper be exactly 15 pages long and a reading

where the maximally authorized length of the paper is equal to 15 pages (an "at least" reading).

- (39) [This draft is 10 pages.] The paper is required to be exactly 5 pages longer than that.

A further problem is that quantifiers in the *than*-phrase do not behave uniformly. We already touched on this point when discussing negative polarity items and other quantifiers above (section "[Uniqueness or Maximality?](#)"). In general, it is problematic for nominal quantifiers to leave the *than*-phrase by quantifier raising from a syntactic point of view. The *than*-phrase is usually considered an island for movement (as well, like conditional clauses). Outscoping the *than*-phrase should not be possible then. But constructions with quantifiers seem to necessitate quantifier movement. (40) means that every girl is such that Jim is taller than she is.

- (40) Jim is taller than every girl is.

Only negative quantifiers remain *in situ* in von Stechow's analysis. Although nominal quantifiers are usually scopally variable they only sometimes show the full range of variability.

Verbal quantifiers are usually scopally fixed. Nevertheless they sometimes show scopal interaction with comparatives. A good example is the difference between *might* and *allowed*, both quantifiers having existential force. (41a) states that the actual temperature exceeds some maximally allowed temperature (narrow scope for the modal) whereas (41b) means that it might be the case that the temperature today exceeds the temperature tomorrow (wide scope for the modal). But we would expect an ambiguity.⁸

- (41) a. It is warmer in here than allowed.
b. It is warmer today than it might be tomorrow.

In addition, there seem to be cross-linguistic differences in the behavior of negative quantifiers in equatives, as already observed by von Stechow (1984b, p. 187) himself. Von Stechow proposes a similar semantics for comparatives and equatives. The construction types only differ in the kind of comparison relation introduced. Whereas the comparative morpheme is associated with a "greater than" relation, the equative is associated with a "greater than or equal to" relation. We already considered the interaction of negation with the comparative. The interaction with the equative is problematic, as well. Von Stechow can account for the unacceptability of negative quantifiers in English in comparatives and equatives.

⁸ Meier (2002) explains these differences by means of pragmatic restrictions on the interpretation of the modal. Krasikova (2008) proposes a similar explanation elaborating on Beck (2010), which was available in 2008 already in an unpublished version.

(42) *Ede is as intelligent as nobody of us.

But German allows for negative quantifiers in the *as*-clause in equatives. Consider (42). The interpretation of the maximality operator and the Russellian *iota*-operator should block the interpretation in German, contrary to intuitions.

(43) Ede ist so klug wie keiner von uns.
 Ede is as intelligent as nobody of us
 “Nobody is as intelligent as Ede.”

Penka (2016), however, discusses a proposal where equatives are not comparison constructions but correlative constructions in German.

Quantifiers and negation may further interact with the degree morpheme: The sentences in (44) and (45) are from Pinkal (1989). (44) means that every girl is at least as tall as Hans and (45) means that Hans is not taller than Karl and Hans is not taller than Peter, giving negation narrow scope in some sense. Von Stechow’s proposal predicts that (44) is true if Hans does not exceed (in height) the maximal height of the girls and (45) could be true if Hans does not exceed (in height) the height of the taller one of Karl and Peter, contrary to our intuitions.

(44) Hans isn’t taller than any girl.

(45) Hans isn’t taller than Karl or Peter.

Pinkal argued that comparative complements are free relative clauses with a universal interpretation.

However, disjunctions in comparatives are problematic from a pragmatic point of view even if unnegated. According to von Stechow, (46) may mean either that Papua New Guinea is smaller than Thailand or that it is smaller than Spain (=ignorance reading, wide scope disjunction). Or it may mean that Papua New Guinea is less big than the smaller one of the two countries (which is Spain). The latter reading is predominant if the speaker is considered competent. She probably knows both country sizes and one wonders why she should mention the bigger country if only the smaller one matters for the truth conditions. In fact such disjunctions are used if the difference in degrees between the disjuncts is irrelevant. They are in one equivalence class.

(46) Papua New Guinea is smaller than Thailand or Spain.

It is interesting to note that disjunctions in comparatives with big differences in size between the disjuncts’ degrees are odd if used by a competent speaker. Consider (47). (47) only has an ignorance reading. The other reading is lost. Von Stechow has no explanation for this difference in judgments.

(47) Papua New Guinea is smaller than Spain or the US.

Impact

Alternative Proposals

We have already mentioned some alternative proposals. One set of proposals revolves around questions of compositionality. Most notably, Heim (1985) reconsidered von Stechow's analysis and pursued an account that moves the morpheme *MORE* in addition to the *than*-phrase. Pinkal's approach to comparatives as free relatives deserves attention, as well. His account is related to Cresswell's account, capturing the comparative construction as universal quantification, and it is related to von Stechow's account in only moving the *than* phrase but not the comparative meaning component. For example, on this analysis (48) means that for all heights d such that Jim has them, there is a difference d^* such that Mary's height is $d + d^*$ (Pinkal, 1989, 1990).

(48) Mary is taller than Jim.

Whether the *than*-clause only or the comparative morpheme is scopally relevant is still discussed from time to time, see Grosu and Horvath (2006), who criticize Bhatt and Pancheva's late merge account.

Schwarzschild and Wilkinson (2002) joined the discussion of scope issues involving quantifiers of all kinds in the *than*-clause of a comparative. They propose an interval approach to comparatives. Consider (49). Informally, the sentence means that there is an interval that covers Jim's height and there is an interval that covers all the heights of the girls and the former interval is situated higher on the scale than the latter one. Schwarzschild & Wilkinson propose a theory that does not need to raise any quantifiers that are embedded within the *than*-clause.

(49) Jim is taller than every girl.

This approach is reformulated by Heim (2006) as what is called the *Pi-theory* today; see Beck (2010), Krasikova (2008), for discussion.

Alrenga and Kennedy (2014) propose that a silent negative differential degree quantifier *no* is part of the semantics of comparative constructions that surfaces in constructions like *no taller than*.

Additional Phenomena

Von Stechow's work triggered a collection of papers and dissertations on selected phenomena in the realm of comparatives. Heim (1985, 1994) is concerned with the denotation of superlatives (see also, e.g., Sharvit, 2010; Romero, 2013; Roelandt, 2016) and the references cited there. Meier (2003) proposed an analysis of excessives and result clauses (see also Hacquard, 2005).

Chomsky (1981, p. 80) mentions multi-head comparatives. The interesting question is how such sentences are constructed and what their interpretation is. Von Stechow proposes the interpretation (50b) for the sentence in (50a). The construction serves von Stechow to test his proposal.

- (50) a. More dogs ate more rats than cats ate mice
 b. “The number of dogs that ate rats is greater than the number of cats that ate mice and the number of rats that were eaten by dogs is greater than the number of mice eaten by cats.”

Further Issues

Hackl (2001) added important work on the semantics of *many* as a comparative element that is a quantificational determiner on many approaches. Lechner (1999) and subsequent work is the standard work on the interpretation of adjectives in attributive position.

More recently, classes of adjectives have been the subject of investigation. Von Stechow did not distinguish any different classes other than antonyms. More in the tradition of Bierwisch (1967), the seminal work on different kinds of adjectives in German, is the work by Kennedy and McNally (2005) on the impact of scale structure and Morzycki (2008) and Sassoon (2010) on different types of adjectives.

Beck et al. (2004) started work on other languages than just English and German, following up on remarks in Schwarzschild (2002); see also the dissertation by Bochnak (2013). The main issue is to find parameters according to which languages vary when expressing comparative constructions.

Conclusion

Constructions of comparison are a well investigated topic thanks to von Stechow’s initiative. He set the boundaries for further investigating a scope-solution within a degree-based account. He brought forward arguments against the delineation account and proposals that use double-indexing in order to explain the behavior of comparatives in the scope of intensional operators. Although his conclusions were not sufficient to explain all data (as he acknowledged himself), his observations (in combination with Bäuerle’s observations) finally led to a new view on world variable binding options and to the insight that these options are independent of issues related to quantifier movement. It is von Stechow’s merit to have set a standard for investigating comparatives to a high level of precision.

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